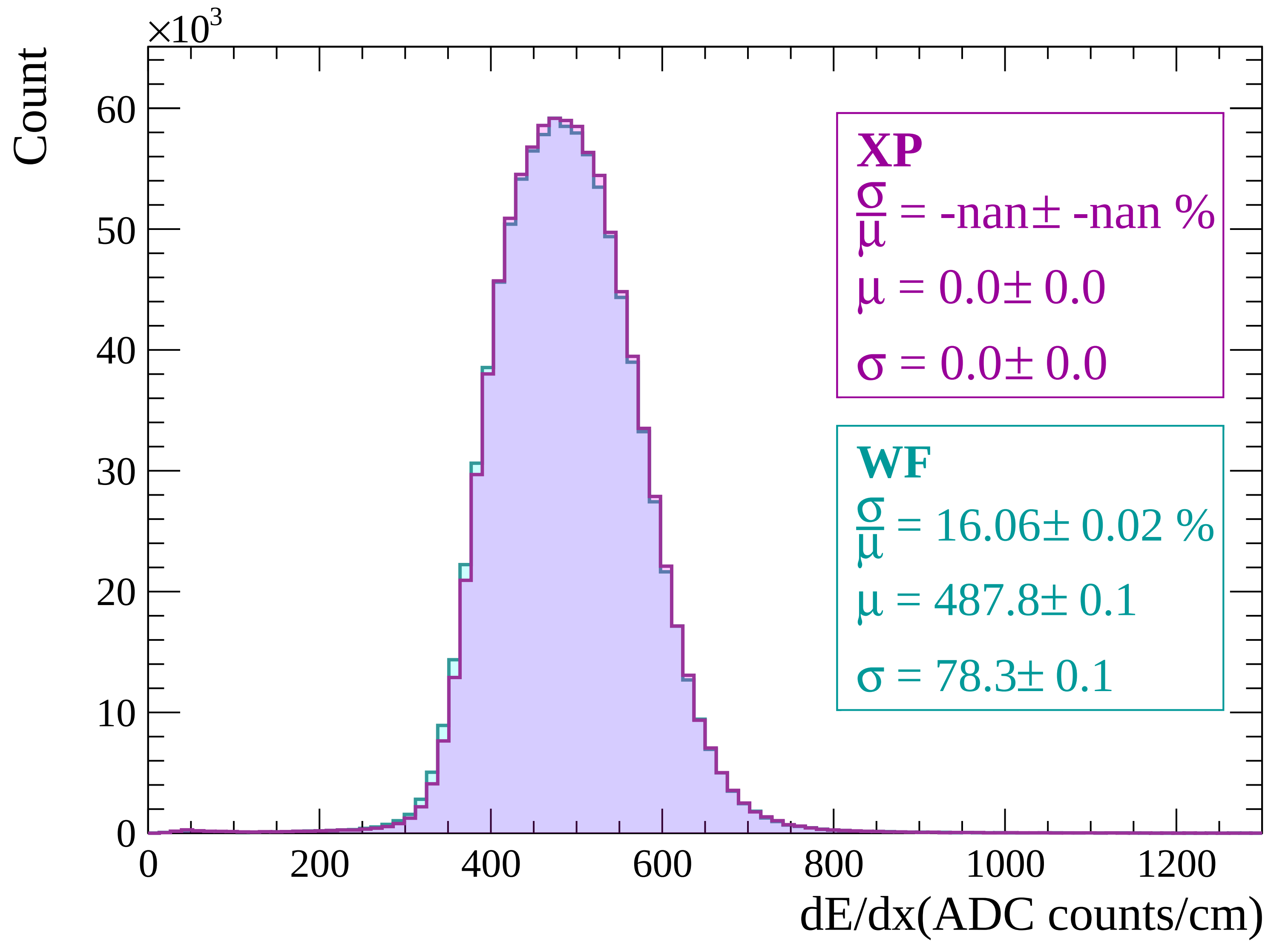
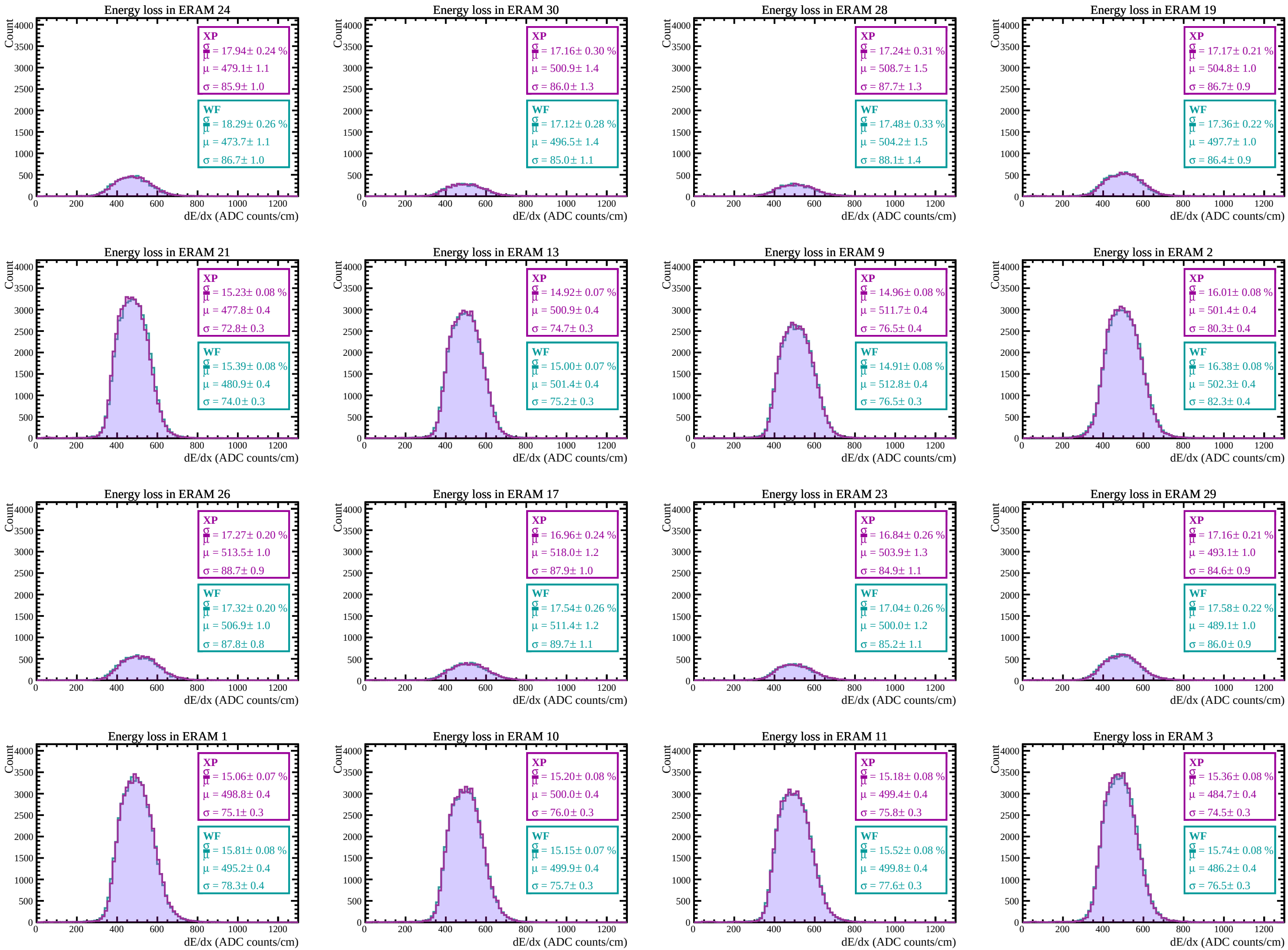
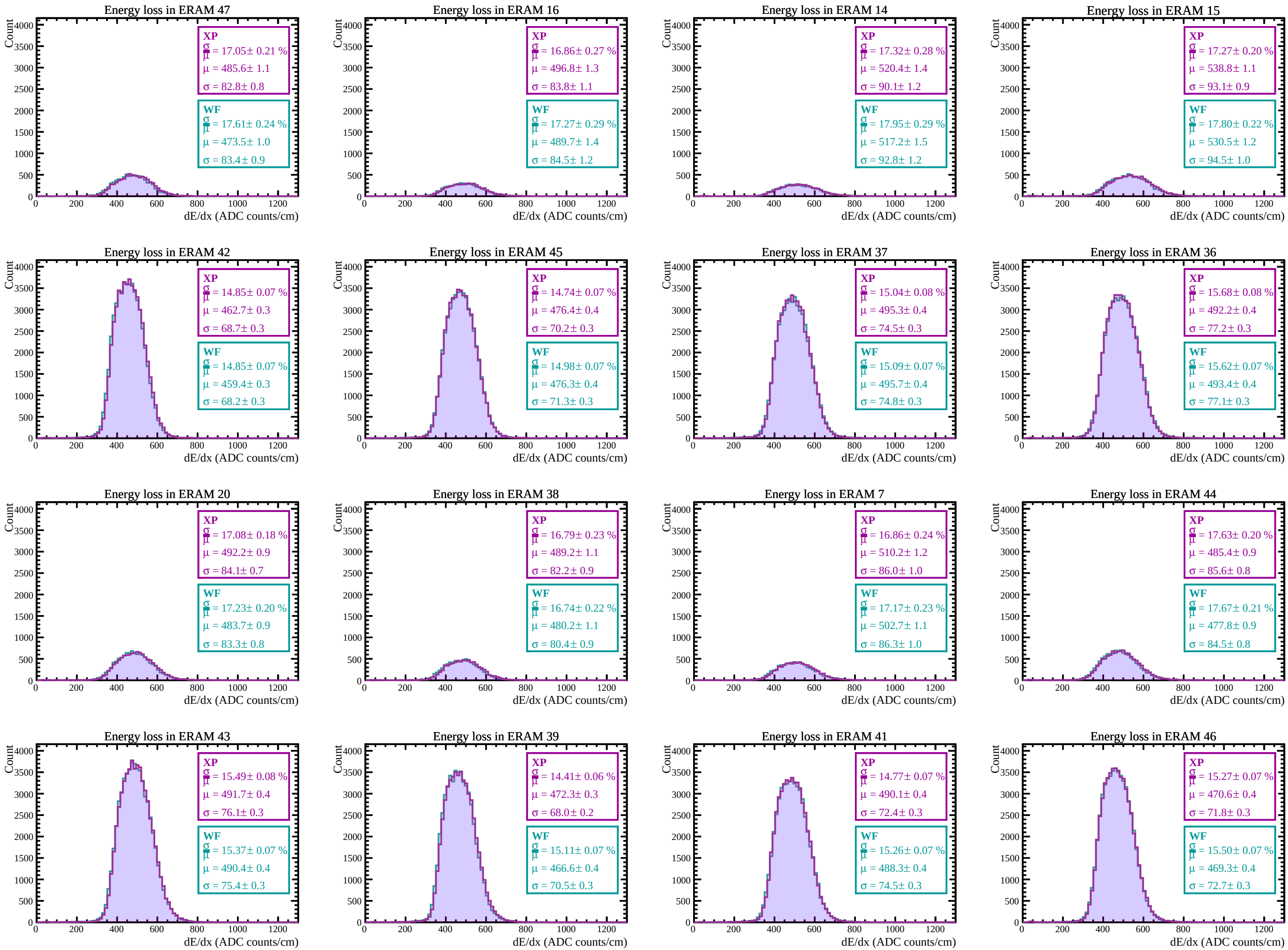
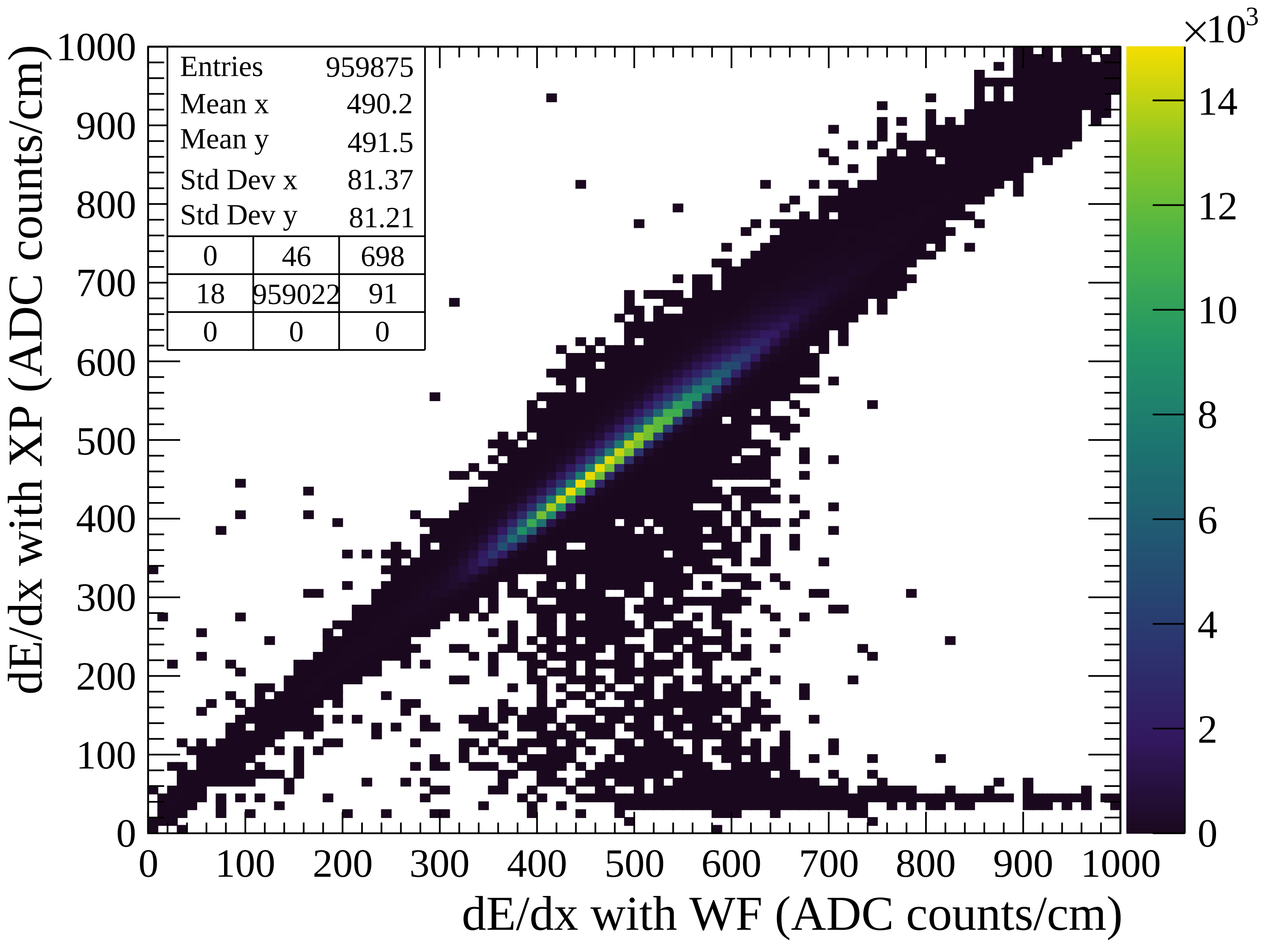


Count
×10^3
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = 16.06± 0.02 %
μ = 487.8± 0.1
σ = 78.3± 0.1

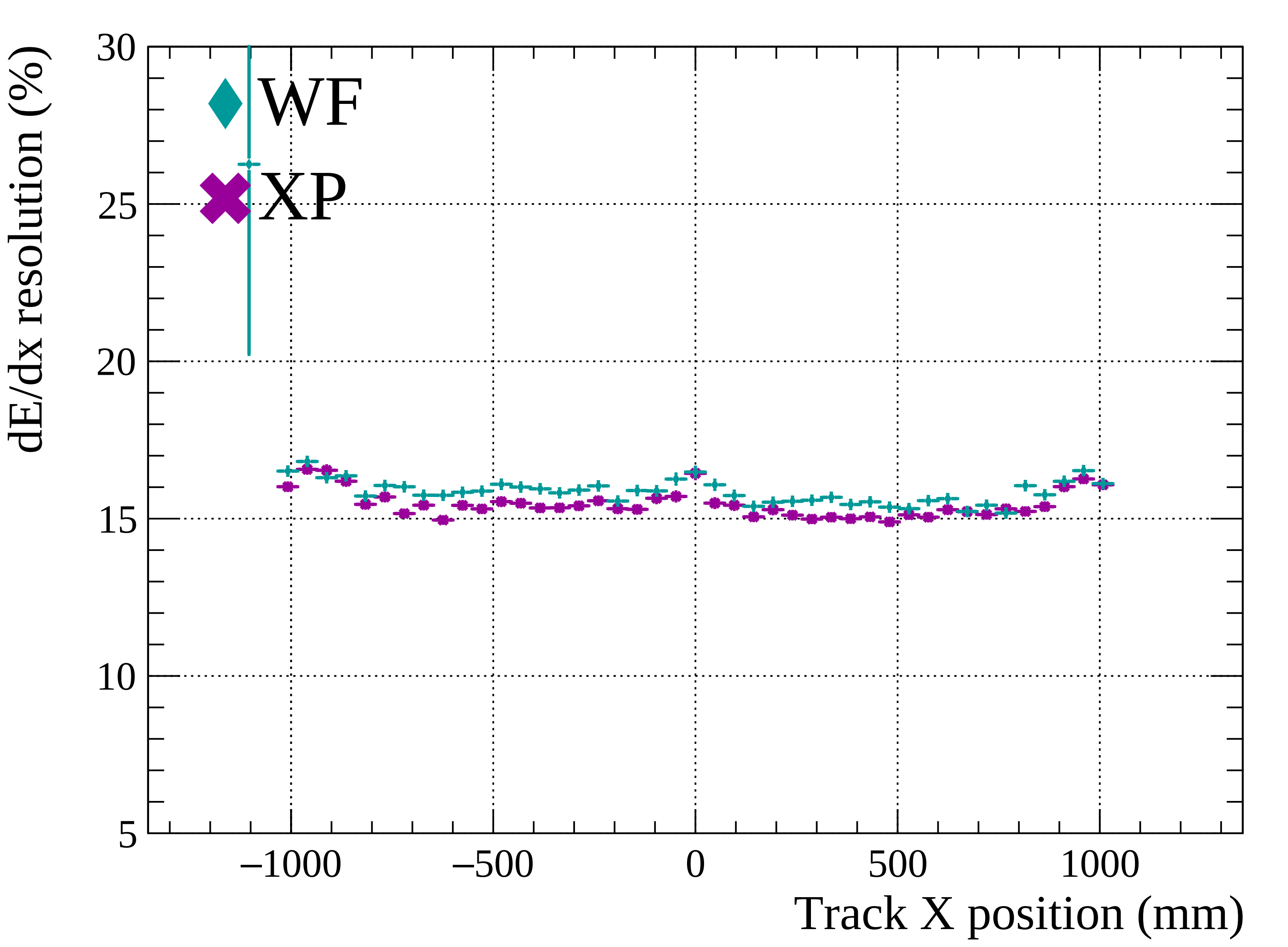


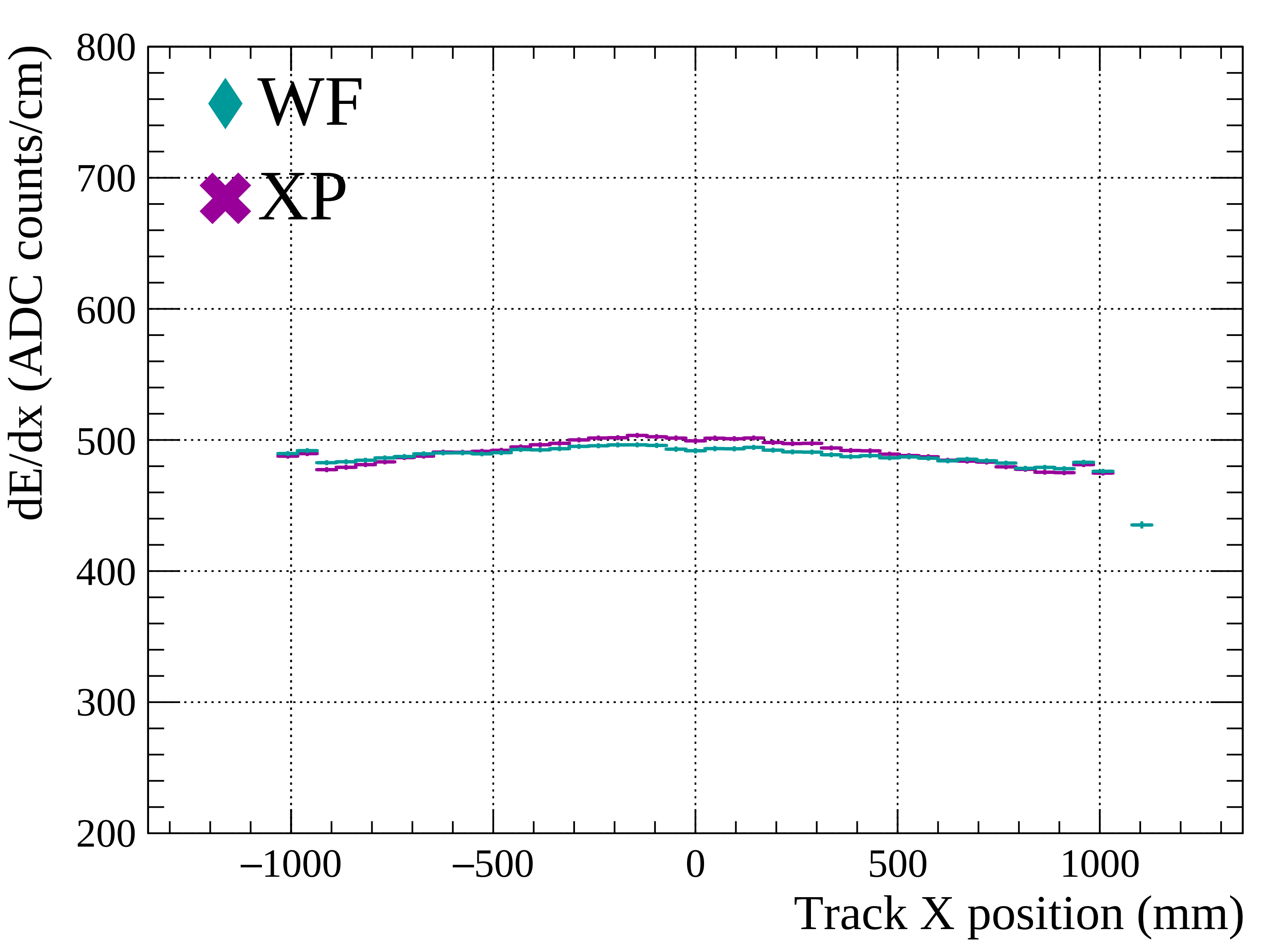
Energy loss in ERAM 24
XP σ/μ = 17.94± 0.24 %
μ = 479.1± 1.1
σ = 85.9± 1.0
WF σ/μ = 18.29± 0.26 %
μ = 473.7± 1.1
σ = 86.7± 1.0
Energy loss in ERAM 30
XP σ/μ = 17.16± 0.30 %
μ = 500.9± 1.4
σ = 86.0± 1.3
WF σ/μ = 17.12± 0.28 %
μ = 496.5± 1.4
σ = 85.0± 1.1
Energy loss in ERAM 28
XP σ/μ = 17.24± 0.31 %
μ = 508.7± 1.5
σ = 87.7± 1.3
WF σ/μ = 17.48± 0.33 %
μ = 504.2± 1.5
σ = 88.1± 1.4
Energy loss in ERAM 19
XP σ/μ = 17.17± 0.21 %
μ = 504.8± 1.0
σ = 86.7± 0.9
WF σ/μ = 17.36± 0.22 %
μ = 497.7± 1.0
σ = 86.4± 0.9
Energy loss in ERAM 21
XP σ/μ = 15.23± 0.08 %
μ = 477.8± 0.4
σ = 72.8± 0.3
WF σ/μ = 15.39± 0.08 %
μ = 480.9± 0.4
σ = 74.0± 0.3
Energy loss in ERAM 13
XP σ/μ = 14.92± 0.07 %
μ = 500.9± 0.4
σ = 74.7± 0.3
WF σ/μ = 15.00± 0.07 %
μ = 501.4± 0.4
σ = 75.2± 0.3
Energy loss in ERAM 9
XP σ/μ = 14.96± 0.08 %
μ = 511.7± 0.4
σ = 76.5± 0.4
WF σ/μ = 14.91± 0.08 %
μ = 512.8± 0.4
σ = 76.5± 0.3
Energy loss in ERAM 2
XP σ/μ = 16.01± 0.08 %
μ = 501.4± 0.4
σ = 80.3± 0.4
WF σ/μ = 16.38± 0.08 %
μ = 502.3± 0.4
σ = 82.3± 0.4
Energy loss in ERAM 26
XP σ/μ = 17.27± 0.20 %
μ = 513.5± 1.0
σ = 88.7± 0.9
WF σ/μ = 17.32± 0.20 %
μ = 506.9± 1.0
σ = 87.8± 0.8
Energy loss in ERAM 17
XP σ/μ = 16.96± 0.24 %
μ = 518.0± 1.2
σ = 87.9± 1.0
WF σ/μ = 17.54± 0.26 %
μ = 511.4± 1.2
σ = 89.7± 1.1
Energy loss in ERAM 23
XP σ/μ = 16.84± 0.26 %
μ = 503.9± 1.3
σ = 84.9± 1.1
WF σ/μ = 17.04± 0.26 %
μ = 500.0± 1.2
σ = 85.2± 1.1
Energy loss in ERAM 29
XP σ/μ = 17.16± 0.21 %
μ = 493.1± 1.0
σ = 84.6± 0.9
WF σ/μ = 17.58± 0.22 %
μ = 489.1± 1.0
σ = 86.0± 0.9
Energy loss in ERAM 1
XP σ/μ = 15.06± 0.07 %
μ = 498.8± 0.4
σ = 75.1± 0.3
WF σ/μ = 15.81± 0.08 %
μ = 495.2± 0.4
σ = 78.3± 0.4
Energy loss in ERAM 10
XP σ/μ = 15.20± 0.08 %
μ = 500.0± 0.4
σ = 76.0± 0.3
WF σ/μ = 15.15± 0.07 %
μ = 499.9± 0.4
σ = 75.7± 0.3
Energy loss in ERAM 11
XP σ/μ = 15.18± 0.08 %
μ = 499.4± 0.4
σ = 75.8± 0.3
WF σ/μ = 15.52± 0.08 %
μ = 499.8± 0.4
σ = 77.6± 0.3
Energy loss in ERAM 3
XP σ/μ = 15.36± 0.08 %
μ = 484.7± 0.4
σ = 74.5± 0.3
WF σ/μ = 15.74± 0.08 %
μ = 486.2± 0.4
σ = 76.5± 0.3
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
XP σ/μ = 17.05± 0.21 % μ = 485.6± 1.1 σ = 82.8± 0.8
WF σ/μ = 17.61± 0.24 % μ = 473.5± 1.0 σ = 83.4± 0.9
Energy loss in ERAM 16
XP σ/μ = 16.86± 0.27 % μ = 496.8± 1.3 σ = 83.8± 1.1
WF σ/μ = 17.27± 0.29 % μ = 489.7± 1.4 σ = 84.5± 1.2
Energy loss in ERAM 14
XP σ/μ = 17.32± 0.28 % μ = 520.4± 1.4 σ = 90.1± 1.2
WF σ/μ = 17.95± 0.29 % μ = 517.2± 1.5 σ = 92.8± 1.2
Energy loss in ERAM 15
XP σ/μ = 17.27± 0.20 % μ = 538.8± 1.1 σ = 93.1± 0.9
WF σ/μ = 17.80± 0.22 % μ = 530.5± 1.2 σ = 94.5± 1.0
Energy loss in ERAM 42
XP σ/μ = 14.85± 0.07 % μ = 462.7± 0.3 σ = 68.7± 0.3
WF σ/μ = 14.85± 0.07 % μ = 459.4± 0.3 σ = 68.2± 0.3
Energy loss in ERAM 45
XP σ/μ = 14.74± 0.07 % μ = 476.4± 0.4 σ = 70.2± 0.3
WF σ/μ = 14.98± 0.07 % μ = 476.3± 0.4 σ = 71.3± 0.3
Energy loss in ERAM 37
XP σ/μ = 15.04± 0.08 % μ = 495.3± 0.4 σ = 74.5± 0.3
WF σ/μ = 15.09± 0.07 % μ = 495.7± 0.4 σ = 74.8± 0.3
Energy loss in ERAM 36
XP σ/μ = 15.68± 0.08 % μ = 492.2± 0.4 σ = 77.2± 0.3
WF σ/μ = 15.62± 0.07 % μ = 493.4± 0.4 σ = 77.1± 0.3
Energy loss in ERAM 20
XP σ/μ = 17.08± 0.18 % μ = 492.2± 0.9 σ = 84.1± 0.7
WF σ/μ = 17.23± 0.20 % μ = 483.7± 0.9 σ = 83.3± 0.8
Energy loss in ERAM 38
XP σ/μ = 16.79± 0.23 % μ = 489.2± 1.1 σ = 82.2± 0.9
WF σ/μ = 16.74± 0.22 % μ = 480.2± 1.1 σ = 80.4± 0.9
Energy loss in ERAM 7
XP σ/μ = 16.86± 0.24 % μ = 510.2± 1.2 σ = 86.0± 1.0
WF σ/μ = 17.17± 0.23 % μ = 502.7± 1.1 σ = 86.3± 1.0
Energy loss in ERAM 44
XP σ/μ = 17.63± 0.20 % μ = 485.4± 0.9 σ = 85.6± 0.8
WF σ/μ = 17.67± 0.21 % μ = 477.8± 0.9 σ = 84.5± 0.8
Energy loss in ERAM 43
XP σ/μ = 15.49± 0.08 % μ = 491.7± 0.4 σ = 76.1± 0.3
WF σ/μ = 15.37± 0.07 % μ = 490.4± 0.4 σ = 75.4± 0.3
Energy loss in ERAM 39
XP σ/μ = 14.41± 0.06 % μ = 472.3± 0.3 σ = 68.0± 0.2
WF σ/μ = 15.11± 0.07 % μ = 466.6± 0.4 σ = 70.5± 0.3
Energy loss in ERAM 41
XP σ/μ = 14.77± 0.07 % μ = 490.1± 0.4 σ = 72.4± 0.3
WF σ/μ = 15.26± 0.07 % μ = 488.3± 0.4 σ = 74.5± 0.3
Energy loss in ERAM 46
XP σ/μ = 15.27± 0.07 % μ = 470.6± 0.4 σ = 71.8± 0.3
WF σ/μ = 15.50± 0.07 % μ = 469.3± 0.4 σ = 72.7± 0.3
Count
dE/dx (ADC counts/cm)

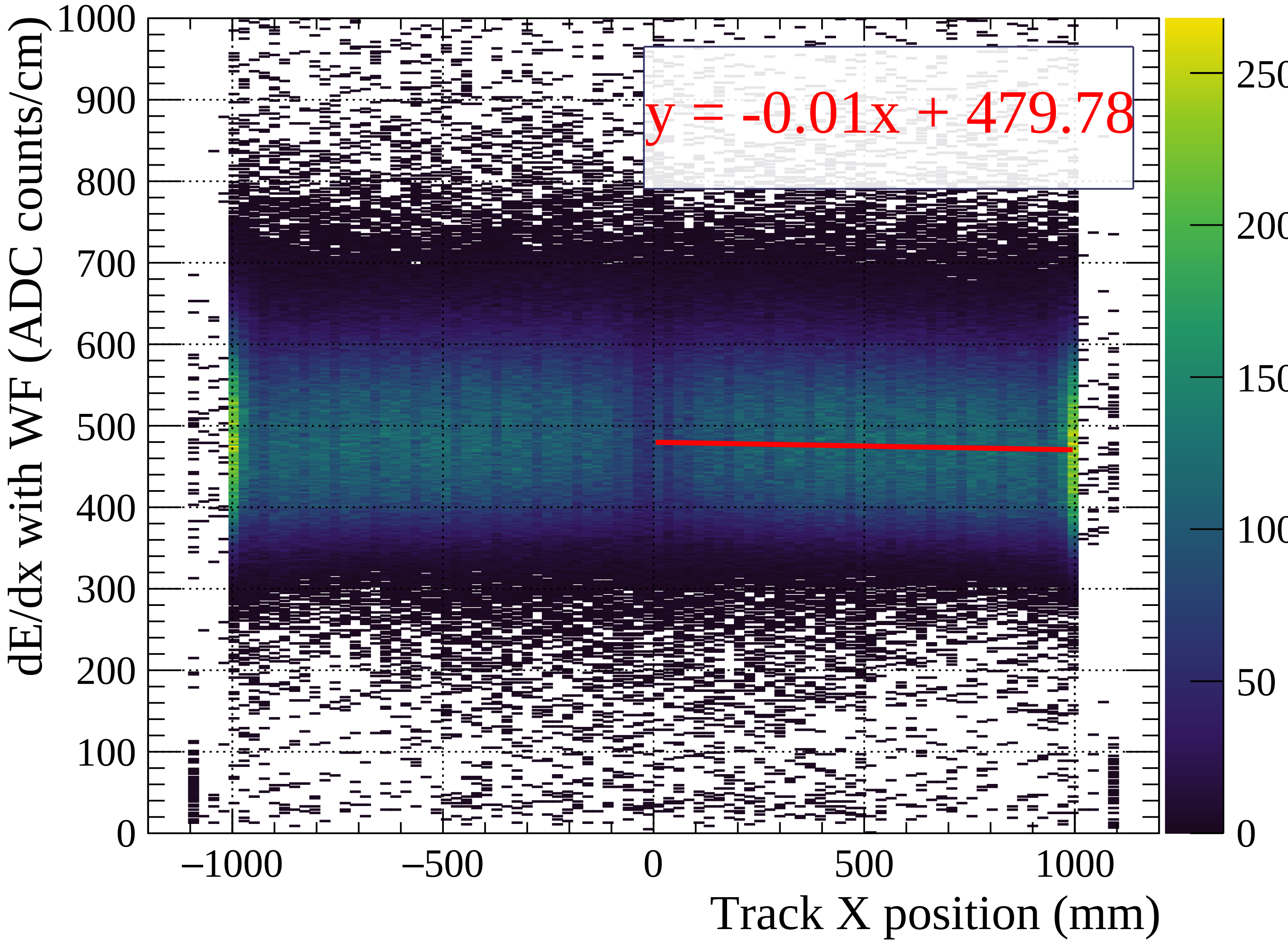


Entries 959875
Mean x 490.2
Mean y 491.5
Std Dev x 81.37
Std Dev y 81.21
0 46 698
18 959022 91
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
2
4
6
8
10
12
14
0
100
200
300
400
500
600
700
800
900
1000

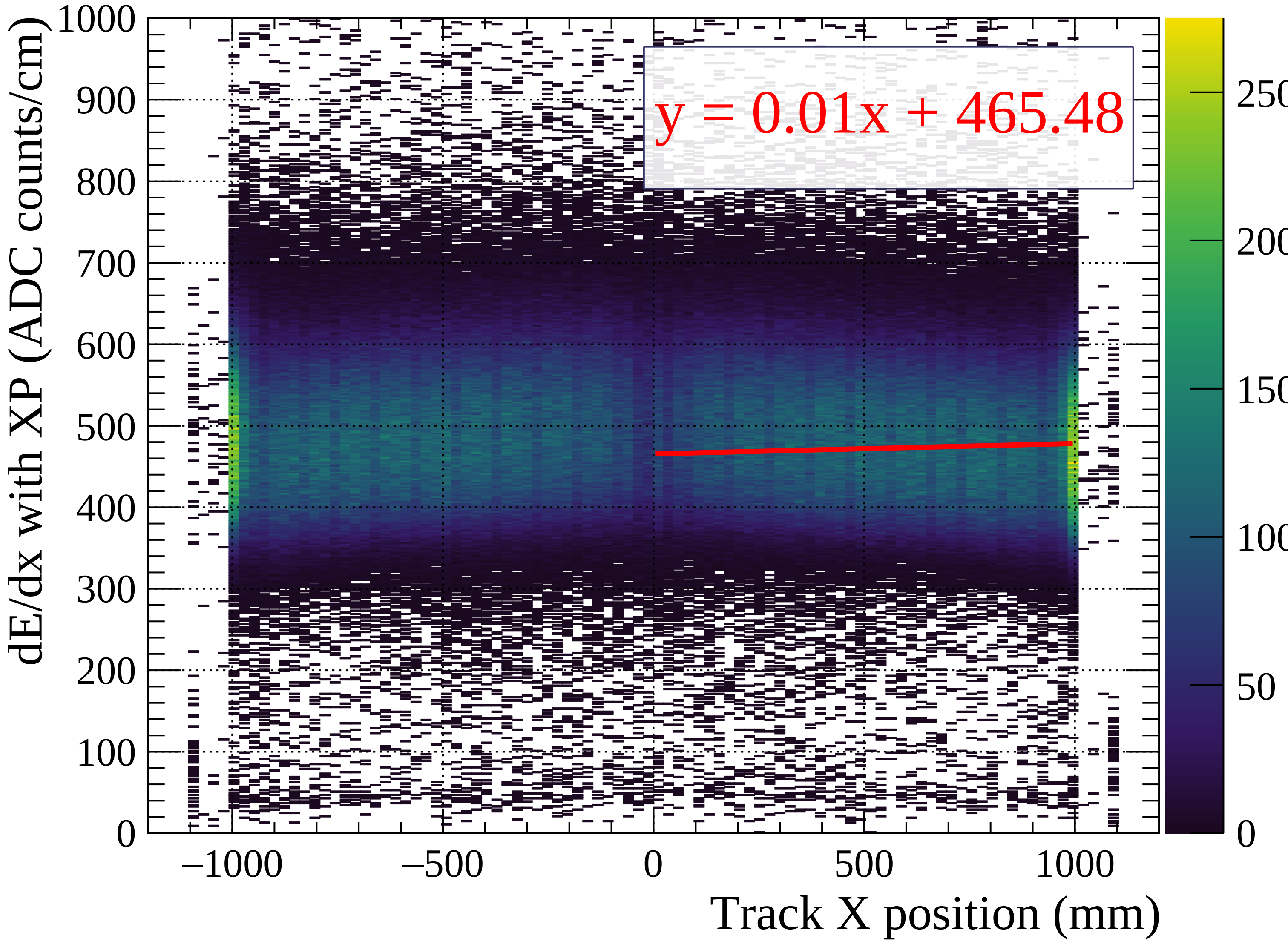

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−1000
−500
0
500
1000
Track X position (mm)



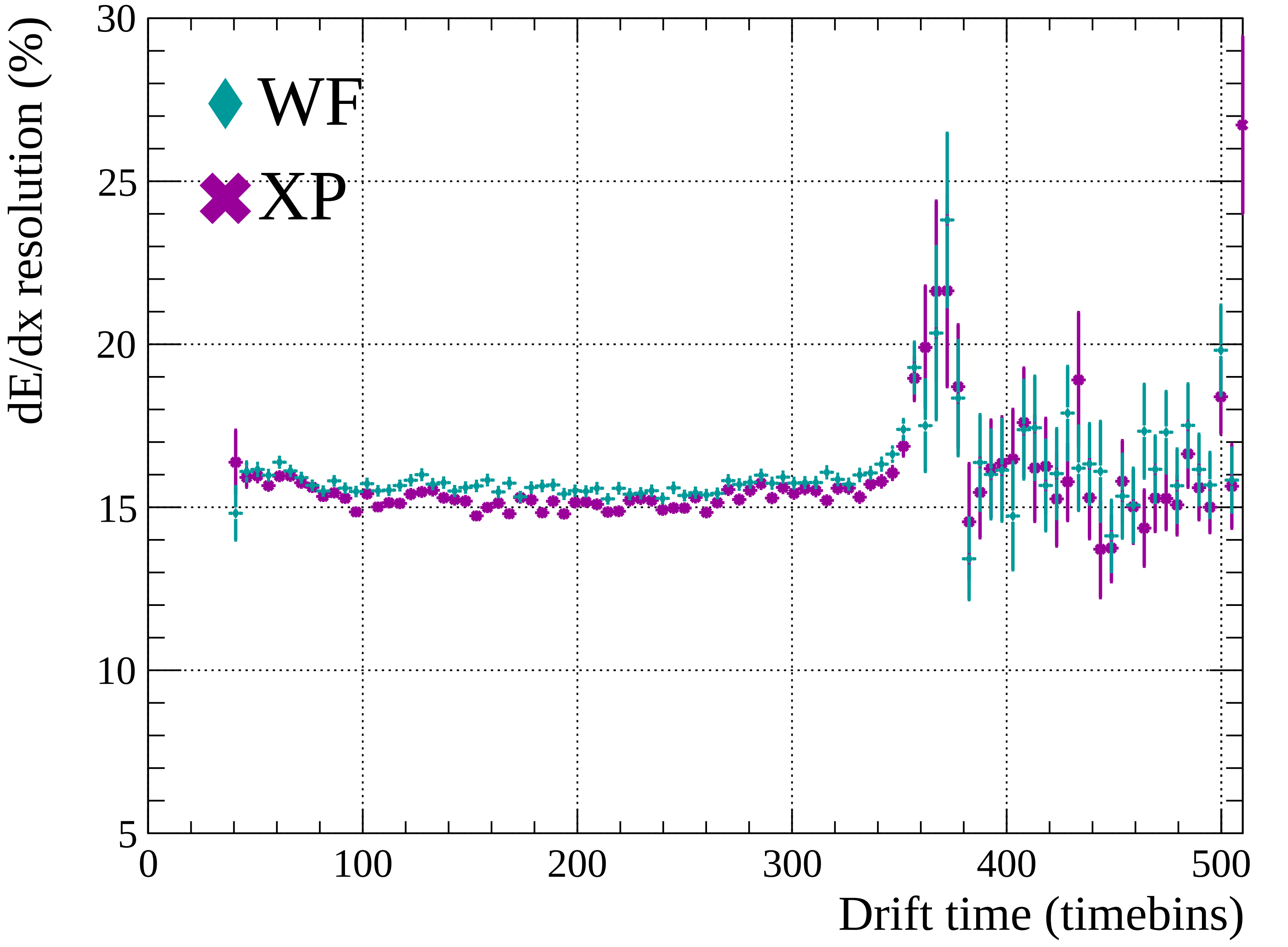
800
700
600
500
400
300
200
dE/dx (ADC counts/cm)
WF
XP
−1000
−500
0
500
1000
Track X position (mm)



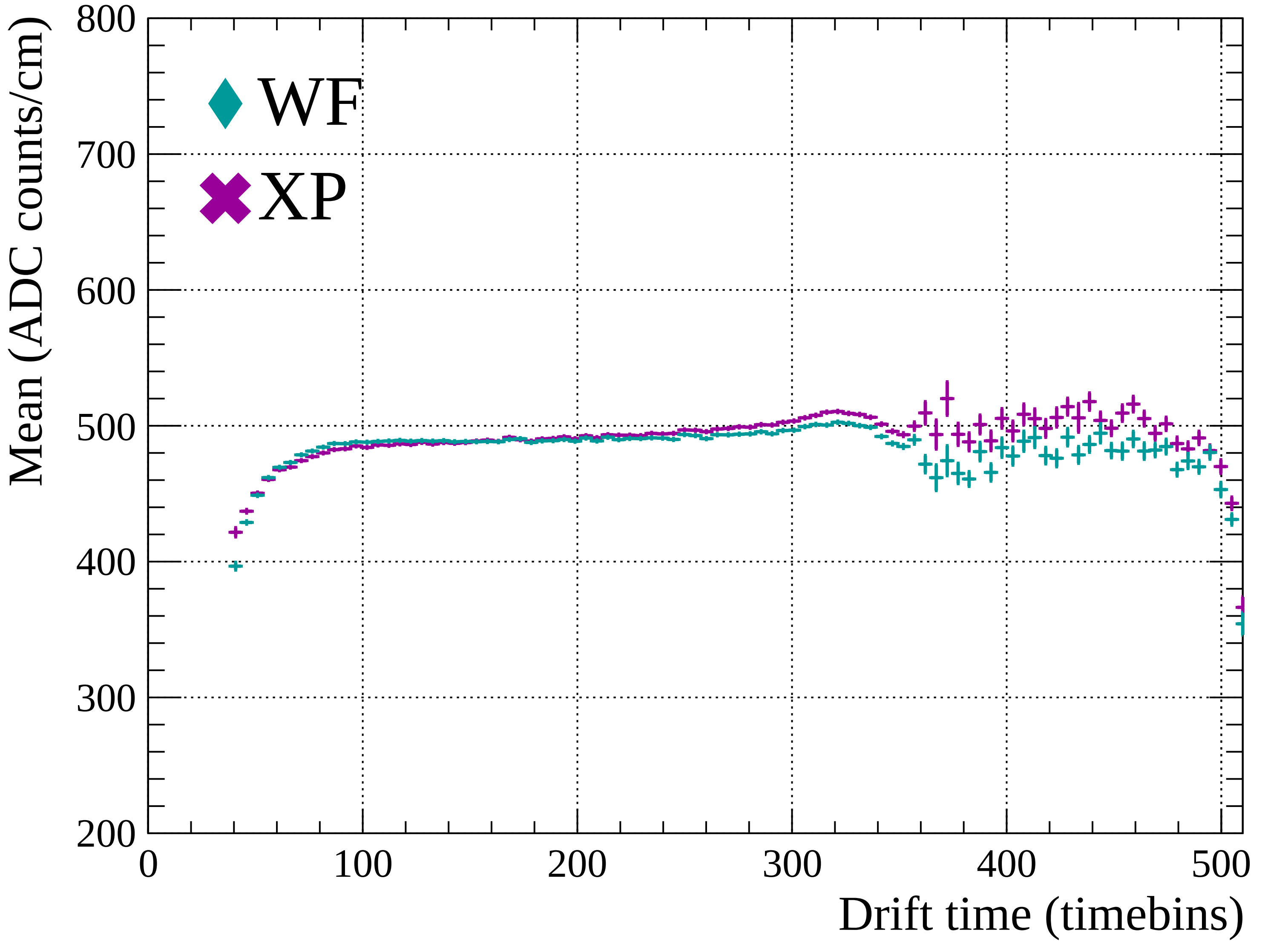
y = -0.01x + 479.78
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
50
100
150
200
250

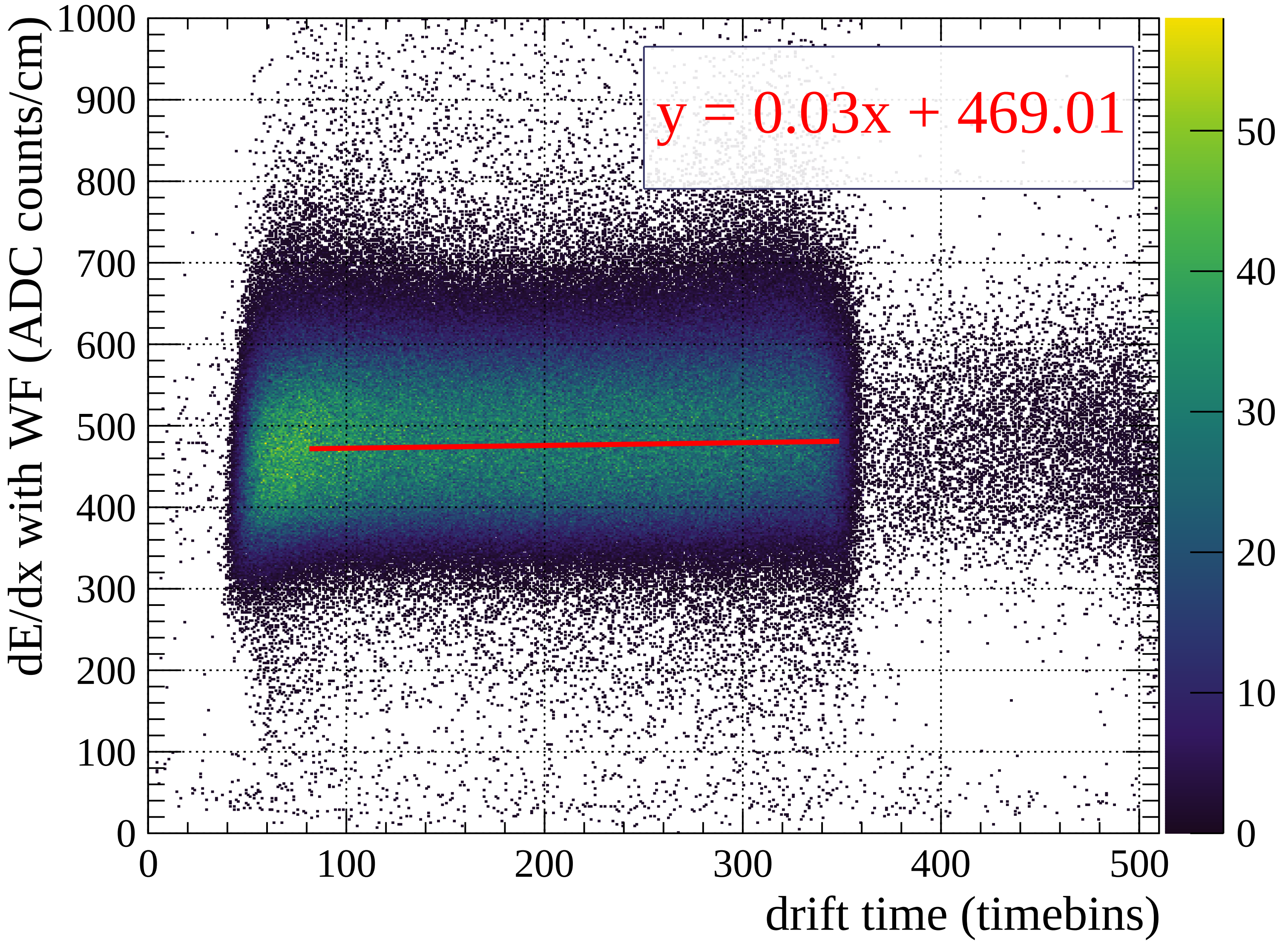


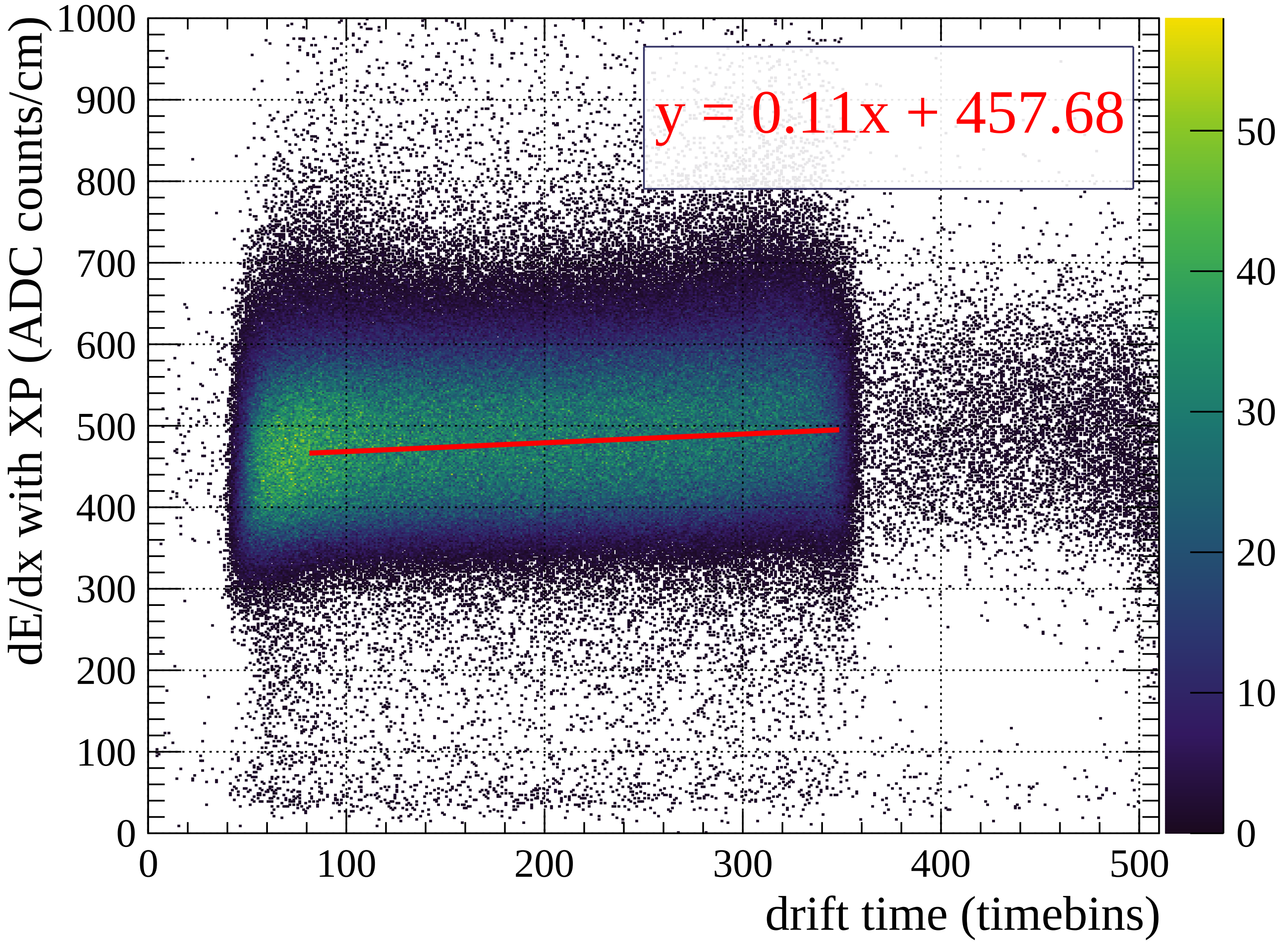
y = 0.01x + 465.48
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
50
100
150
200
250



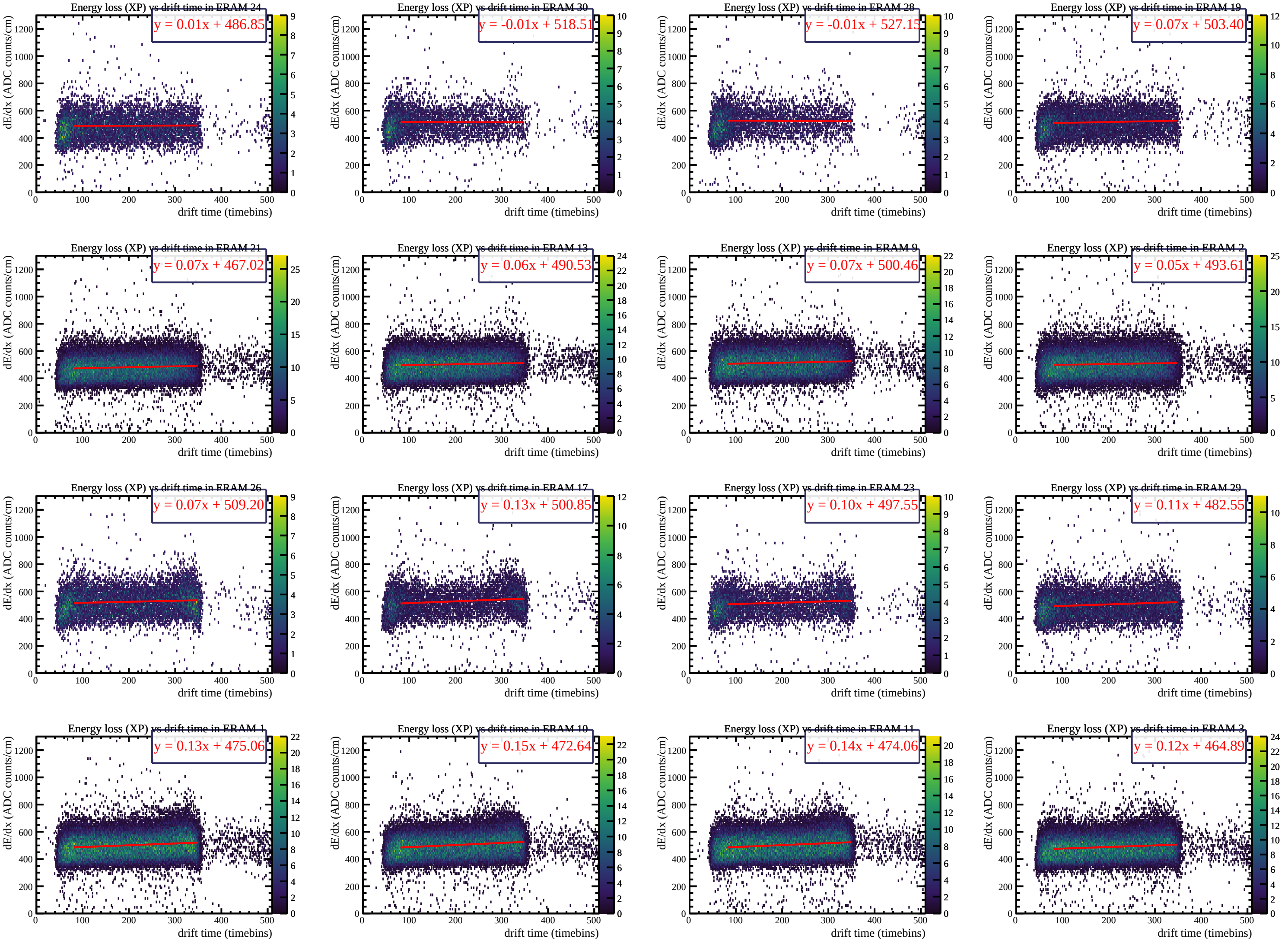
WF
XP
dE/dx resolution (%)
Drift time (timebins)
5
10
15
20
25
30
0
100
200
300
400
500

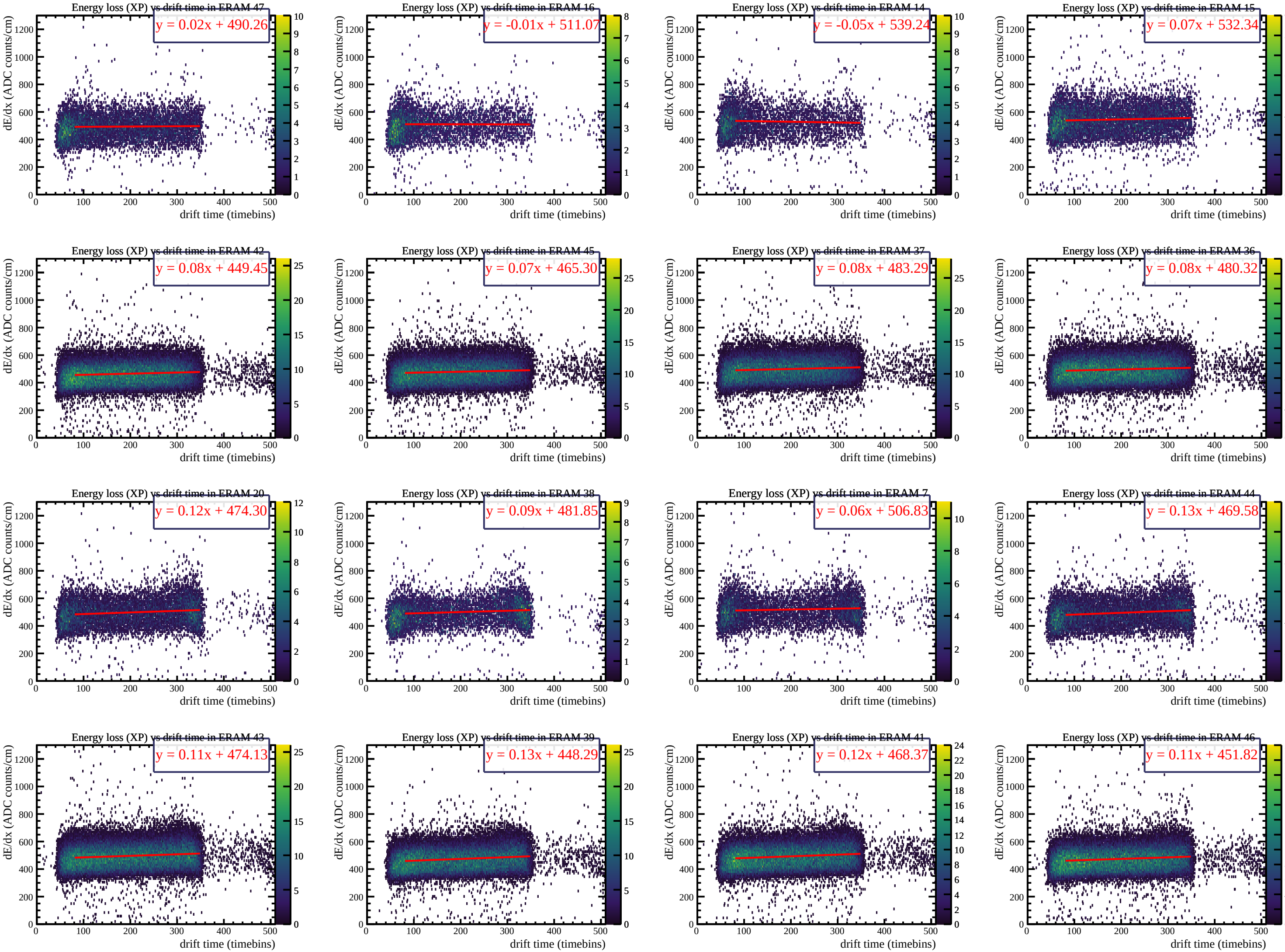

Mean (ADC counts/cm)
800
700
600
500
400
300
200
0
100
200
300
400
500
Drift time (timebins)
WF
XP

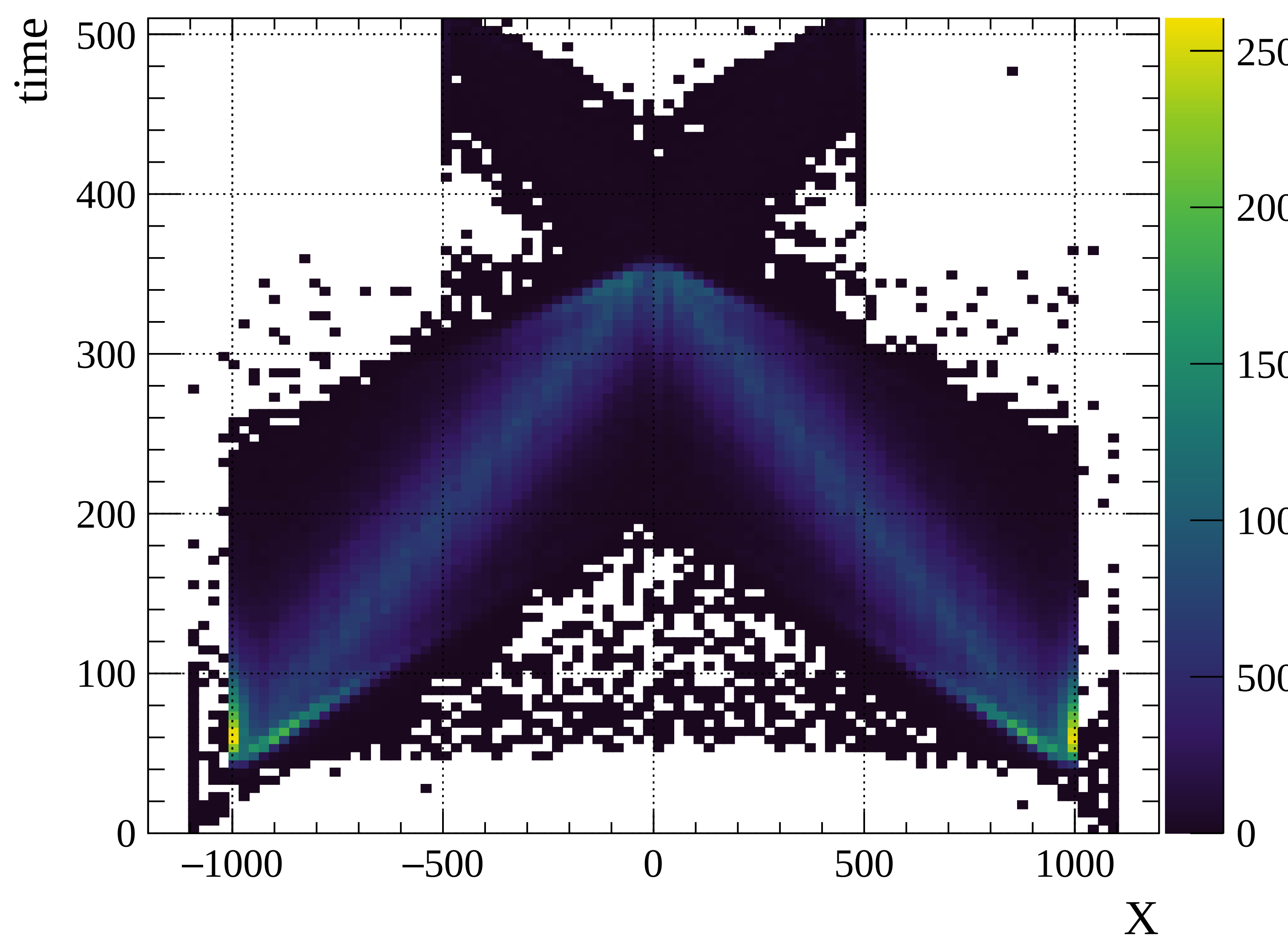

y = 0.03x + 469.01
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50



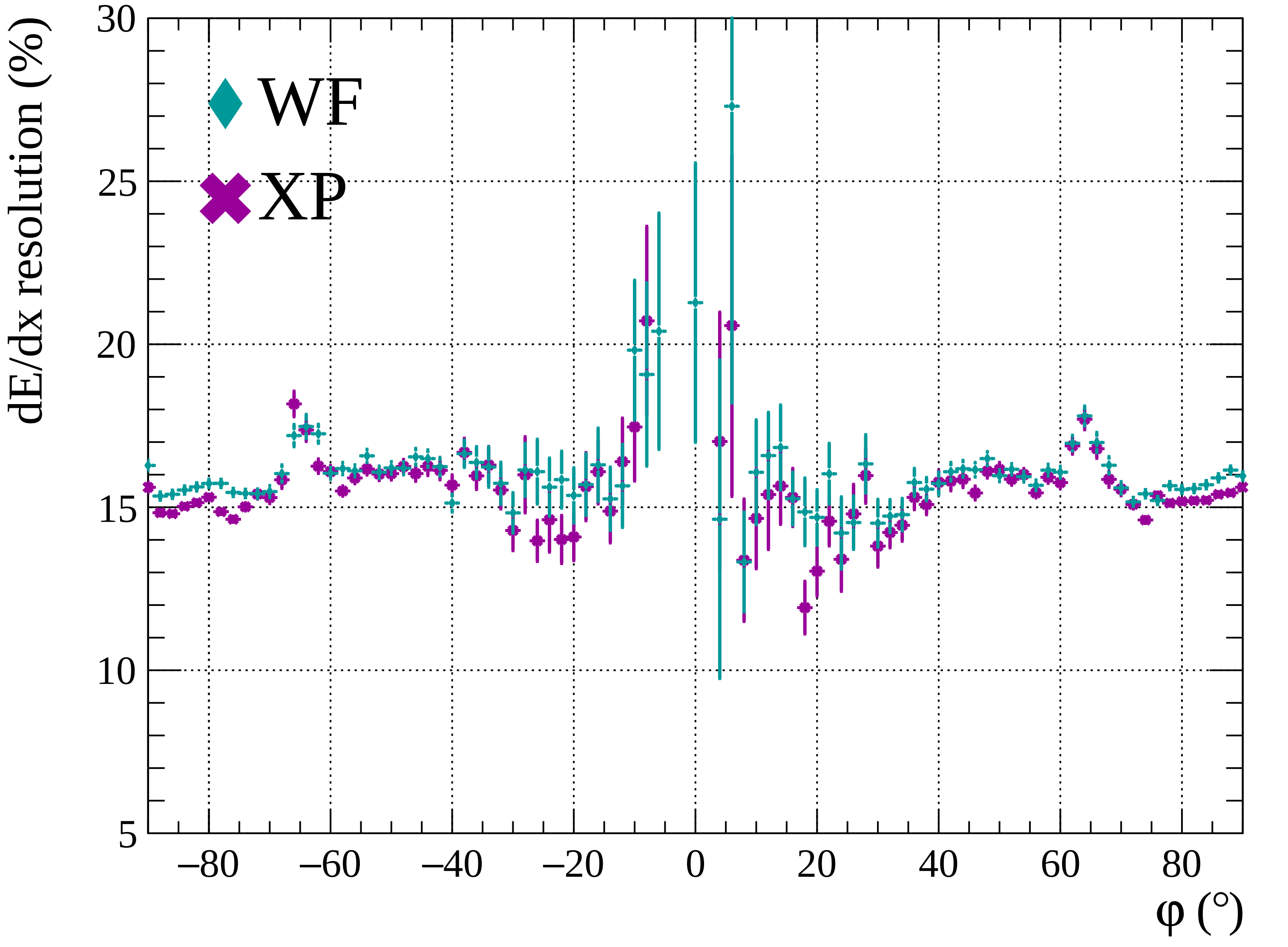
y = 0.11x + 457.68
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50


Energy loss (XP) vs drift time in ERAM 24
y = 0.01x + 486.85
Energy loss (XP) vs drift time in ERAM 30
y = -0.01x + 518.51
Energy loss (XP) vs drift time in ERAM 28
y = -0.01x + 527.15
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 503.40
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 467.02
Energy loss (XP) vs drift time in ERAM 13
y = 0.06x + 490.53
Energy loss (XP) vs drift time in ERAM 9
y = 0.07x + 500.46
Energy loss (XP) vs drift time in ERAM 2
y = 0.05x + 493.61
Energy loss (XP) vs drift time in ERAM 26
y = 0.07x + 509.20
Energy loss (XP) vs drift time in ERAM 17
y = 0.13x + 500.85
Energy loss (XP) vs drift time in ERAM 23
y = 0.10x + 497.55
Energy loss (XP) vs drift time in ERAM 29
y = 0.11x + 482.55
Energy loss (XP) vs drift time in ERAM 1
y = 0.13x + 475.06
Energy loss (XP) vs drift time in ERAM 10
y = 0.15x + 472.64
Energy loss (XP) vs drift time in ERAM 11
y = 0.14x + 474.06
Energy loss (XP) vs drift time in ERAM 3
y = 0.12x + 464.89
dE/dx (ADC counts/cm)
drift time (timebins)

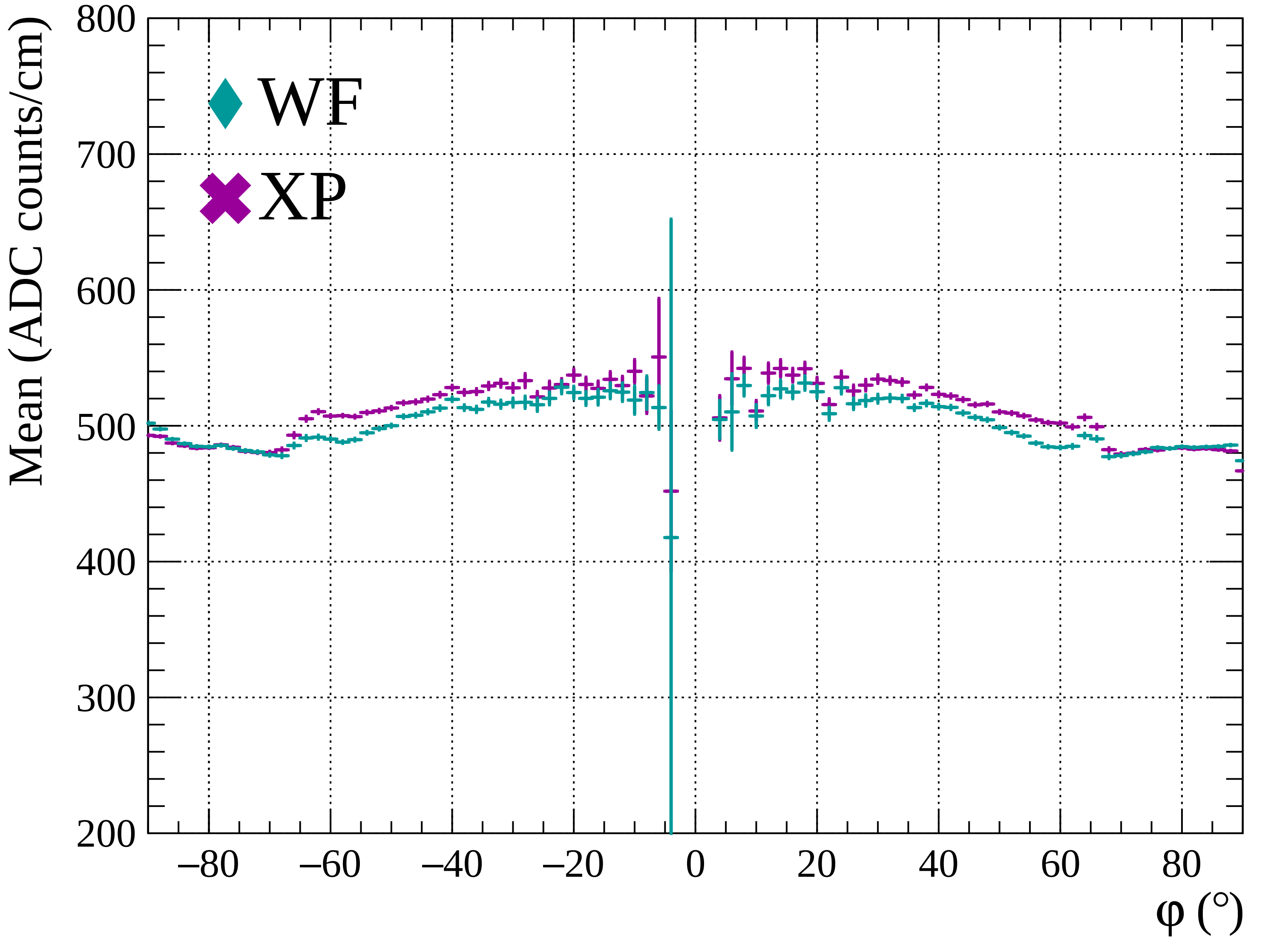

Energy loss (XP) vs drift time in ERAM 47
y = 0.02x + 490.26
Energy loss (XP) vs drift time in ERAM 16
y = -0.01x + 511.07
Energy loss (XP) vs drift time in ERAM 14
y = -0.05x + 539.24
Energy loss (XP) vs drift time in ERAM 15
y = 0.07x + 532.34
Energy loss (XP) vs drift time in ERAM 42
y = 0.08x + 449.45
Energy loss (XP) vs drift time in ERAM 45
y = 0.07x + 465.30
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 483.29
Energy loss (XP) vs drift time in ERAM 36
y = 0.08x + 480.32
Energy loss (XP) vs drift time in ERAM 20
y = 0.12x + 474.30
Energy loss (XP) vs drift time in ERAM 38
y = 0.09x + 481.85
Energy loss (XP) vs drift time in ERAM 7
y = 0.06x + 506.83
Energy loss (XP) vs drift time in ERAM 44
y = 0.13x + 469.58
Energy loss (XP) vs drift time in ERAM 43
y = 0.11x + 474.13
Energy loss (XP) vs drift time in ERAM 39
y = 0.13x + 448.29
Energy loss (XP) vs drift time in ERAM 41
y = 0.12x + 468.37
Energy loss (XP) vs drift time in ERAM 46
y = 0.11x + 451.82
dE/dx (ADC counts/cm)
drift time (timebins)



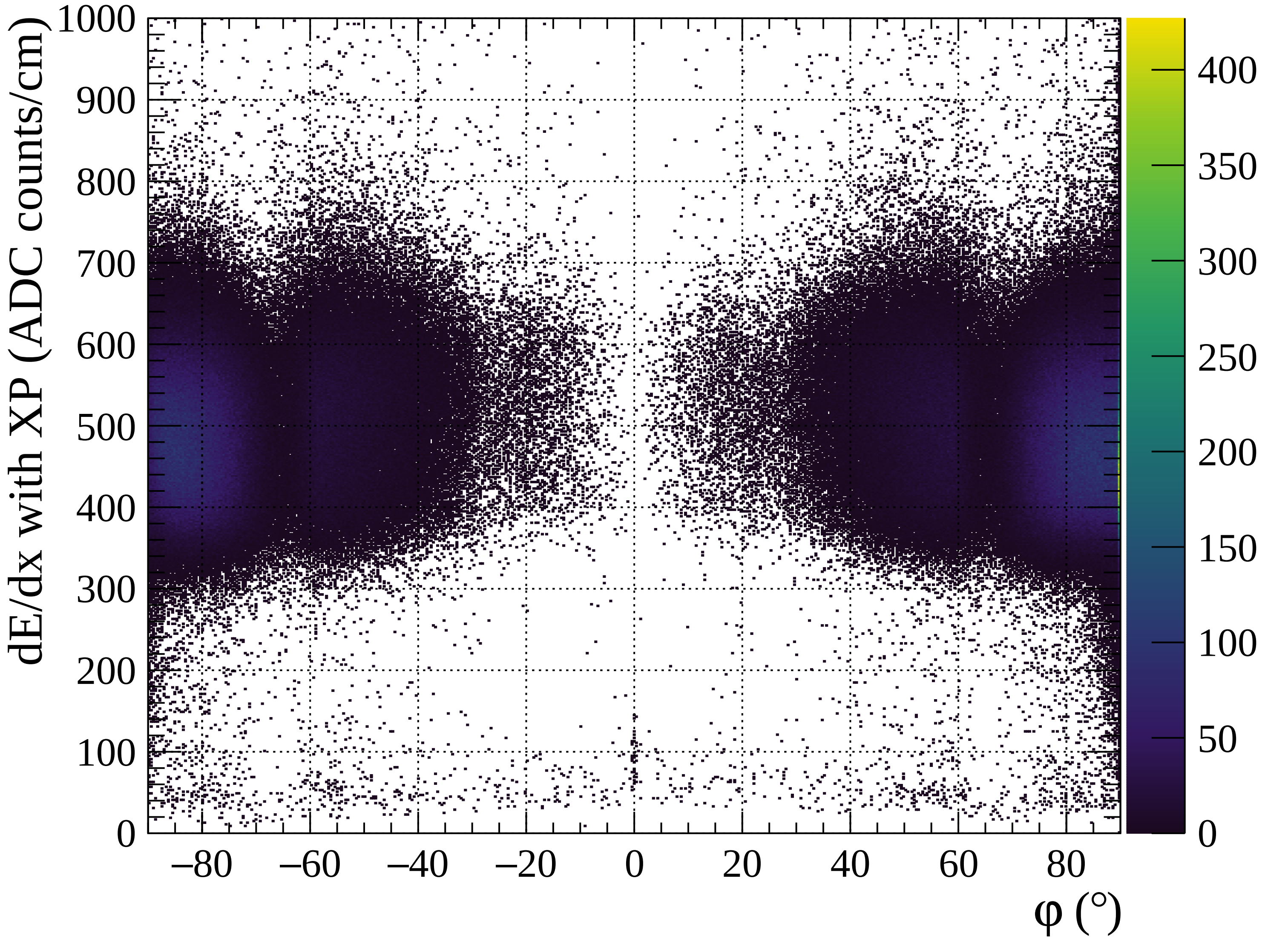
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
250
200
150
100
500
0



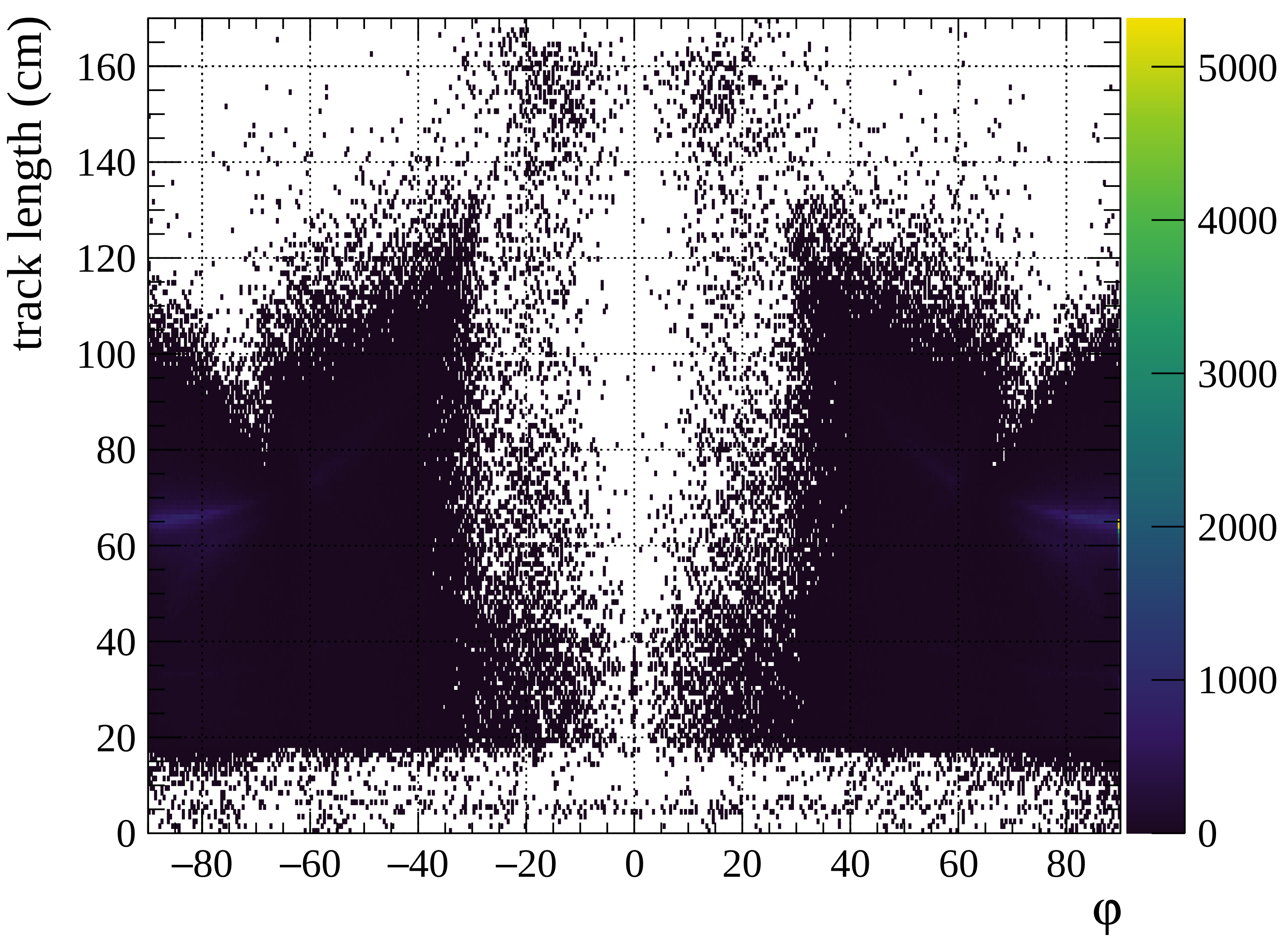
dE/dx resolution (%)
φ (°)
WF
XP
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



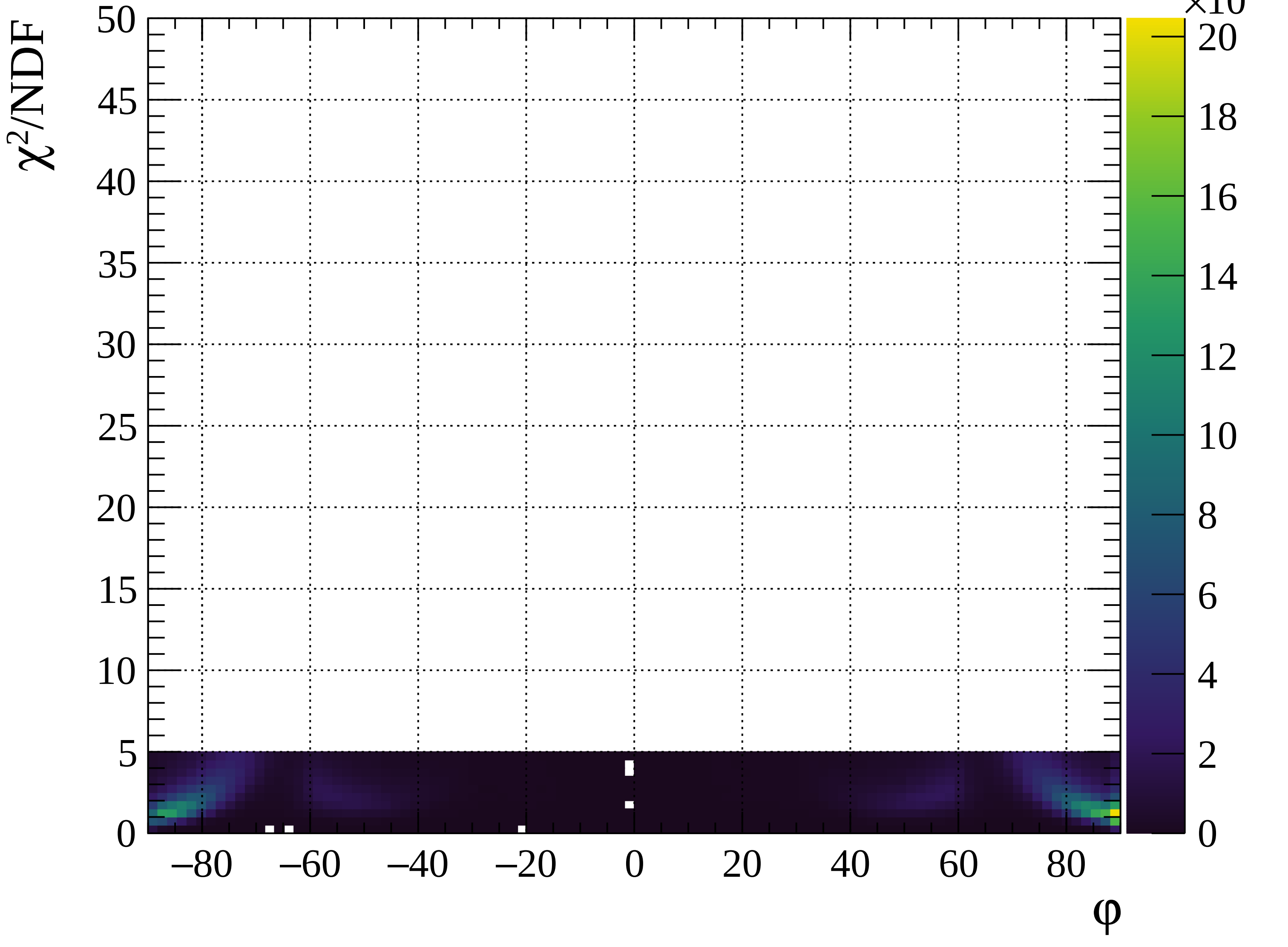
WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)



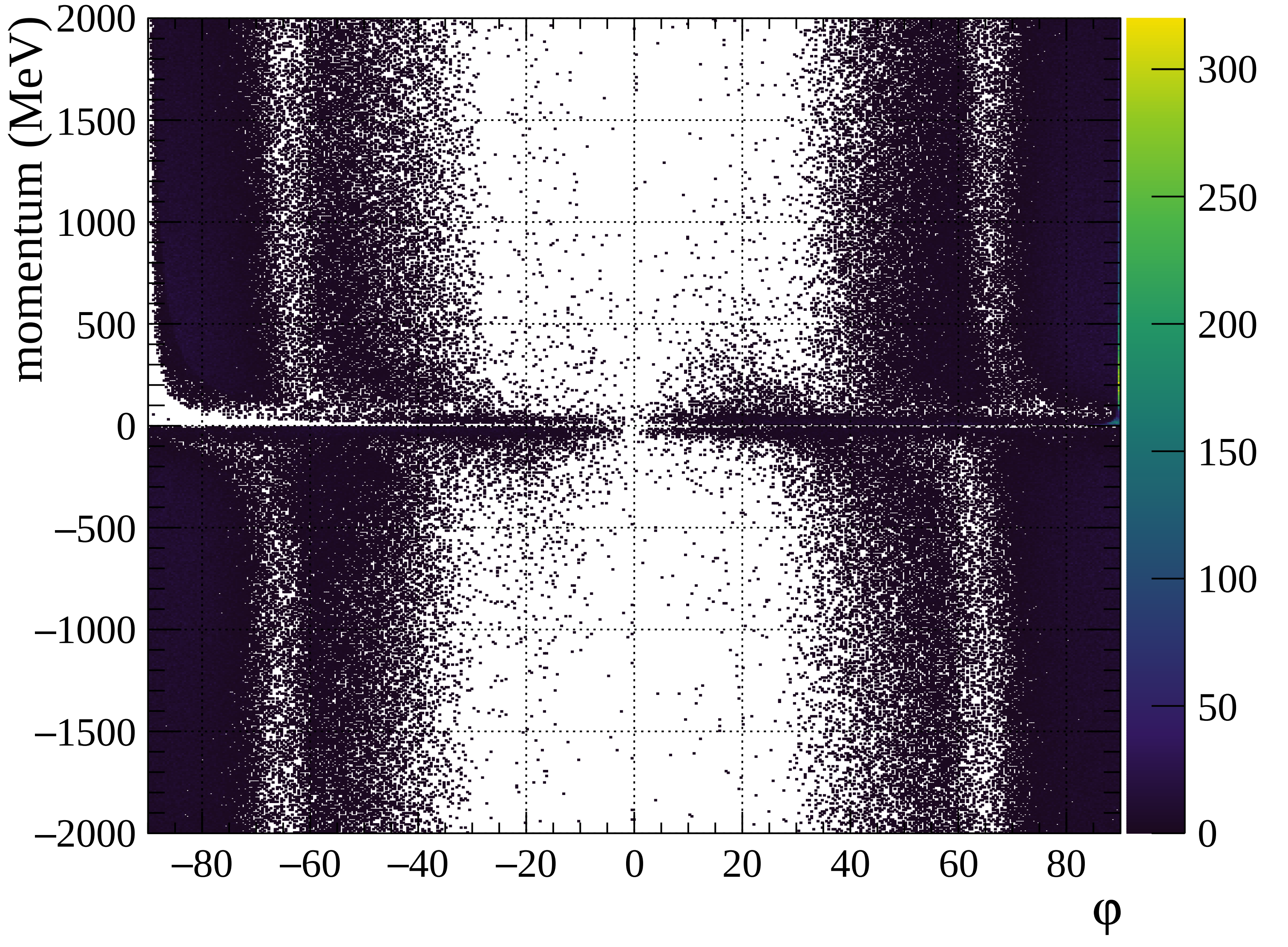
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
400
350
300
250
200
150
100
50
0

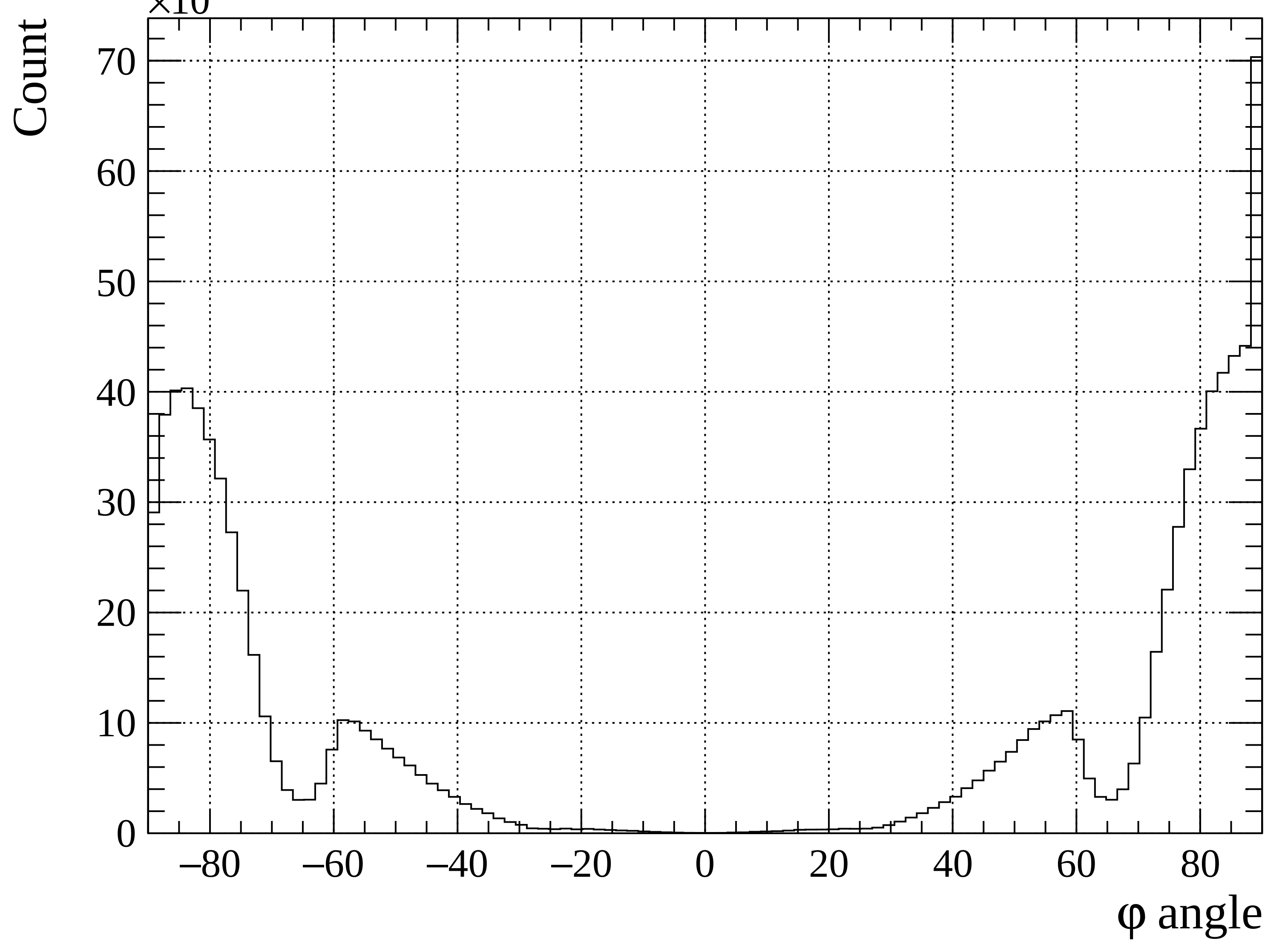


track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
1000
2000
3000
4000
5000

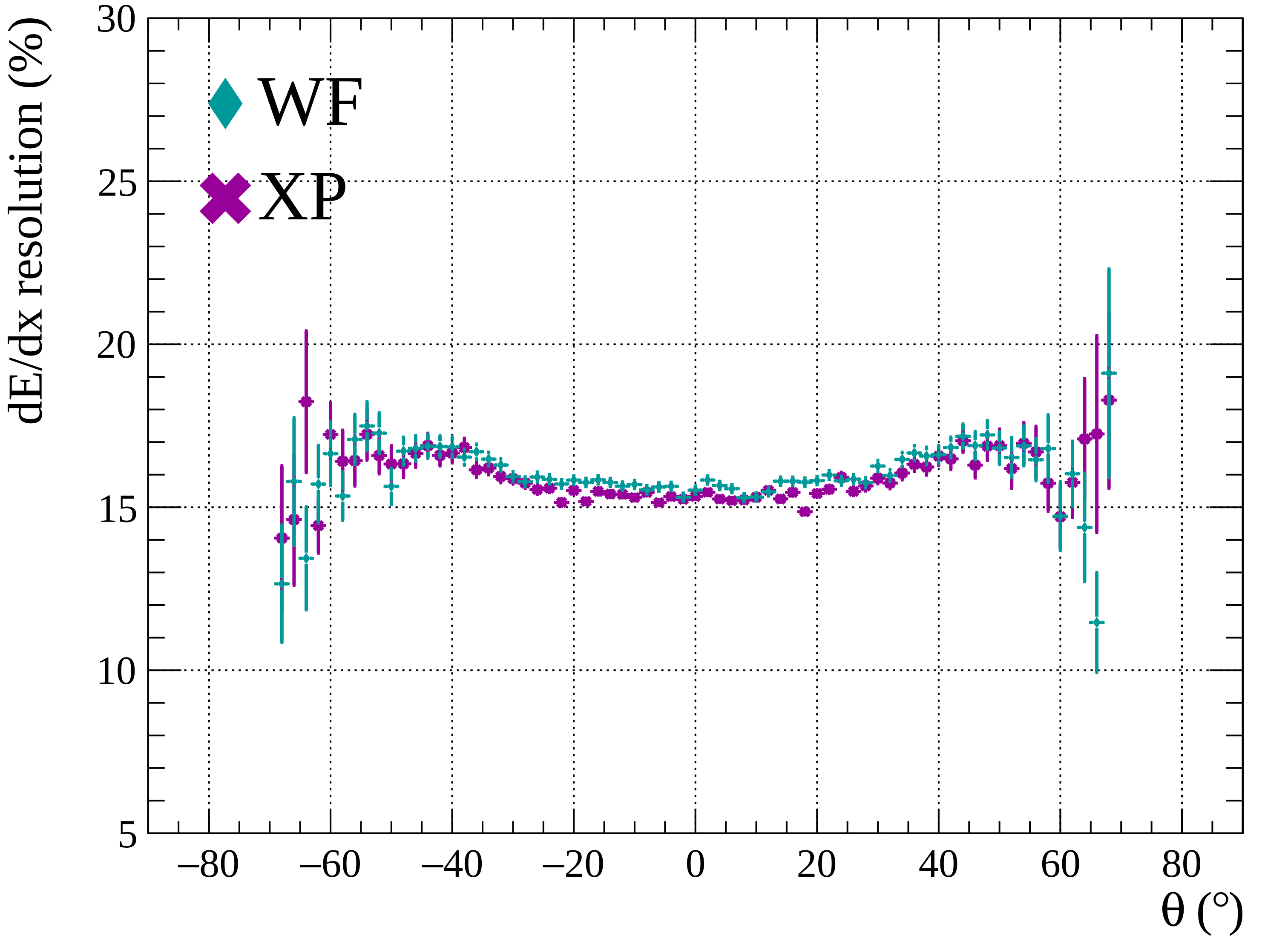


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
$\times 10$
20
18
16
14
12
10
8
6
4
2
0

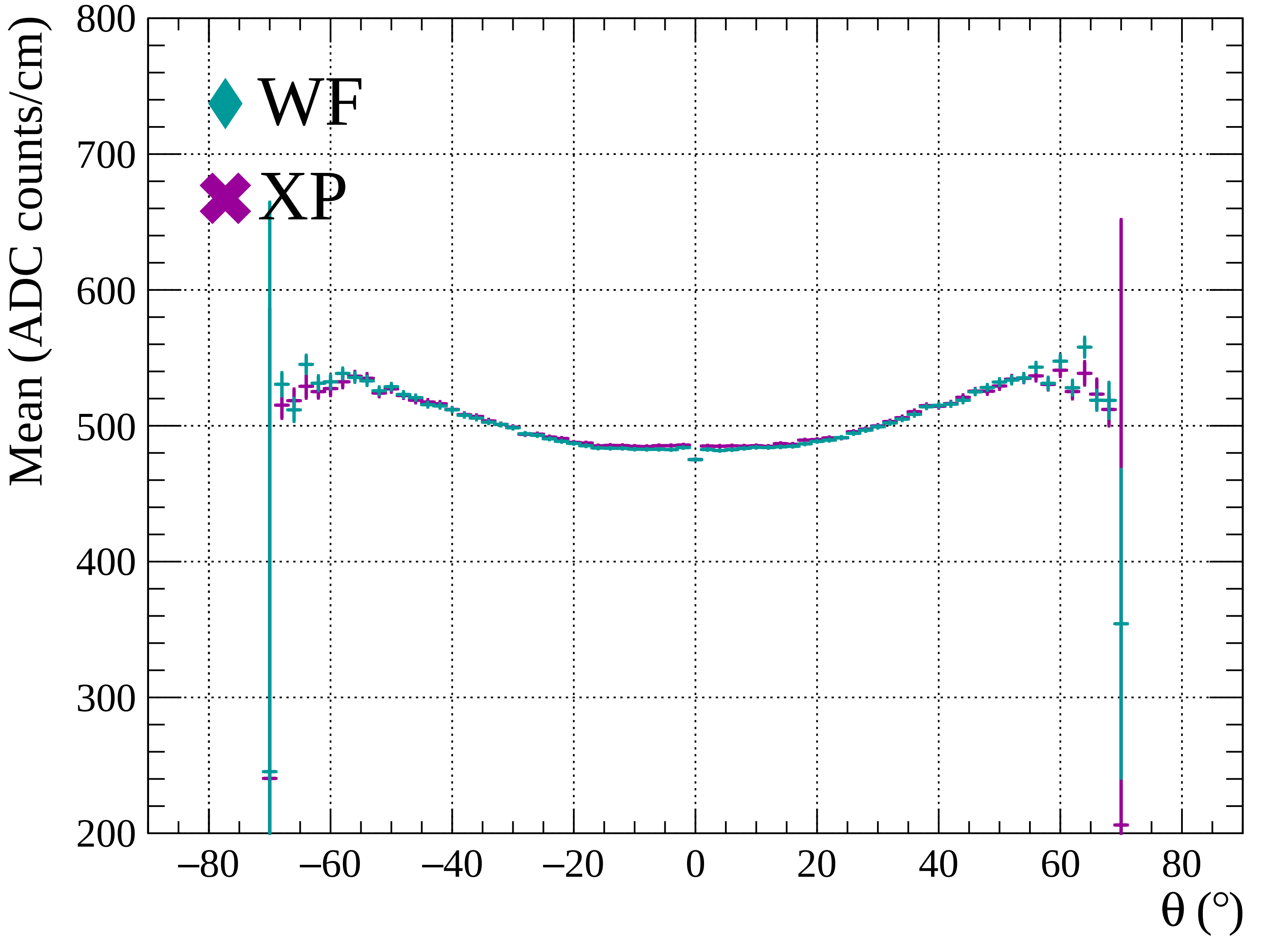

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0

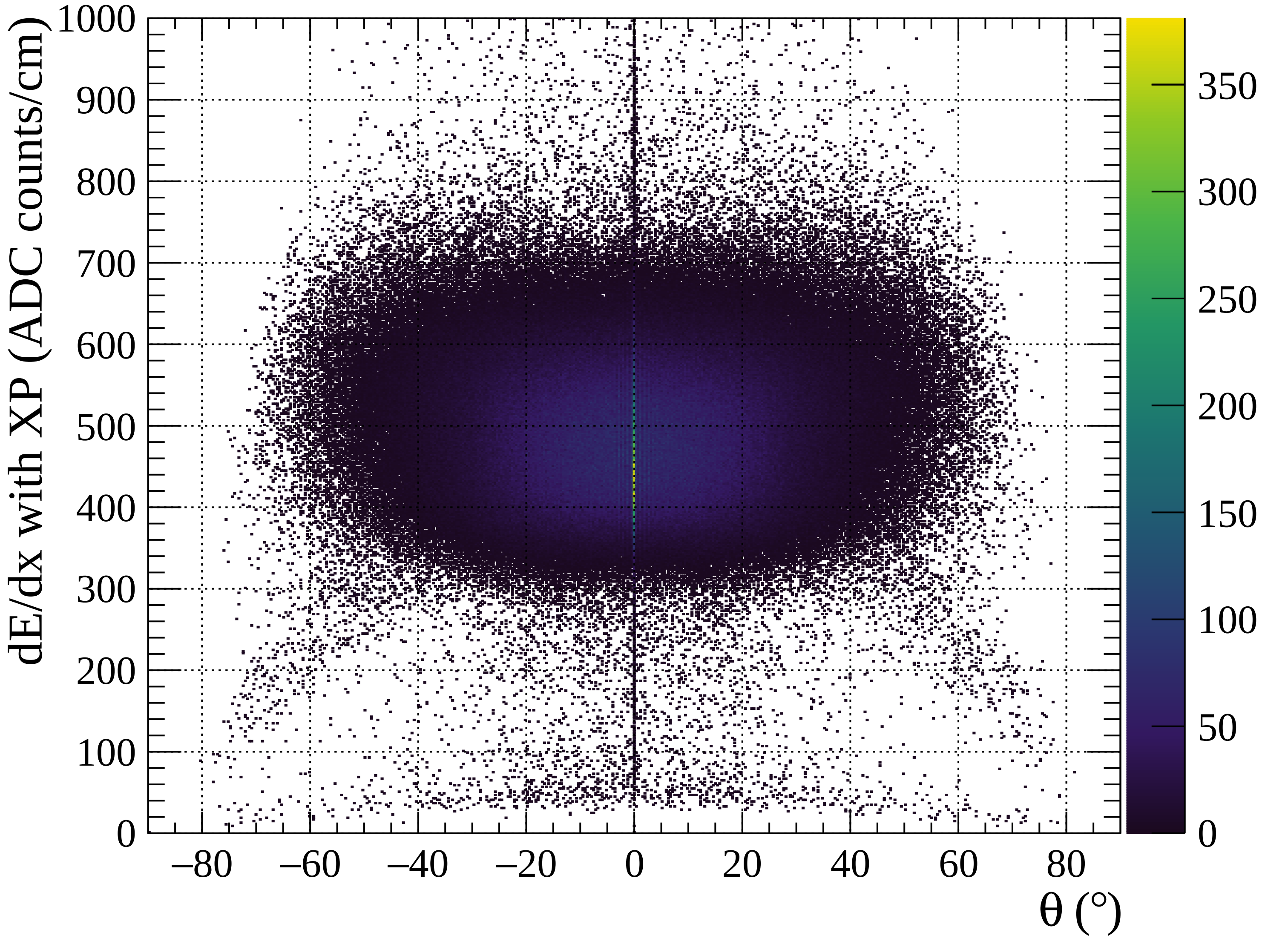


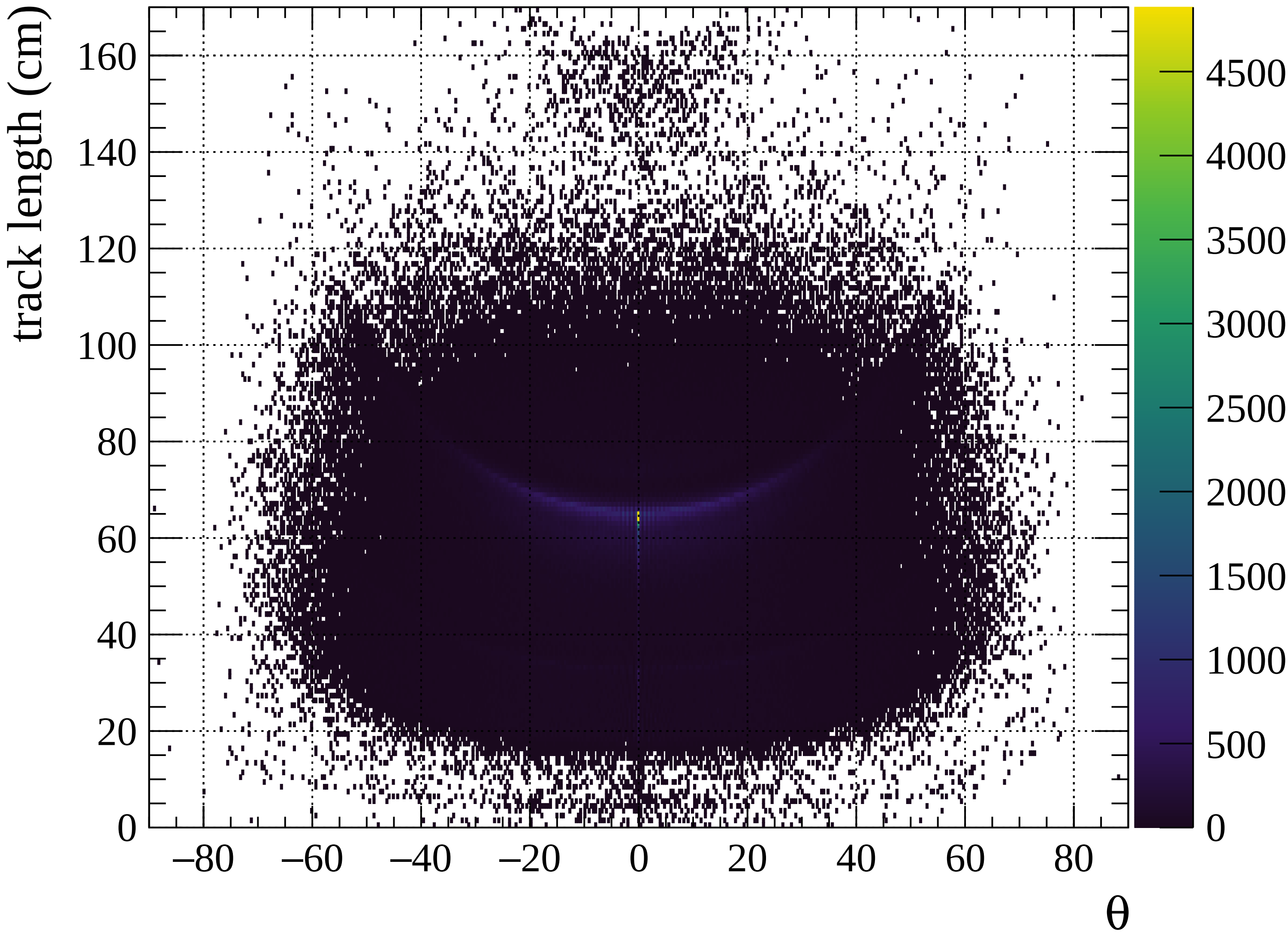
Count
×10
0
10
20
30
40
50
60
70
−80
−60
−40
−20
0
20
40
60
80
φ angle



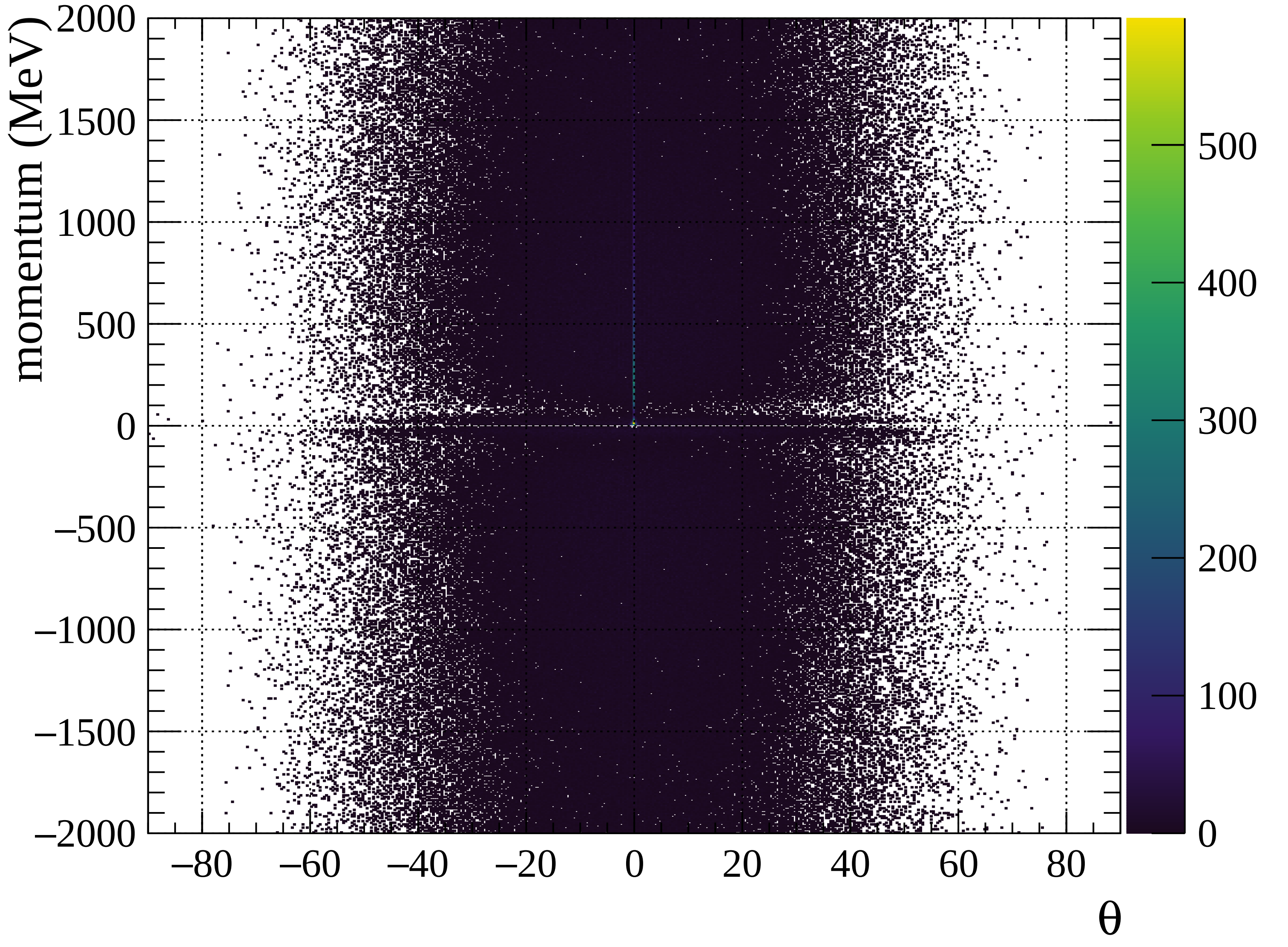
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80


WF
XP
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
θ (°)

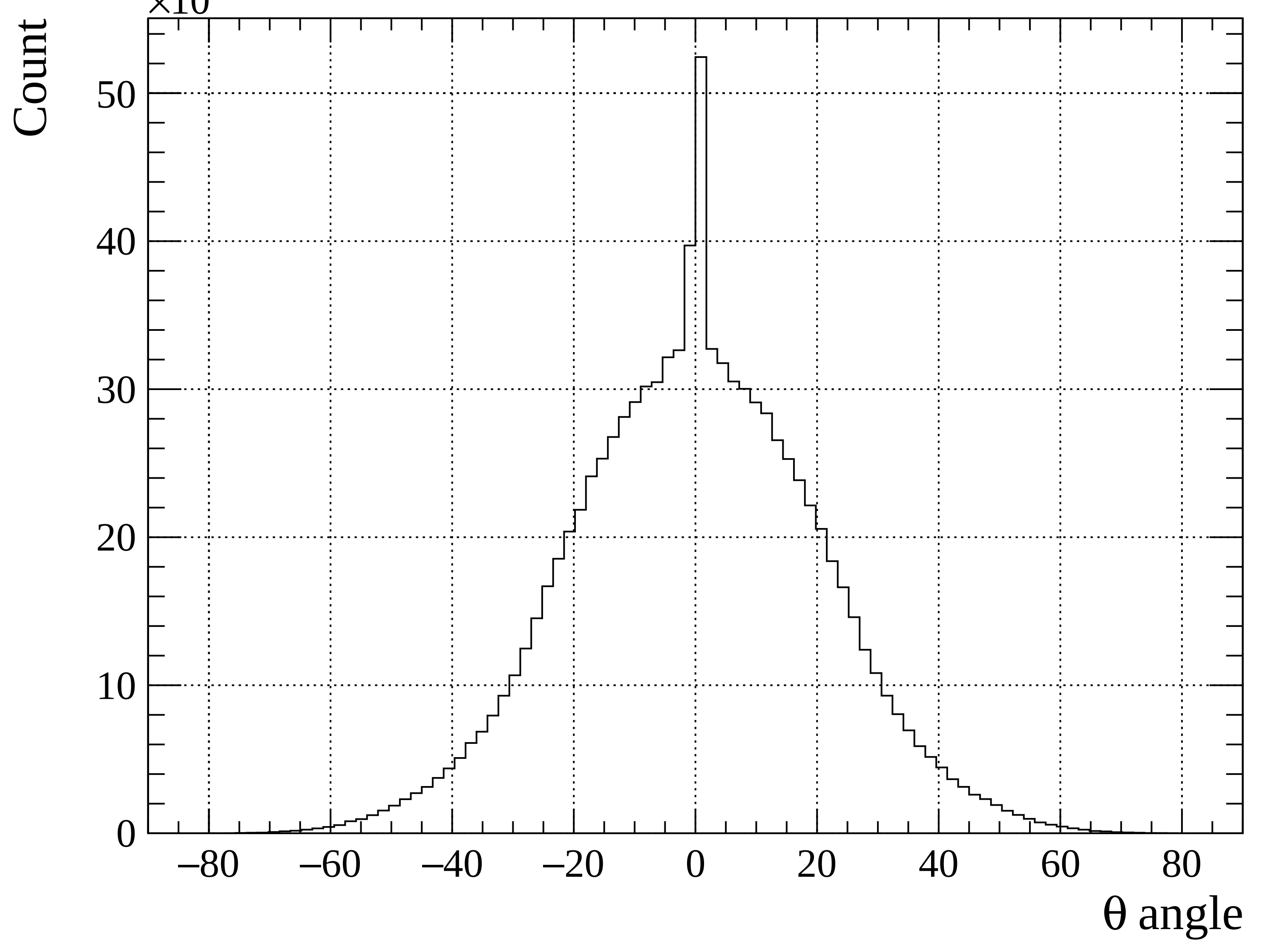

dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300
350

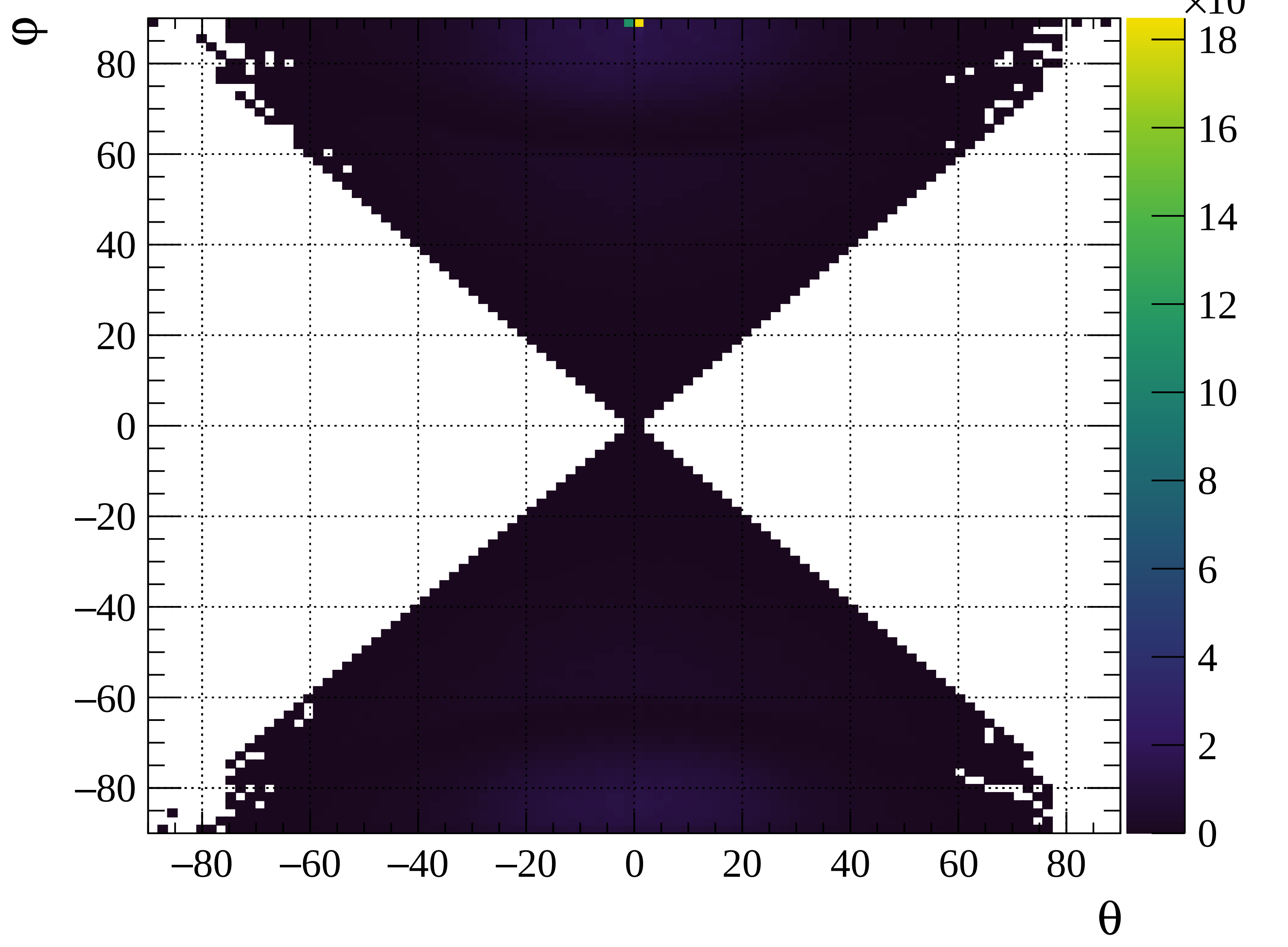


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
4500
4000
3500
3000
2500
2000
1500
1000
500
0

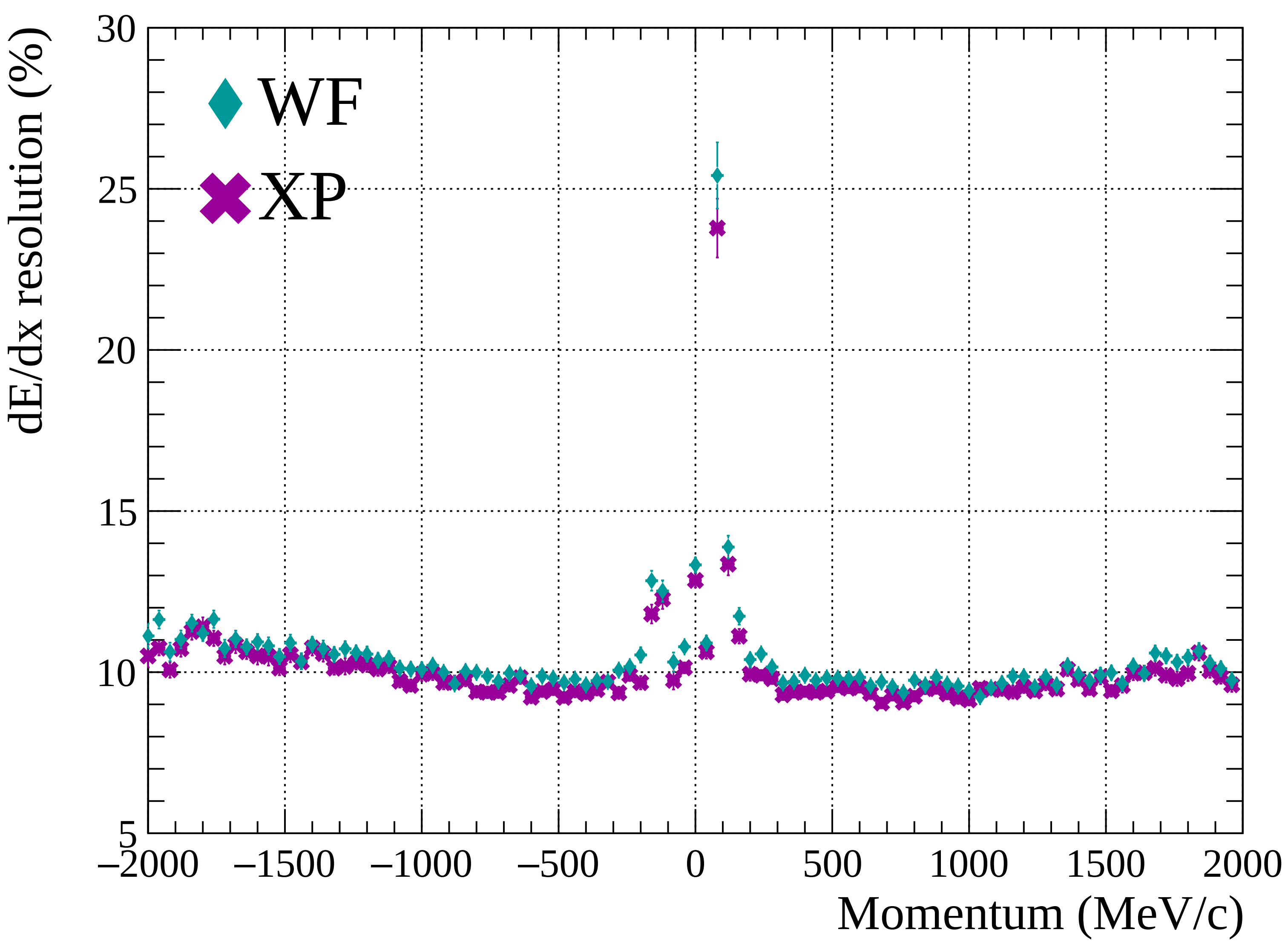


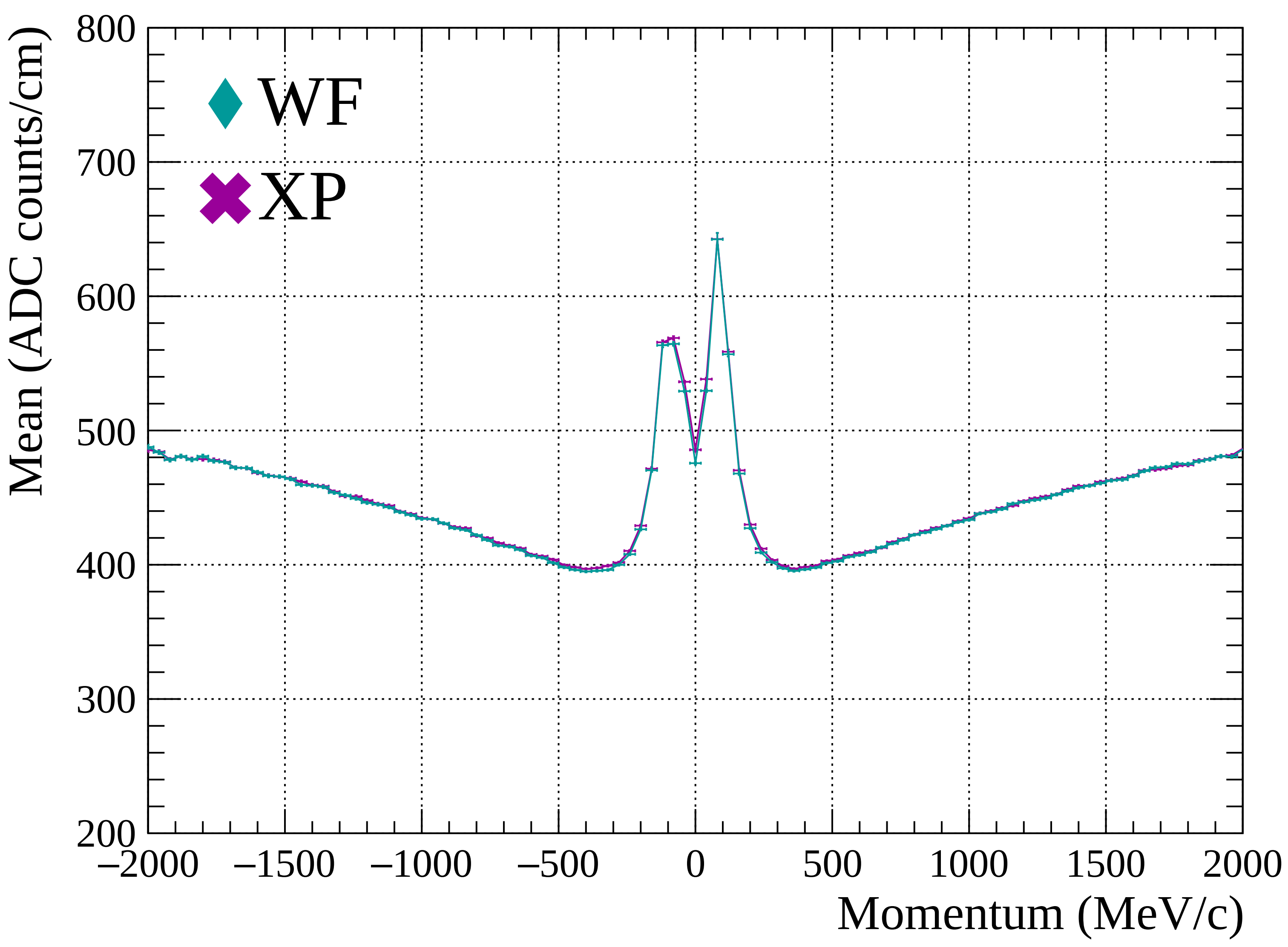
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
500
400
300
200
100
0


×10
Count
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

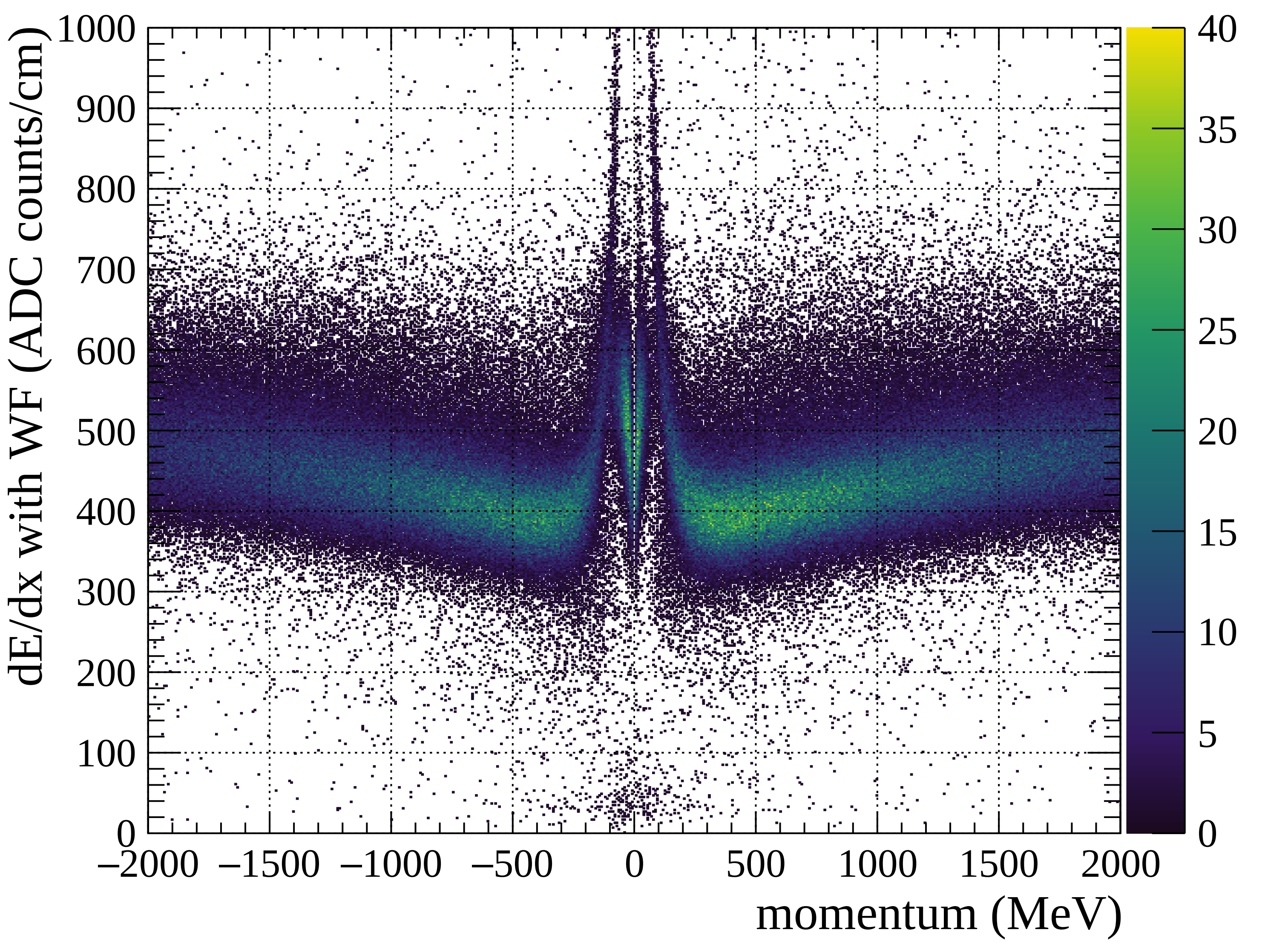


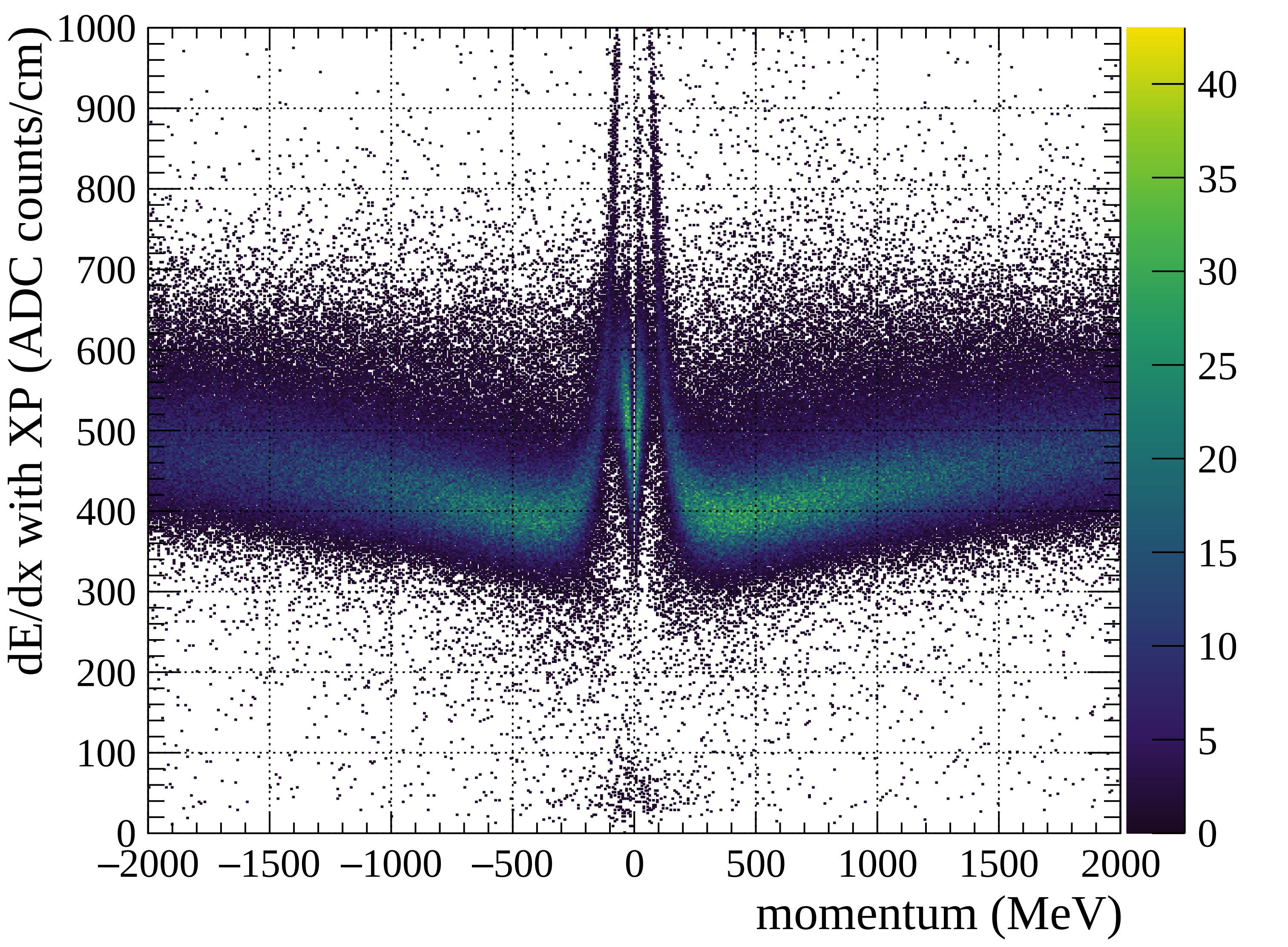
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
×10
18
16
14
12
10
8
6
4
2
0


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

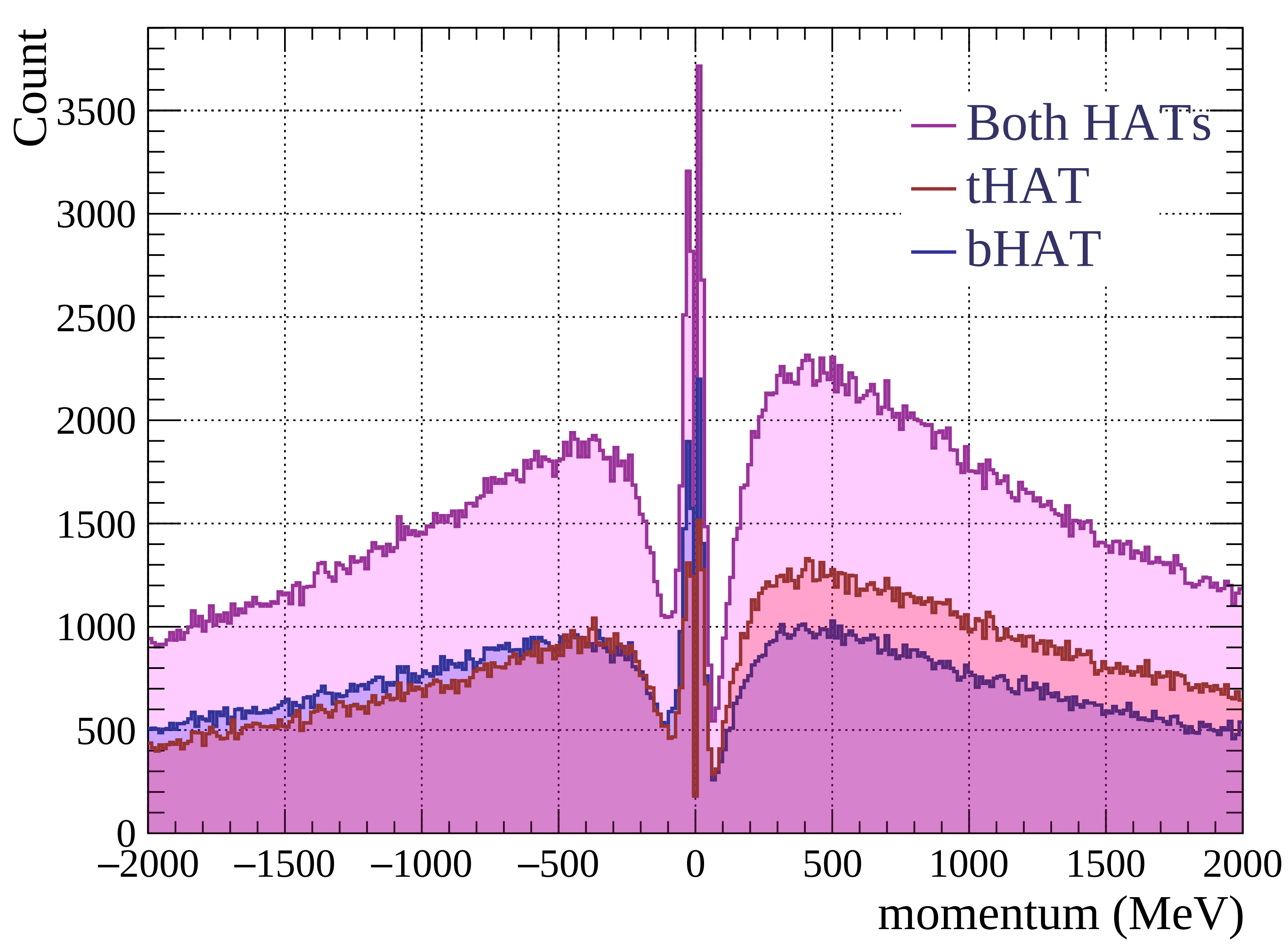


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000

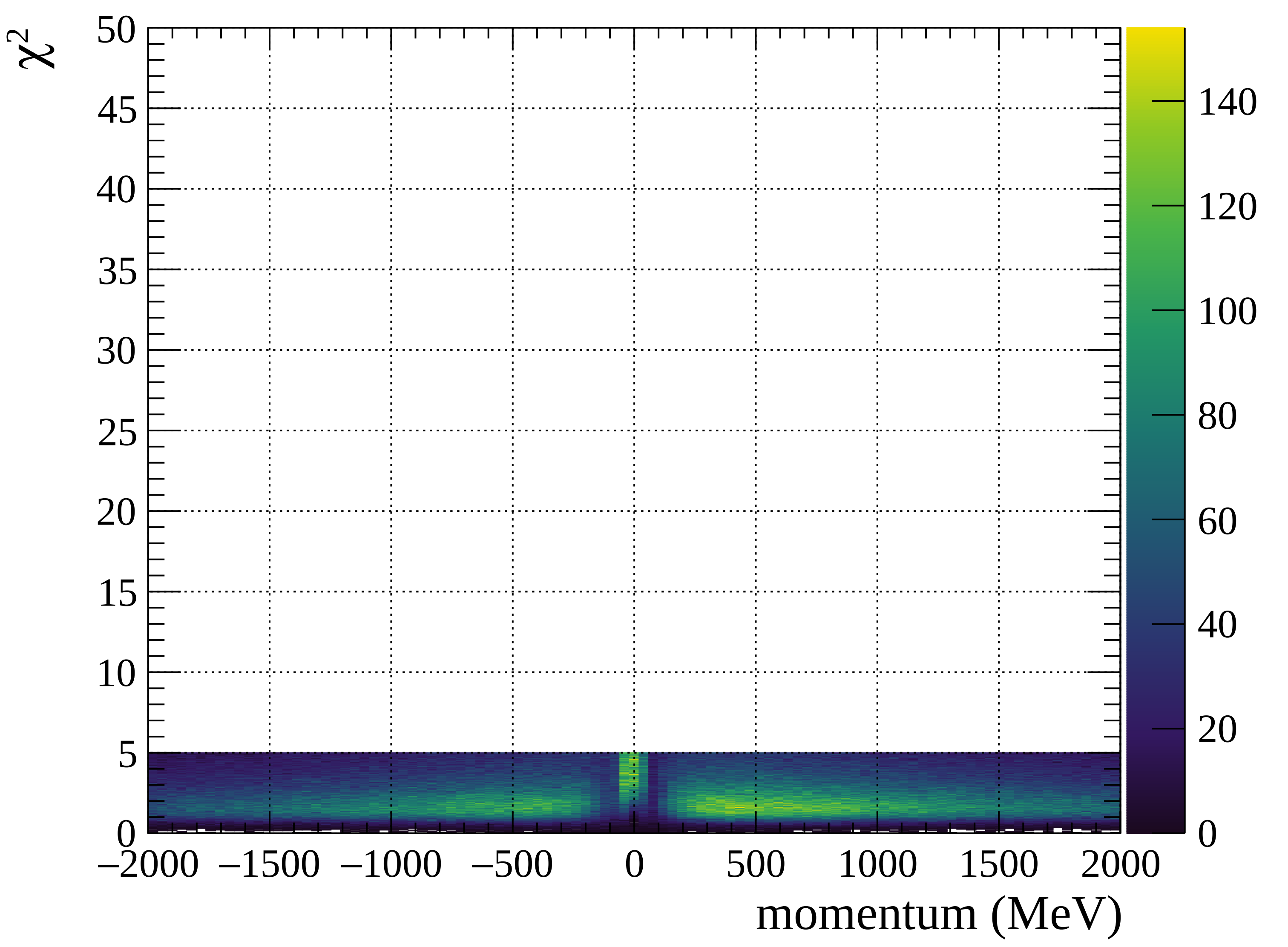

dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
5
10
15
20
25
30
35
40



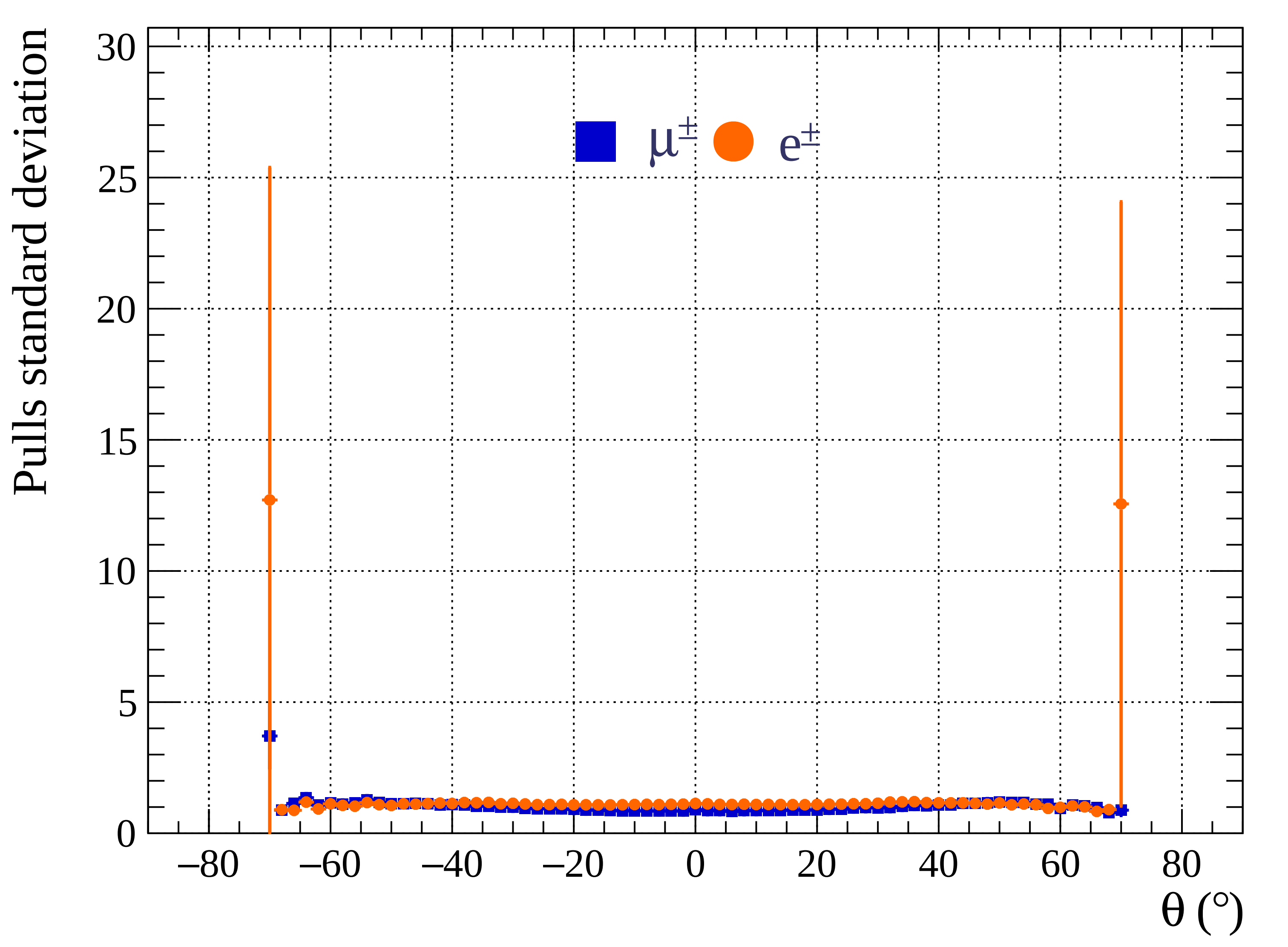
dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
40
35
30
25
20
15
10
5
0



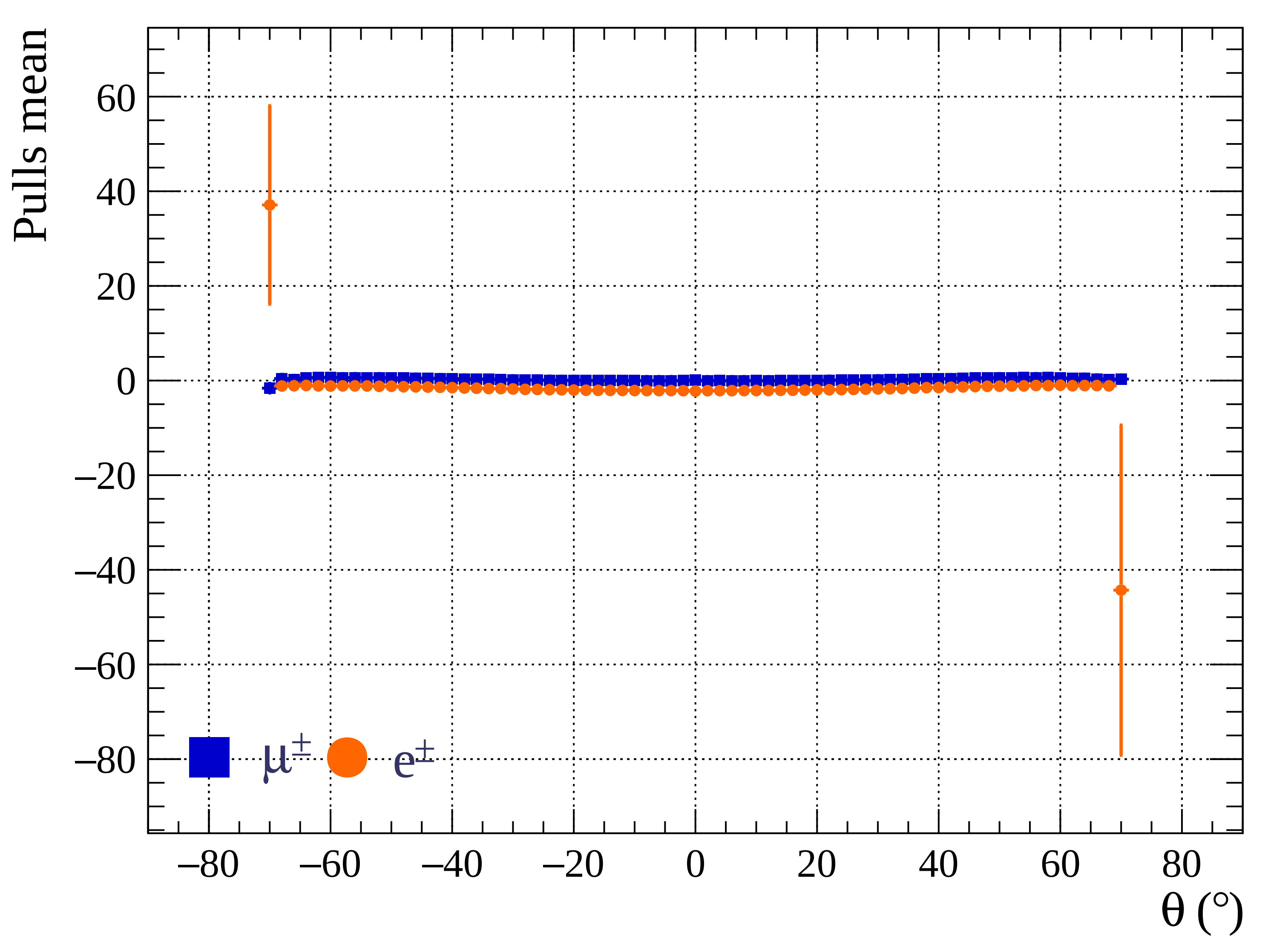
Count
0
500
1000
1500
2000
2500
3000
3500
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



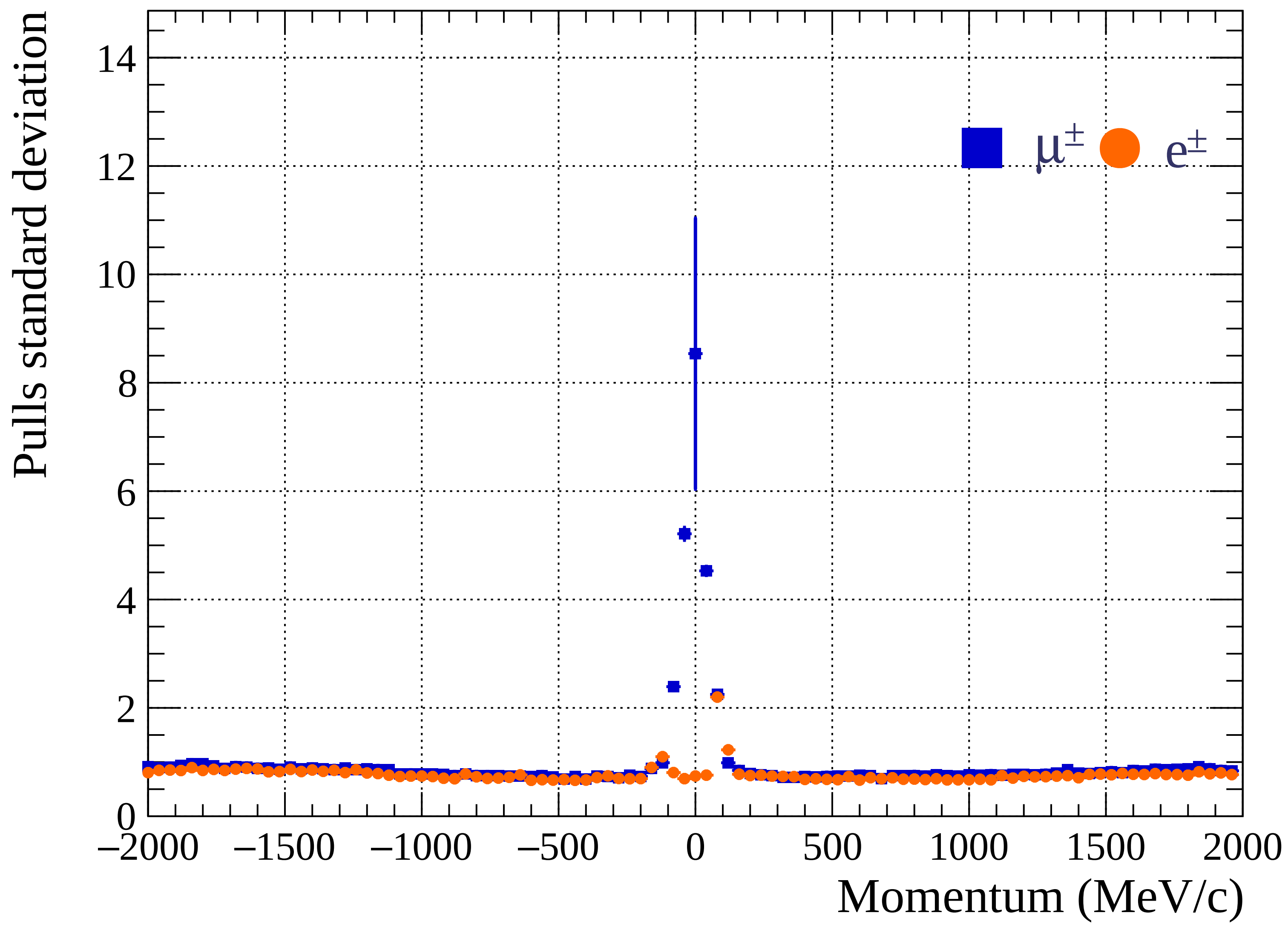
χ^2
0
5
10
15
20
25
30
35
40
45
50
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
20
40
60
80
100
120
140



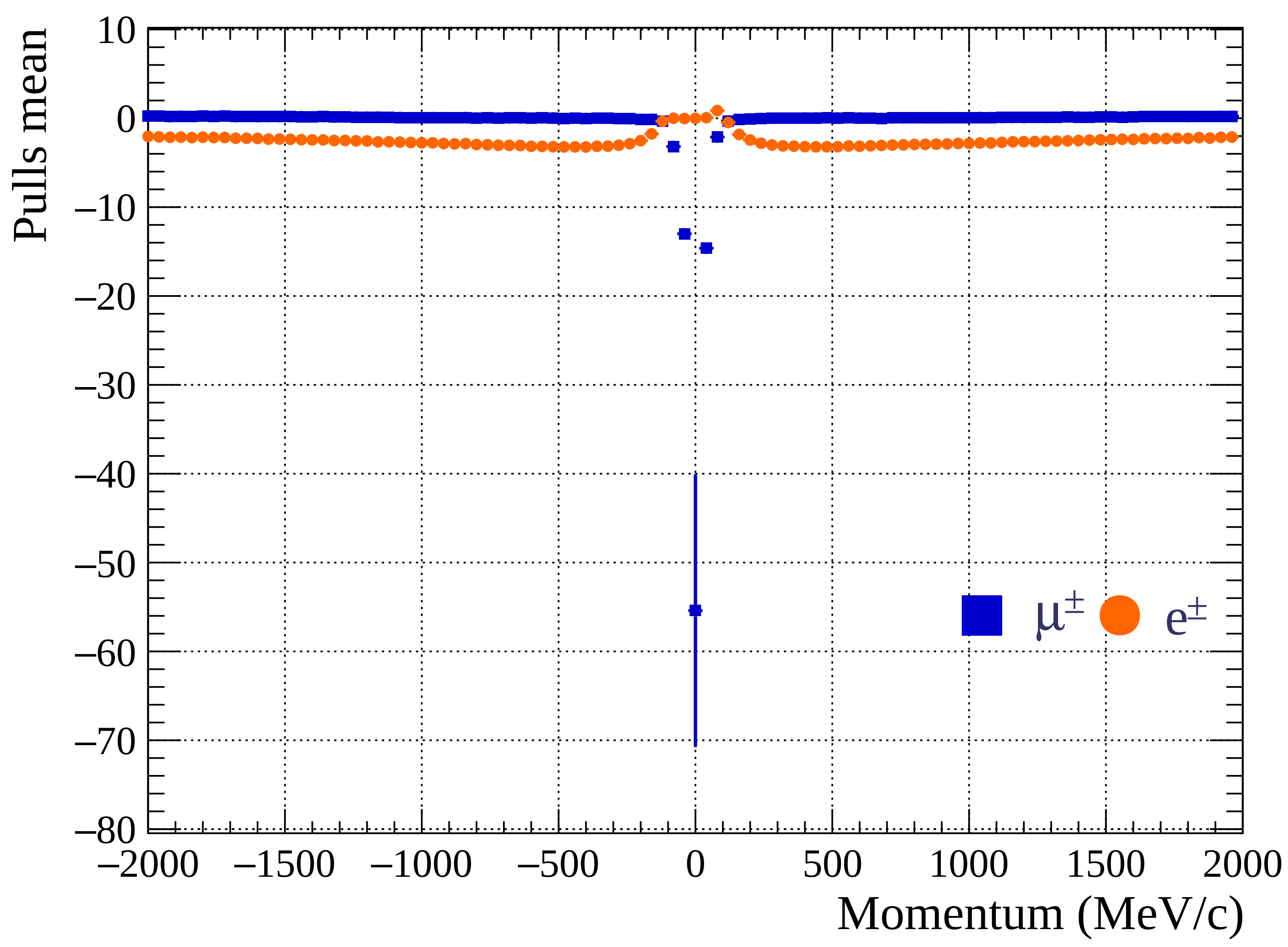
Pulls standard deviation
$\mu^{\pm}$
$e^{\pm}$
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)



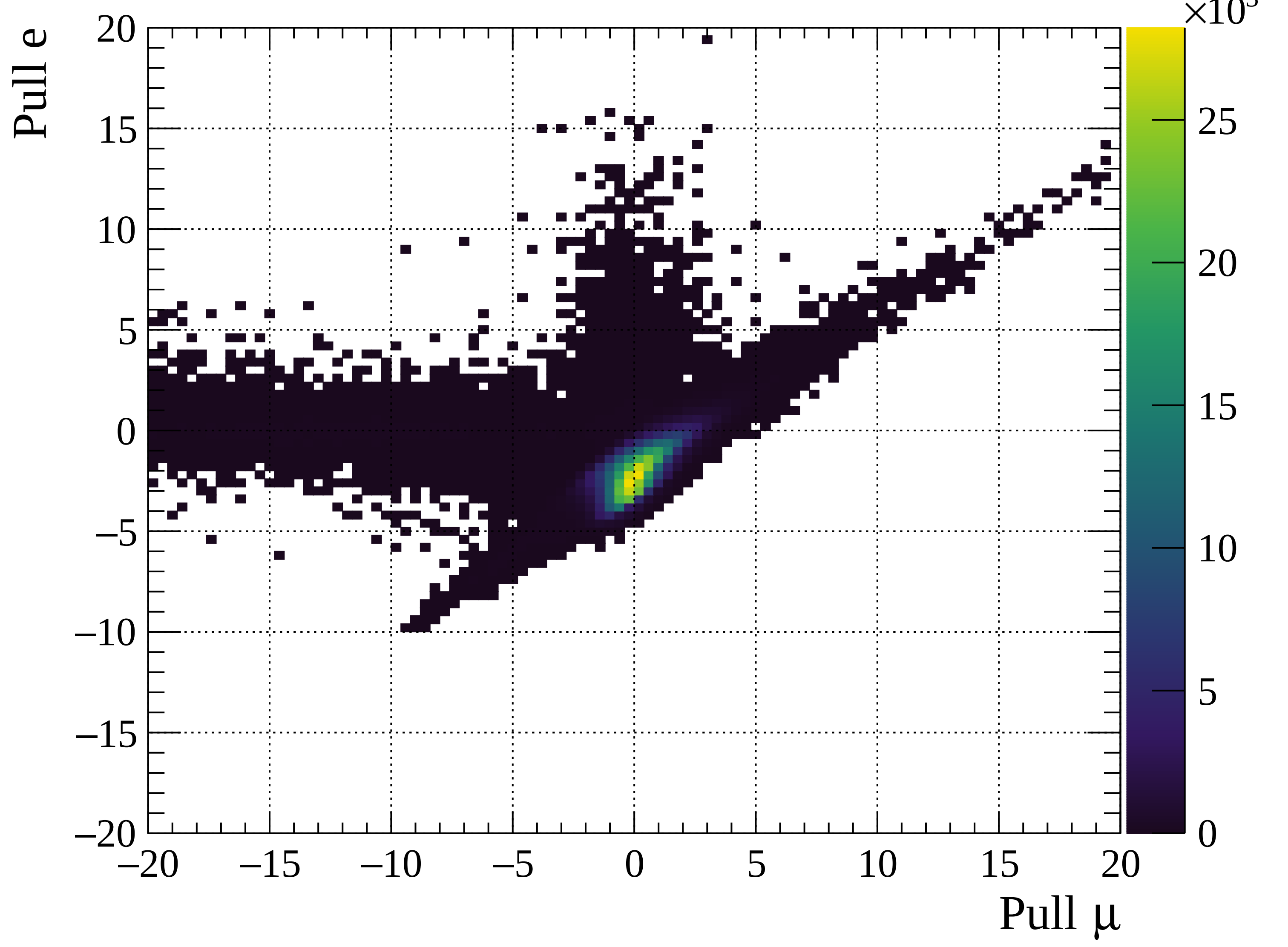
Pulls mean
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ (°)
$\mu^{\pm}$
$e^{\pm}$



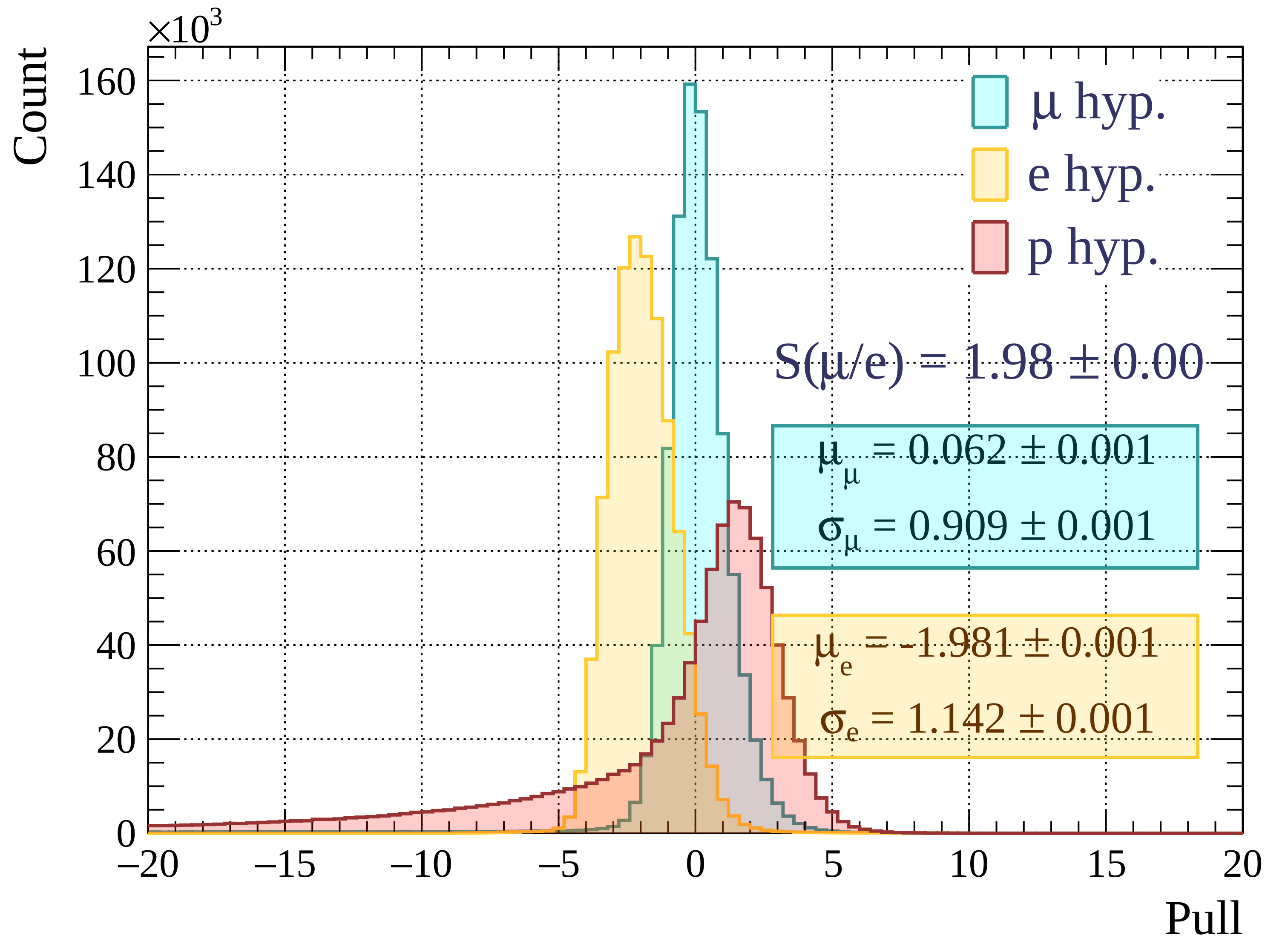
Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
2
4
6
8
10
12
14
−2000
−1500
−1000
−500
0
500
1000
1500
2000



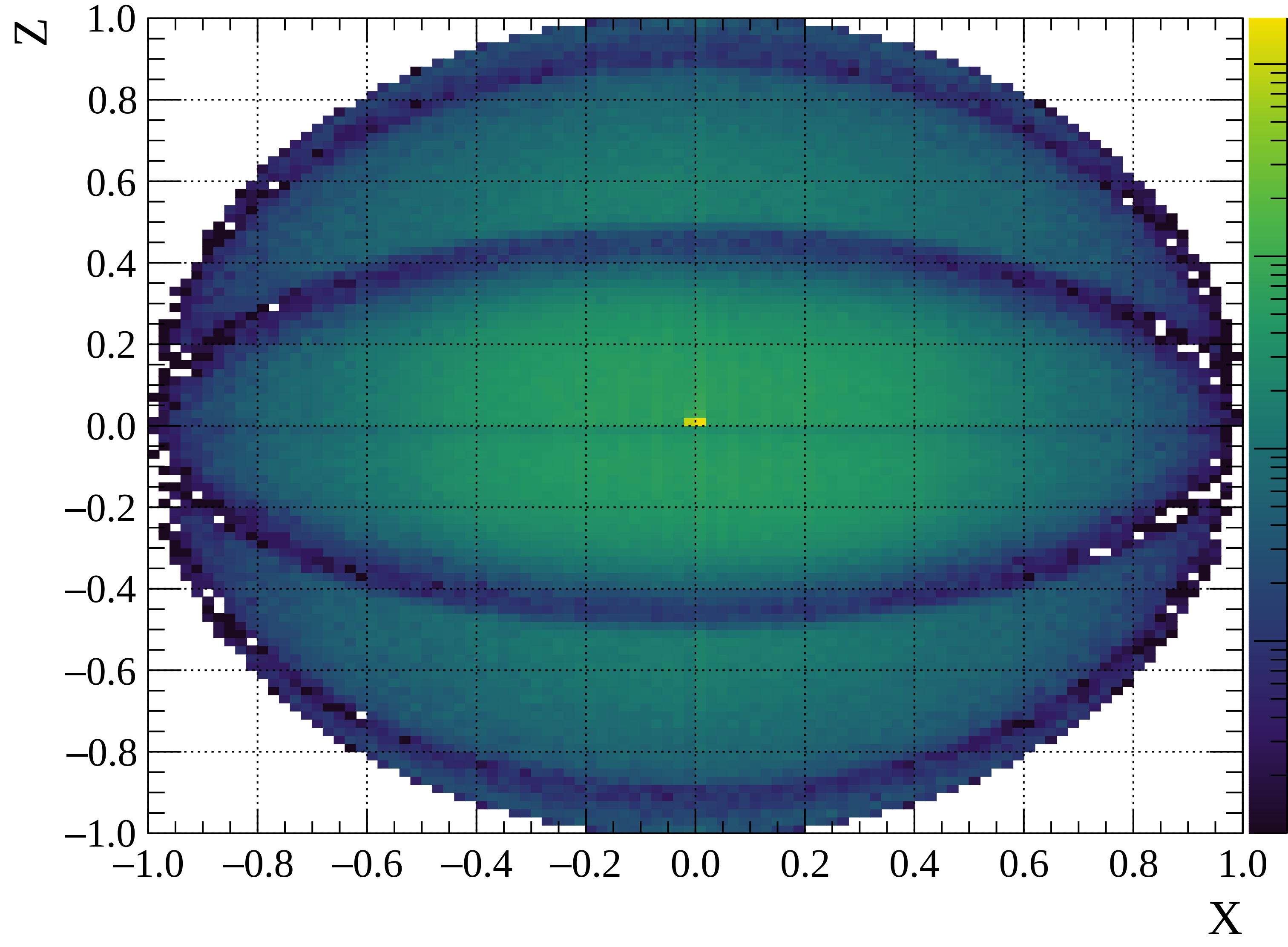
Pulls mean
10
0
−10
−20
−30
−40
−50
−60
−70
−80
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



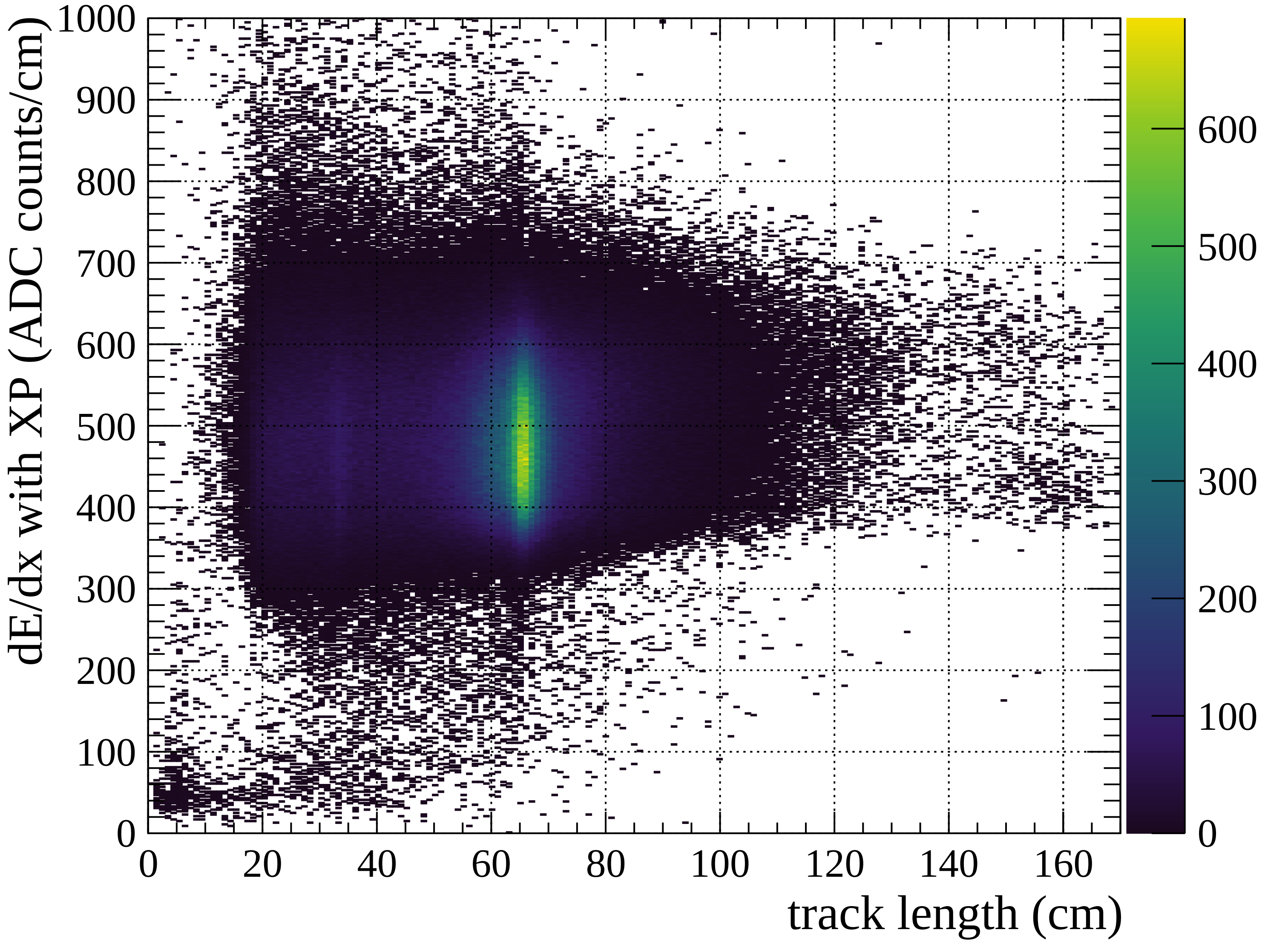
Pull e
Pull μ
$\times 10^3$
20
15
10
5
0
−5
−10
−15
−20
25

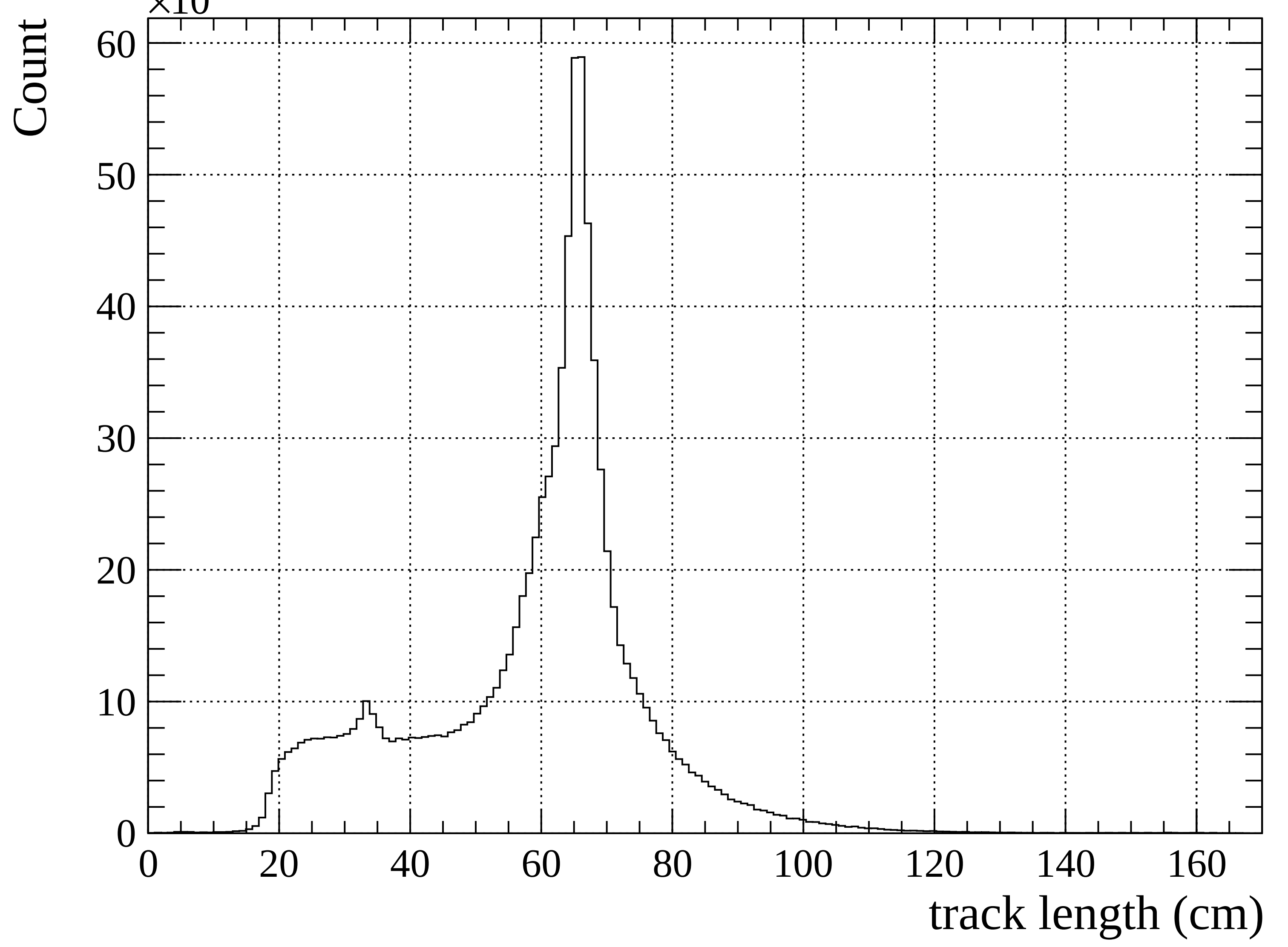


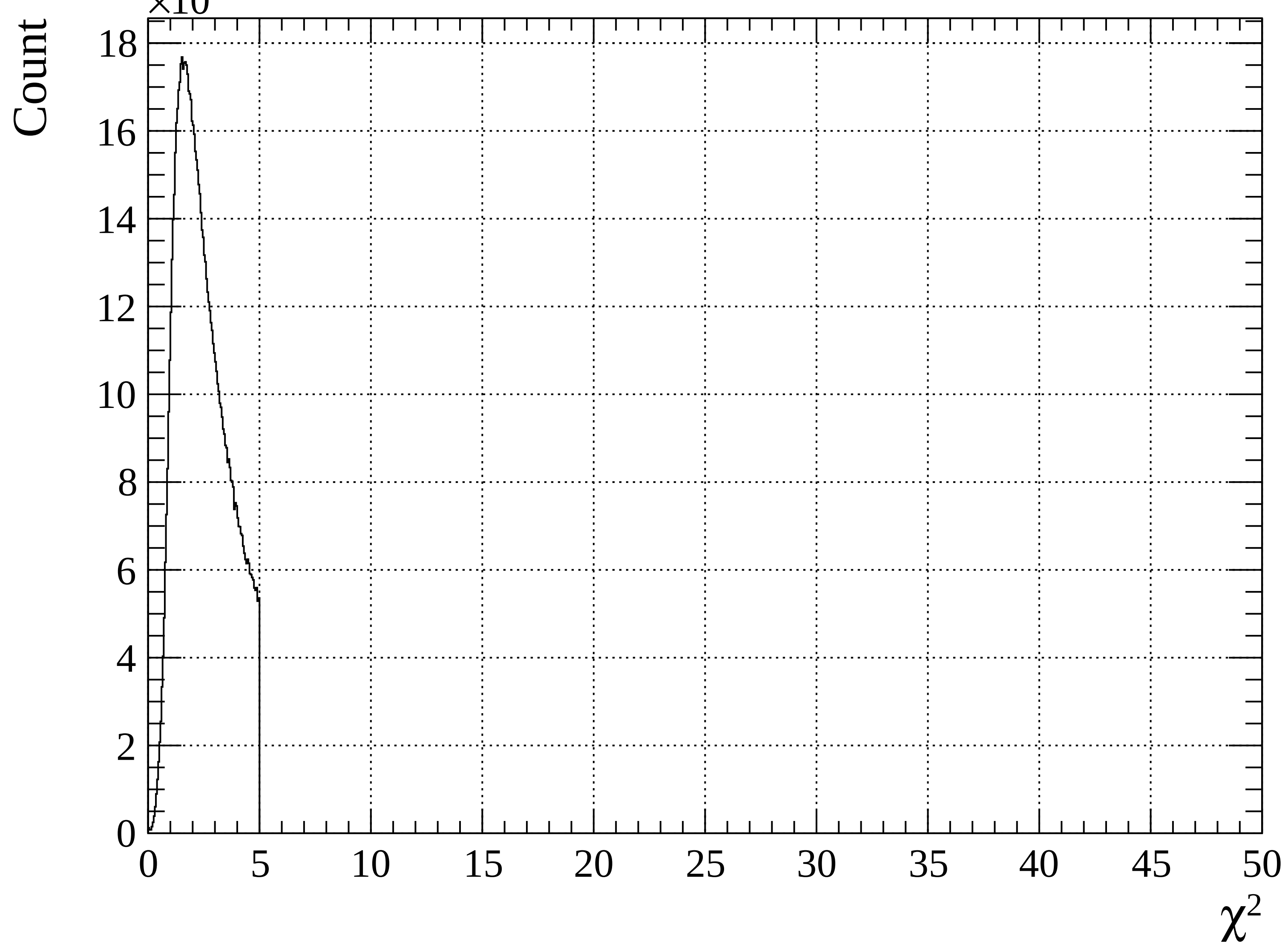
×10^3
Count
160
140
120
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 1.98 ± 0.00
μ_μ = 0.062 ± 0.001
σ_μ = 0.909 ± 0.001
μ_e = -1.981 ± 0.001
σ_e = 1.142 ± 0.001

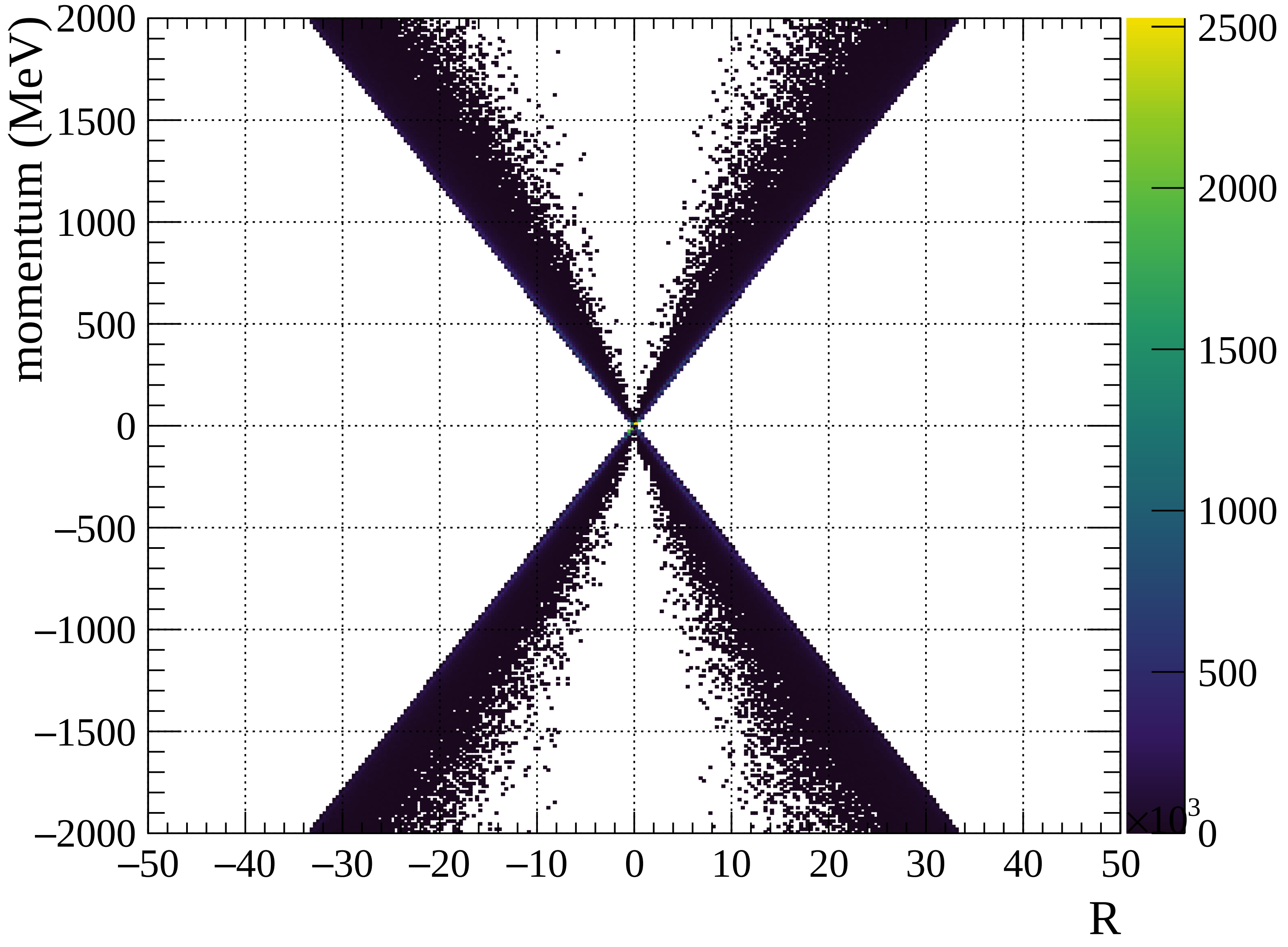


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

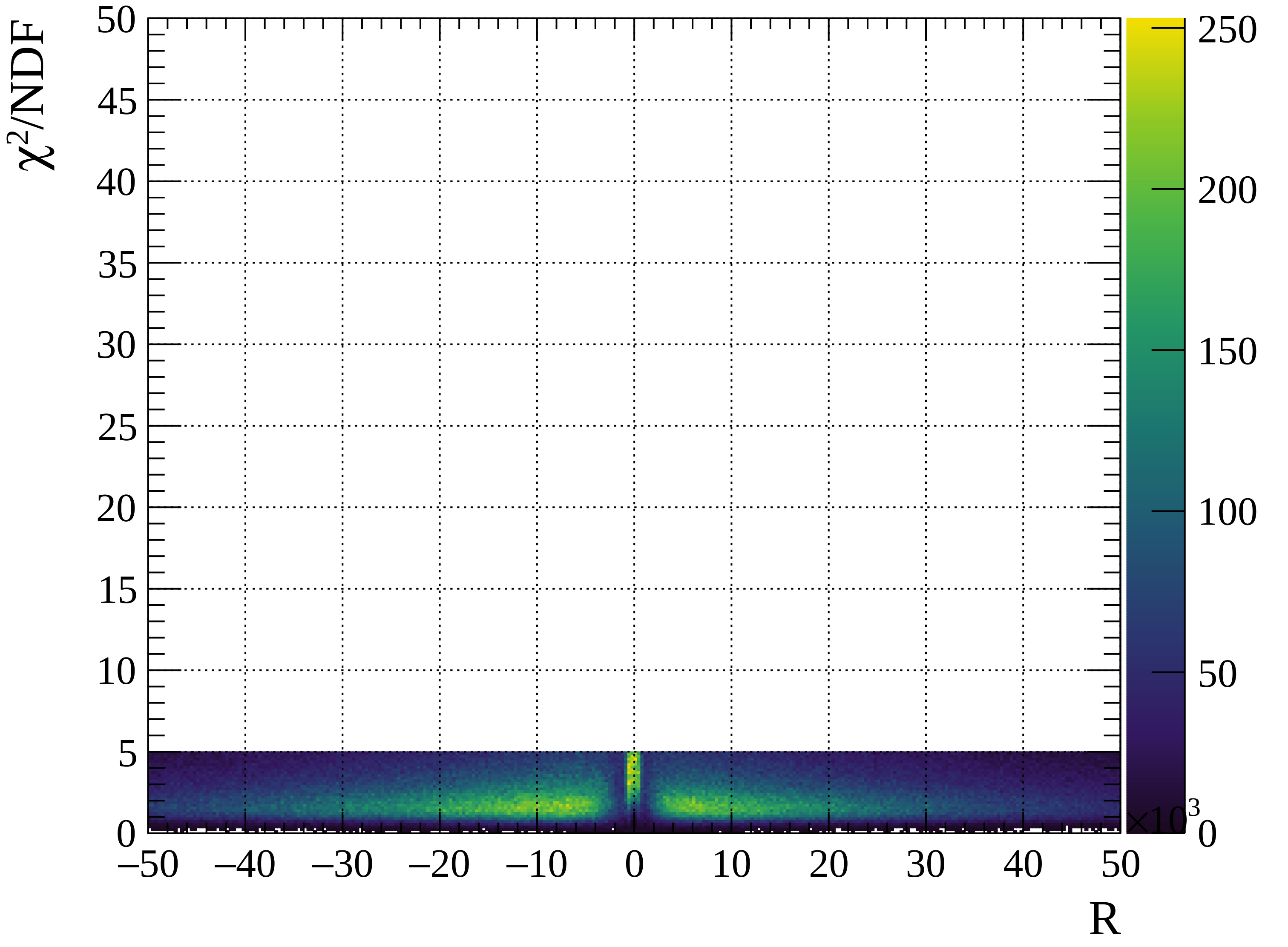

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600


×10
Count
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)

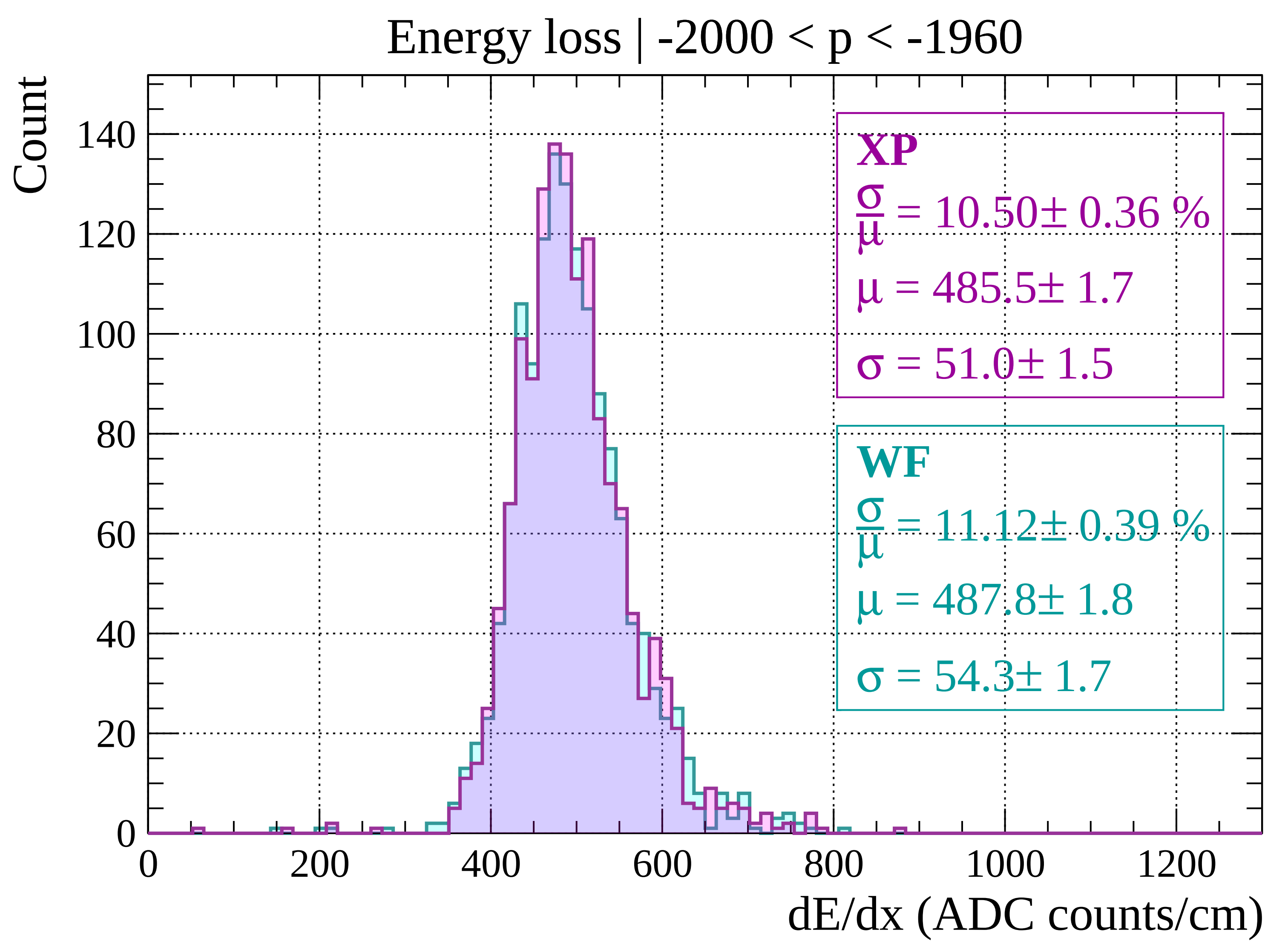

Count
×10
0
2
4
6
8
10
12
14
16
18
0
5
10
15
20
25
30
35
40
45
50
χ^2



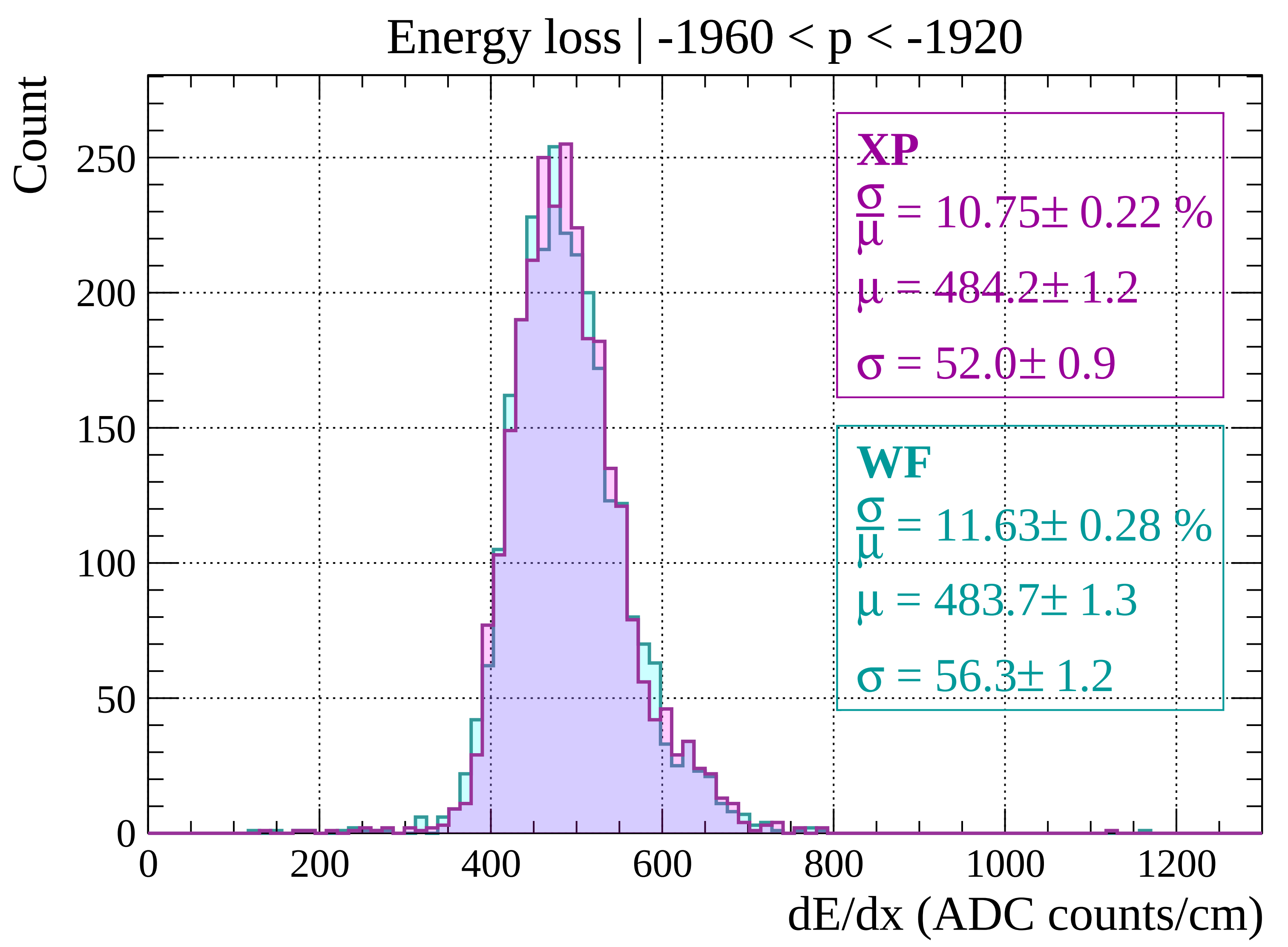
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
2500
2000
1500
1000
500
0
×10³



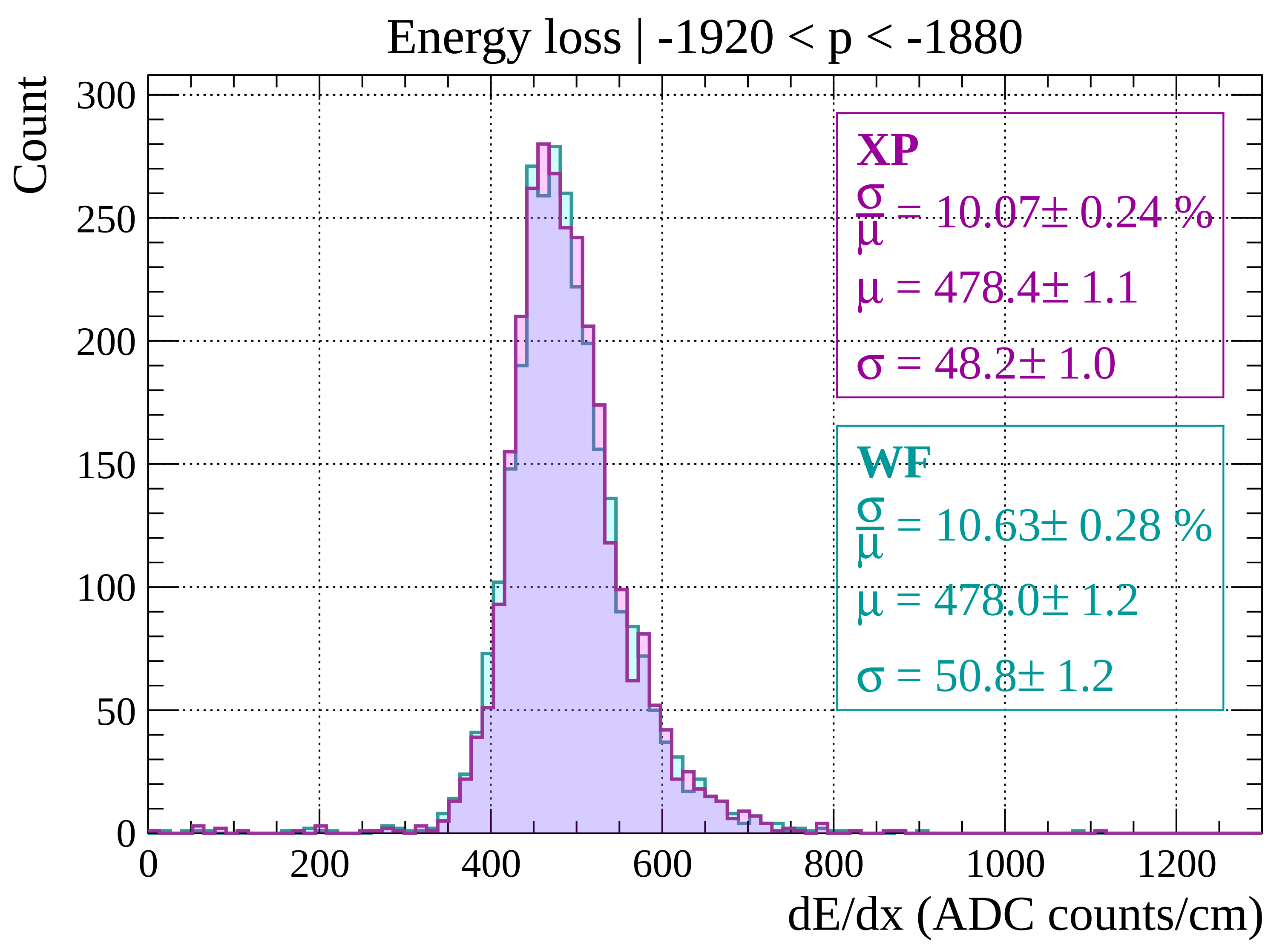
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
250
200
150
100
50
0
$\times 10^3$



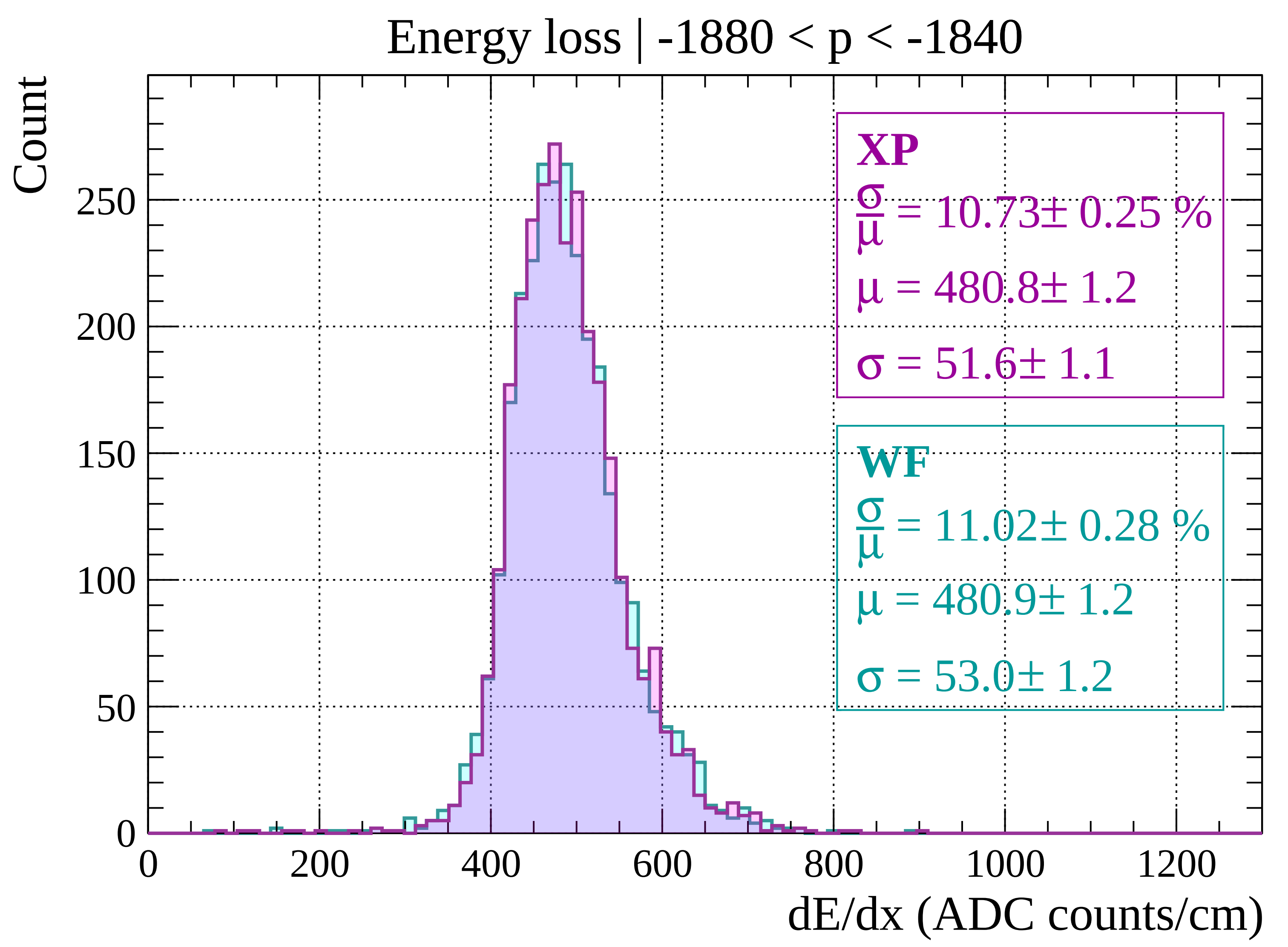
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 10.50± 0.36 %
μ = 485.5± 1.7
σ = 51.0± 1.5
WF
σ/μ = 11.12± 0.39 %
μ = 487.8± 1.8
σ = 54.3± 1.7



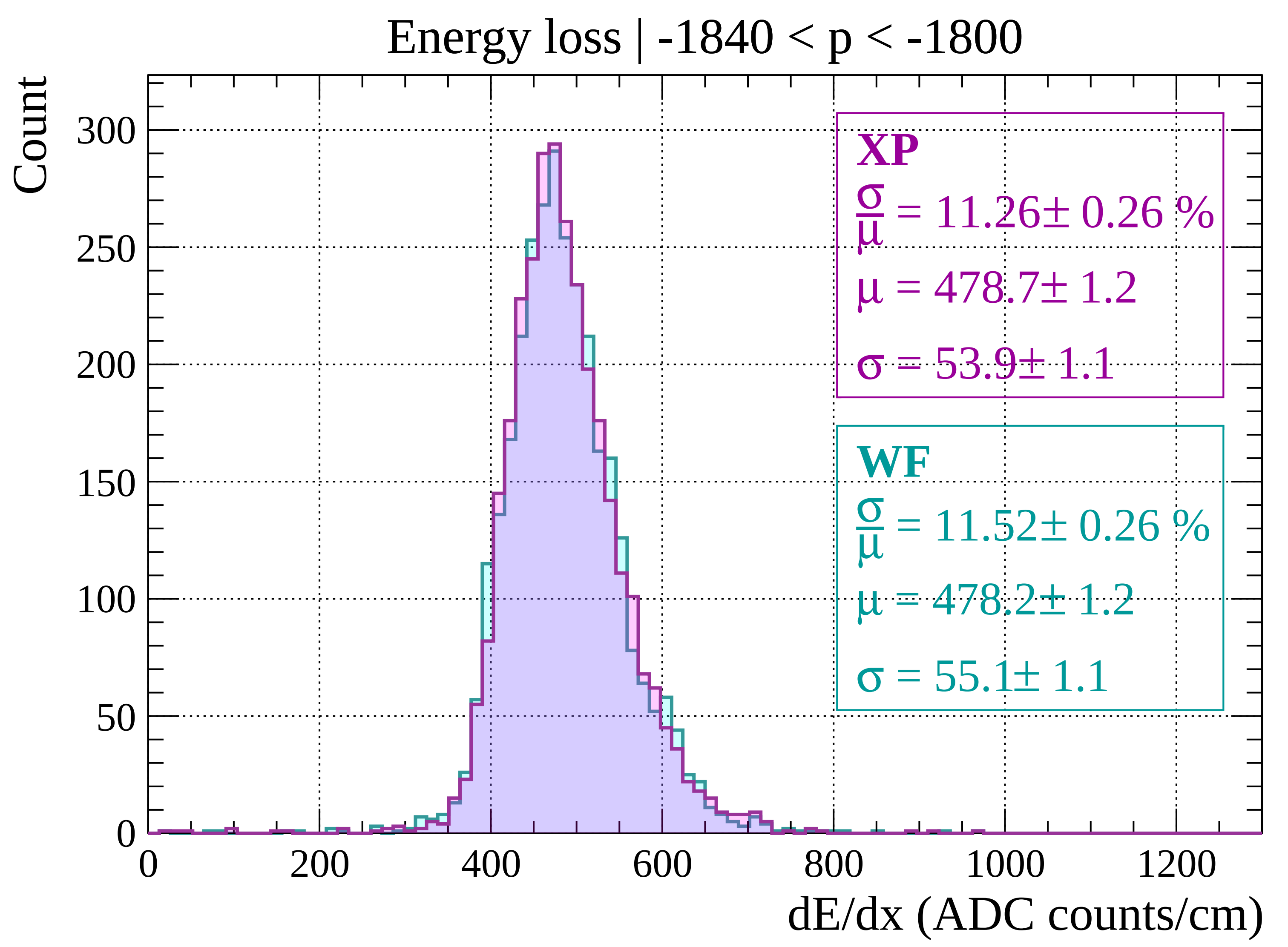
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 10.75± 0.22 %
μ = 484.2± 1.2
σ = 52.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 11.63± 0.28 %
μ = 483.7± 1.3
σ = 56.3± 1.2



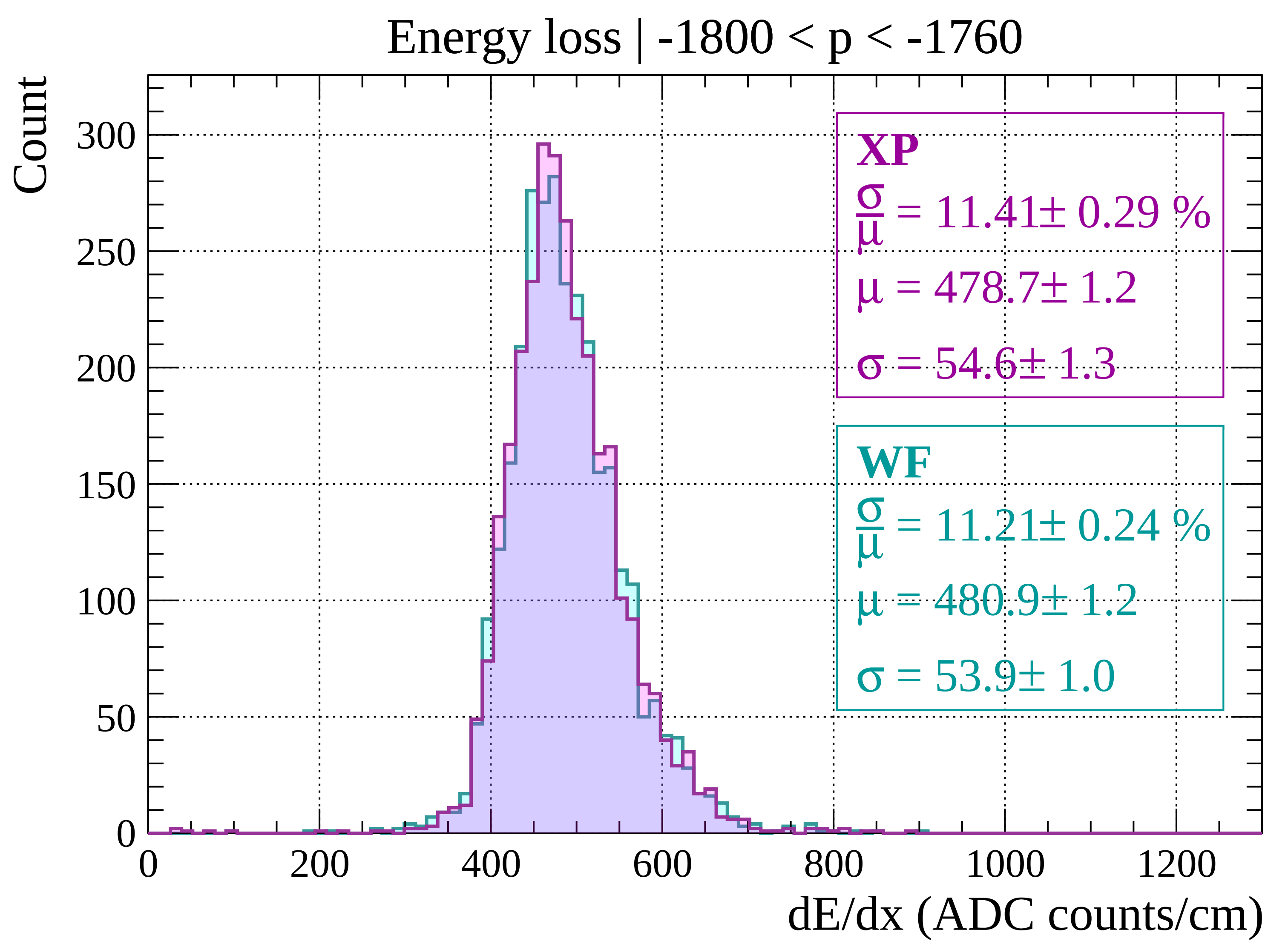
Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
σ/μ = 10.07± 0.24 %
μ = 478.4± 1.1
σ = 48.2± 1.0
WF
σ/μ = 10.63± 0.28 %
μ = 478.0± 1.2
σ = 50.8± 1.2



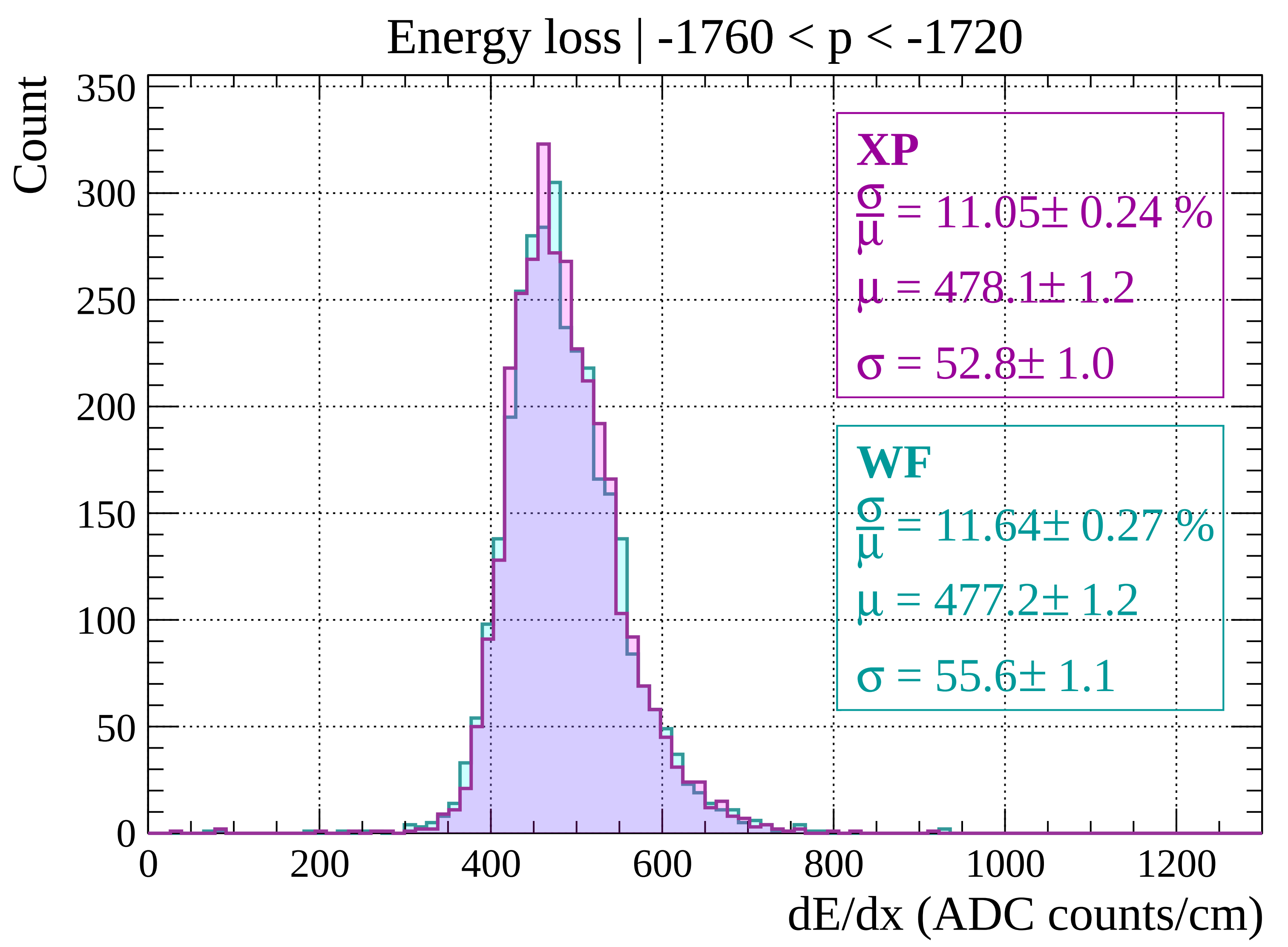
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.73± 0.25 %
μ = 480.8± 1.2
σ = 51.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.02± 0.28 %
μ = 480.9± 1.2
σ = 53.0± 1.2



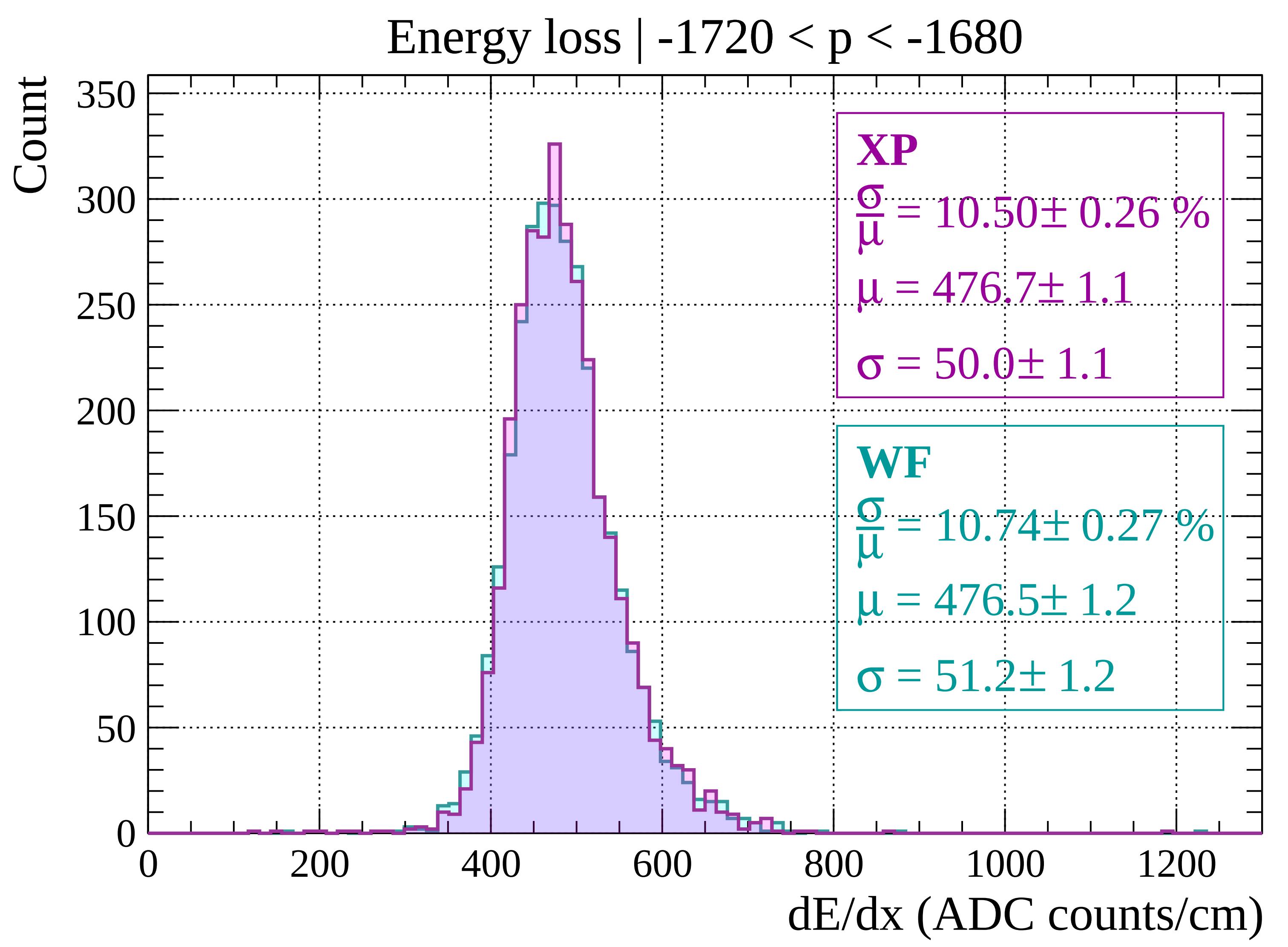
Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.26± 0.26 %
μ = 478.7± 1.2
σ = 53.9± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.52± 0.26 %
μ = 478.2± 1.2
σ = 55.1± 1.1



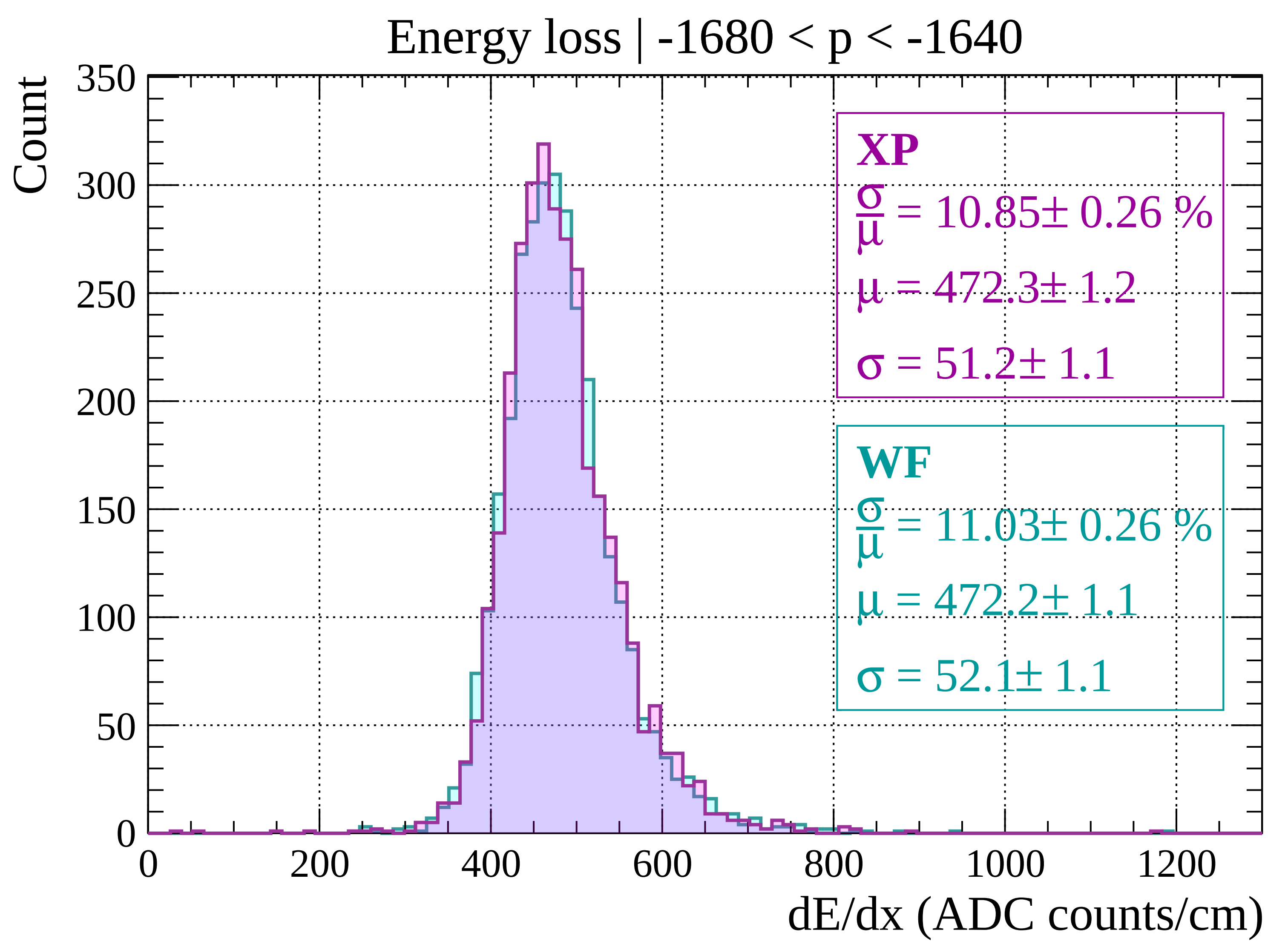
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.41± 0.29 %
μ = 478.7± 1.2
σ = 54.6± 1.3
WF
σ/μ = 11.21± 0.24 %
μ = 480.9± 1.2
σ = 53.9± 1.0



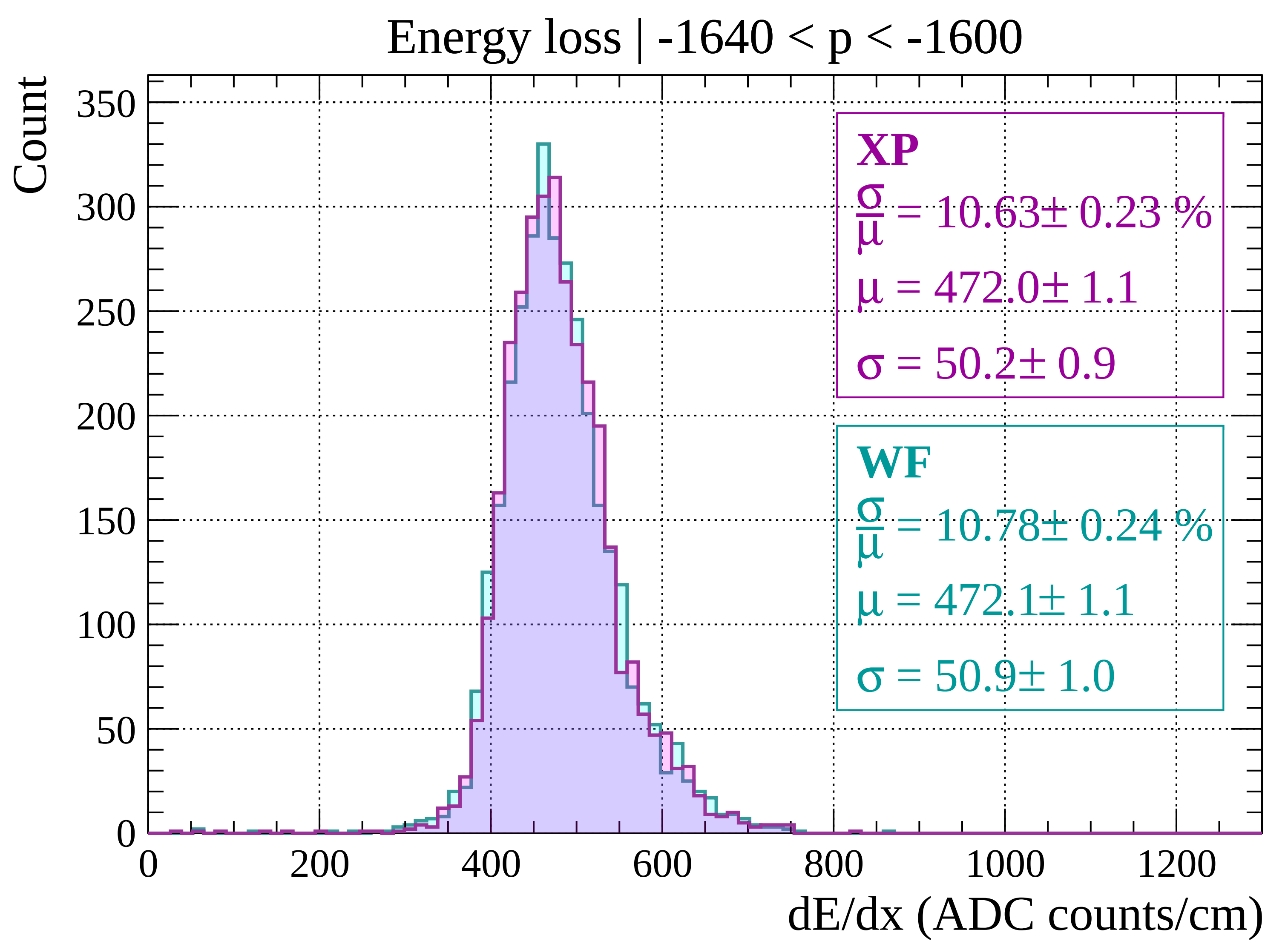
Energy loss | -1760 < p < -1720
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.05± 0.24 %
μ = 478.1± 1.2
σ = 52.8± 1.0
WF
σ/μ = 11.64± 0.27 %
μ = 477.2± 1.2
σ = 55.6± 1.1



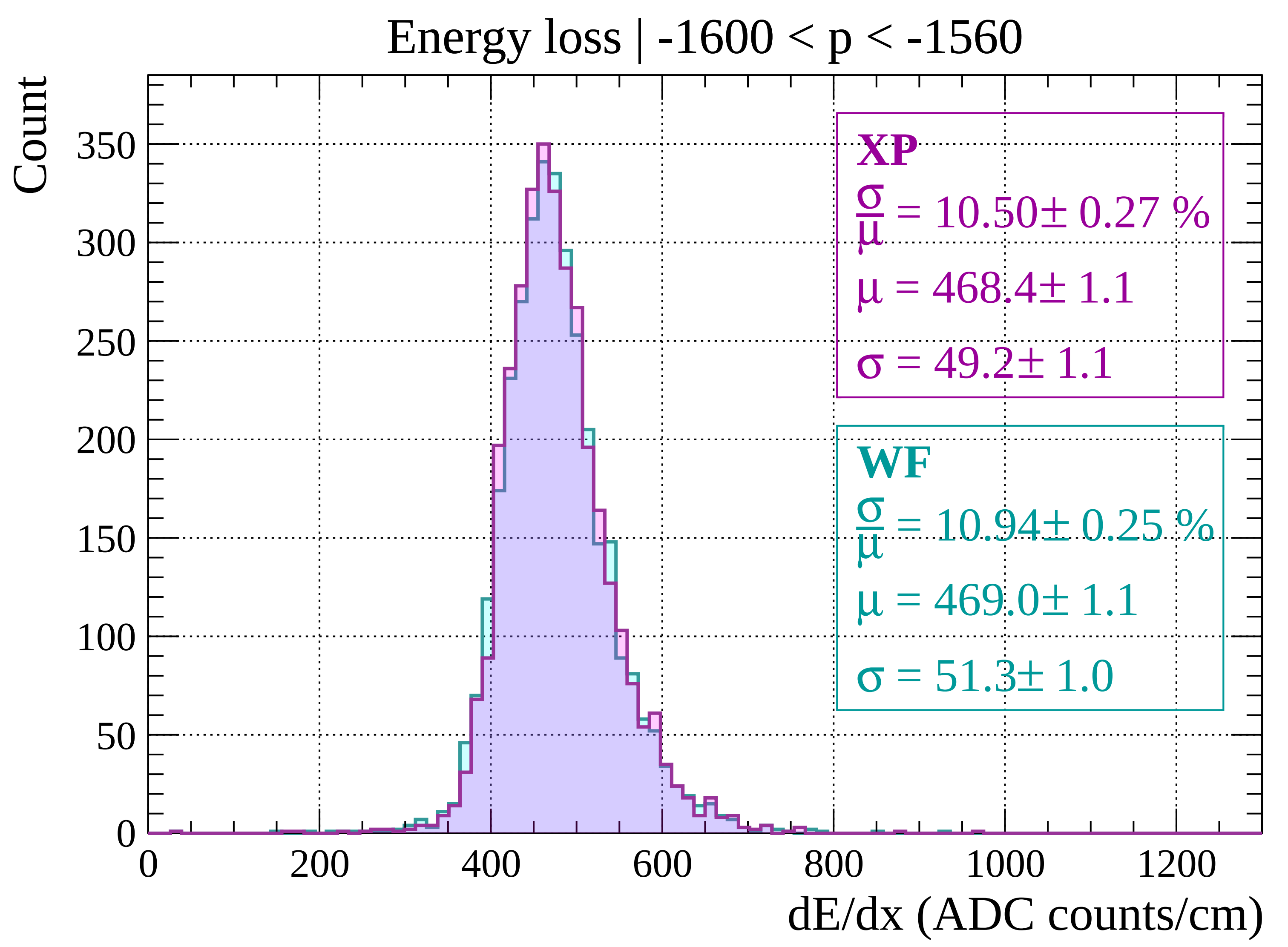
Energy loss | -1720 < p < -1680
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.50± 0.26 %
μ = 476.7± 1.1
σ = 50.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.74± 0.27 %
μ = 476.5± 1.2
σ = 51.2± 1.2



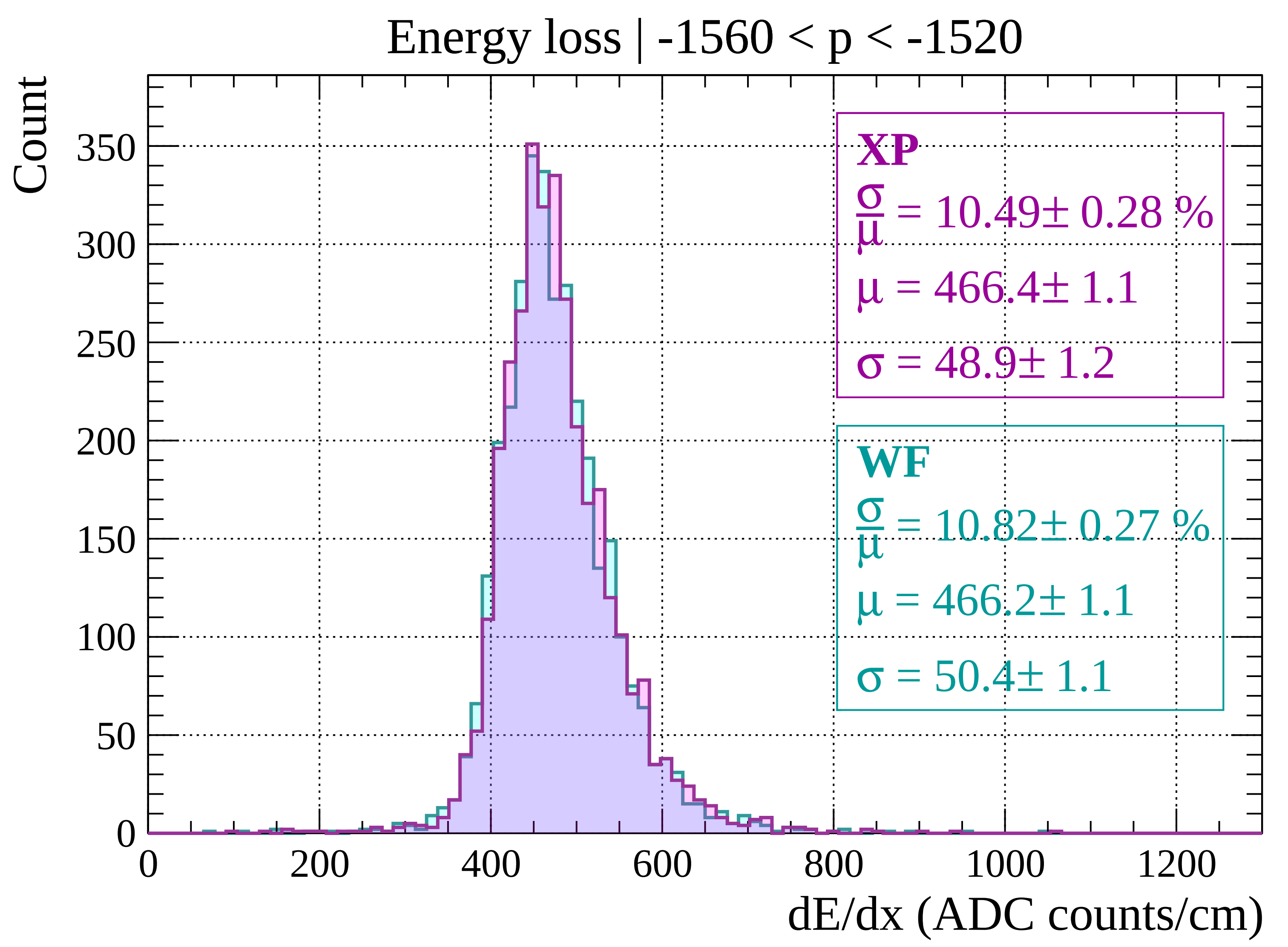
Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 10.85± 0.26 %
μ = 472.3± 1.2
σ = 51.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.03± 0.26 %
μ = 472.2± 1.1
σ = 52.1± 1.1



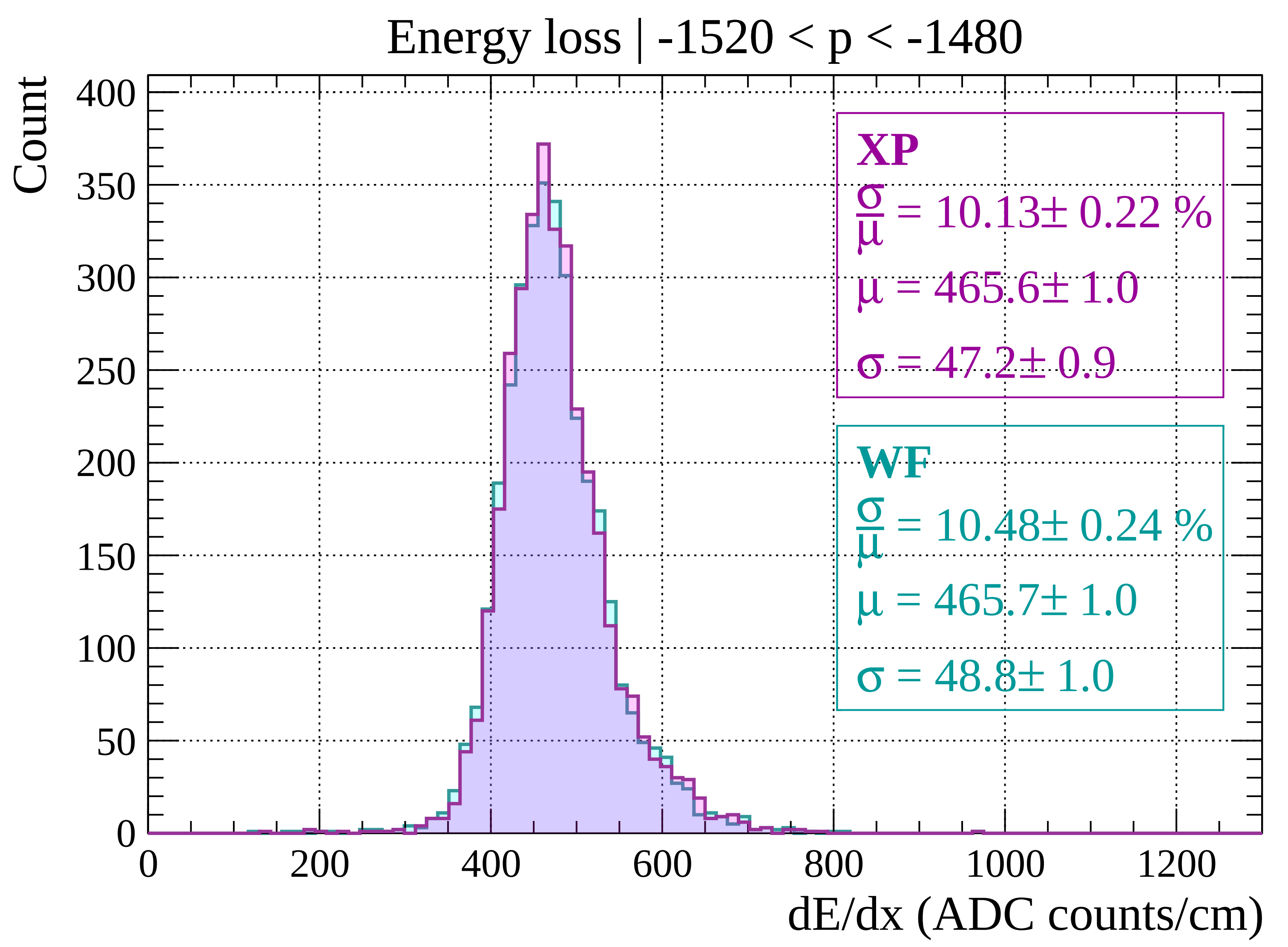
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 10.63± 0.23 %
μ = 472.0± 1.1
σ = 50.2± 0.9
WF
σ/μ = 10.78± 0.24 %
μ = 472.1± 1.1
σ = 50.9± 1.0

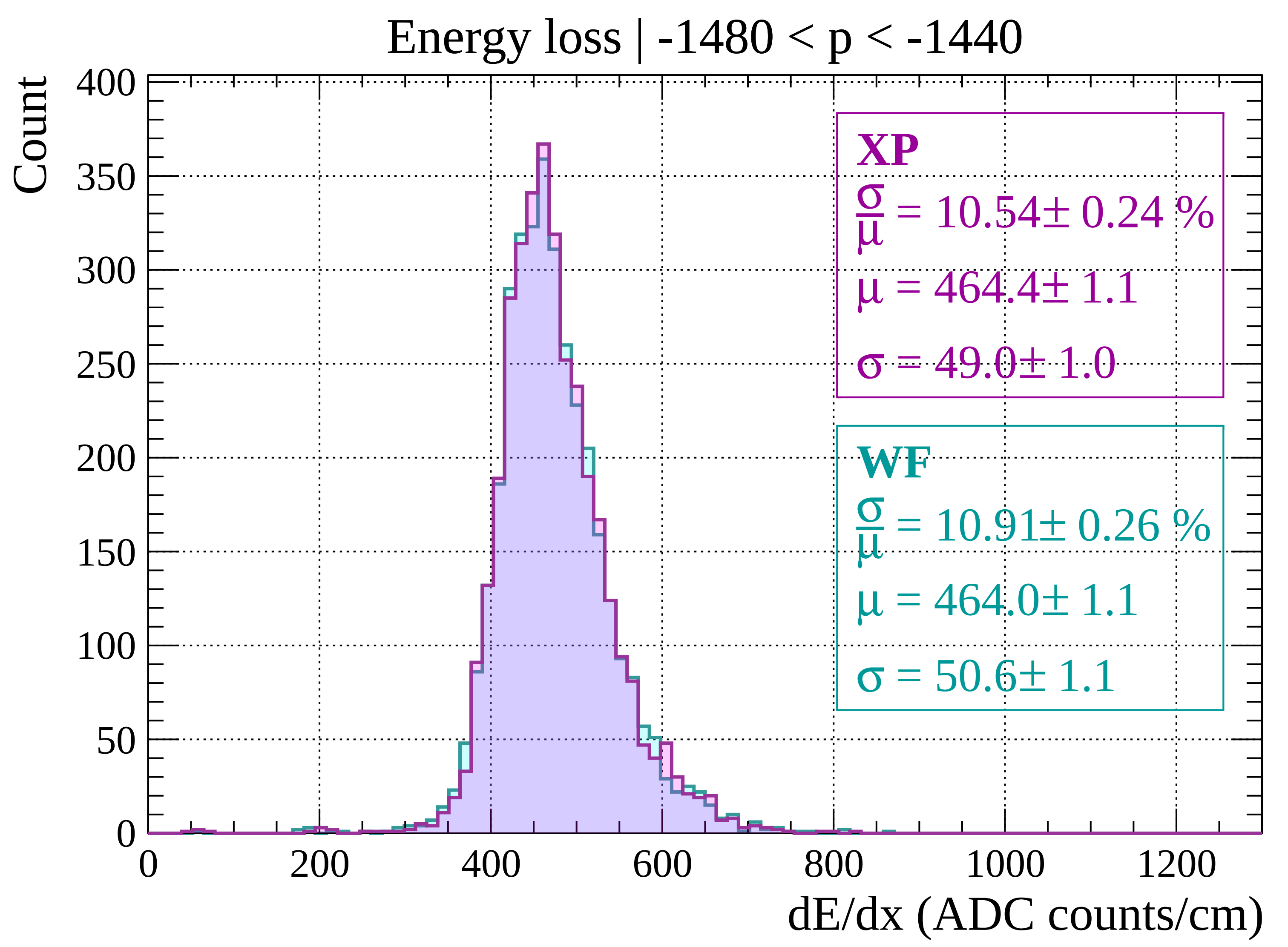


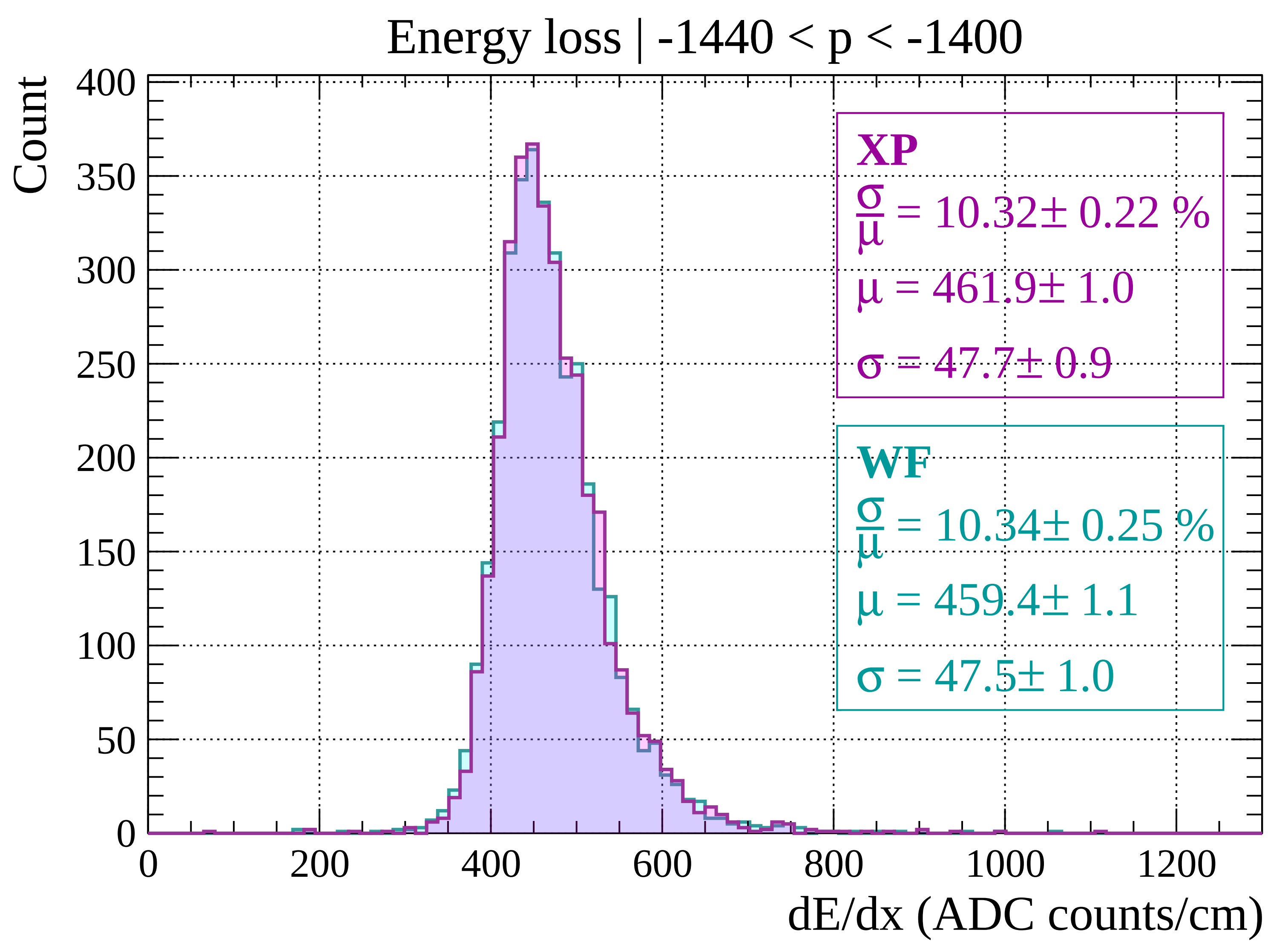
Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 10.50± 0.27 %
μ = 468.4± 1.1
σ = 49.2± 1.1
WF
σ/μ = 10.94± 0.25 %
μ = 469.0± 1.1
σ = 51.3± 1.0

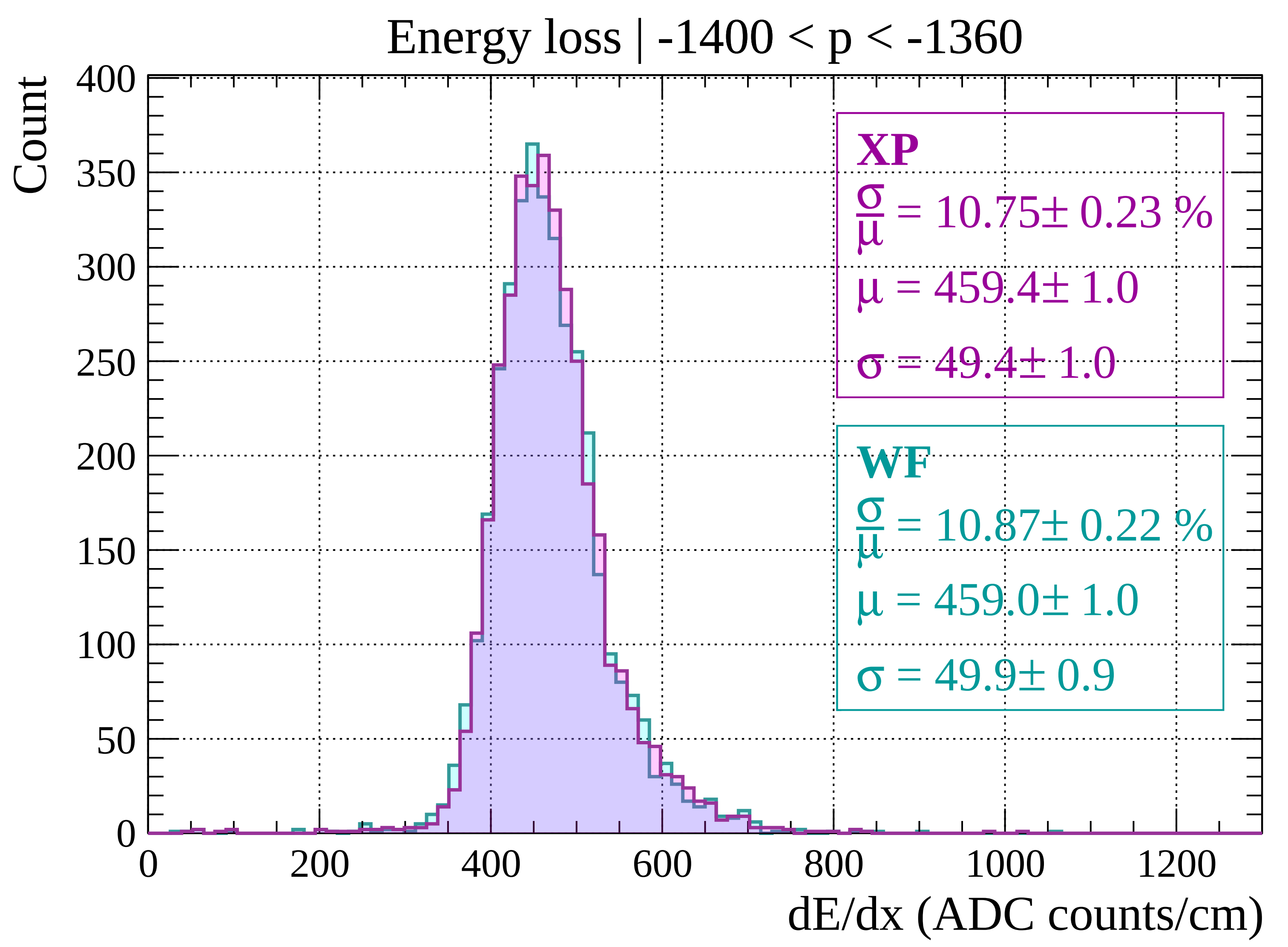


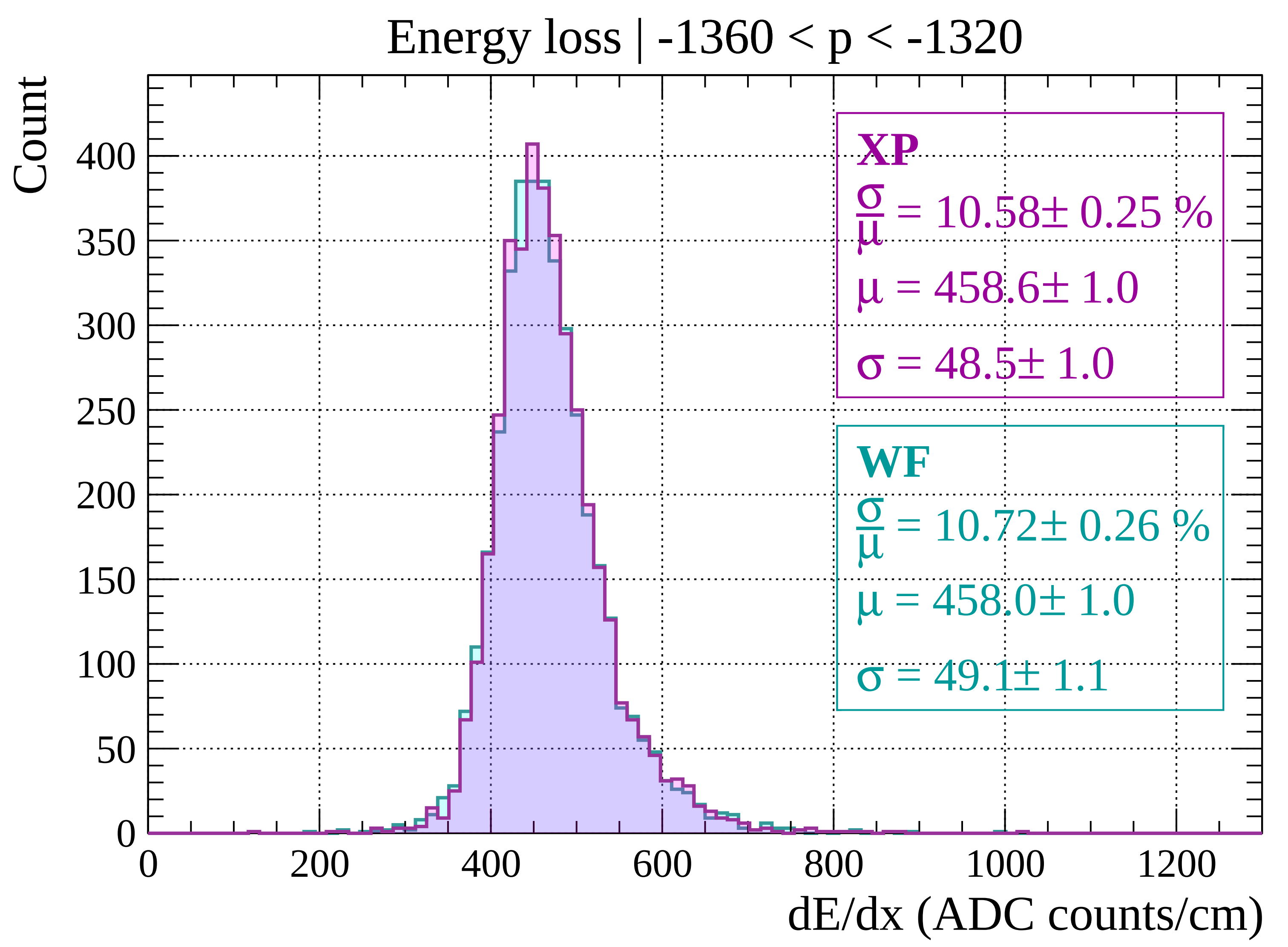
Energy loss | -1560 < p < -1520
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.49± 0.28 %
μ = 466.4± 1.1
σ = 48.9± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.82± 0.27 %
μ = 466.2± 1.1
σ = 50.4± 1.1

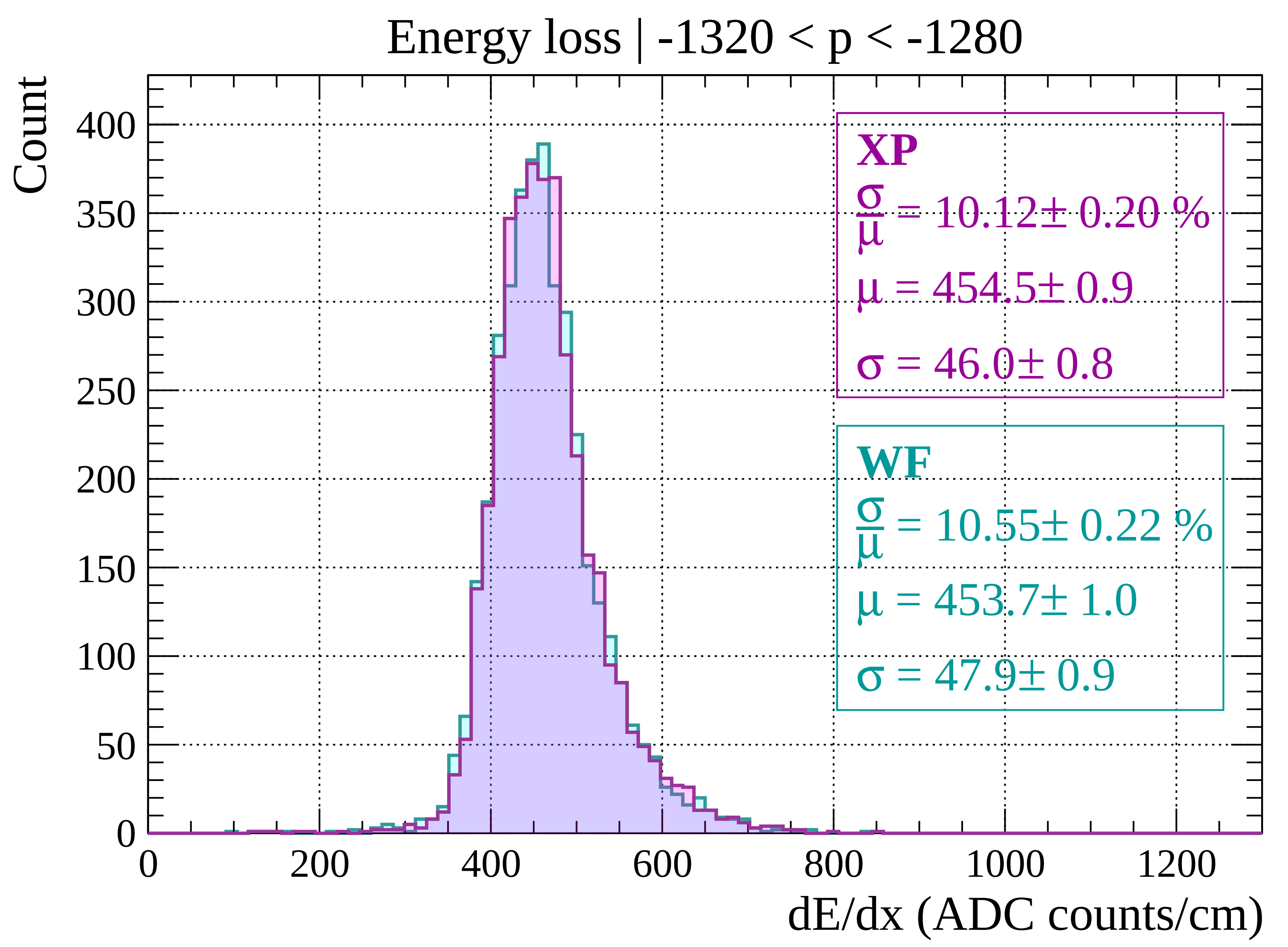

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.13± 0.22 %
μ = 465.6± 1.0
σ = 47.2± 0.9
WF
σ/μ = 10.48± 0.24 %
μ = 465.7± 1.0
σ = 48.8± 1.0

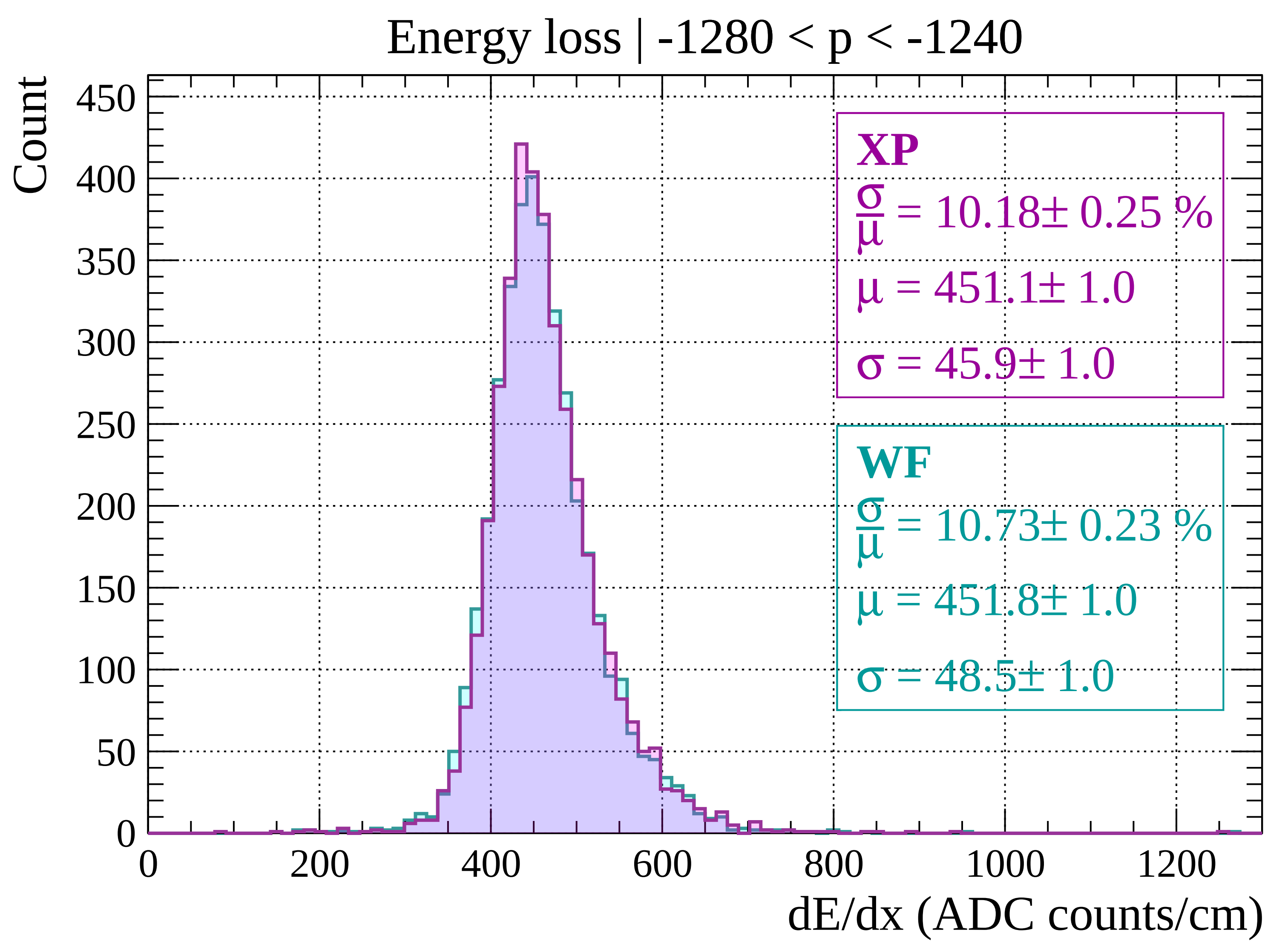

Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 10.54± 0.24 %
μ = 464.4± 1.1
σ = 49.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.91± 0.26 %
μ = 464.0± 1.1
σ = 50.6± 1.1


Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.32± 0.22 %
μ = 461.9± 1.0
σ = 47.7± 0.9
WF
σ/μ = 10.34± 0.25 %
μ = 459.4± 1.1
σ = 47.5± 1.0

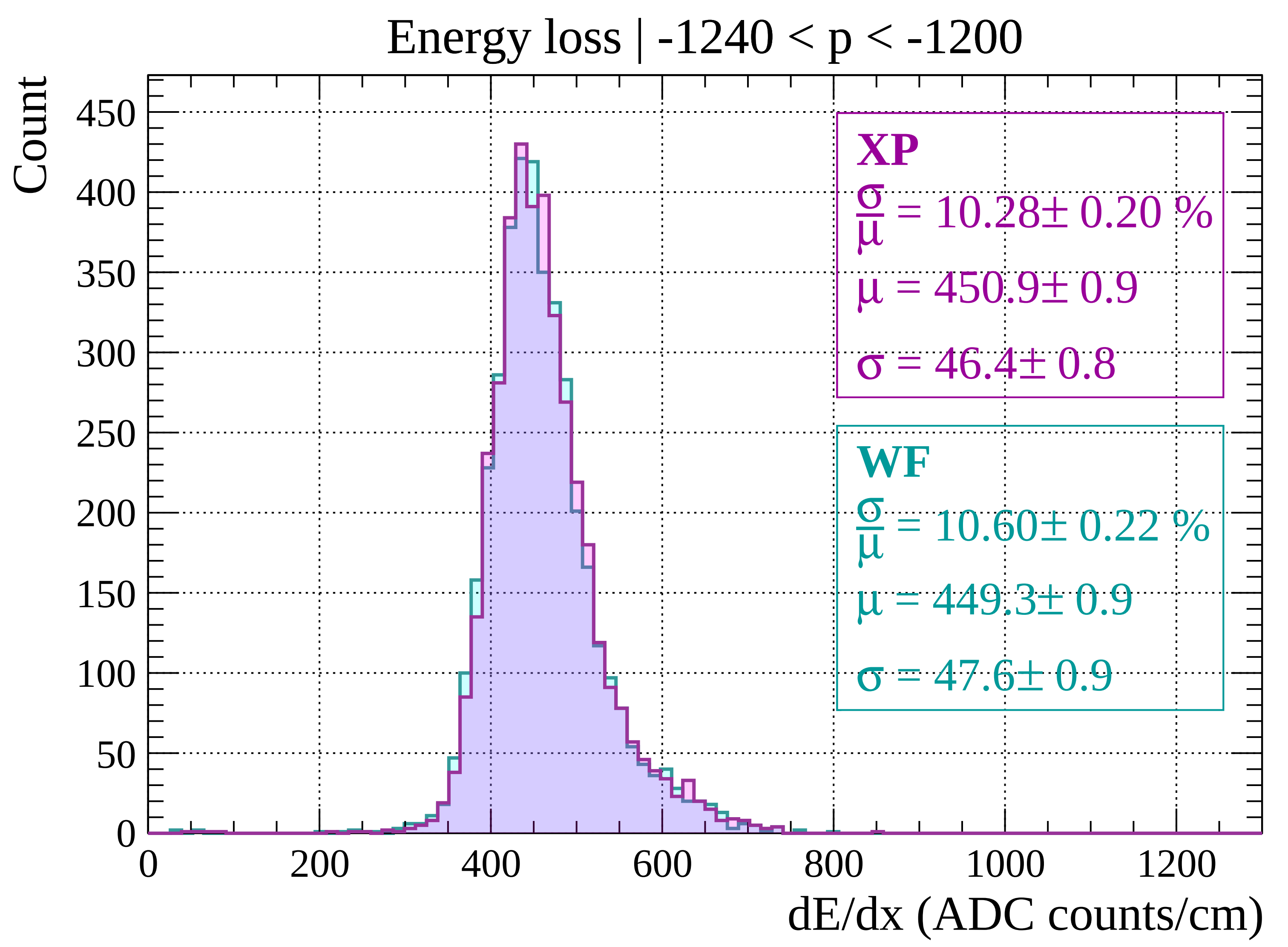

Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.75± 0.23 %
μ = 459.4± 1.0
σ = 49.4± 1.0
WF
σ/μ = 10.87± 0.22 %
μ = 459.0± 1.0
σ = 49.9± 0.9


Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.58± 0.25 %
μ = 458.6± 1.0
σ = 48.5± 1.0
WF
σ/μ = 10.72± 0.26 %
μ = 458.0± 1.0
σ = 49.1± 1.1

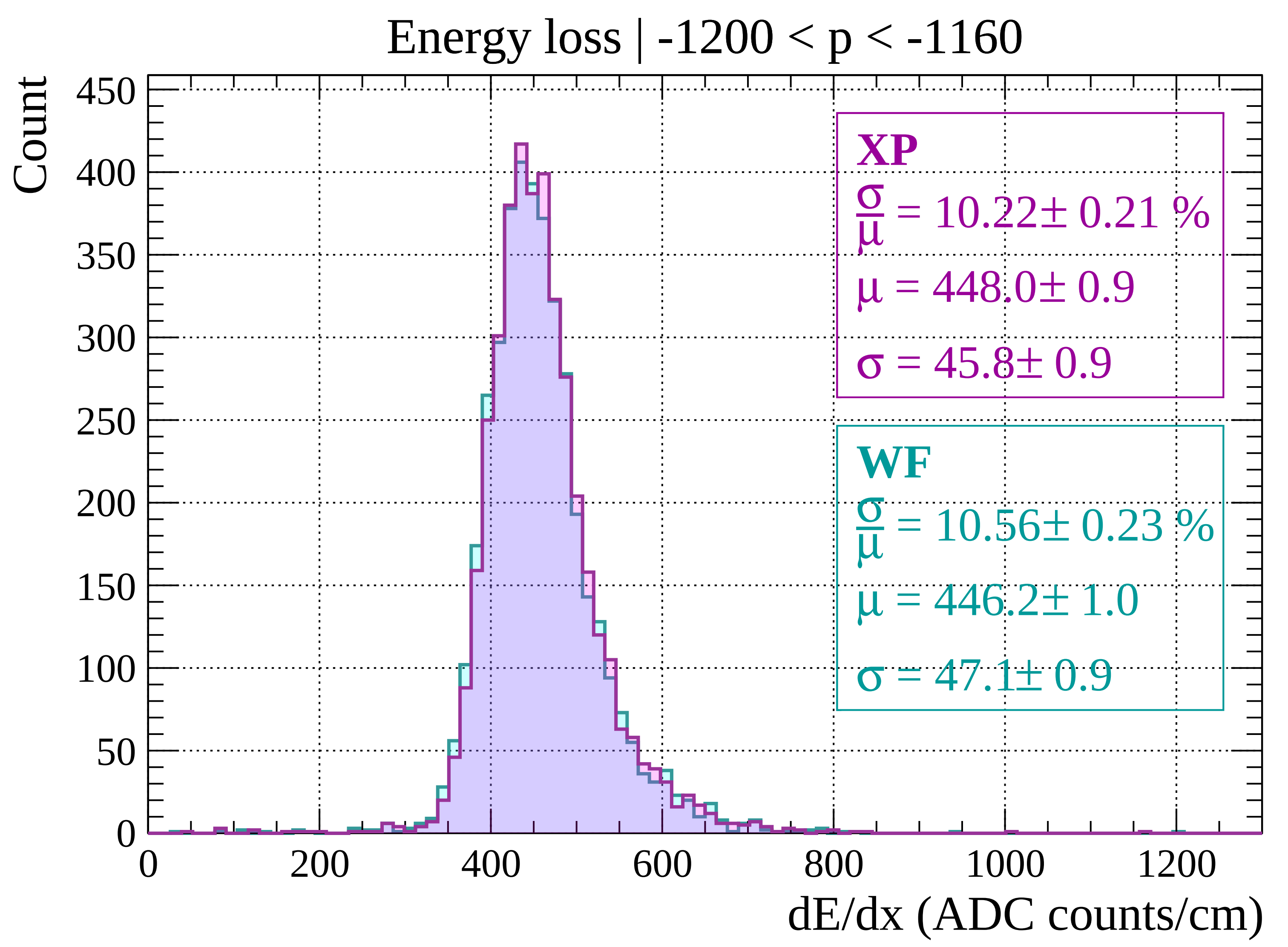

Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.12± 0.20 %
μ = 454.5± 0.9
σ = 46.0± 0.8
WF
σ/μ = 10.55± 0.22 %
μ = 453.7± 1.0
σ = 47.9± 0.9



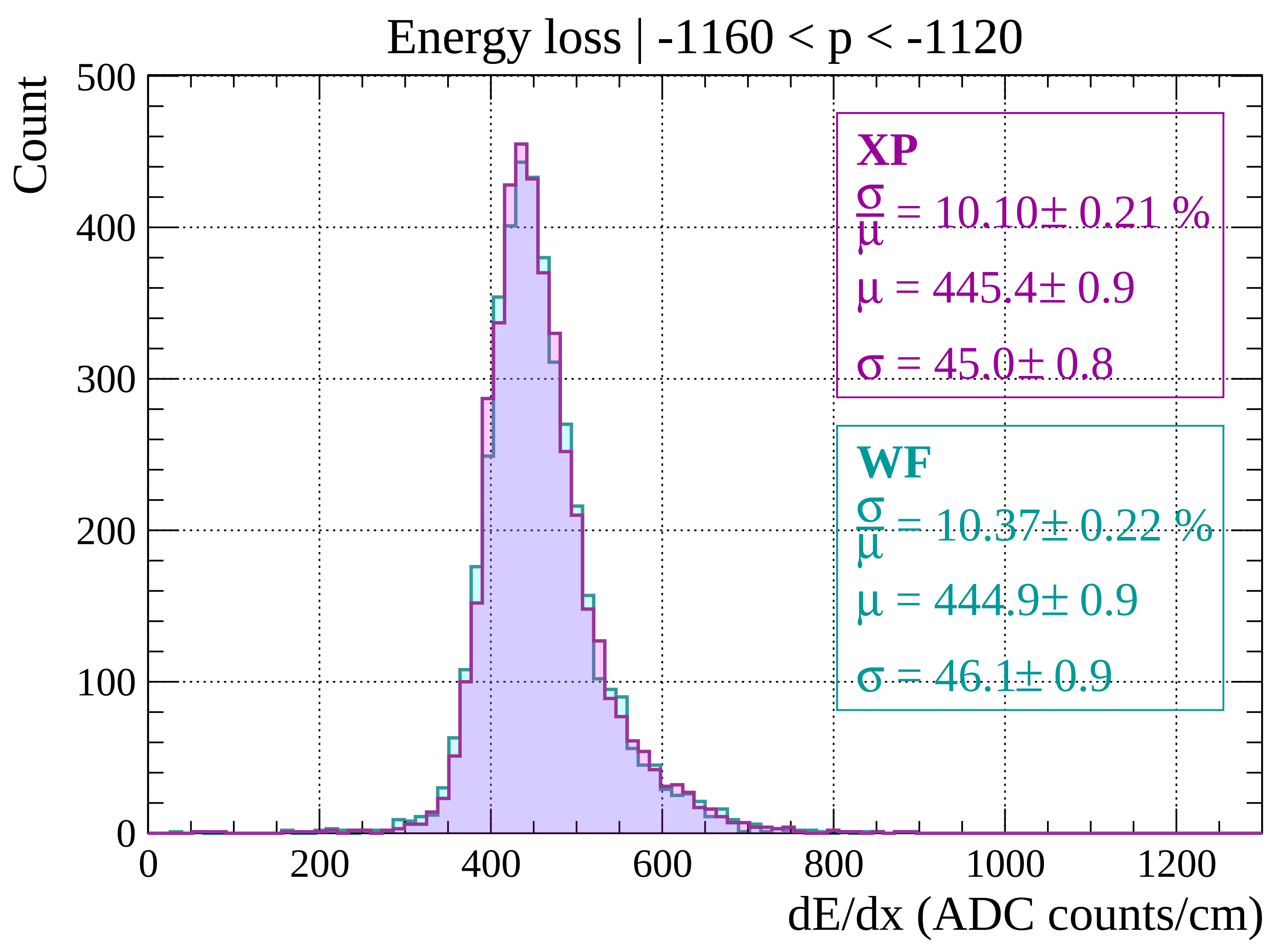
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 10.18± 0.25 %
μ = 451.1± 1.0
σ = 45.9± 1.0
WF
σ/μ = 10.73± 0.23 %
μ = 451.8± 1.0
σ = 48.5± 1.0



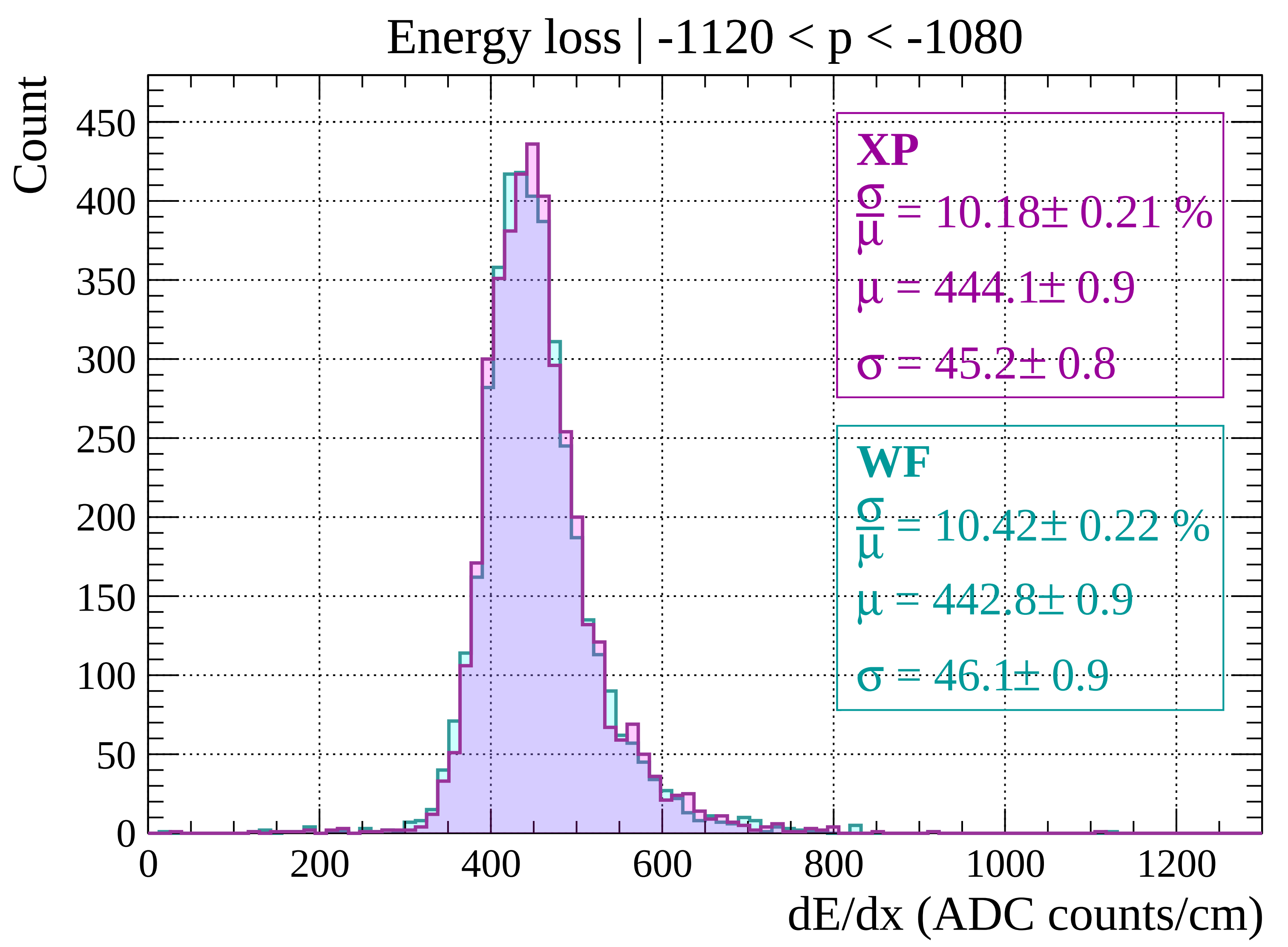
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 10.28± 0.20 %
μ = 450.9± 0.9
σ = 46.4± 0.8
WF
σ/μ = 10.60± 0.22 %
μ = 449.3± 0.9
σ = 47.6± 0.9

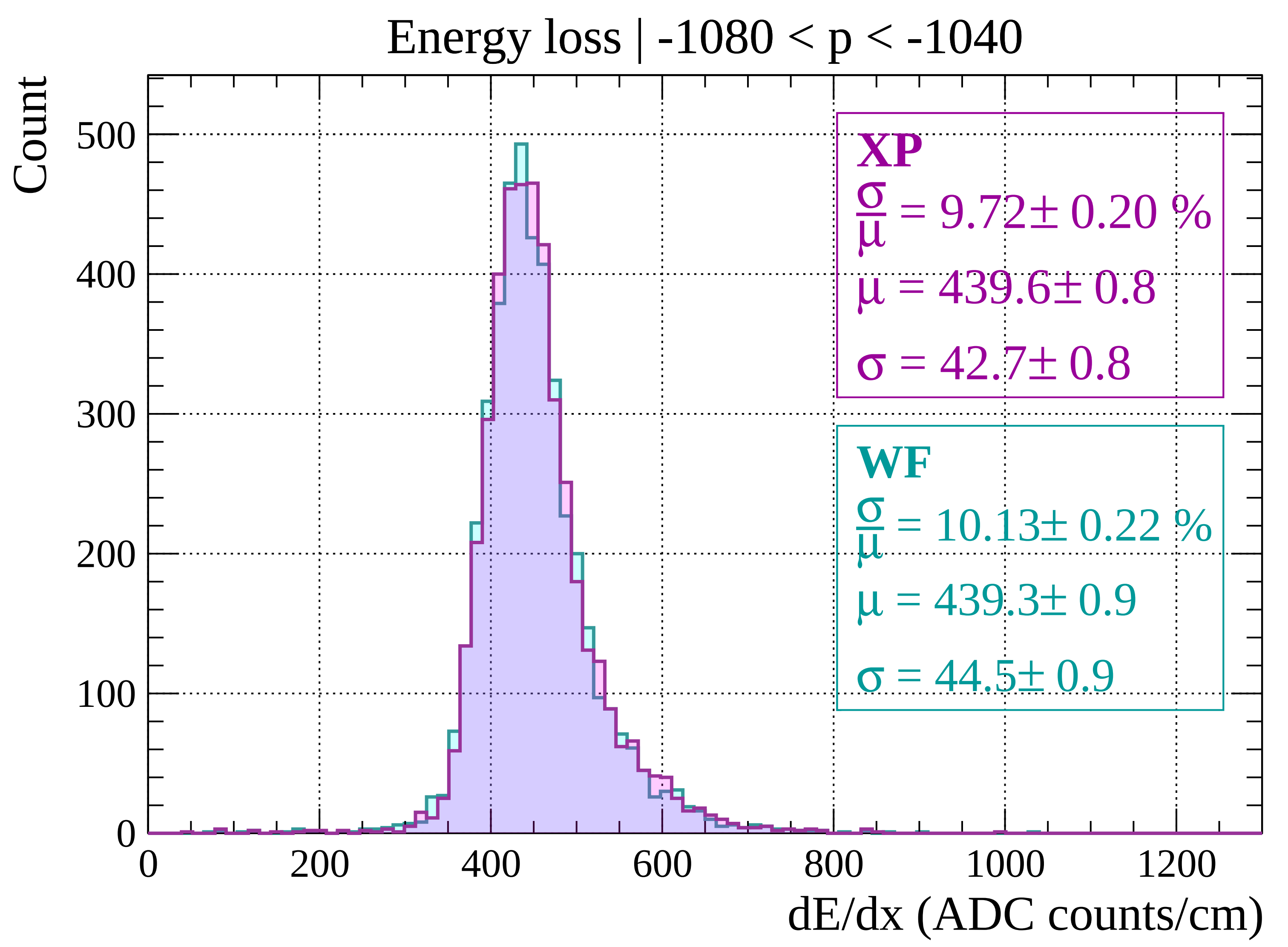


Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.22± 0.21 %
μ = 448.0± 0.9
σ = 45.8± 0.9
WF
σ/μ = 10.56± 0.23 %
μ = 446.2± 1.0
σ = 47.1± 0.9

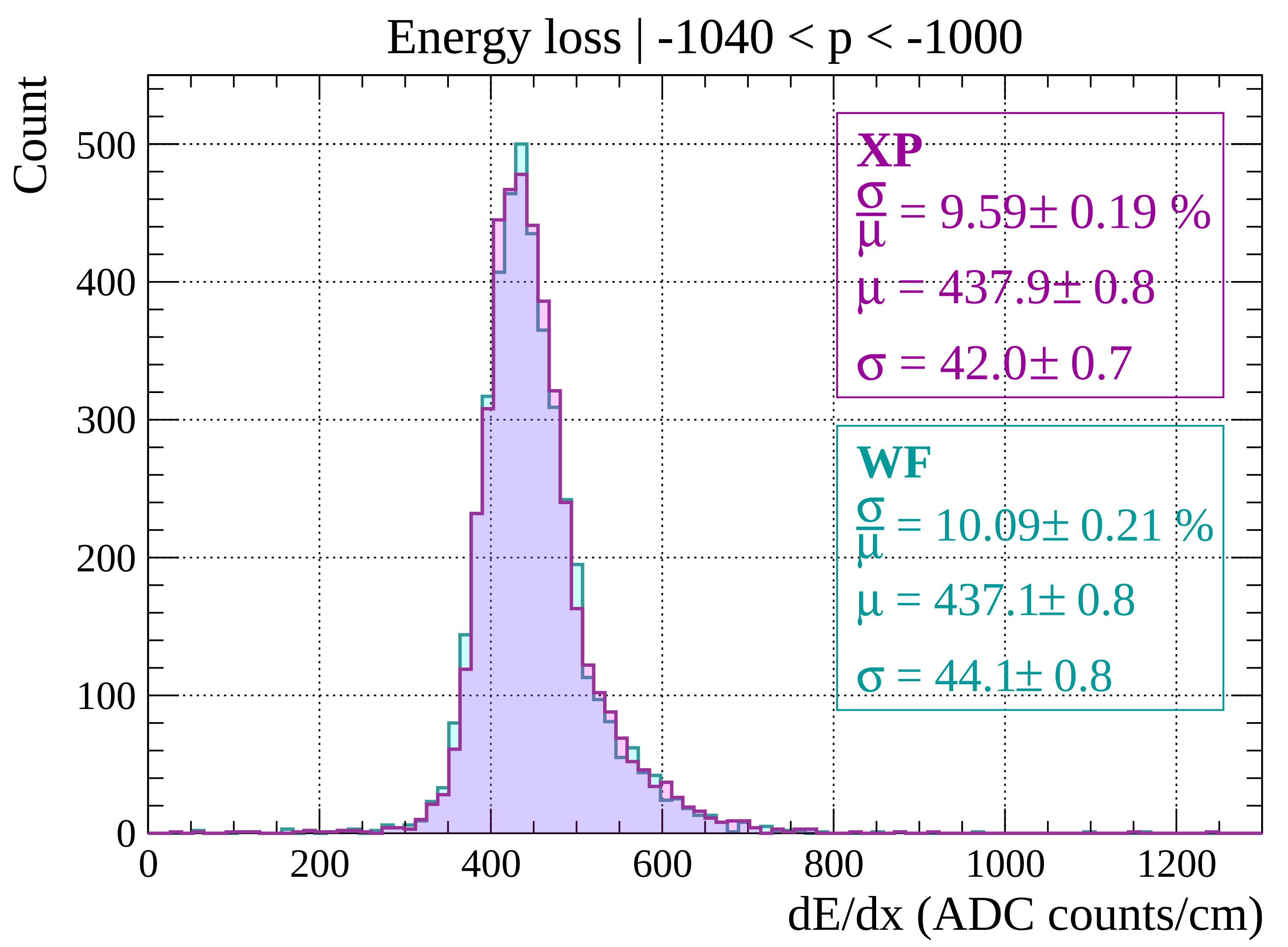


Energy loss | -1160 < p < -1120
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.10± 0.21 %
μ = 445.4± 0.9
σ = 45.0± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.37± 0.22 %
μ = 444.9± 0.9
σ = 46.1± 0.9

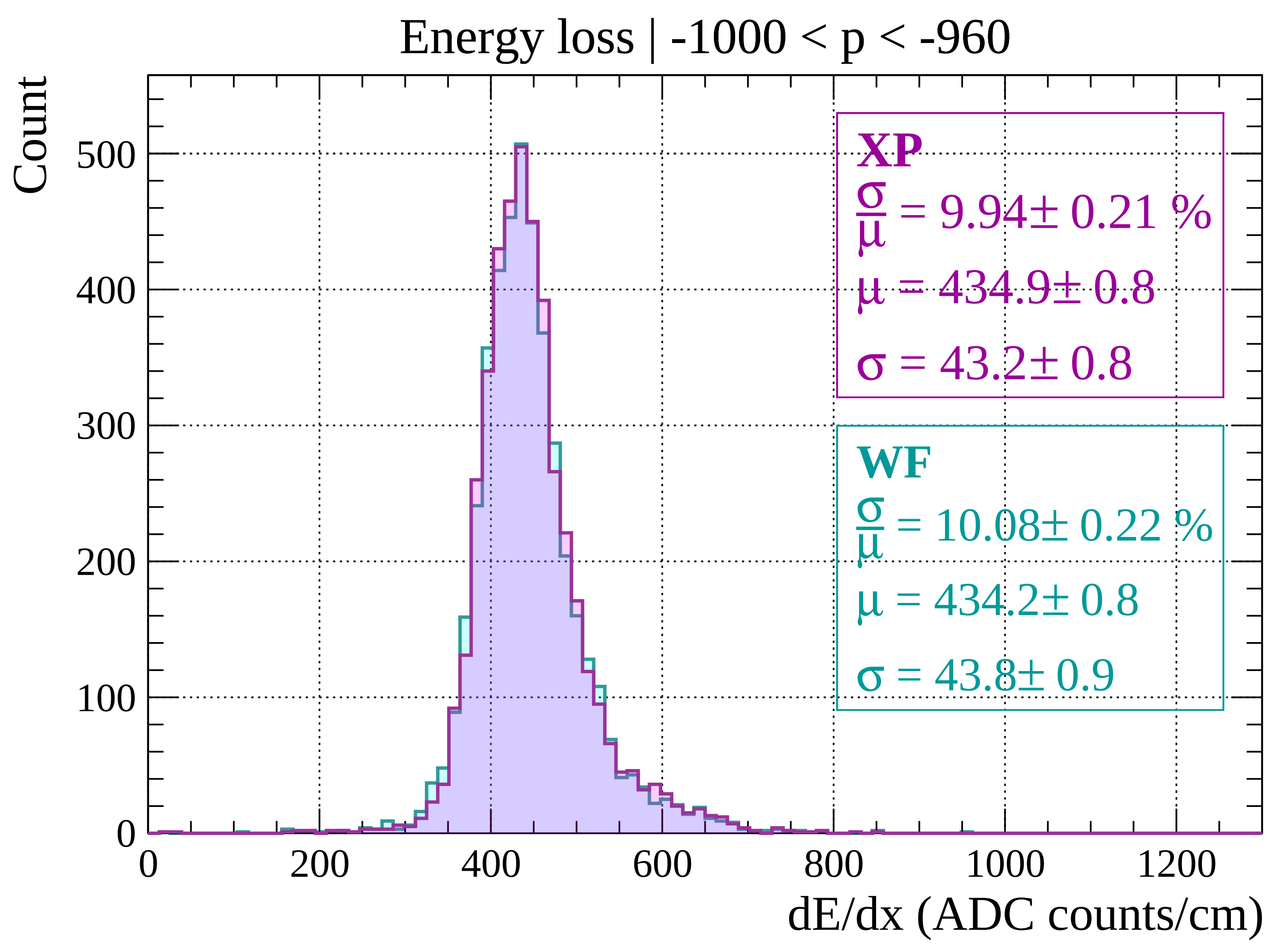

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.18± 0.21 %
μ = 444.1± 0.9
σ = 45.2± 0.8
WF
σ/μ = 10.42± 0.22 %
μ = 442.8± 0.9
σ = 46.1± 0.9



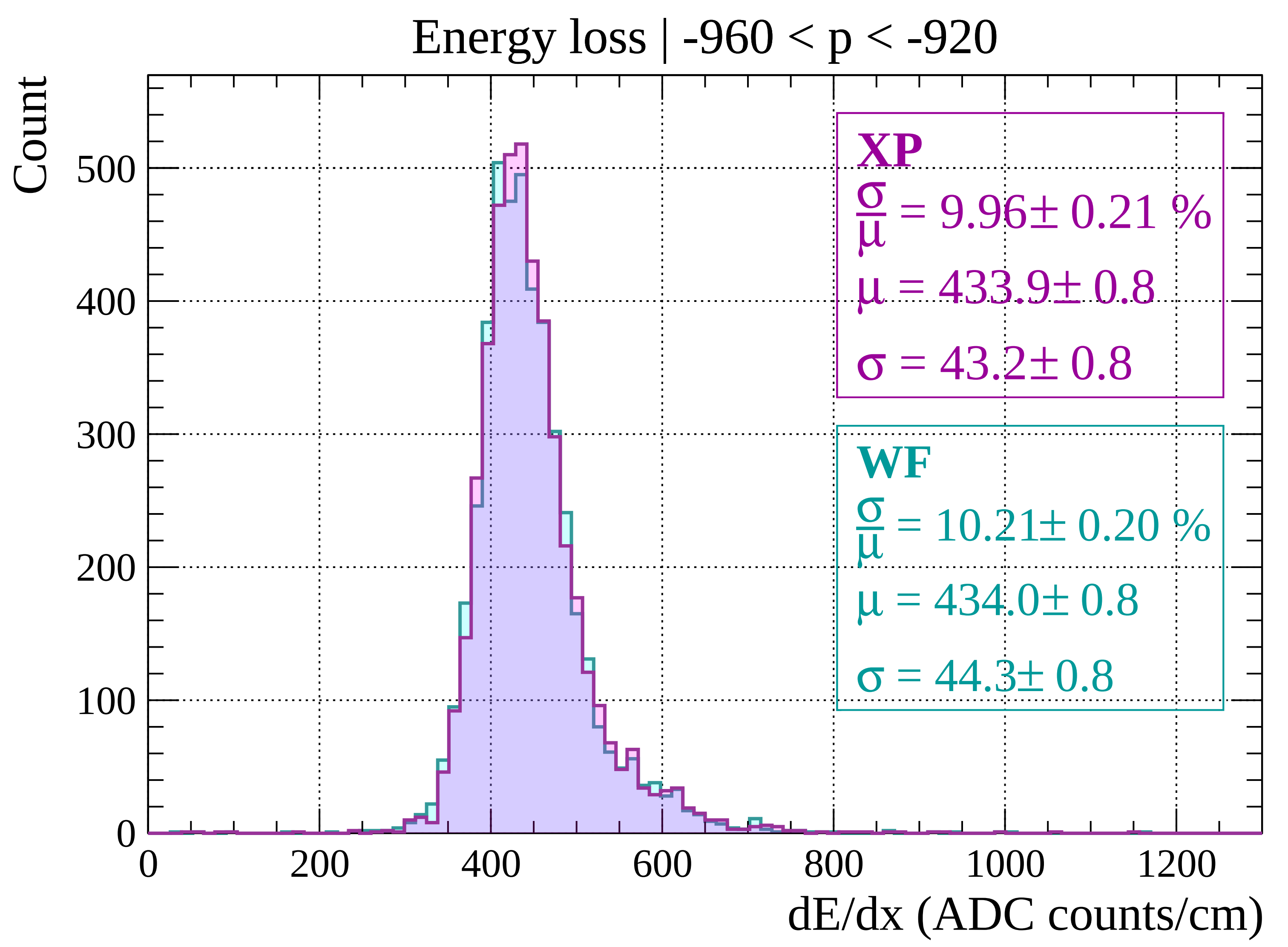
Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.72± 0.20 %
μ = 439.6± 0.8
σ = 42.7± 0.8
WF
σ/μ = 10.13± 0.22 %
μ = 439.3± 0.9
σ = 44.5± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200



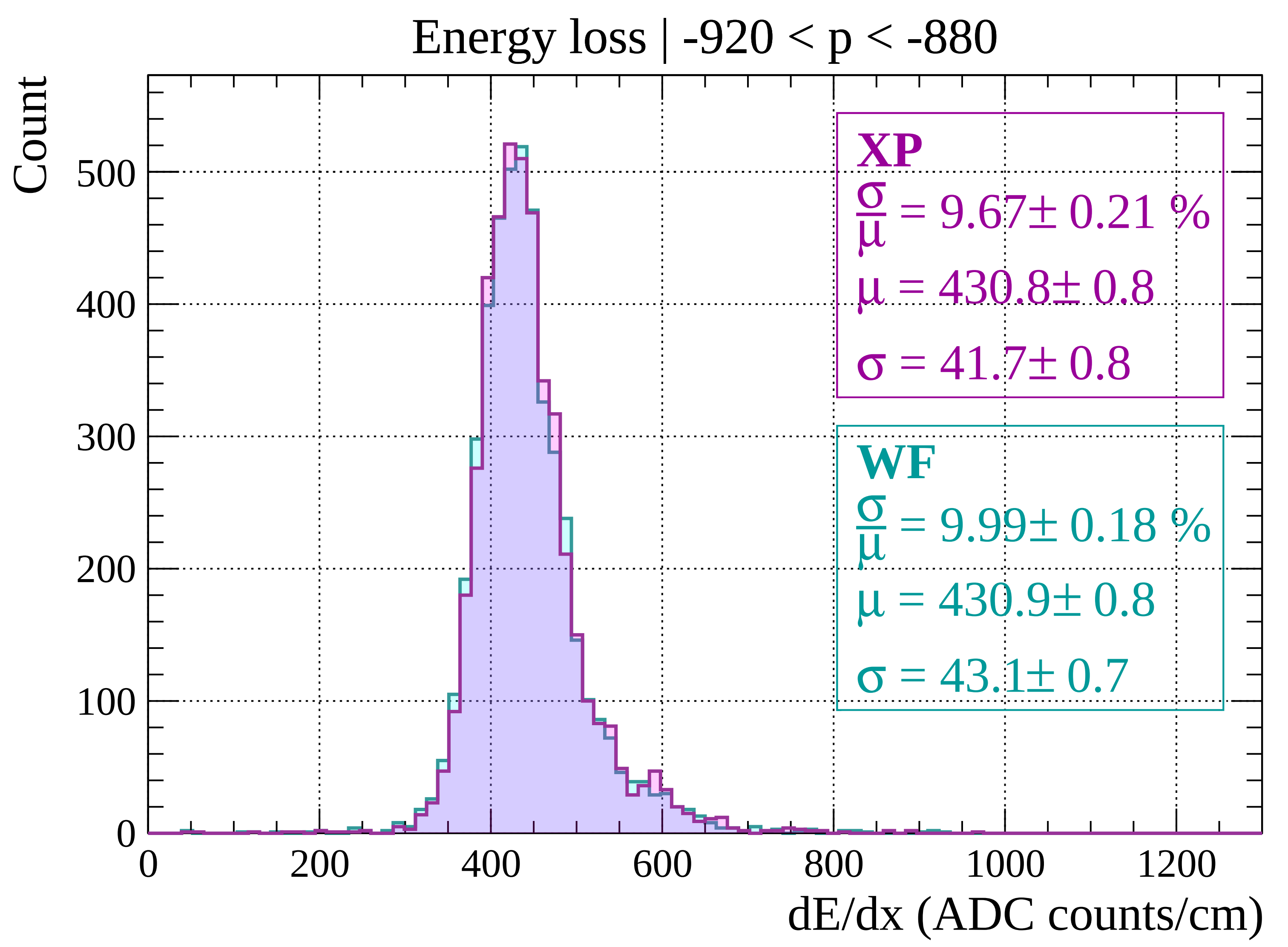
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.59± 0.19 %
μ = 437.9± 0.8
σ = 42.0± 0.7
WF
σ/μ = 10.09± 0.21 %
μ = 437.1± 0.8
σ = 44.1± 0.8

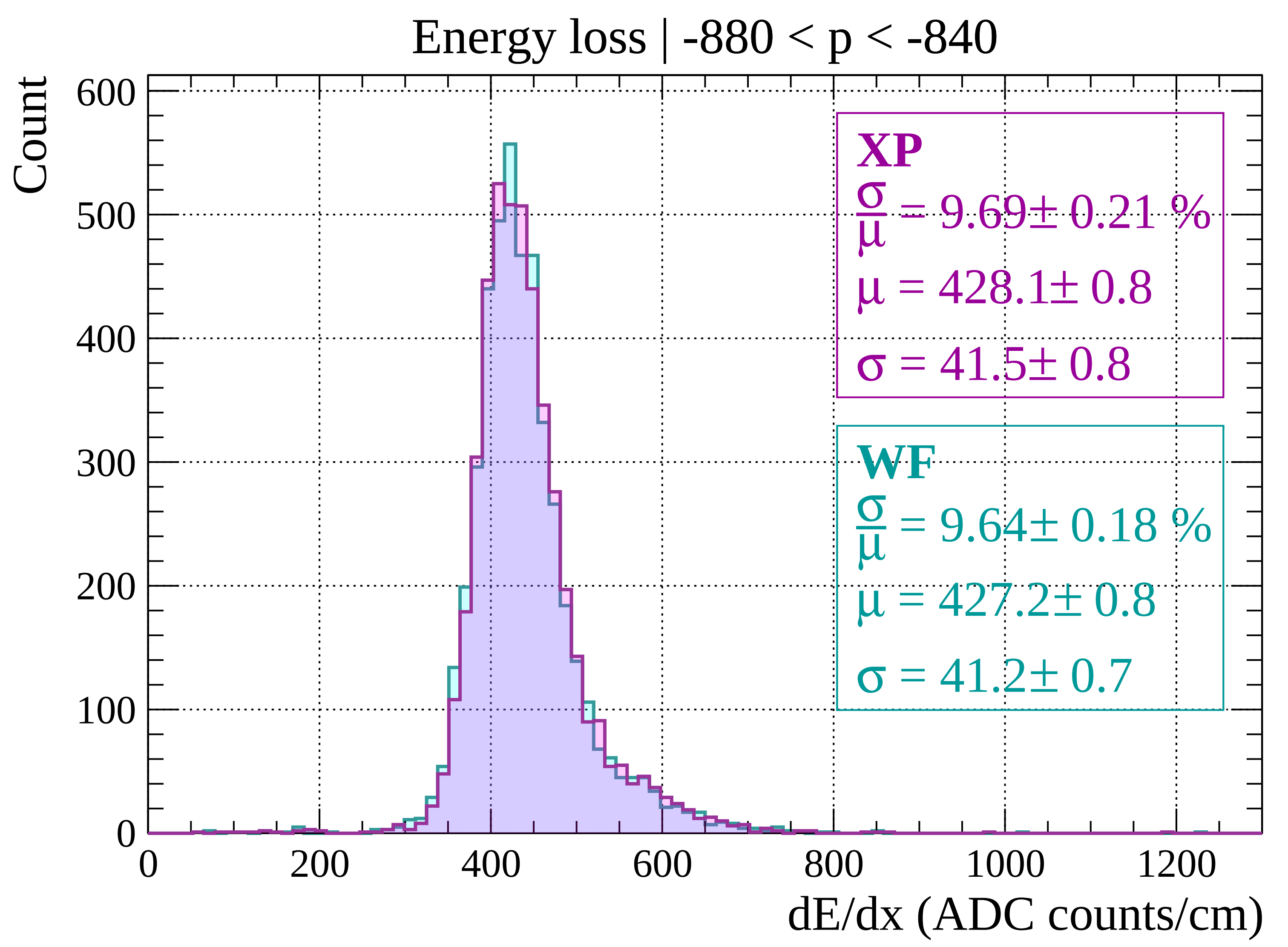


Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.94± 0.21 %
μ = 434.9± 0.8
σ = 43.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.22 %
μ = 434.2± 0.8
σ = 43.8± 0.9

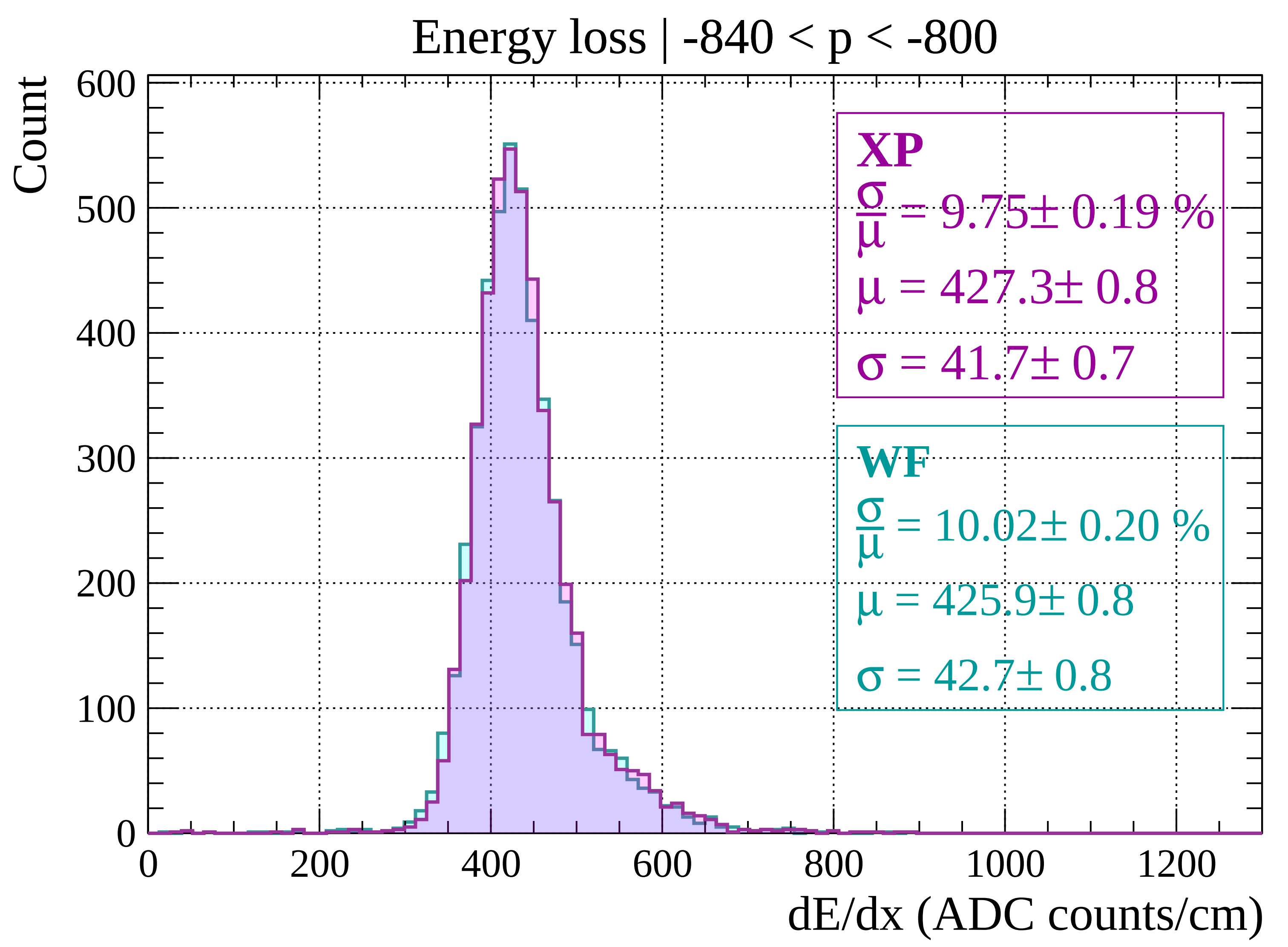


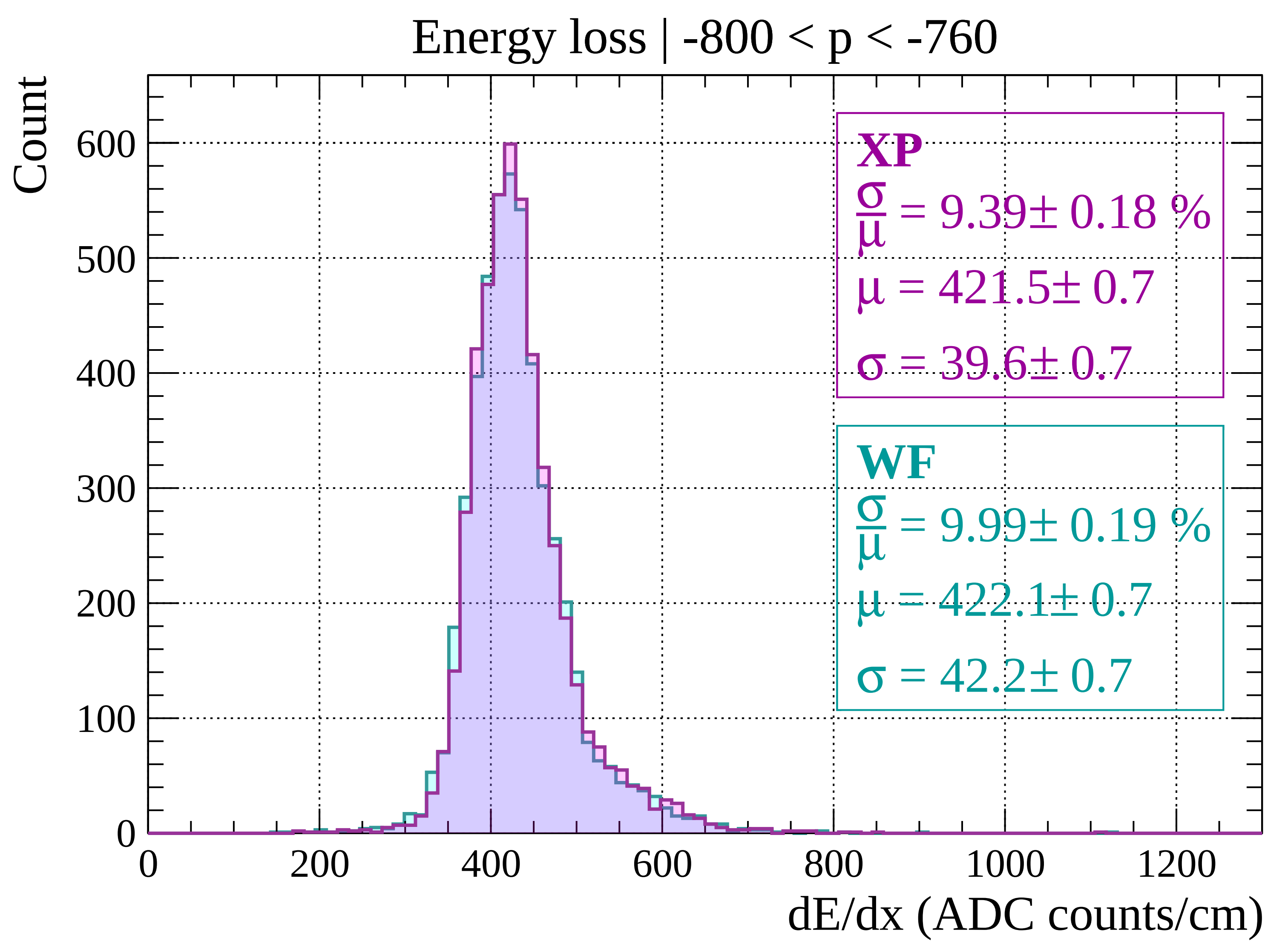
Energy loss | -960 < p < -920
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.96± 0.21 %
μ = 433.9± 0.8
σ = 43.2± 0.8
WF
σ/μ = 10.21± 0.20 %
μ = 434.0± 0.8
σ = 44.3± 0.8


Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.67± 0.21 %
μ = 430.8± 0.8
σ = 41.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.99± 0.18 %
μ = 430.9± 0.8
σ = 43.1± 0.7
0
100
200
300
400
500
0
200
400
600
800
1000
1200

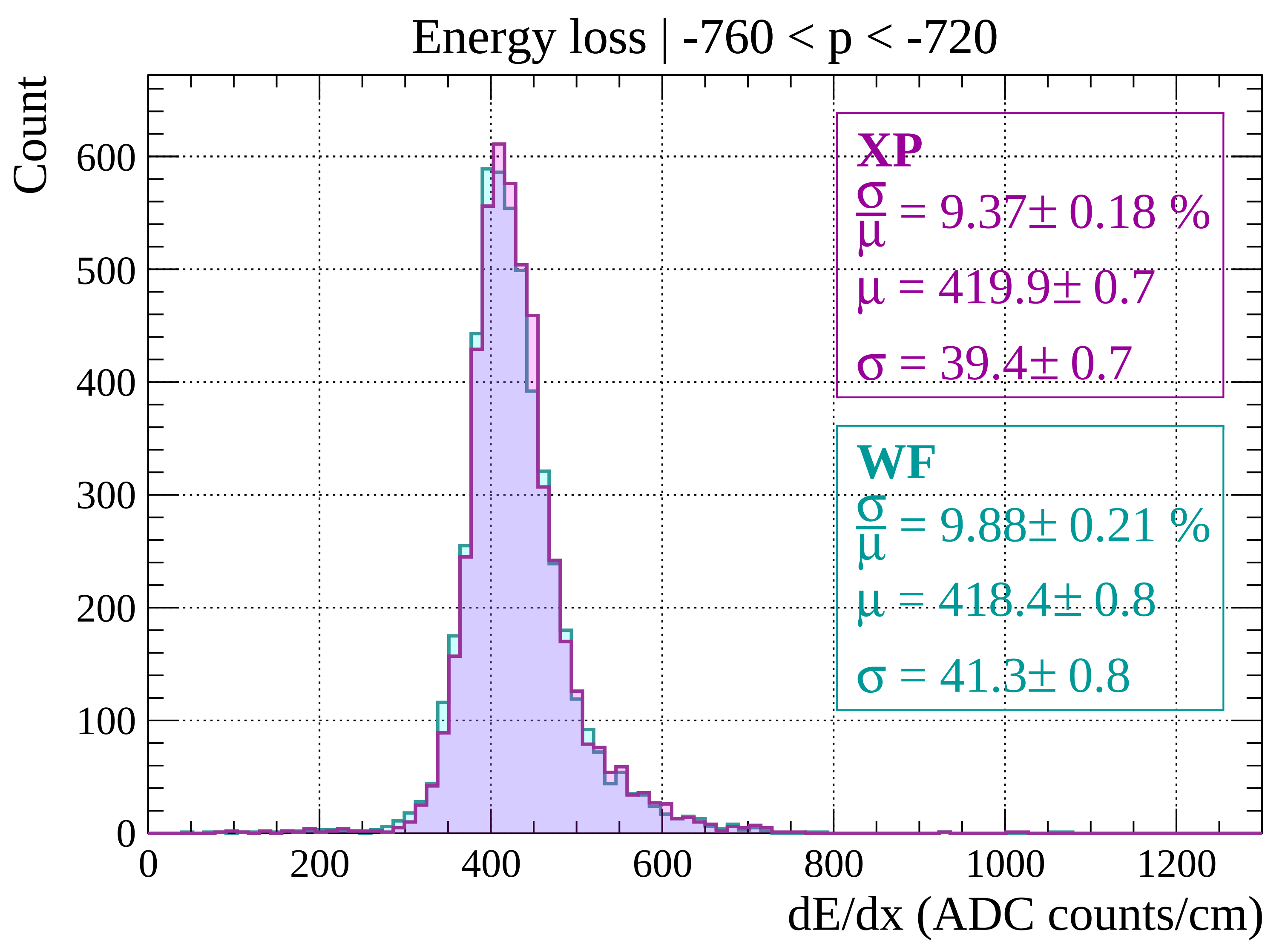


Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.69 ± 0.21 %
μ = 428.1 ± 0.8
σ = 41.5 ± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.64 ± 0.18 %
μ = 427.2 ± 0.8
σ = 41.2 ± 0.7

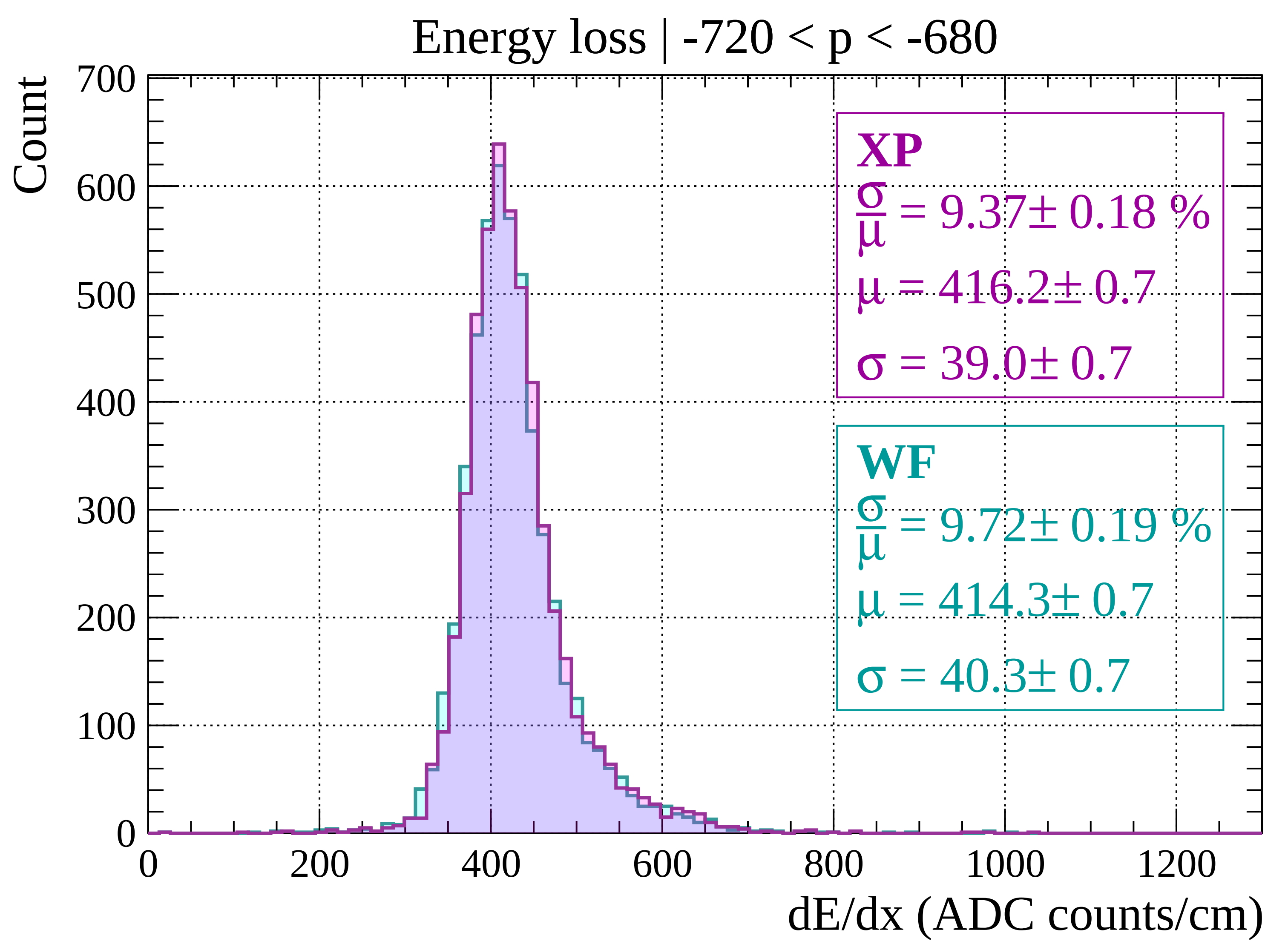

Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.75± 0.19 %
μ = 427.3± 0.8
σ = 41.7± 0.7
WF
σ/μ = 10.02± 0.20 %
μ = 425.9± 0.8
σ = 42.7± 0.8



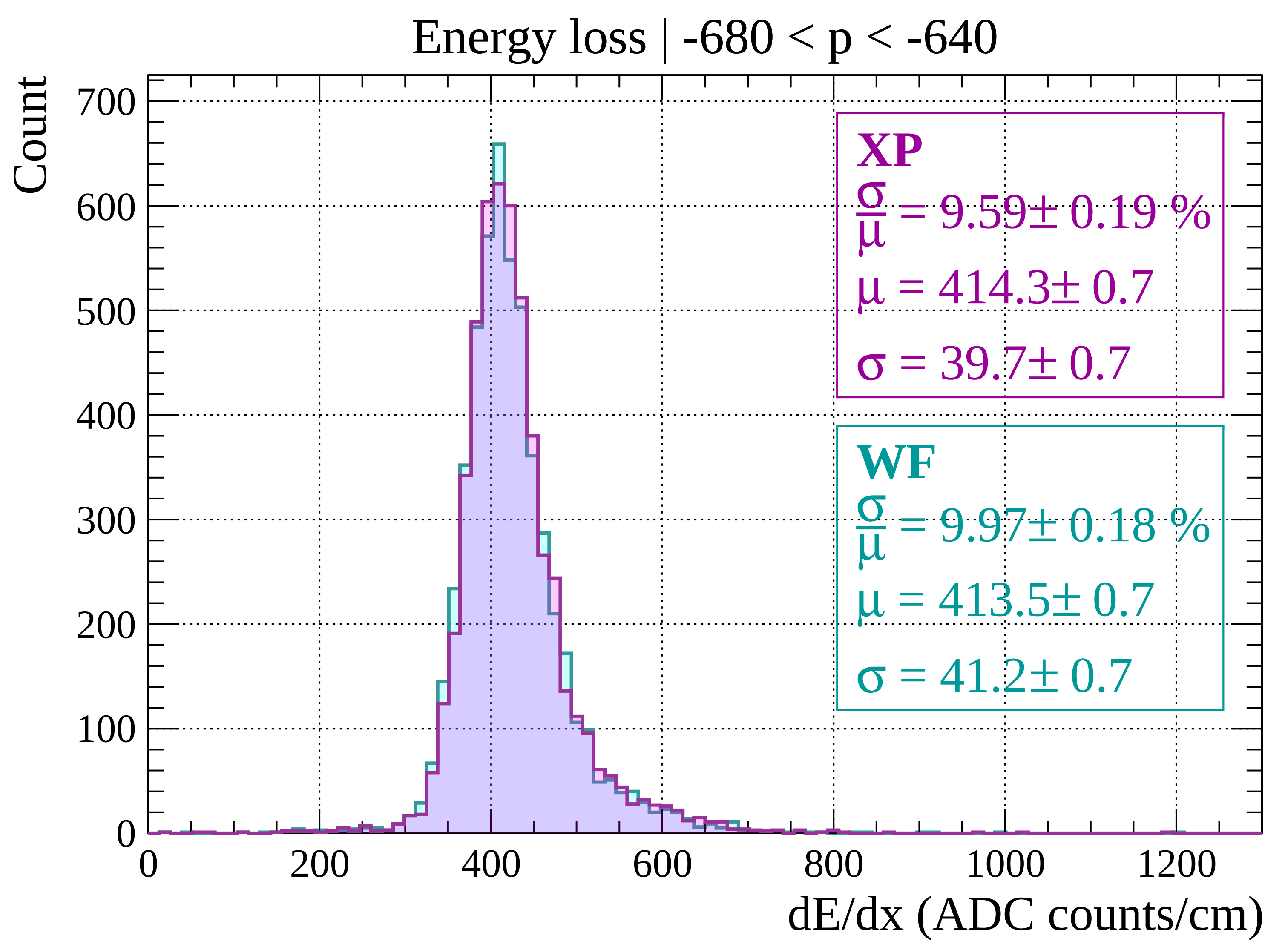
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.39± 0.18 %
μ = 421.5± 0.7
σ = 39.6± 0.7
WF
σ/μ = 9.99± 0.19 %
μ = 422.1± 0.7
σ = 42.2± 0.7



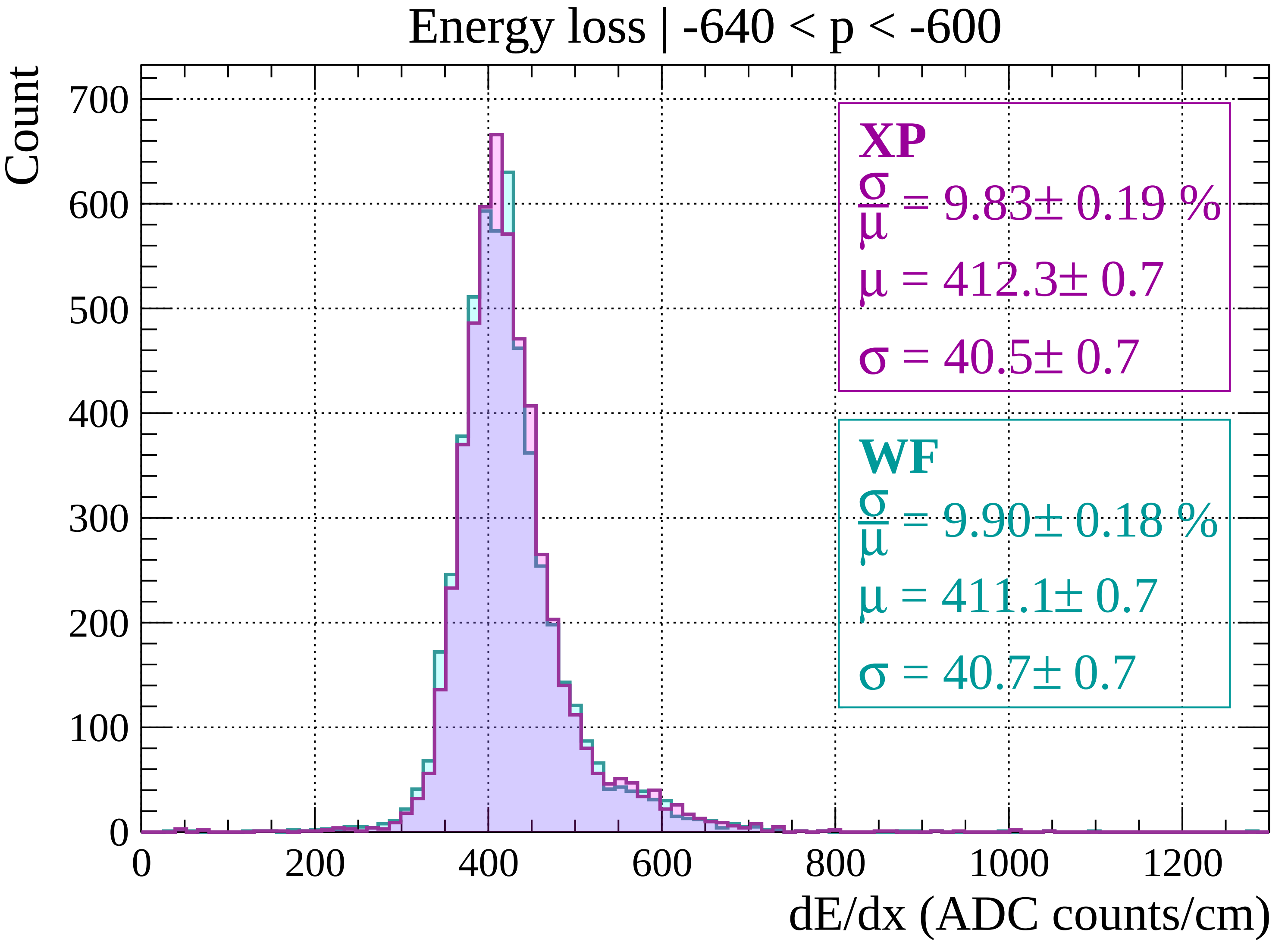
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.37± 0.18 %
μ = 419.9± 0.7
σ = 39.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.88± 0.21 %
μ = 418.4± 0.8
σ = 41.3± 0.8



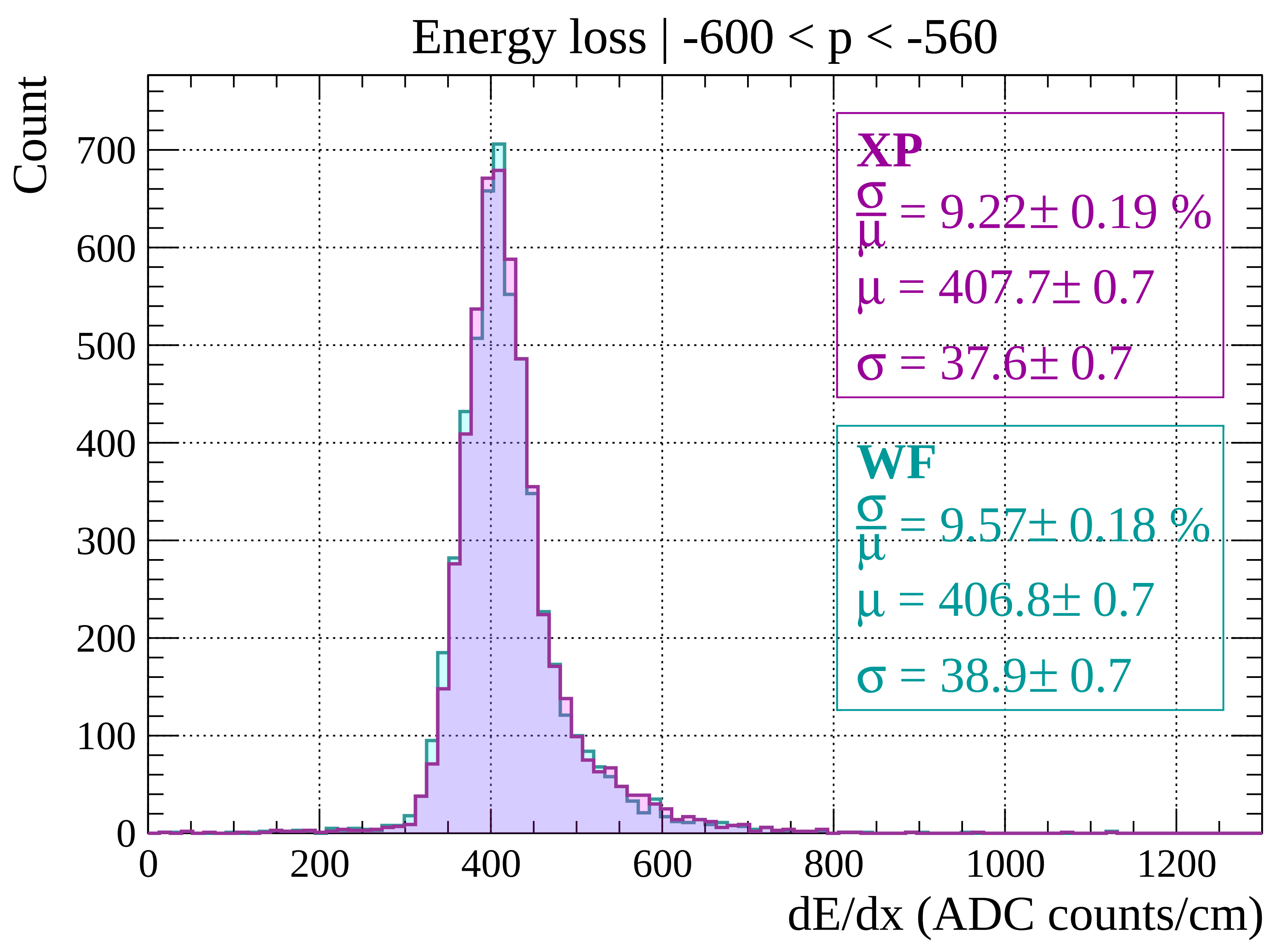
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.37± 0.18 %
μ = 416.2± 0.7
σ = 39.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.72± 0.19 %
μ = 414.3± 0.7
σ = 40.3± 0.7



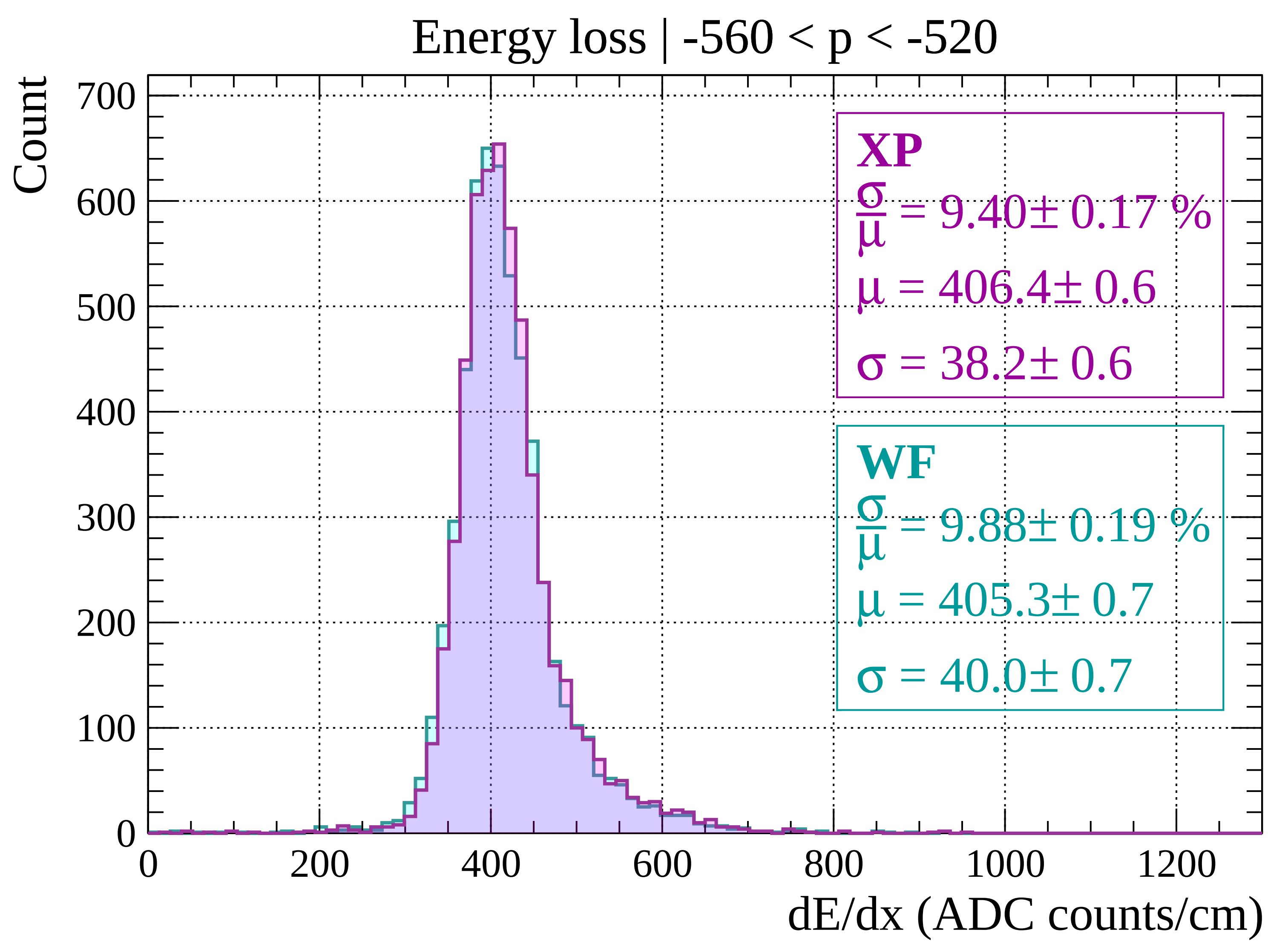
Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.59± 0.19 %
μ = 414.3± 0.7
σ = 39.7± 0.7
WF
σ/μ = 9.97± 0.18 %
μ = 413.5± 0.7
σ = 41.2± 0.7



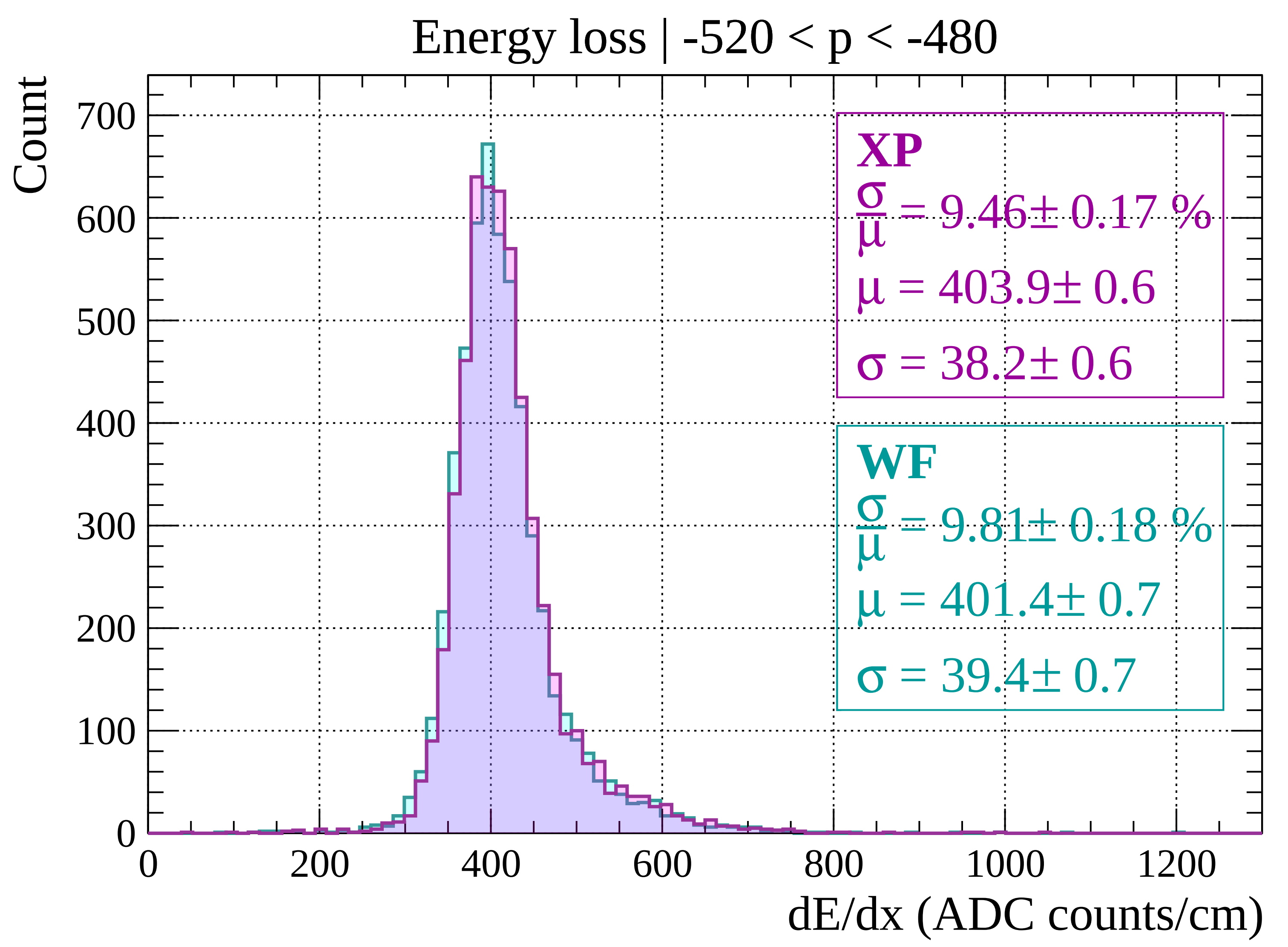
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.83± 0.19 %
μ = 412.3± 0.7
σ = 40.5± 0.7
WF
σ/μ = 9.90± 0.18 %
μ = 411.1± 0.7
σ = 40.7± 0.7



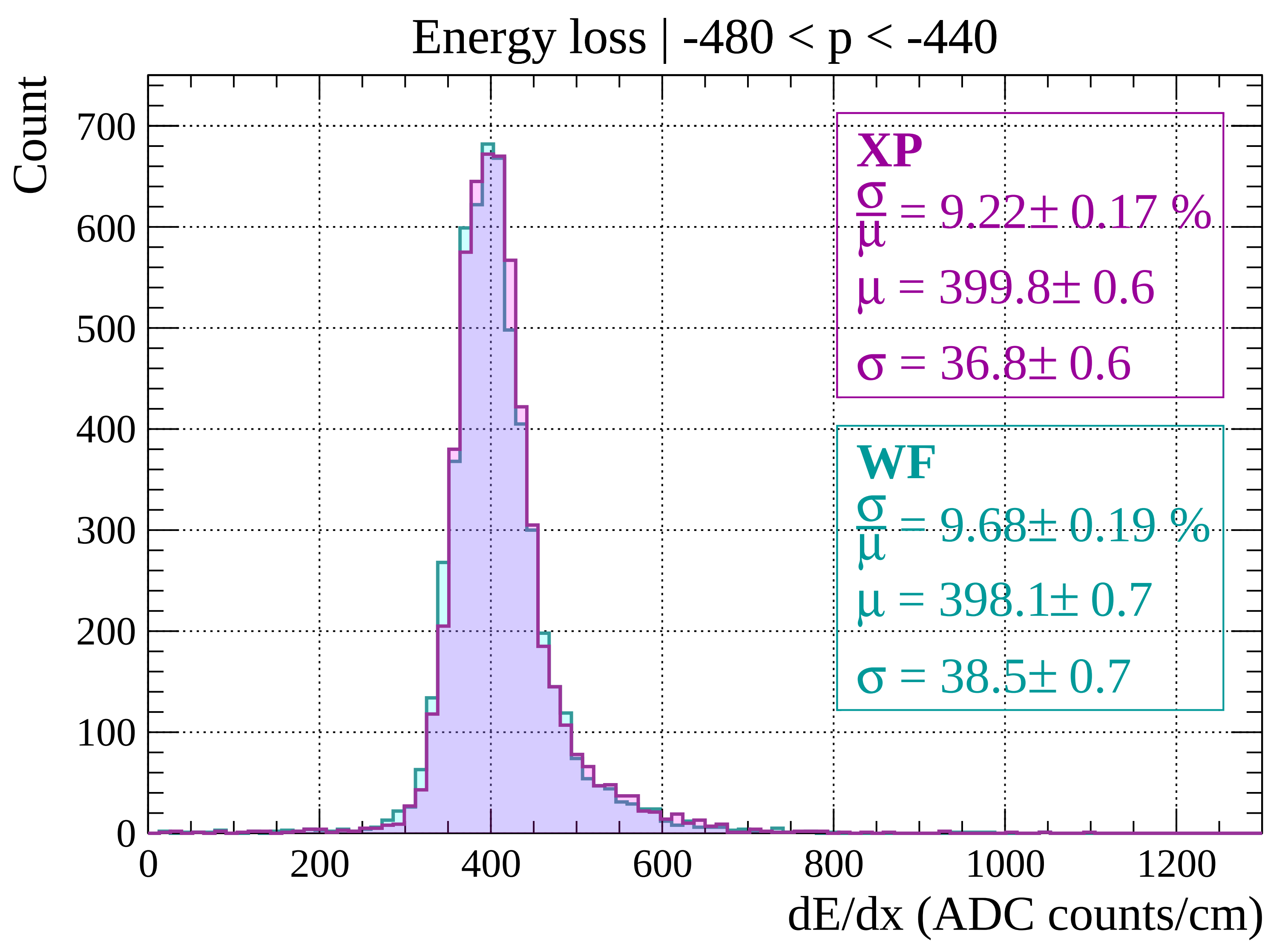
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.22± 0.19 %
μ = 407.7± 0.7
σ = 37.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.57± 0.18 %
μ = 406.8± 0.7
σ = 38.9± 0.7



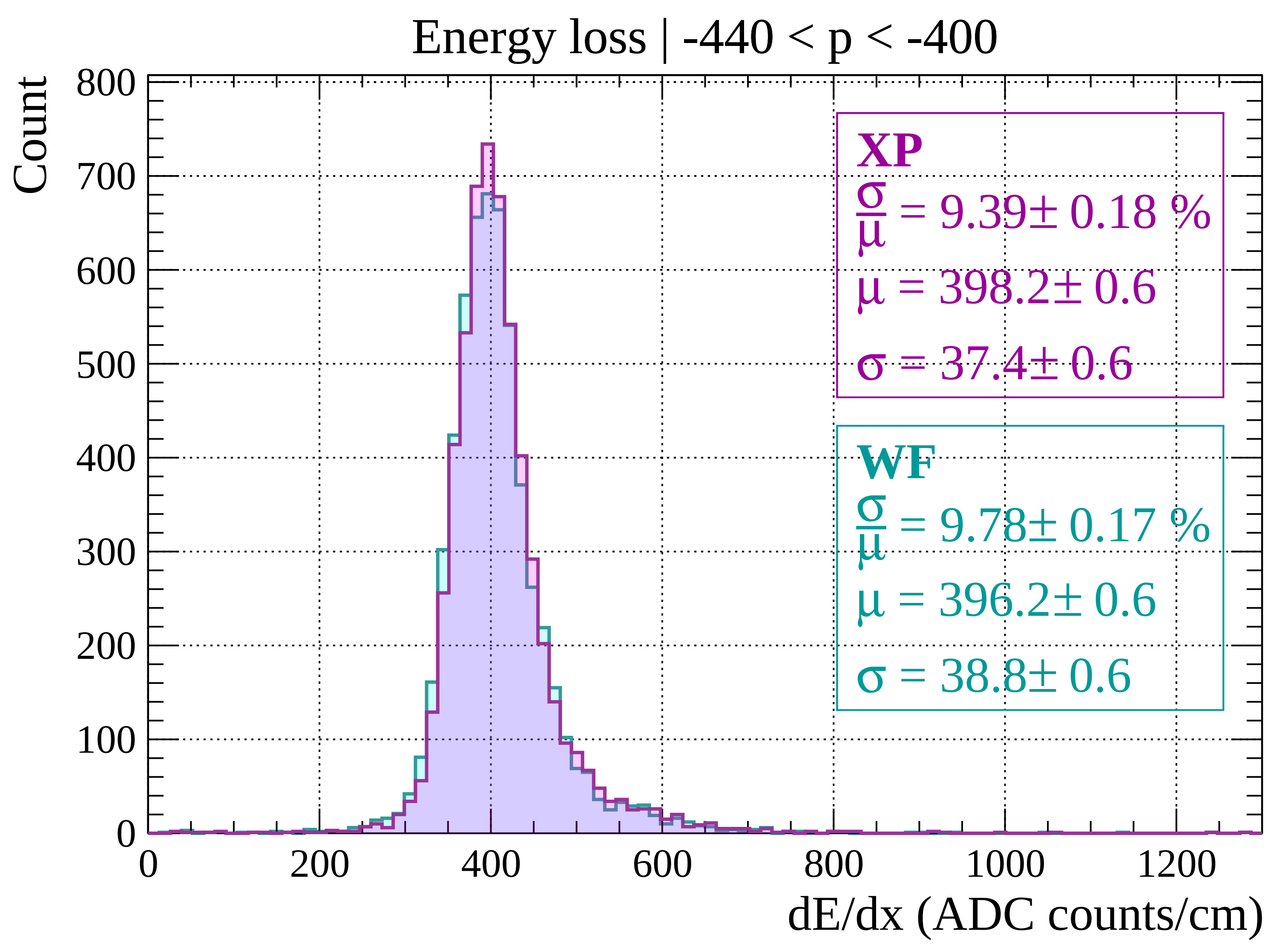
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.40± 0.17 %
μ = 406.4± 0.6
σ = 38.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.88± 0.19 %
μ = 405.3± 0.7
σ = 40.0± 0.7



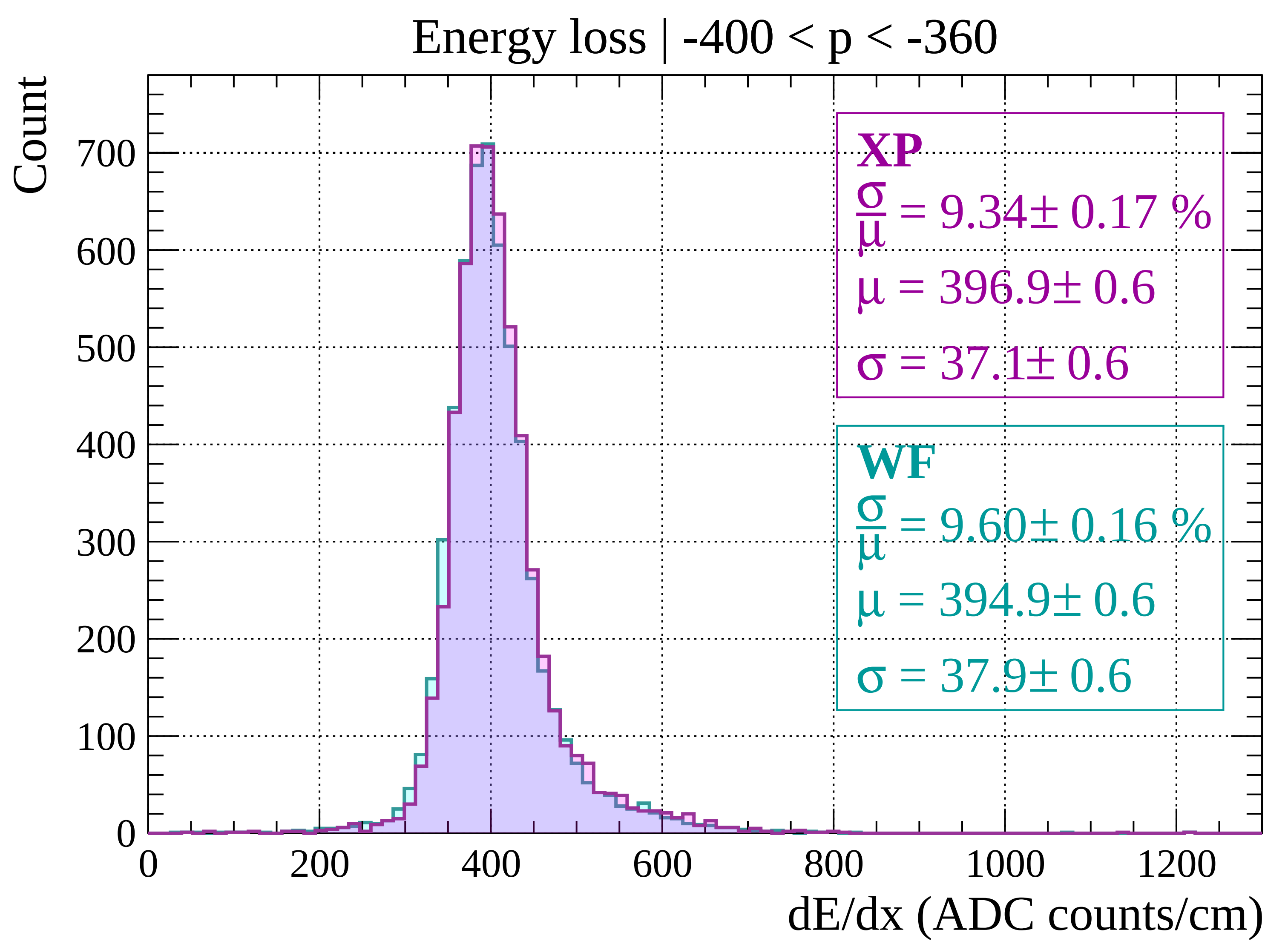
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.46± 0.17 %
μ = 403.9± 0.6
σ = 38.2± 0.6
WF
σ/μ = 9.81± 0.18 %
μ = 401.4± 0.7
σ = 39.4± 0.7



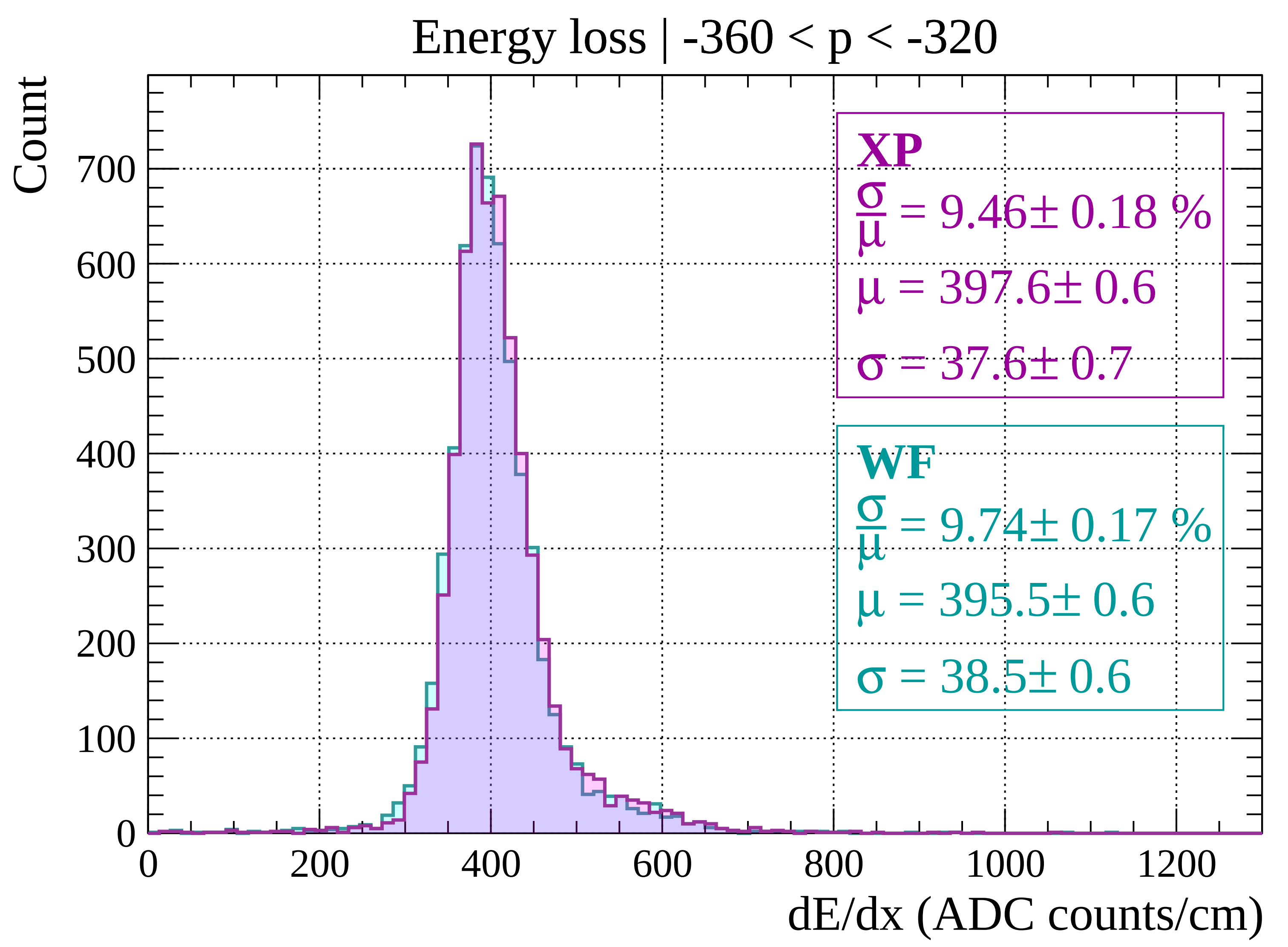
Energy loss | -480 < p < -440
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.22± 0.17 %
μ = 399.8± 0.6
σ = 36.8± 0.6
WF
σ/μ = 9.68± 0.19 %
μ = 398.1± 0.7
σ = 38.5± 0.7



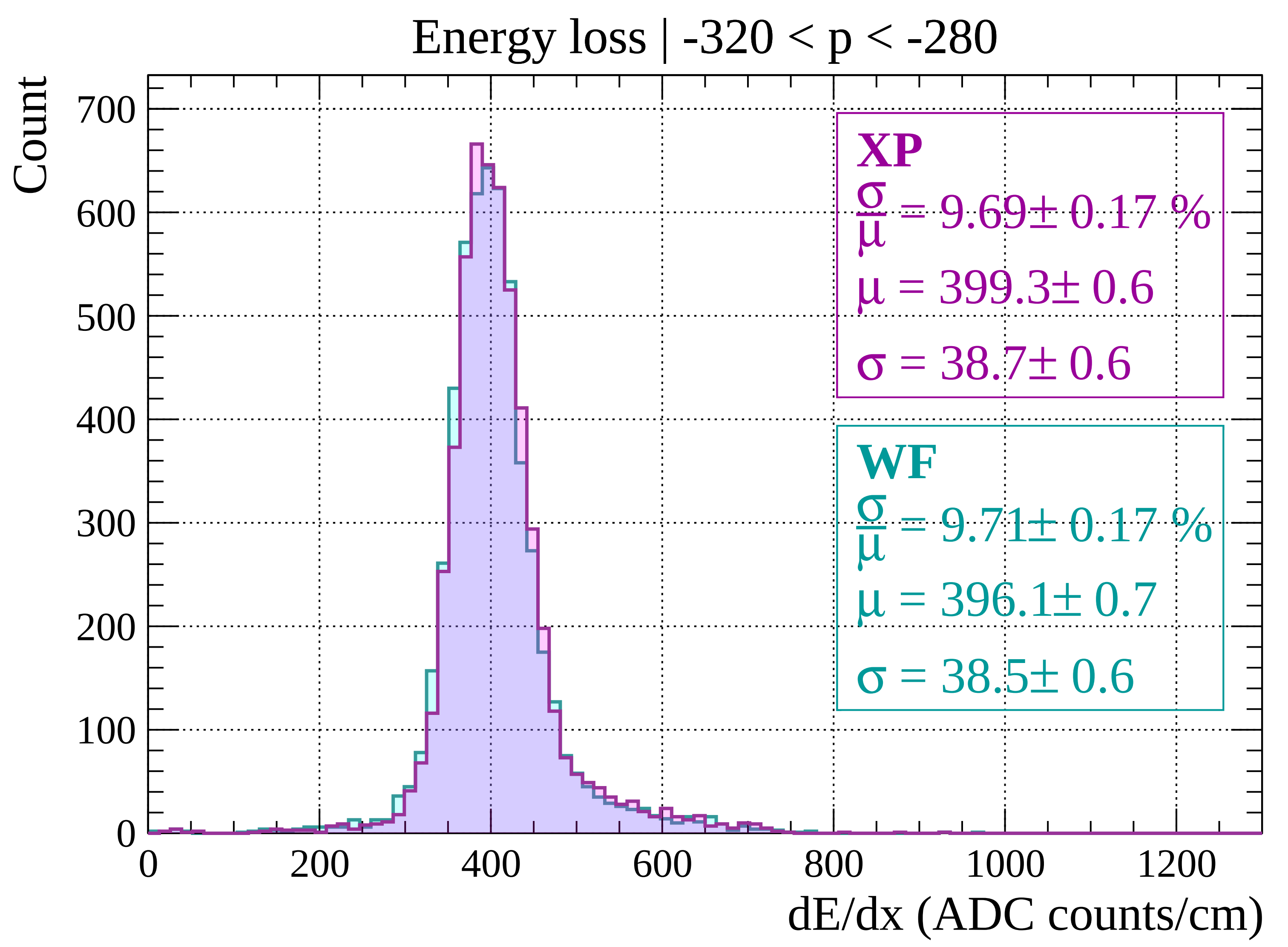
Energy loss | -440 < p < -400
Count
XP
σ/μ = 9.39± 0.18 %
μ = 398.2± 0.6
σ = 37.4± 0.6
WF
σ/μ = 9.78± 0.17 %
μ = 396.2± 0.6
σ = 38.8± 0.6
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
1000
1200



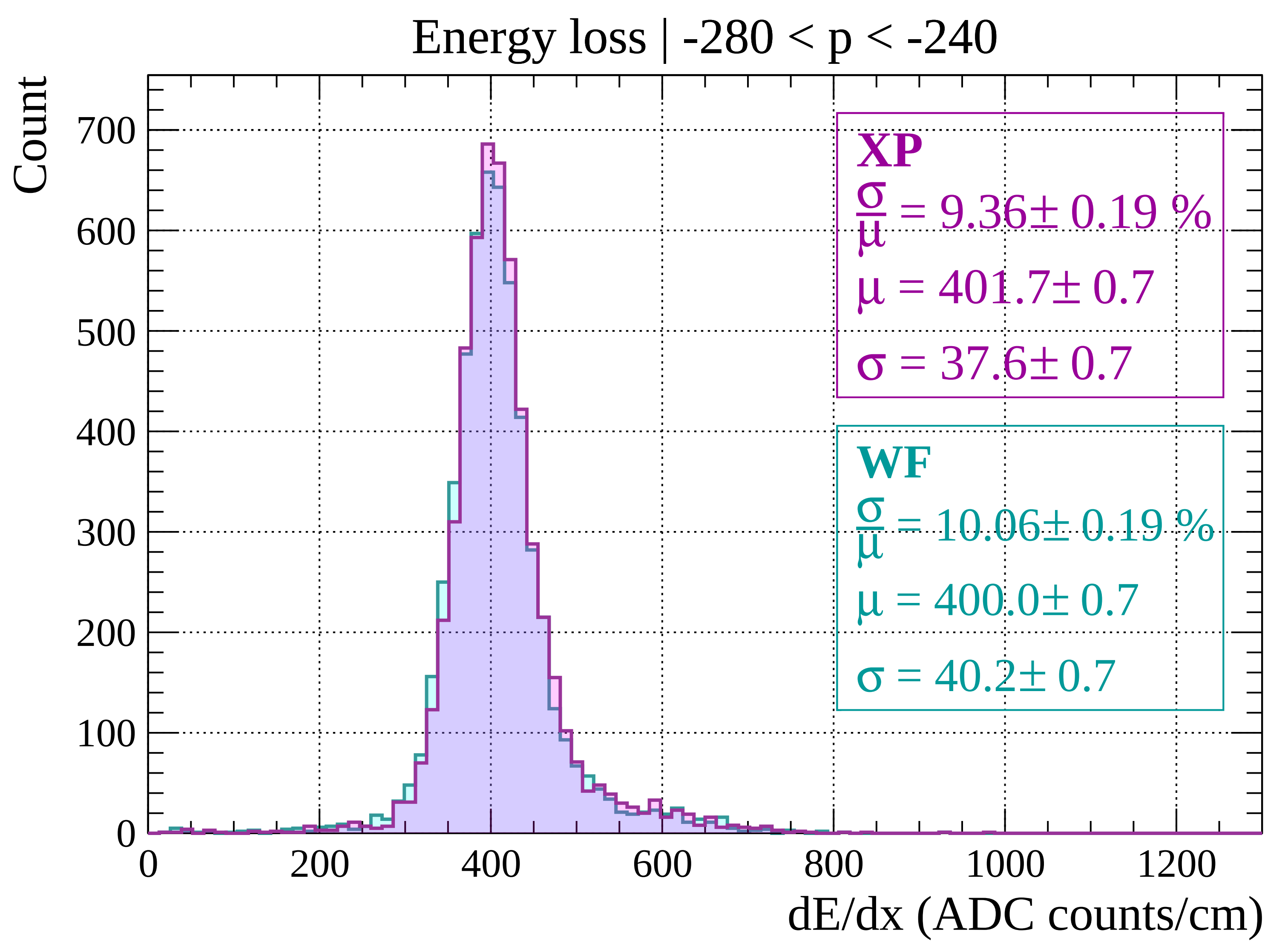
Energy loss | -400 < p < -360
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.34 \pm 0.17\ \%$
$\mu = 396.9 \pm 0.6$
$\sigma = 37.1 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.60 \pm 0.16\ \%$
$\mu = 394.9 \pm 0.6$
$\sigma = 37.9 \pm 0.6$

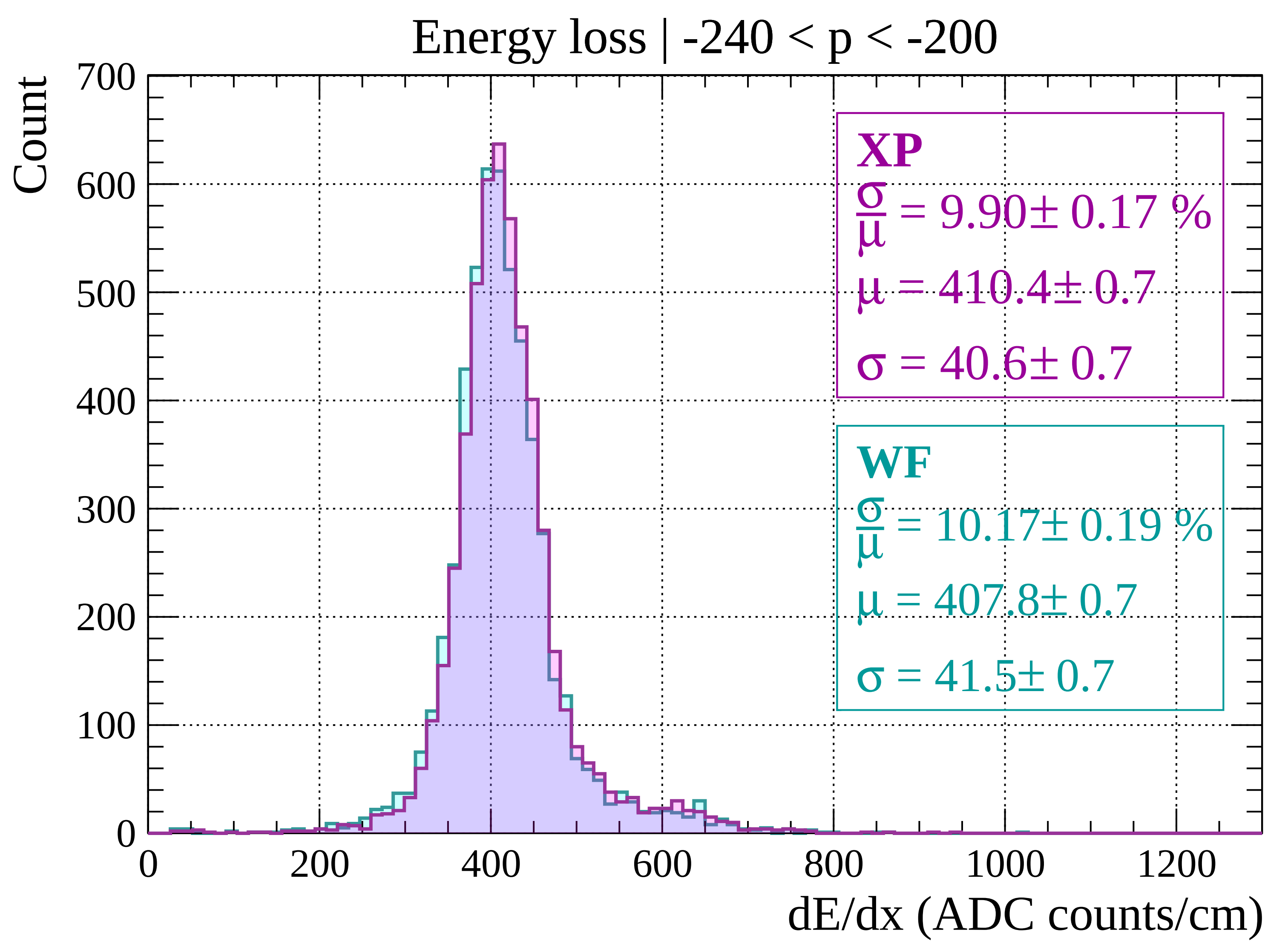


Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.18 %
μ = 397.6± 0.6
σ = 37.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.74± 0.17 %
μ = 395.5± 0.6
σ = 38.5± 0.6

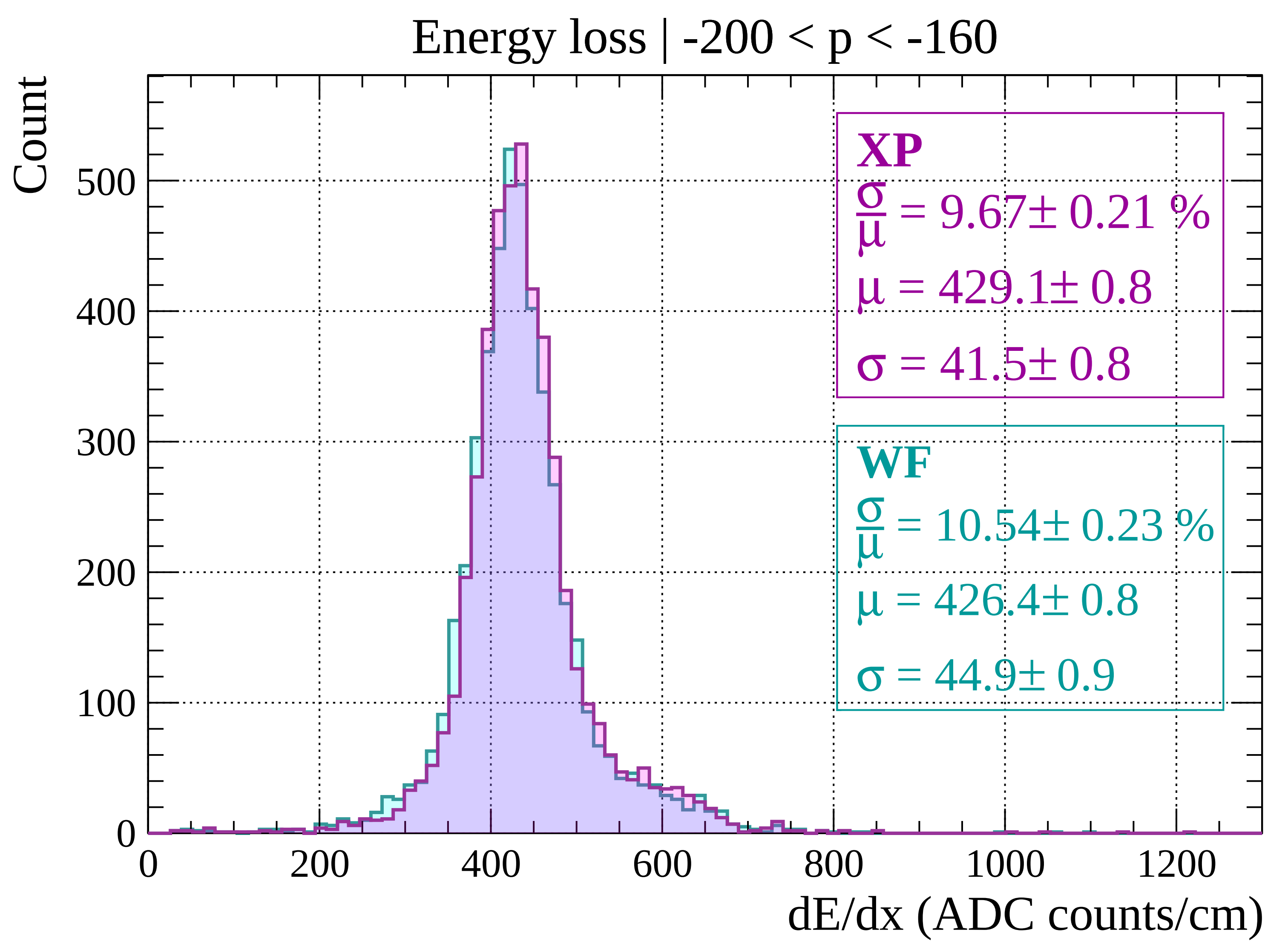


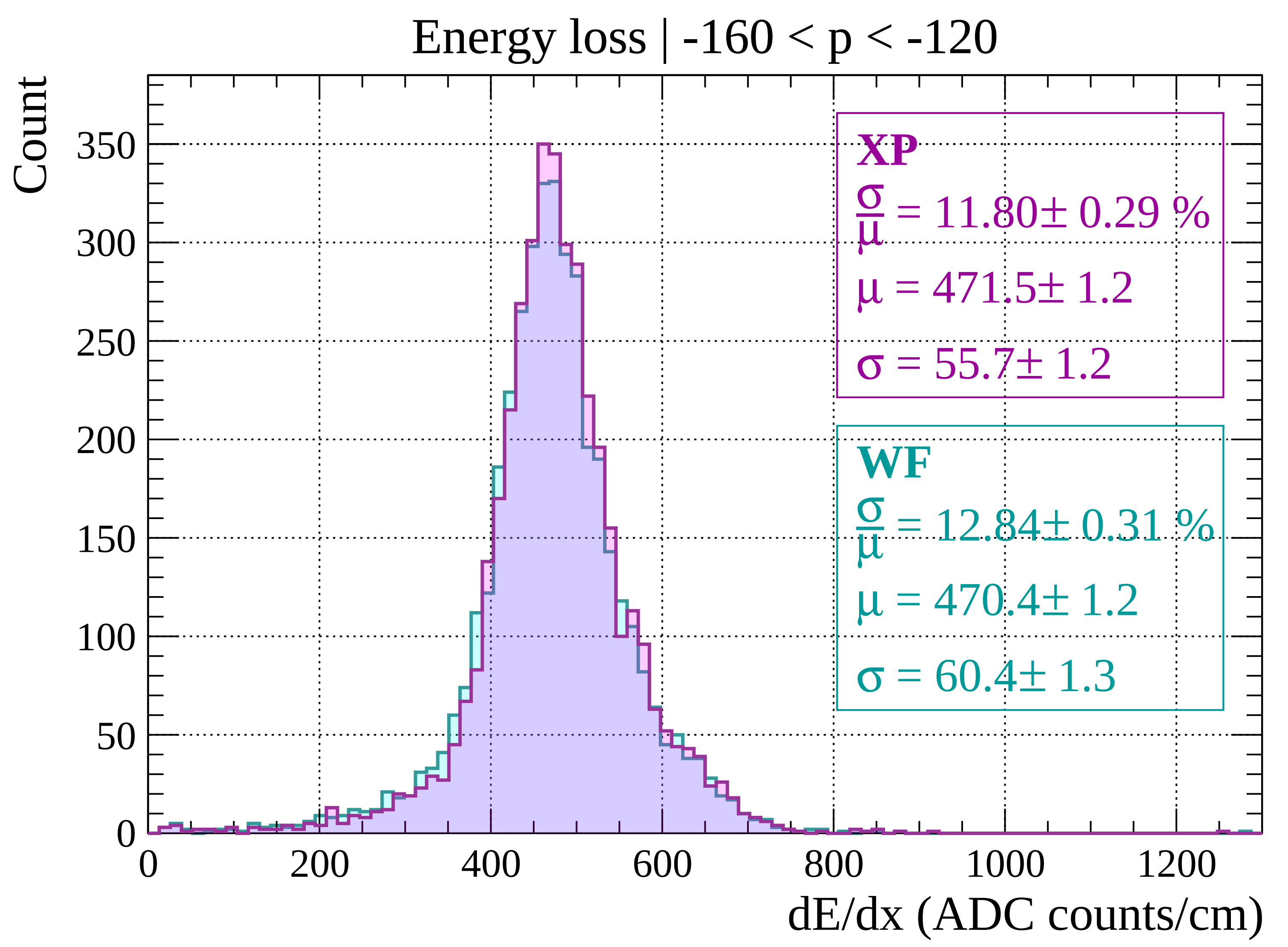
Energy loss | -320 < p < -280
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.69± 0.17 %
μ = 399.3± 0.6
σ = 38.7± 0.6
WF
σ/μ = 9.71± 0.17 %
μ = 396.1± 0.7
σ = 38.5± 0.6


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.36± 0.19 %
μ = 401.7± 0.7
σ = 37.6± 0.7
WF
σ/μ = 10.06± 0.19 %
μ = 400.0± 0.7
σ = 40.2± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

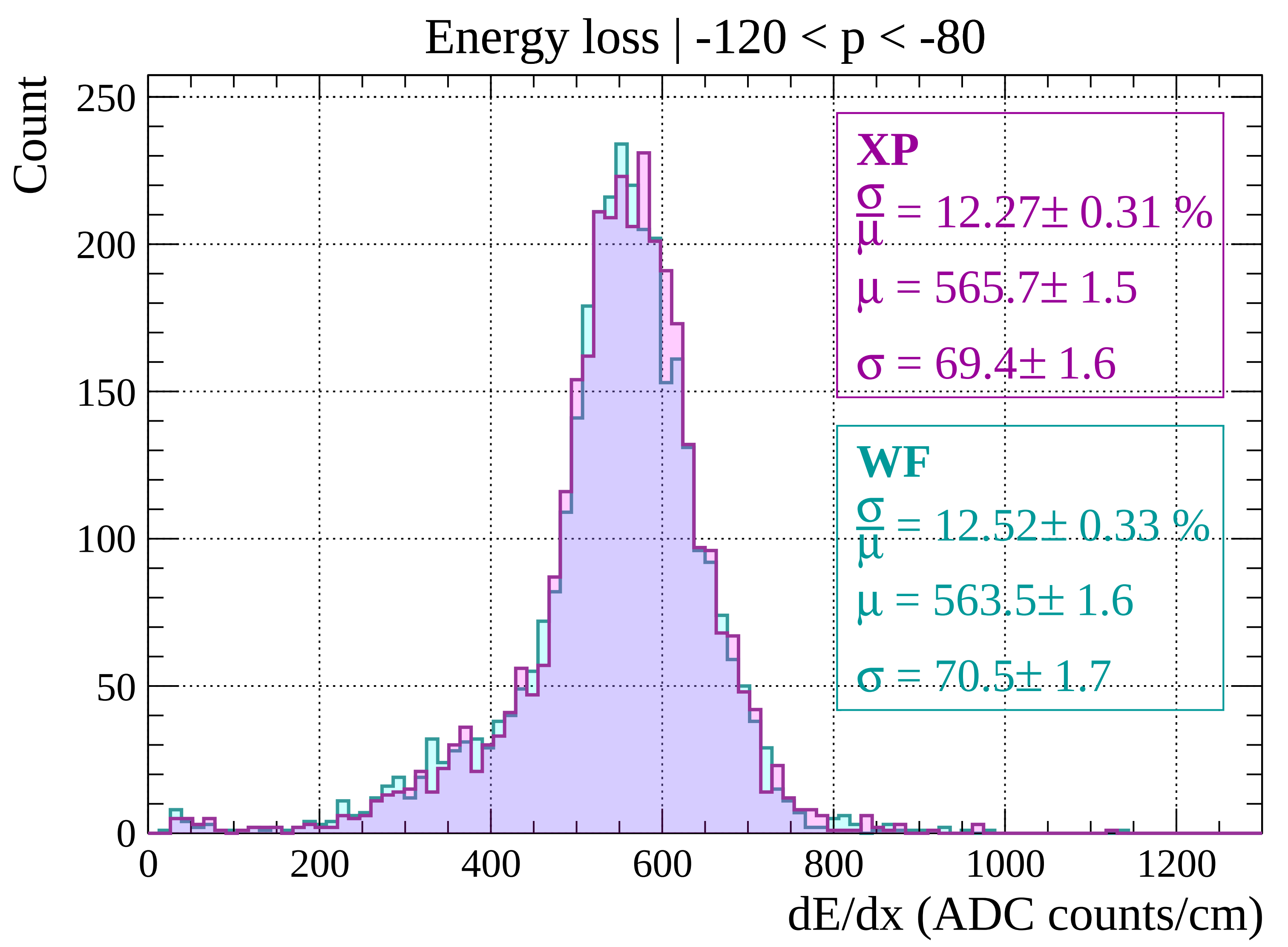


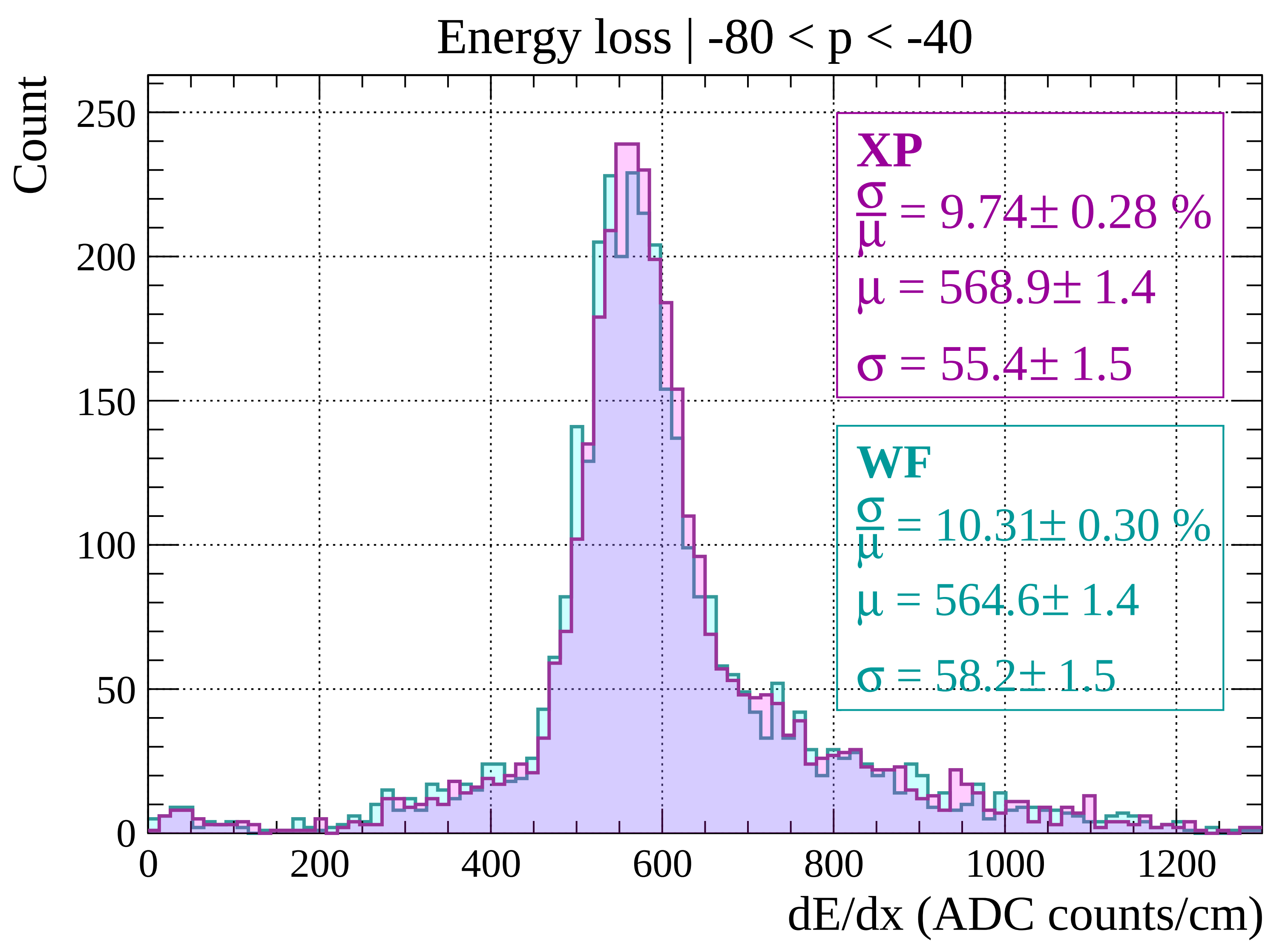
Energy loss | -240 < p < -200
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.90± 0.17 %
μ = 410.4± 0.7
σ = 40.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 10.17± 0.19 %
μ = 407.8± 0.7
σ = 41.5± 0.7


Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.67± 0.21 %
μ = 429.1± 0.8
σ = 41.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.54± 0.23 %
μ = 426.4± 0.8
σ = 44.9± 0.9

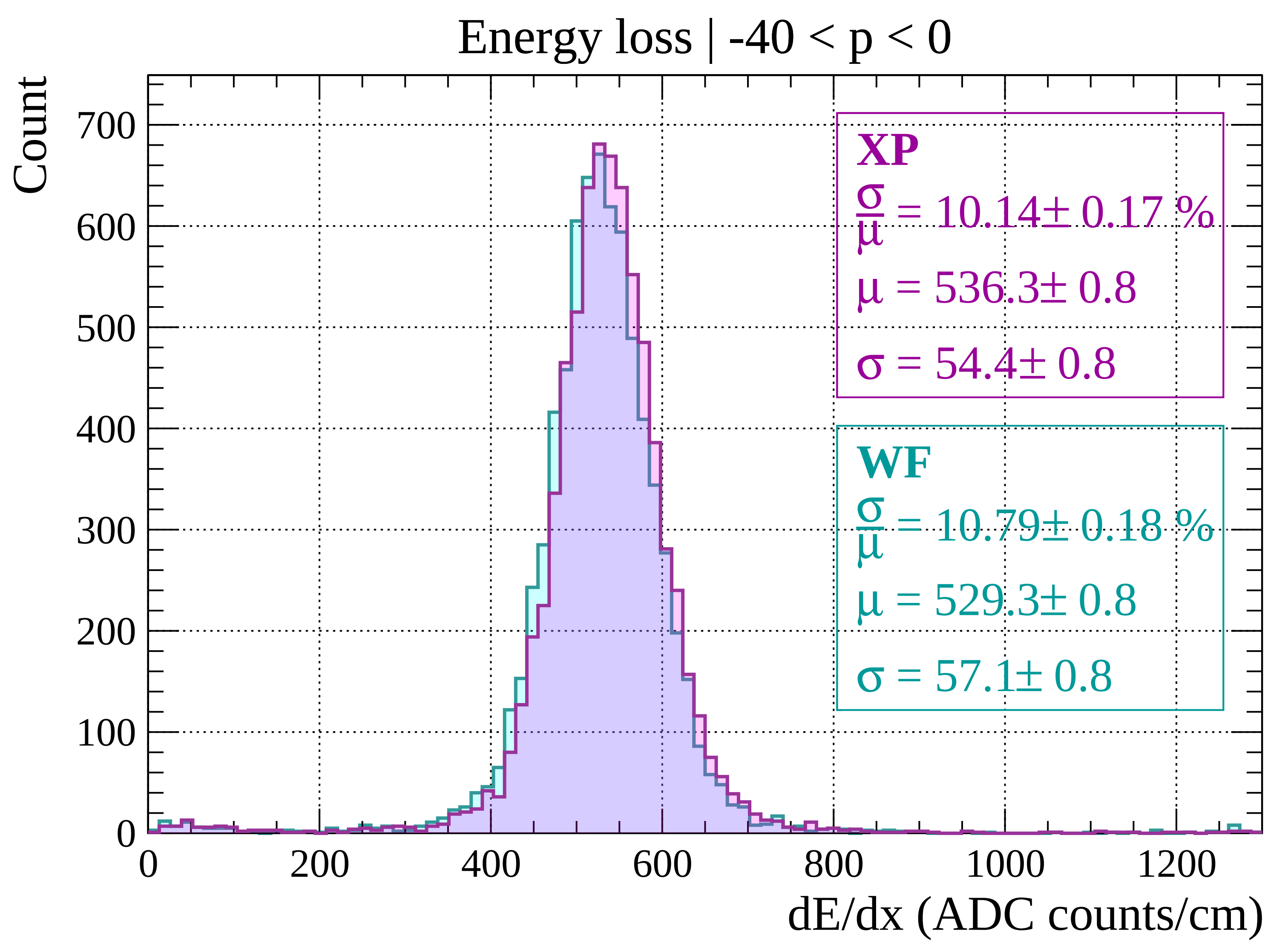


Energy loss | -160 < p < -120
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.80± 0.29 %
μ = 471.5± 1.2
σ = 55.7± 1.2
WF
σ/μ = 12.84± 0.31 %
μ = 470.4± 1.2
σ = 60.4± 1.3

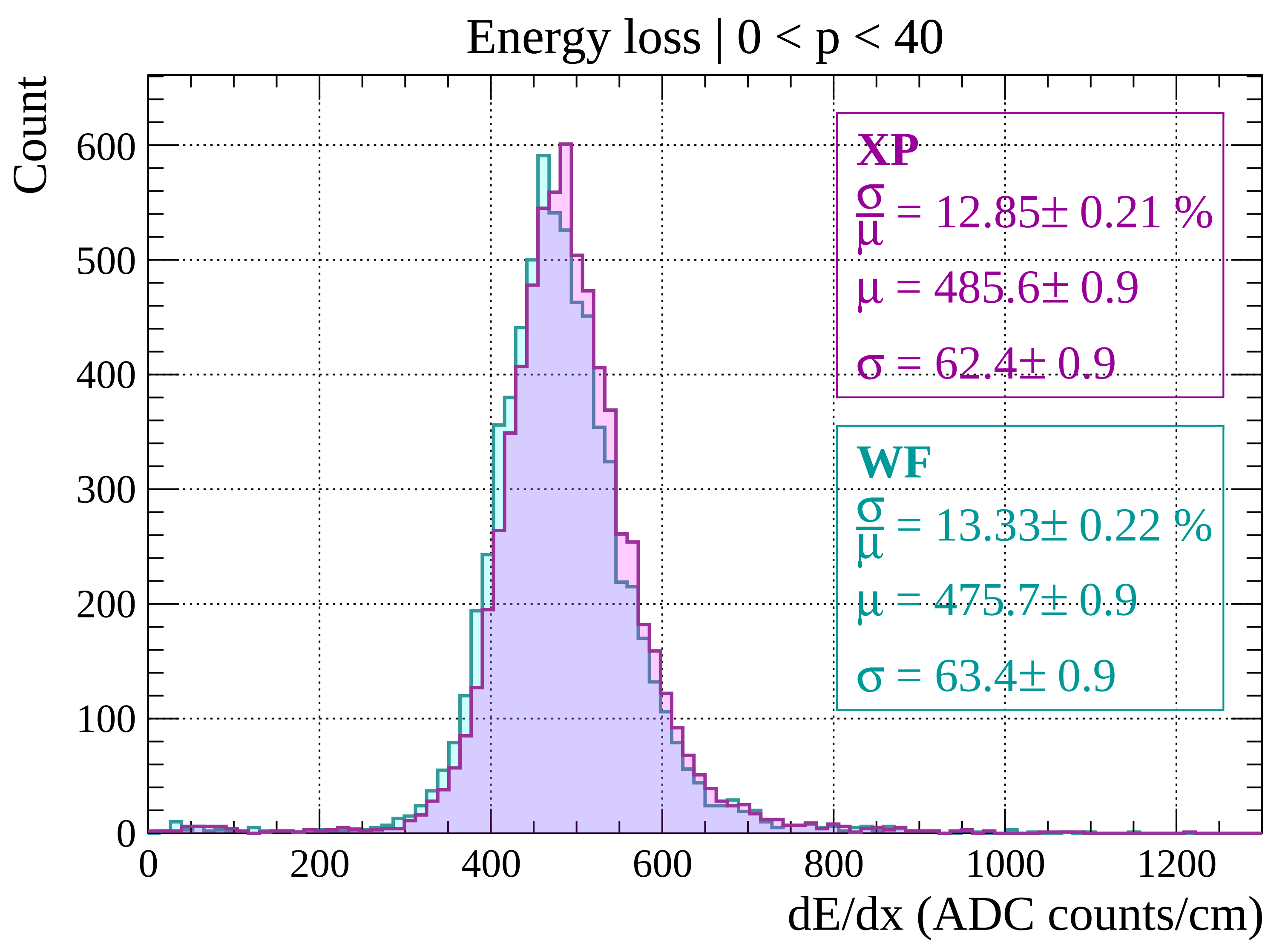

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 12.27± 0.31 %
μ = 565.7± 1.5
σ = 69.4± 1.6
WF
σ/μ = 12.52± 0.33 %
μ = 563.5± 1.6
σ = 70.5± 1.7



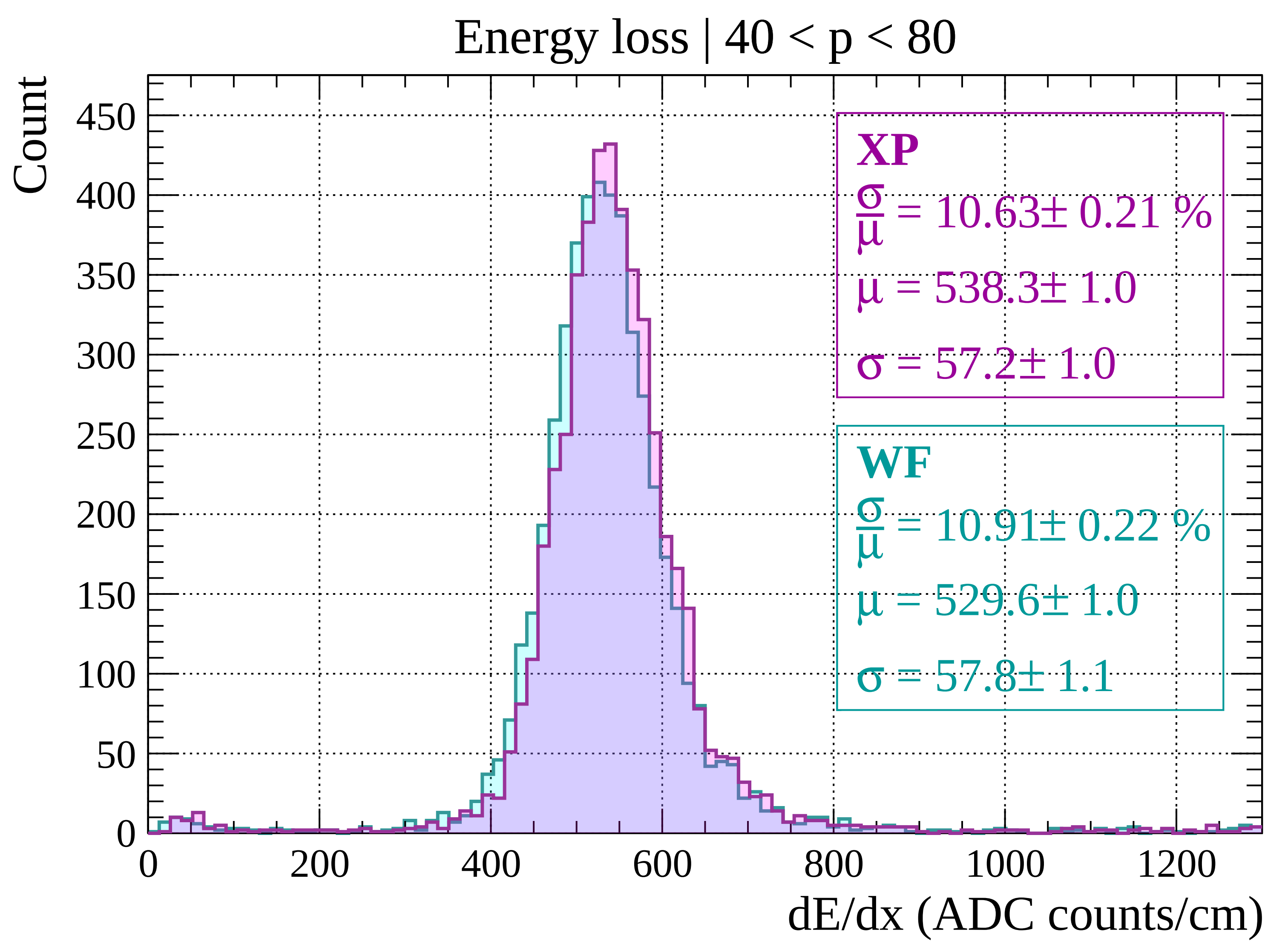
Energy loss | -80 < p < -40
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.74± 0.28 %
μ = 568.9± 1.4
σ = 55.4± 1.5
WF
σ/μ = 10.31± 0.30 %
μ = 564.6± 1.4
σ = 58.2± 1.5



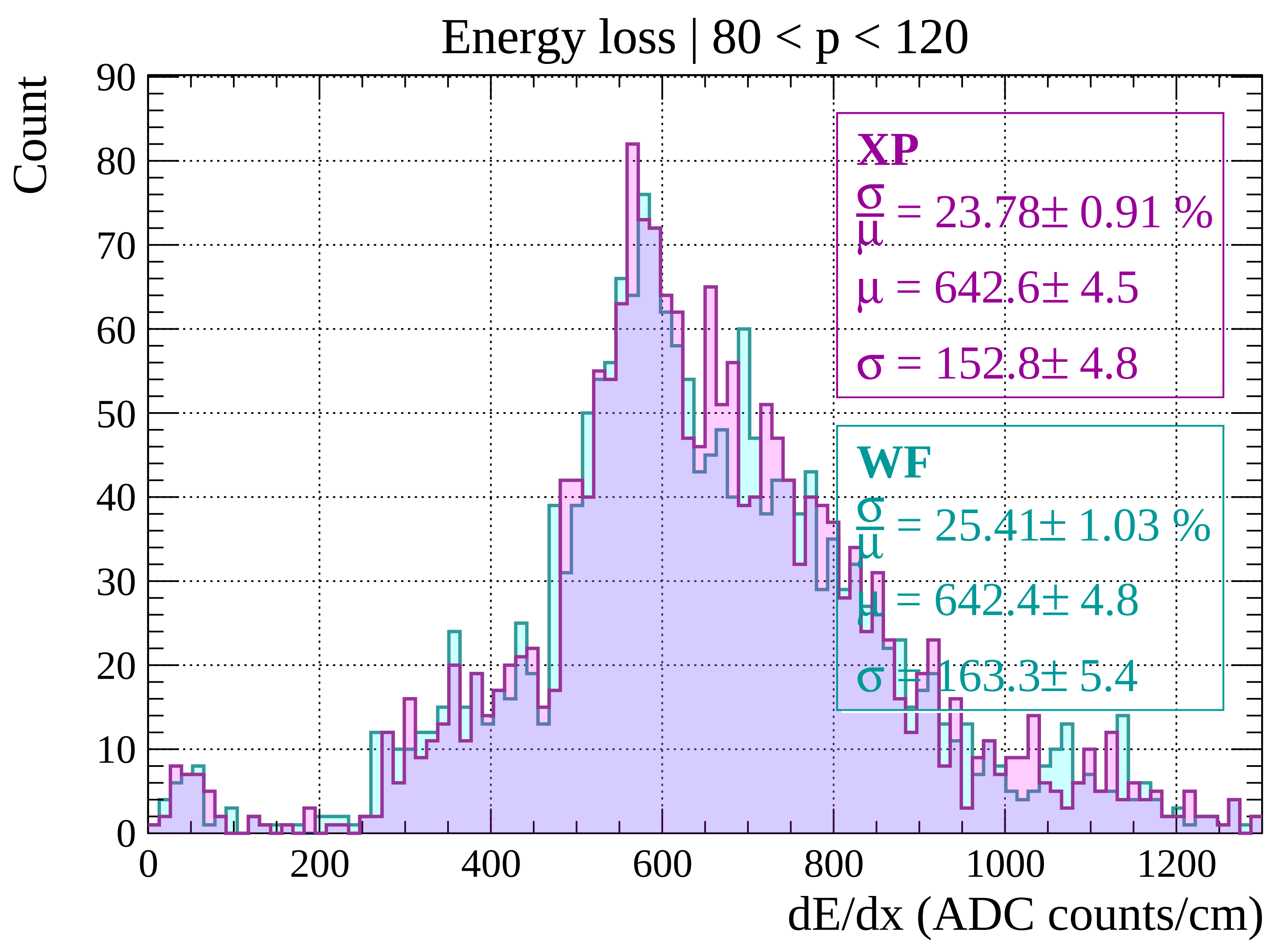
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 10.14± 0.17 %
μ = 536.3± 0.8
σ = 54.4± 0.8
WF
σ/μ = 10.79± 0.18 %
μ = 529.3± 0.8
σ = 57.1± 0.8

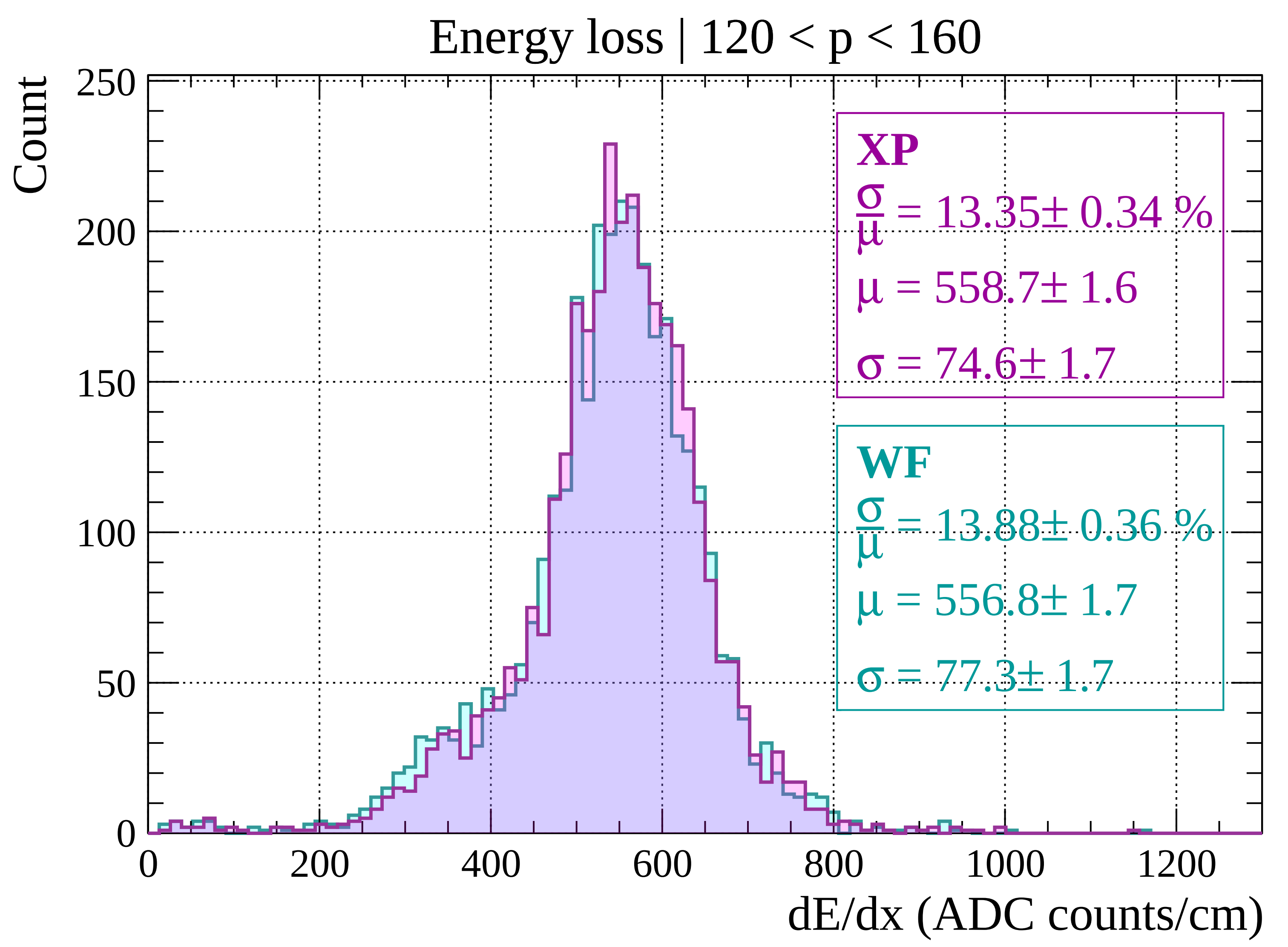


Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
σ/μ = 12.85± 0.21 %
μ = 485.6± 0.9
σ = 62.4± 0.9
WF
σ/μ = 13.33± 0.22 %
μ = 475.7± 0.9
σ = 63.4± 0.9

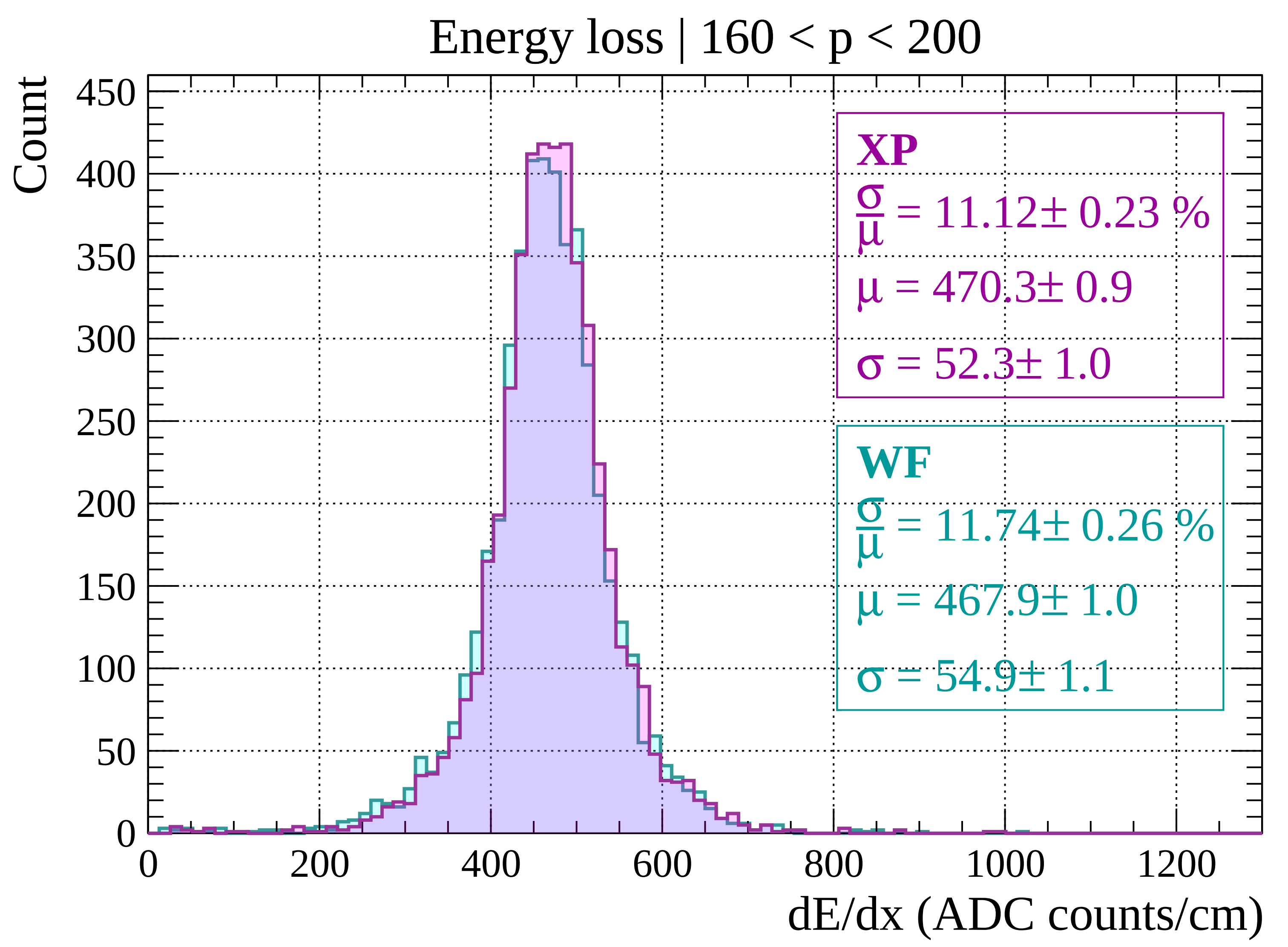


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.63± 0.21 %
μ = 538.3± 1.0
σ = 57.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.91± 0.22 %
μ = 529.6± 1.0
σ = 57.8± 1.1

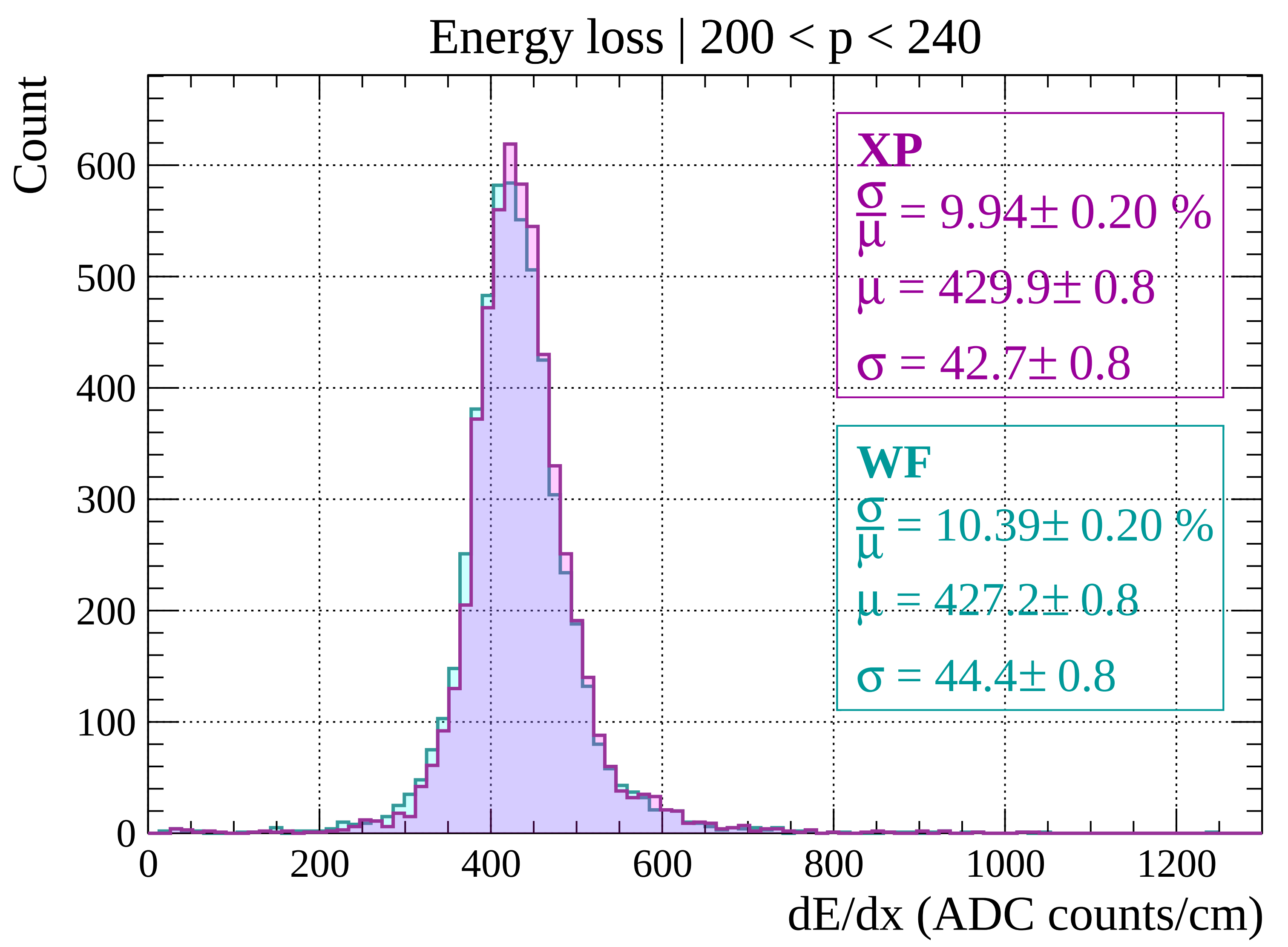

Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 23.78± 0.91 %
μ = 642.6± 4.5
σ = 152.8± 4.8
WF
σ/μ = 25.41± 1.03 %
μ = 642.4± 4.8
σ = 163.3± 5.4



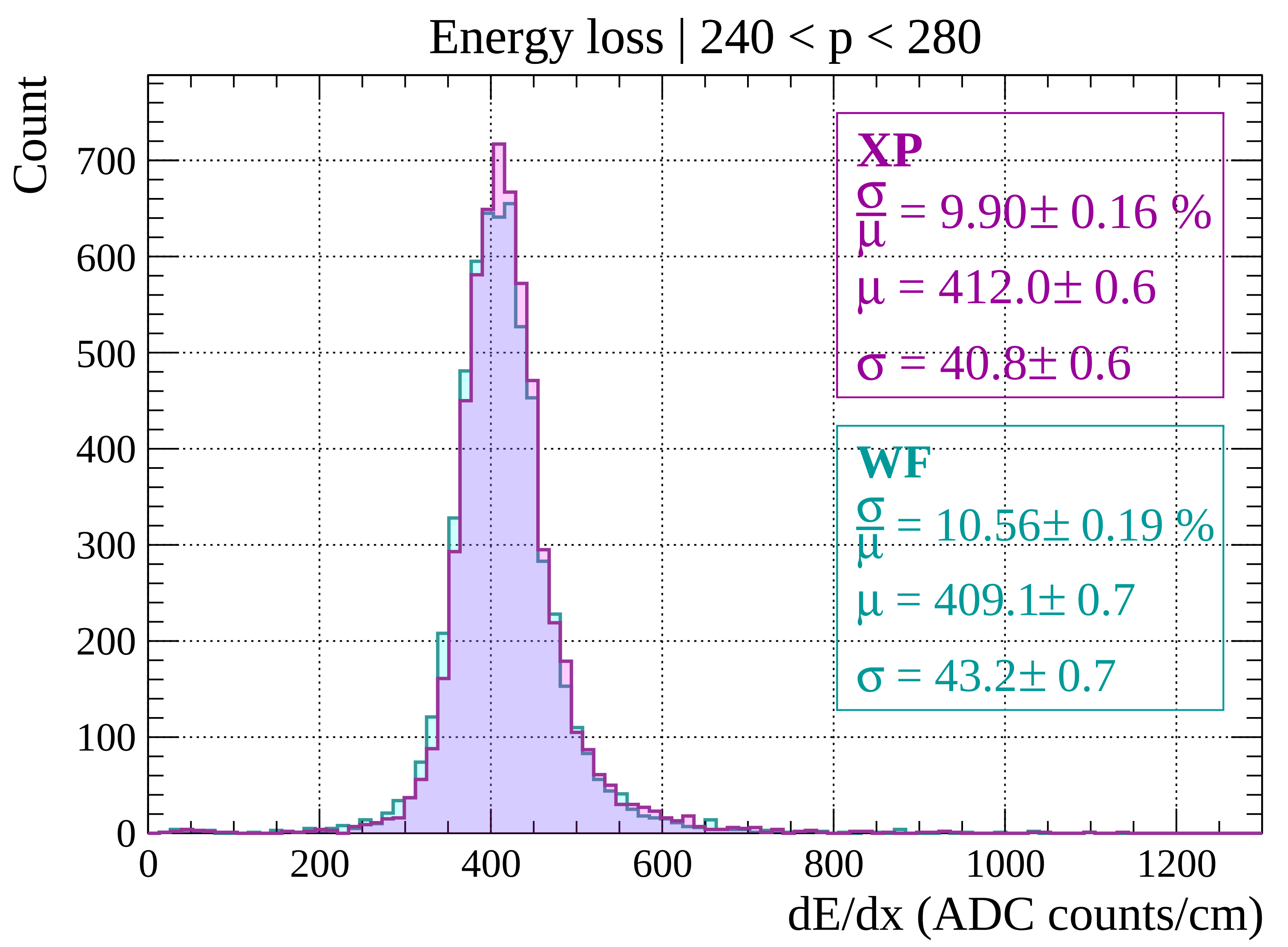
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.35± 0.34 %
μ = 558.7± 1.6
σ = 74.6± 1.7
WF
$\frac{\sigma}{\mu}$ = 13.88± 0.36 %
μ = 556.8± 1.7
σ = 77.3± 1.7

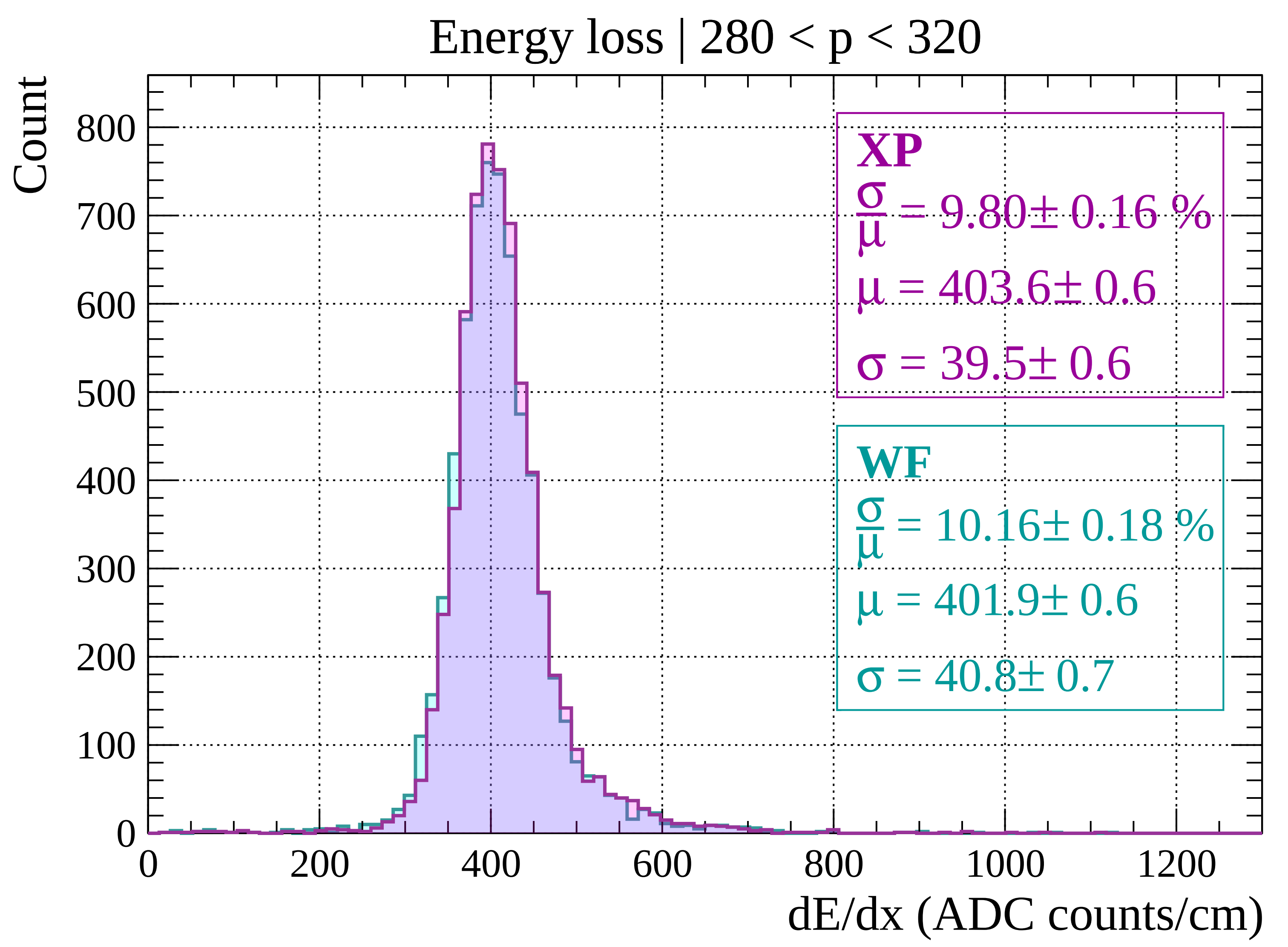


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 11.12± 0.23 %
μ = 470.3± 0.9
σ = 52.3± 1.0
WF
σ/μ = 11.74± 0.26 %
μ = 467.9± 1.0
σ = 54.9± 1.1

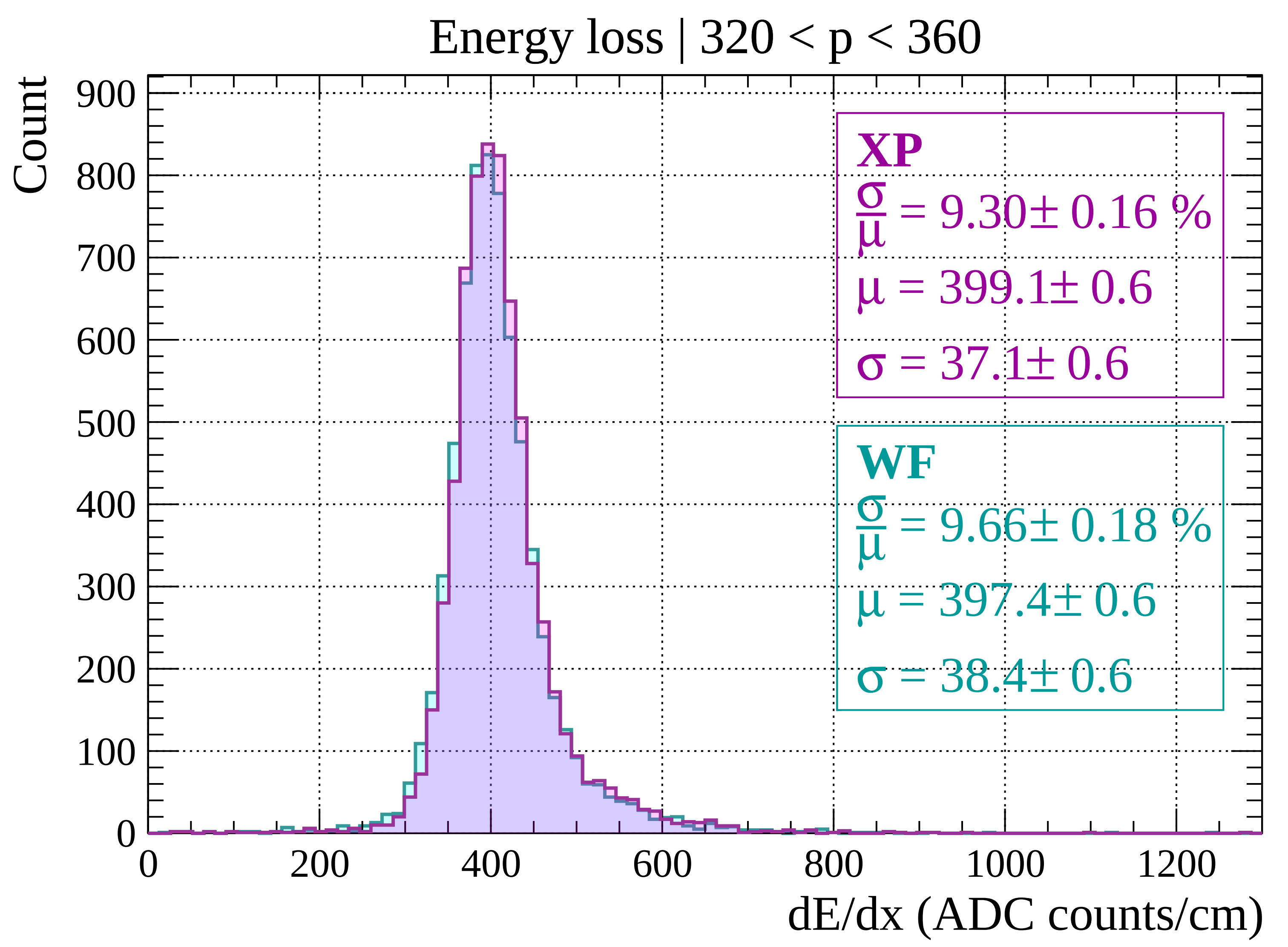


Energy loss | 200 < p < 240
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.94± 0.20 %
μ = 429.9± 0.8
σ = 42.7± 0.8
WF
σ/μ = 10.39± 0.20 %
μ = 427.2± 0.8
σ = 44.4± 0.8

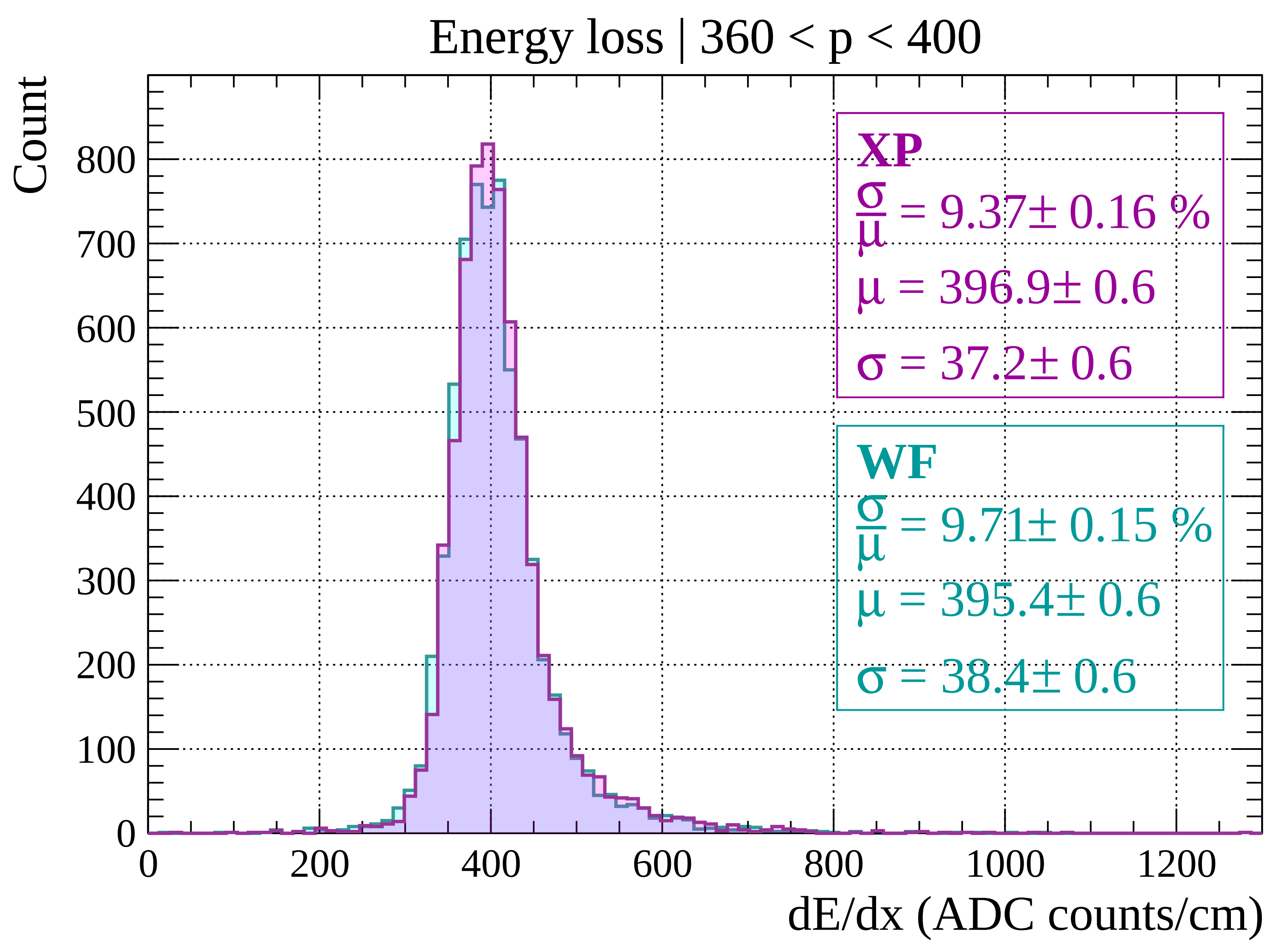

Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.90± 0.16 %
μ = 412.0± 0.6
σ = 40.8± 0.6
WF
σ/μ = 10.56± 0.19 %
μ = 409.1± 0.7
σ = 43.2± 0.7

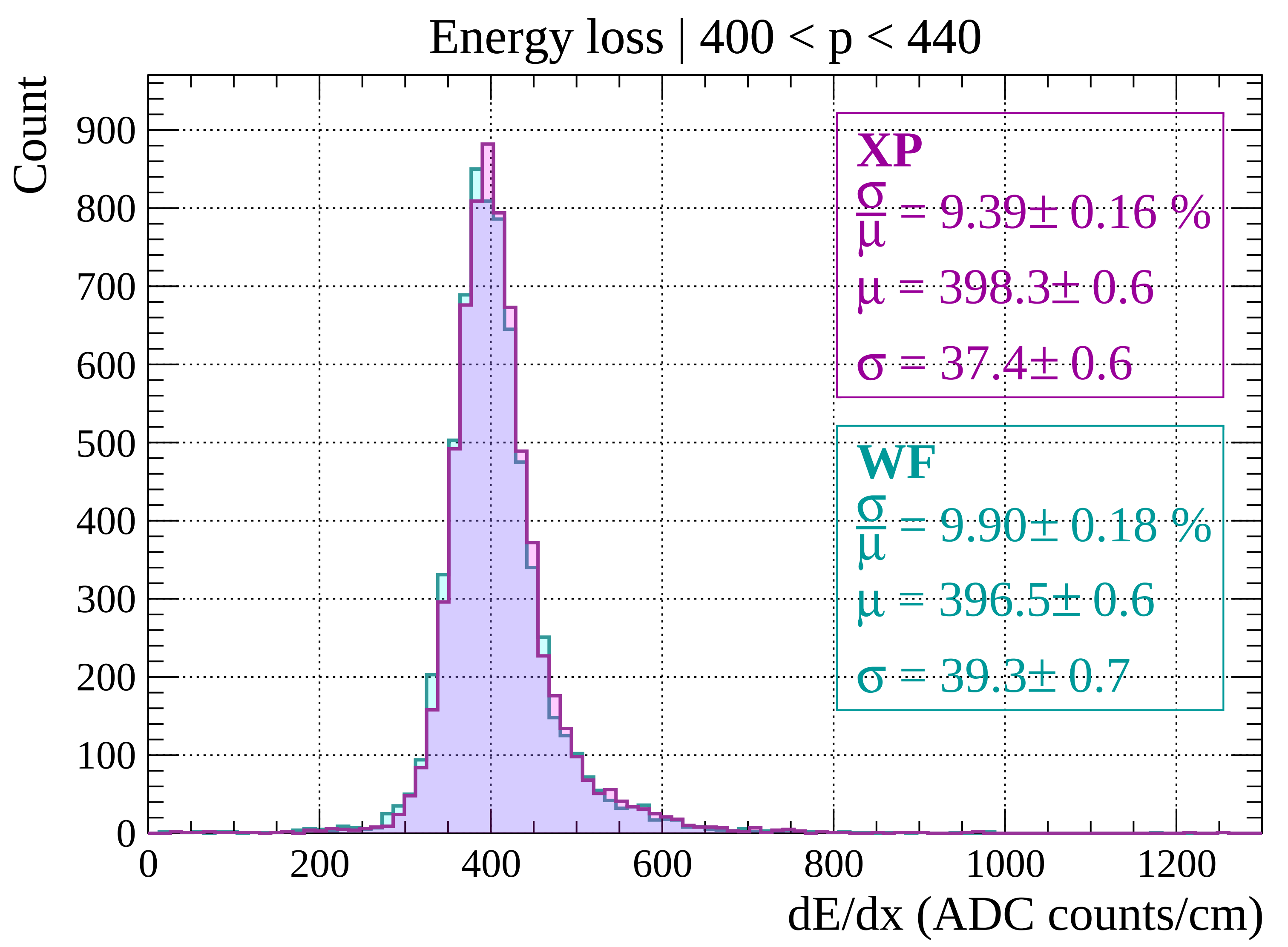


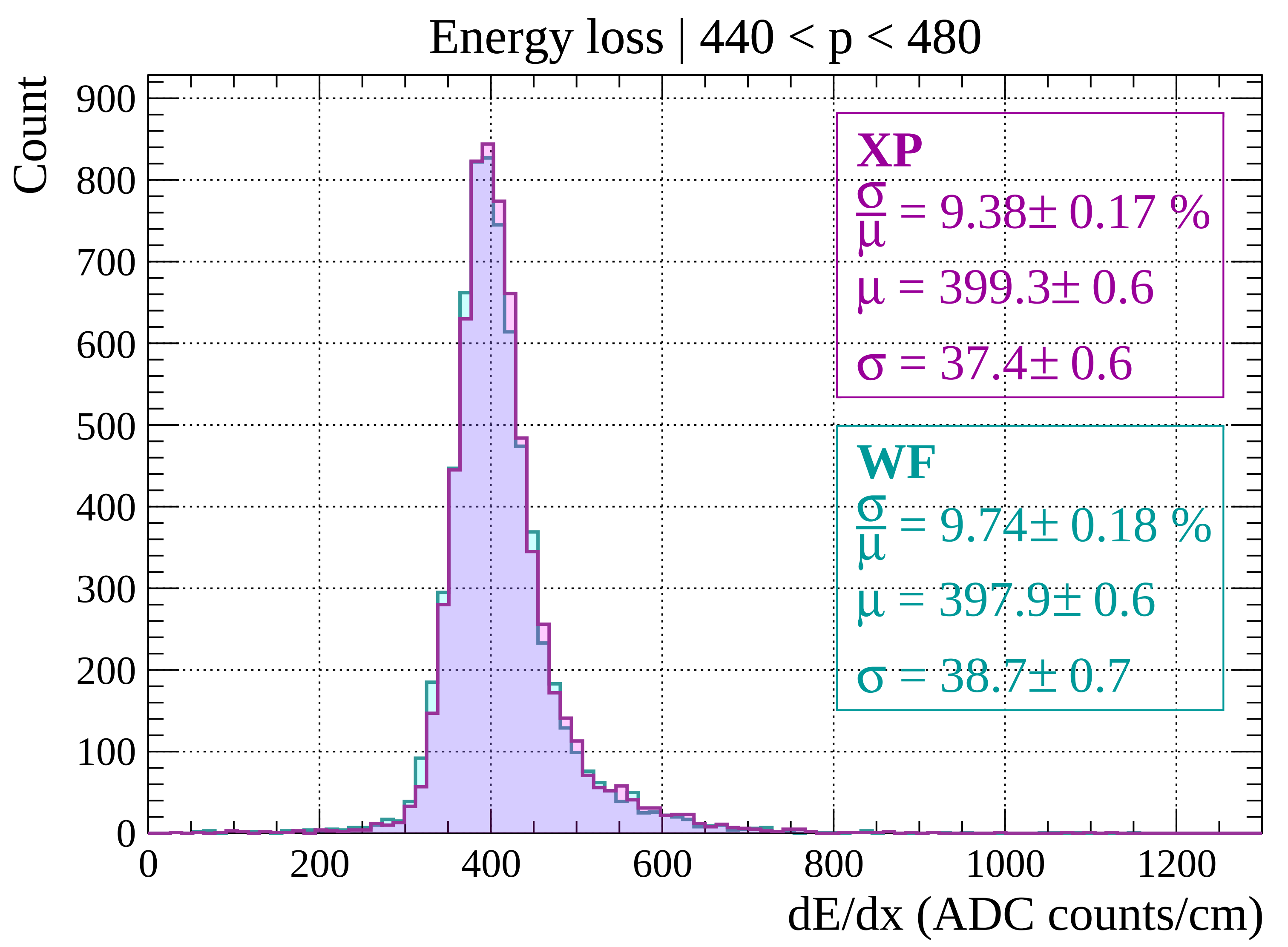
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.80 \pm 0.16$ %
$\mu = 403.6 \pm 0.6$
$\sigma = 39.5 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 10.16 \pm 0.18$ %
$\mu = 401.9 \pm 0.6$
$\sigma = 40.8 \pm 0.7$

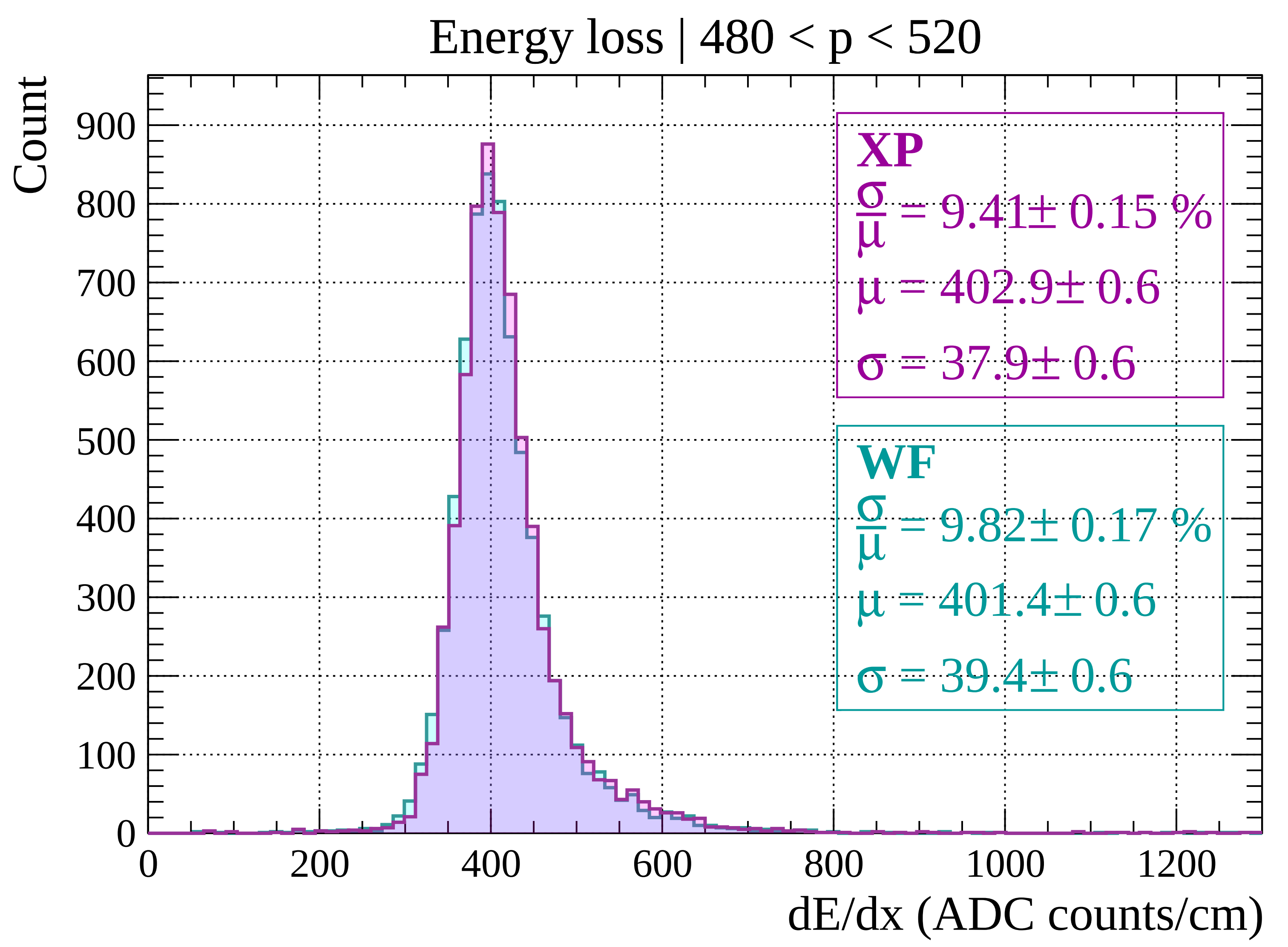


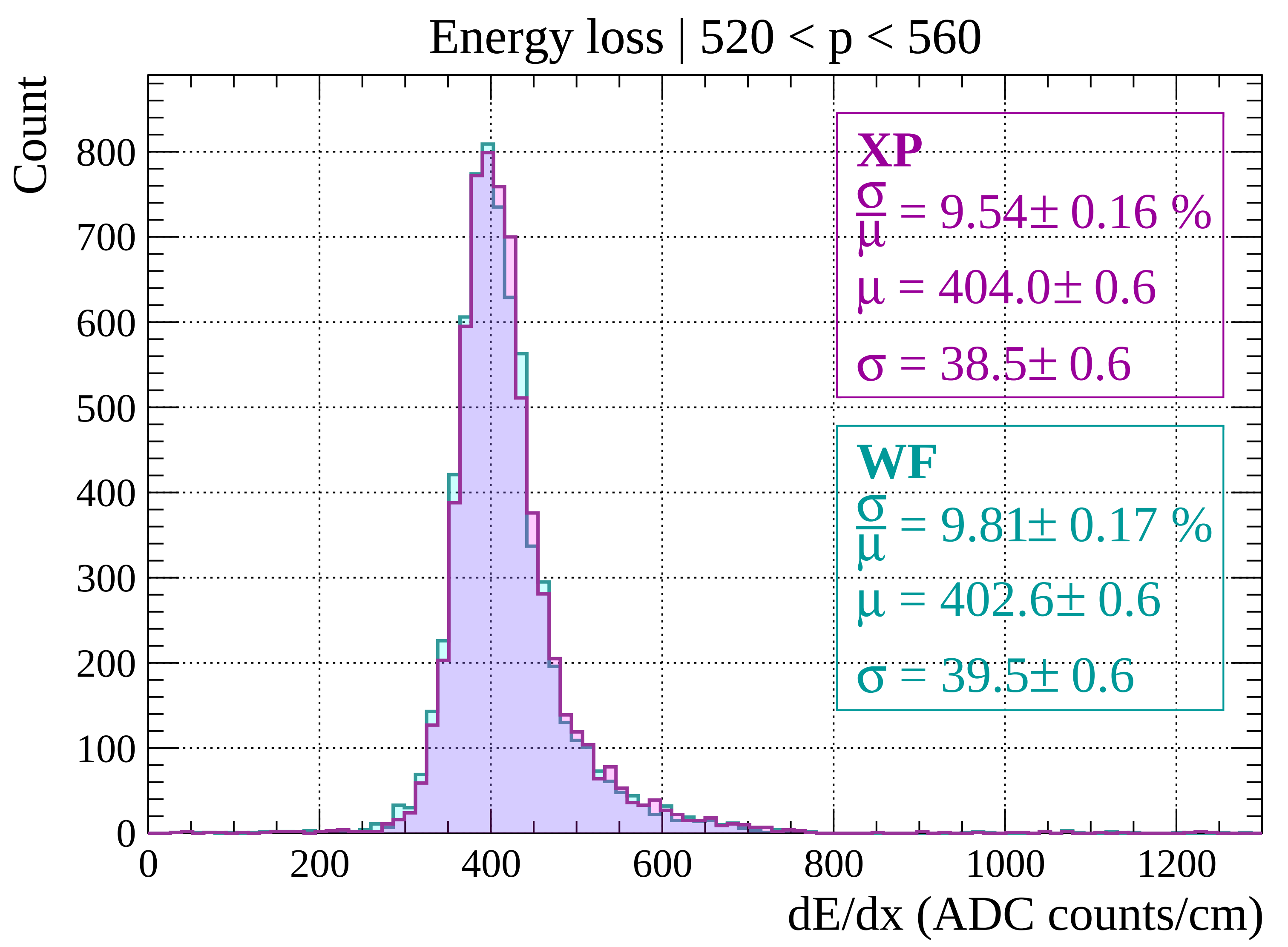
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.30± 0.16 %
μ = 399.1± 0.6
σ = 37.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.66± 0.18 %
μ = 397.4± 0.6
σ = 38.4± 0.6


Energy loss | 360 < p < 400
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.37± 0.16 %
μ = 396.9± 0.6
σ = 37.2± 0.6
WF
σ/μ = 9.71± 0.15 %
μ = 395.4± 0.6
σ = 38.4± 0.6

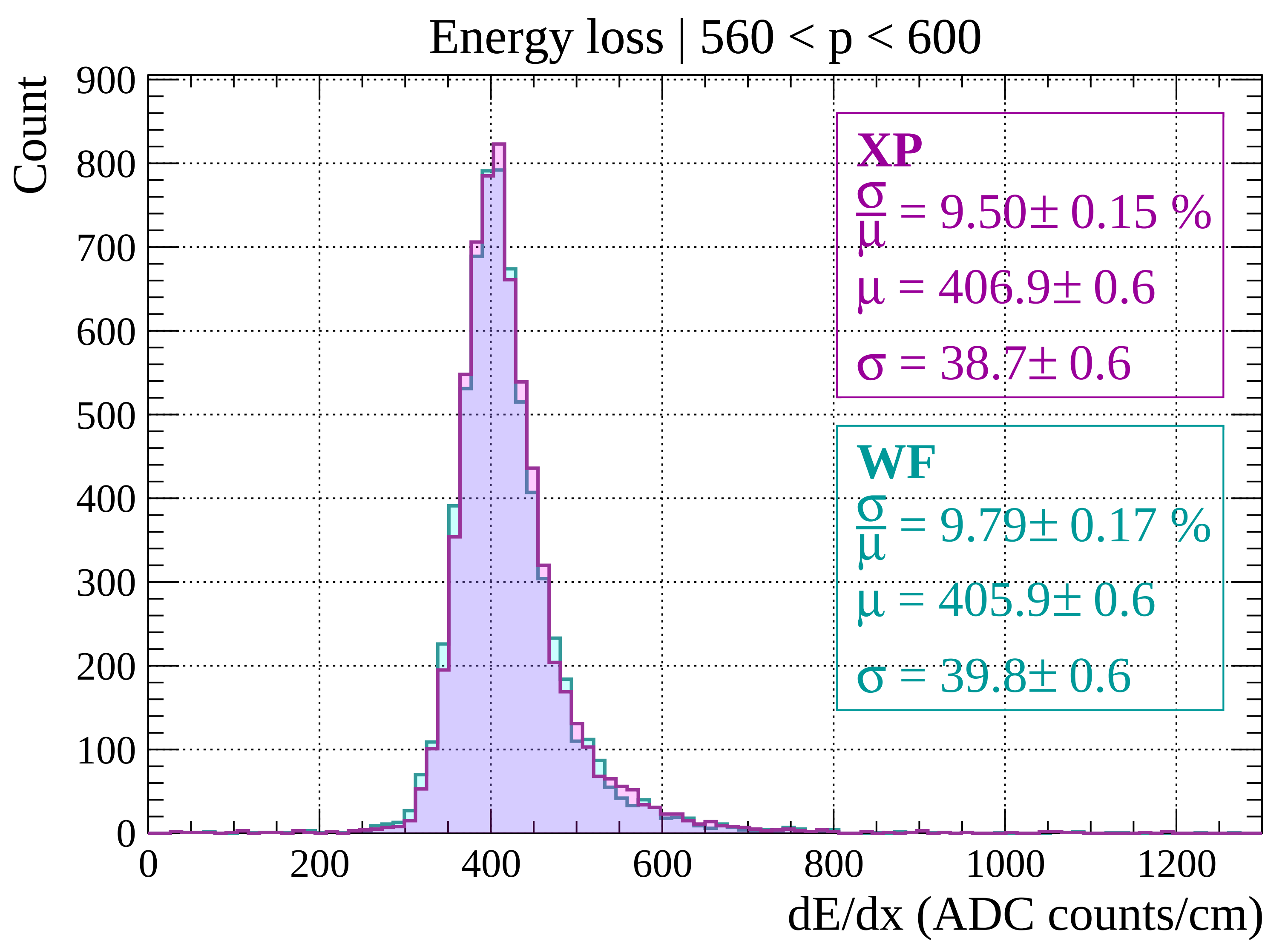

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.39± 0.16 %
μ = 398.3± 0.6
σ = 37.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.90± 0.18 %
μ = 396.5± 0.6
σ = 39.3± 0.7
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.38± 0.17 %
μ = 399.3± 0.6
σ = 37.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.74± 0.18 %
μ = 397.9± 0.6
σ = 38.7± 0.7
0
100
200
300
400
500
600
700
800
900
1000
1200

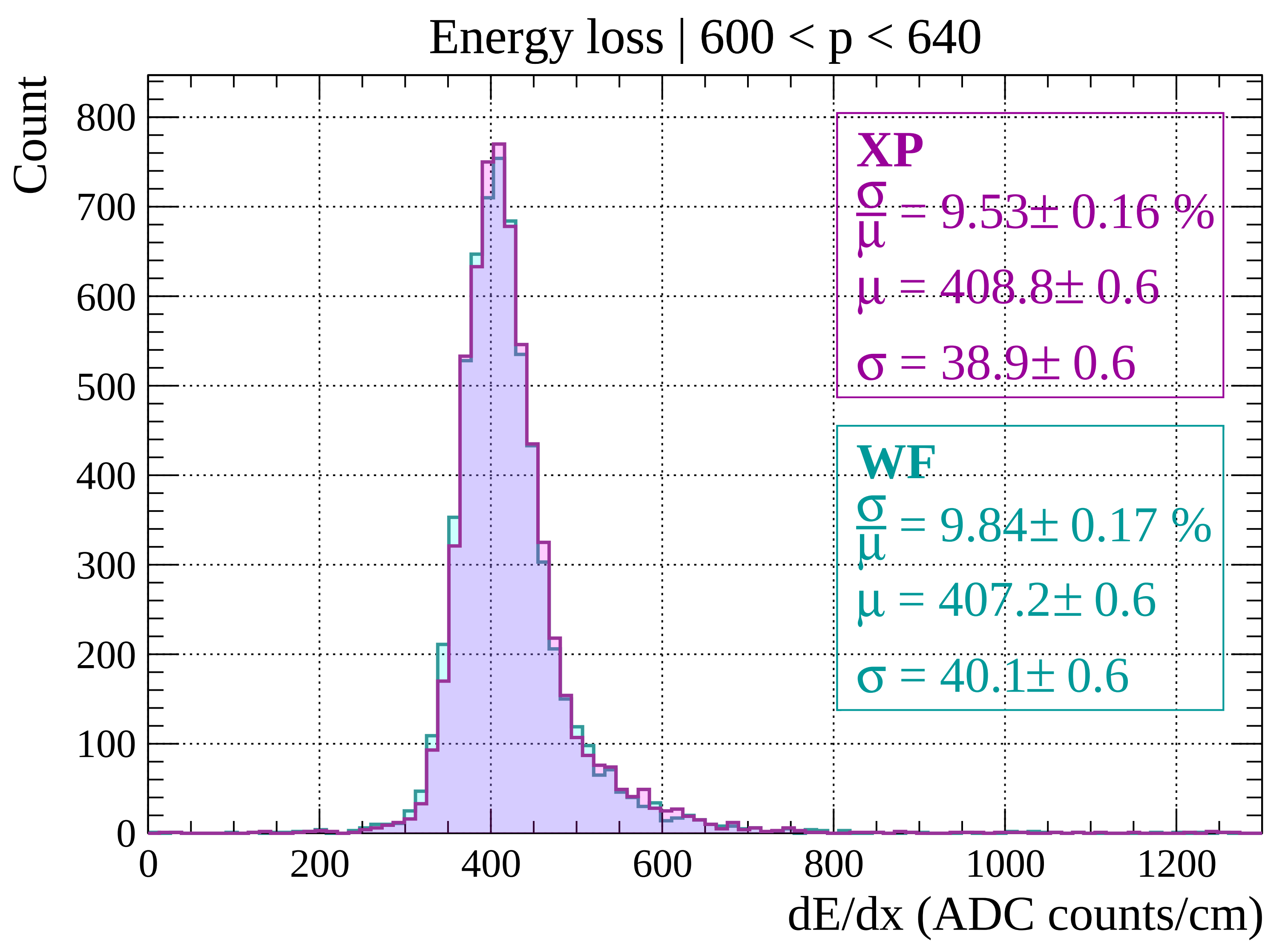

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.41± 0.15 %
μ = 402.9± 0.6
σ = 37.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.82± 0.17 %
μ = 401.4± 0.6
σ = 39.4± 0.6

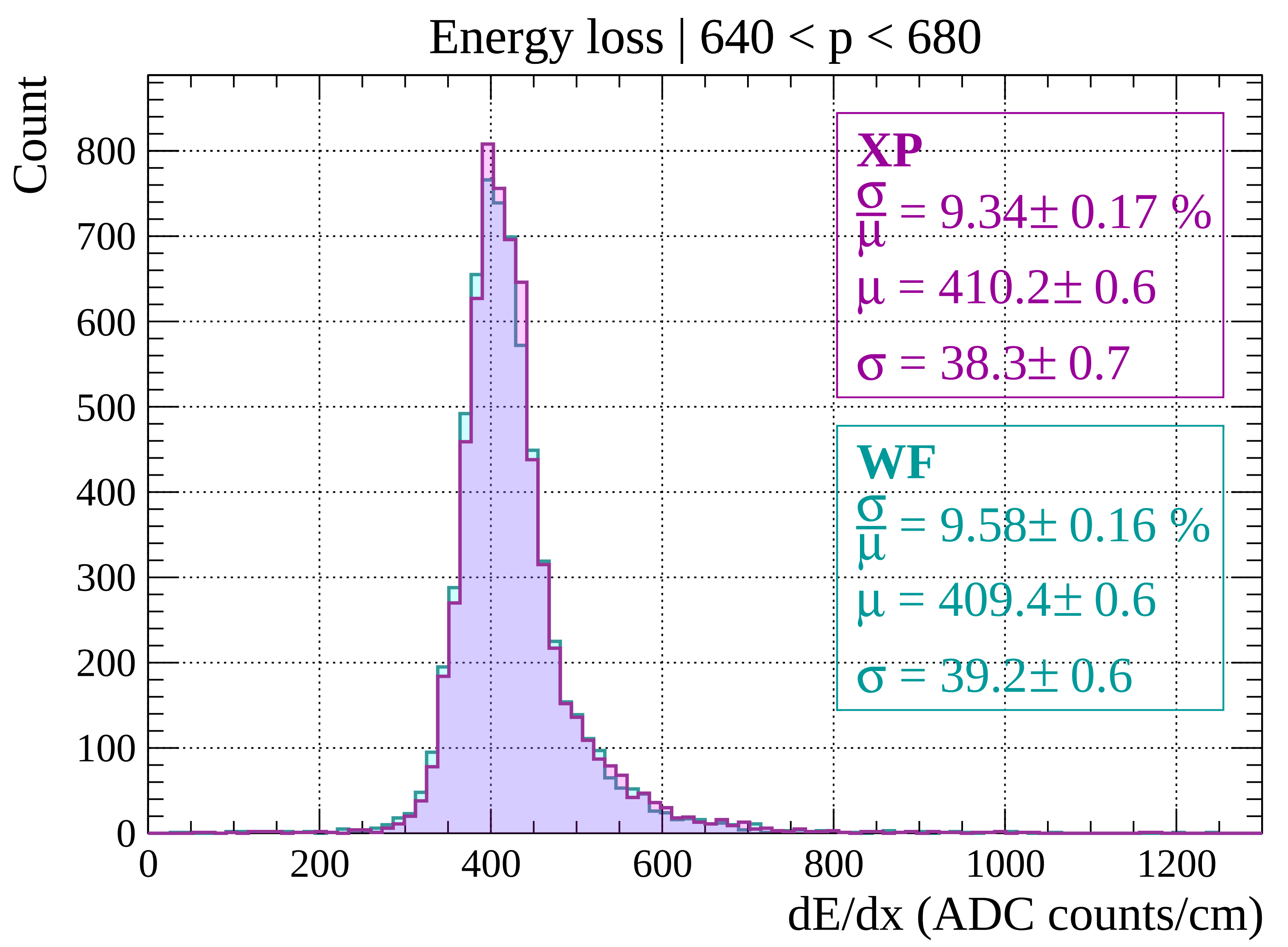


Energy loss | 520 < p < 560
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.54± 0.16 %
μ = 404.0± 0.6
σ = 38.5± 0.6
WF
σ/μ = 9.81± 0.17 %
μ = 402.6± 0.6
σ = 39.5± 0.6

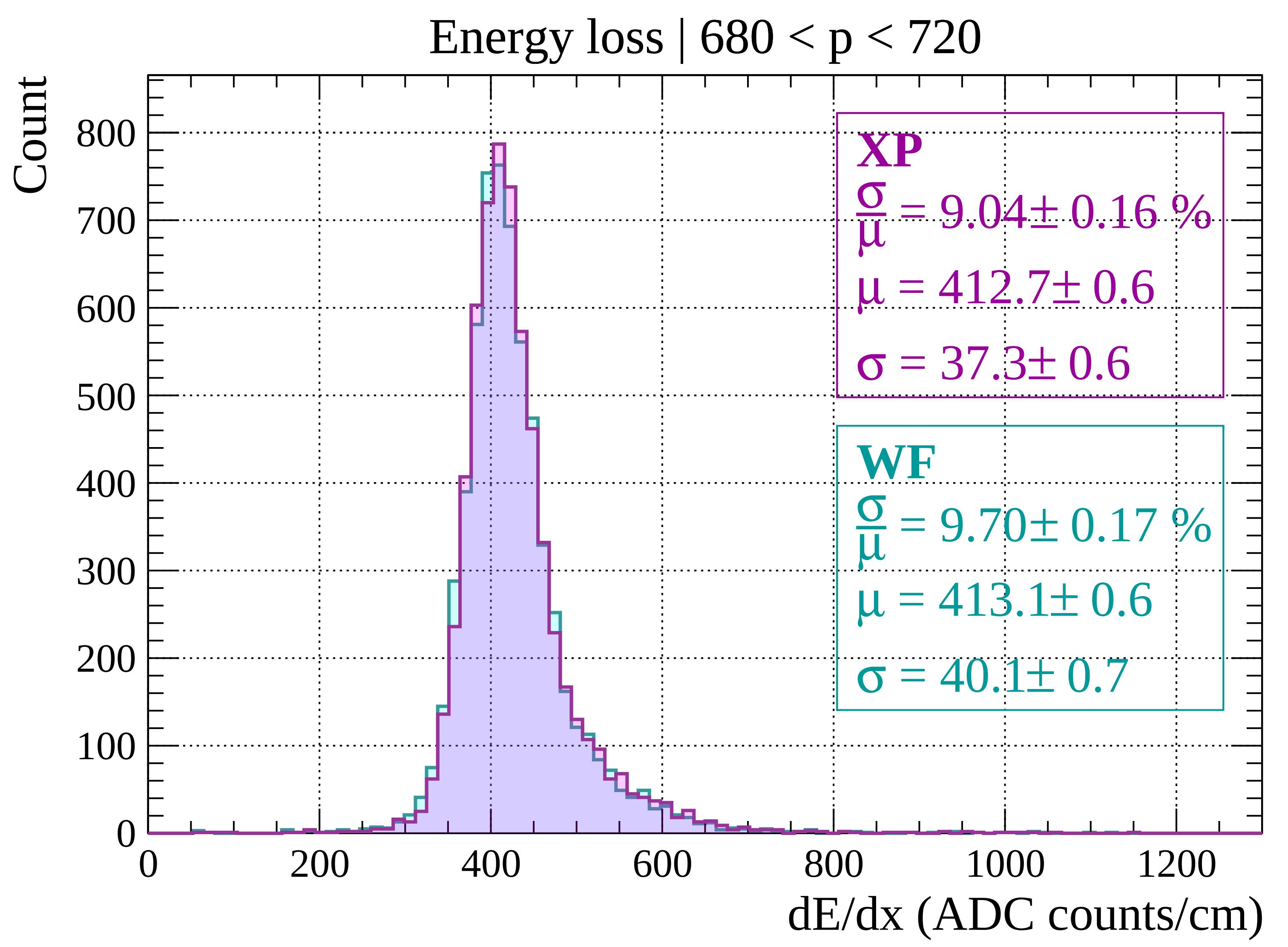


Energy loss | 560 < p < 600
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.50± 0.15 %
μ = 406.9± 0.6
σ = 38.7± 0.6
WF
σ/μ = 9.79± 0.17 %
μ = 405.9± 0.6
σ = 39.8± 0.6

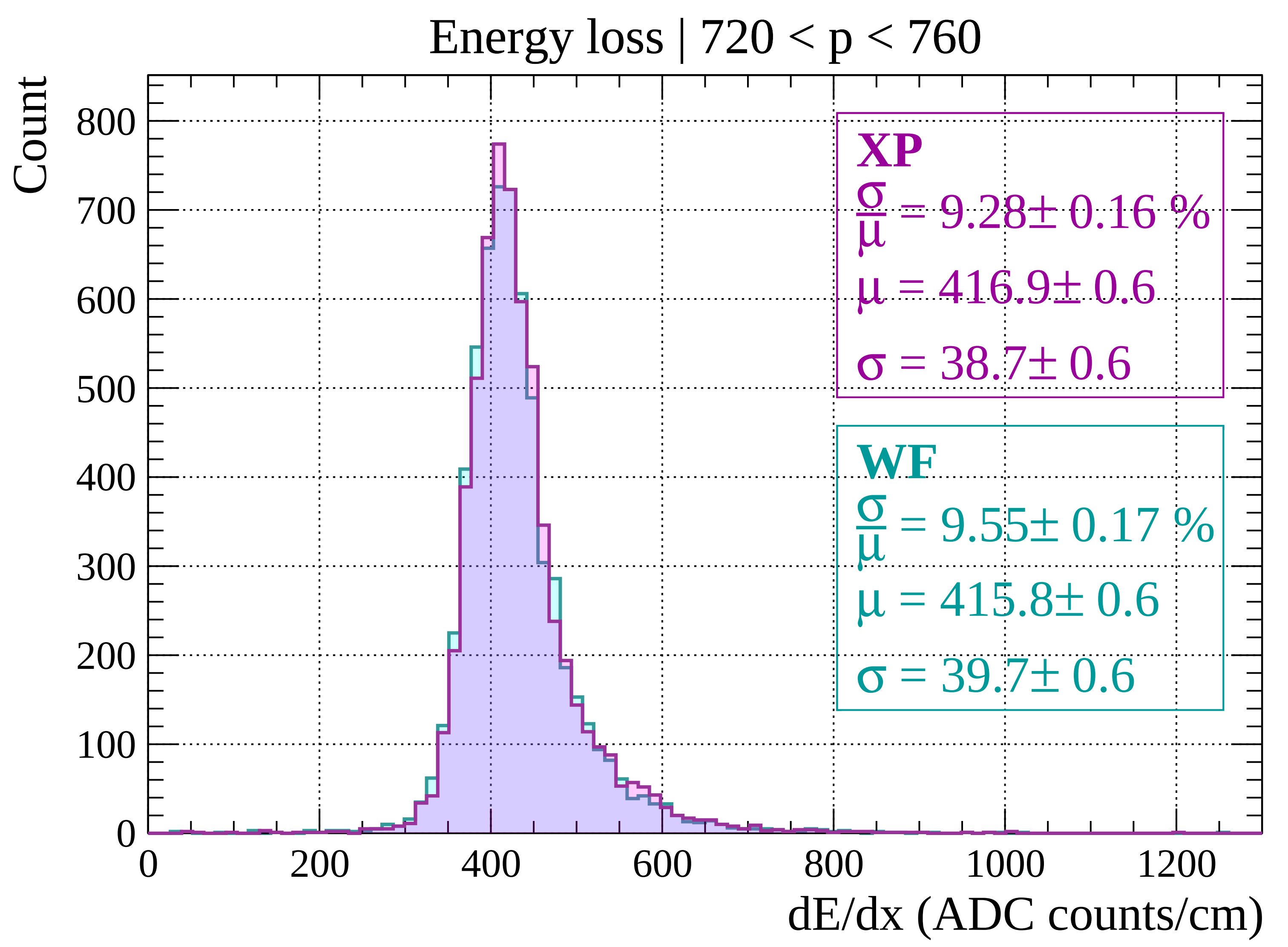

Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.53± 0.16 %
μ = 408.8± 0.6
σ = 38.9± 0.6
WF
σ/μ = 9.84± 0.17 %
μ = 407.2± 0.6
σ = 40.1± 0.6



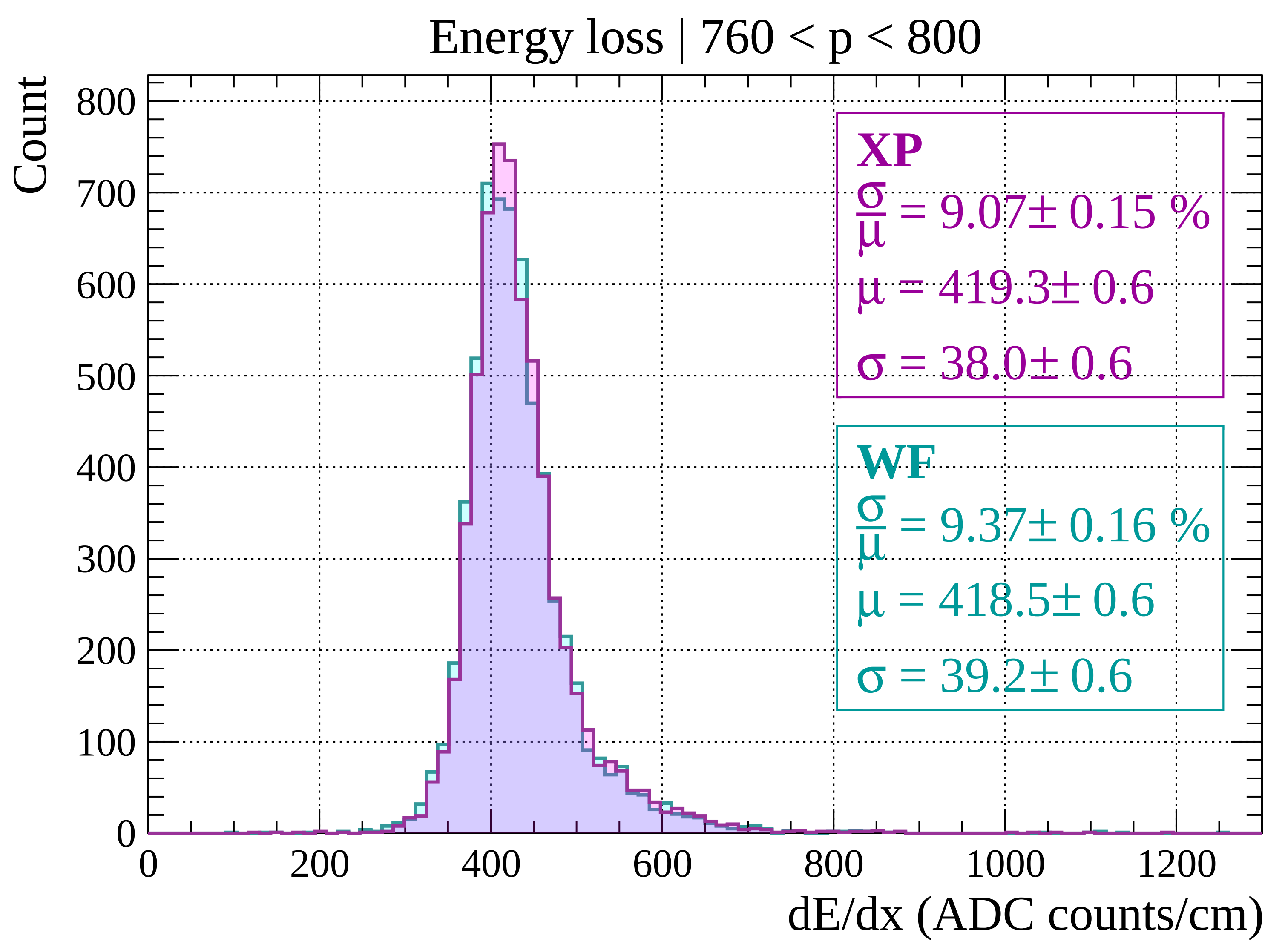
Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.34± 0.17 %
μ = 410.2± 0.6
σ = 38.3± 0.7
WF
σ/μ = 9.58± 0.16 %
μ = 409.4± 0.6
σ = 39.2± 0.6



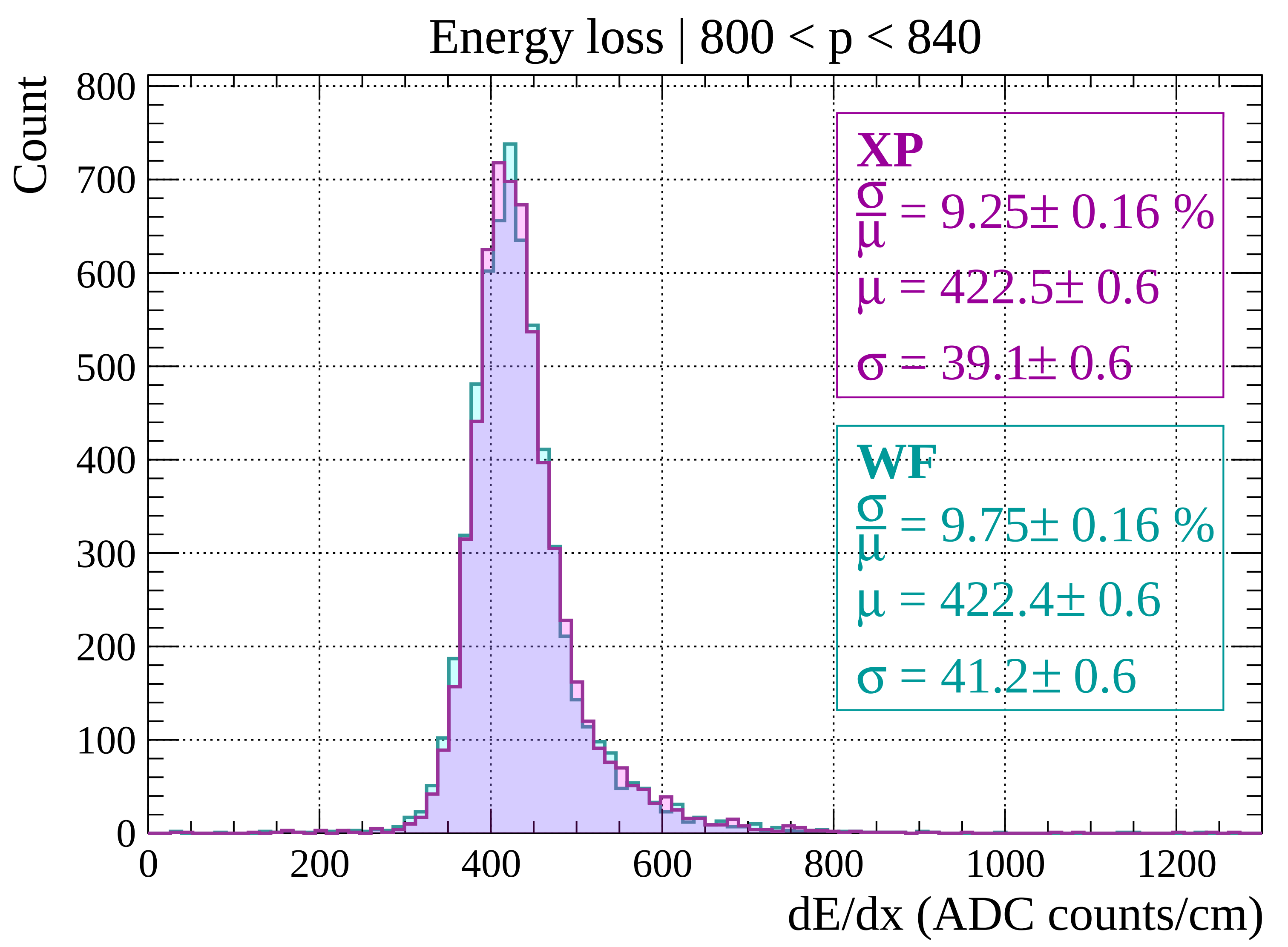
Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.04± 0.16 %
μ = 412.7± 0.6
σ = 37.3± 0.6
WF
σ/μ = 9.70± 0.17 %
μ = 413.1± 0.6
σ = 40.1± 0.7



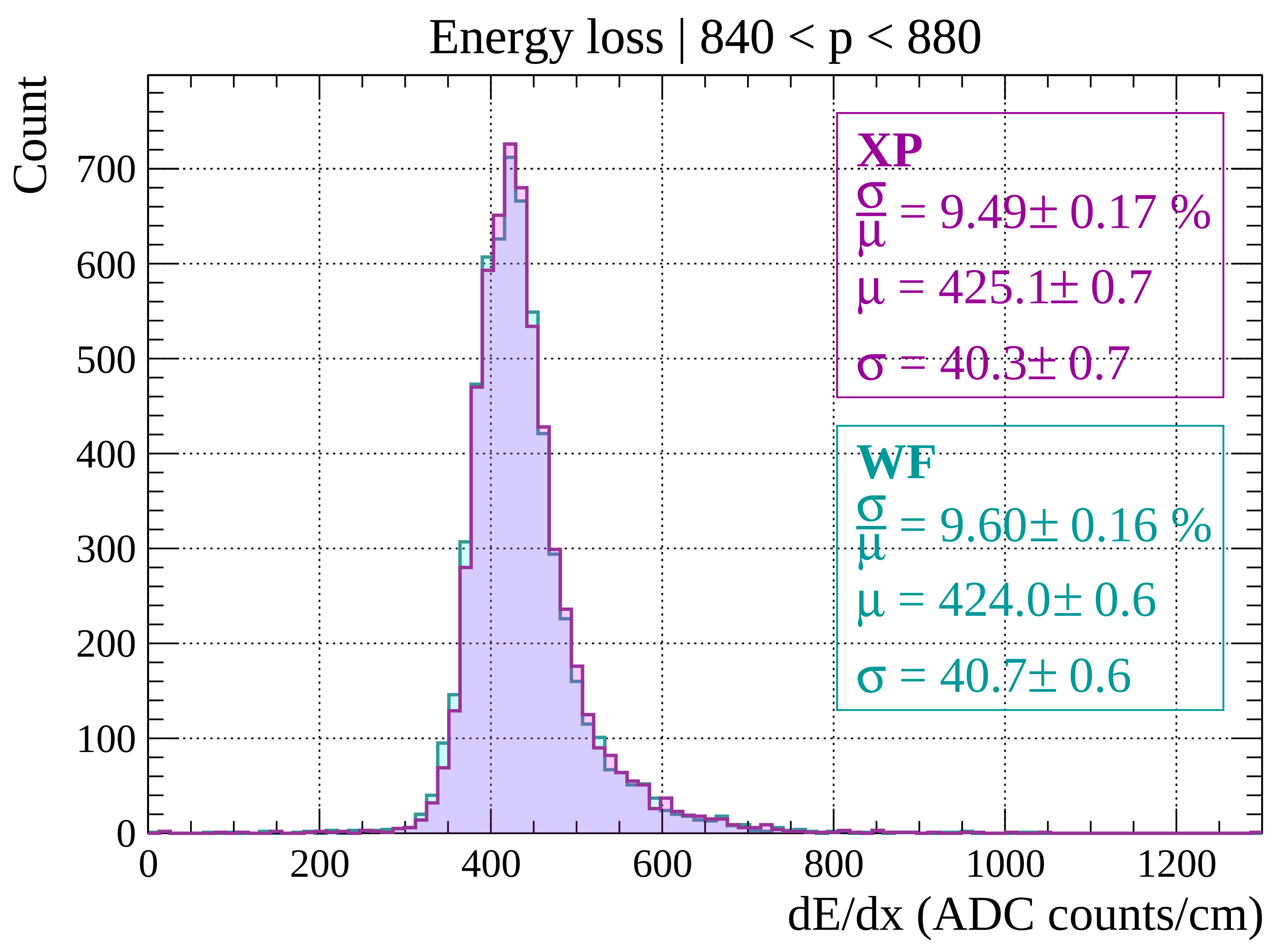
Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
$\frac{\sigma}{\mu} = 9.28 \pm 0.16$ %
$\mu = 416.9 \pm 0.6$
$\sigma = 38.7 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.55 \pm 0.17$ %
$\mu = 415.8 \pm 0.6$
$\sigma = 39.7 \pm 0.6$

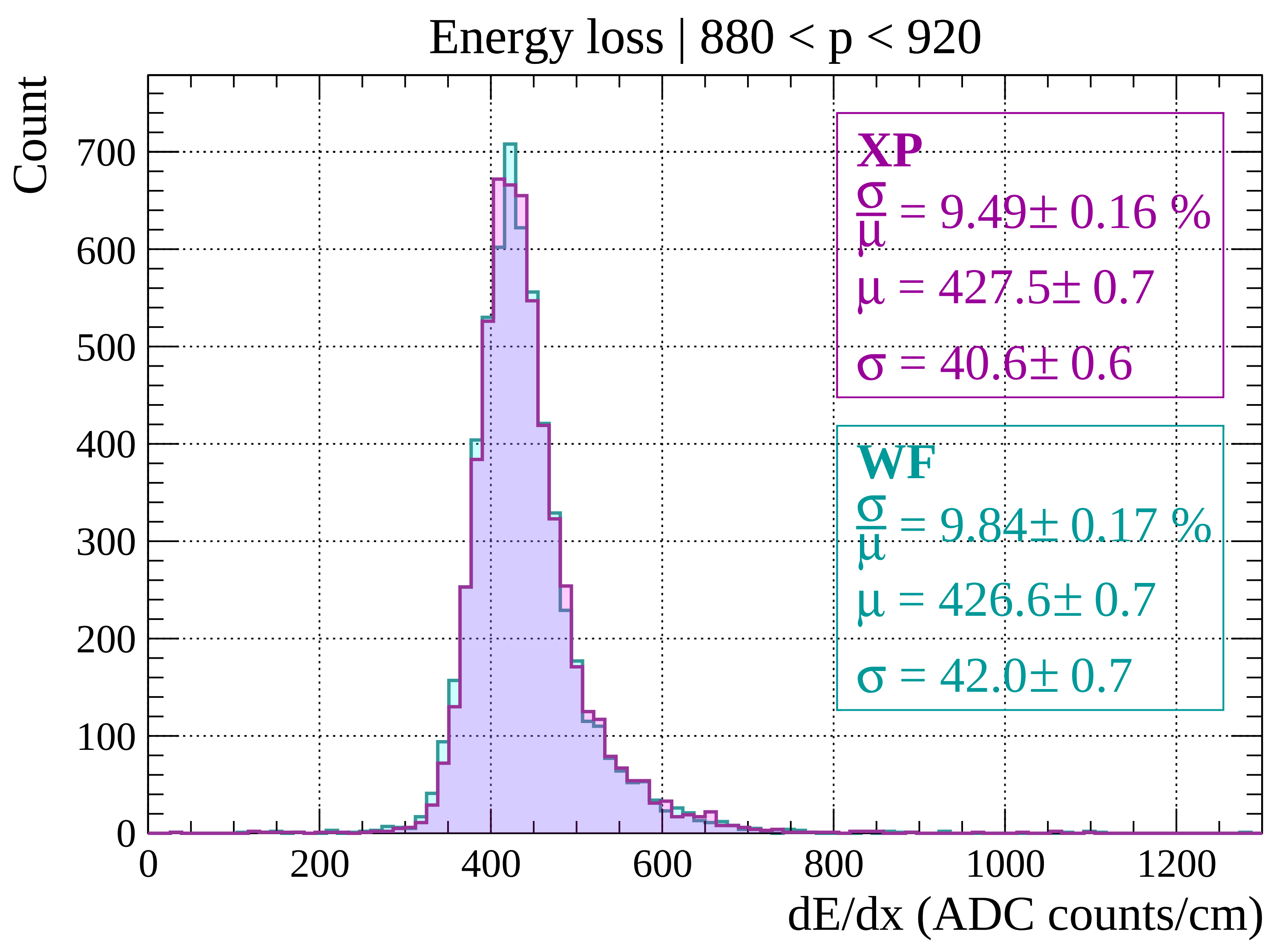


Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.07± 0.15 %
μ = 419.3± 0.6
σ = 38.0± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.37± 0.16 %
μ = 418.5± 0.6
σ = 39.2± 0.6

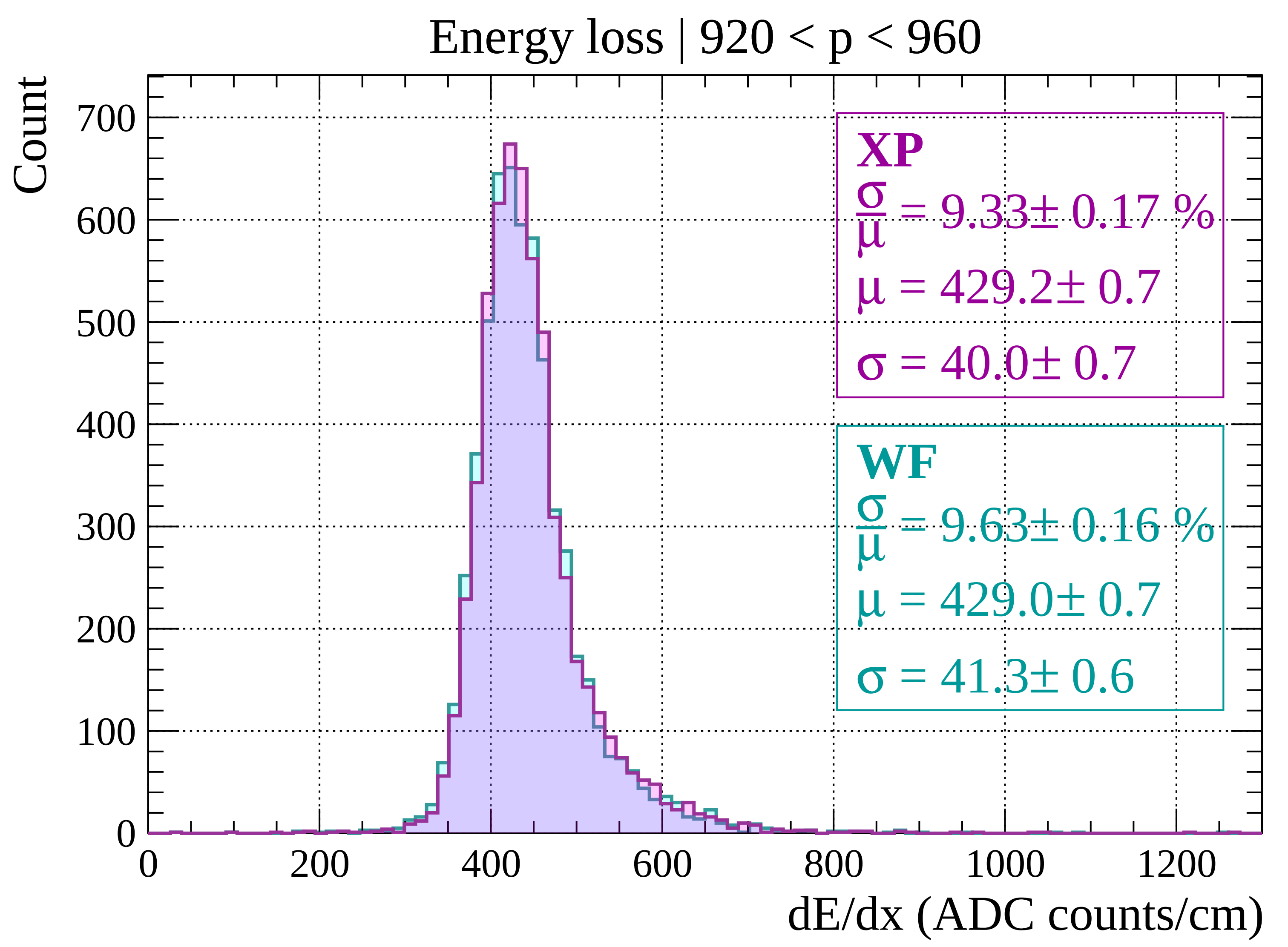


Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
σ/μ = 9.25± 0.16 %
μ = 422.5± 0.6
σ = 39.1± 0.6
WF
σ/μ = 9.75± 0.16 %
μ = 422.4± 0.6
σ = 41.2± 0.6

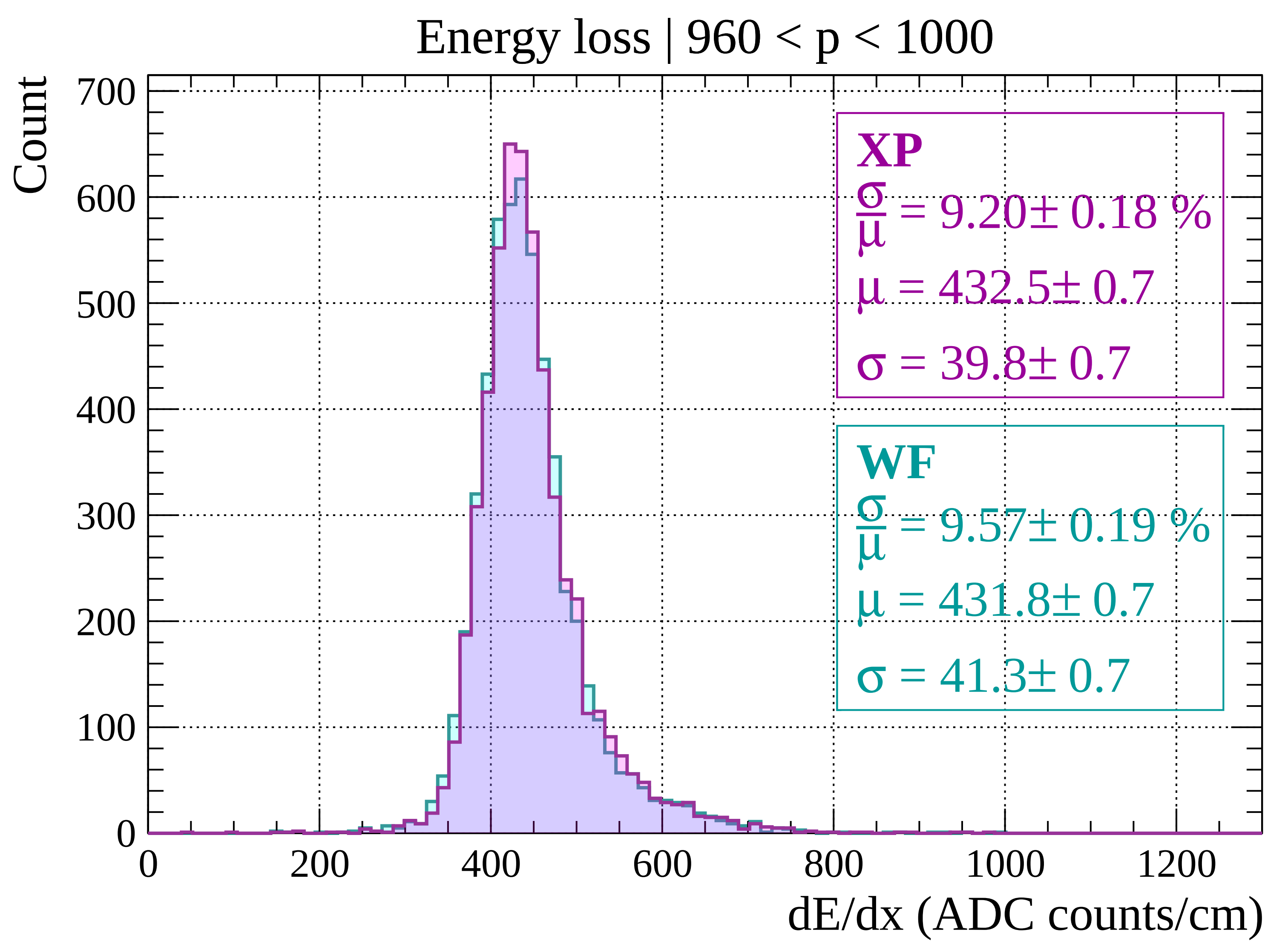

Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.17 %
μ = 425.1± 0.7
σ = 40.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.60± 0.16 %
μ = 424.0± 0.6
σ = 40.7± 0.6



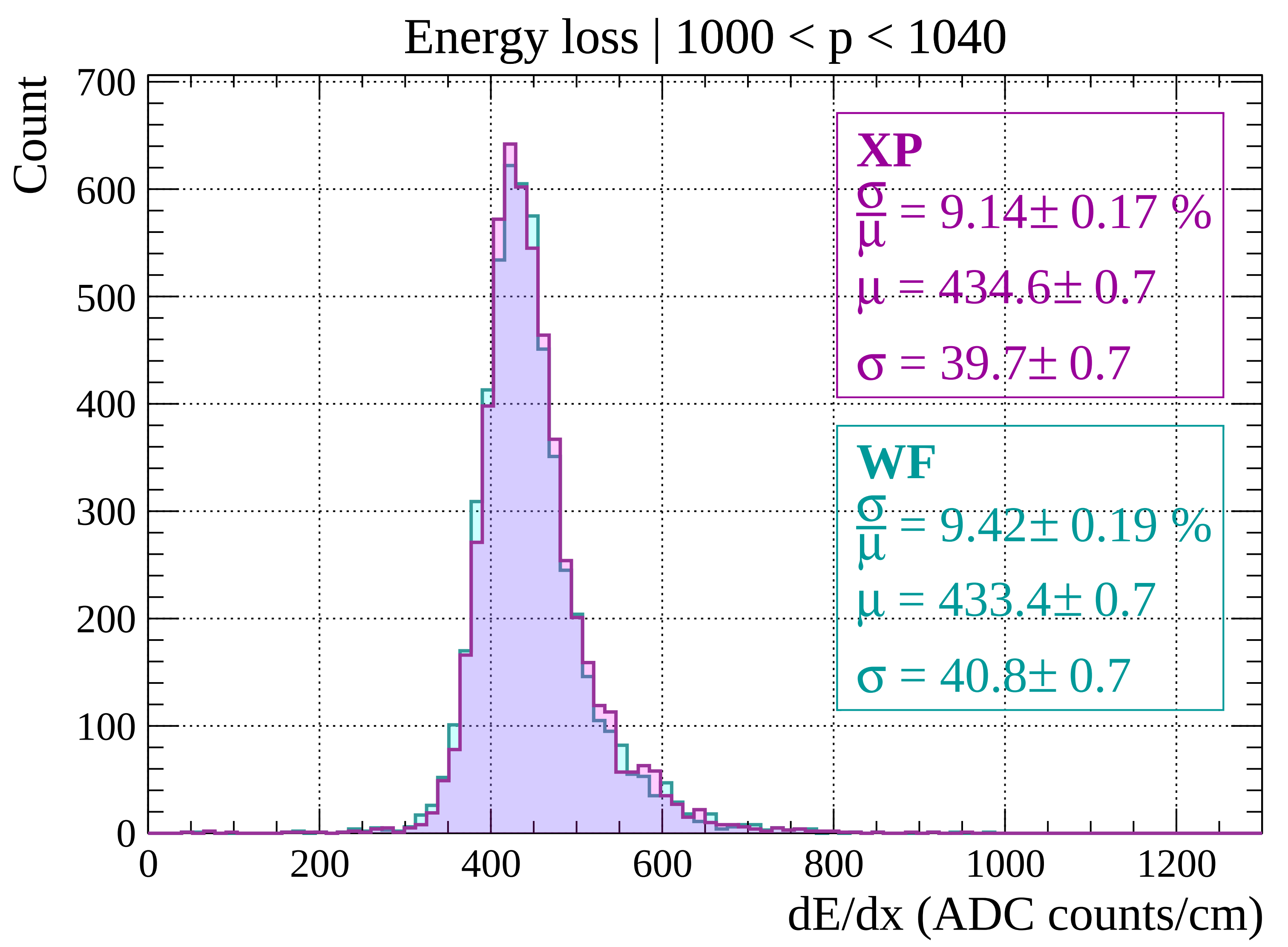
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.16 %
μ = 427.5± 0.7
σ = 40.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.84± 0.17 %
μ = 426.6± 0.7
σ = 42.0± 0.7



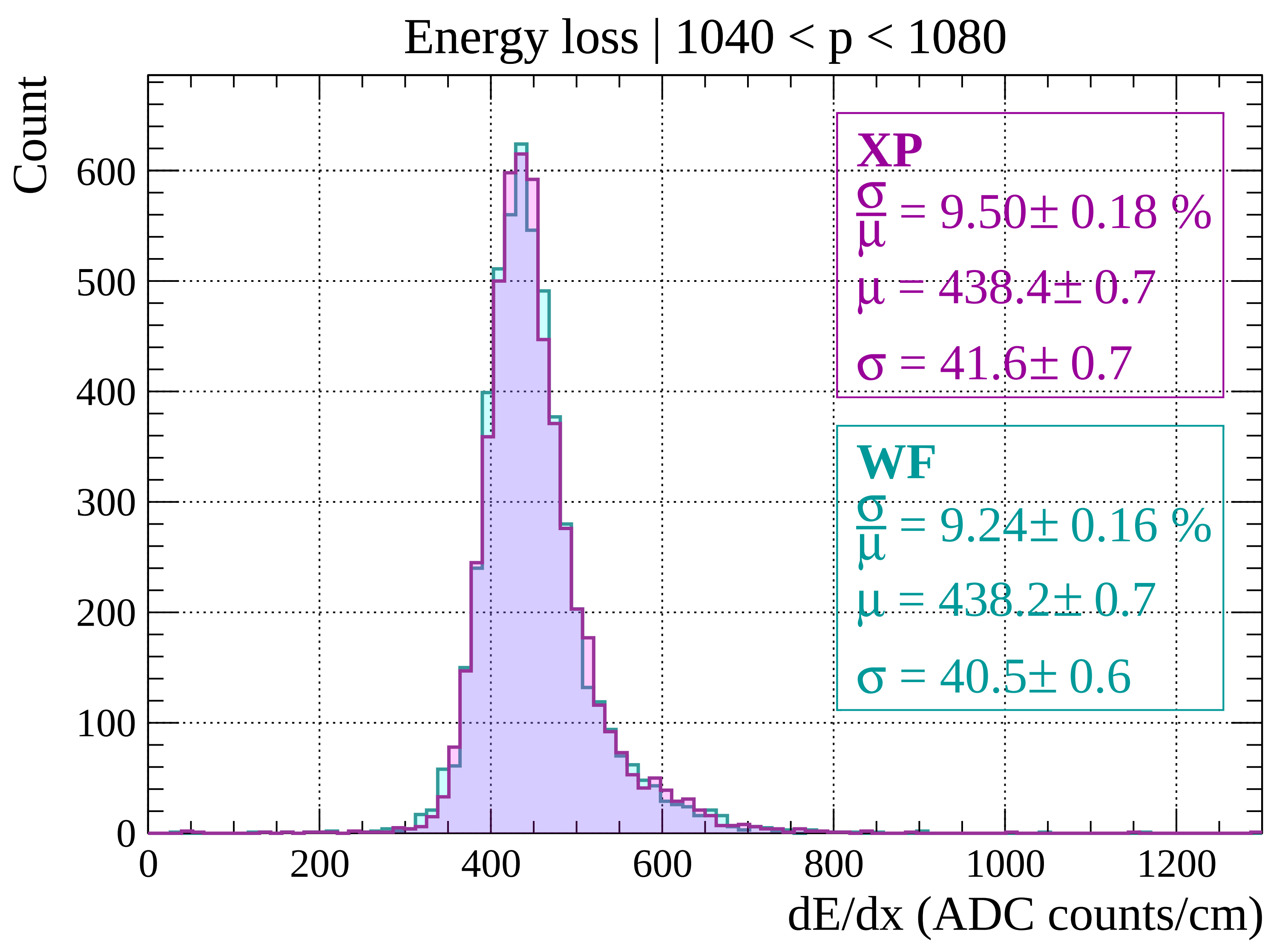
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.33 \pm 0.17$ %
$\mu = 429.2 \pm 0.7$
$\sigma = 40.0 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.63 \pm 0.16$ %
$\mu = 429.0 \pm 0.7$
$\sigma = 41.3 \pm 0.6$
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

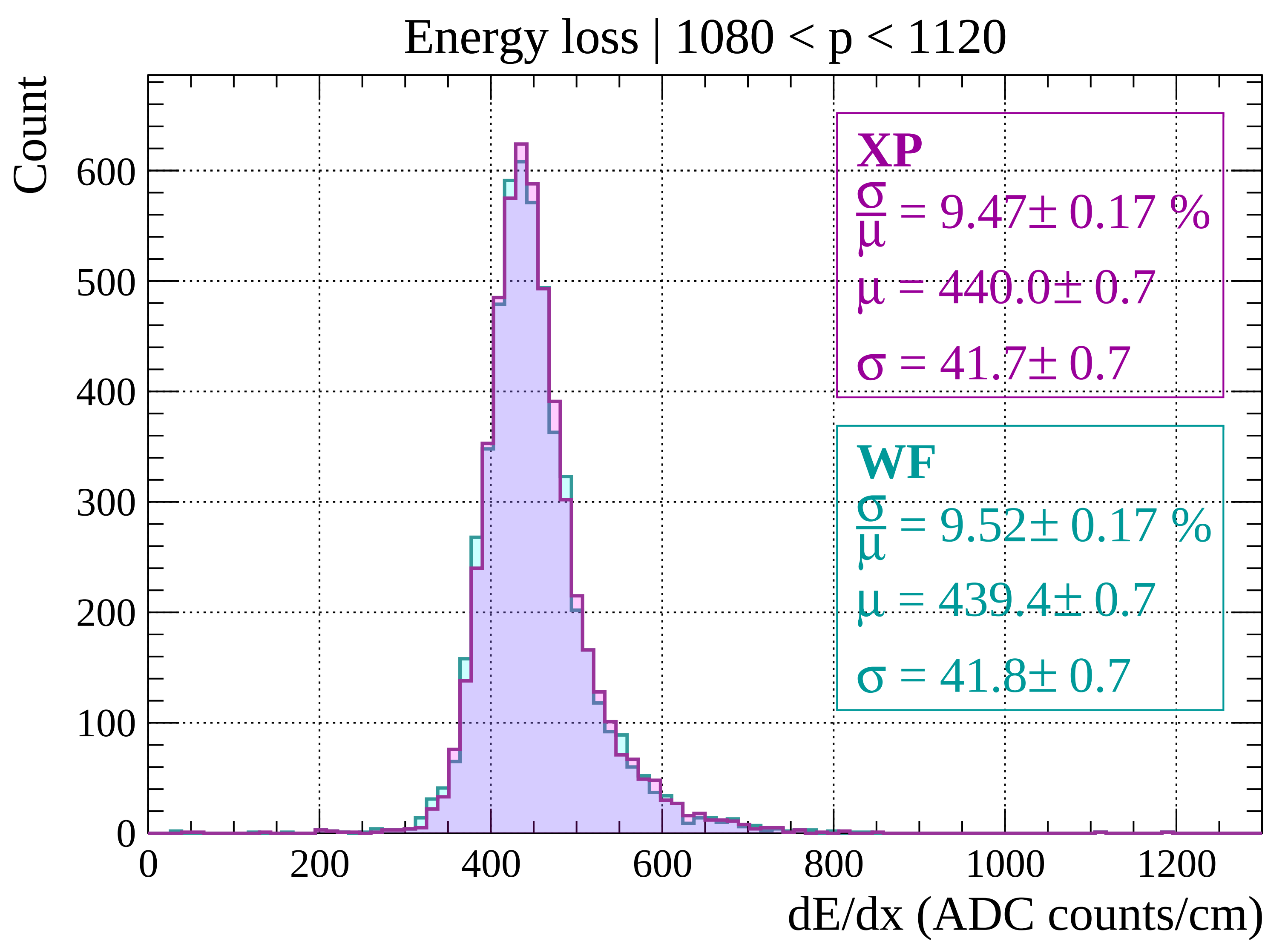


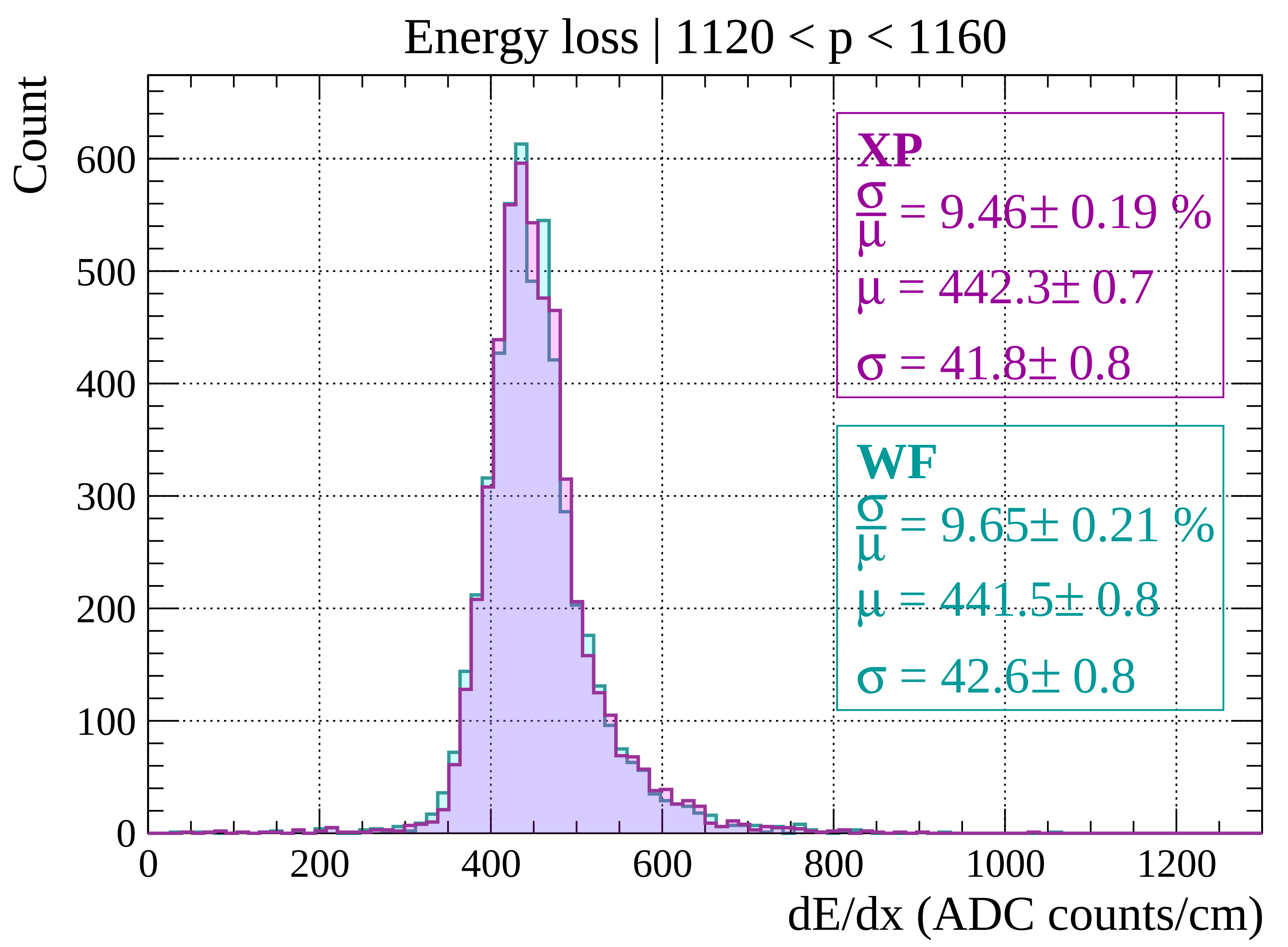
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.20± 0.18 %
μ = 432.5± 0.7
σ = 39.8± 0.7
WF
σ/μ = 9.57± 0.19 %
μ = 431.8± 0.7
σ = 41.3± 0.7



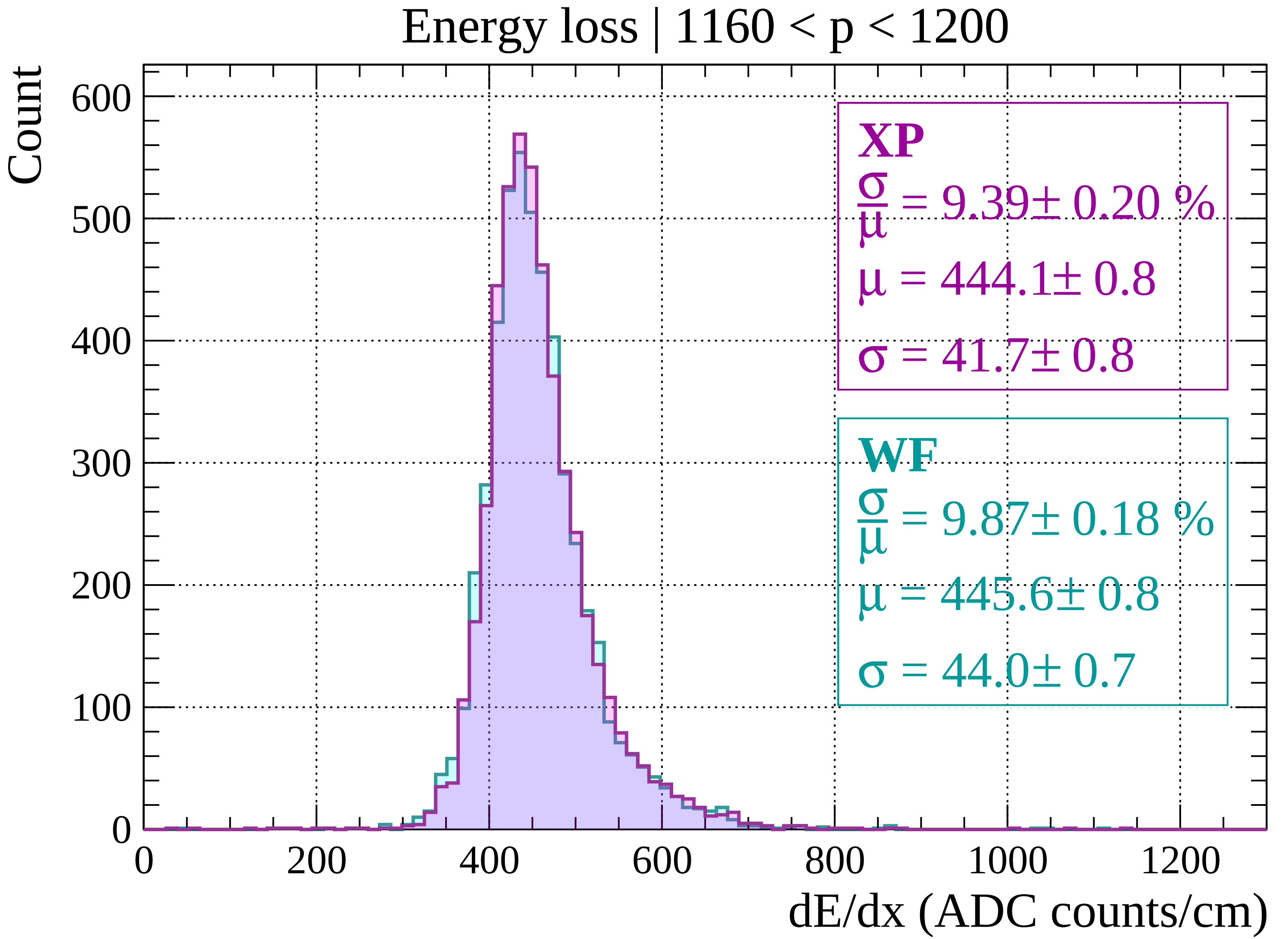
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.14± 0.17 %
μ = 434.6± 0.7
σ = 39.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.42± 0.19 %
μ = 433.4± 0.7
σ = 40.8± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

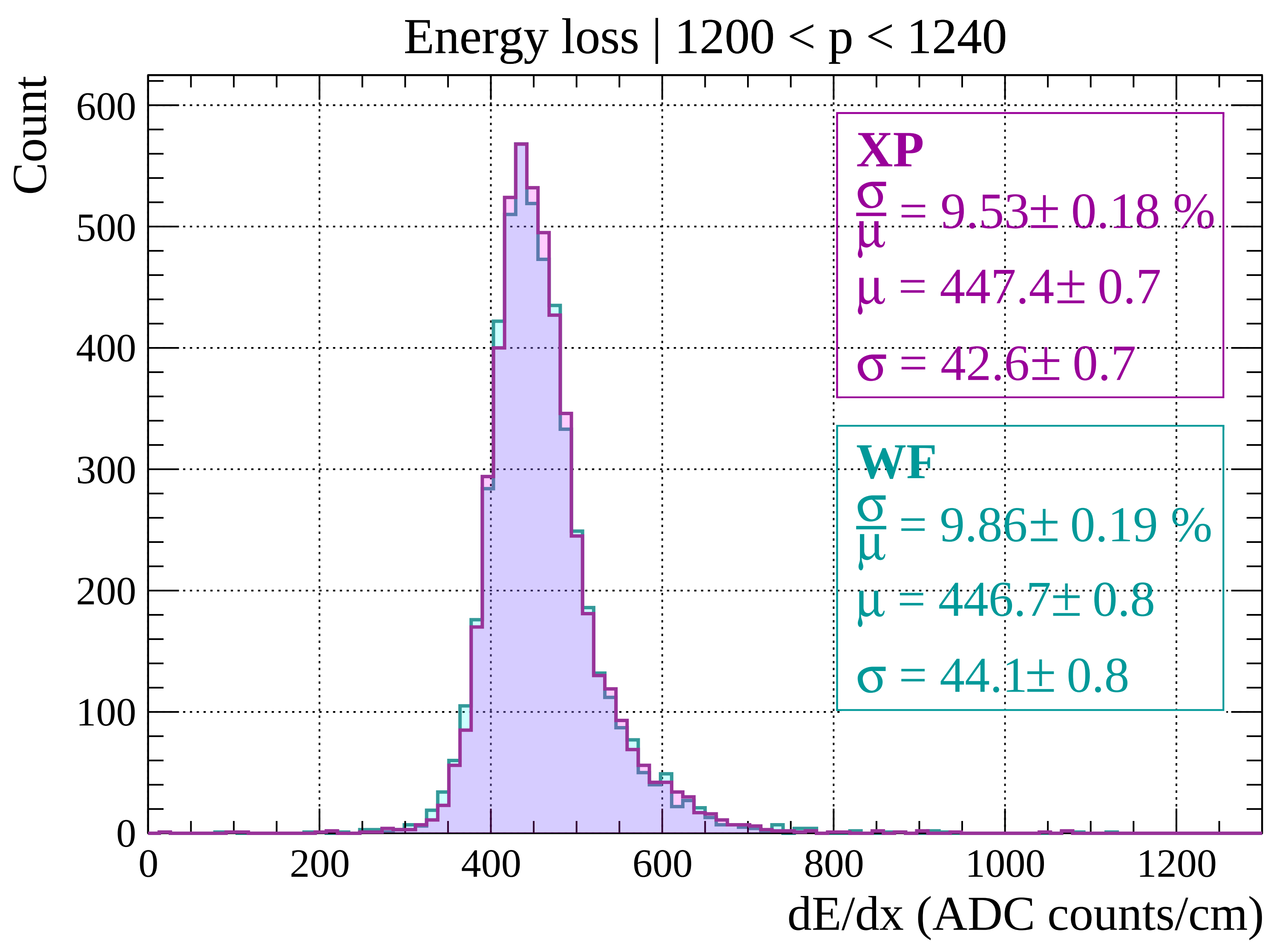

Energy loss | 1040 < p < 1080
Count
600
500
400
300
200
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.50± 0.18 %
μ = 438.4± 0.7
σ = 41.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.24± 0.16 %
μ = 438.2± 0.7
σ = 40.5± 0.6


Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.47± 0.17 %
μ = 440.0± 0.7
σ = 41.7± 0.7
WF
σ/μ = 9.52± 0.17 %
μ = 439.4± 0.7
σ = 41.8± 0.7

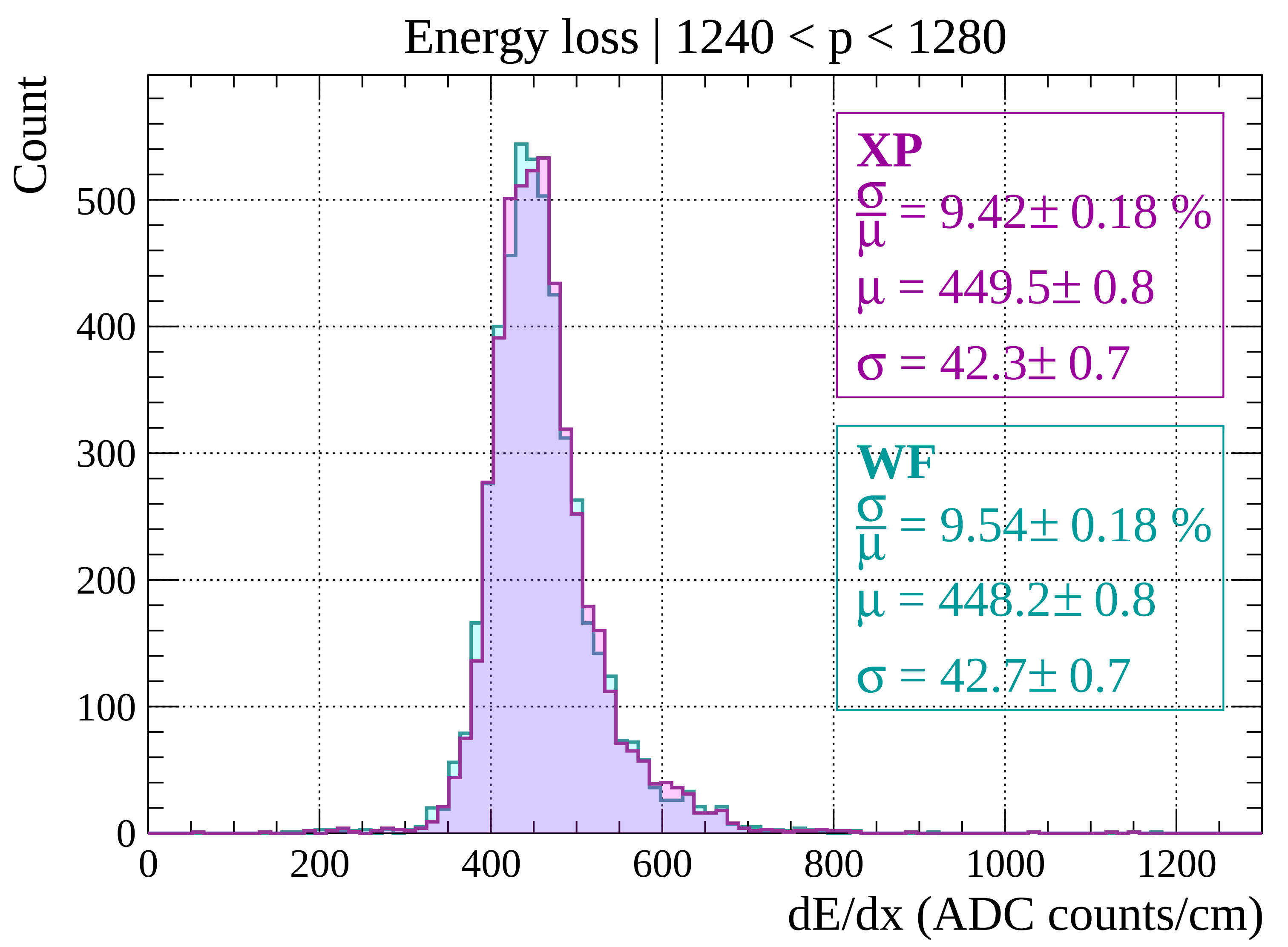


Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.19 %
μ = 442.3± 0.7
σ = 41.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.21 %
μ = 441.5± 0.8
σ = 42.6± 0.8

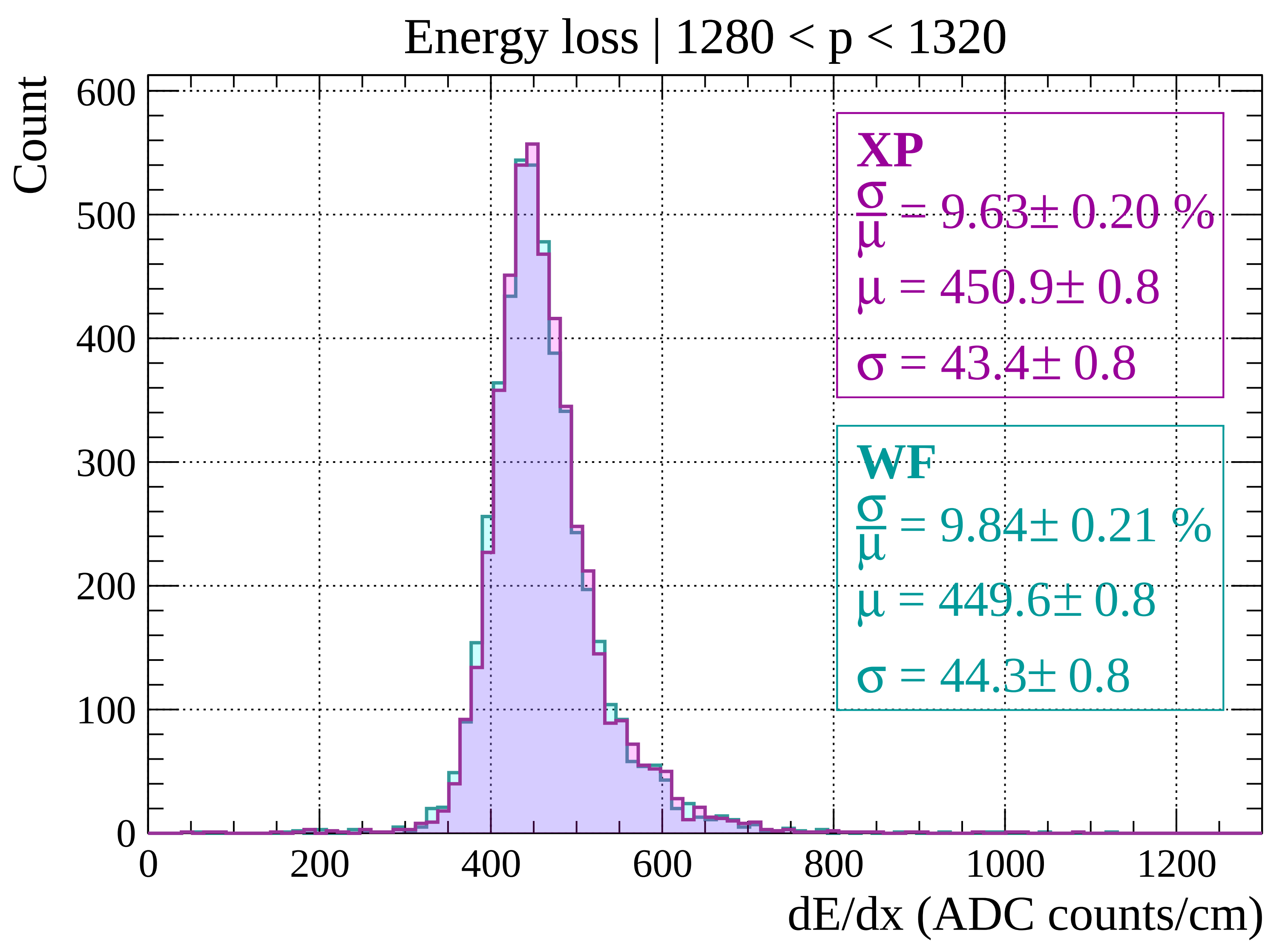

Energy loss | 1160 < p < 1200
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.39± 0.20 %
μ = 444.1± 0.8
σ = 41.7± 0.8
WF
σ/μ = 9.87± 0.18 %
μ = 445.6± 0.8
σ = 44.0± 0.7

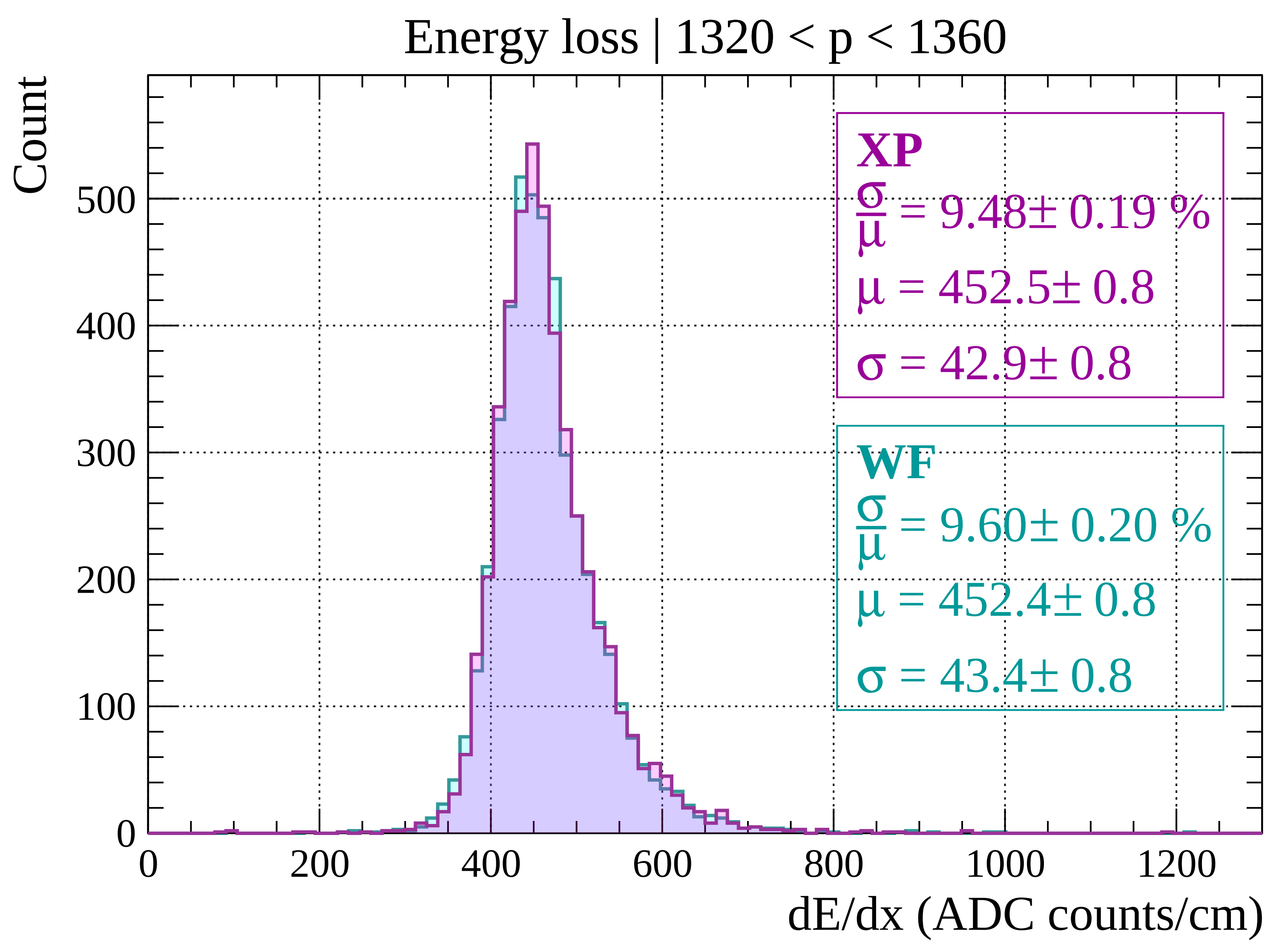


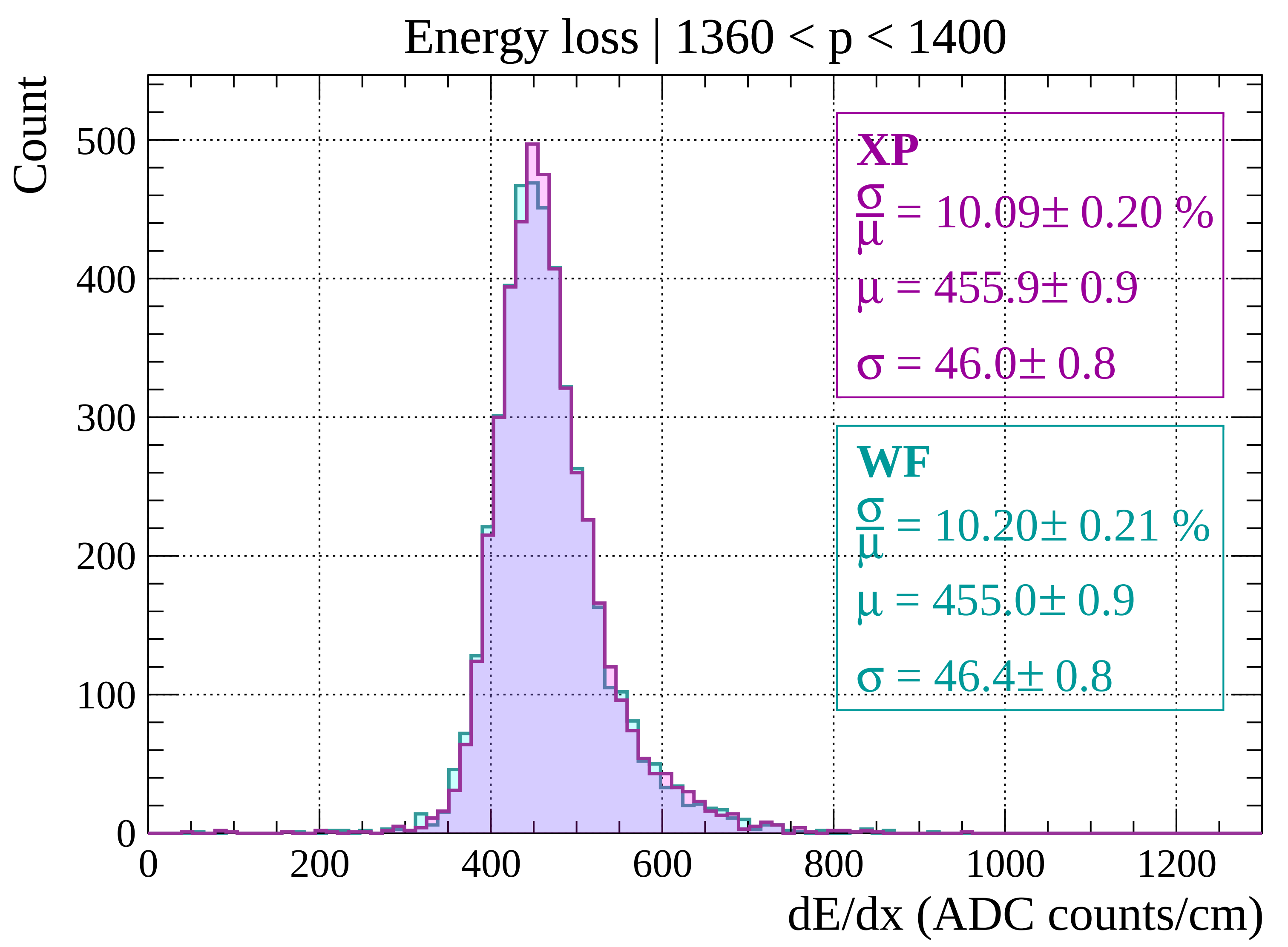
Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.53± 0.18 %
μ = 447.4± 0.7
σ = 42.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.86± 0.19 %
μ = 446.7± 0.8
σ = 44.1± 0.8



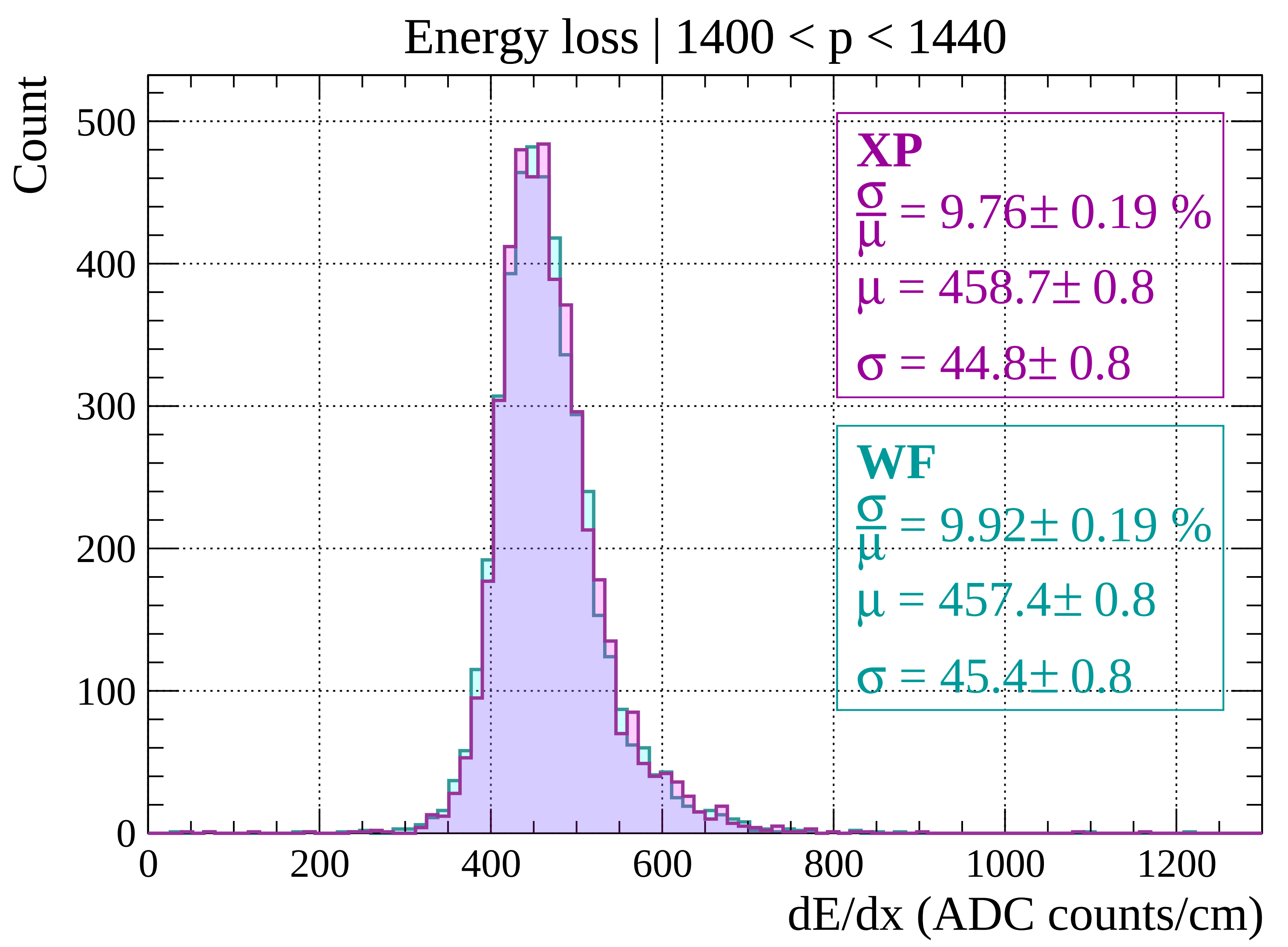
Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.42± 0.18 %
μ = 449.5± 0.8
σ = 42.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.54± 0.18 %
μ = 448.2± 0.8
σ = 42.7± 0.7


Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.63± 0.20 %
μ = 450.9± 0.8
σ = 43.4± 0.8
WF
σ/μ = 9.84± 0.21 %
μ = 449.6± 0.8
σ = 44.3± 0.8

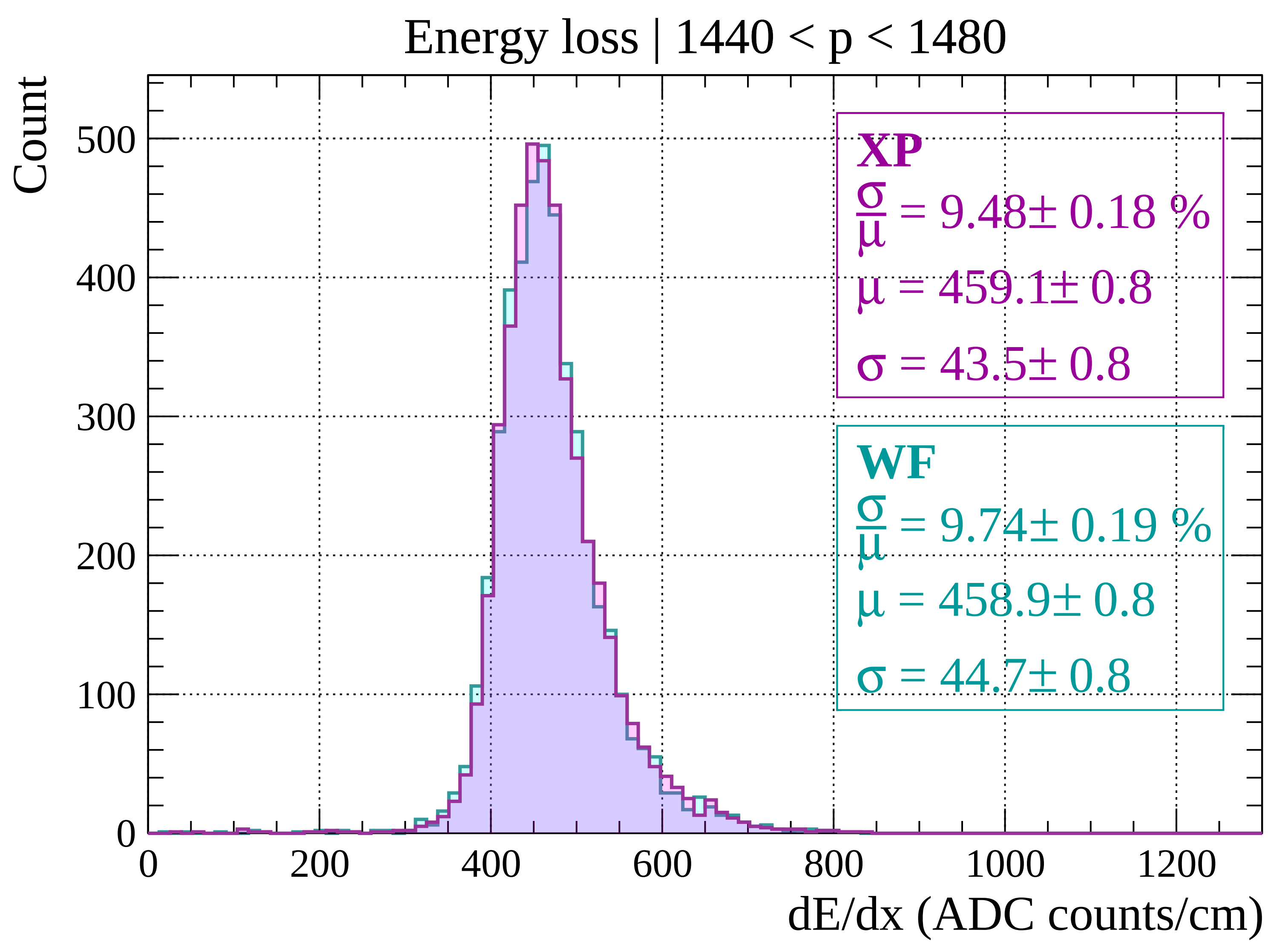

Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.48± 0.19 %
μ = 452.5± 0.8
σ = 42.9± 0.8
WF
σ/μ = 9.60± 0.20 %
μ = 452.4± 0.8
σ = 43.4± 0.8



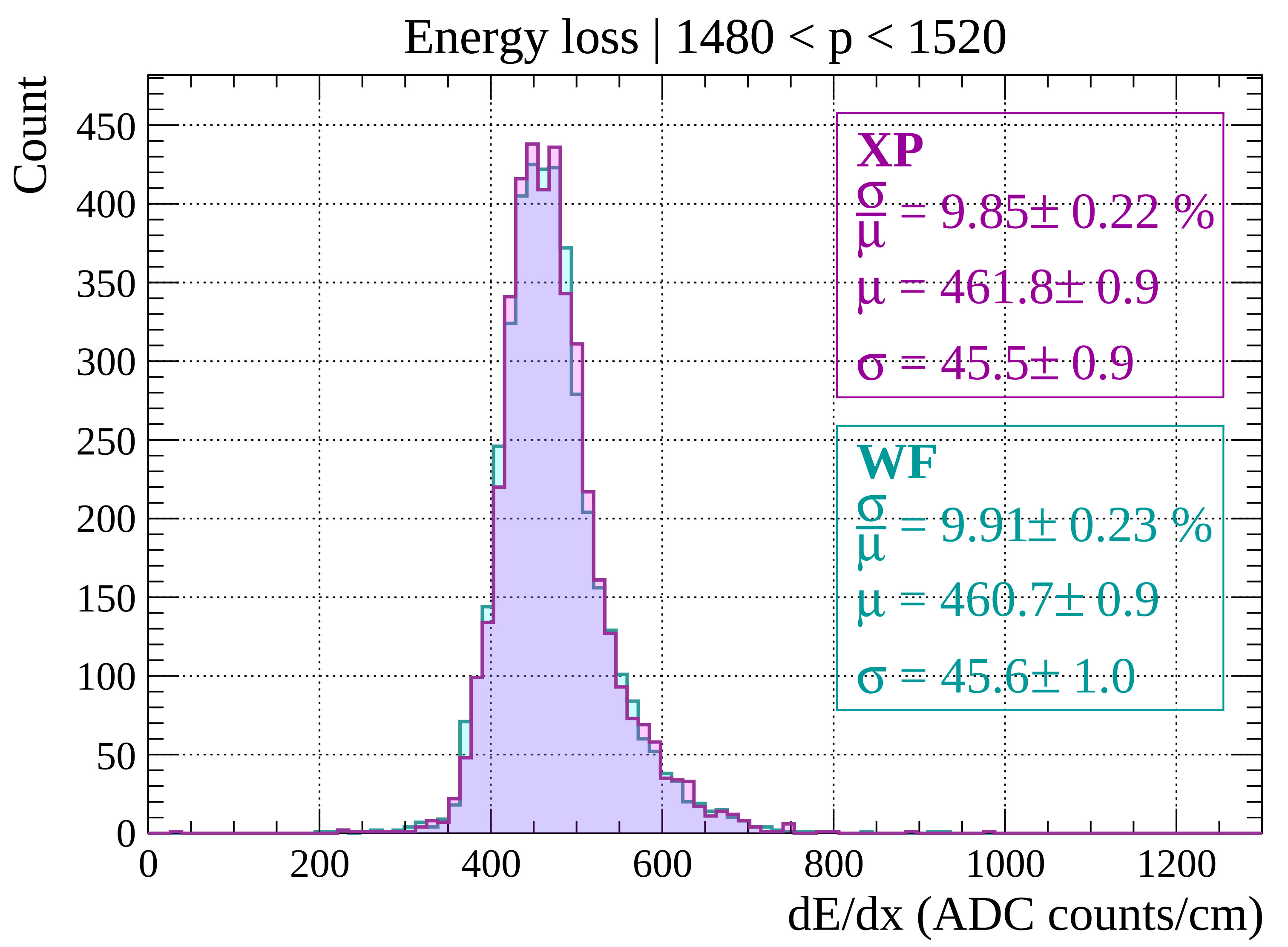
Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.09± 0.20 %
μ = 455.9± 0.9
σ = 46.0± 0.8
WF
σ/μ = 10.20± 0.21 %
μ = 455.0± 0.9
σ = 46.4± 0.8

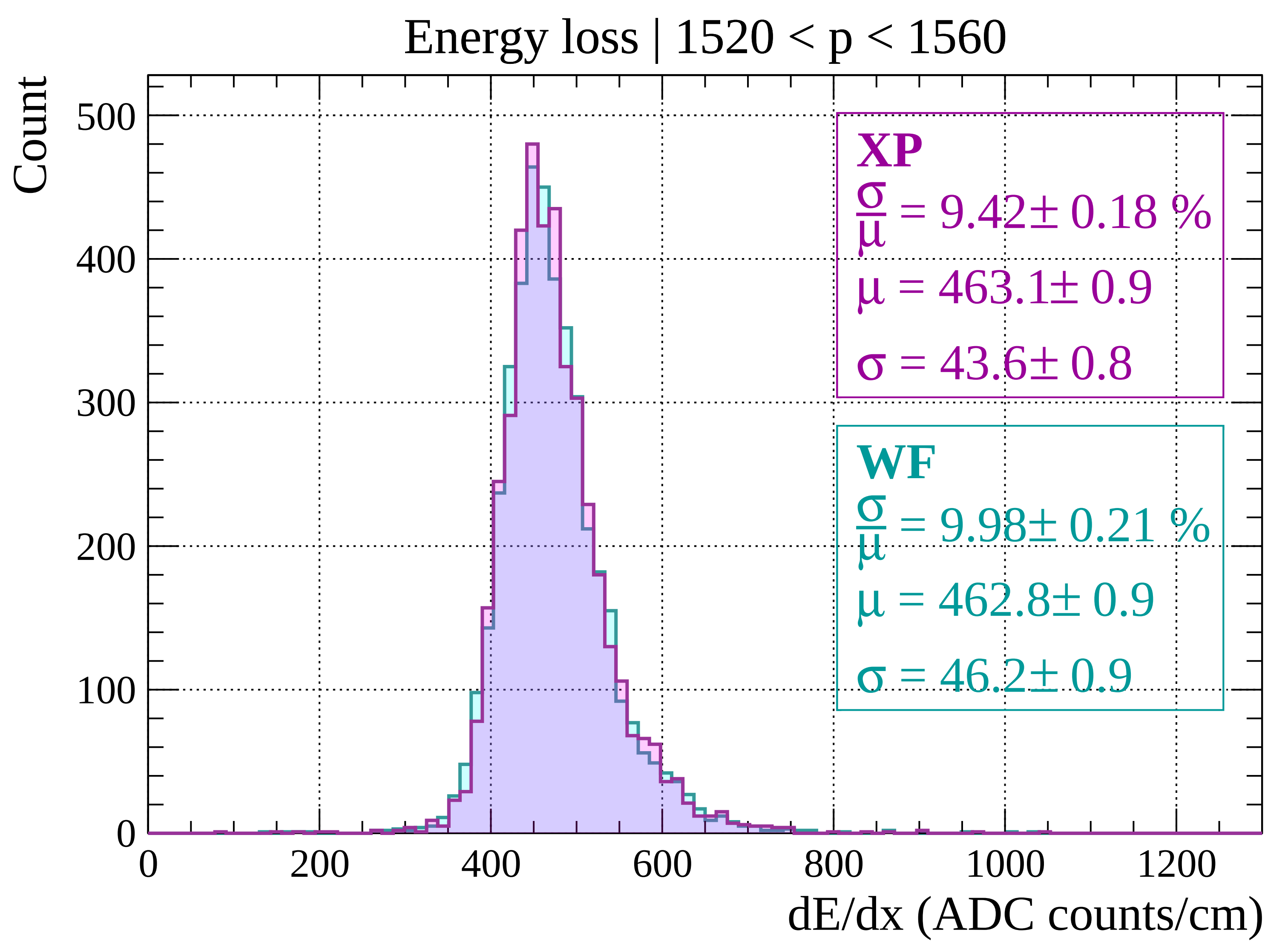


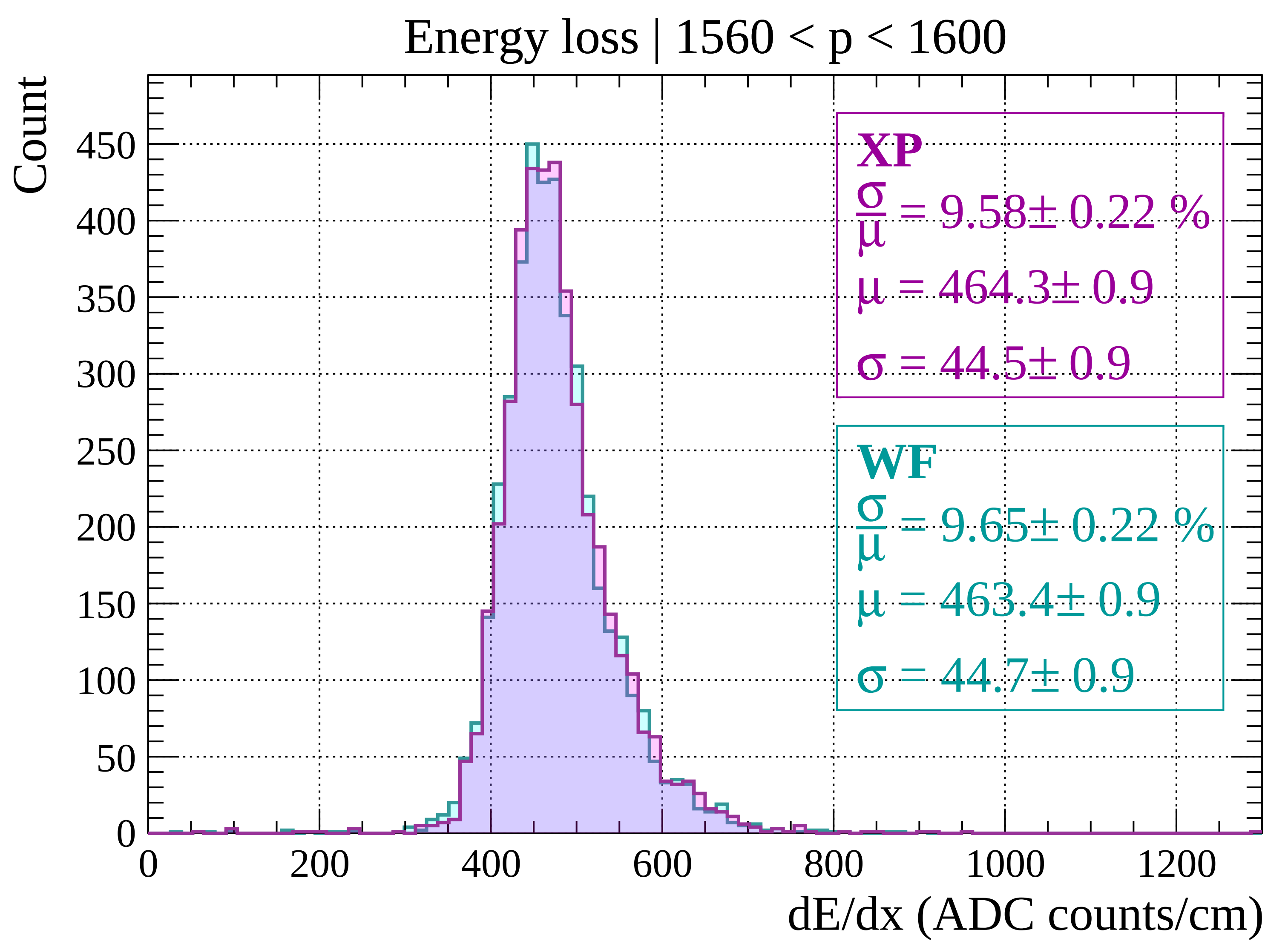
Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.76± 0.19 %
μ = 458.7± 0.8
σ = 44.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.92± 0.19 %
μ = 457.4± 0.8
σ = 45.4± 0.8



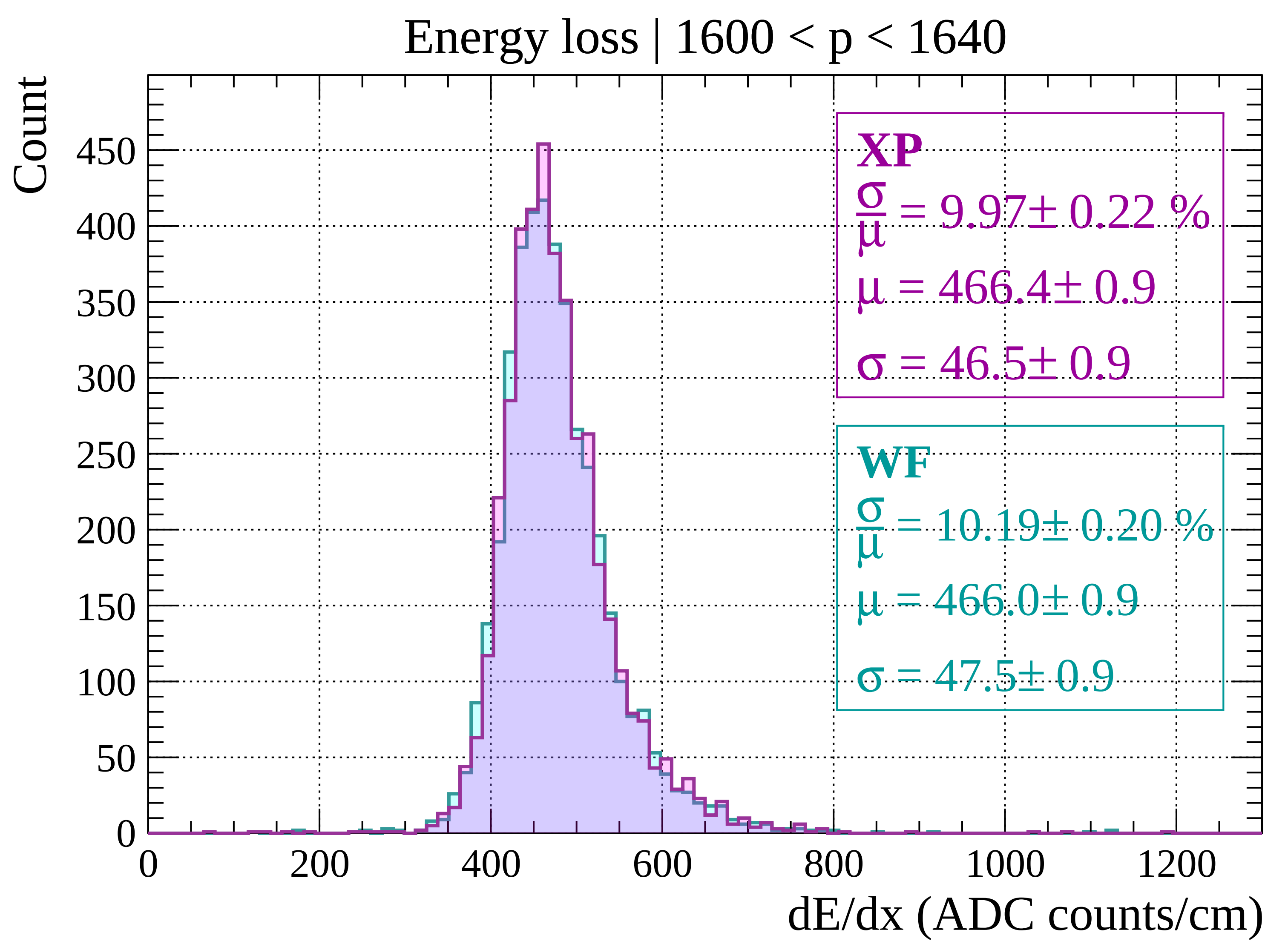
Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.48± 0.18 %
μ = 459.1± 0.8
σ = 43.5± 0.8
WF
σ/μ = 9.74± 0.19 %
μ = 458.9± 0.8
σ = 44.7± 0.8

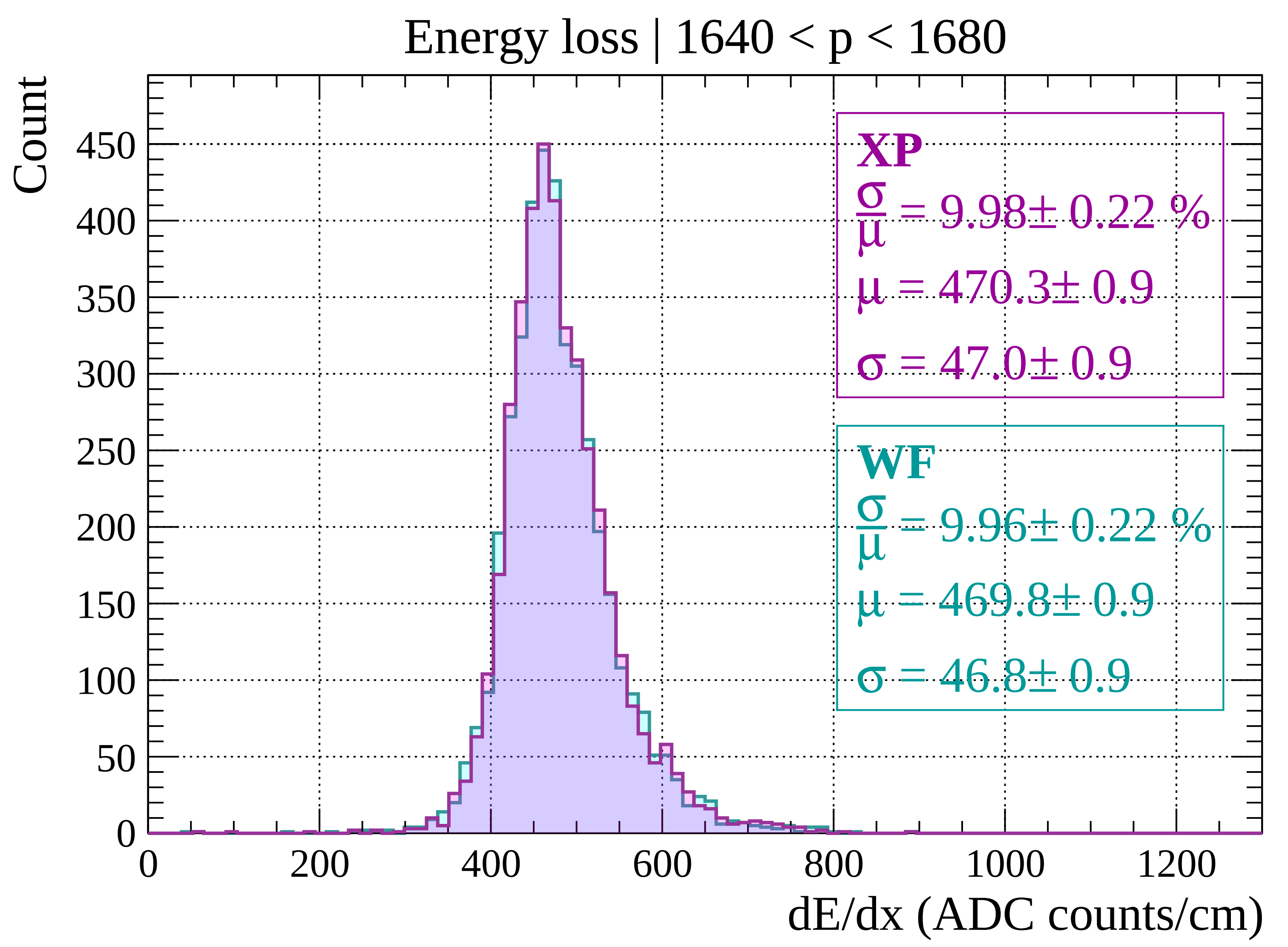

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.85 \pm 0.22$ %
$\mu = 461.8 \pm 0.9$
$\sigma = 45.5 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 9.91 \pm 0.23$ %
$\mu = 460.7 \pm 0.9$
$\sigma = 45.6 \pm 1.0$

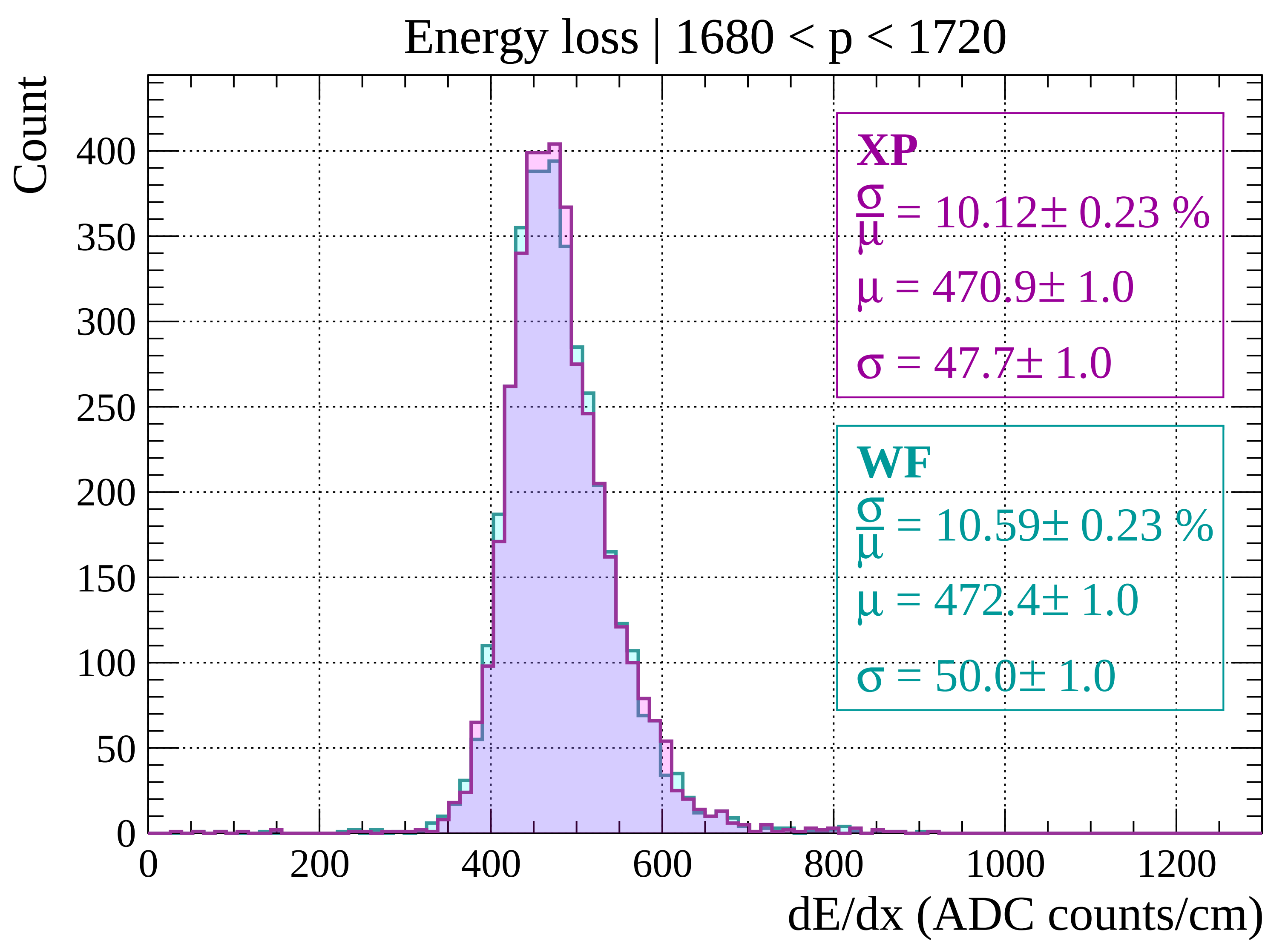

Energy loss | 1520 < p < 1560
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.42± 0.18 %
μ = 463.1± 0.9
σ = 43.6± 0.8
WF
σ/μ = 9.98± 0.21 %
μ = 462.8± 0.9
σ = 46.2± 0.9

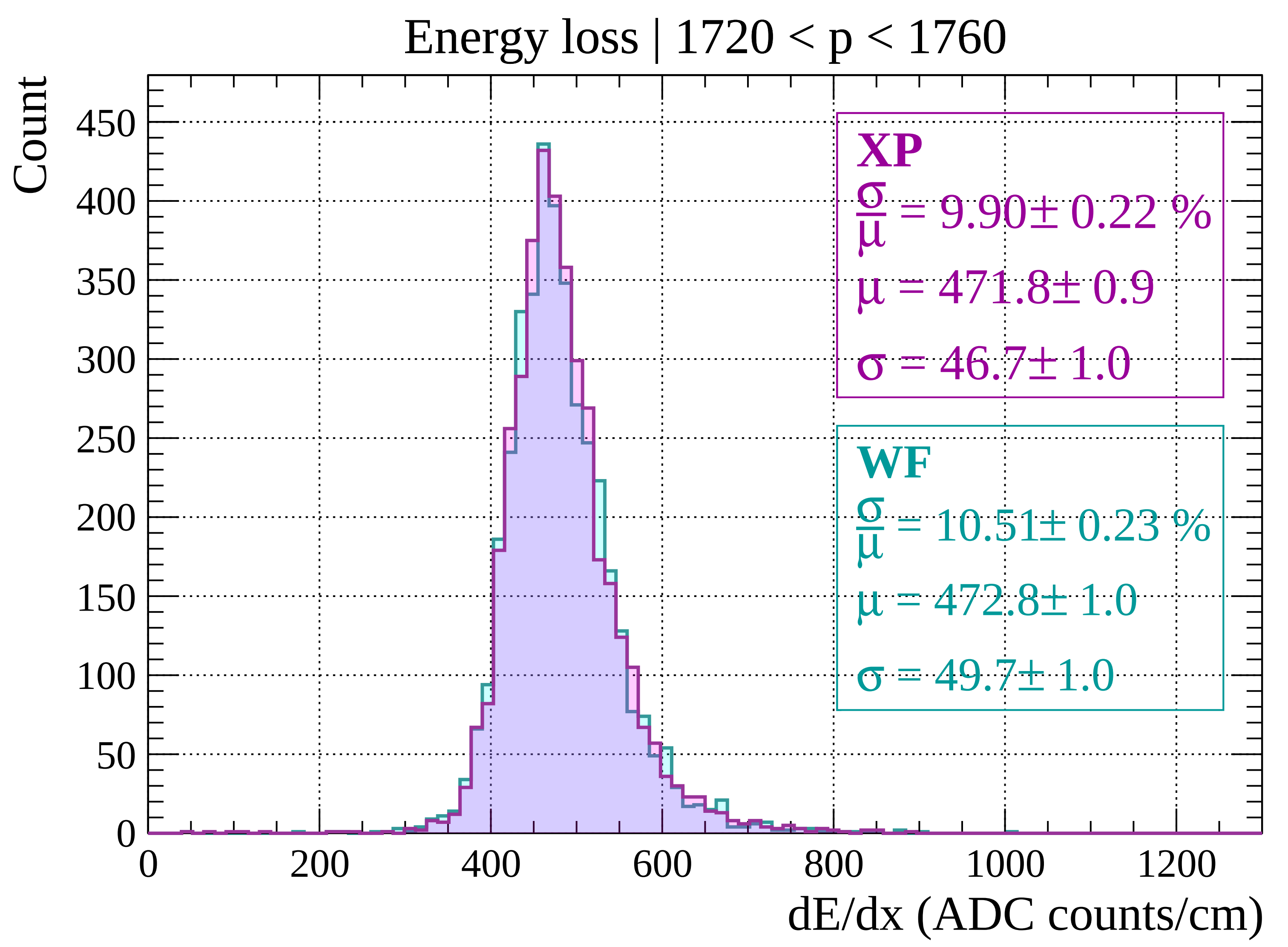


Energy loss | 1560 < p < 1600
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.58± 0.22 %
μ = 464.3± 0.9
σ = 44.5± 0.9
WF
σ/μ = 9.65± 0.22 %
μ = 463.4± 0.9
σ = 44.7± 0.9

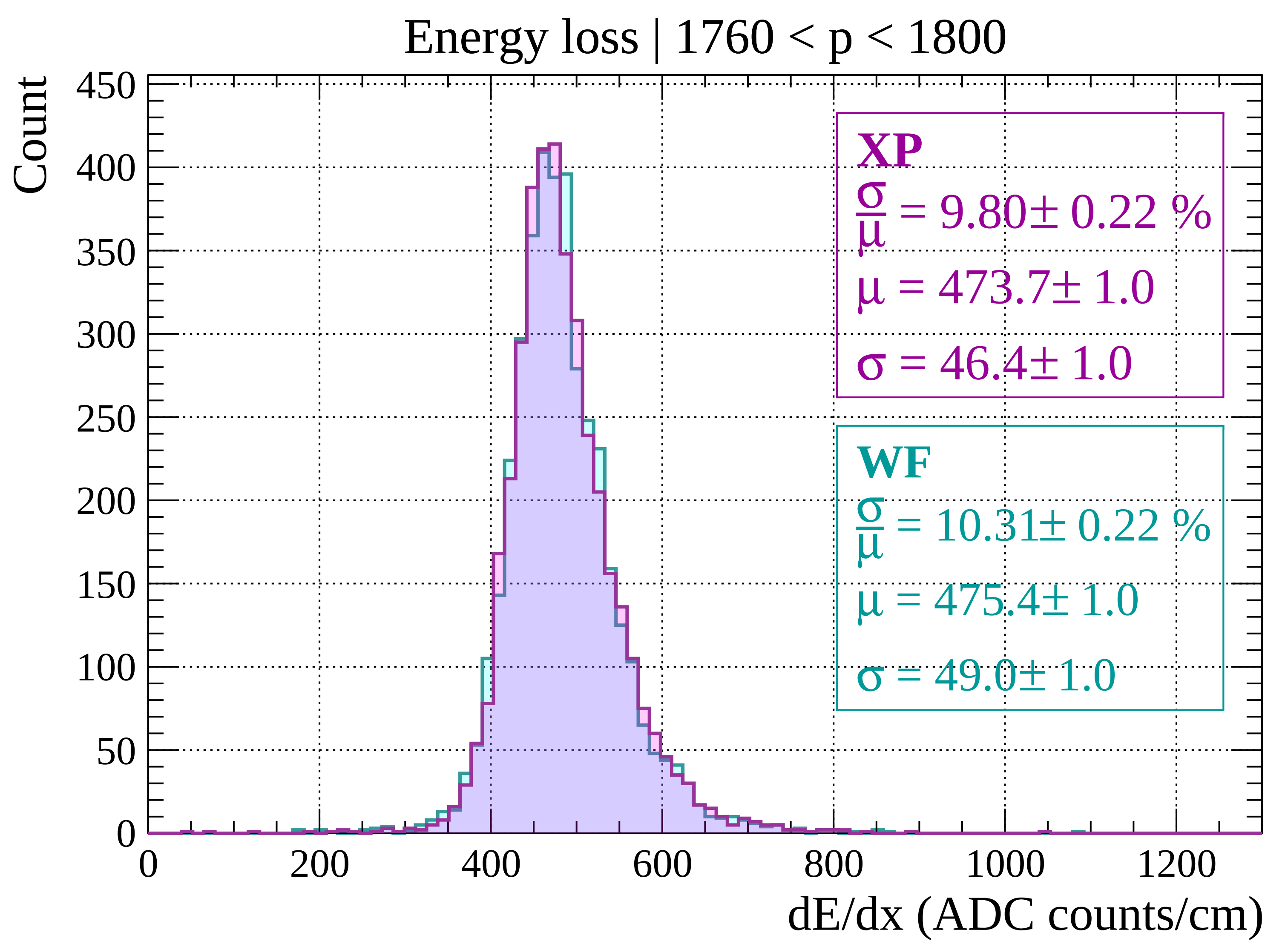

Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.97 \pm 0.22$ %
$\mu = 466.4 \pm 0.9$
$\sigma = 46.5 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 10.19 \pm 0.20$ %
$\mu = 466.0 \pm 0.9$
$\sigma = 47.5 \pm 0.9$


Energy loss | 1640 < p < 1680
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.98± 0.22 %
μ = 470.3± 0.9
σ = 47.0± 0.9
WF
σ/μ = 9.96± 0.22 %
μ = 469.8± 0.9
σ = 46.8± 0.9

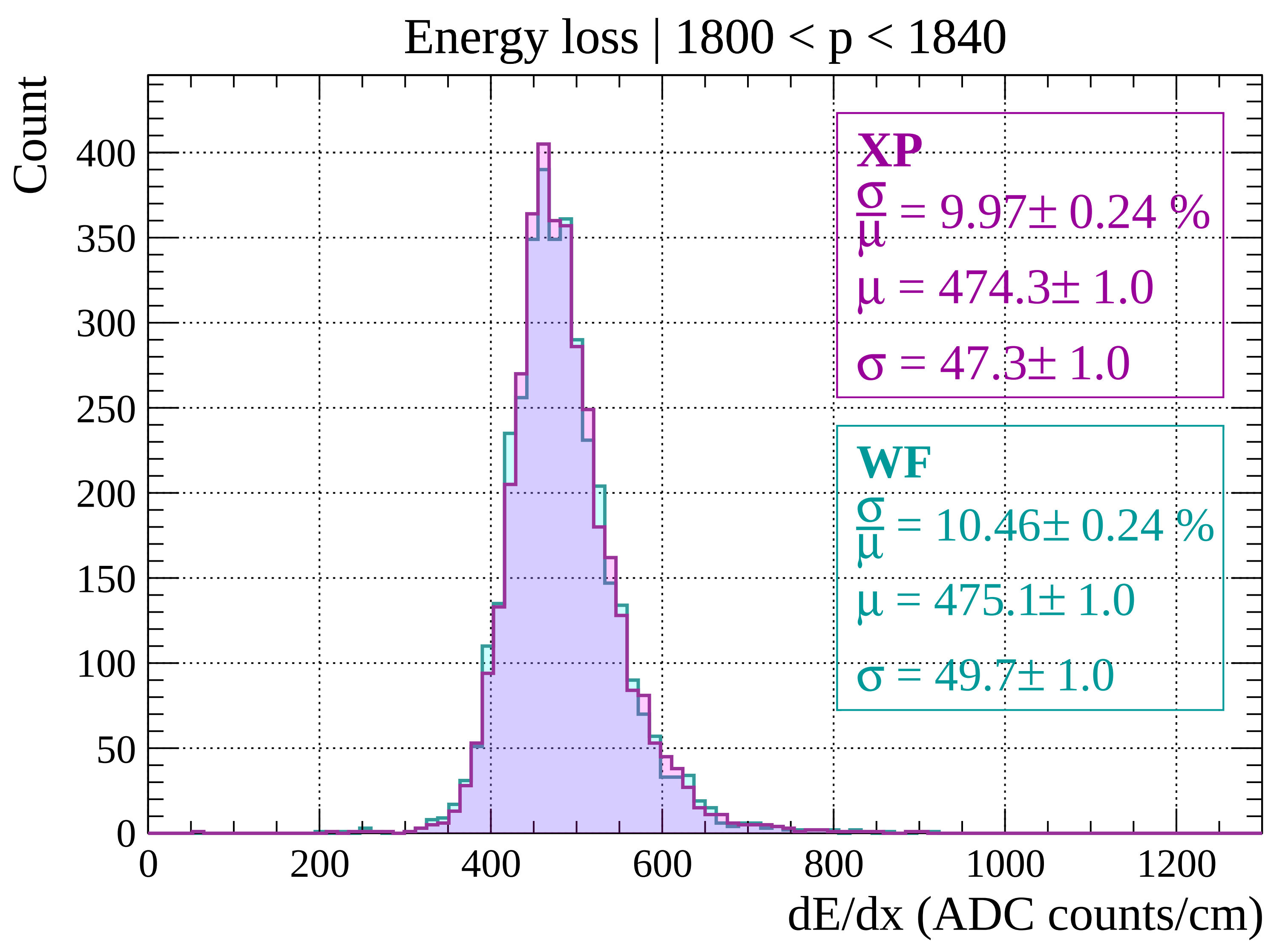

Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 10.12± 0.23 %
μ = 470.9± 1.0
σ = 47.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.59± 0.23 %
μ = 472.4± 1.0
σ = 50.0± 1.0



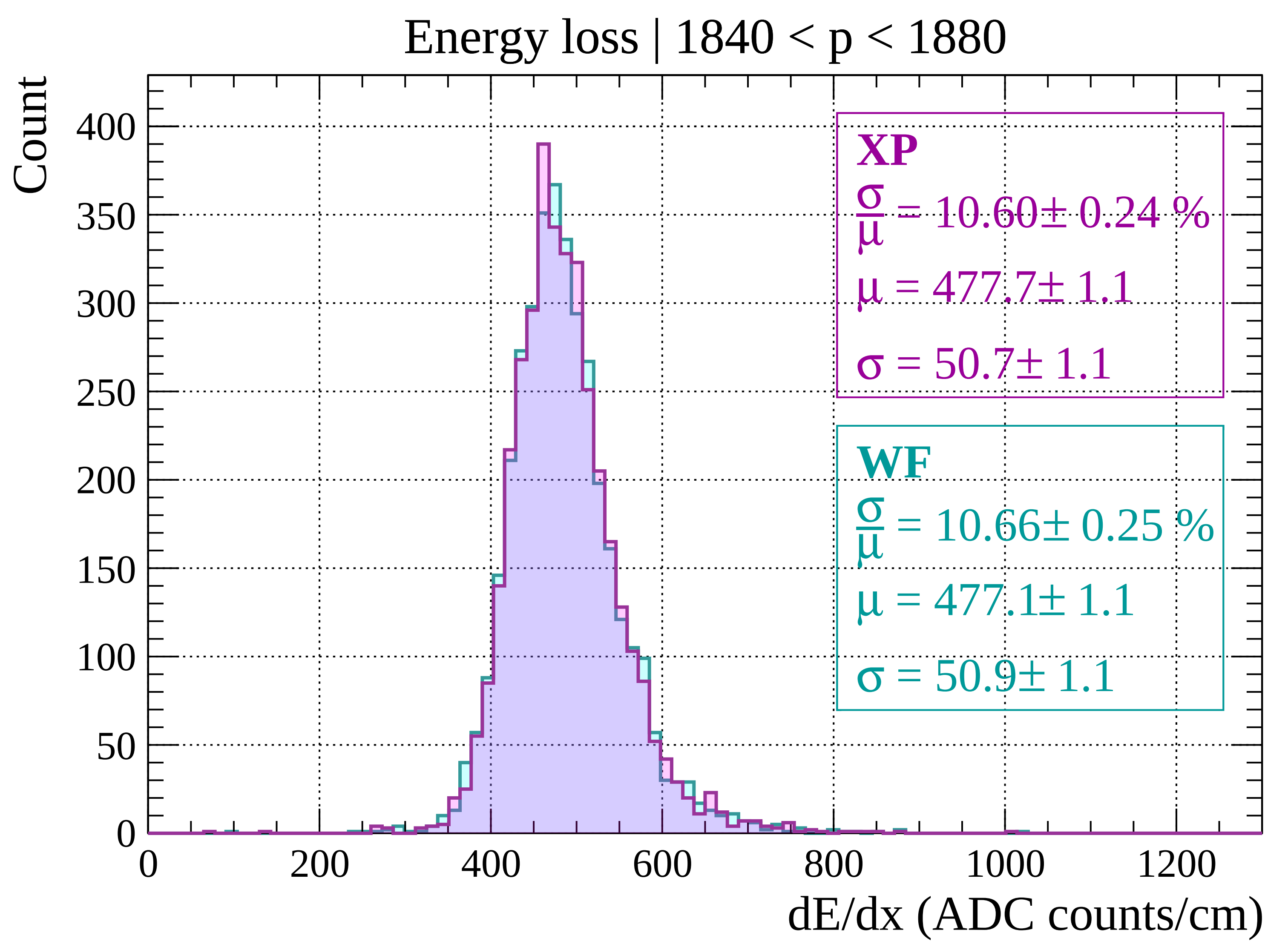
Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.90 ± 0.22 %
μ = 471.8 ± 0.9
σ = 46.7 ± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.51 ± 0.23 %
μ = 472.8 ± 1.0
σ = 49.7 ± 1.0



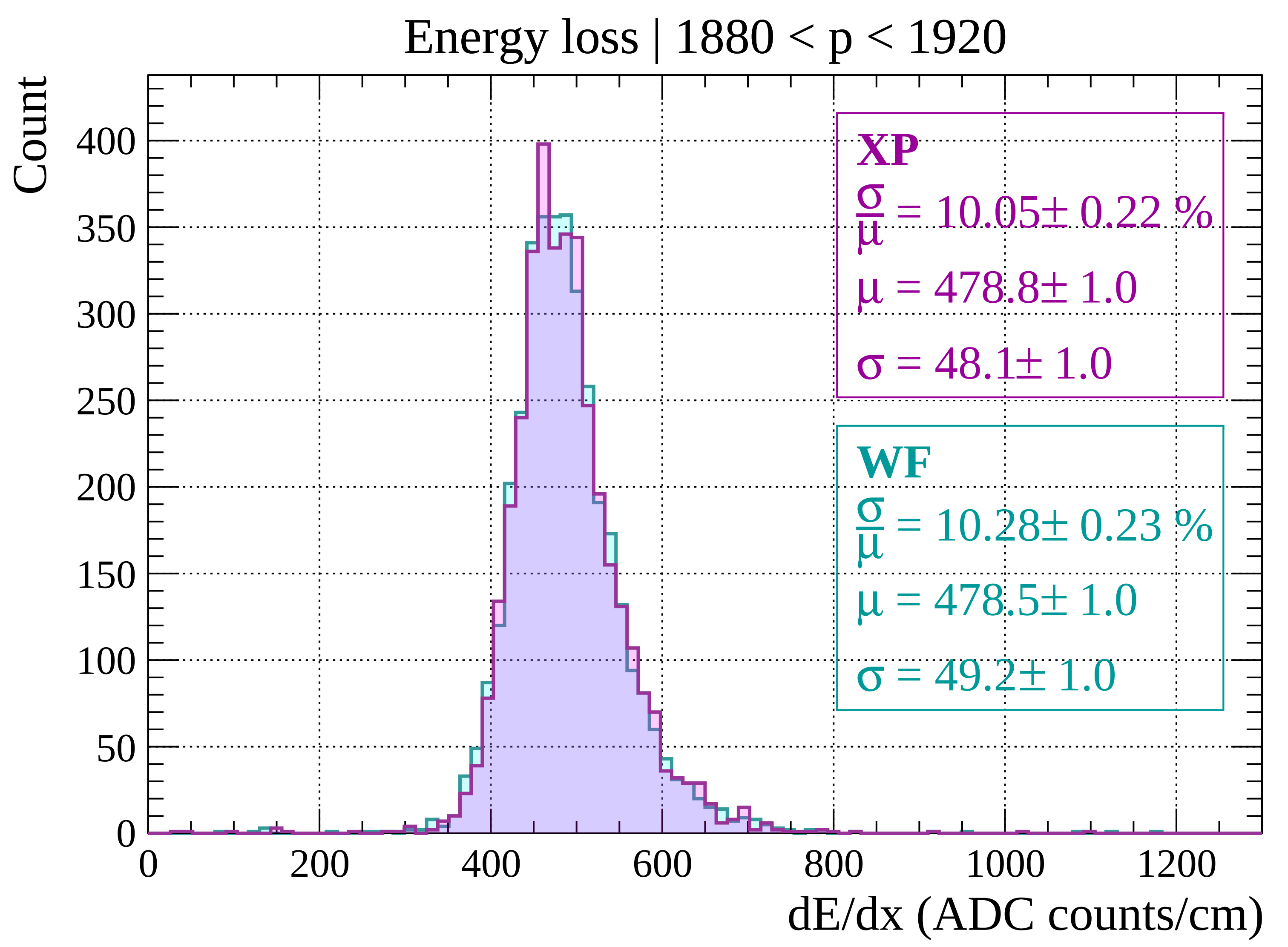
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 9.80± 0.22 %
μ = 473.7± 1.0
σ = 46.4± 1.0
WF
σ/μ = 10.31± 0.22 %
μ = 475.4± 1.0
σ = 49.0± 1.0



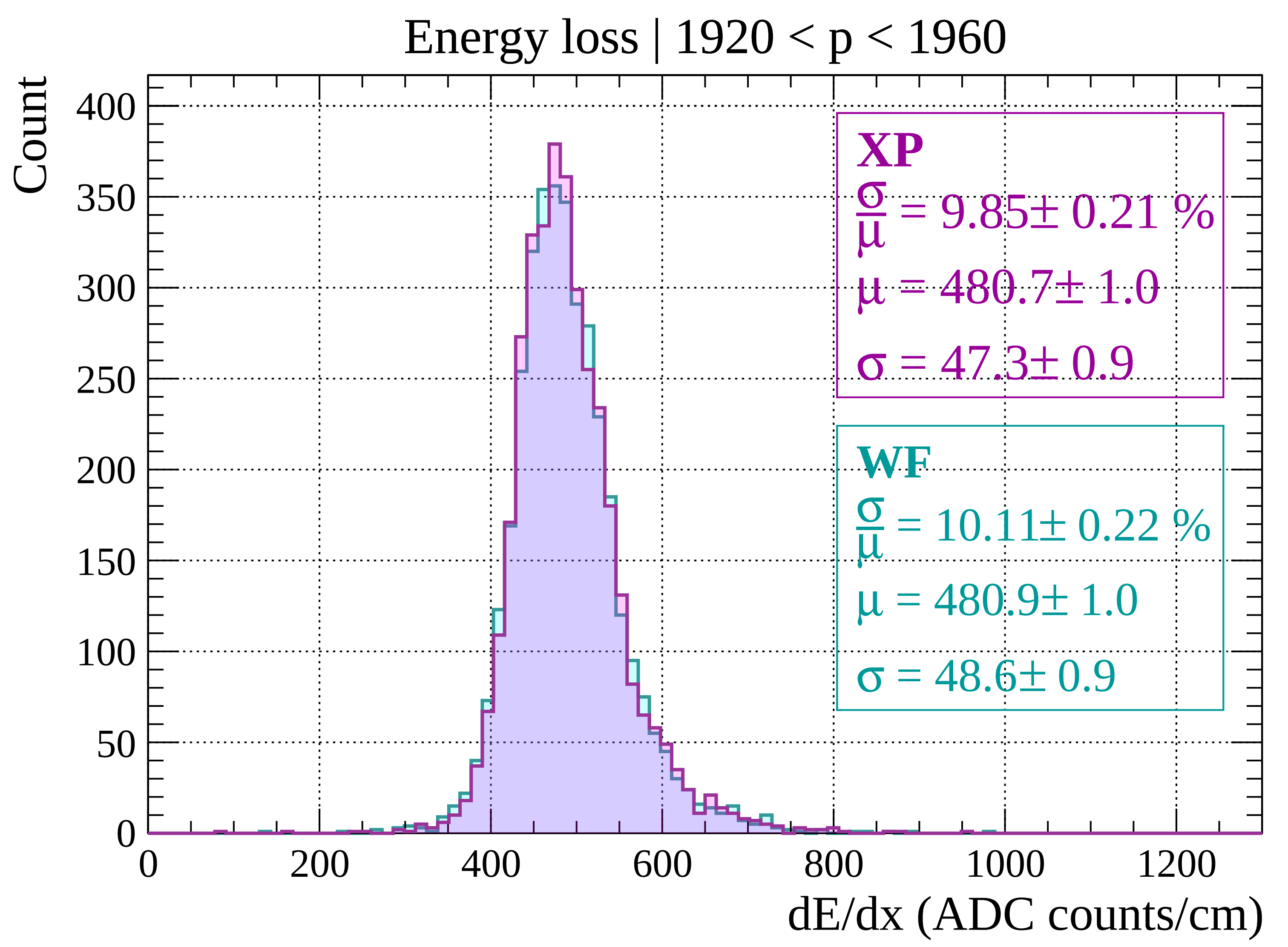
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.97± 0.24 %
μ = 474.3± 1.0
σ = 47.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.46± 0.24 %
μ = 475.1± 1.0
σ = 49.7± 1.0



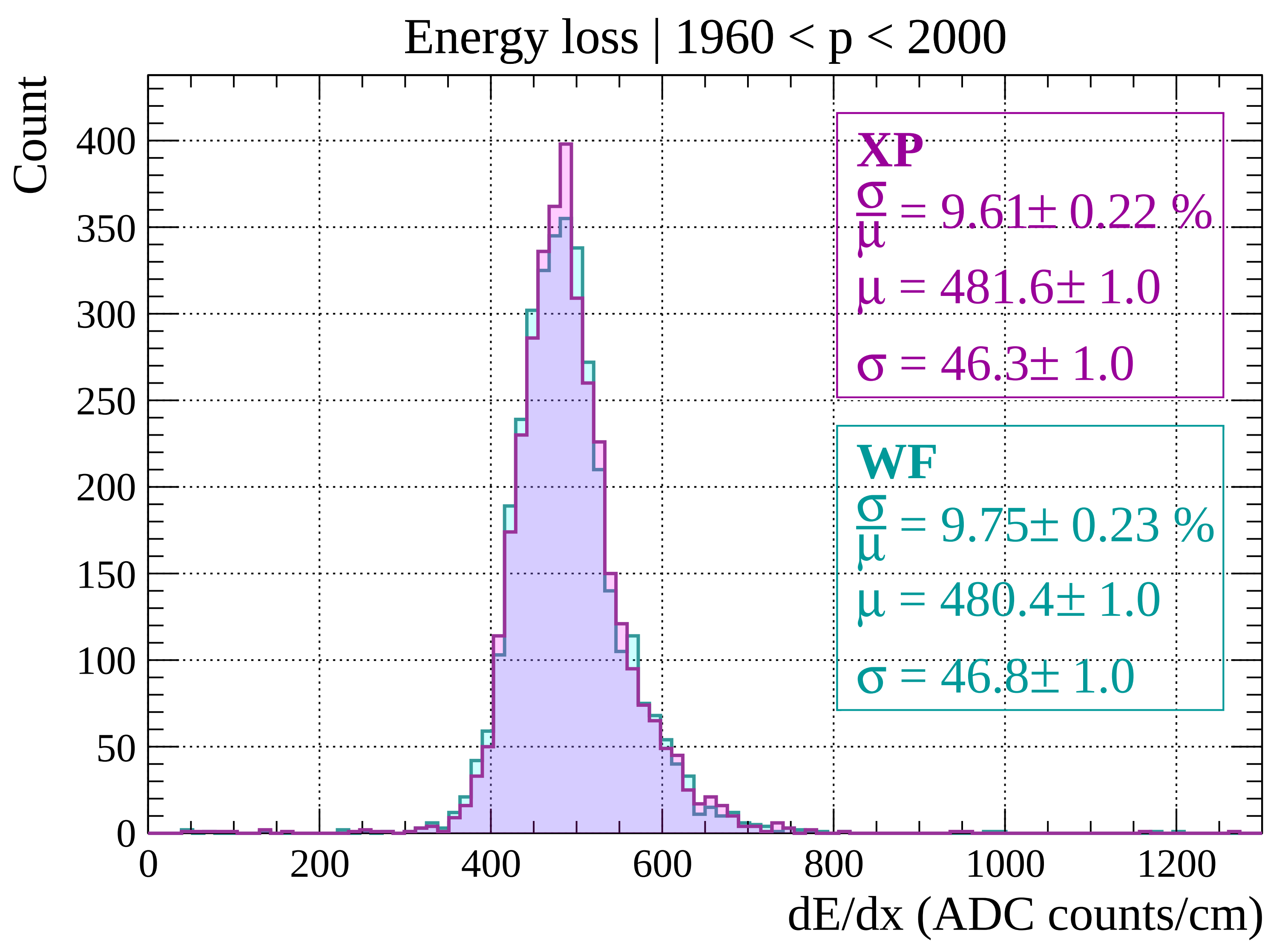
Energy loss | 1840 < p < 1880
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.60± 0.24 %
μ = 477.7± 1.1
σ = 50.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.66± 0.25 %
μ = 477.1± 1.1
σ = 50.9± 1.1



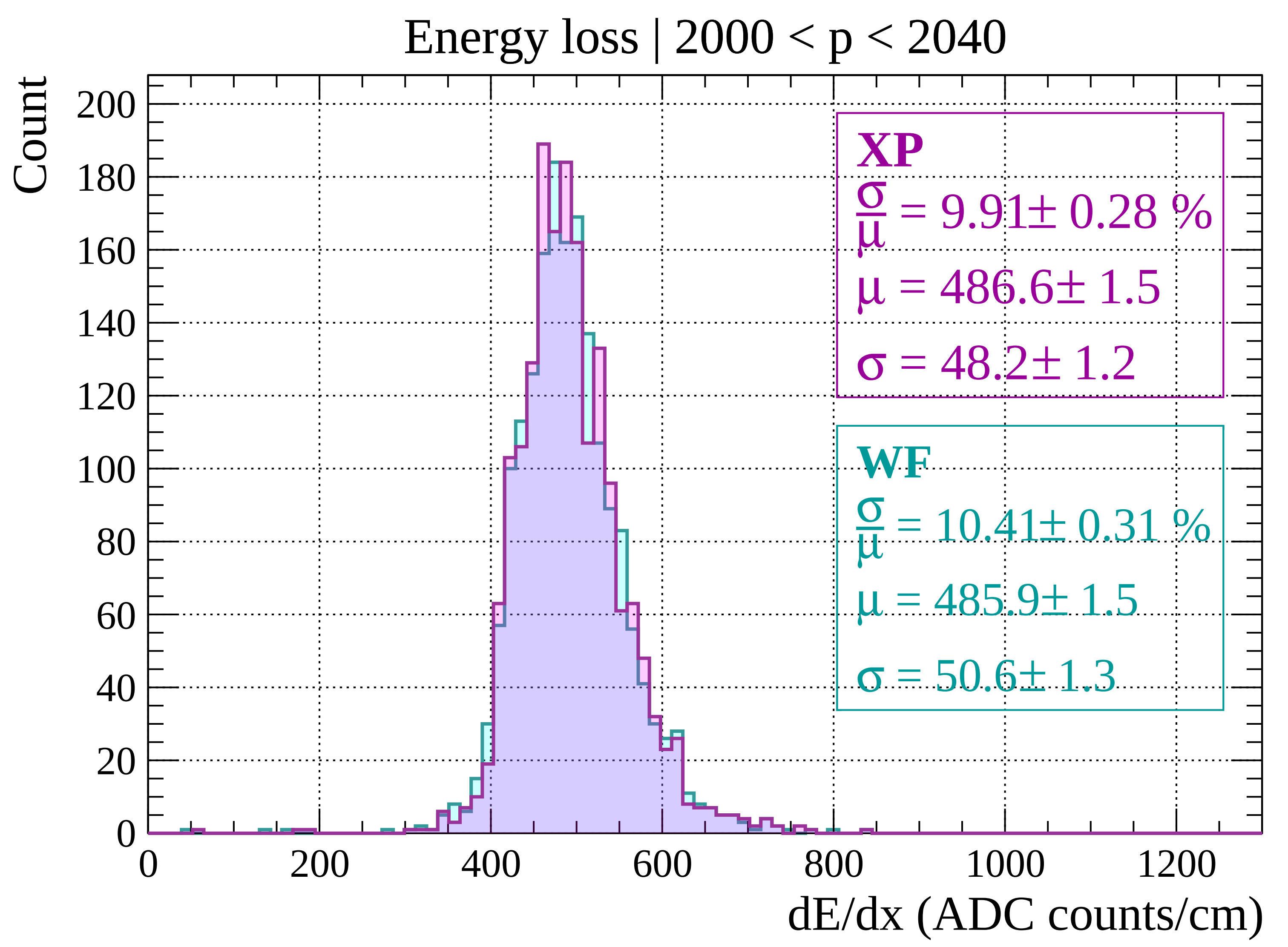
Energy loss | 1880 < p < 1920
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.05± 0.22 %
μ = 478.8± 1.0
σ = 48.1± 1.0
WF
σ/μ = 10.28± 0.23 %
μ = 478.5± 1.0
σ = 49.2± 1.0



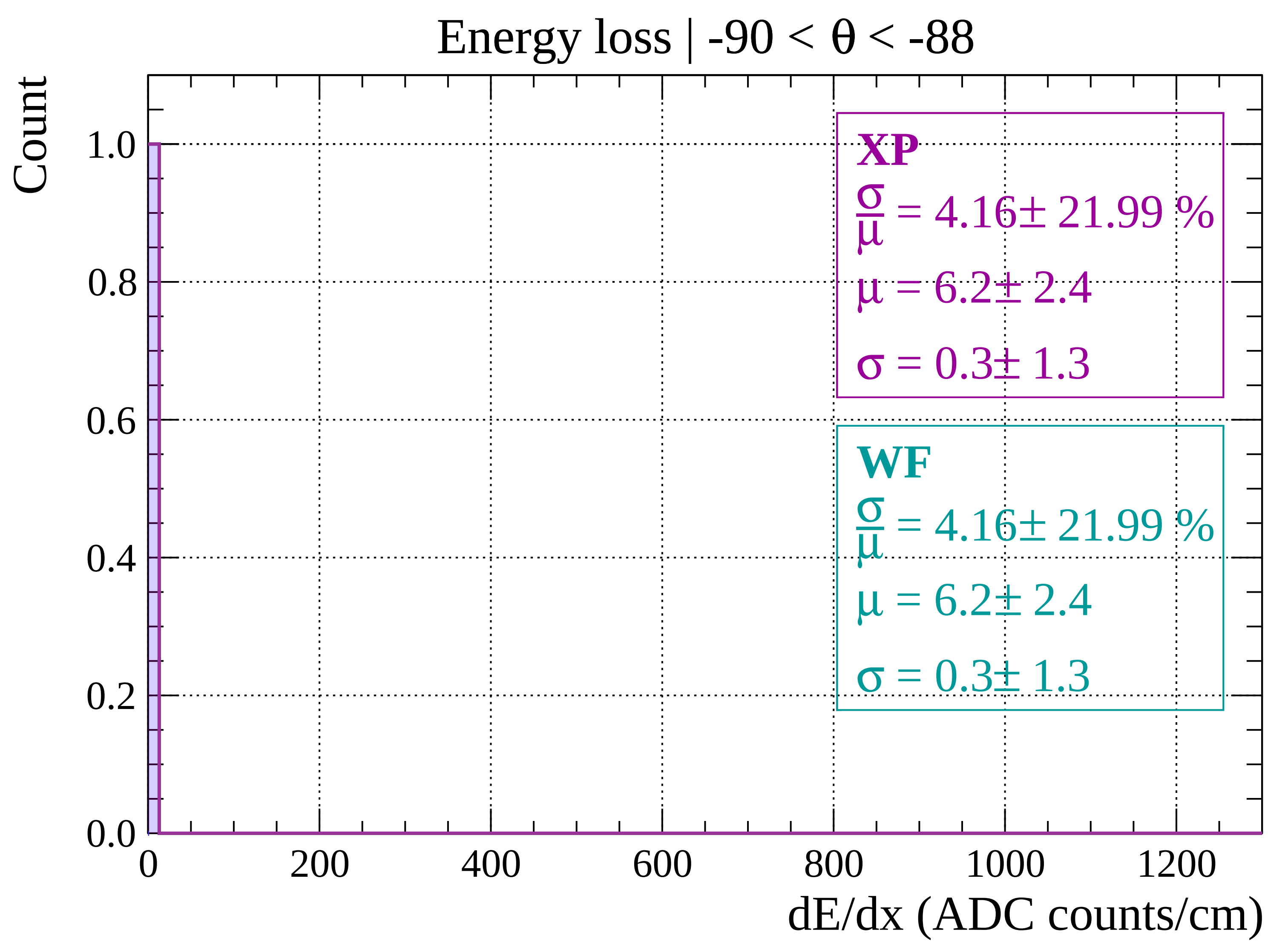
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.85± 0.21 %
μ = 480.7± 1.0
σ = 47.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.11± 0.22 %
μ = 480.9± 1.0
σ = 48.6± 0.9



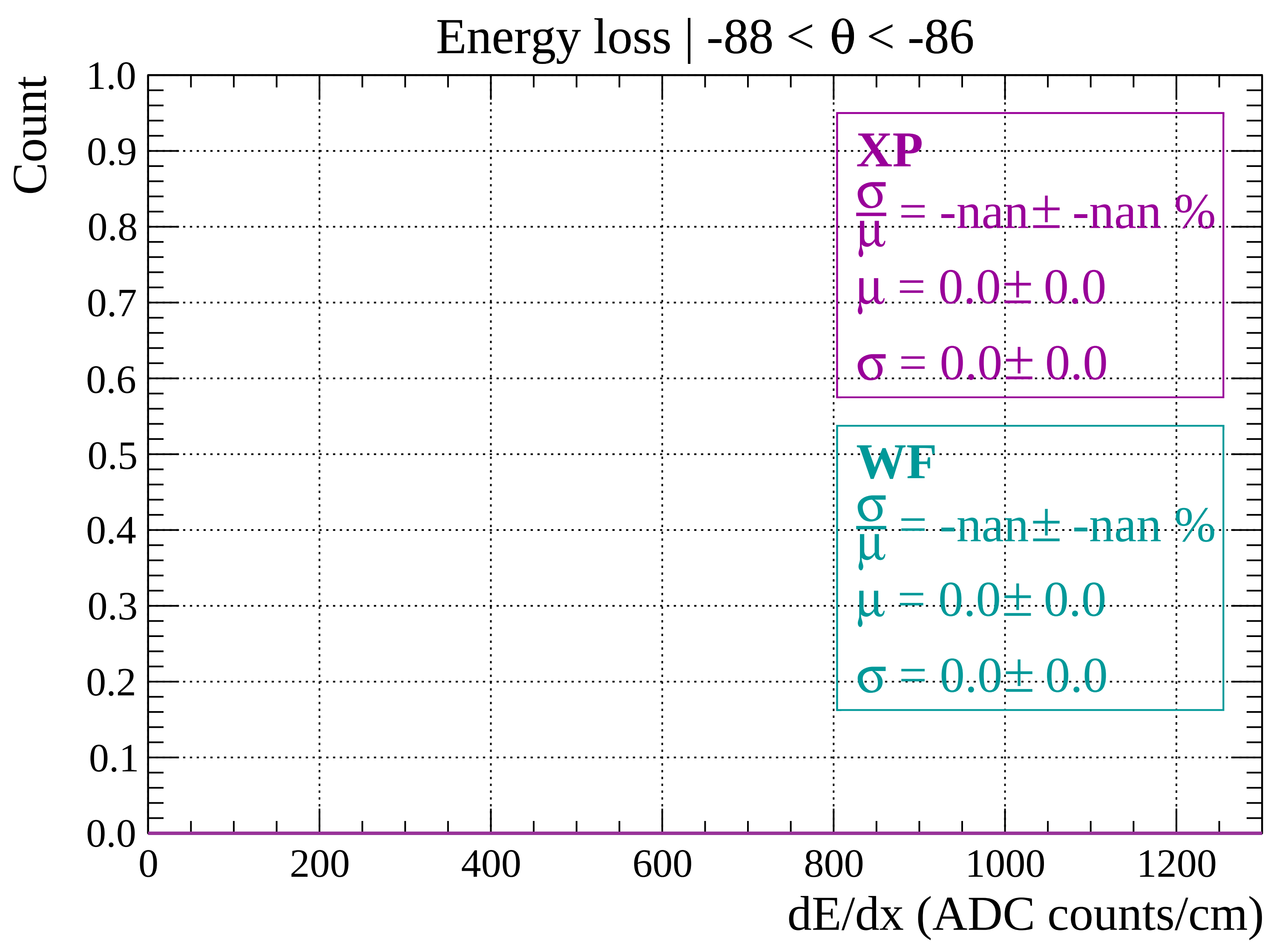
Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 9.61± 0.22 %
μ = 481.6± 1.0
σ = 46.3± 1.0
WF
σ/μ = 9.75± 0.23 %
μ = 480.4± 1.0
σ = 46.8± 1.0



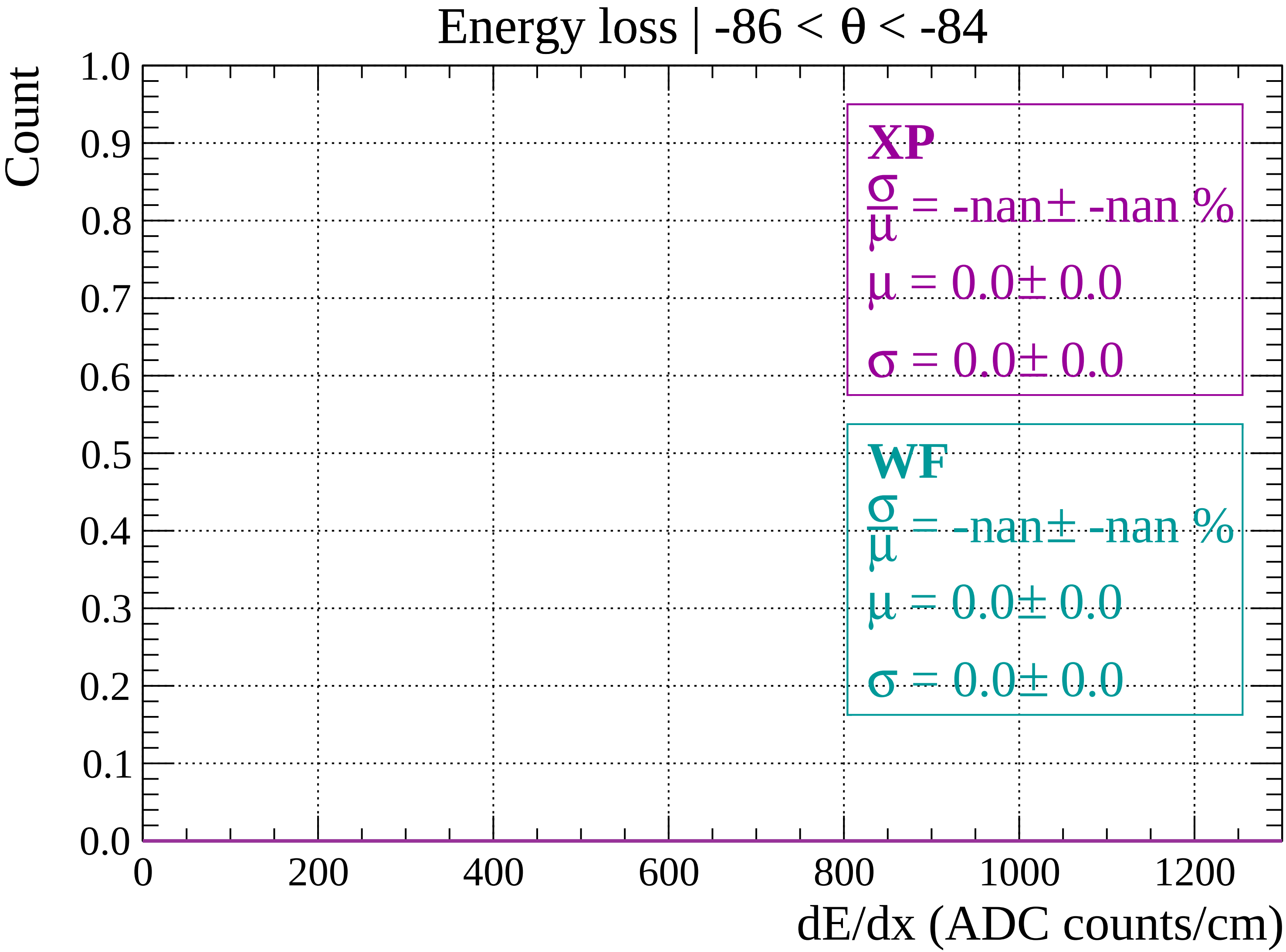
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 9.91± 0.28 %
μ = 486.6± 1.5
σ = 48.2± 1.2
WF
σ/μ = 10.41± 0.31 %
μ = 485.9± 1.5
σ = 50.6± 1.3



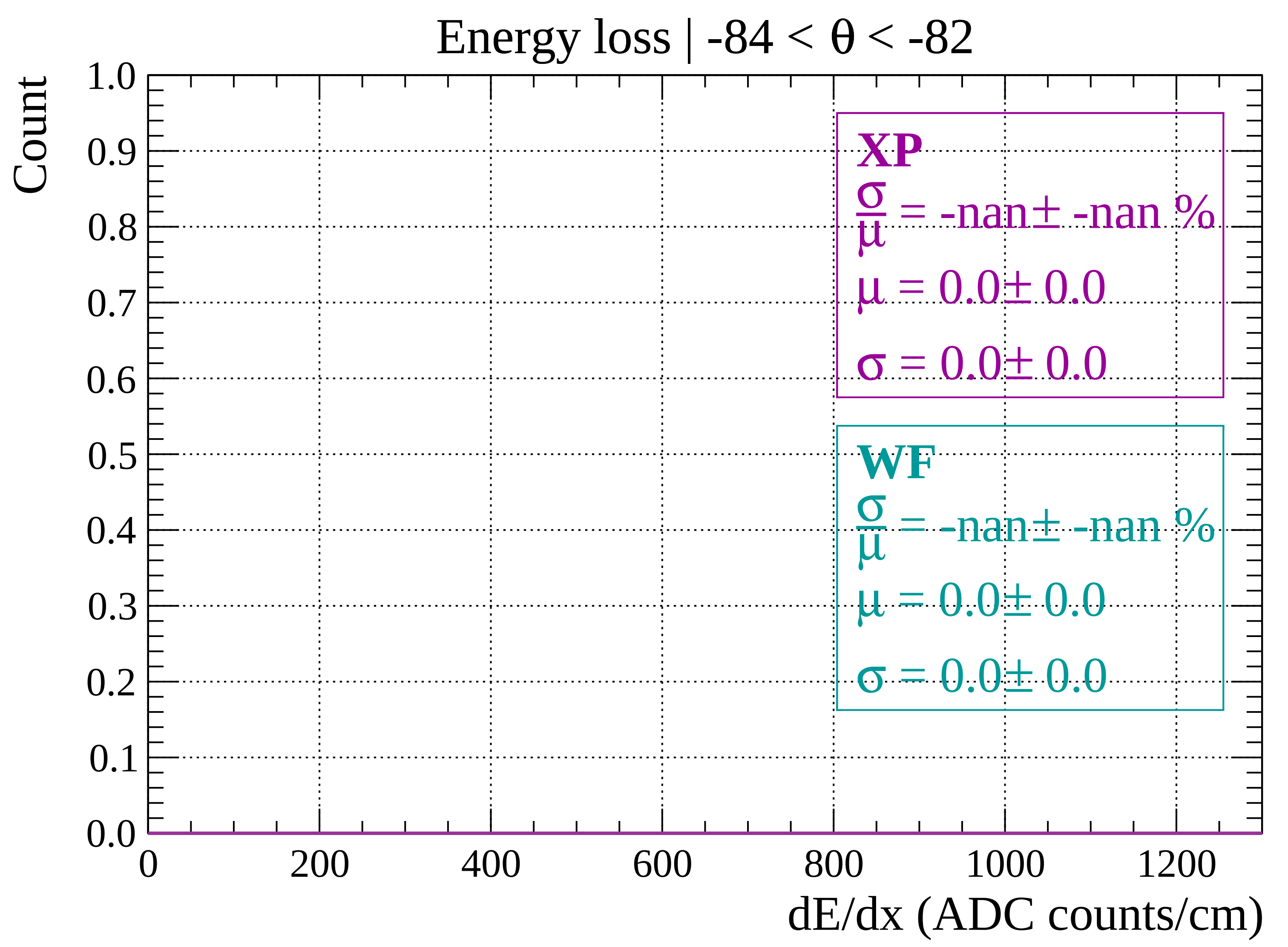
Energy loss | -90 < θ < -88
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3
WF
σ/μ = 4.16± 21.99 %
μ = 6.2± 2.4
σ = 0.3± 1.3



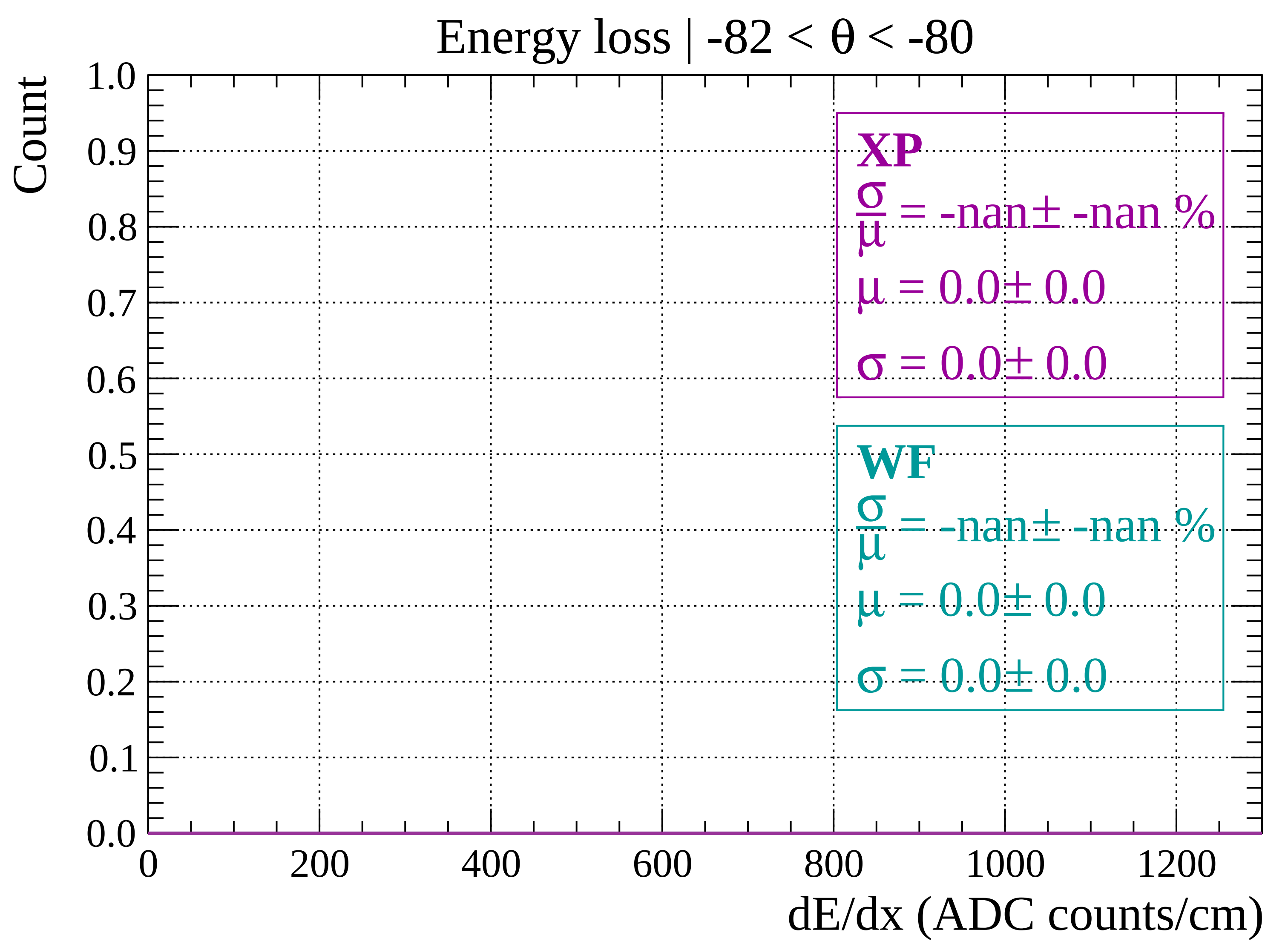
Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



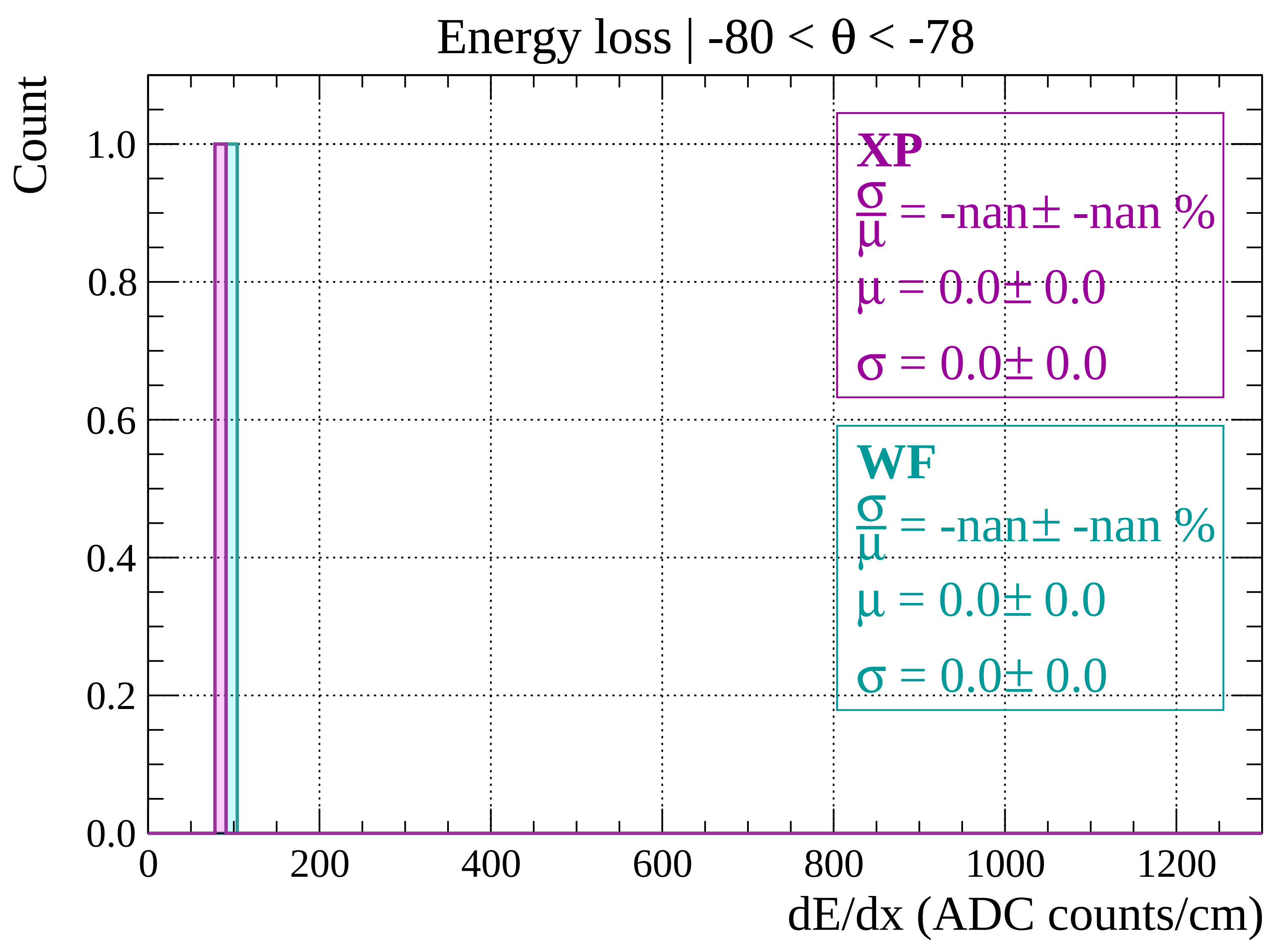
Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



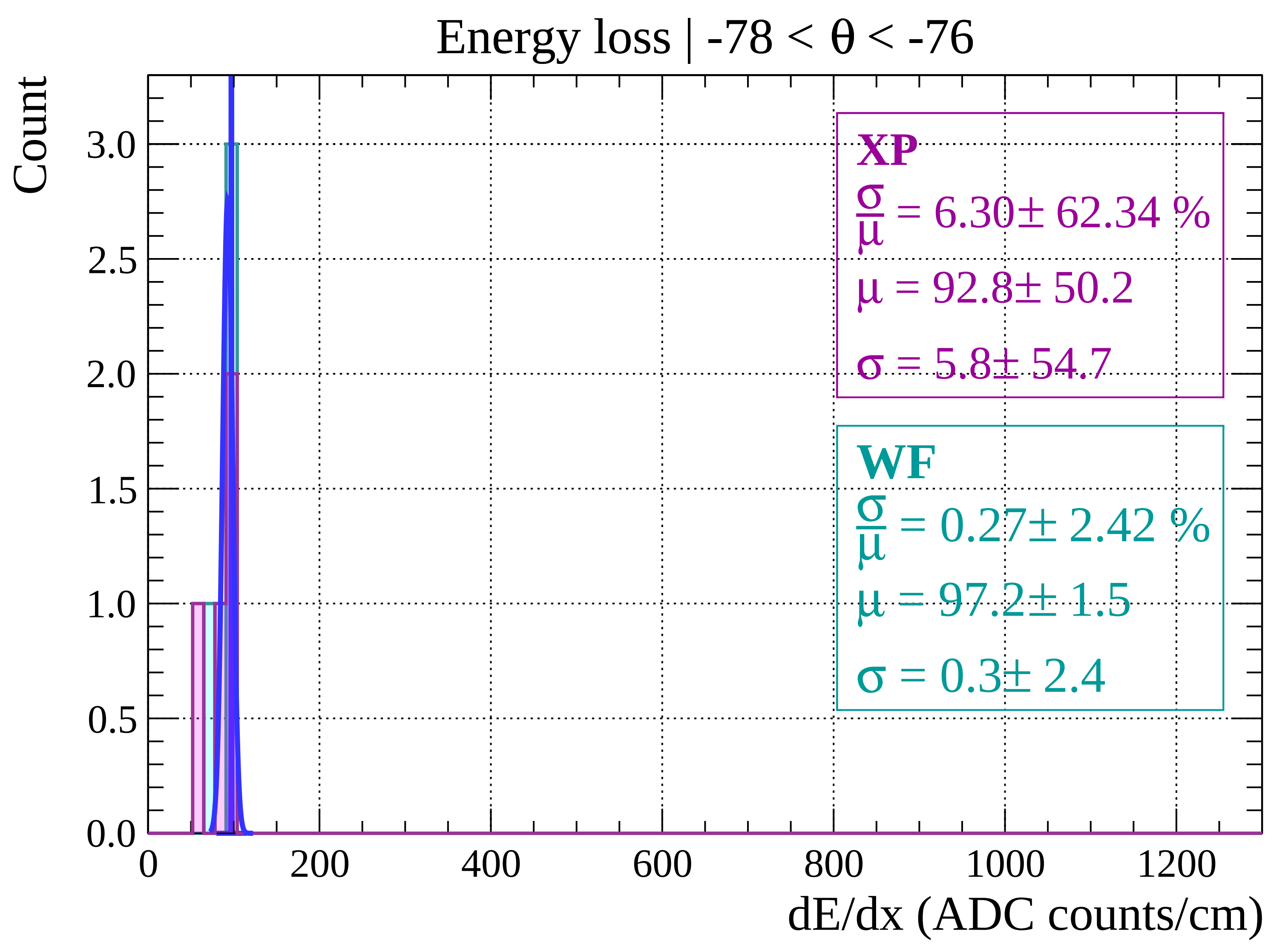
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



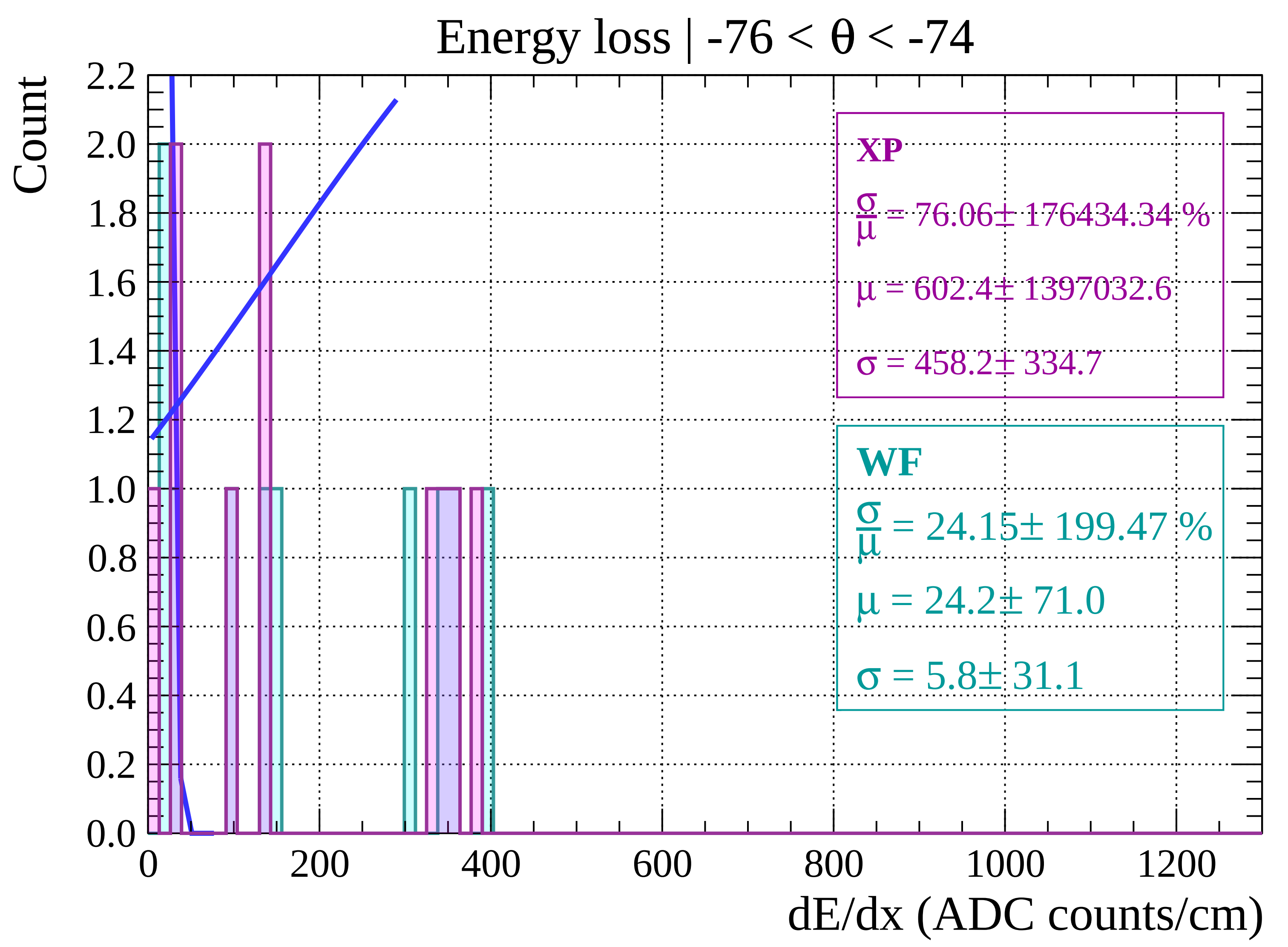
Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



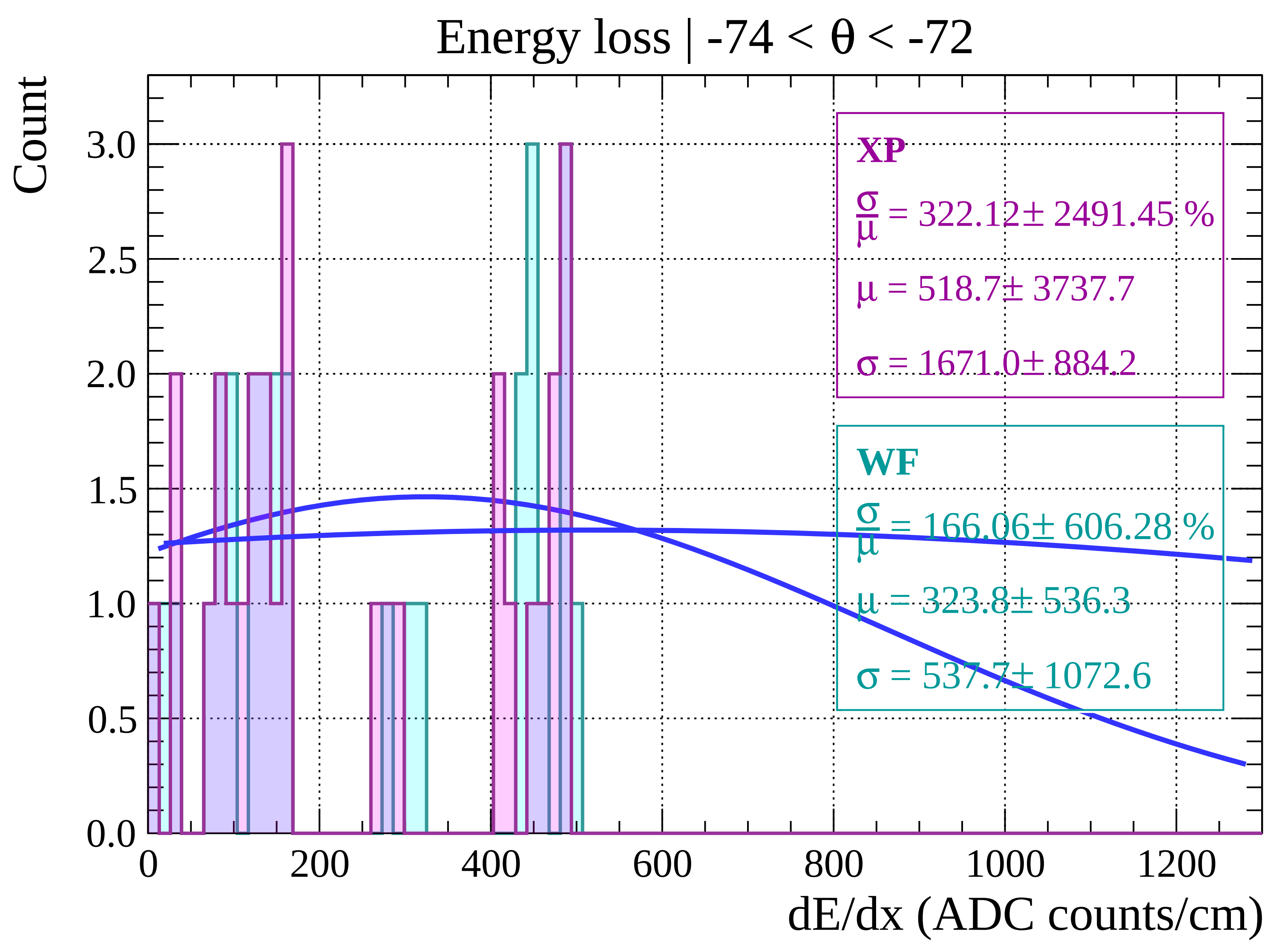
Energy loss | -80 < θ < -78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



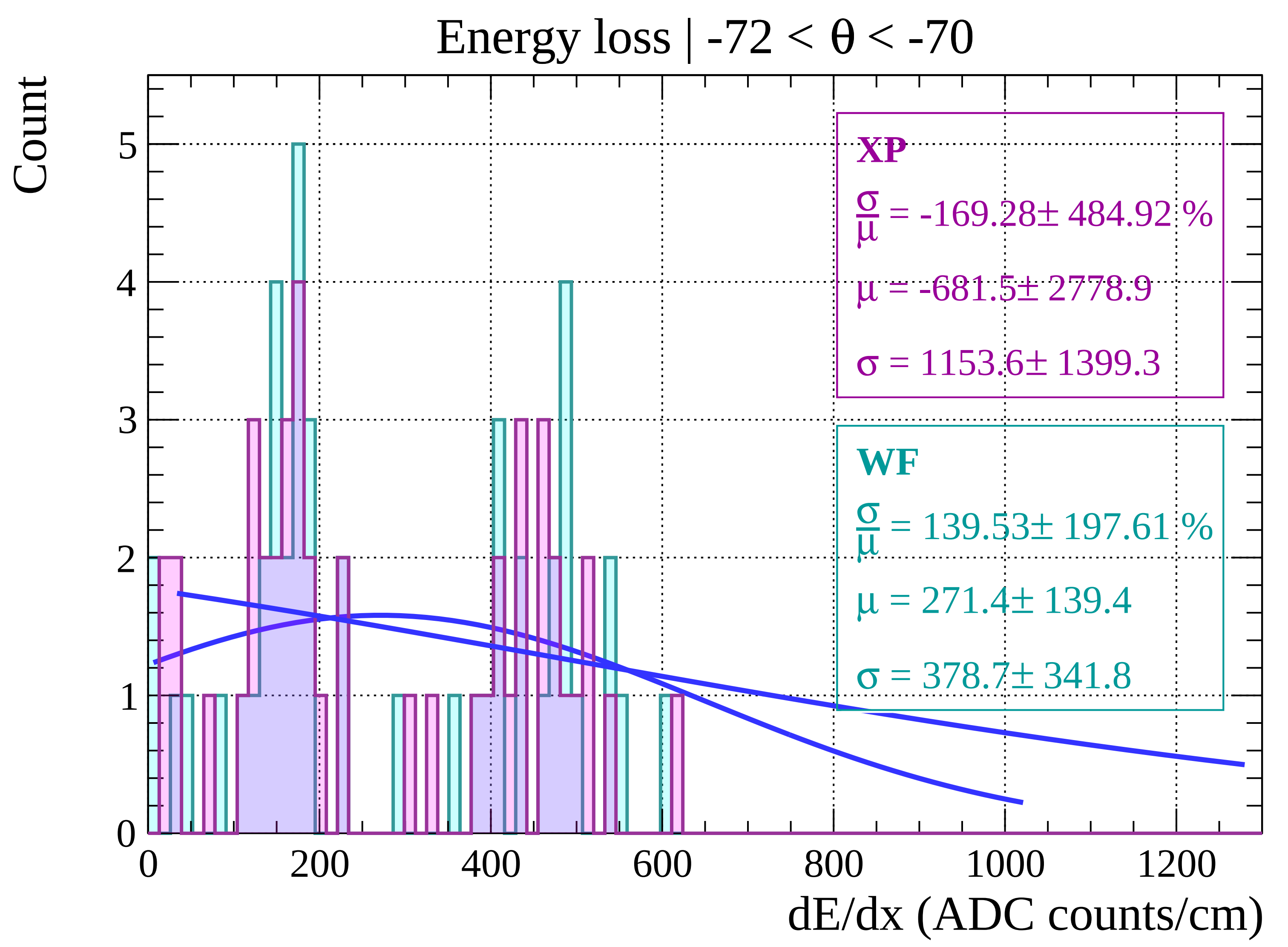
Energy loss | -78 < θ < -76
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.30± 62.34 %
μ = 92.8± 50.2
σ = 5.8± 54.7
WF
σ/μ = 0.27± 2.42 %
μ = 97.2± 1.5
σ = 0.3± 2.4

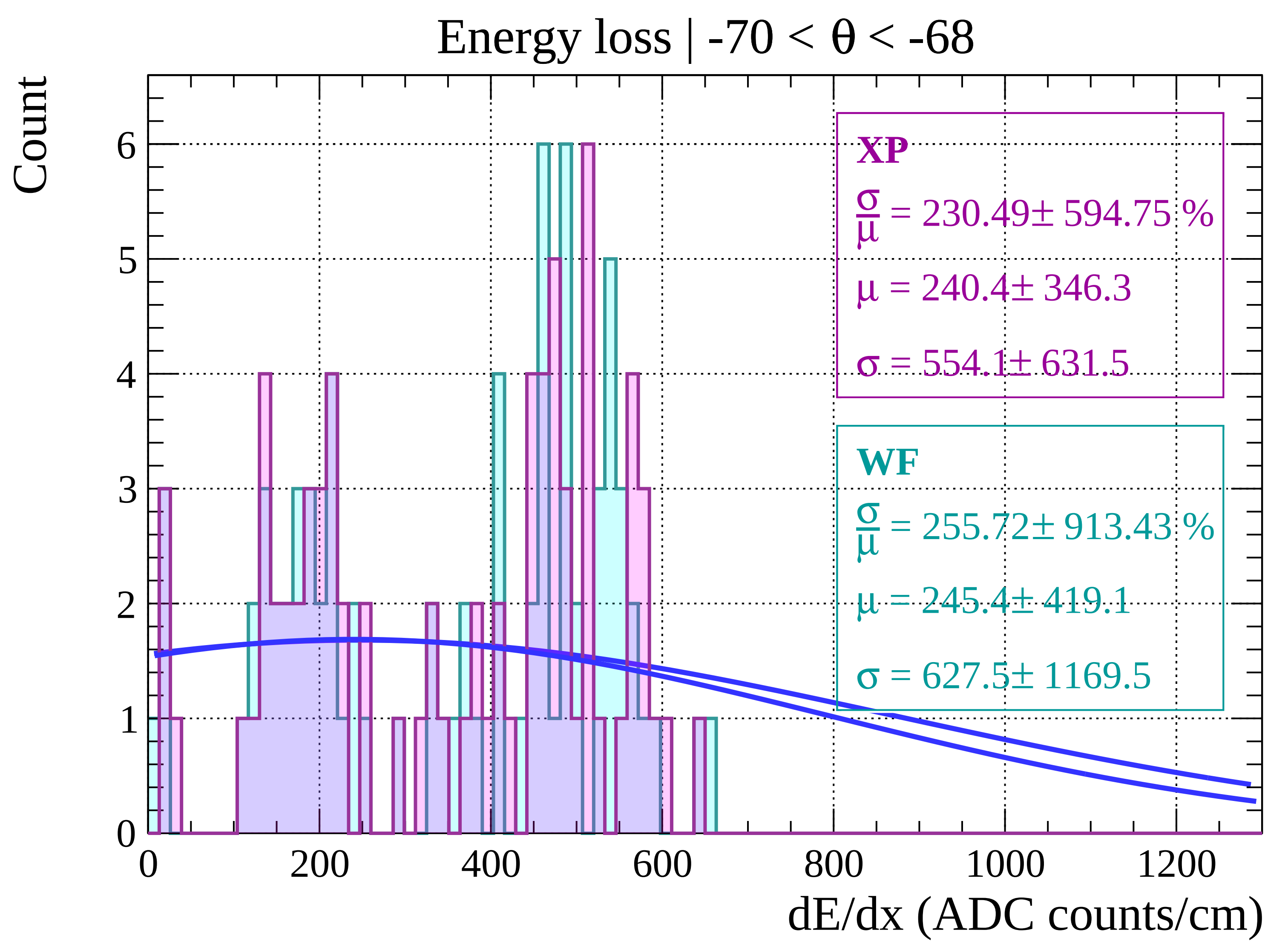


Energy loss | -76 < θ < -74
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 76.06± 176434.34 %
μ = 602.4± 1397032.6
σ = 458.2± 334.7
WF
σ/μ = 24.15± 199.47 %
μ = 24.2± 71.0
σ = 5.8± 31.1

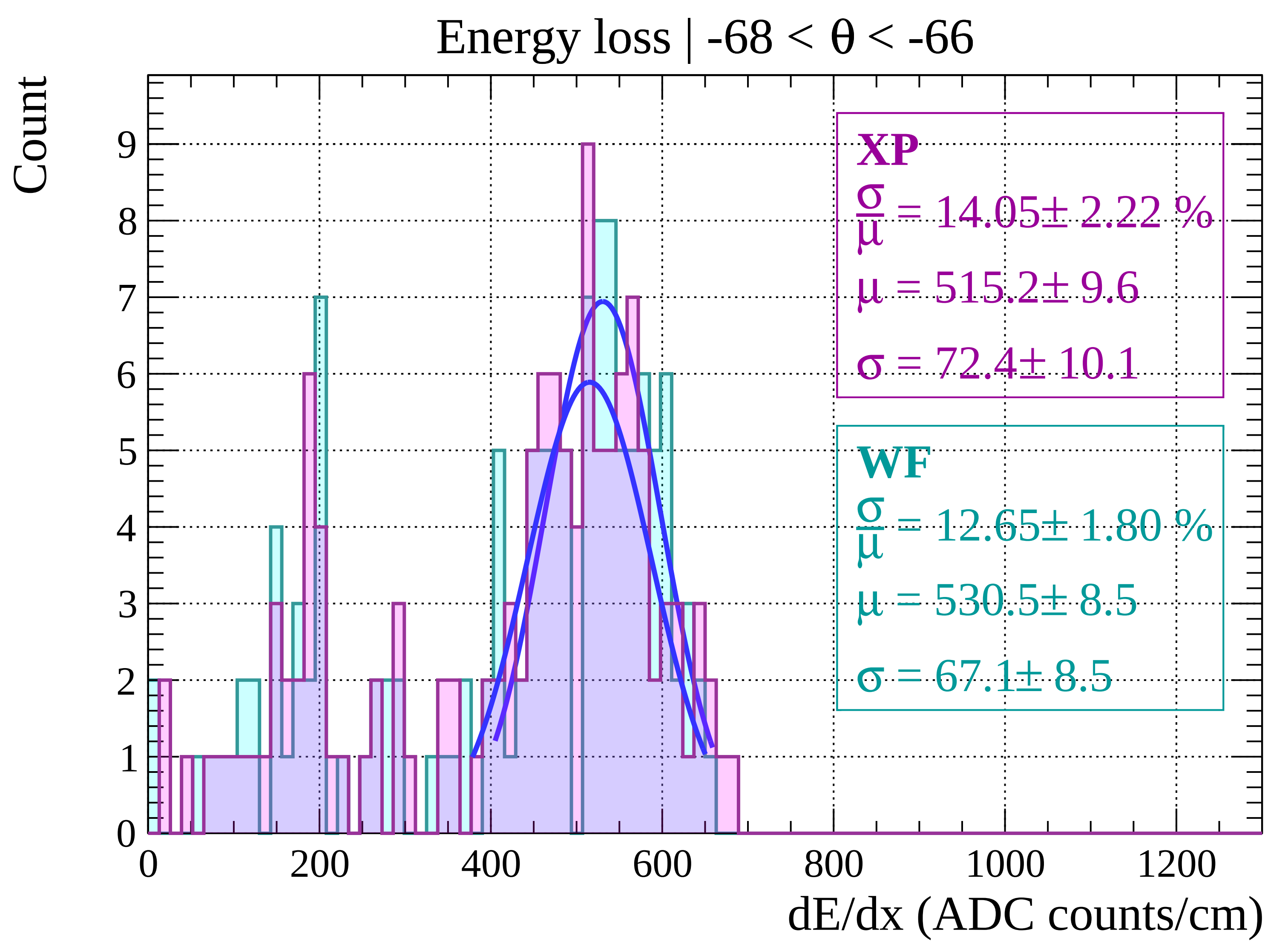


Energy loss | -74 < θ < -72
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 322.12± 2491.45 %
μ = 518.7± 3737.7
σ = 1671.0± 884.2
WF
σ/μ = 166.06± 606.28 %
μ = 323.8± 536.3
σ = 537.7± 1072.6

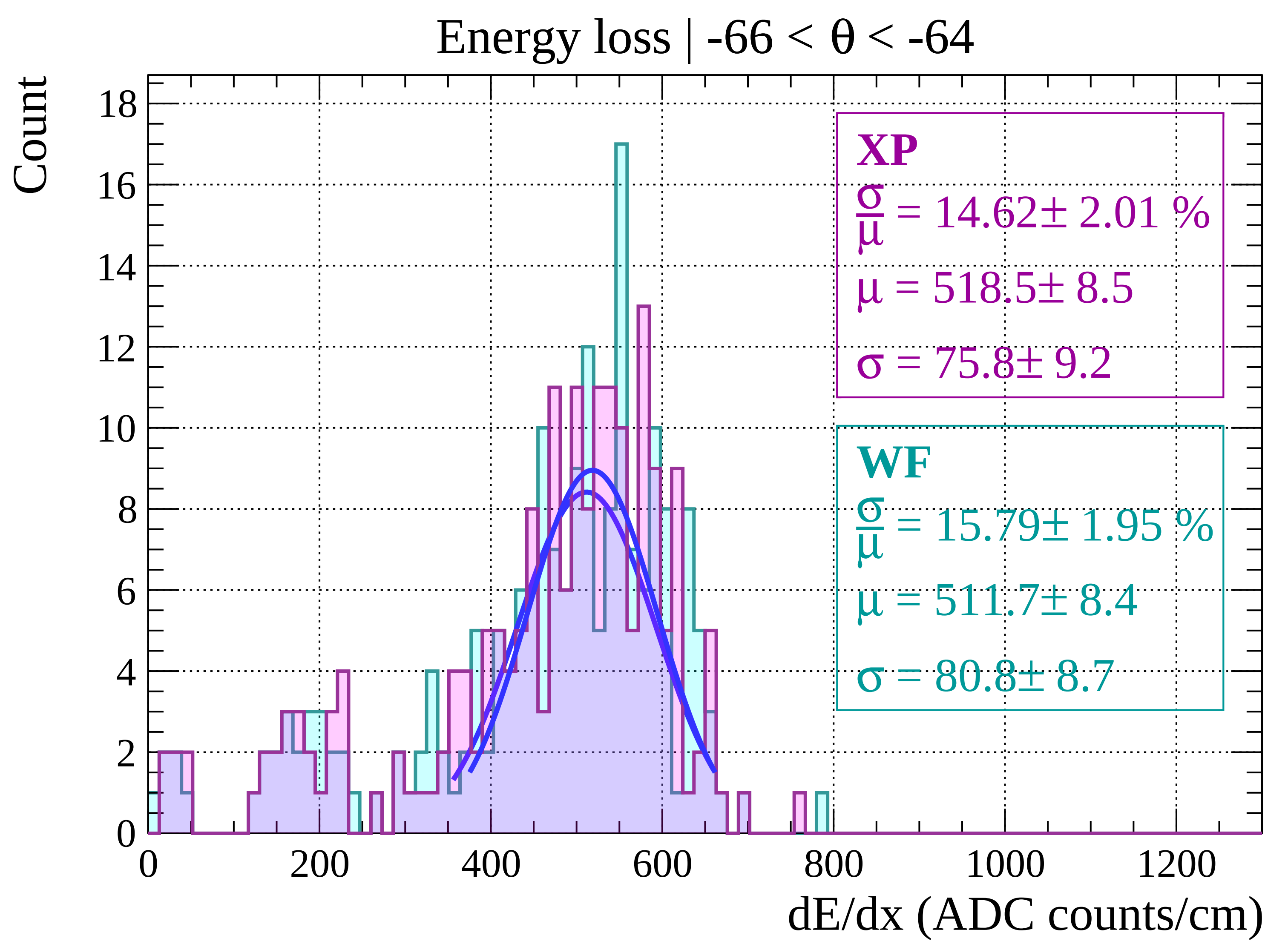

Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -169.28± 484.92 %
μ = -681.5± 2778.9
σ = 1153.6± 1399.3
WF
$\frac{\sigma}{\mu}$ = 139.53± 197.61 %
μ = 271.4± 139.4
σ = 378.7± 341.8



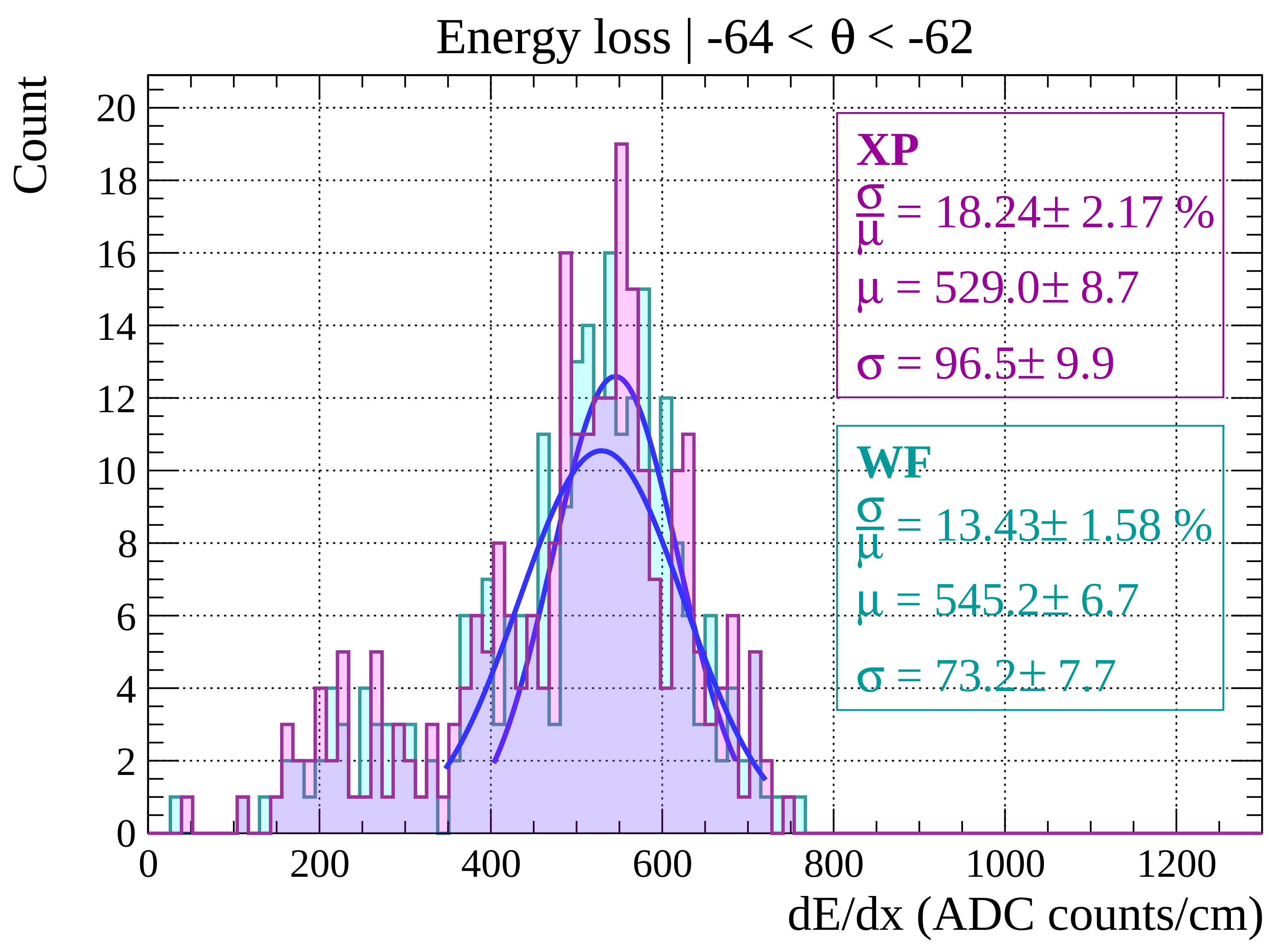
Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP
σ/μ = 230.49± 594.75 %
μ = 240.4± 346.3
σ = 554.1± 631.5
WF
σ/μ = 255.72± 913.43 %
μ = 245.4± 419.1
σ = 627.5± 1169.5



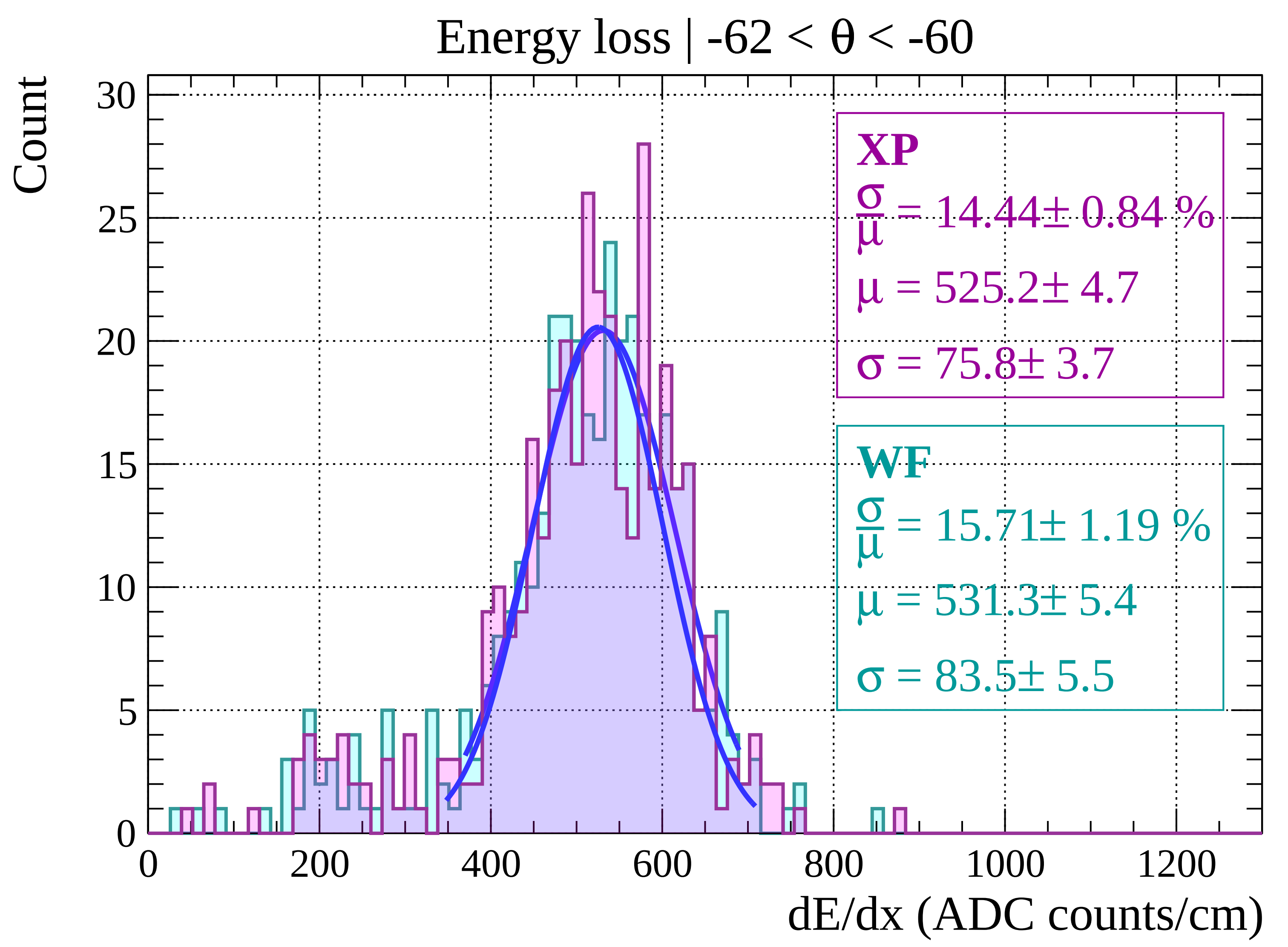
Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
8
9
0
200
400
600
800
1000
1200
XP
σ/μ = 14.05± 2.22 %
μ = 515.2± 9.6
σ = 72.4± 10.1
WF
σ/μ = 12.65± 1.80 %
μ = 530.5± 8.5
σ = 67.1± 8.5

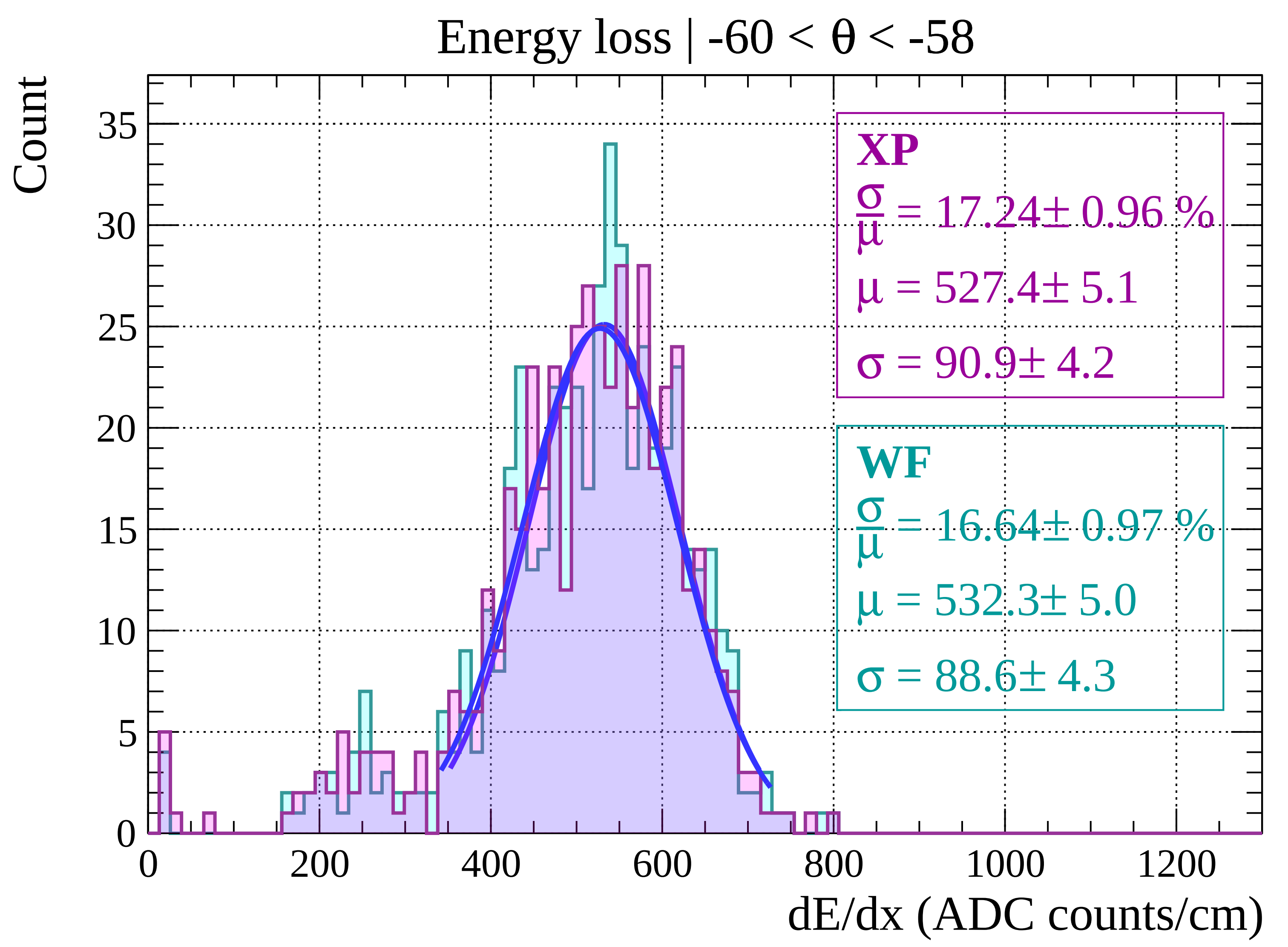


Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
200
400
600
800
1000
1200
XP
σ/μ = 14.62± 2.01 %
μ = 518.5± 8.5
σ = 75.8± 9.2
WF
σ/μ = 15.79± 1.95 %
μ = 511.7± 8.4
σ = 80.8± 8.7

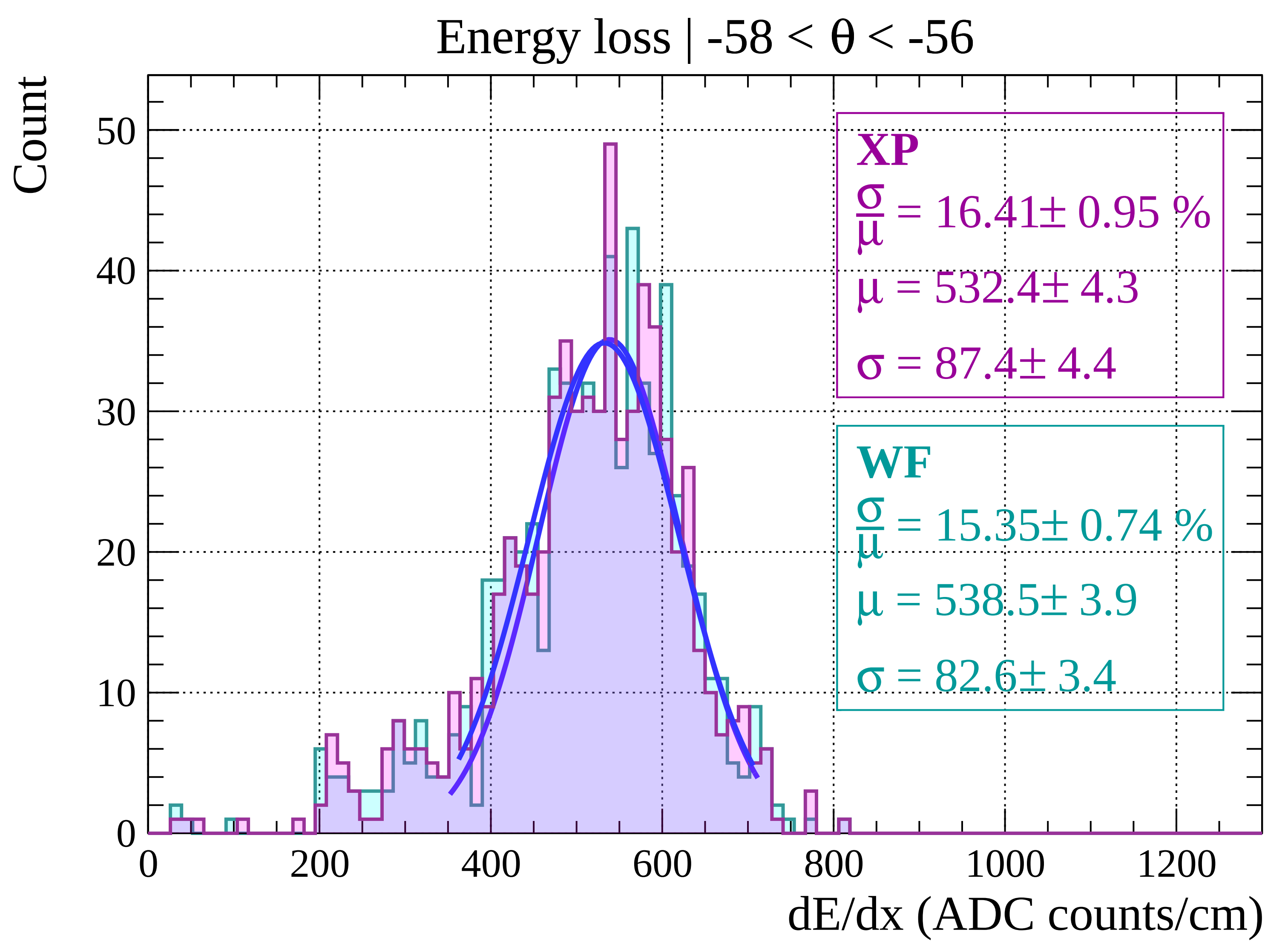


Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.24± 2.17 %
μ = 529.0± 8.7
σ = 96.5± 9.9
WF
σ/μ = 13.43± 1.58 %
μ = 545.2± 6.7
σ = 73.2± 7.7

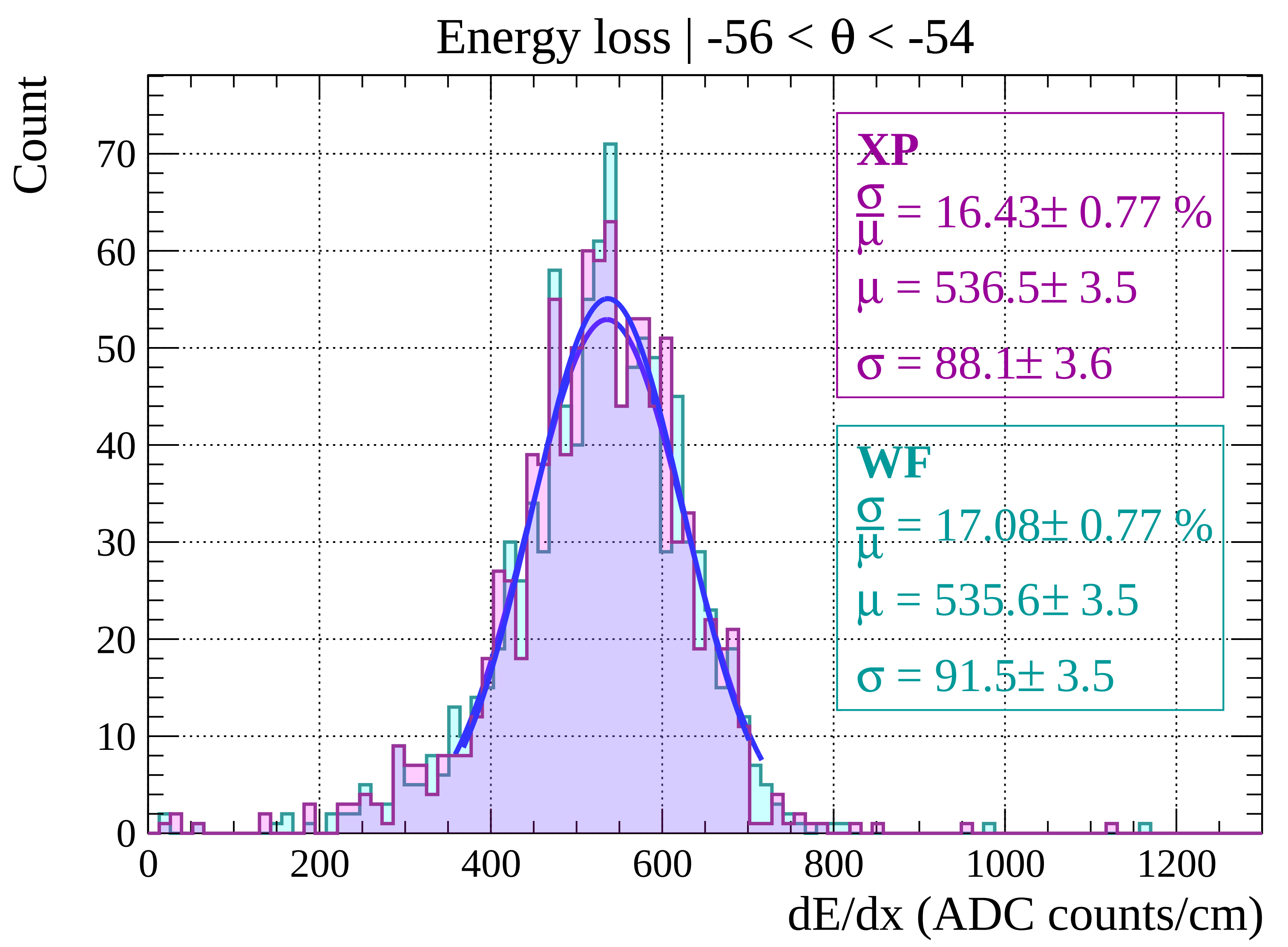

Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 14.44± 0.84 %
μ = 525.2± 4.7
σ = 75.8± 3.7
WF
σ/μ = 15.71± 1.19 %
μ = 531.3± 5.4
σ = 83.5± 5.5

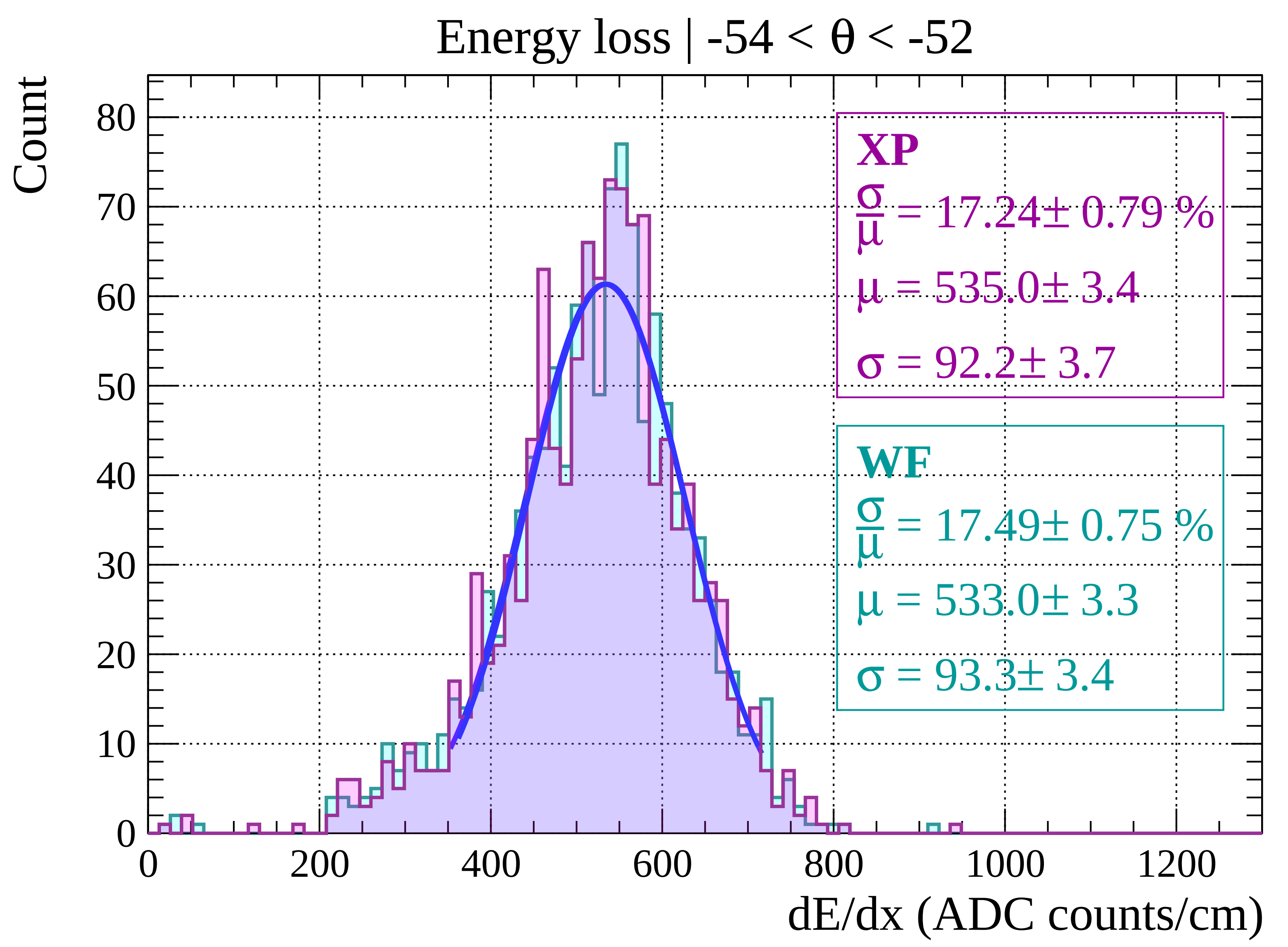


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.24± 0.96 %
μ = 527.4± 5.1
σ = 90.9± 4.2
WF
σ/μ = 16.64± 0.97 %
μ = 532.3± 5.0
σ = 88.6± 4.3
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200

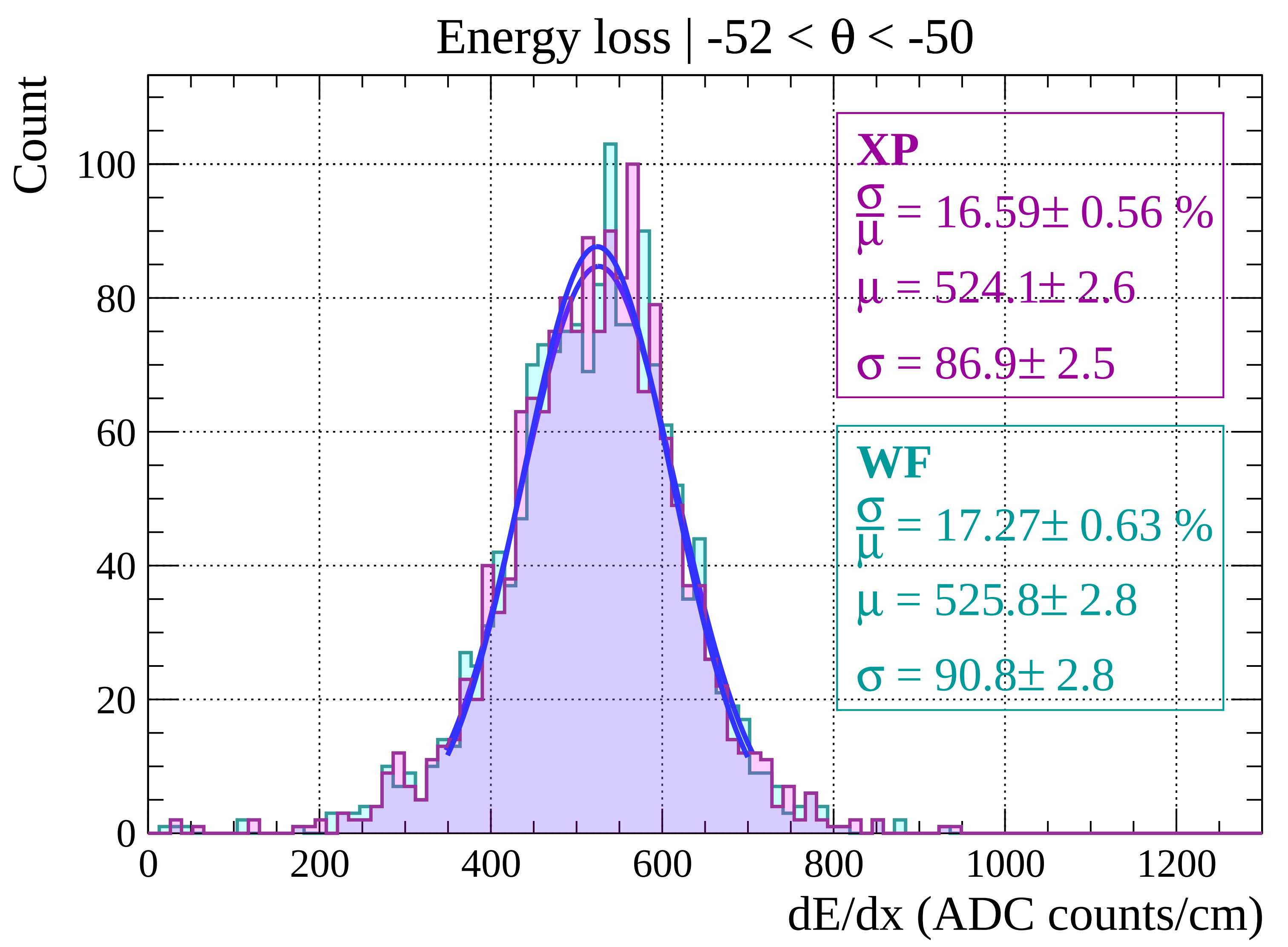


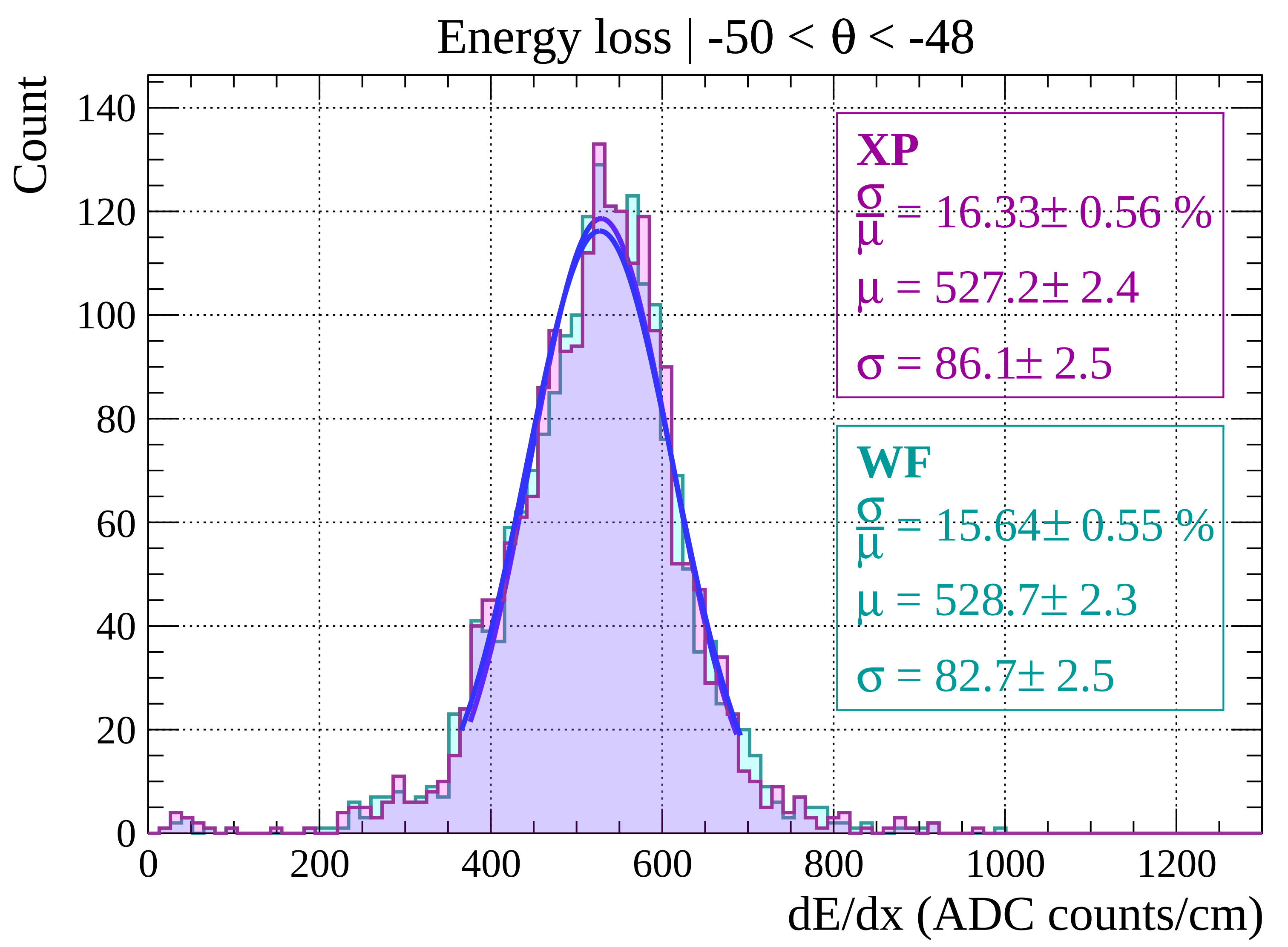
Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.41± 0.95 %
μ = 532.4± 4.3
σ = 87.4± 4.4
WF
$\frac{\sigma}{\mu}$ = 15.35± 0.74 %
μ = 538.5± 3.9
σ = 82.6± 3.4
0
10
20
30
40
50
0
200
400
600
800
1000
1200


Energy loss | -56 < θ < -54
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.43± 0.77 %
μ = 536.5± 3.5
σ = 88.1± 3.6
WF
σ/μ = 17.08± 0.77 %
μ = 535.6± 3.5
σ = 91.5± 3.5

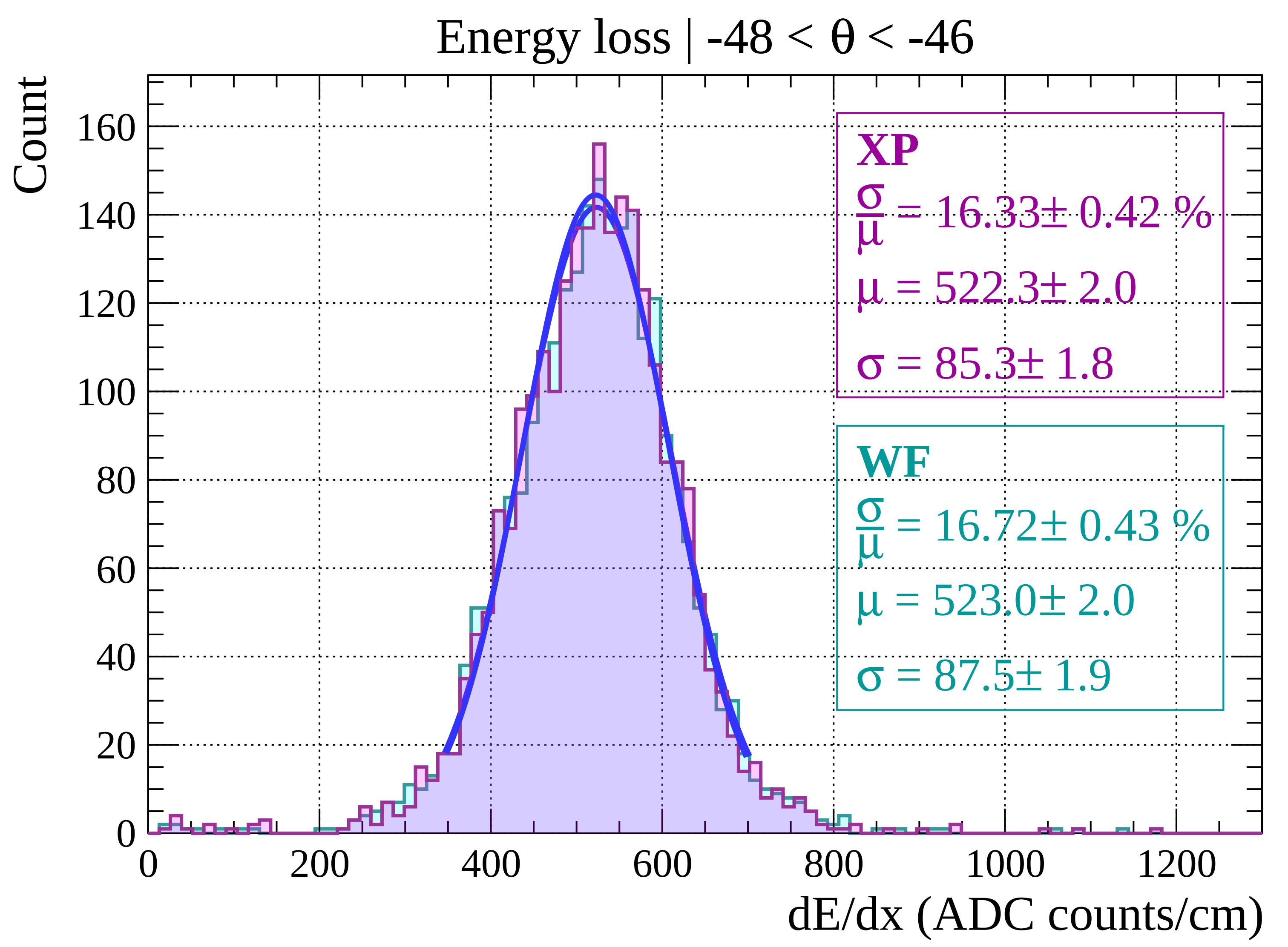


Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 17.24± 0.79 %
μ = 535.0± 3.4
σ = 92.2± 3.7
WF
σ/μ = 17.49± 0.75 %
μ = 533.0± 3.3
σ = 93.3± 3.4

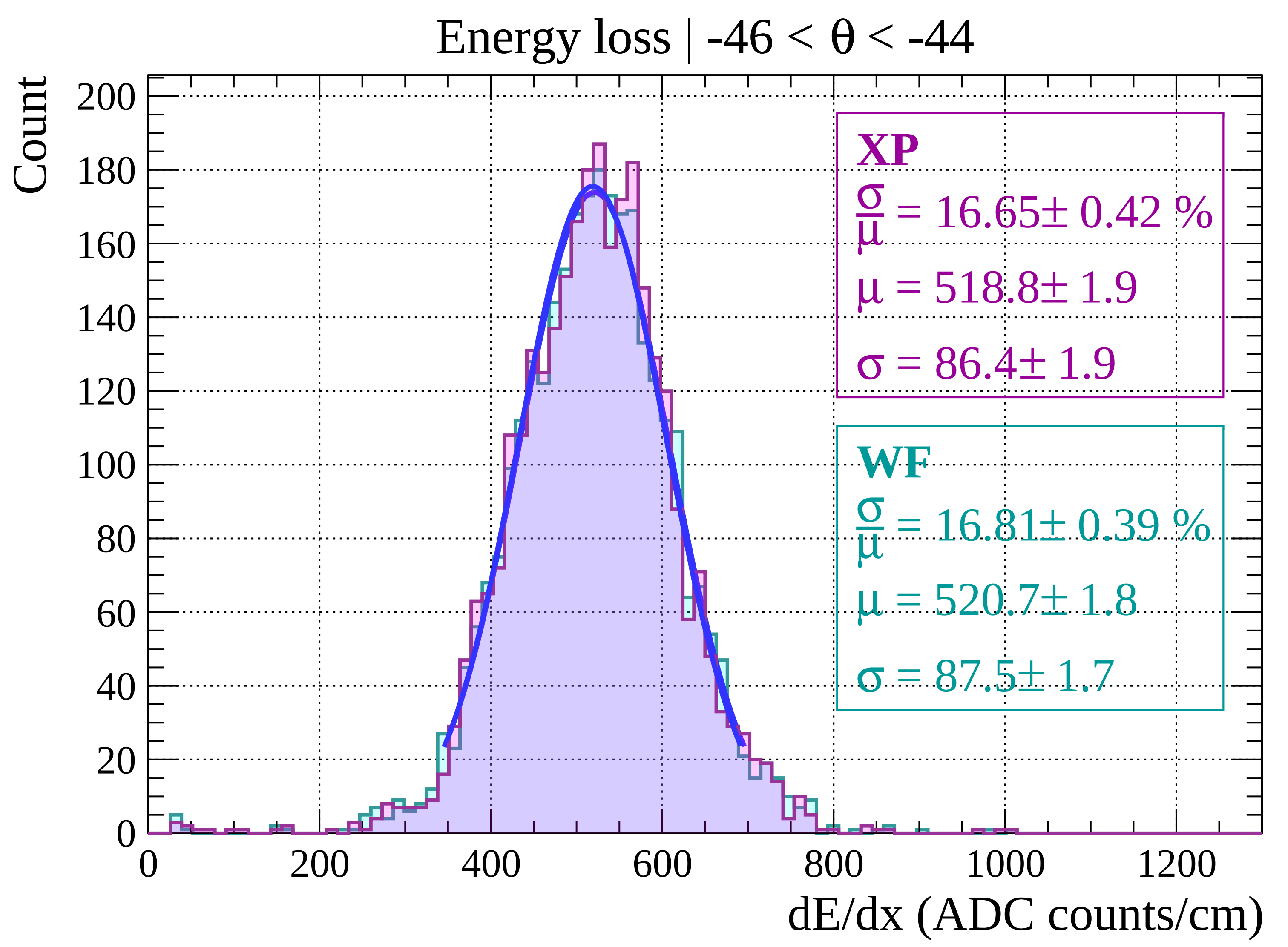

Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.59± 0.56 %
μ = 524.1± 2.6
σ = 86.9± 2.5
WF
$\frac{\sigma}{\mu}$ = 17.27± 0.63 %
μ = 525.8± 2.8
σ = 90.8± 2.8



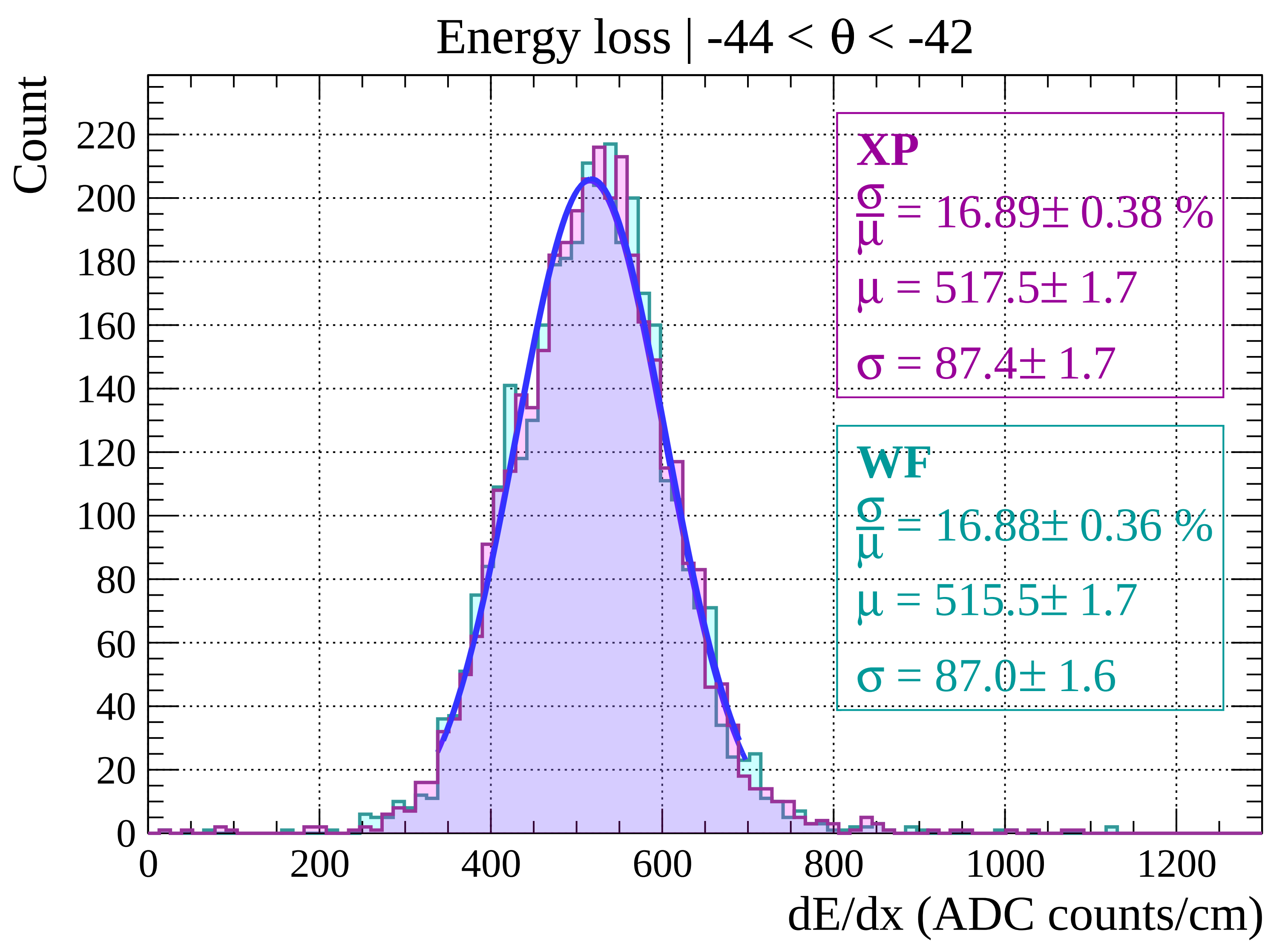
Energy loss | -50 < θ < -48
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.33± 0.56 %
μ = 527.2± 2.4
σ = 86.1± 2.5
WF
$\frac{\sigma}{\mu}$ = 15.64± 0.55 %
μ = 528.7± 2.3
σ = 82.7± 2.5

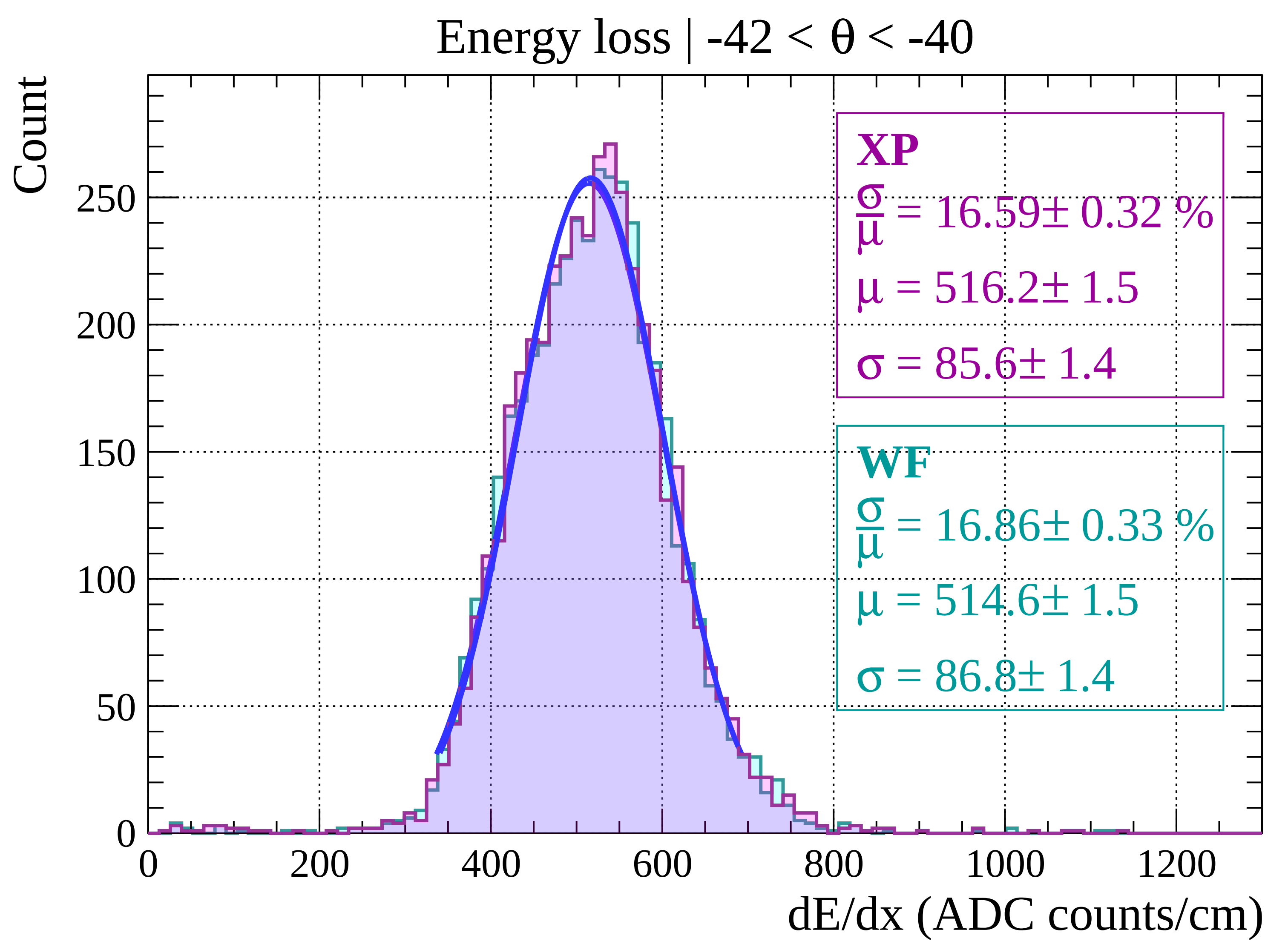


Energy loss | -48 < θ < -46
Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.33± 0.42 %
μ = 522.3± 2.0
σ = 85.3± 1.8
WF
σ/μ = 16.72± 0.43 %
μ = 523.0± 2.0
σ = 87.5± 1.9

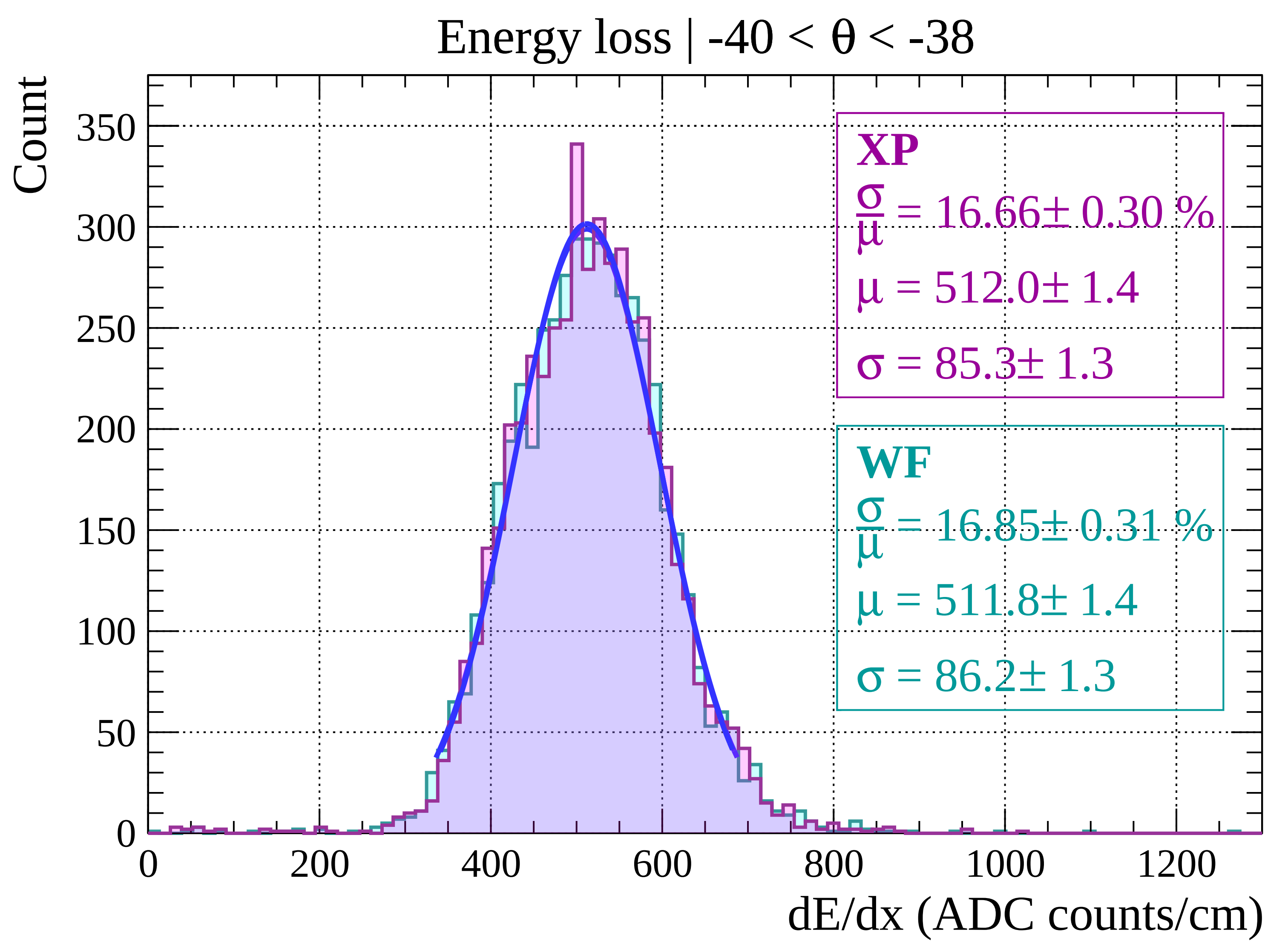


Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 16.65± 0.42 %
μ = 518.8± 1.9
σ = 86.4± 1.9
WF
σ/μ = 16.81± 0.39 %
μ = 520.7± 1.8
σ = 87.5± 1.7

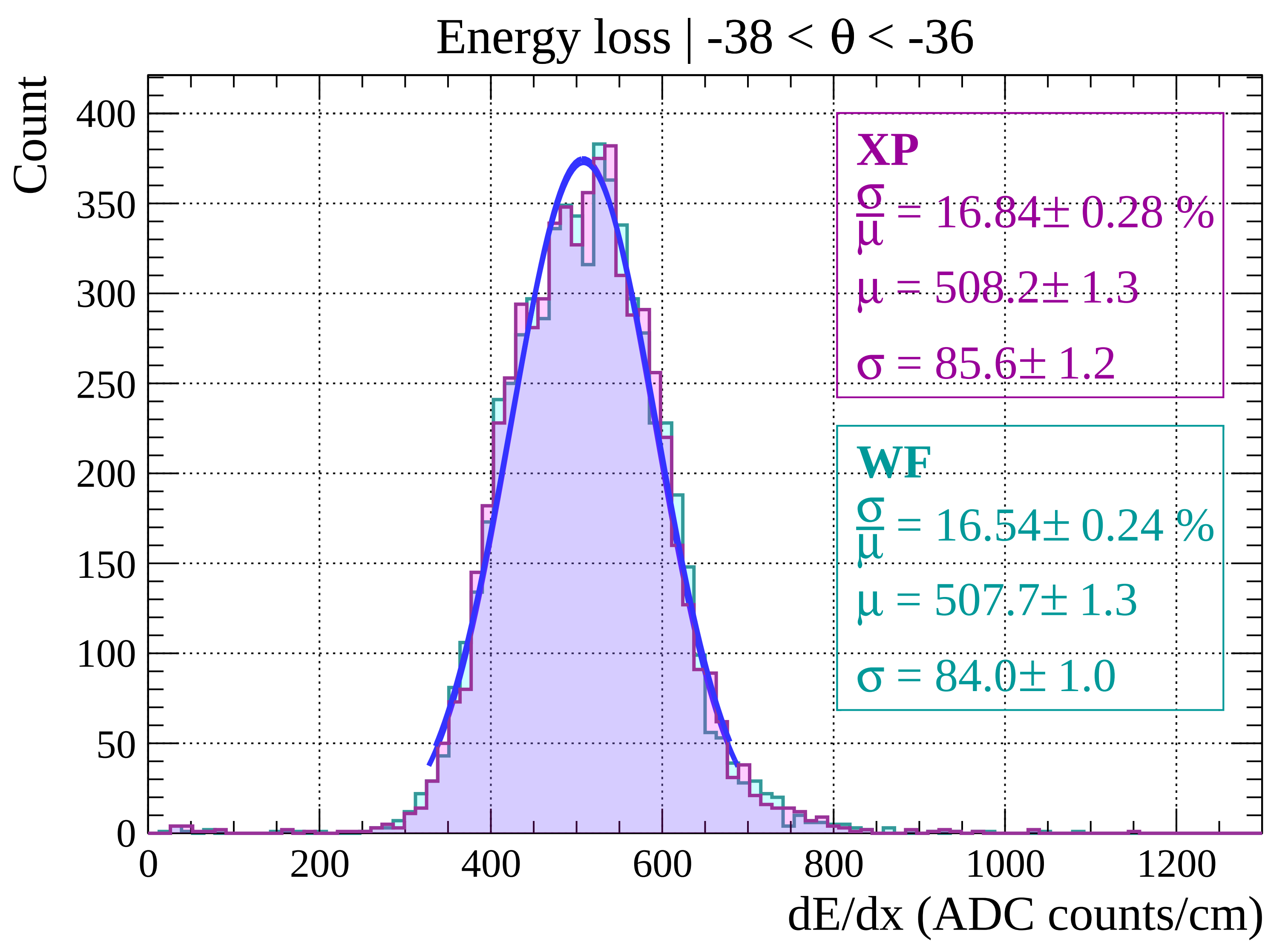

Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.89± 0.38 %
μ = 517.5± 1.7
σ = 87.4± 1.7
WF
$\frac{\sigma}{\mu}$ = 16.88± 0.36 %
μ = 515.5± 1.7
σ = 87.0± 1.6



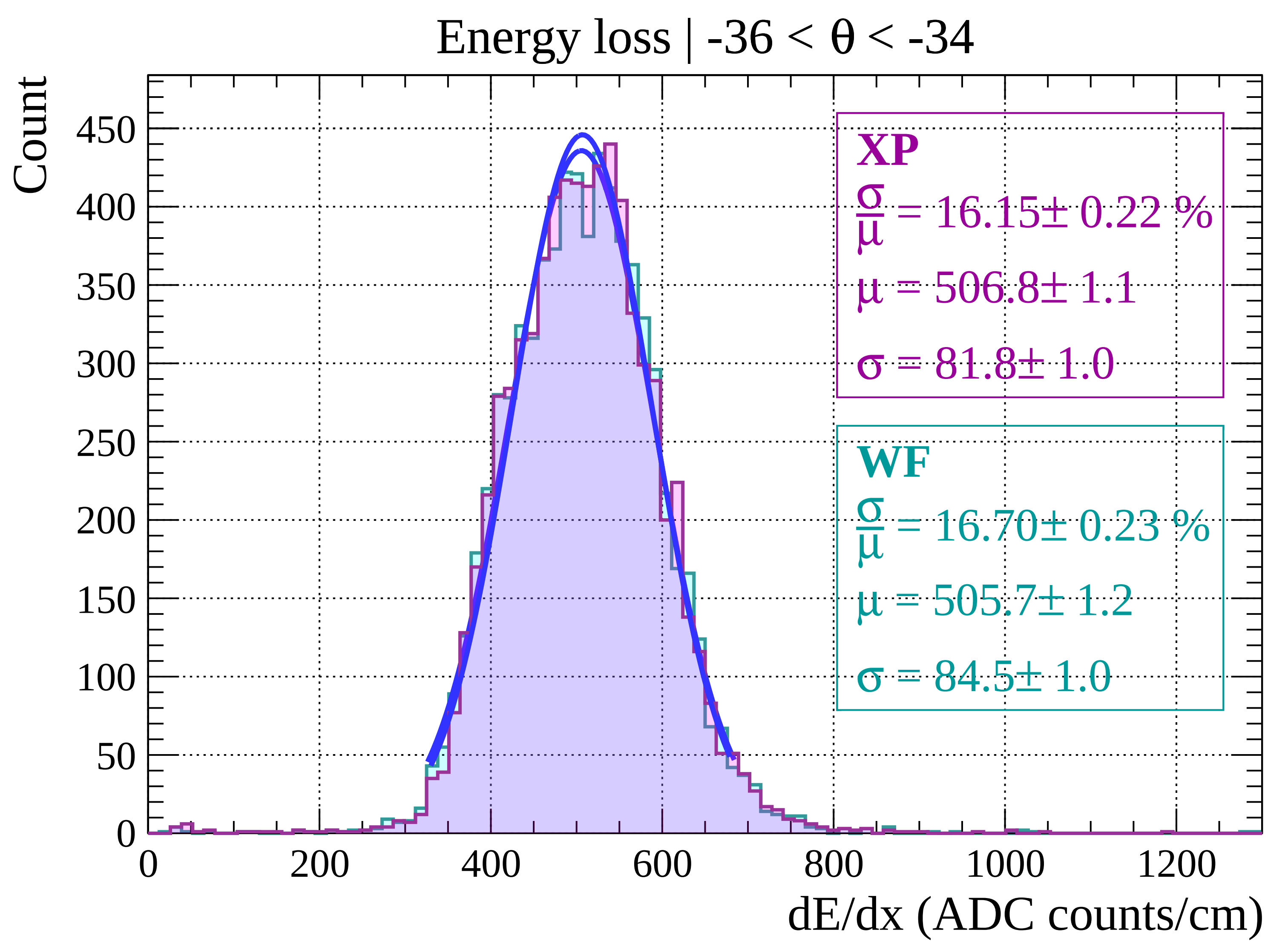
Energy loss | -42 < θ < -40
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.59± 0.32 %
μ = 516.2± 1.5
σ = 85.6± 1.4
WF
σ/μ = 16.86± 0.33 %
μ = 514.6± 1.5
σ = 86.8± 1.4



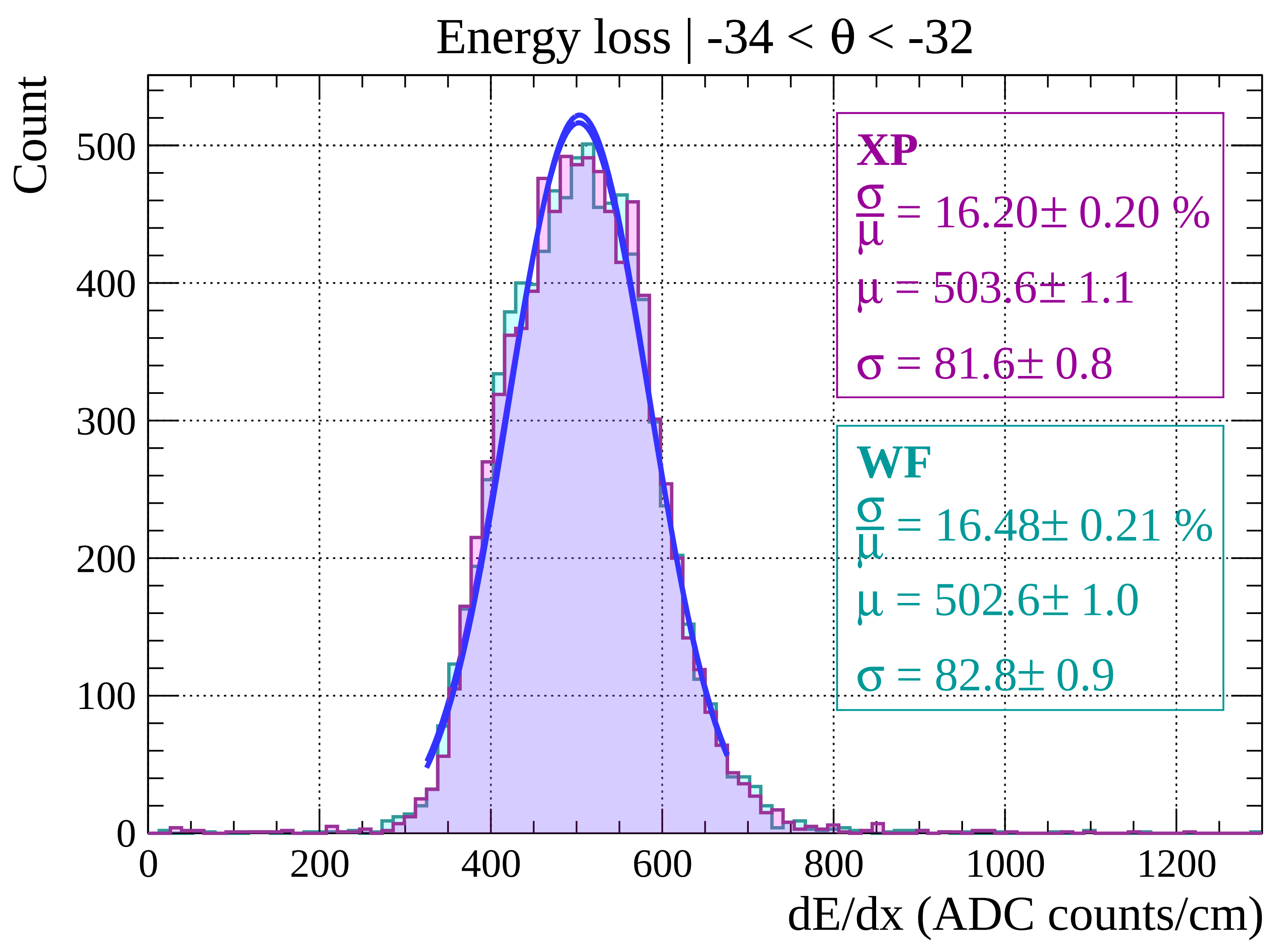
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 16.66± 0.30 %
μ = 512.0± 1.4
σ = 85.3± 1.3
WF
σ/μ = 16.85± 0.31 %
μ = 511.8± 1.4
σ = 86.2± 1.3

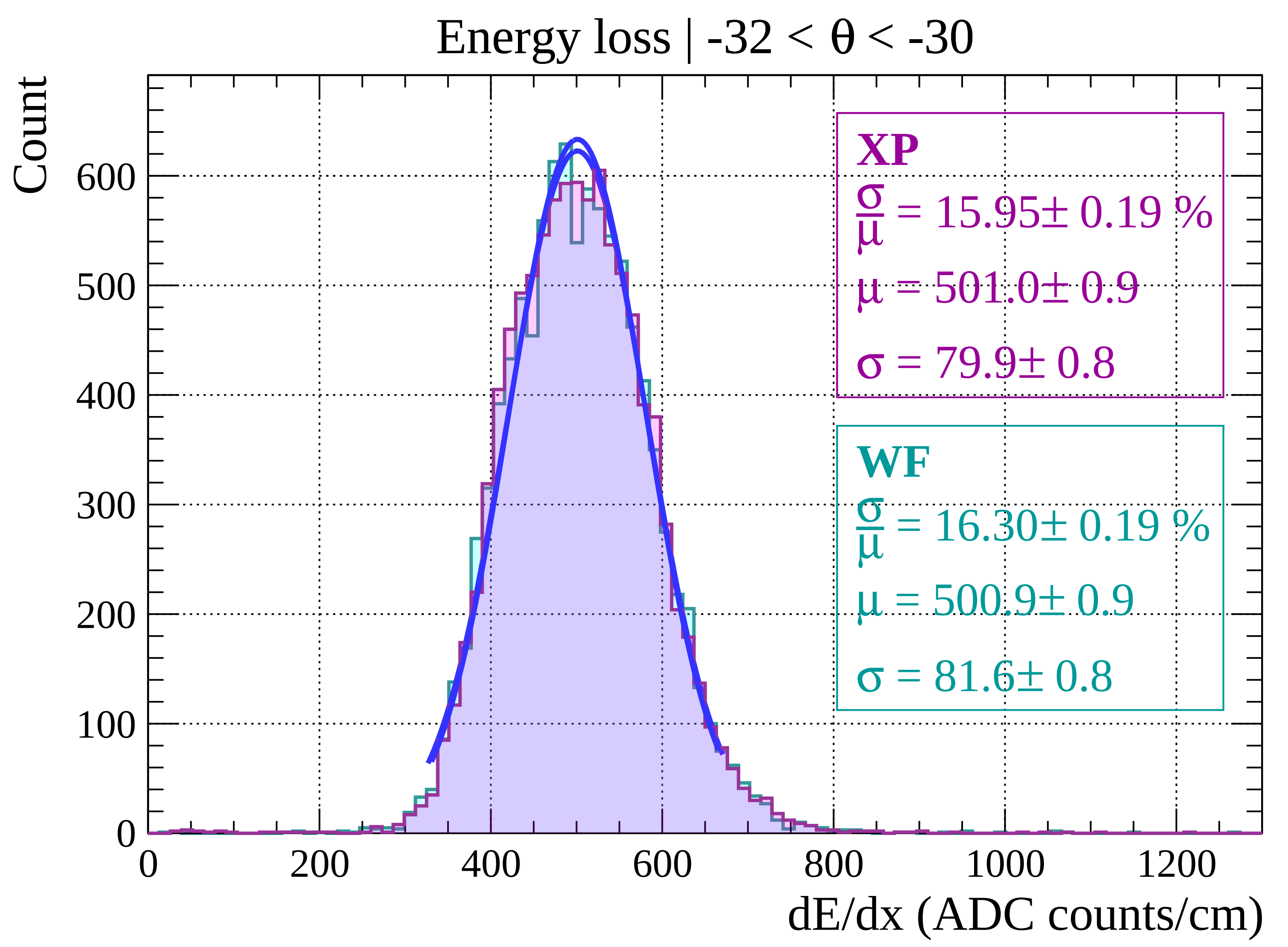


Energy loss | -38 < θ < -36
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.84± 0.28 %
μ = 508.2± 1.3
σ = 85.6± 1.2
WF
σ/μ = 16.54± 0.24 %
μ = 507.7± 1.3
σ = 84.0± 1.0

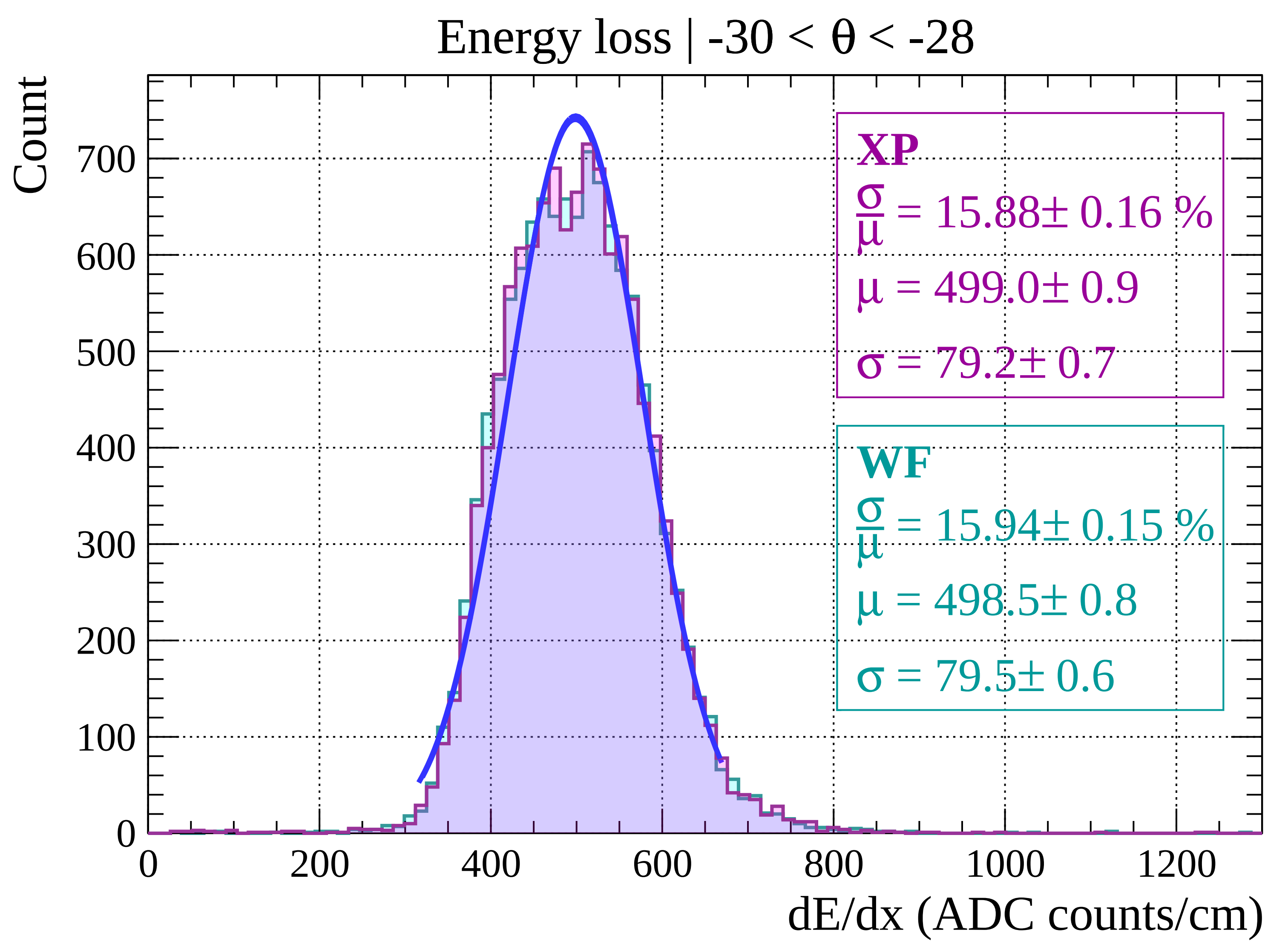


Energy loss | -36 < θ < -34
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.15± 0.22 %
μ = 506.8± 1.1
σ = 81.8± 1.0
WF
σ/μ = 16.70± 0.23 %
μ = 505.7± 1.2
σ = 84.5± 1.0

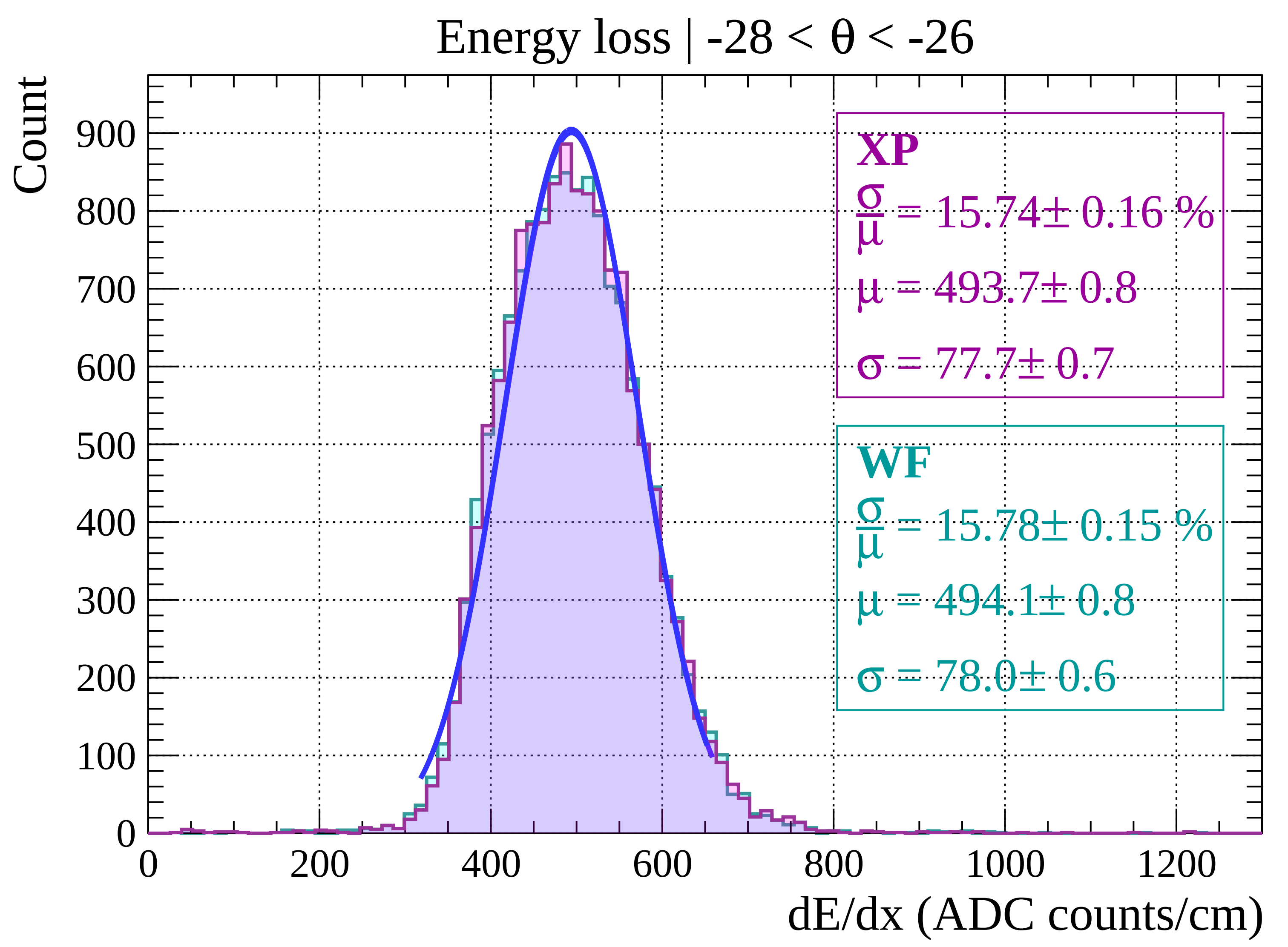

Energy loss | -34 < θ < -32
Count
XP
σ/μ = 16.20± 0.20 %
μ = 503.6± 1.1
σ = 81.6± 0.8
WF
σ/μ = 16.48± 0.21 %
μ = 502.6± 1.0
σ = 82.8± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



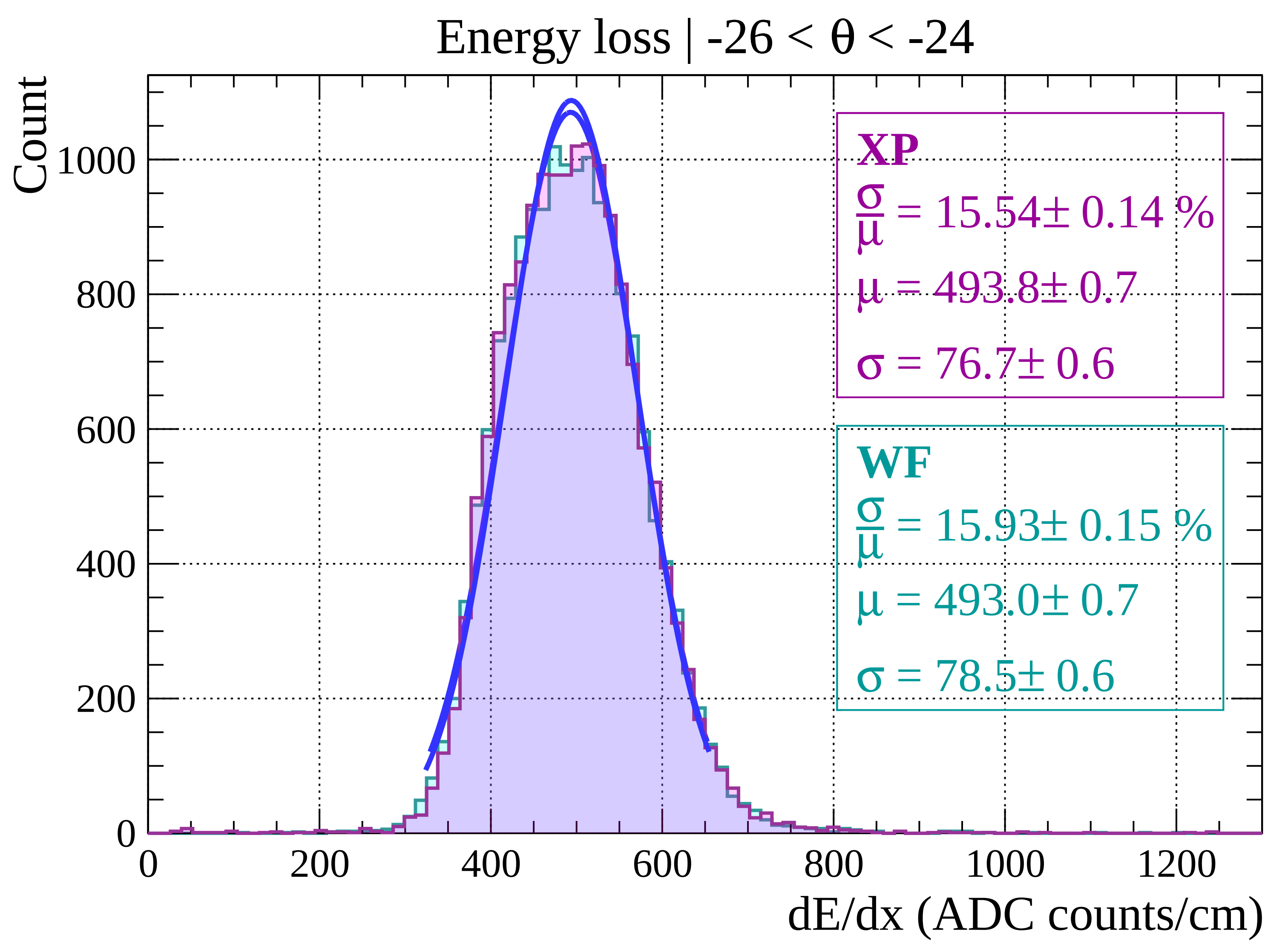
Energy loss | -32 < θ < -30
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.95± 0.19 %
μ = 501.0± 0.9
σ = 79.9± 0.8
WF
σ/μ = 16.30± 0.19 %
μ = 500.9± 0.9
σ = 81.6± 0.8



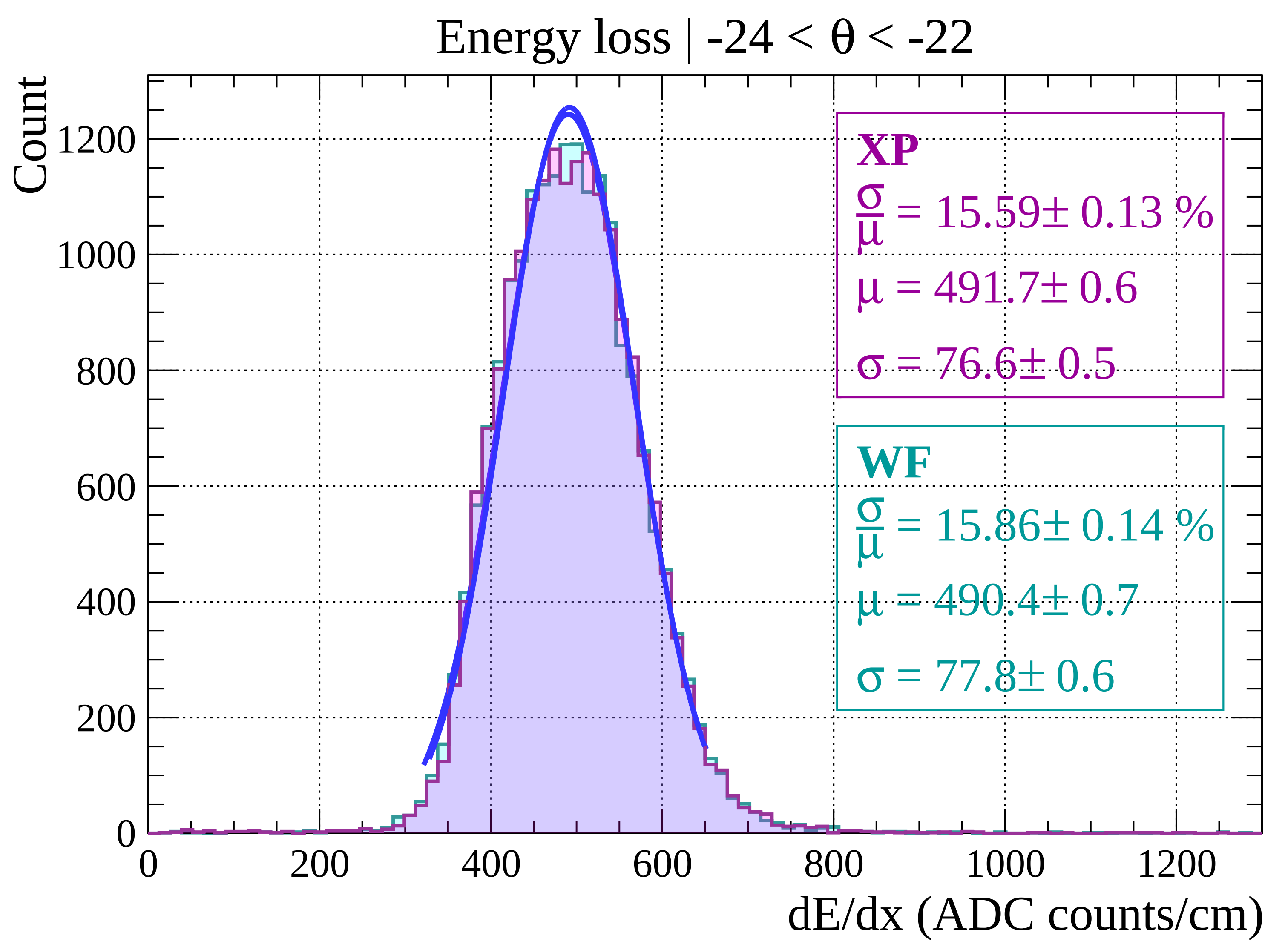
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.88± 0.16 %
μ = 499.0± 0.9
σ = 79.2± 0.7
WF
σ/μ = 15.94± 0.15 %
μ = 498.5± 0.8
σ = 79.5± 0.6



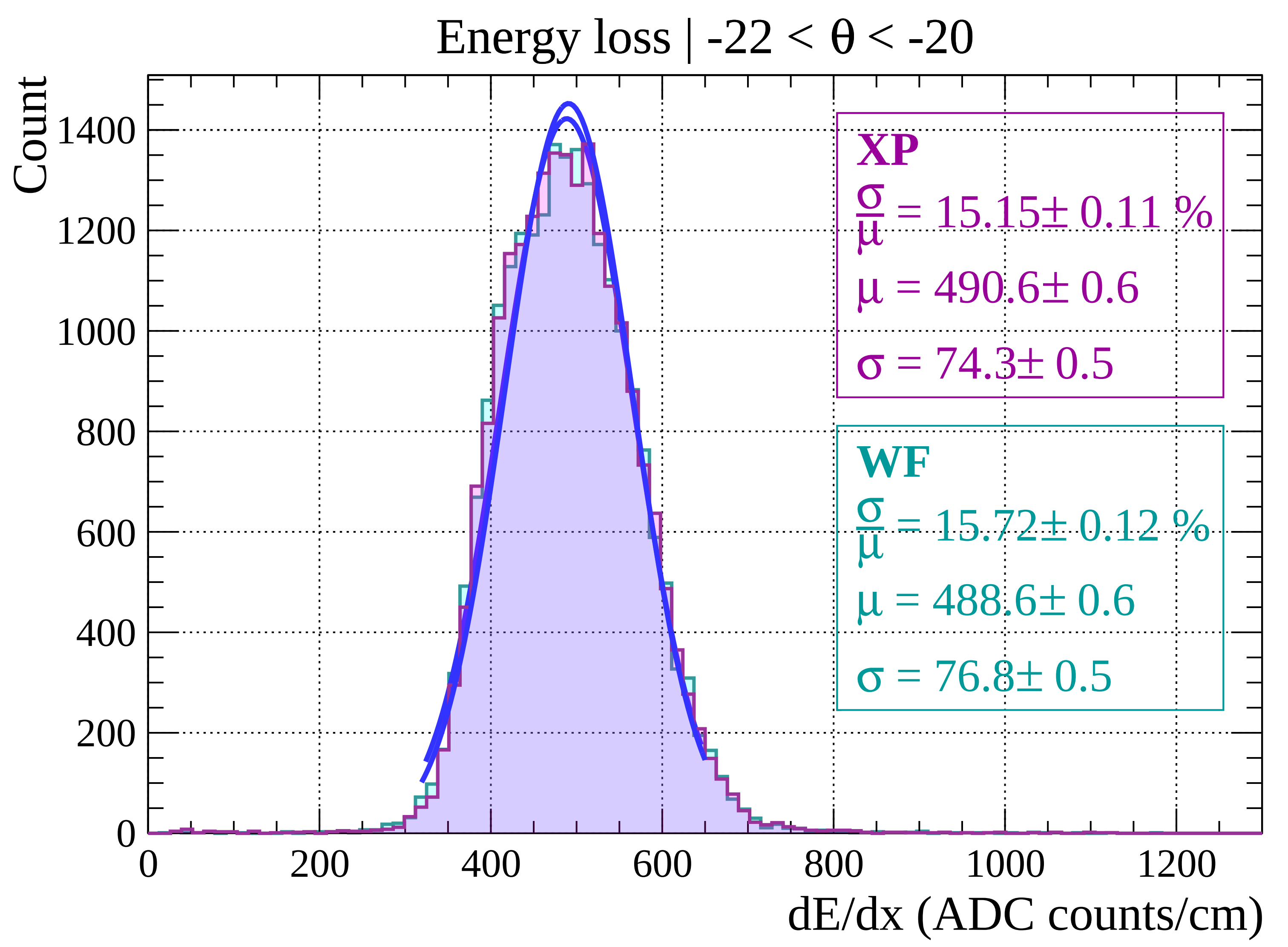
Energy loss | -28 < θ < -26
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.74± 0.16 %
μ = 493.7± 0.8
σ = 77.7± 0.7
WF
σ/μ = 15.78± 0.15 %
μ = 494.1± 0.8
σ = 78.0± 0.6



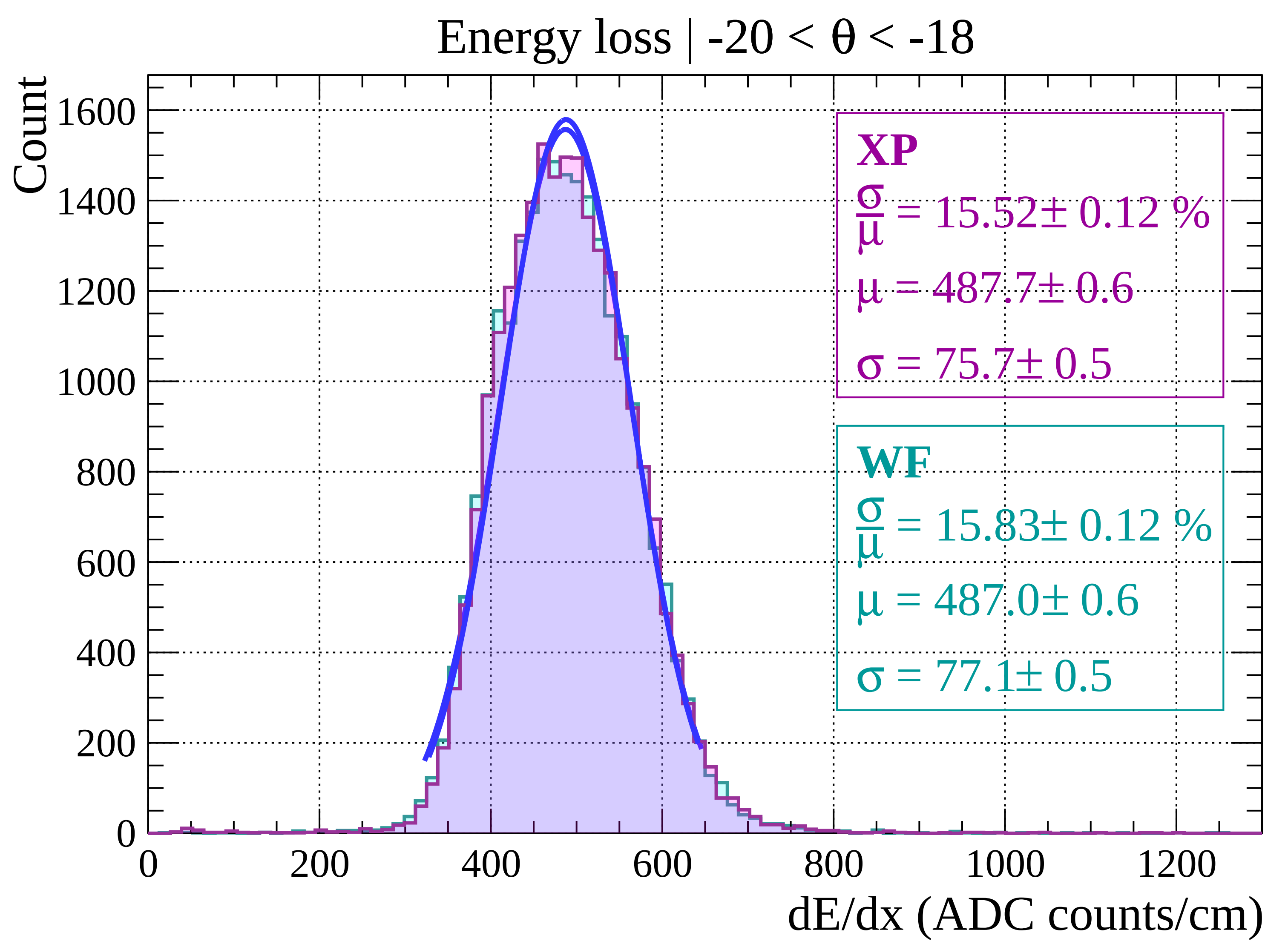
Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.54± 0.14 %
μ = 493.8± 0.7
σ = 76.7± 0.6
WF
σ/μ = 15.93± 0.15 %
μ = 493.0± 0.7
σ = 78.5± 0.6



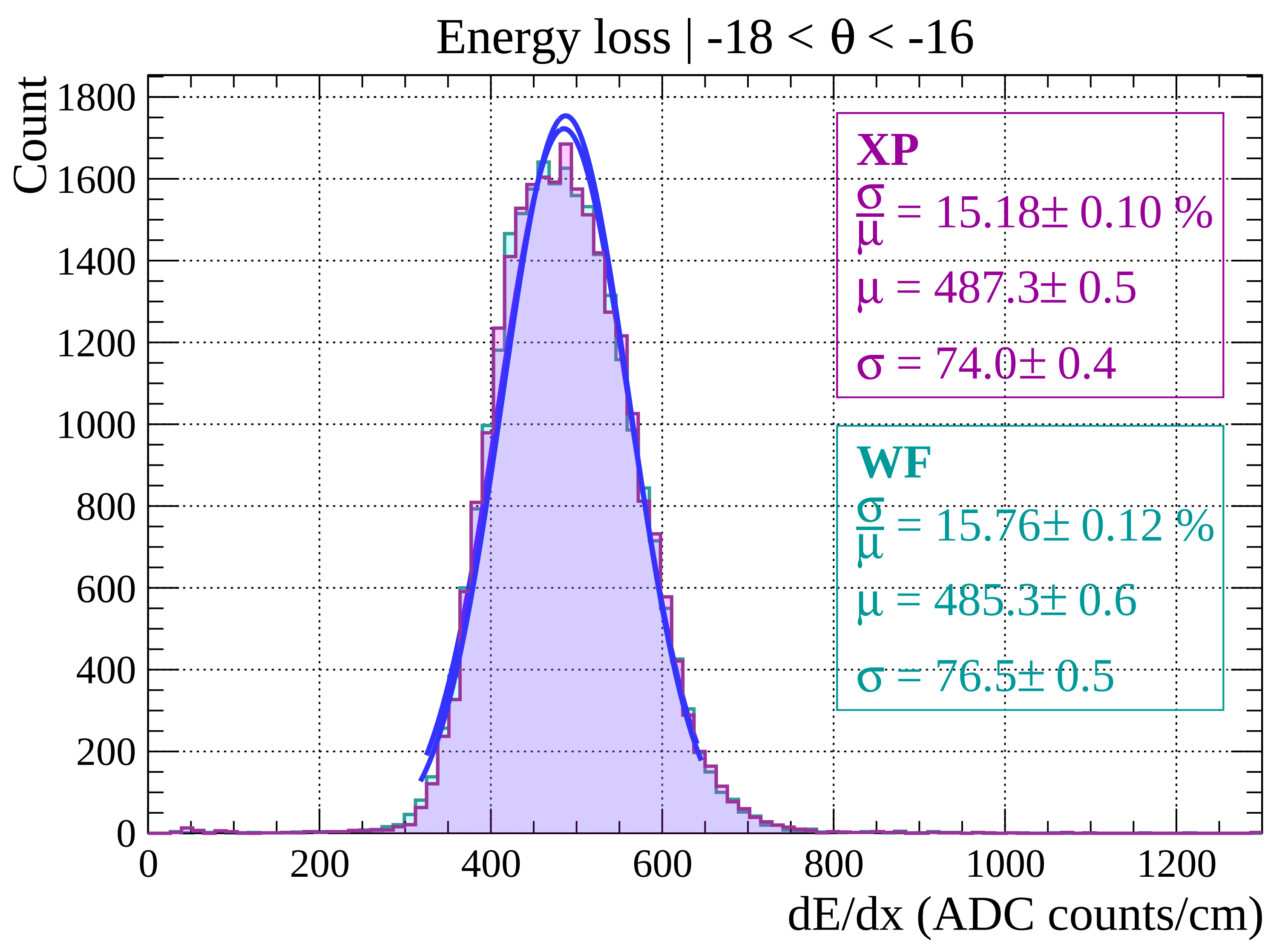
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.59± 0.13 %
μ = 491.7± 0.6
σ = 76.6± 0.5
WF
σ/μ = 15.86± 0.14 %
μ = 490.4± 0.7
σ = 77.8± 0.6



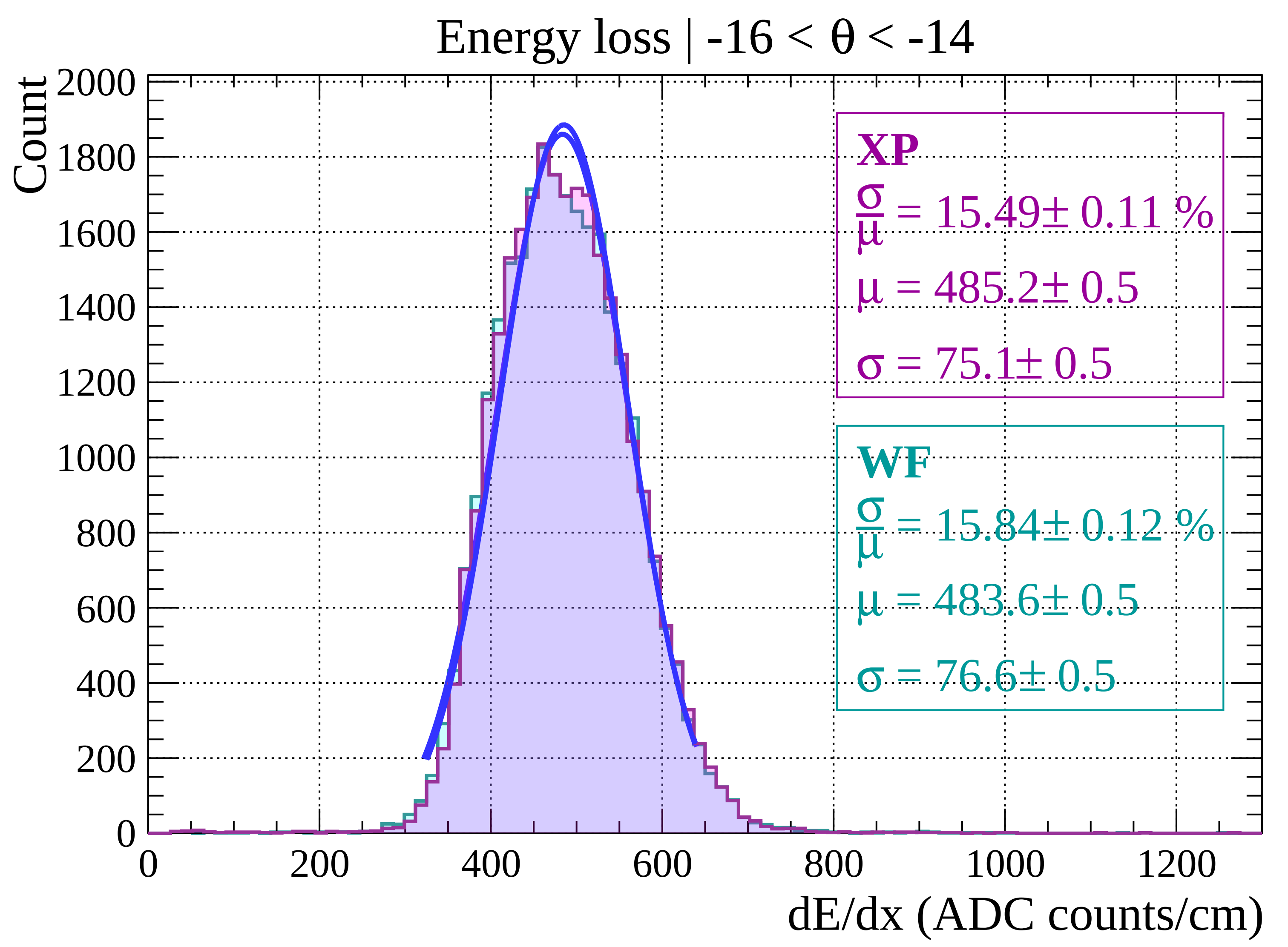
Energy loss | -22 < θ < -20
Count
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 15.15± 0.11 %
μ = 490.6± 0.6
σ = 74.3± 0.5
WF
σ/μ = 15.72± 0.12 %
μ = 488.6± 0.6
σ = 76.8± 0.5
dE/dx (ADC counts/cm)



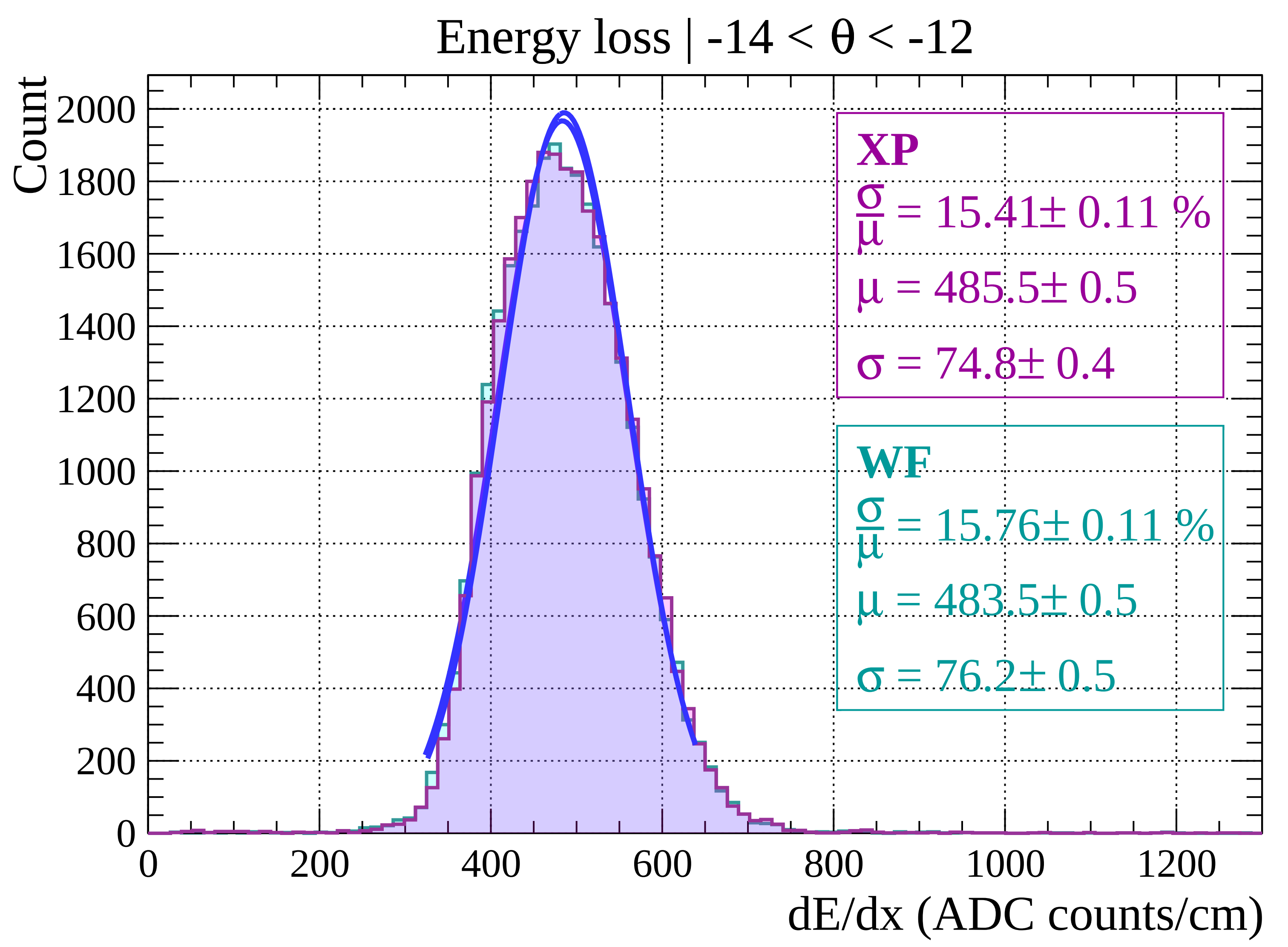
Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.52± 0.12 %
μ = 487.7± 0.6
σ = 75.7± 0.5
WF
σ/μ = 15.83± 0.12 %
μ = 487.0± 0.6
σ = 77.1± 0.5
0
200
400
600
800
1000
1200
1400
1600



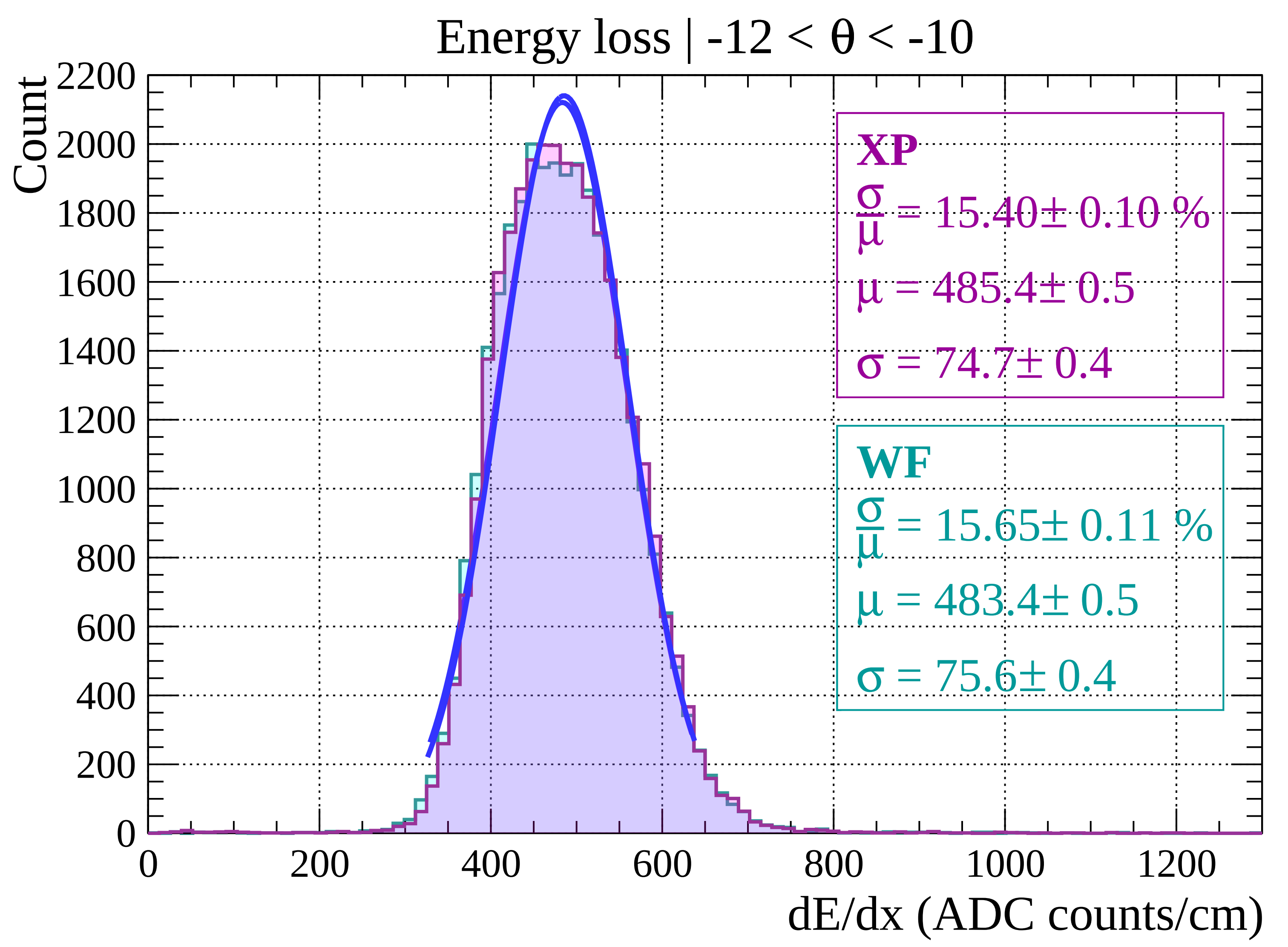
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP
σ/μ = 15.18± 0.10 %
μ = 487.3± 0.5
σ = 74.0± 0.4
WF
σ/μ = 15.76± 0.12 %
μ = 485.3± 0.6
σ = 76.5± 0.5



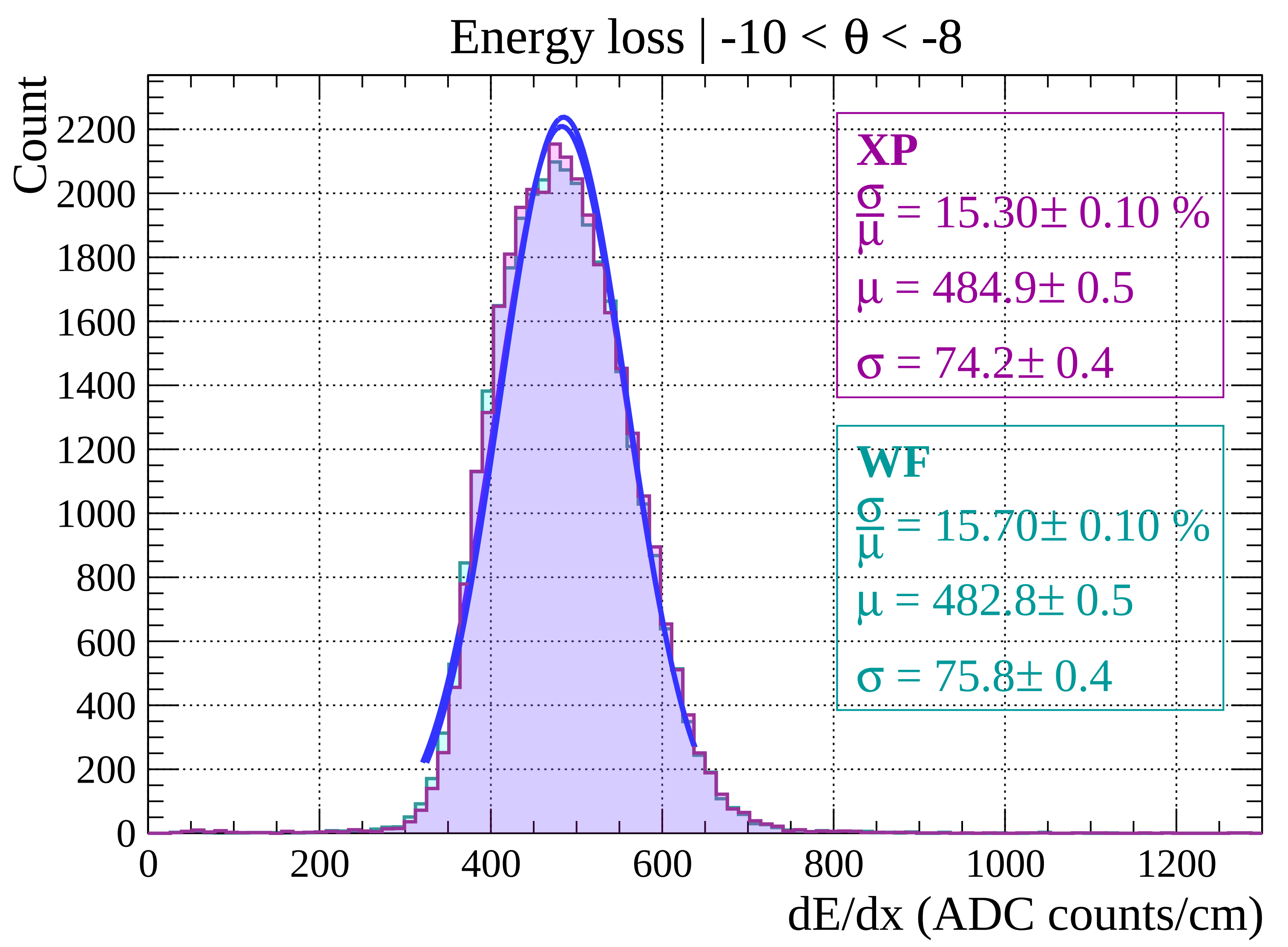
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.49± 0.11 %
μ = 485.2± 0.5
σ = 75.1± 0.5
WF
σ/μ = 15.84± 0.12 %
μ = 483.6± 0.5
σ = 76.6± 0.5

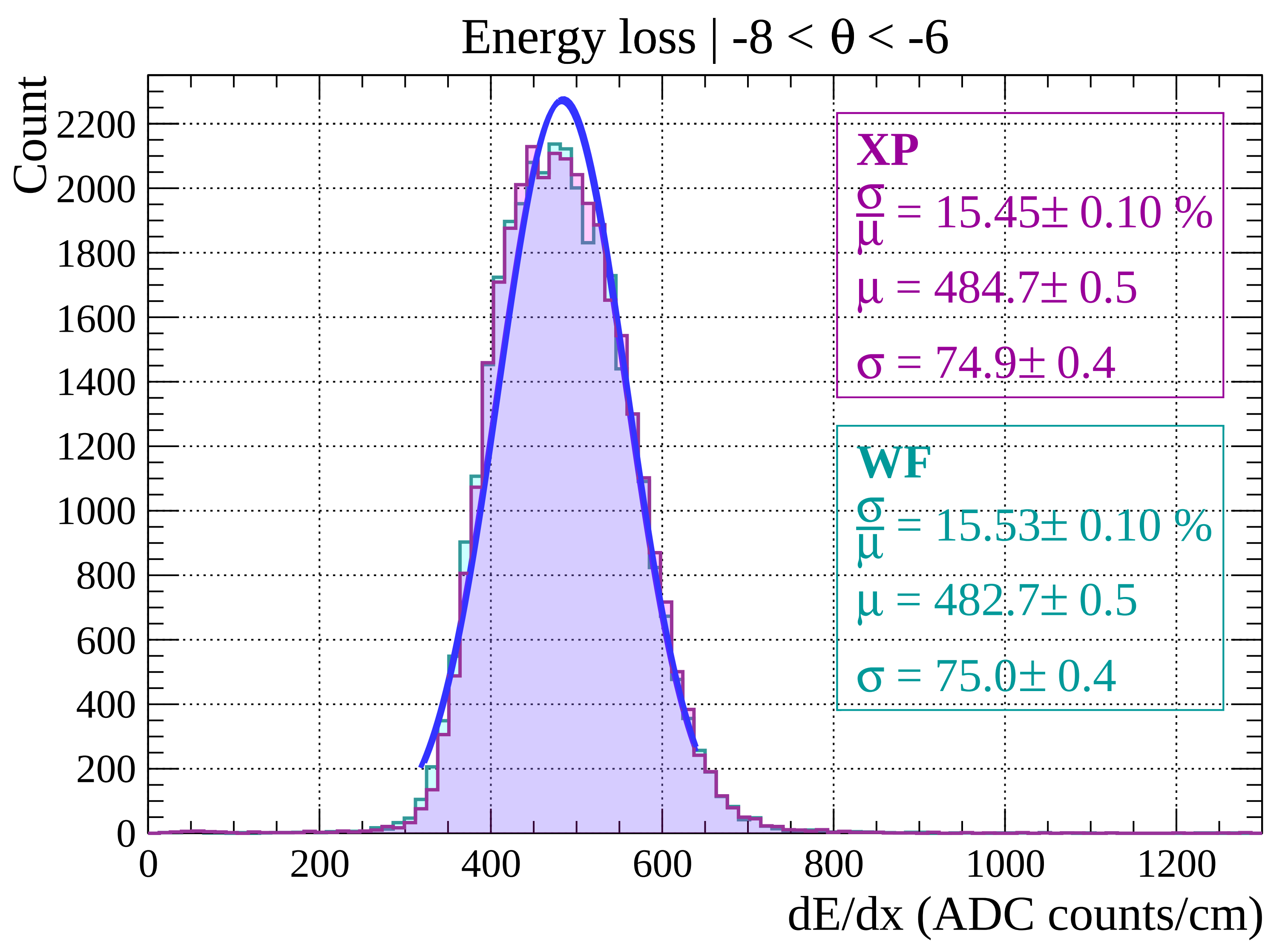


Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.41± 0.11 %
μ = 485.5± 0.5
σ = 74.8± 0.4
WF
σ/μ = 15.76± 0.11 %
μ = 483.5± 0.5
σ = 76.2± 0.5

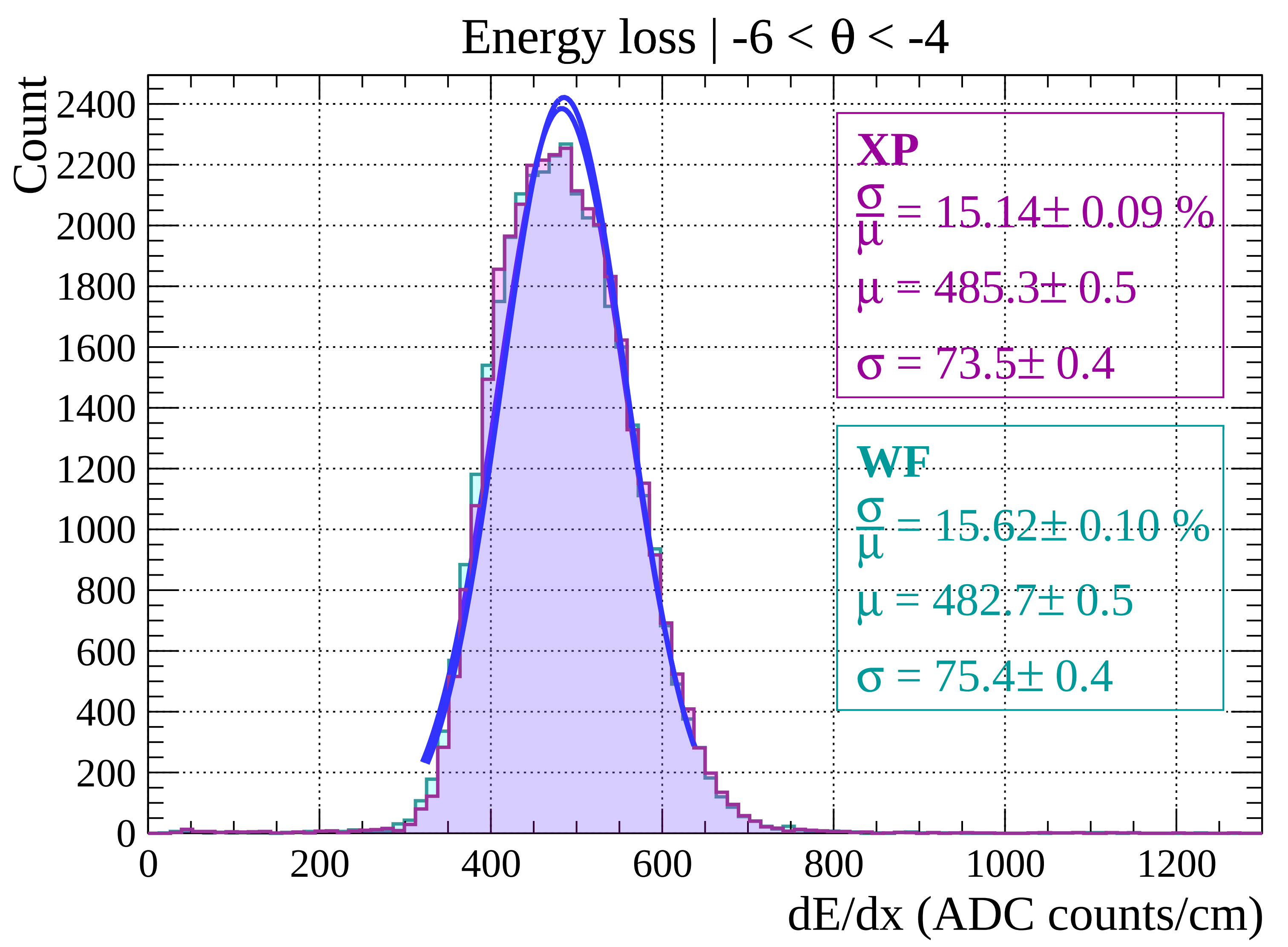


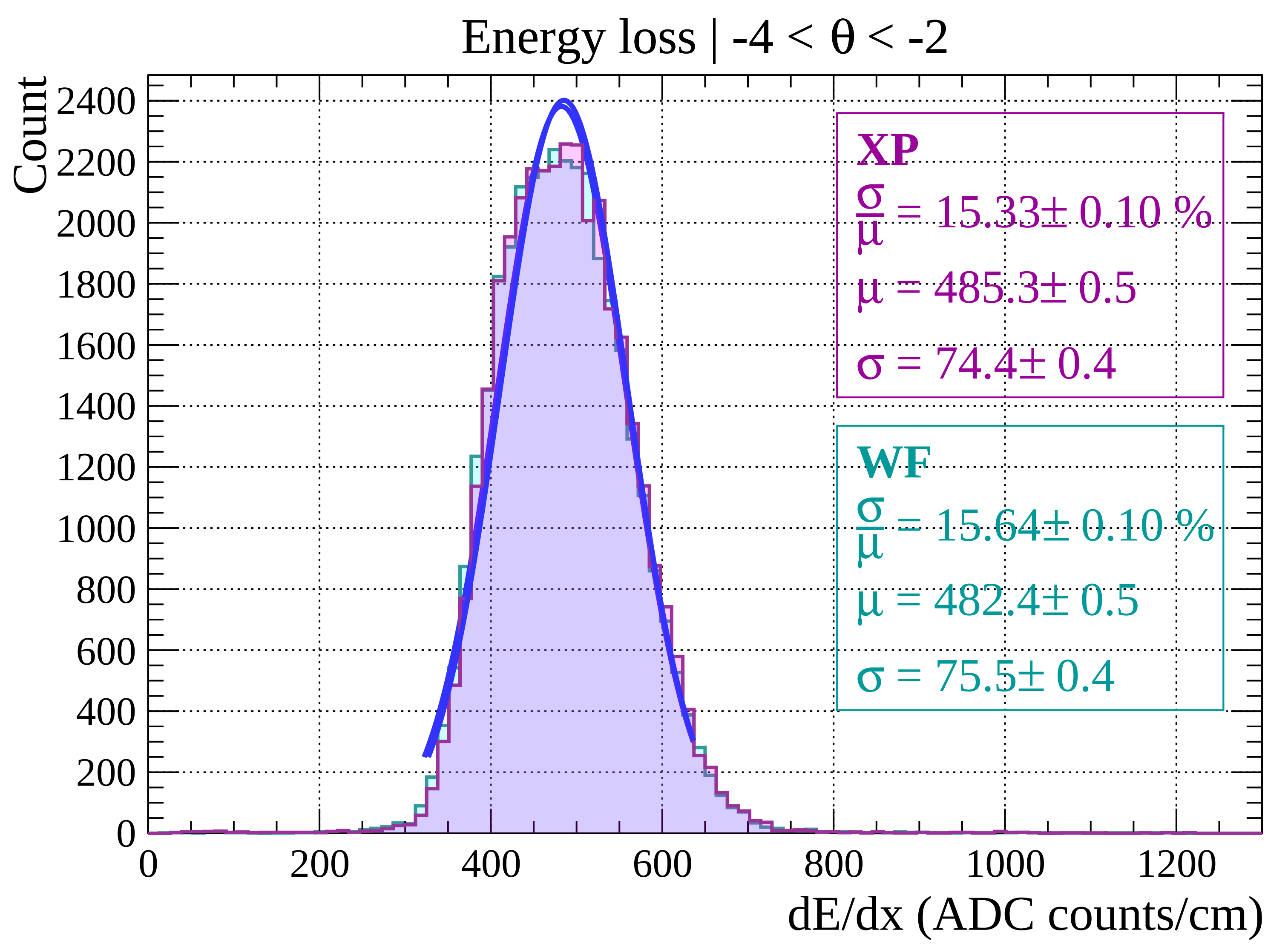
Energy loss | -12 < θ < -10
Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.40± 0.10 %
μ = 485.4± 0.5
σ = 74.7± 0.4
WF
σ/μ = 15.65± 0.11 %
μ = 483.4± 0.5
σ = 75.6± 0.4

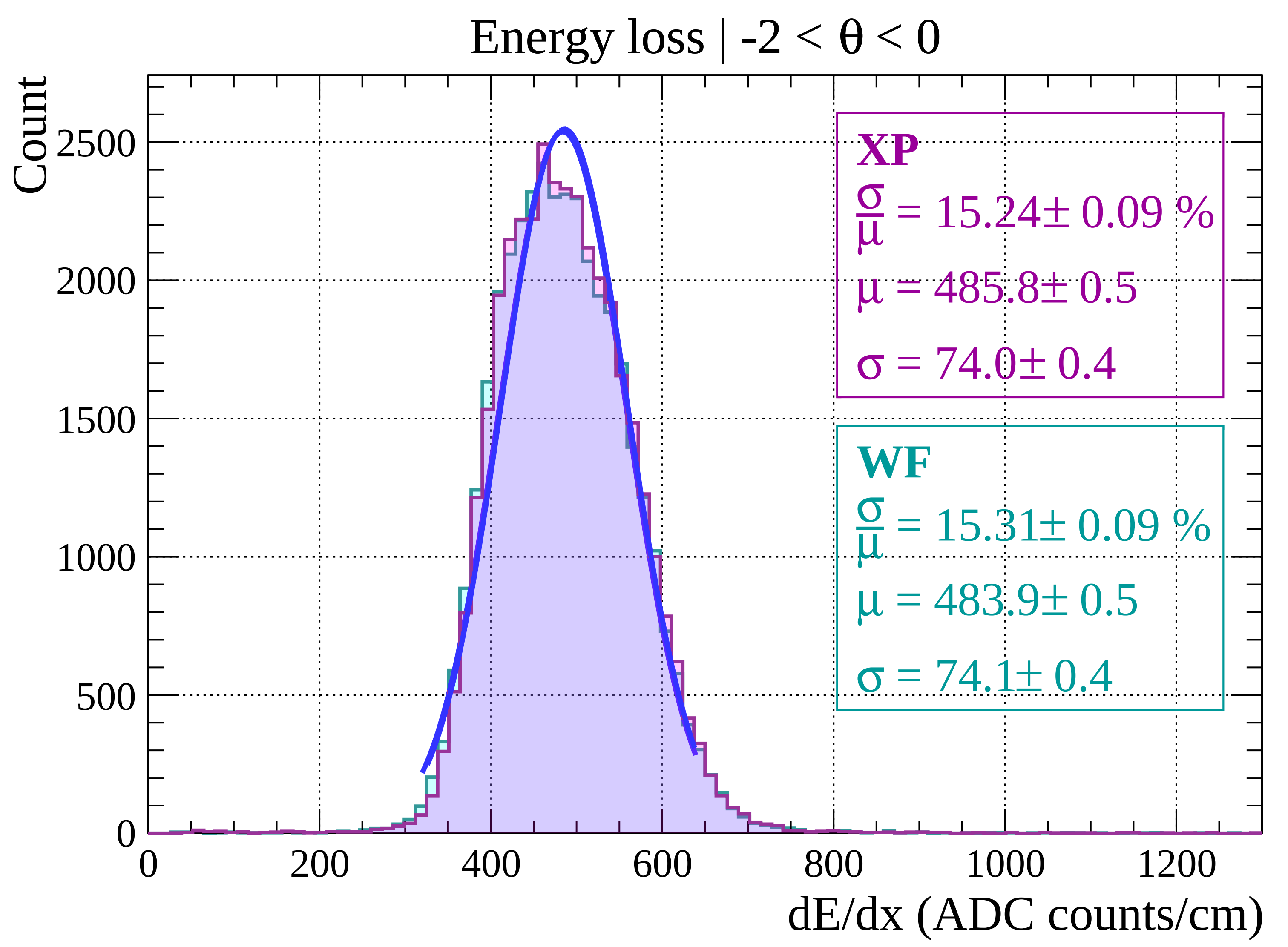

Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 15.30± 0.10 %
μ = 484.9± 0.5
σ = 74.2± 0.4
WF
σ/μ = 15.70± 0.10 %
μ = 482.8± 0.5
σ = 75.8± 0.4



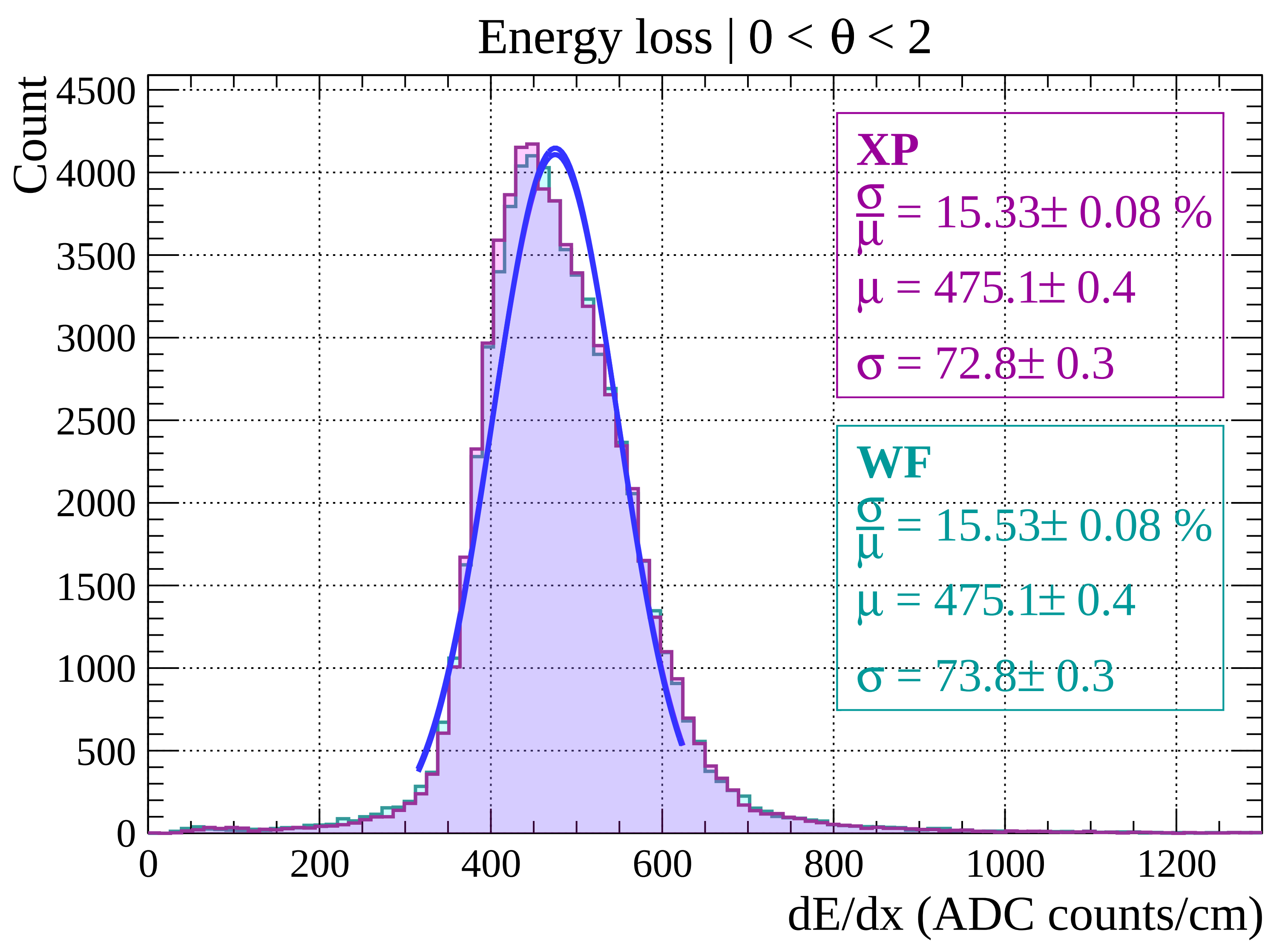
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.45± 0.10 %
μ = 484.7± 0.5
σ = 74.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.53± 0.10 %
μ = 482.7± 0.5
σ = 75.0± 0.4


Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.14± 0.09 %
μ = 485.3± 0.5
σ = 73.5± 0.4
WF
σ/μ = 15.62± 0.10 %
μ = 482.7± 0.5
σ = 75.4± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400

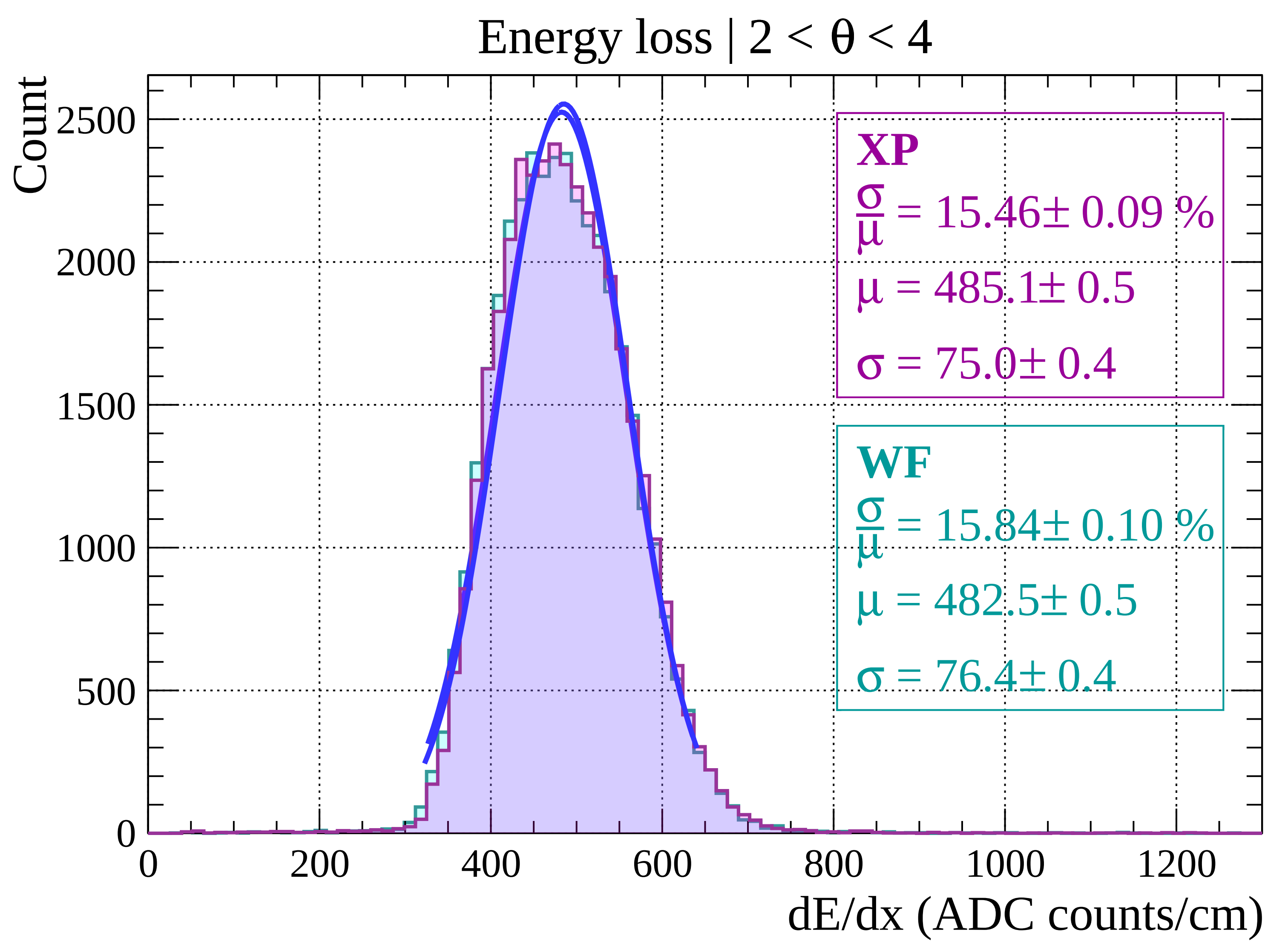

Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.33± 0.10 %
μ = 485.3± 0.5
σ = 74.4± 0.4
WF
σ/μ = 15.64± 0.10 %
μ = 482.4± 0.5
σ = 75.5± 0.4

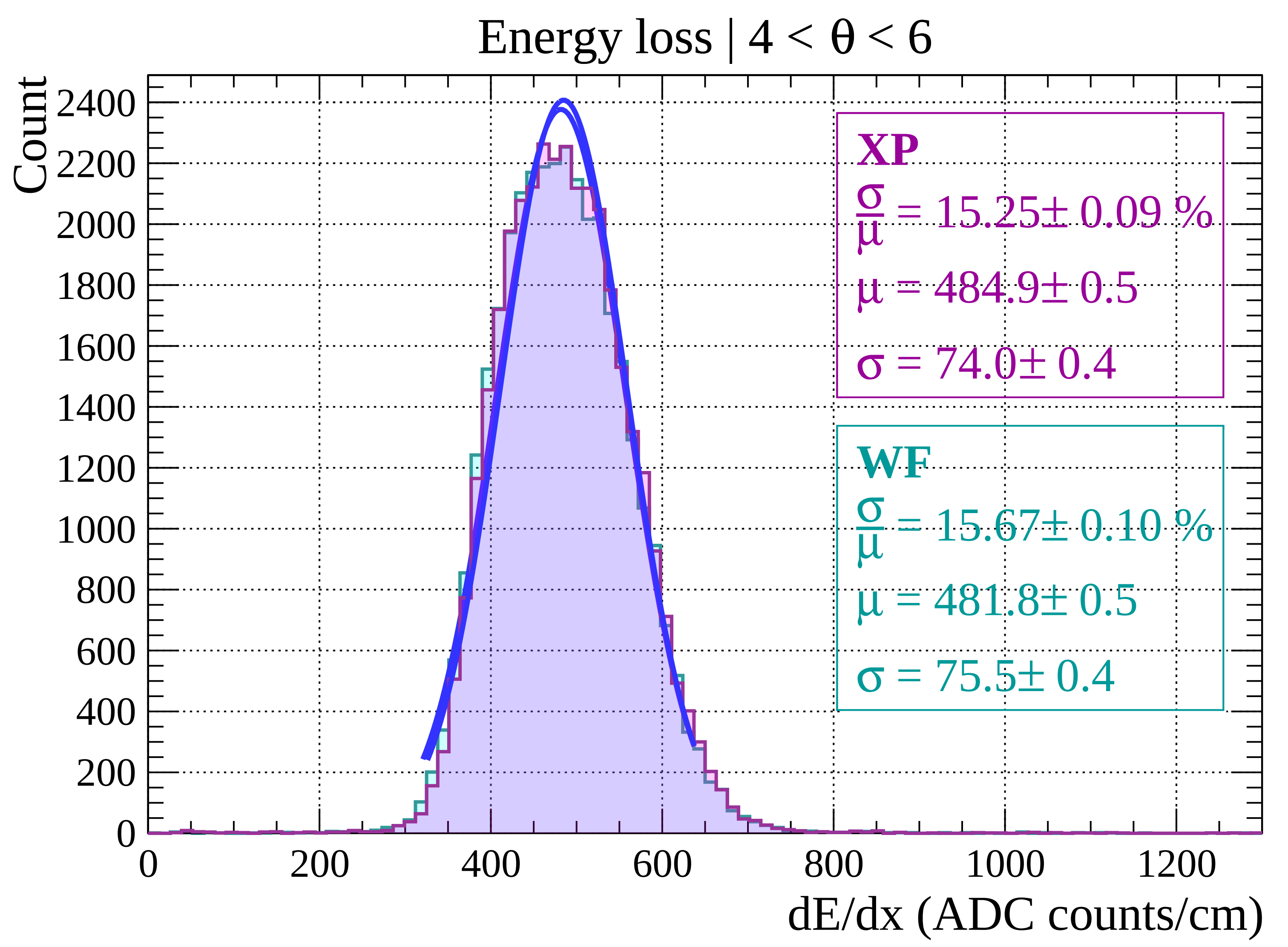


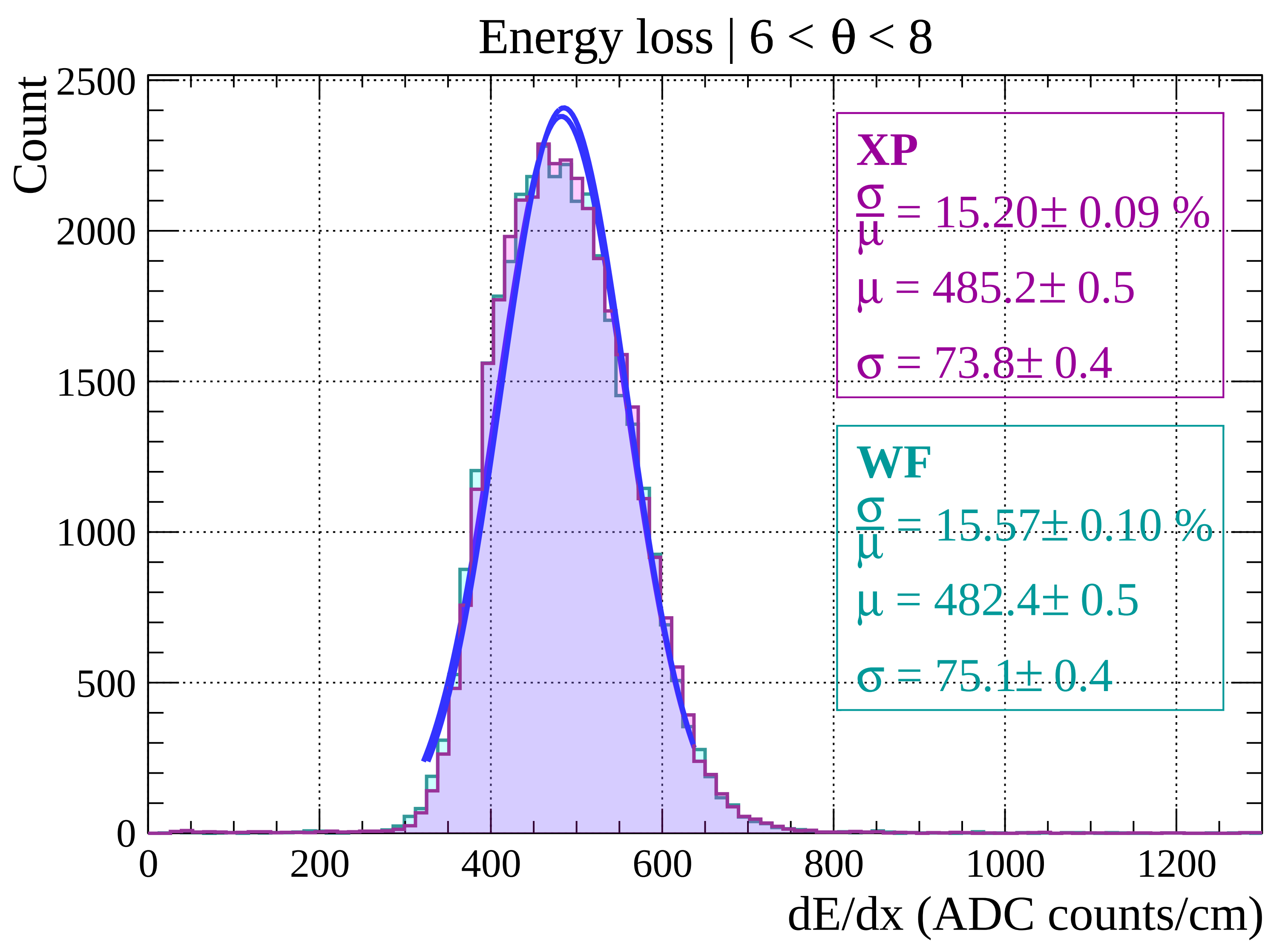
Energy loss | -2 < θ < 0
Count
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.24± 0.09 %
μ = 485.8± 0.5
σ = 74.0± 0.4
WF
σ/μ = 15.31± 0.09 %
μ = 483.9± 0.5
σ = 74.1± 0.4



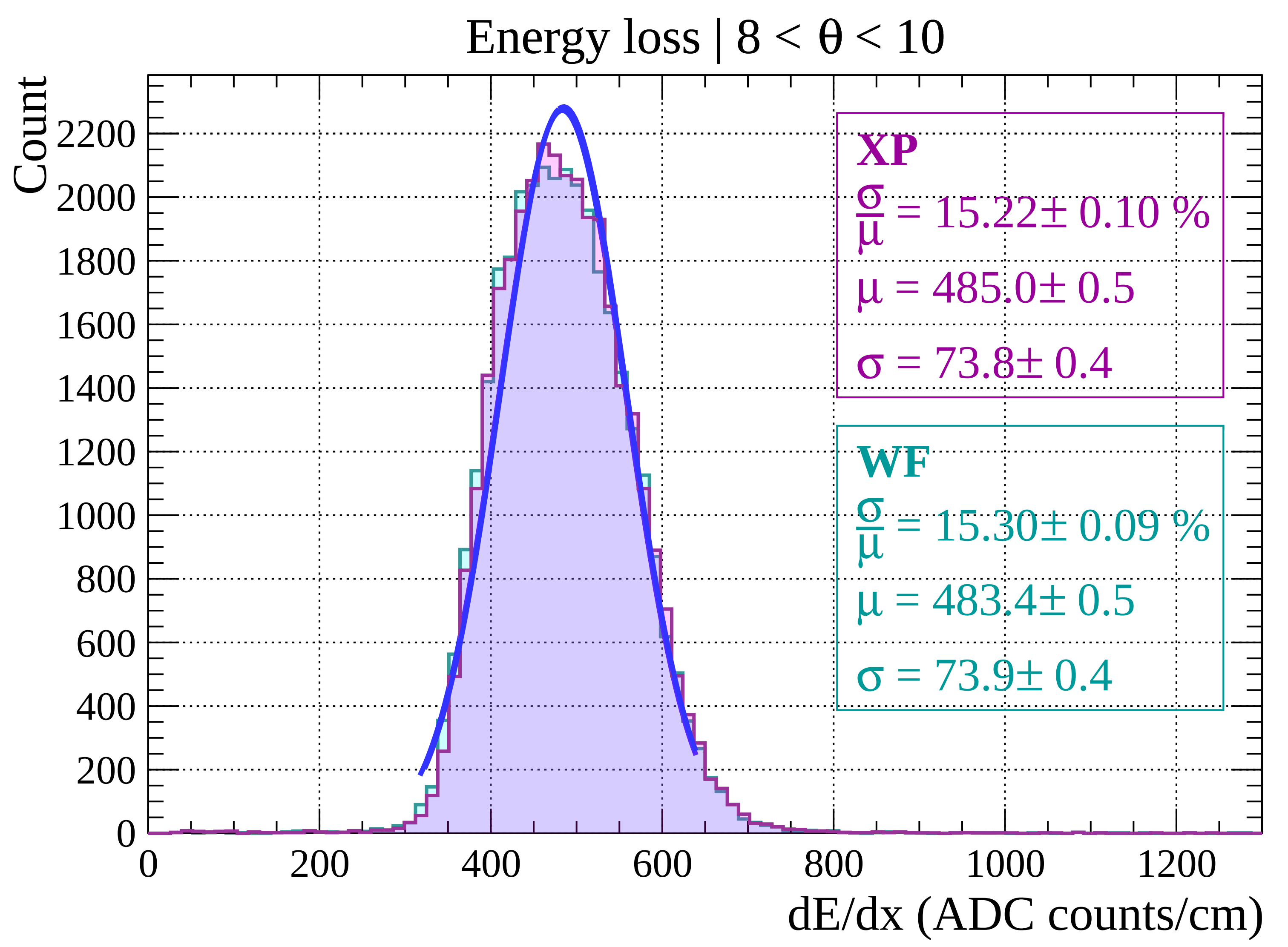
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
3000
3500
4000
4500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.33± 0.08 %
μ = 475.1± 0.4
σ = 72.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 15.53± 0.08 %
μ = 475.1± 0.4
σ = 73.8± 0.3


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.46± 0.09 %
μ = 485.1± 0.5
σ = 75.0± 0.4
WF
σ/μ = 15.84± 0.10 %
μ = 482.5± 0.5
σ = 76.4± 0.4
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200

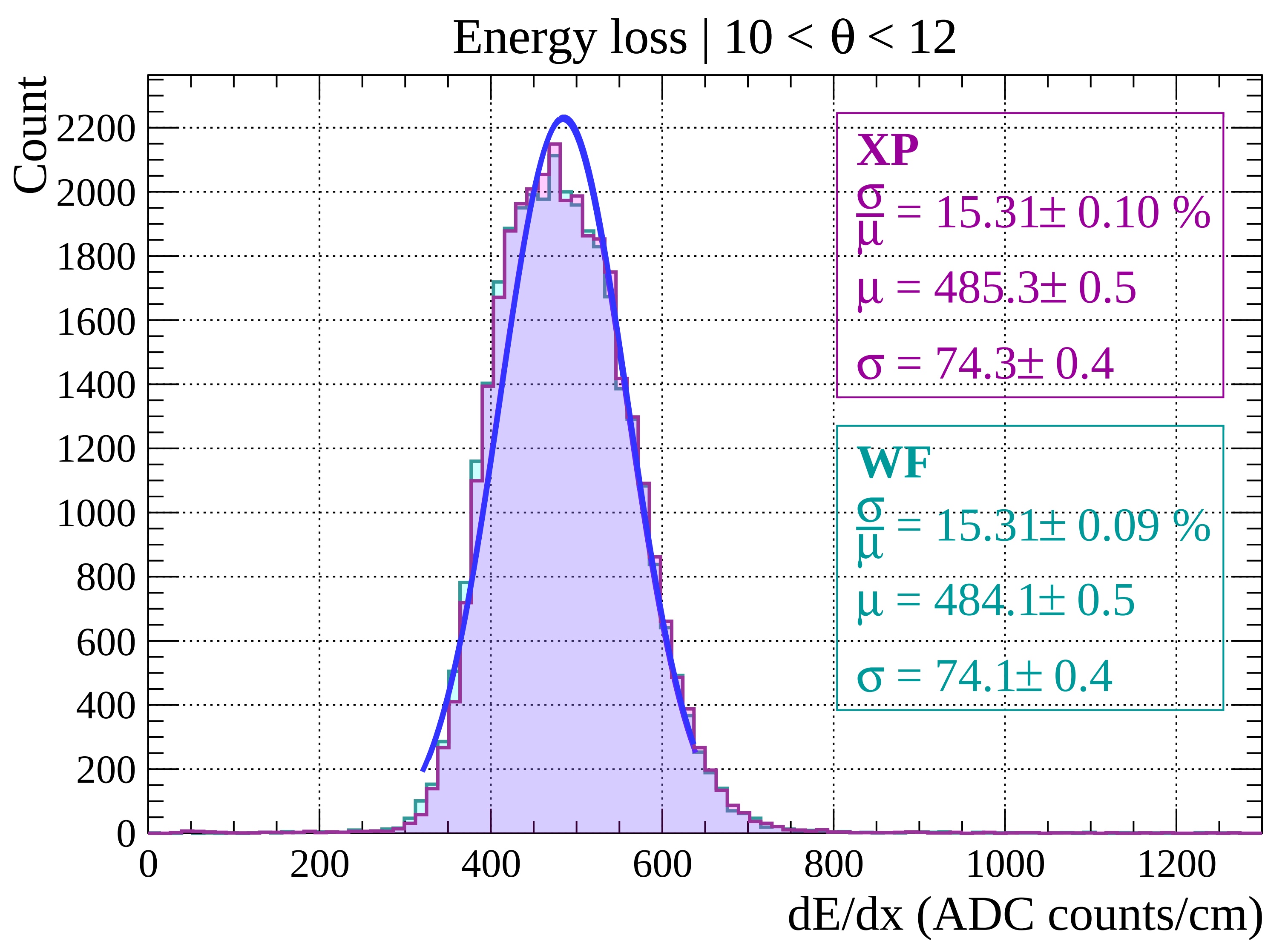

Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
0
200
400
600
800
1000
1200
XP
σ/μ = 15.25± 0.09 %
μ = 484.9± 0.5
σ = 74.0± 0.4
WF
σ/μ = 15.67± 0.10 %
μ = 481.8± 0.5
σ = 75.5± 0.4



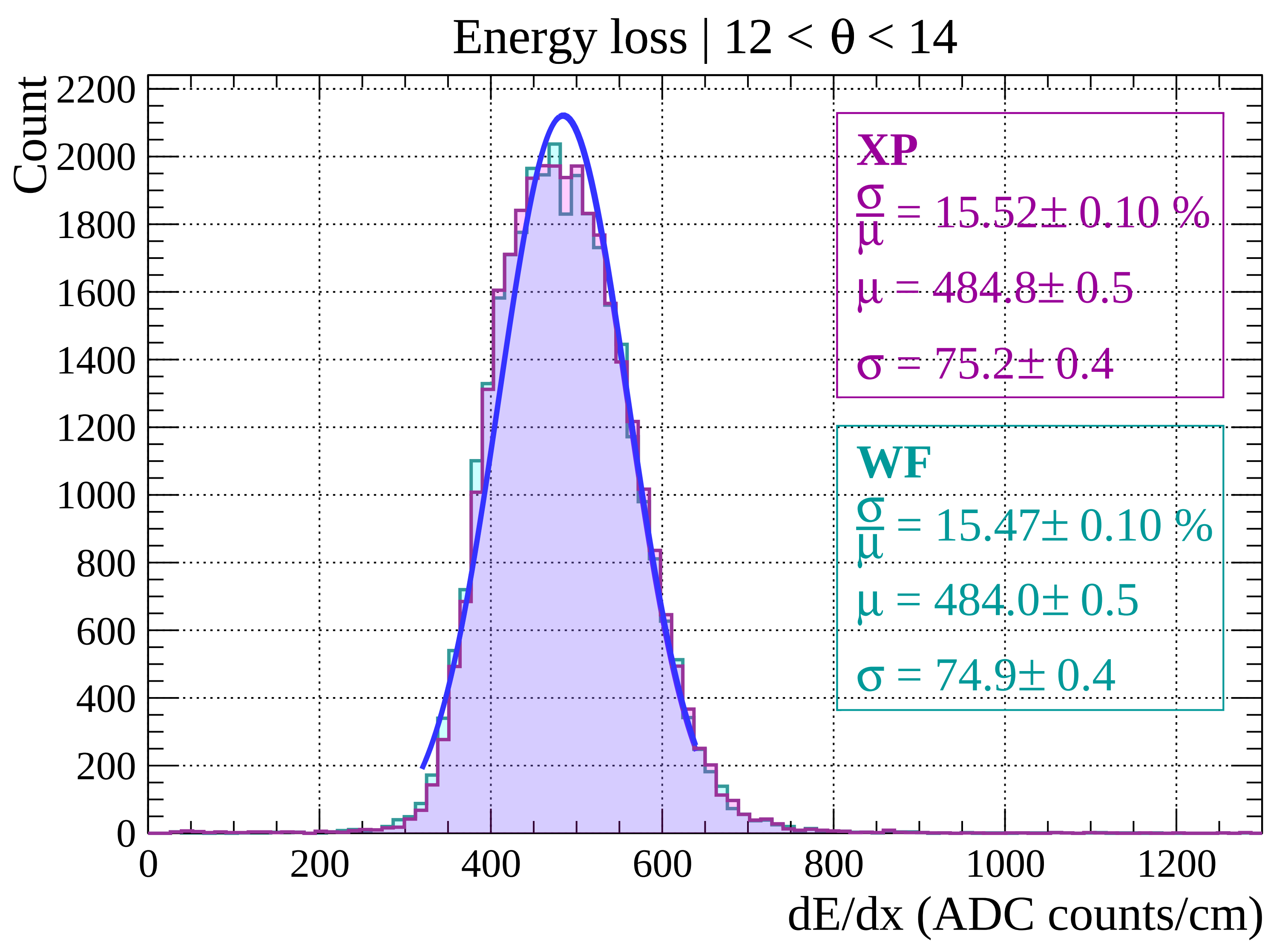
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
500
1000
1500
2000
2500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.20± 0.09 %
μ = 485.2± 0.5
σ = 73.8± 0.4
WF
σ/μ = 15.57± 0.10 %
μ = 482.4± 0.5
σ = 75.1± 0.4

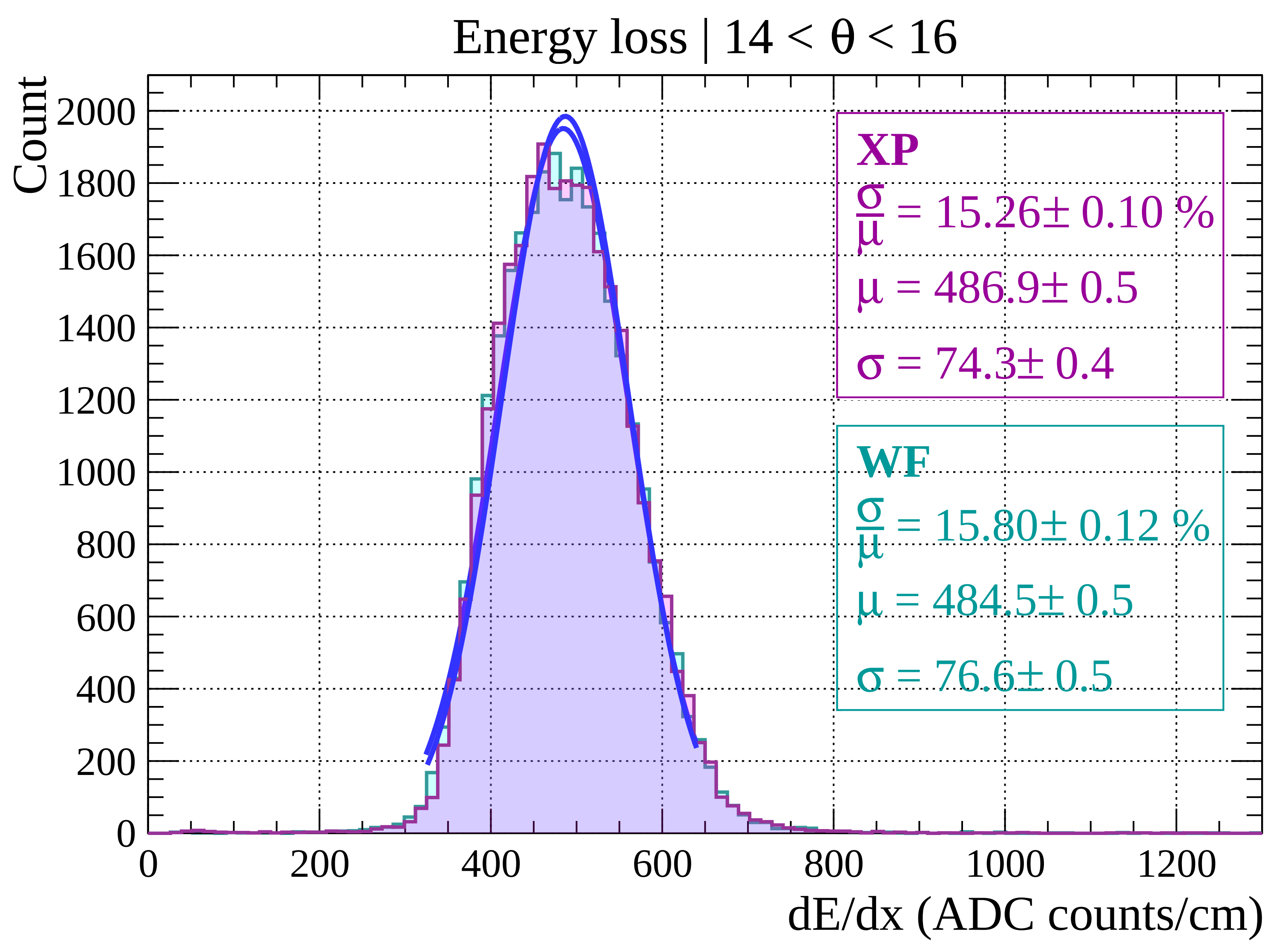


Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 15.22± 0.10 %
μ = 485.0± 0.5
σ = 73.8± 0.4
WF
σ/μ = 15.30± 0.09 %
μ = 483.4± 0.5
σ = 73.9± 0.4

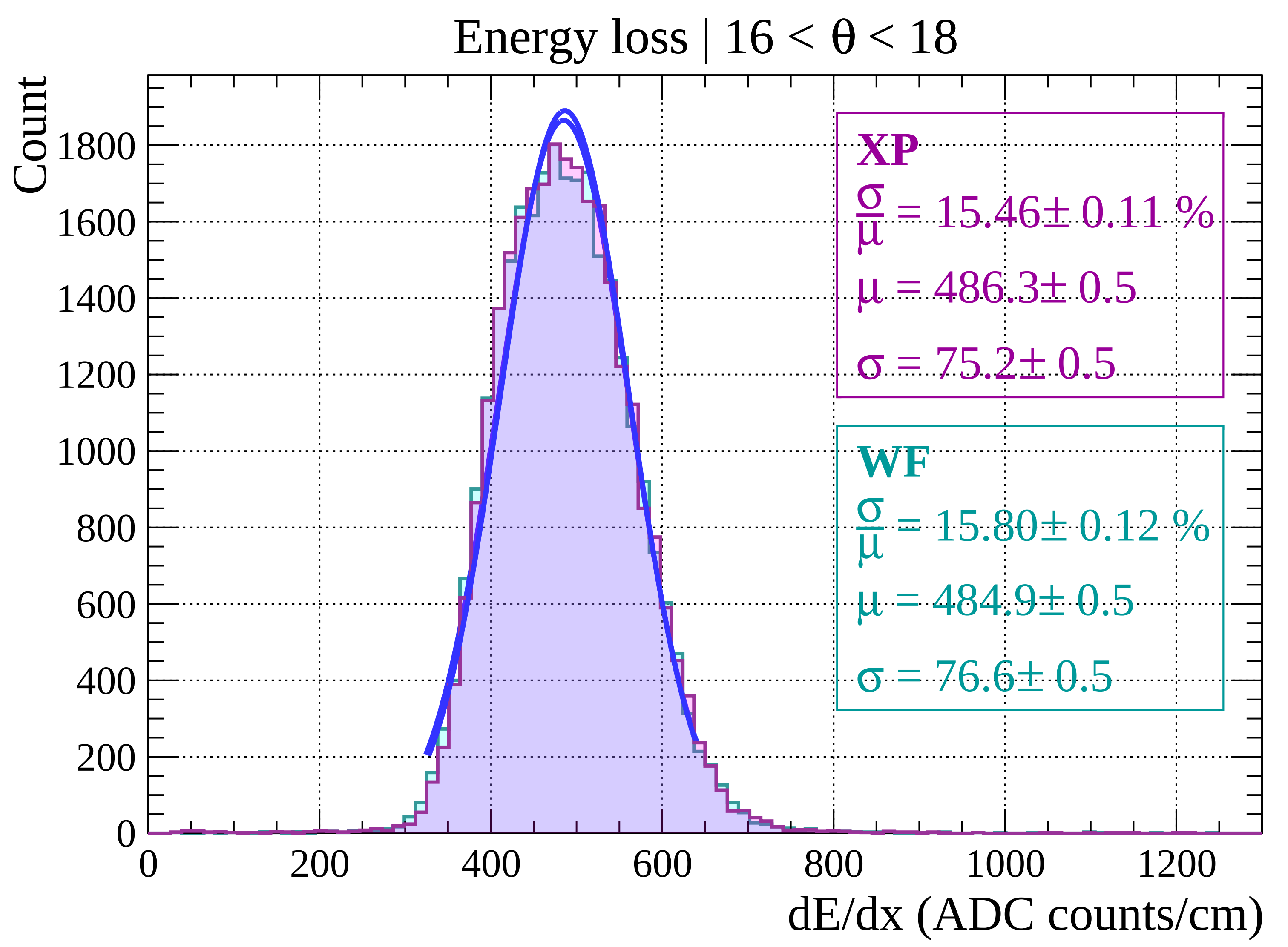


Energy loss | 10 < θ < 12
Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.31± 0.10 %
μ = 485.3± 0.5
σ = 74.3± 0.4
WF
σ/μ = 15.31± 0.09 %
μ = 484.1± 0.5
σ = 74.1± 0.4

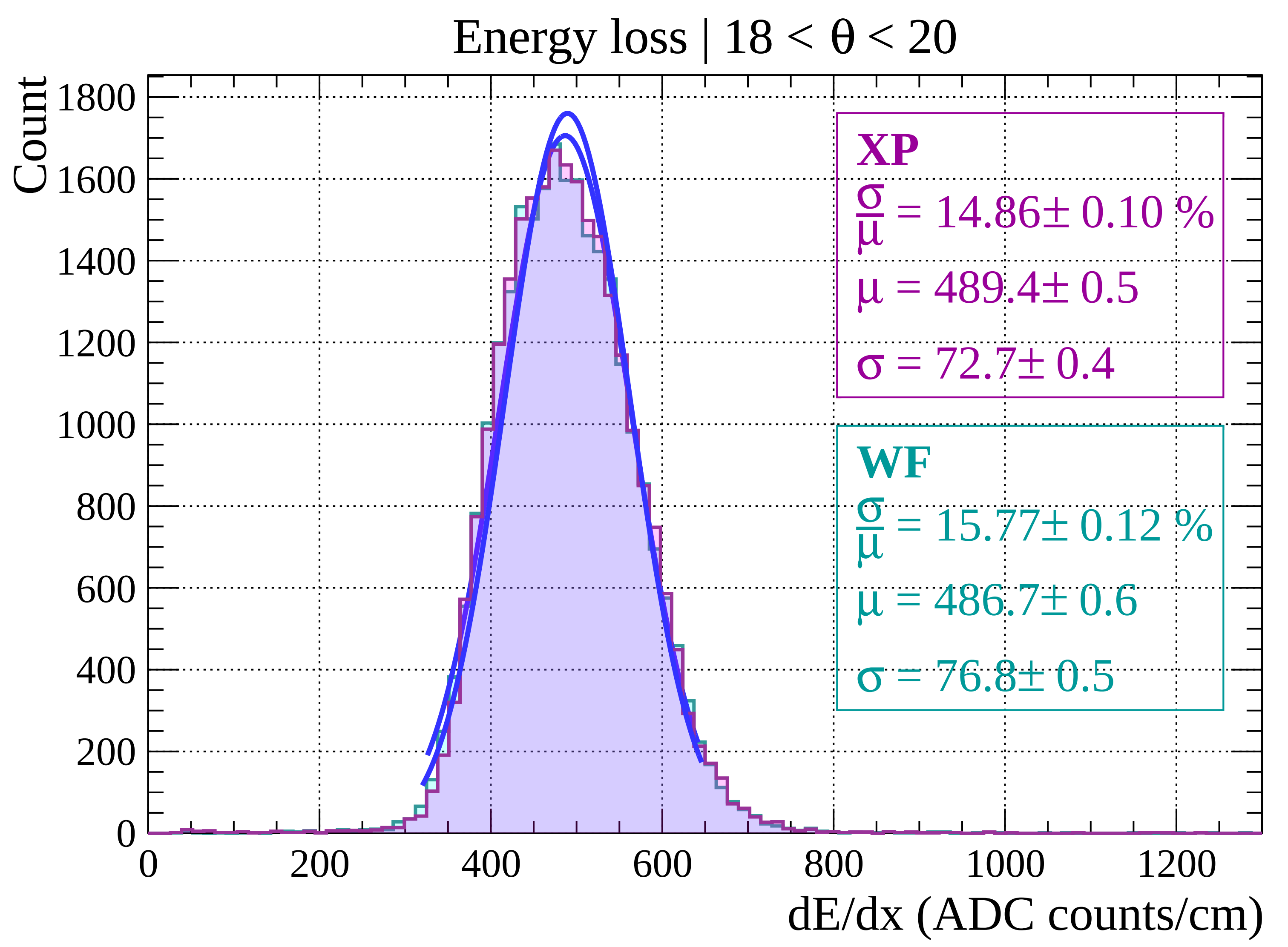

Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.52± 0.10 %
μ = 484.8± 0.5
σ = 75.2± 0.4
WF
σ/μ = 15.47± 0.10 %
μ = 484.0± 0.5
σ = 74.9± 0.4



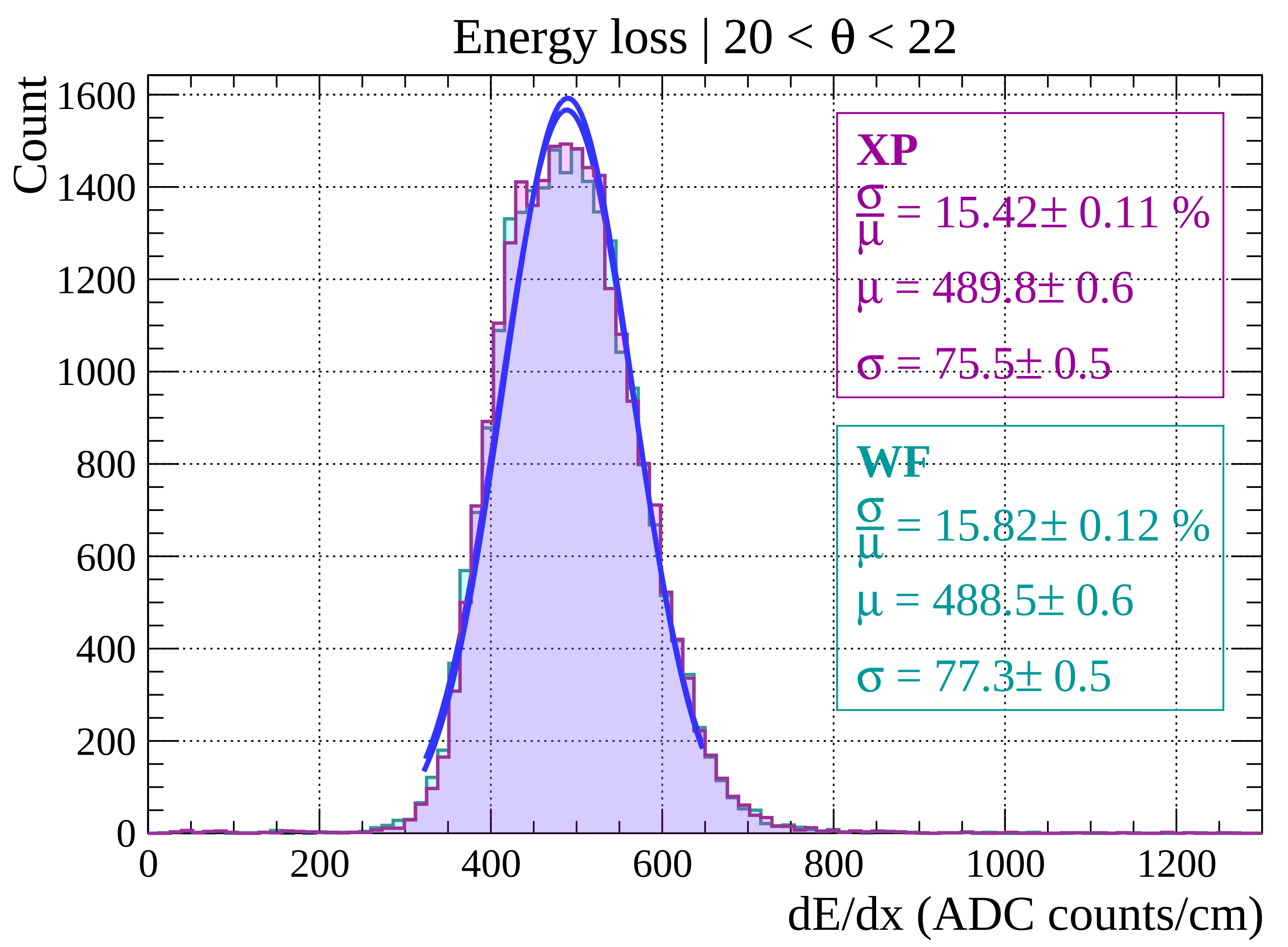
Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.26± 0.10 %
μ = 486.9± 0.5
σ = 74.3± 0.4
WF
σ/μ = 15.80± 0.12 %
μ = 484.5± 0.5
σ = 76.6± 0.5



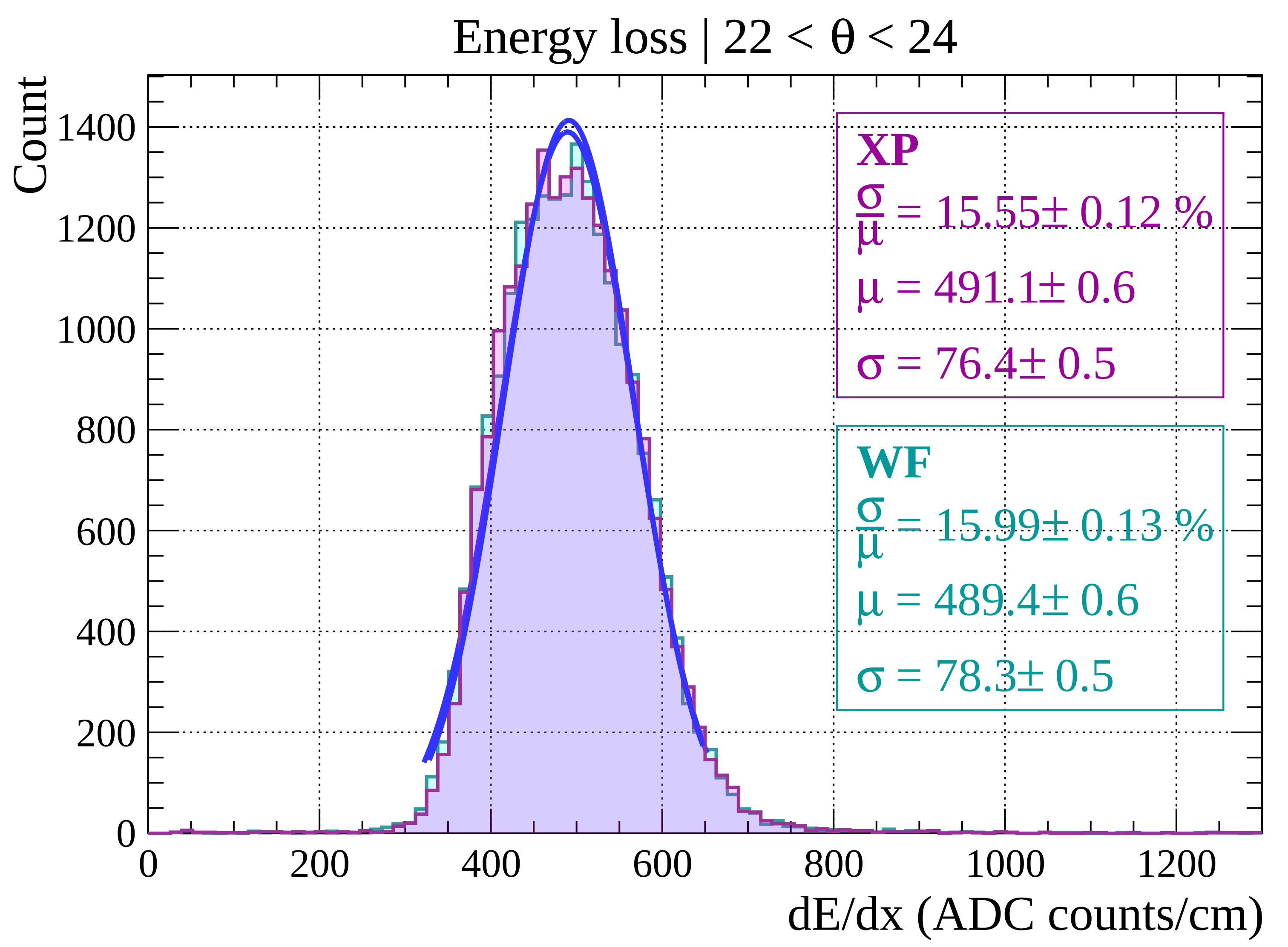
Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.46± 0.11 %
μ = 486.3± 0.5
σ = 75.2± 0.5
WF
σ/μ = 15.80± 0.12 %
μ = 484.9± 0.5
σ = 76.6± 0.5



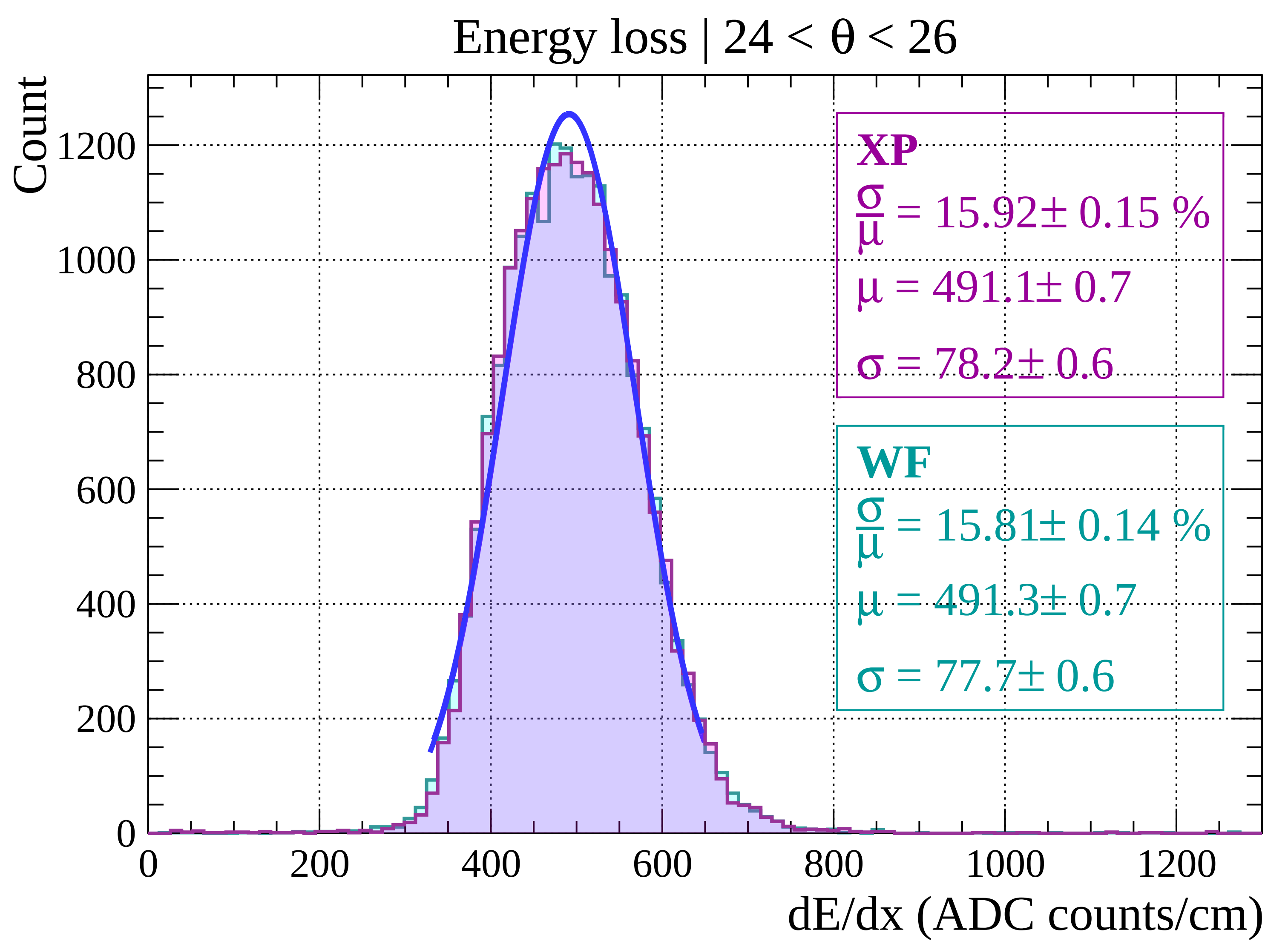
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 14.86± 0.10 %
μ = 489.4± 0.5
σ = 72.7± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.77± 0.12 %
μ = 486.7± 0.6
σ = 76.8± 0.5
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200



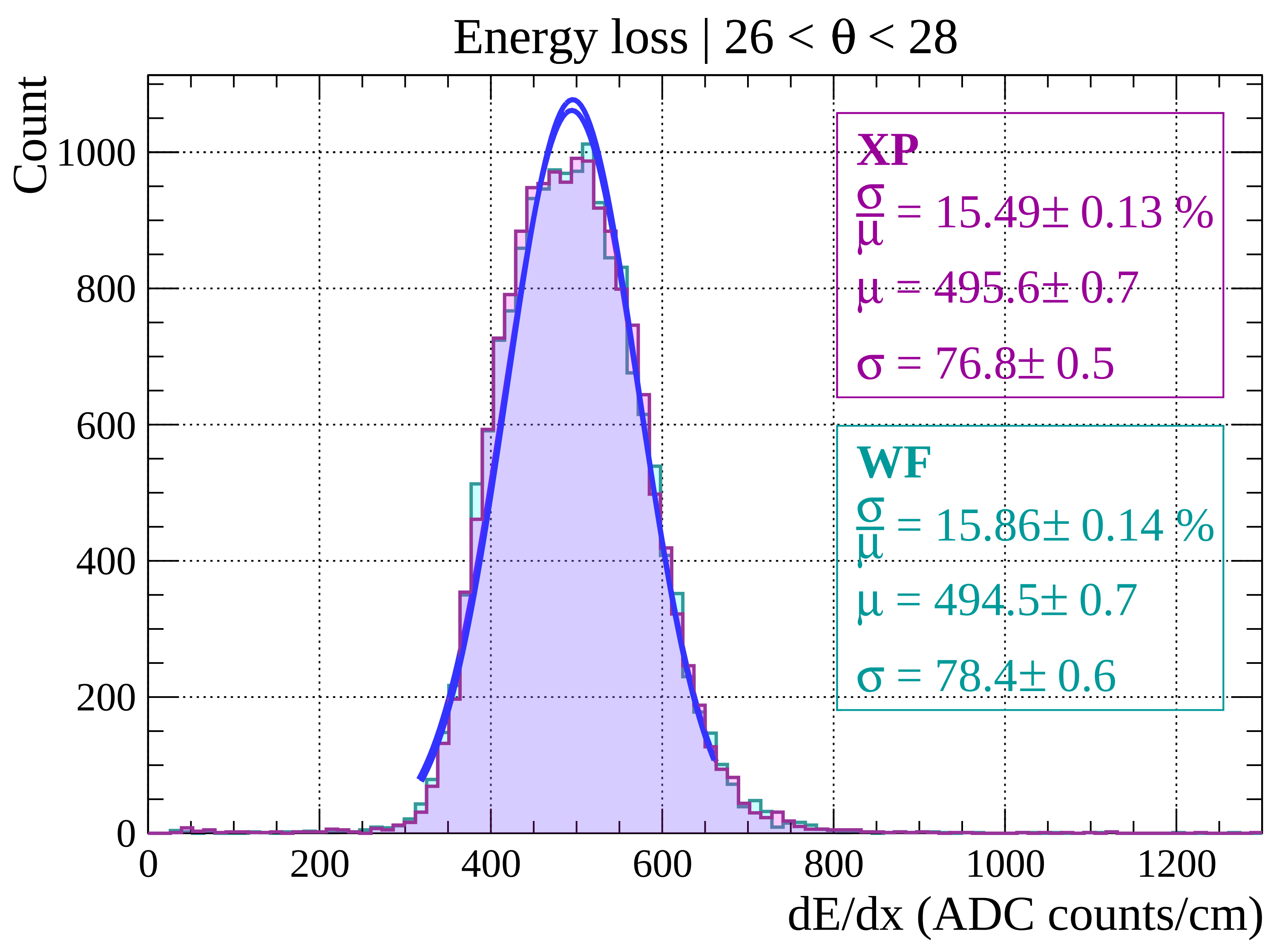
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.42± 0.11 %
μ = 489.8± 0.6
σ = 75.5± 0.5
WF
σ/μ = 15.82± 0.12 %
μ = 488.5± 0.6
σ = 77.3± 0.5
0
200
400
600
800
1000
1200
1400
1600



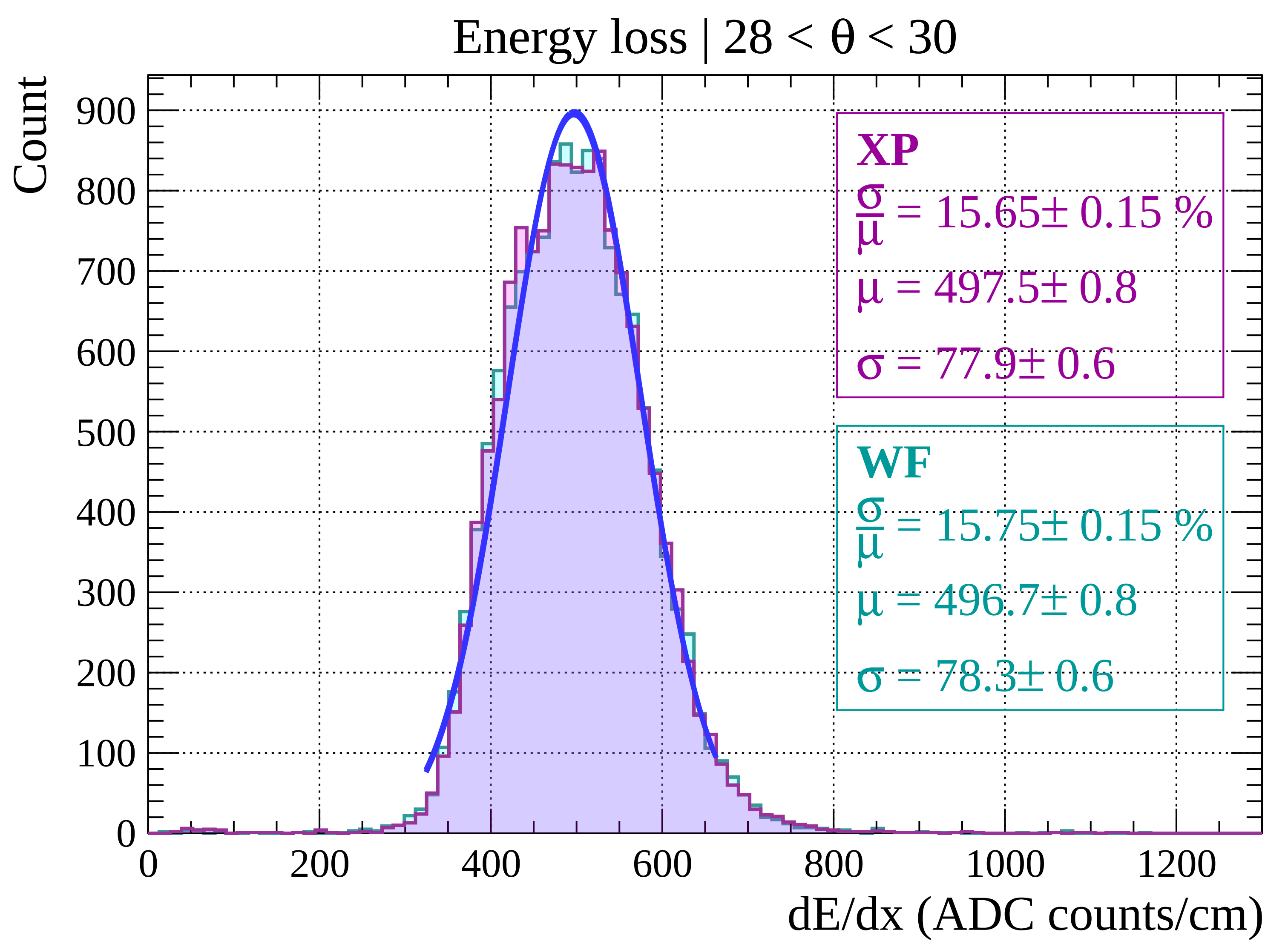
Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.55± 0.12 %
μ = 491.1± 0.6
σ = 76.4± 0.5
WF
σ/μ = 15.99± 0.13 %
μ = 489.4± 0.6
σ = 78.3± 0.5
0
200
400
600
800
1000
1200
1400



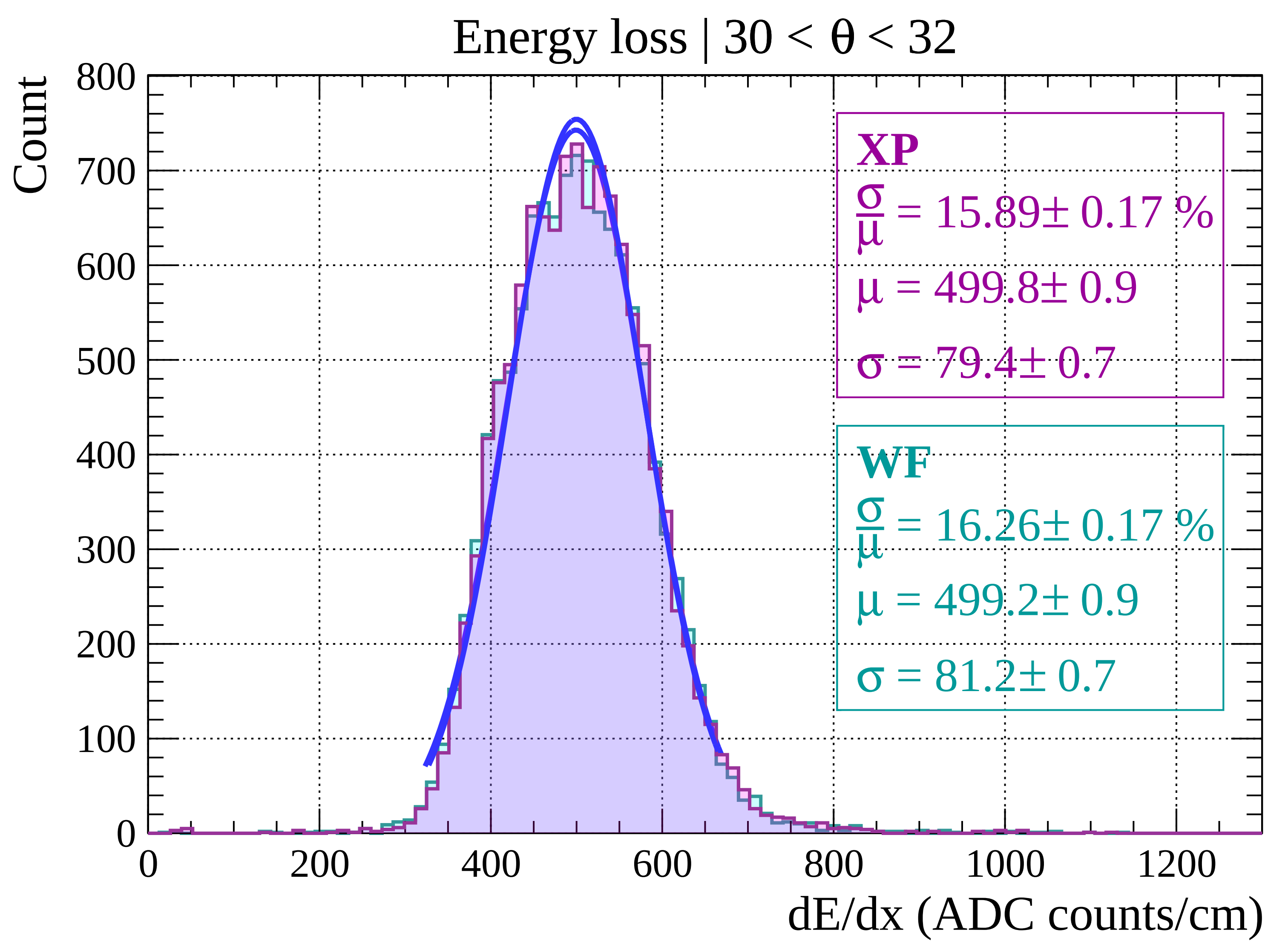
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.92± 0.15 %
μ = 491.1± 0.7
σ = 78.2± 0.6
WF
σ/μ = 15.81± 0.14 %
μ = 491.3± 0.7
σ = 77.7± 0.6



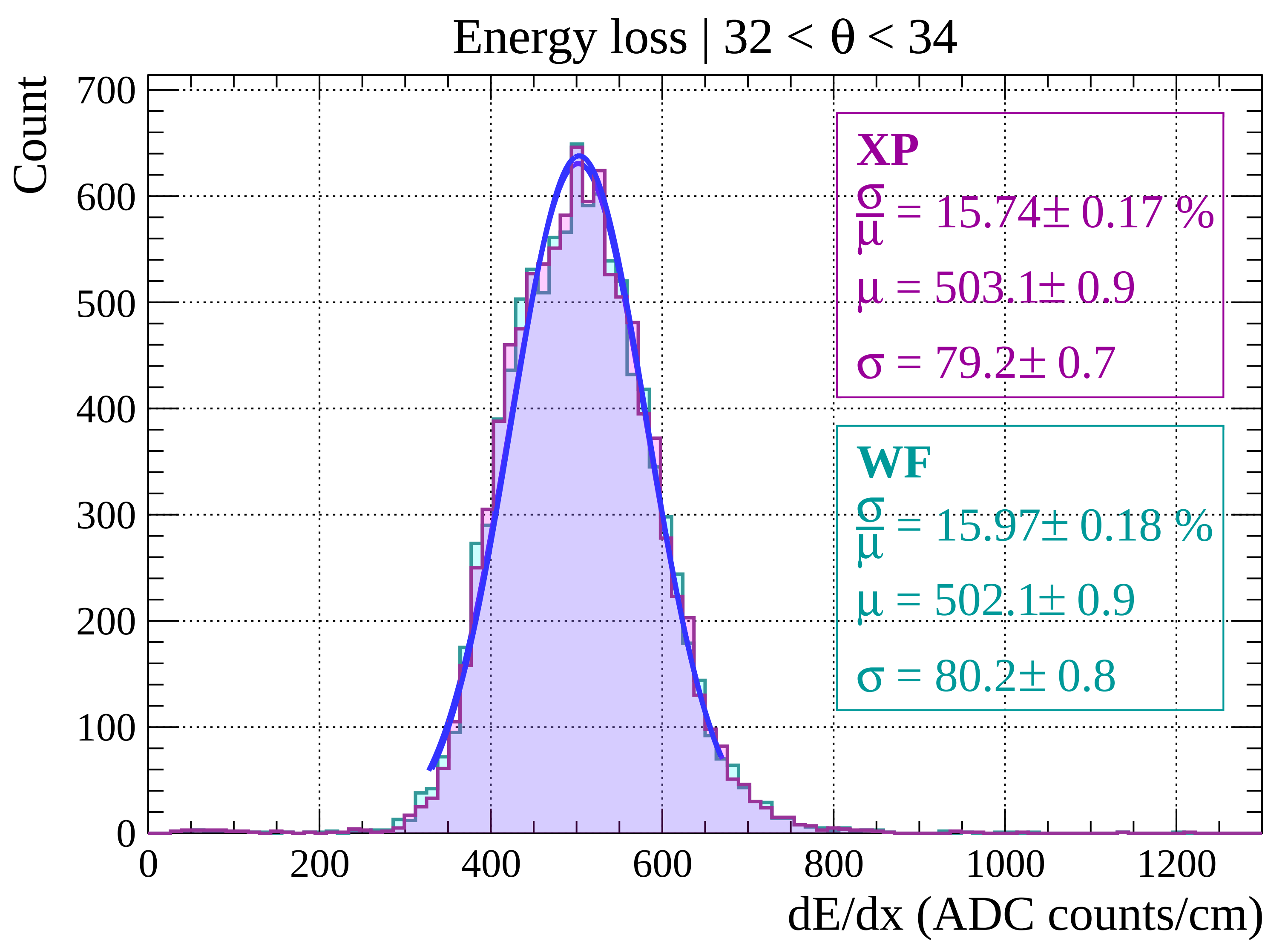
Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.49± 0.13 %
μ = 495.6± 0.7
σ = 76.8± 0.5
WF
$\frac{\sigma}{\mu}$ = 15.86± 0.14 %
μ = 494.5± 0.7
σ = 78.4± 0.6

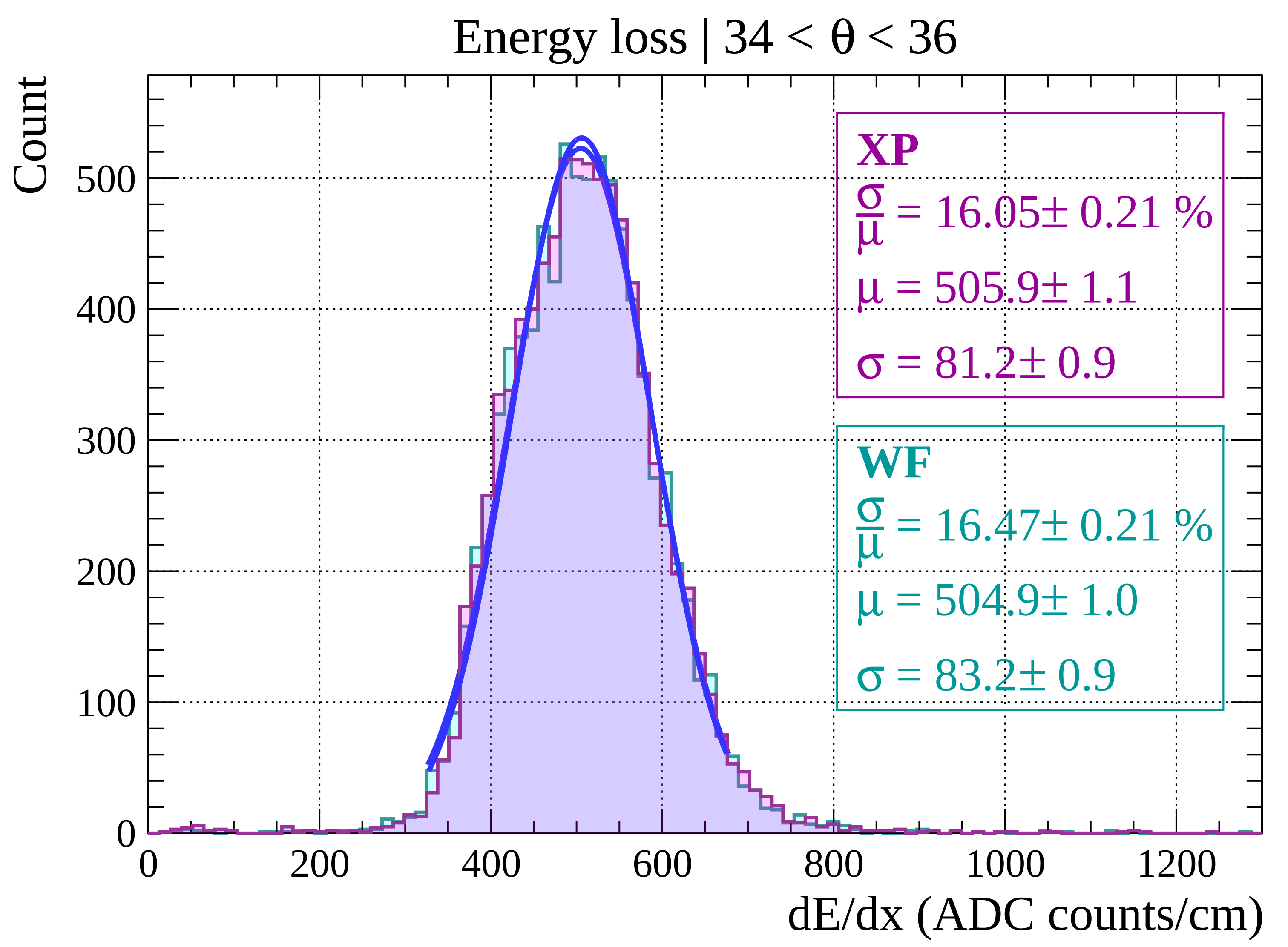


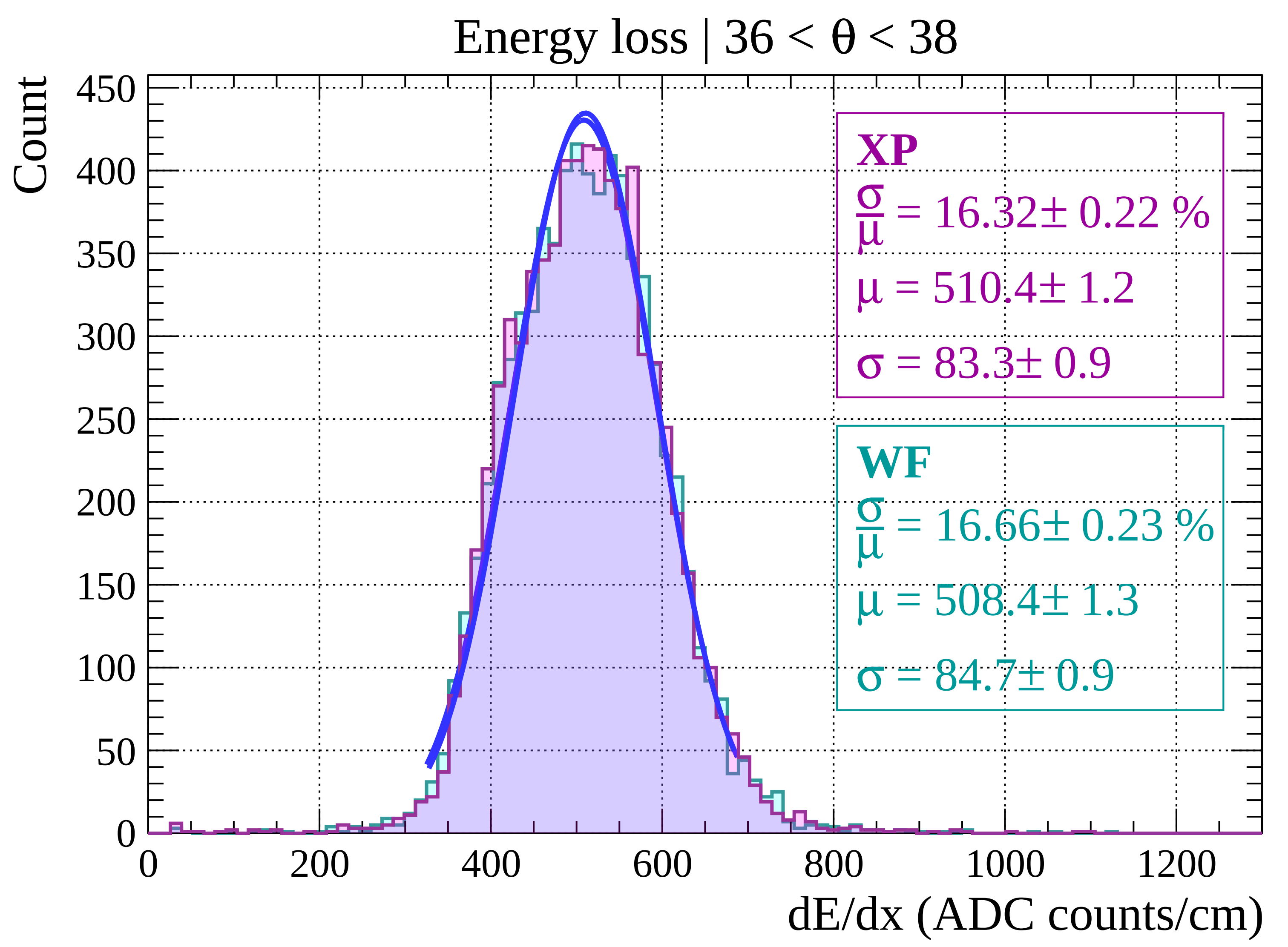
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.65± 0.15 %
μ = 497.5± 0.8
σ = 77.9± 0.6
WF
σ/μ = 15.75± 0.15 %
μ = 496.7± 0.8
σ = 78.3± 0.6



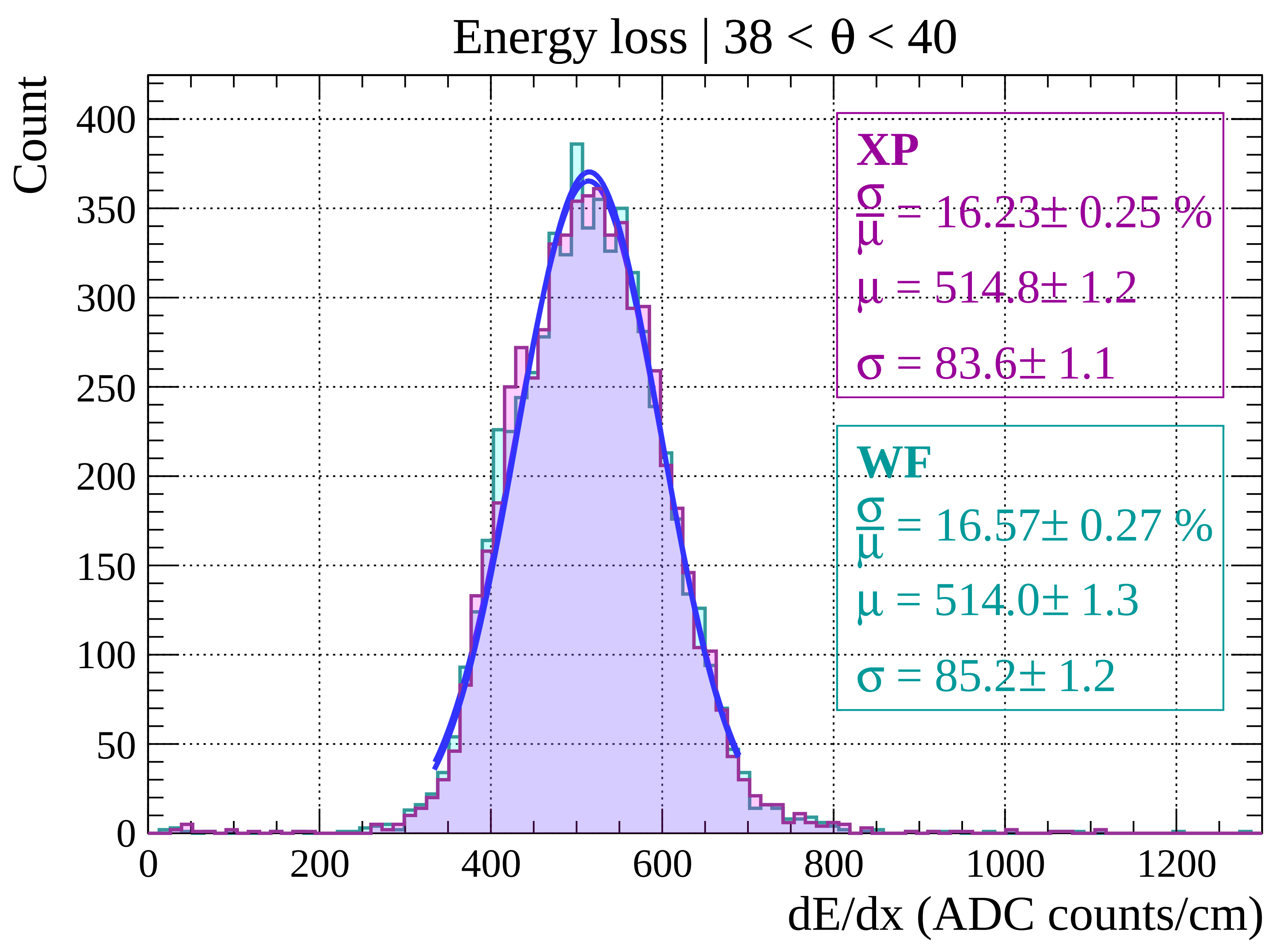
Energy loss | 30 < θ < 32
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.89± 0.17 %
μ = 499.8± 0.9
σ = 79.4± 0.7
WF
σ/μ = 16.26± 0.17 %
μ = 499.2± 0.9
σ = 81.2± 0.7


Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
σ/μ = 15.74± 0.17 %
μ = 503.1± 0.9
σ = 79.2± 0.7
WF
σ/μ = 15.97± 0.18 %
μ = 502.1± 0.9
σ = 80.2± 0.8

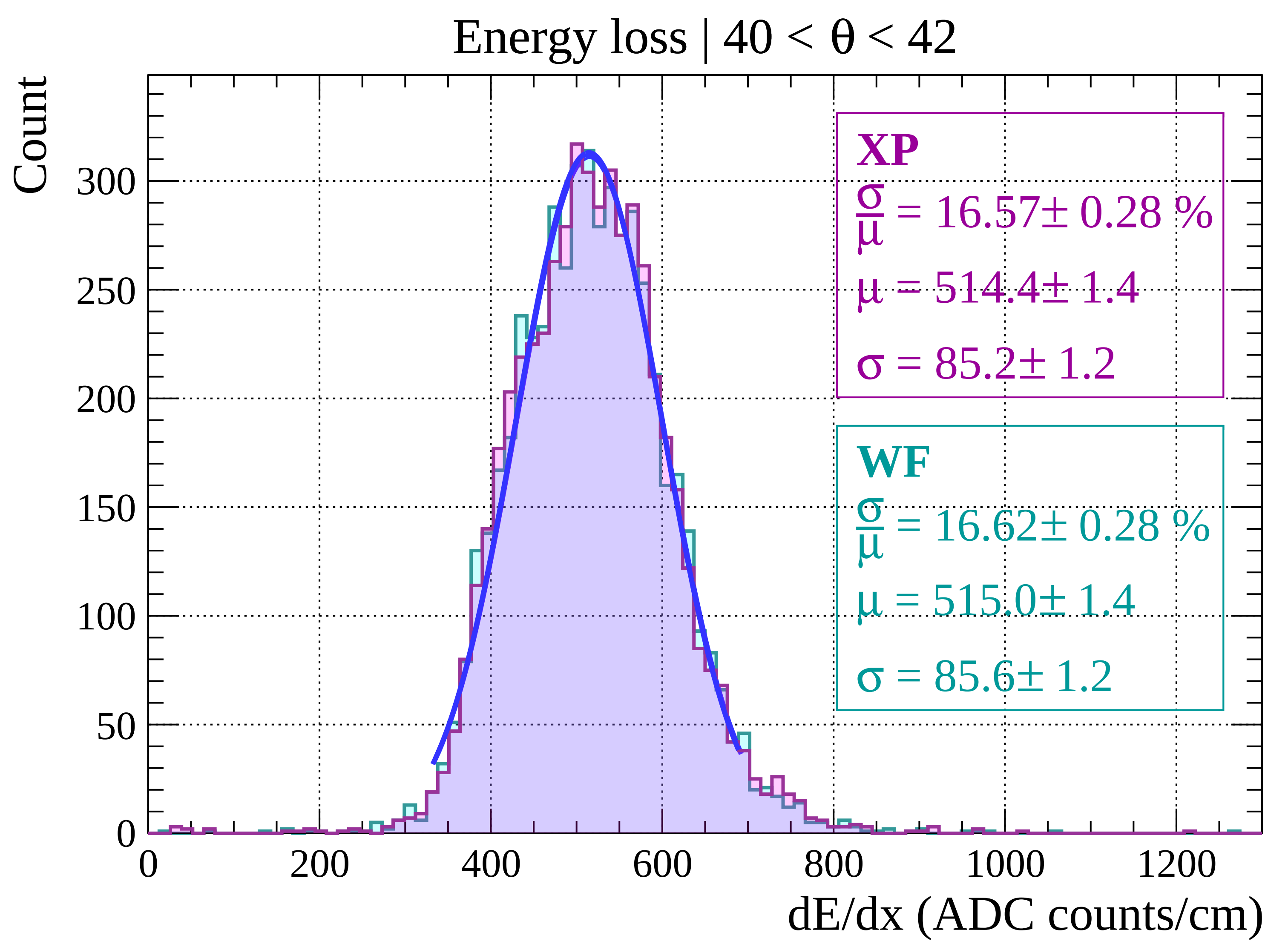

Energy loss | 34 < θ < 36
Count
XP
σ/μ = 16.05± 0.21 %
μ = 505.9± 1.1
σ = 81.2± 0.9
WF
σ/μ = 16.47± 0.21 %
μ = 504.9± 1.0
σ = 83.2± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

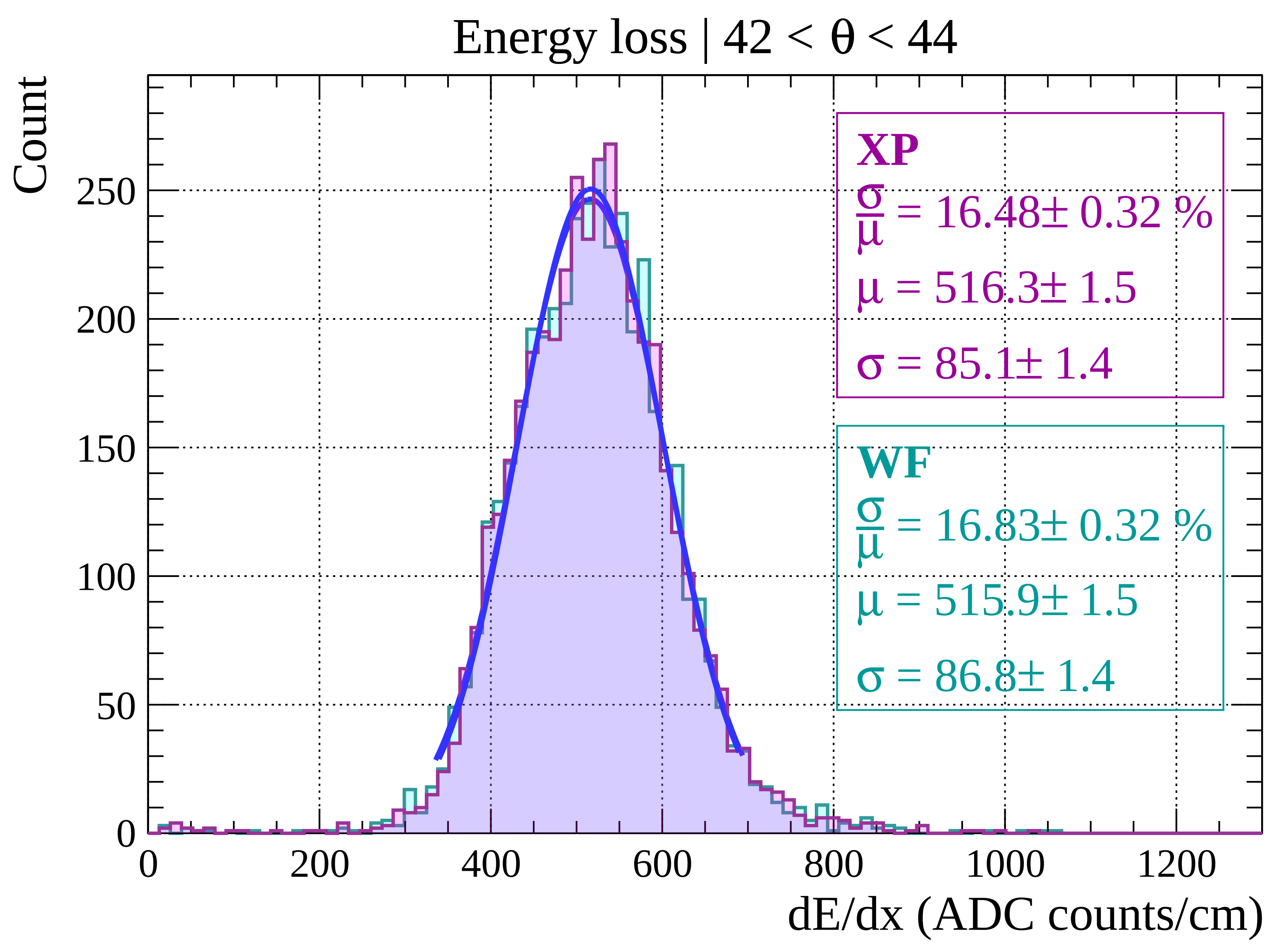


Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.32± 0.22 %
μ = 510.4± 1.2
σ = 83.3± 0.9
WF
σ/μ = 16.66± 0.23 %
μ = 508.4± 1.3
σ = 84.7± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

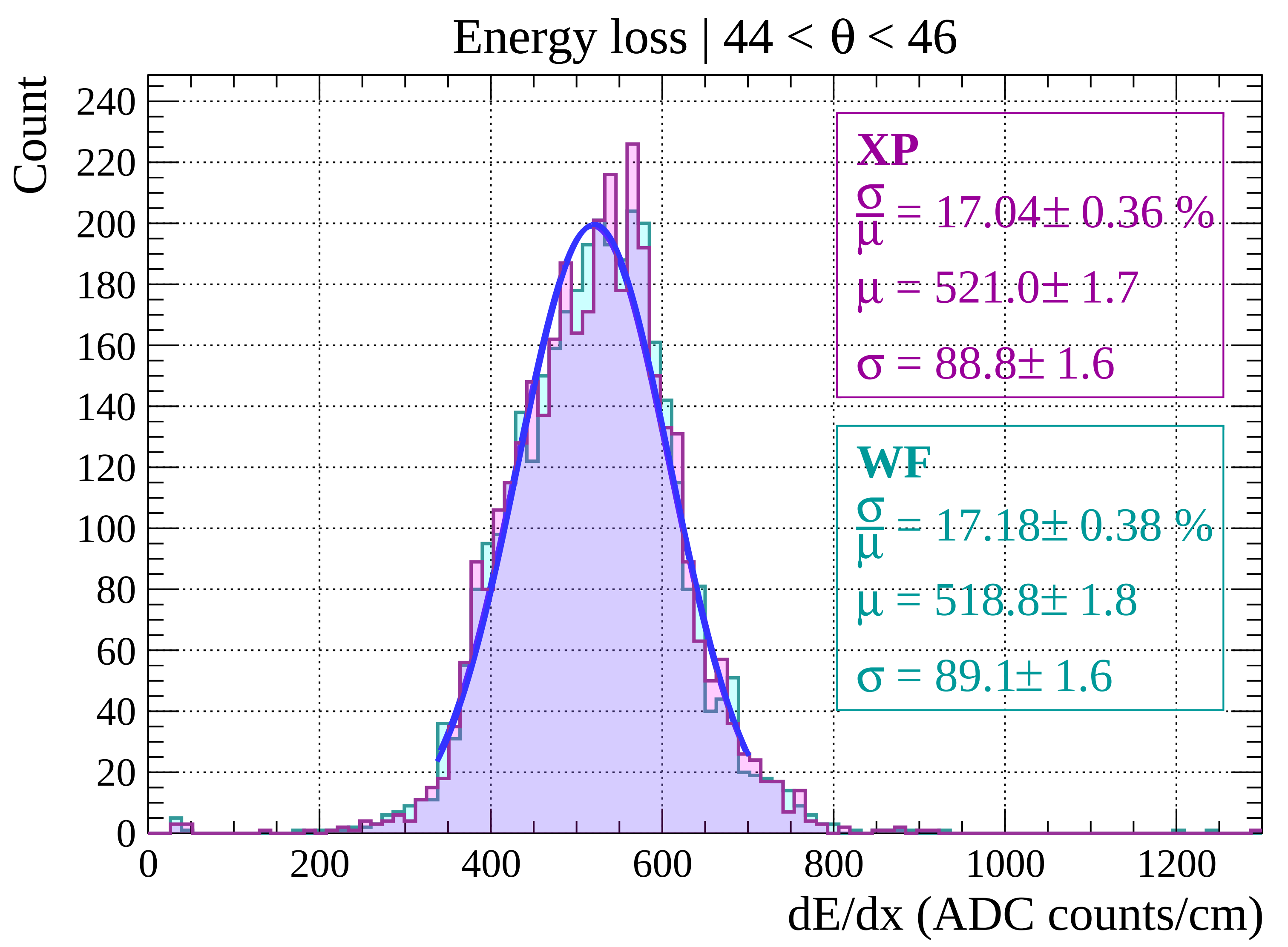


Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.23± 0.25 %
μ = 514.8± 1.2
σ = 83.6± 1.1
WF
σ/μ = 16.57± 0.27 %
μ = 514.0± 1.3
σ = 85.2± 1.2

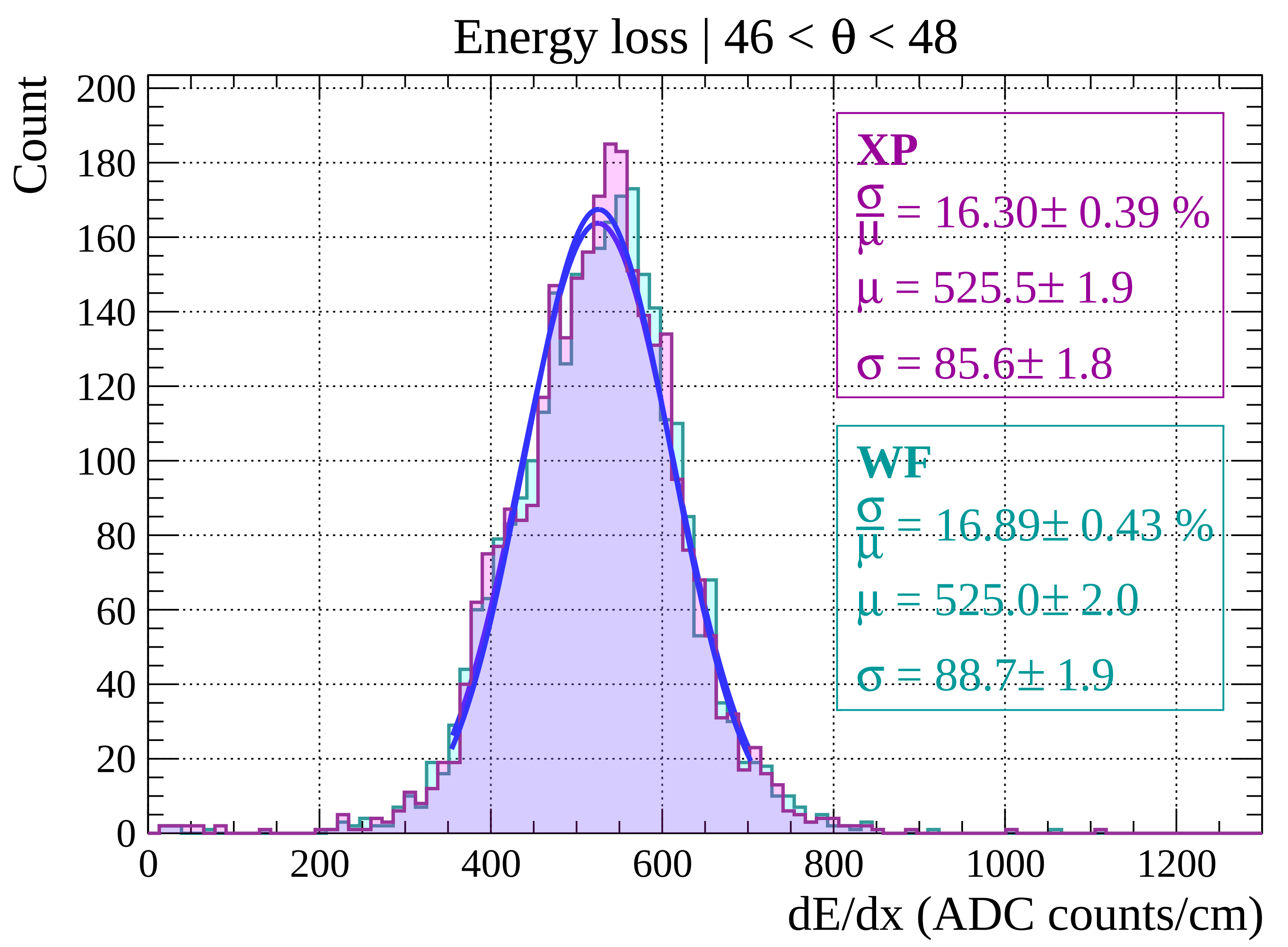

Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.57± 0.28 %
μ = 514.4± 1.4
σ = 85.2± 1.2
WF
σ/μ = 16.62± 0.28 %
μ = 515.0± 1.4
σ = 85.6± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

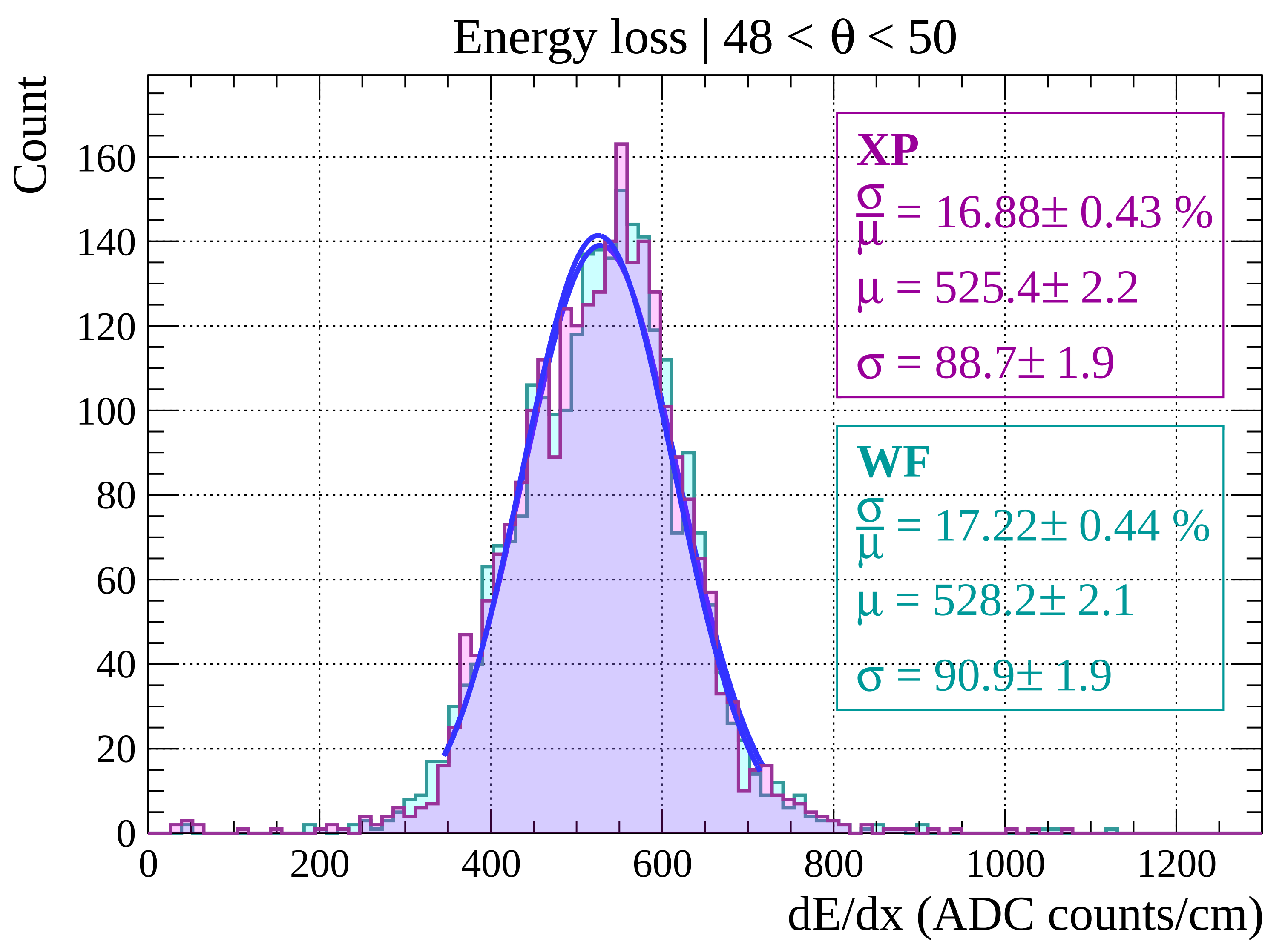


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 16.48± 0.32 %
μ = 516.3± 1.5
σ = 85.1± 1.4
WF
σ/μ = 16.83± 0.32 %
μ = 515.9± 1.5
σ = 86.8± 1.4

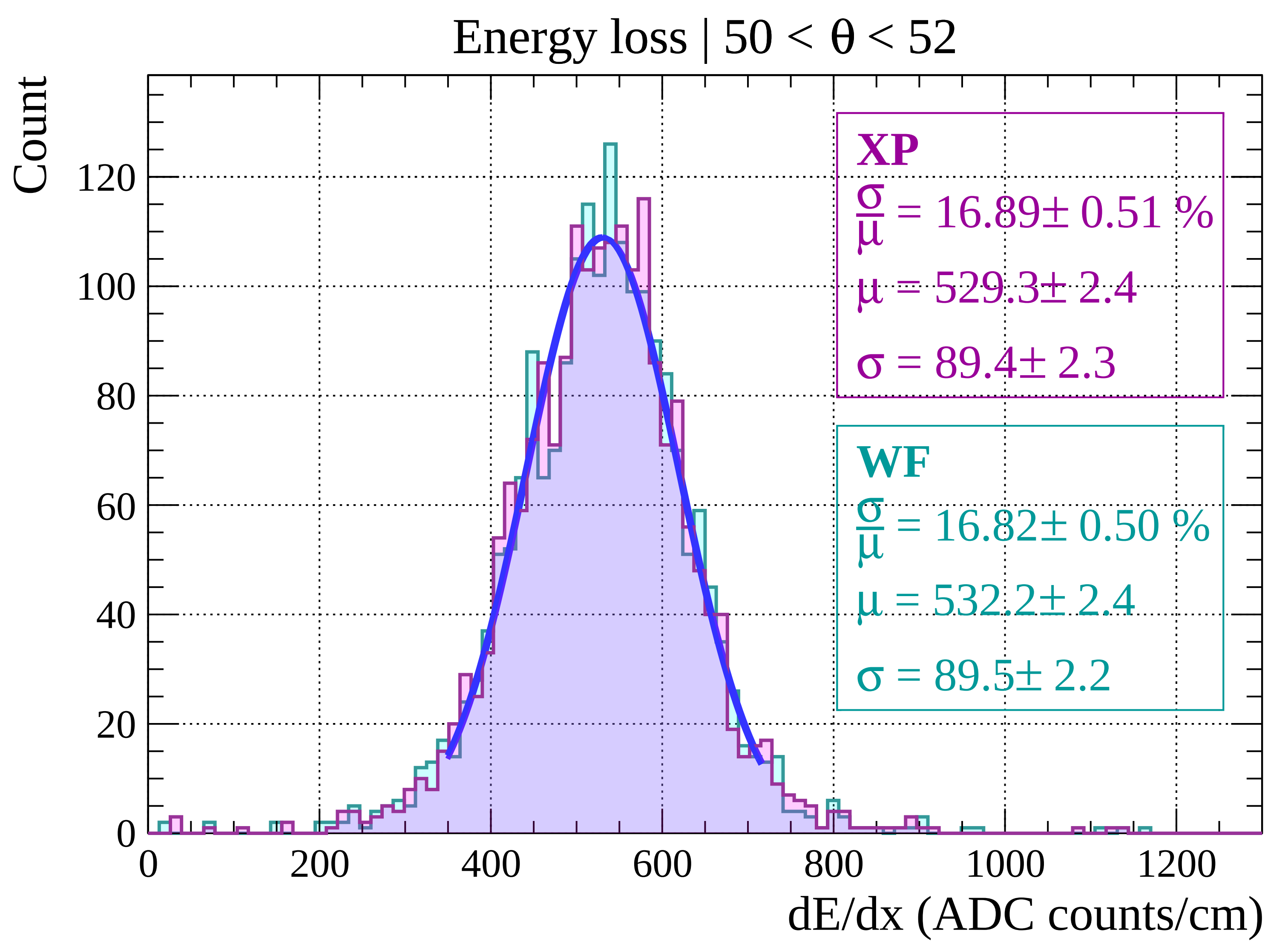


Energy loss | 44 < θ < 46
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 17.04± 0.36 %
μ = 521.0± 1.7
σ = 88.8± 1.6
WF
σ/μ = 17.18± 0.38 %
μ = 518.8± 1.8
σ = 89.1± 1.6

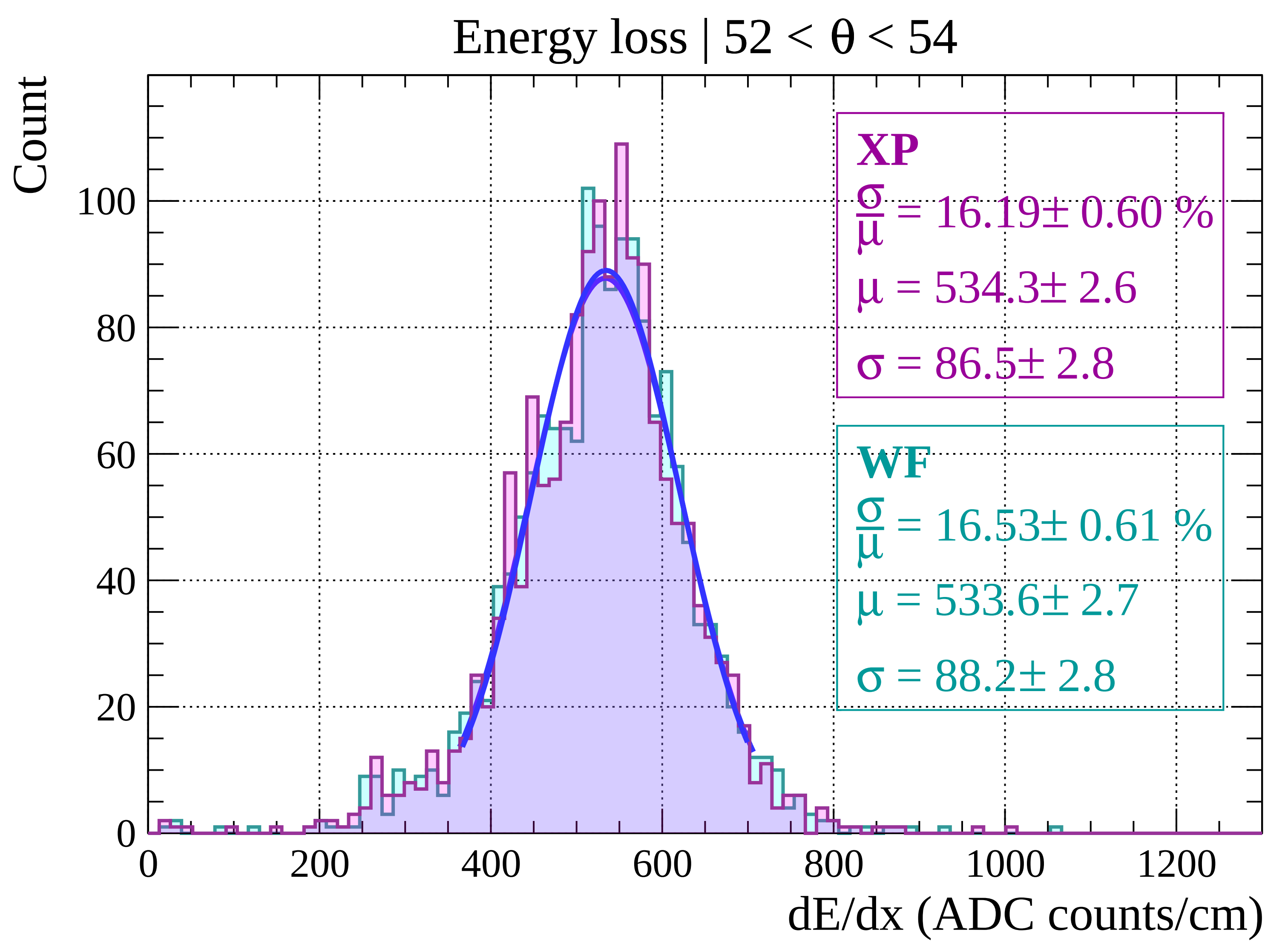

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.30± 0.39 %
μ = 525.5± 1.9
σ = 85.6± 1.8
WF
σ/μ = 16.89± 0.43 %
μ = 525.0± 2.0
σ = 88.7± 1.9
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



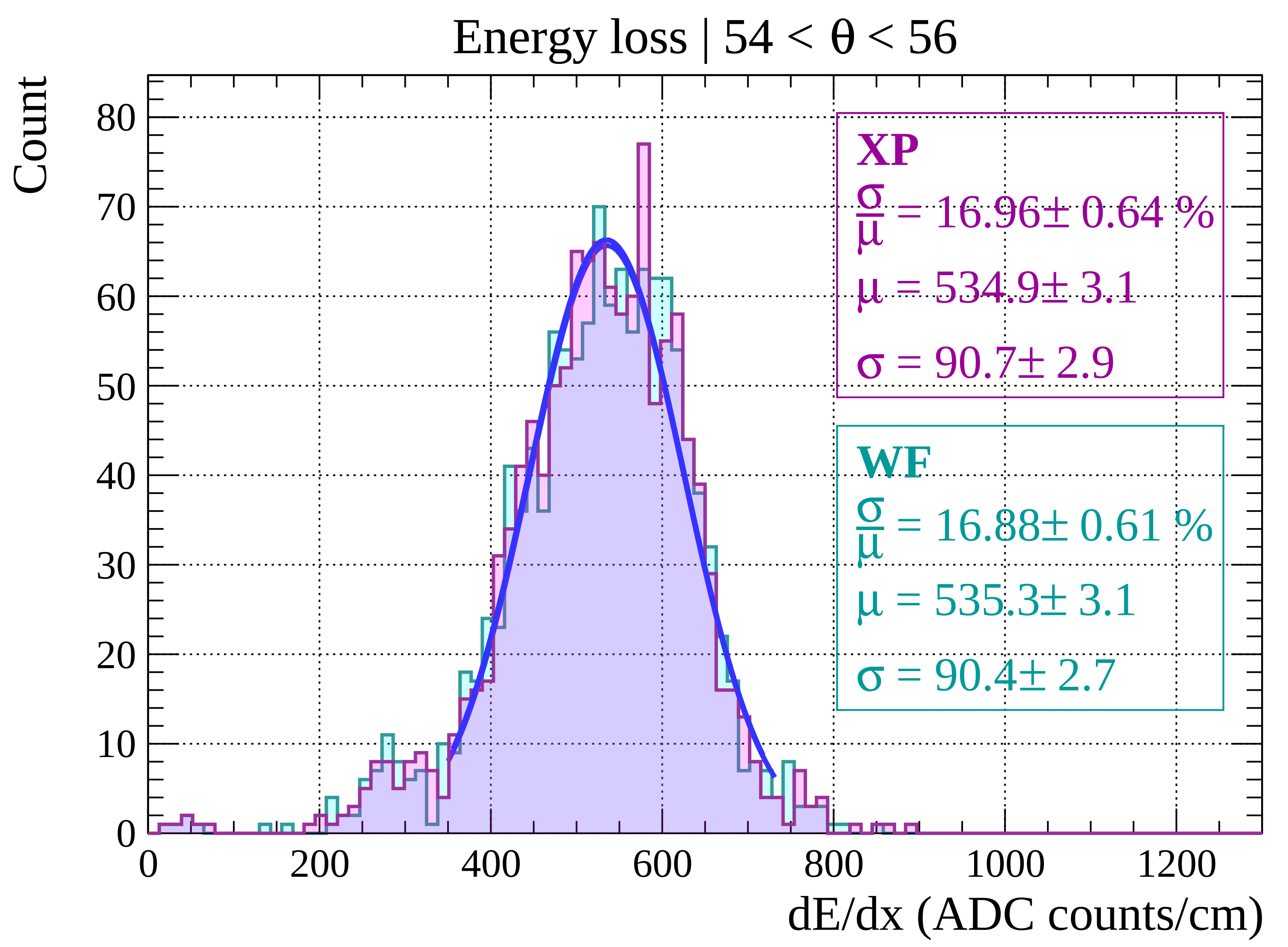
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 16.88± 0.43 %
μ = 525.4± 2.2
σ = 88.7± 1.9
WF
σ/μ = 17.22± 0.44 %
μ = 528.2± 2.1
σ = 90.9± 1.9



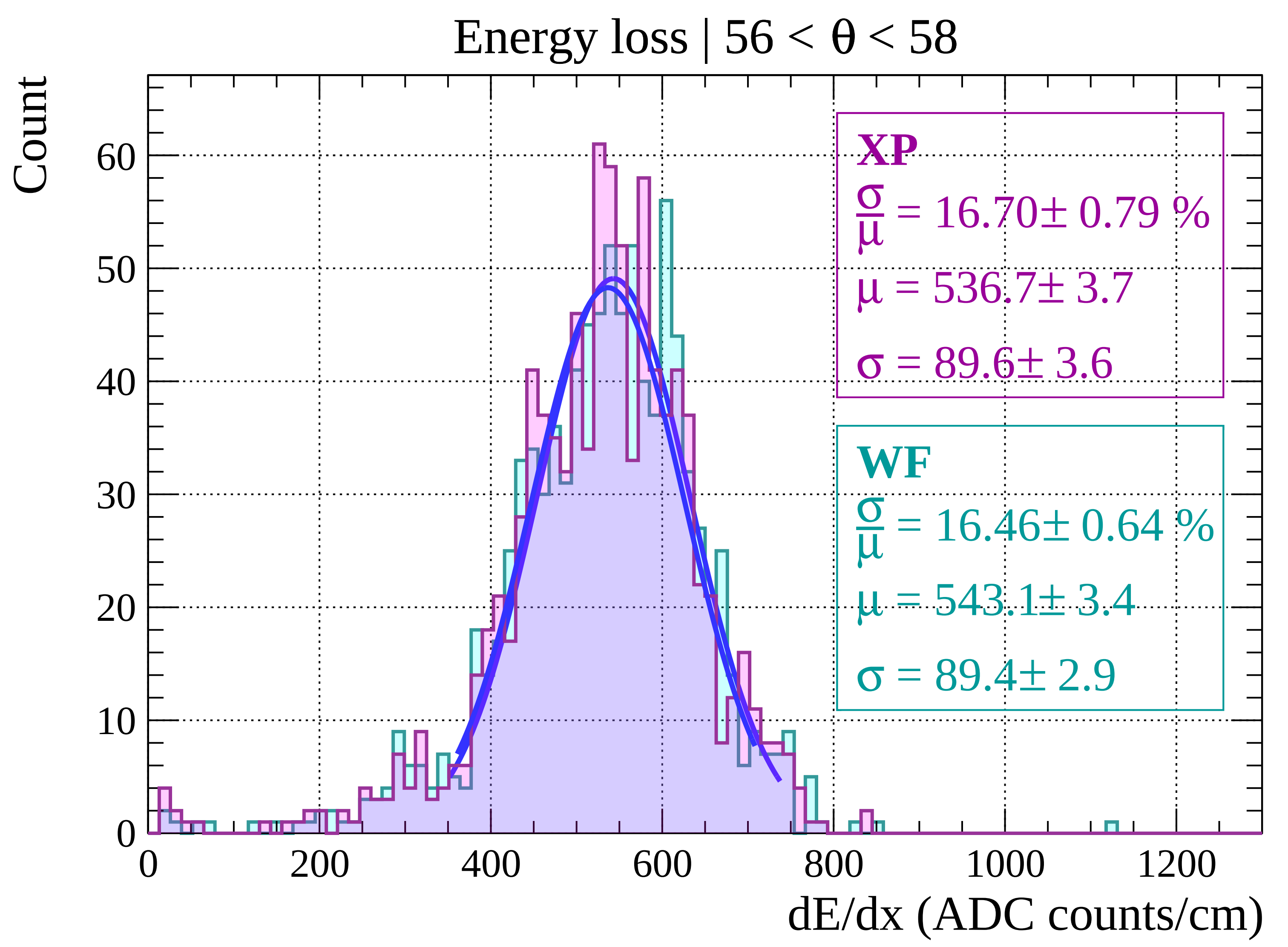
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 16.89± 0.51 %
μ = 529.3± 2.4
σ = 89.4± 2.3
WF
σ/μ = 16.82± 0.50 %
μ = 532.2± 2.4
σ = 89.5± 2.2



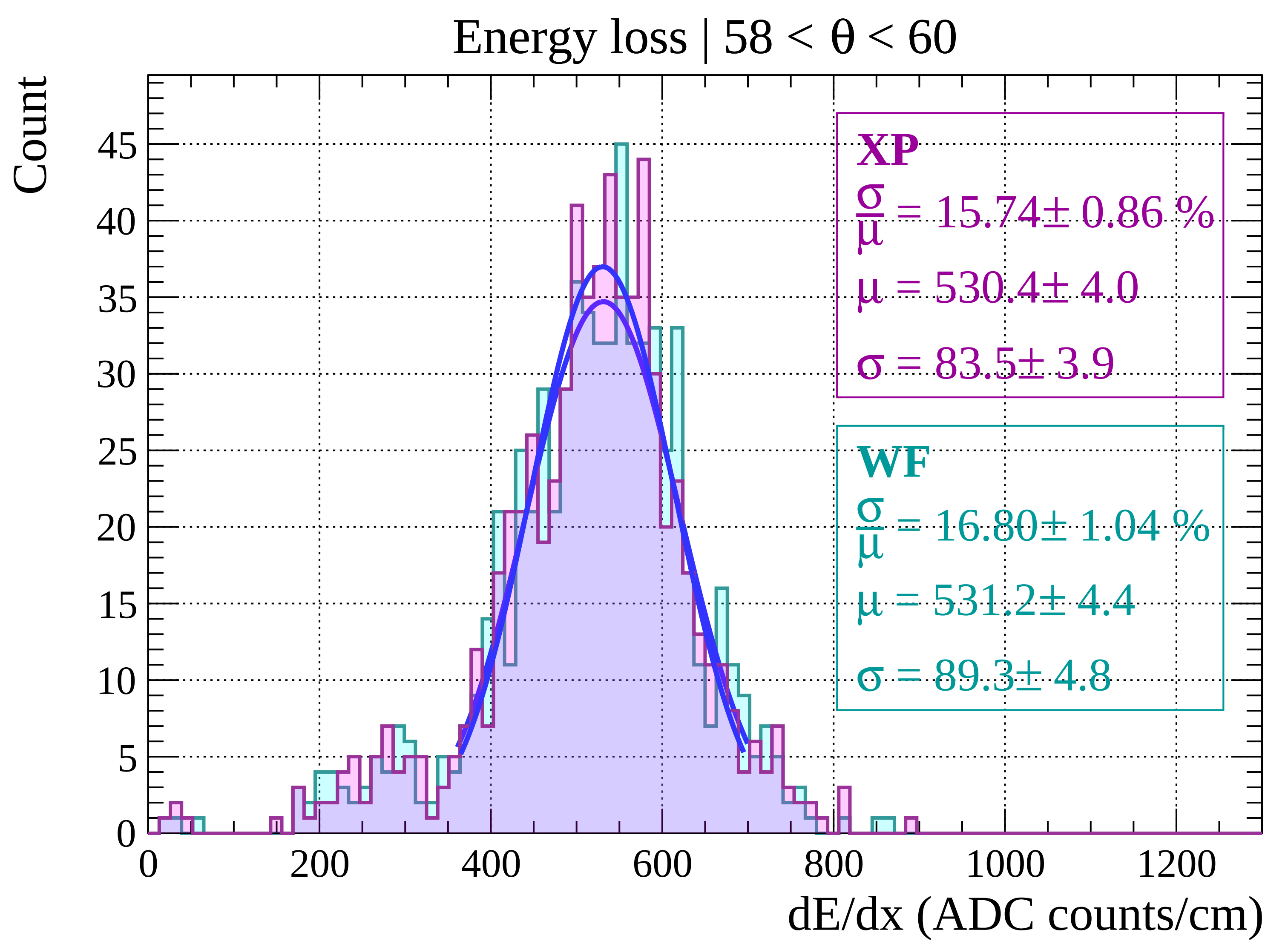
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.19± 0.60 %
μ = 534.3± 2.6
σ = 86.5± 2.8
WF
$\frac{\sigma}{\mu}$ = 16.53± 0.61 %
μ = 533.6± 2.7
σ = 88.2± 2.8



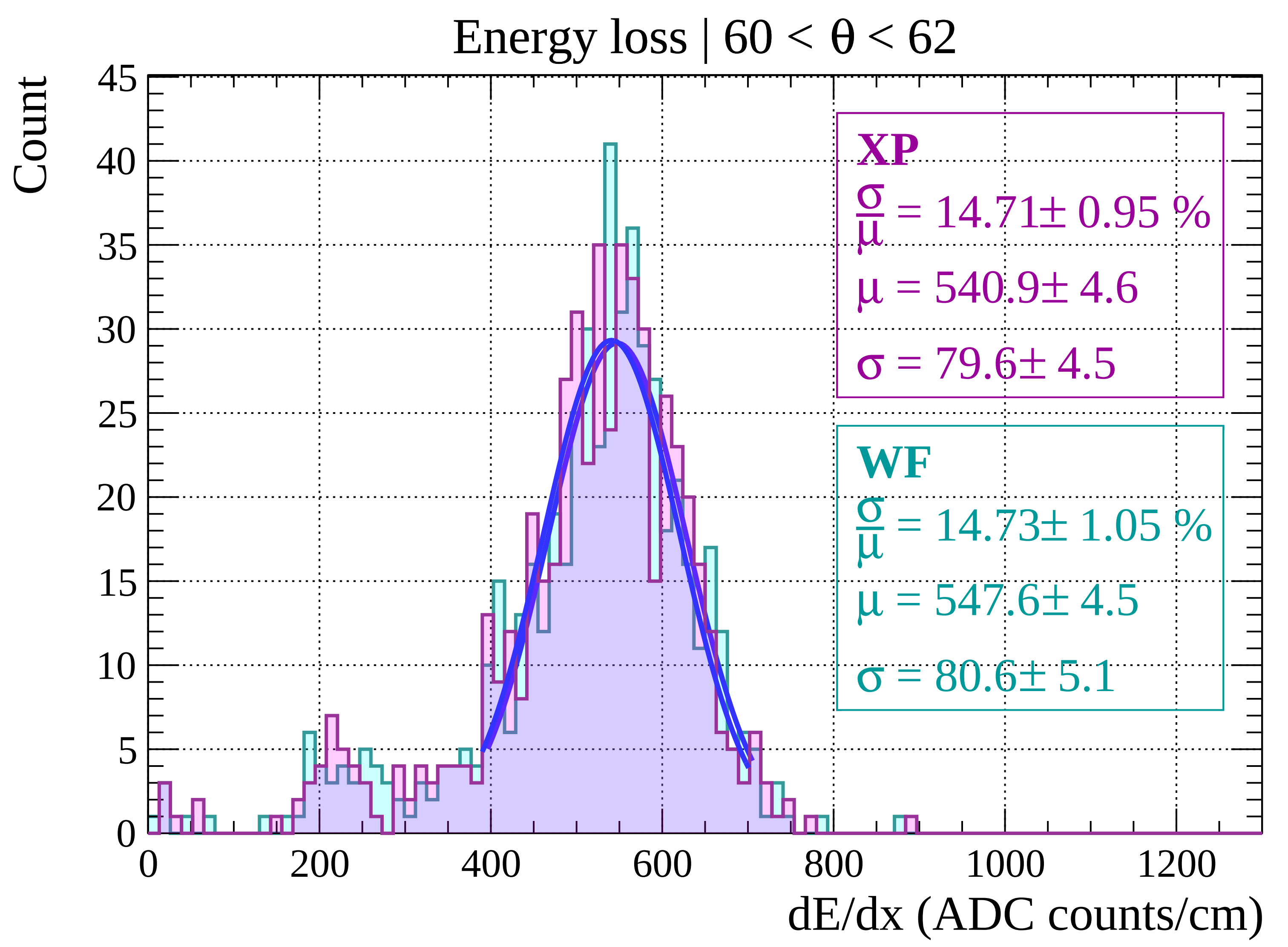
Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.96± 0.64 %
μ = 534.9± 3.1
σ = 90.7± 2.9
WF
σ/μ = 16.88± 0.61 %
μ = 535.3± 3.1
σ = 90.4± 2.7
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



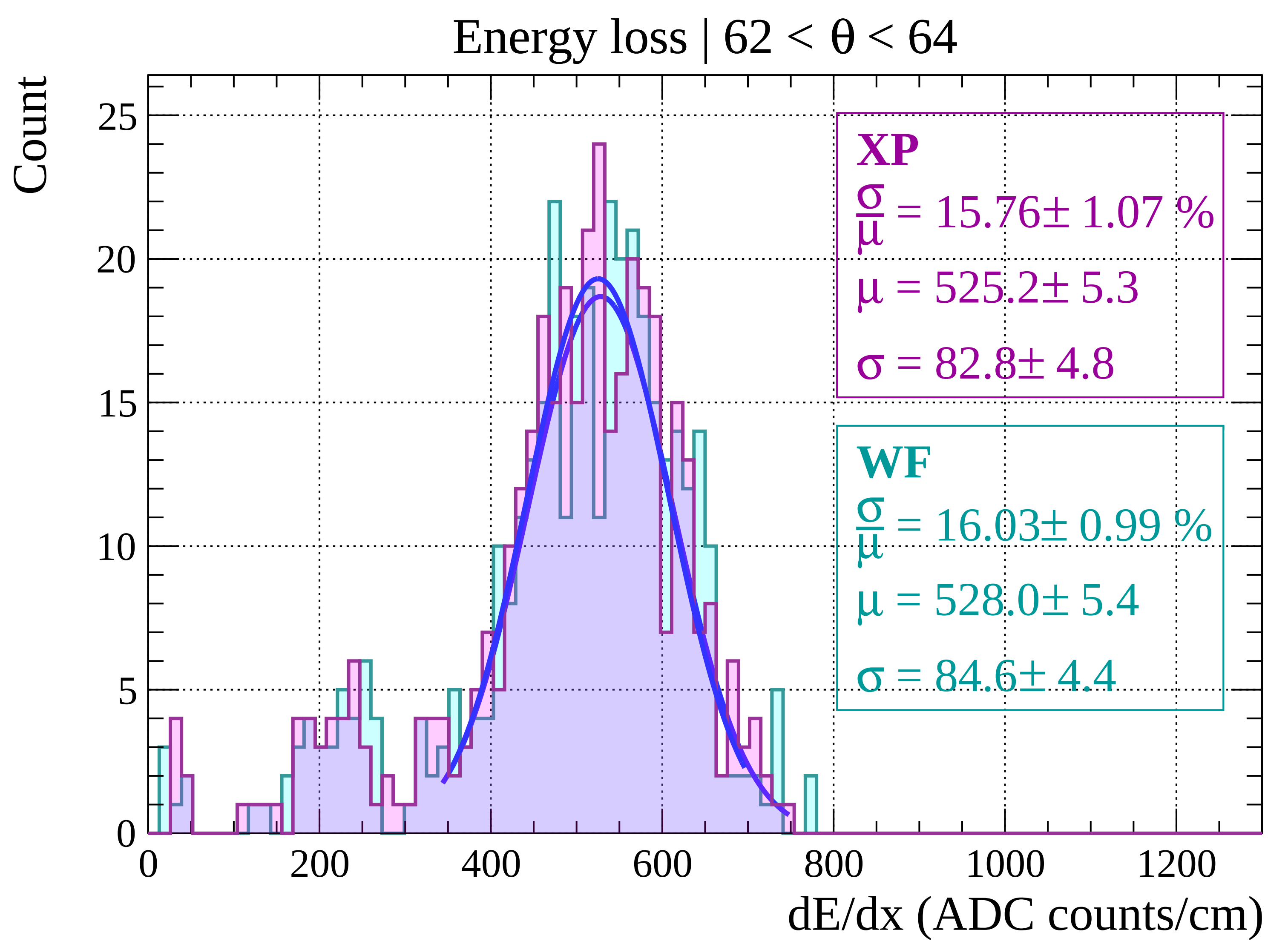
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 16.70± 0.79 %
μ = 536.7± 3.7
σ = 89.6± 3.6
WF
σ/μ = 16.46± 0.64 %
μ = 543.1± 3.4
σ = 89.4± 2.9



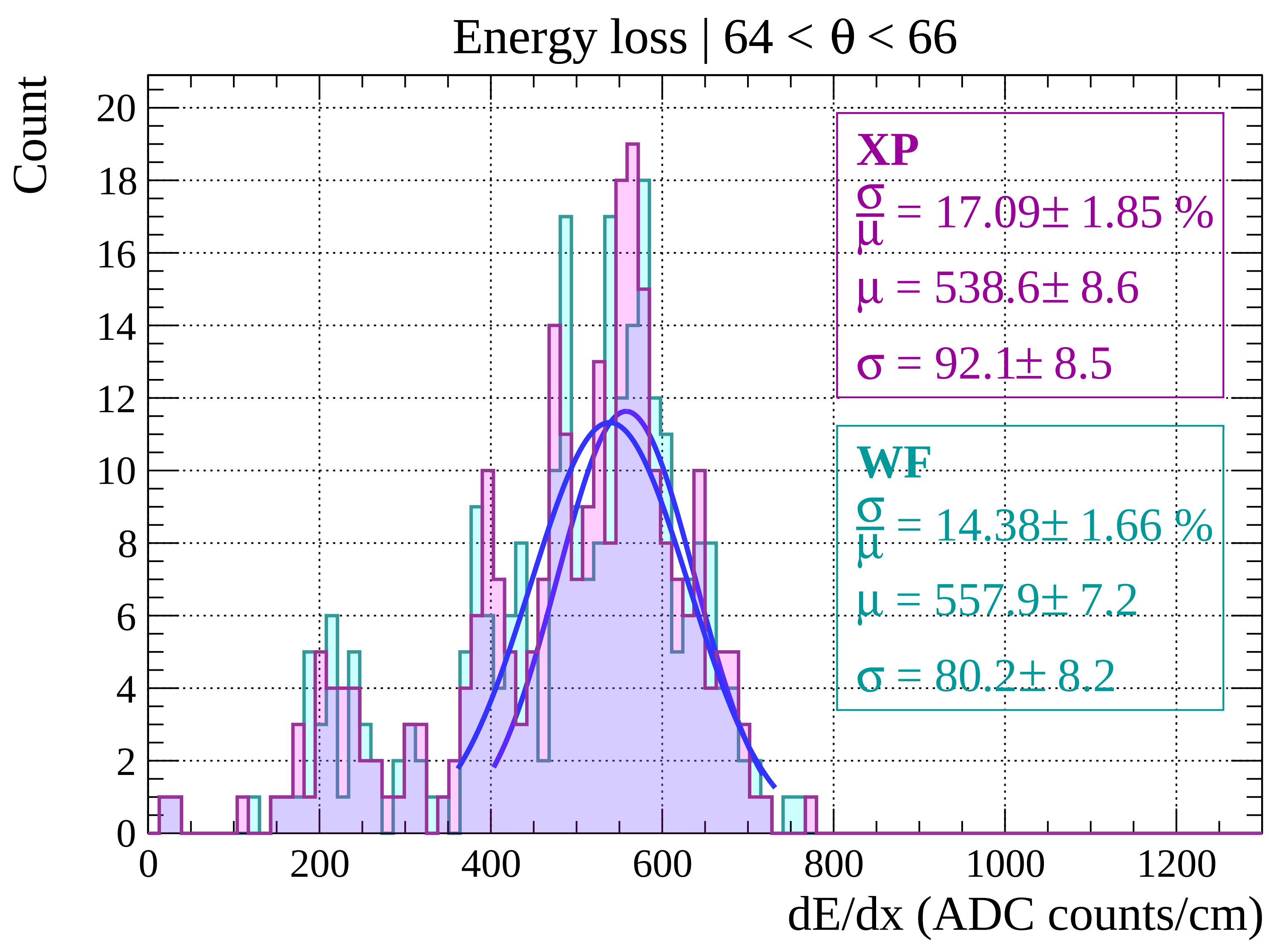
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 15.74 ± 0.86 %
μ = 530.4 ± 4.0
σ = 83.5 ± 3.9
WF
σ/μ = 16.80 ± 1.04 %
μ = 531.2 ± 4.4
σ = 89.3 ± 4.8

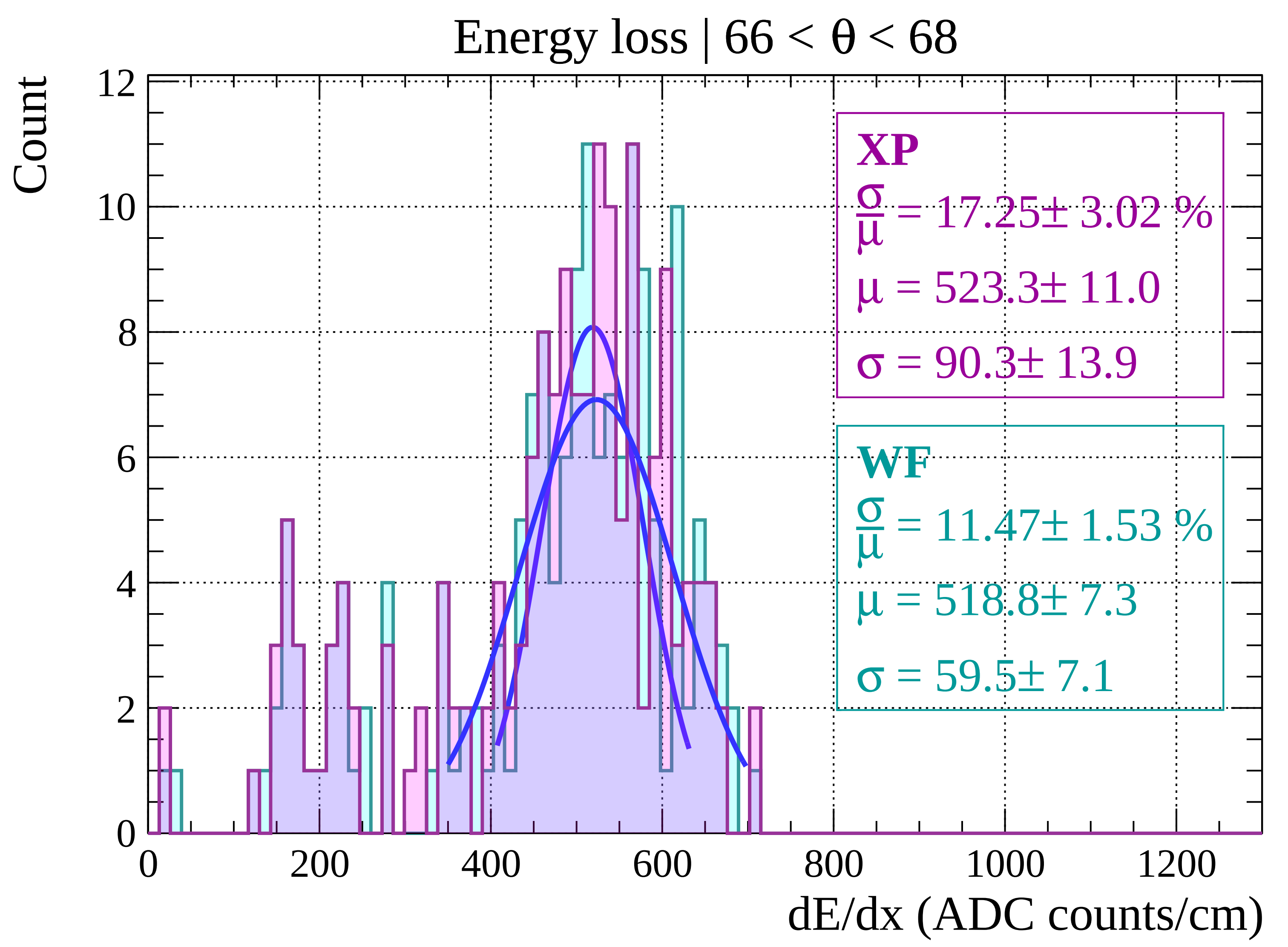


Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 14.71± 0.95 %
μ = 540.9± 4.6
σ = 79.6± 4.5
WF
σ/μ = 14.73± 1.05 %
μ = 547.6± 4.5
σ = 80.6± 5.1

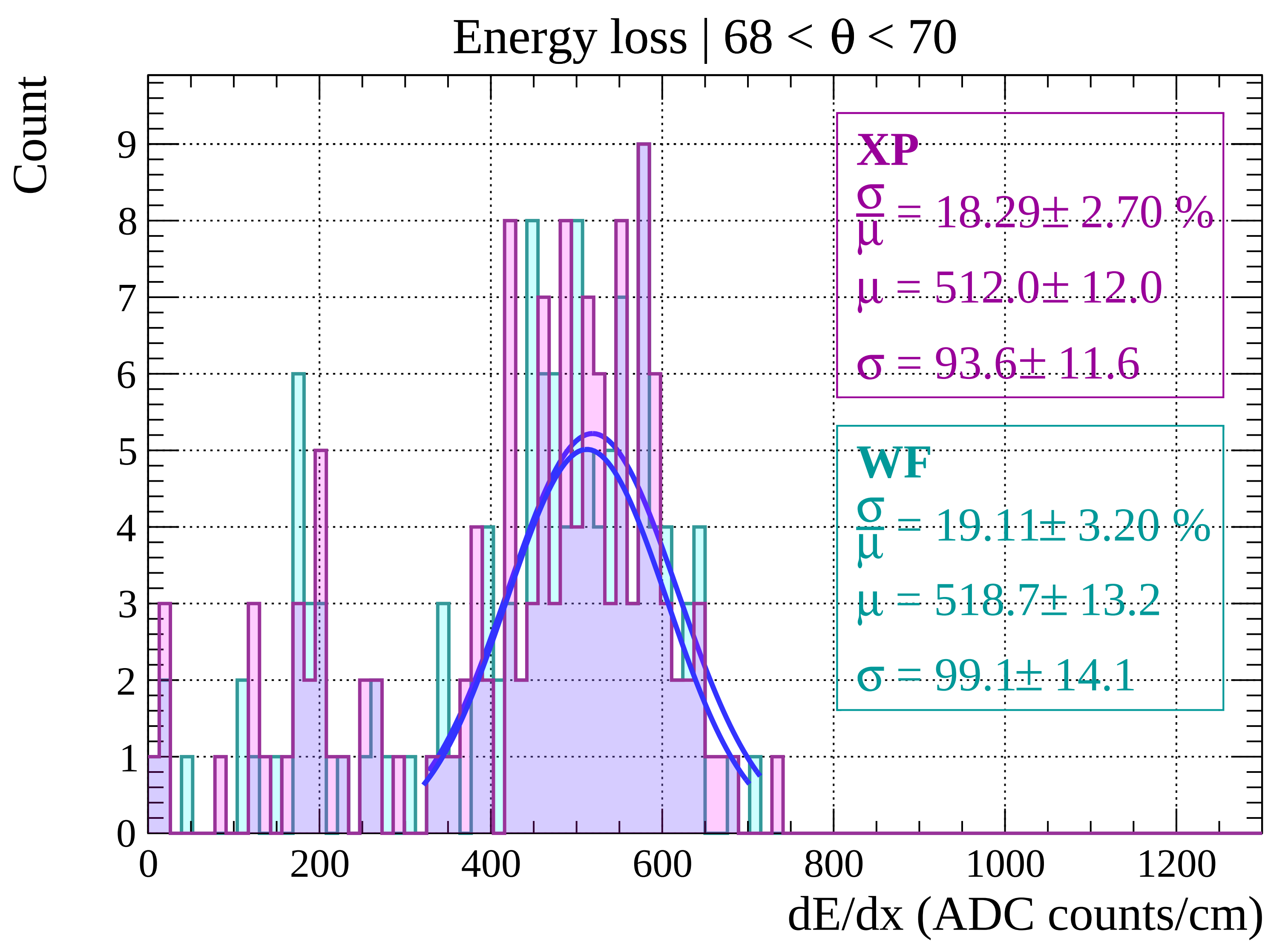


Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 15.76± 1.07 %
μ = 525.2± 5.3
σ = 82.8± 4.8
WF
σ/μ = 16.03± 0.99 %
μ = 528.0± 5.4
σ = 84.6± 4.4

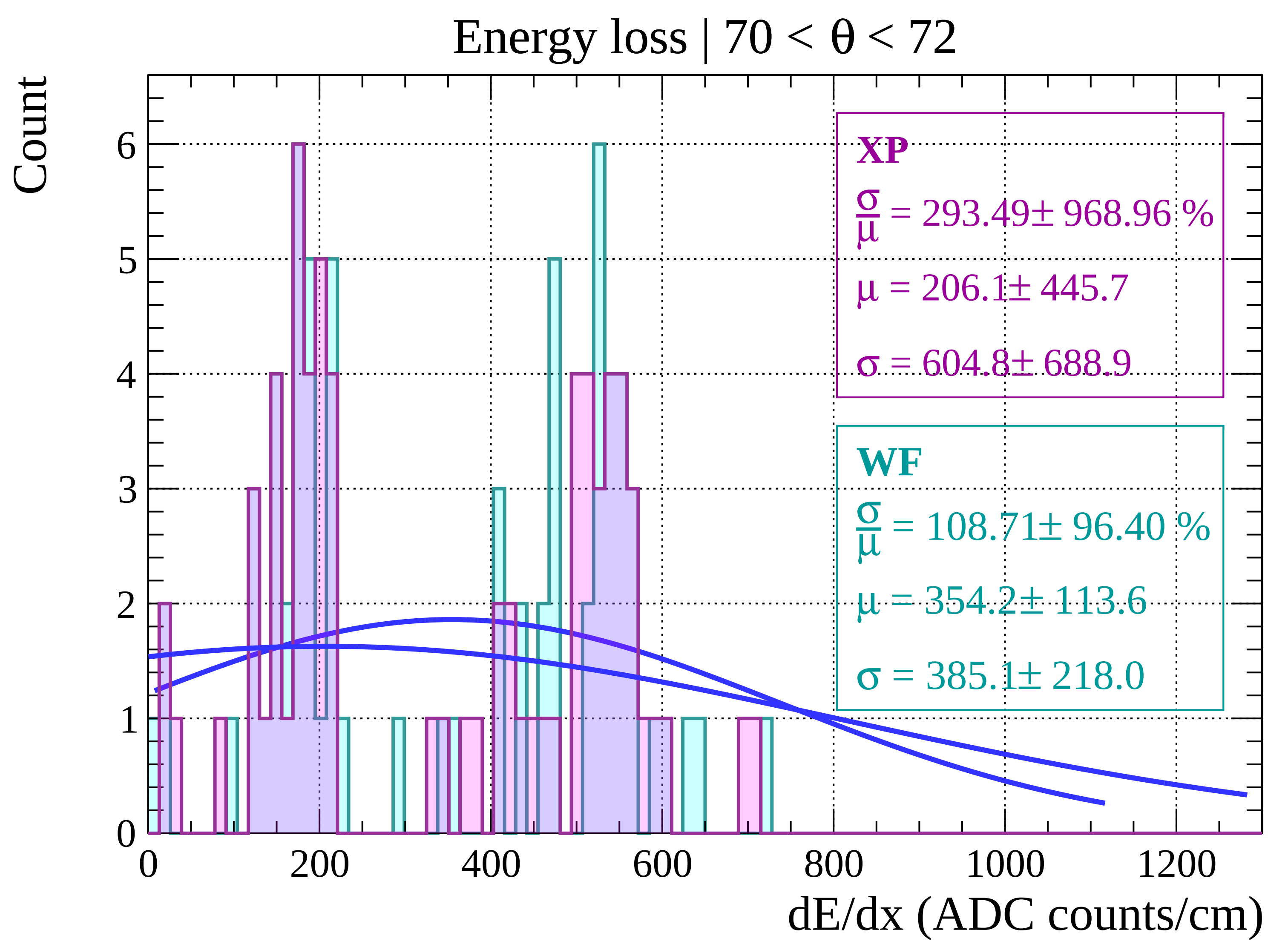

Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200
XP
σ/μ = 17.09± 1.85 %
μ = 538.6± 8.6
σ = 92.1± 8.5
WF
σ/μ = 14.38± 1.66 %
μ = 557.9± 7.2
σ = 80.2± 8.2



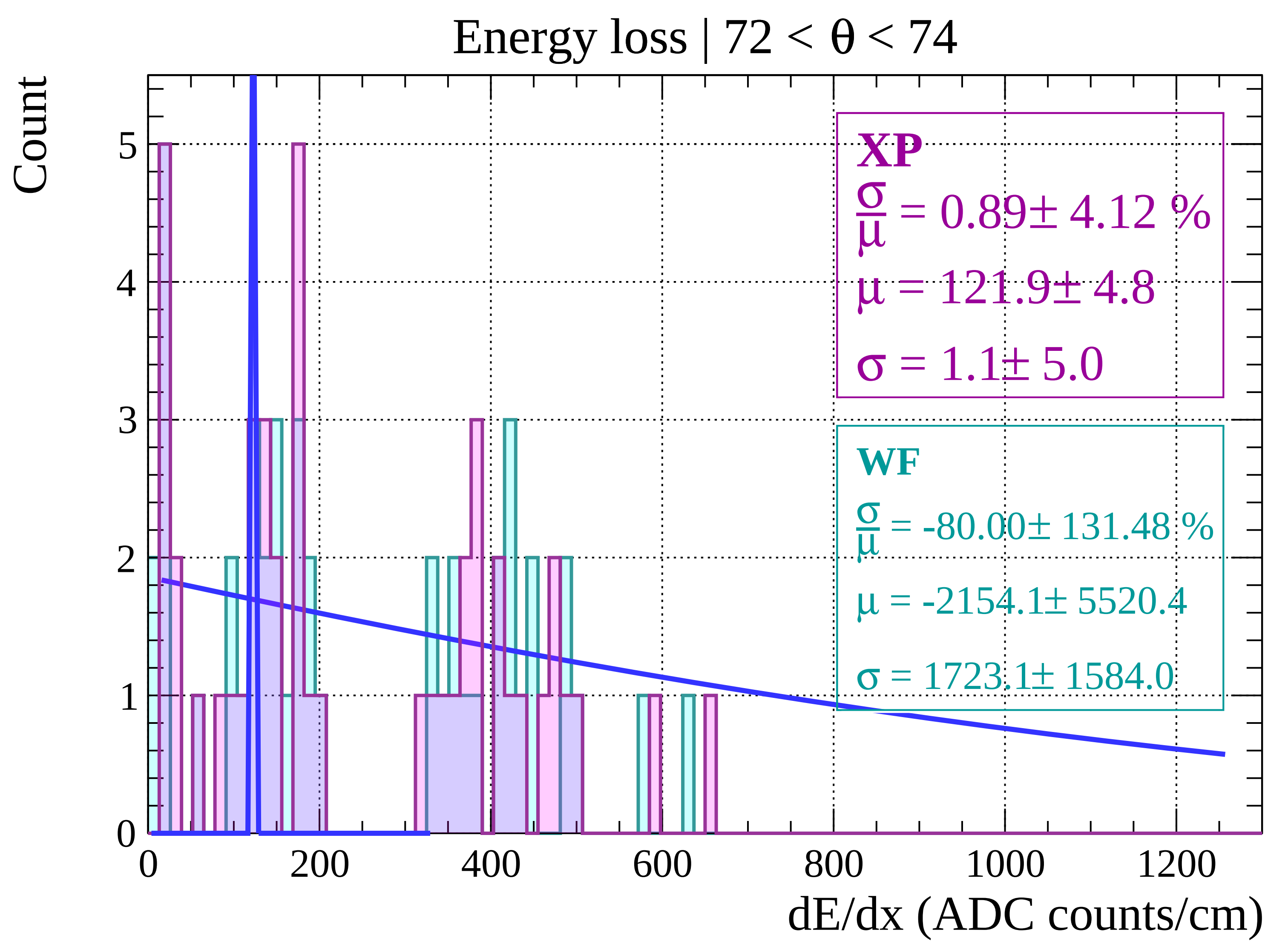
Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 17.25± 3.02 %
μ = 523.3± 11.0
σ = 90.3± 13.9
WF
σ/μ = 11.47± 1.53 %
μ = 518.8± 7.3
σ = 59.5± 7.1



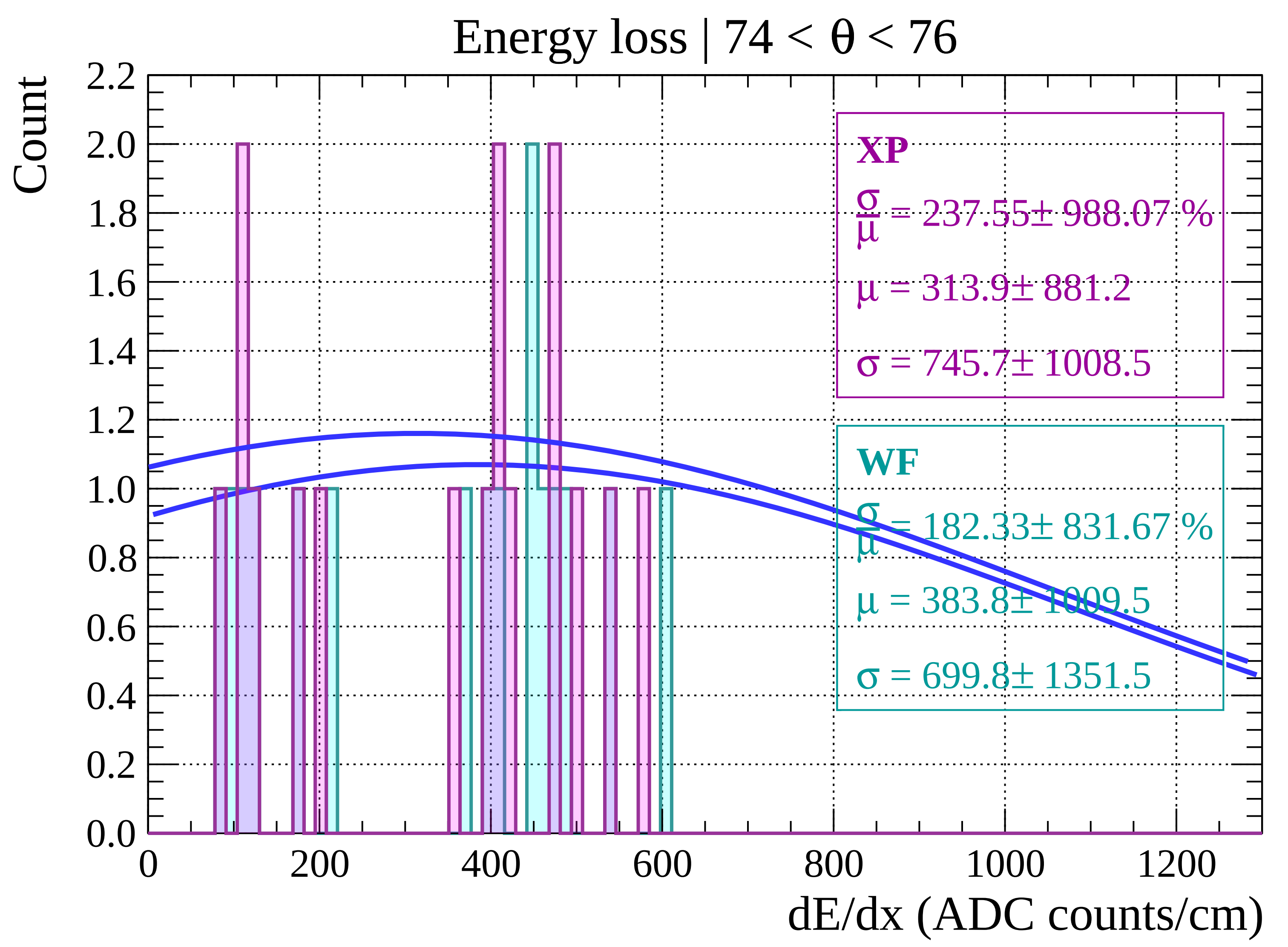
Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 18.29± 2.70 %
μ = 512.0± 12.0
σ = 93.6± 11.6
WF
σ/μ = 19.11± 3.20 %
μ = 518.7± 13.2
σ = 99.1± 14.1



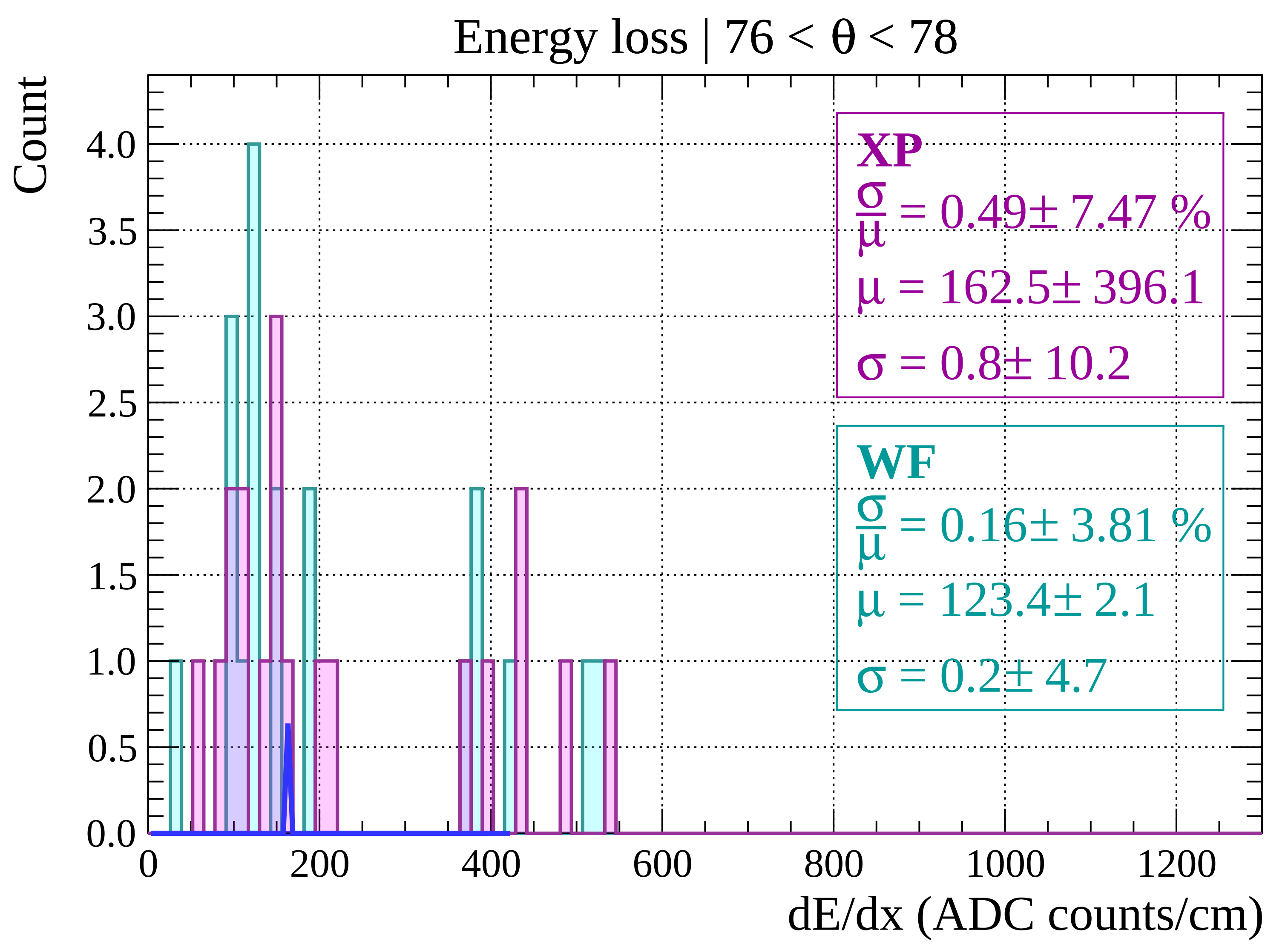
Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 293.49± 968.96 %
μ = 206.1± 445.7
σ = 604.8± 688.9
WF
σ/μ = 108.71± 96.40 %
μ = 354.2± 113.6
σ = 385.1± 218.0



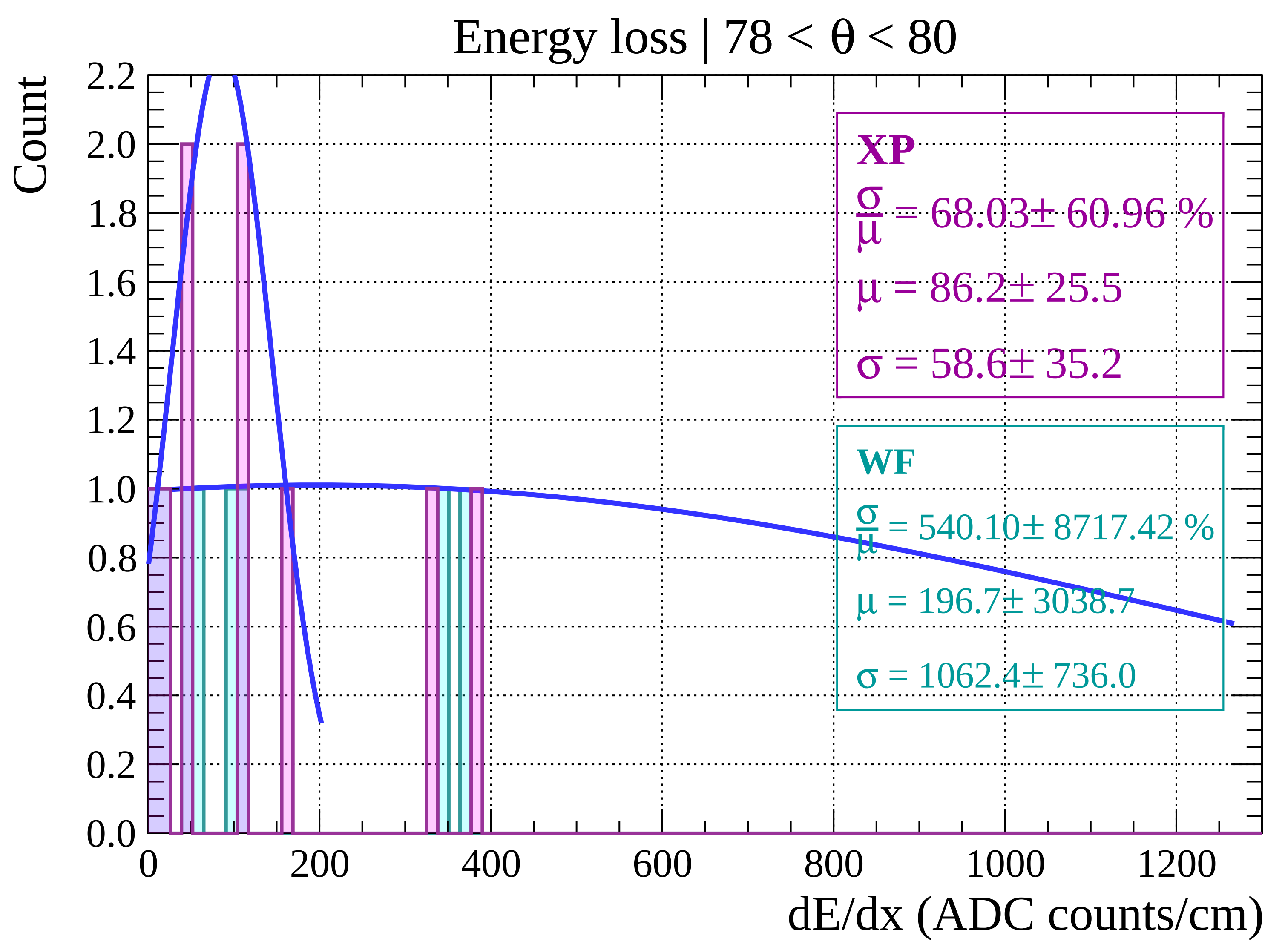
Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.89± 4.12 %
μ = 121.9± 4.8
σ = 1.1± 5.0
WF
$\frac{\sigma}{\mu}$ = -80.00± 131.48 %
μ = -2154.1± 5520.4
σ = 1723.1± 1584.0



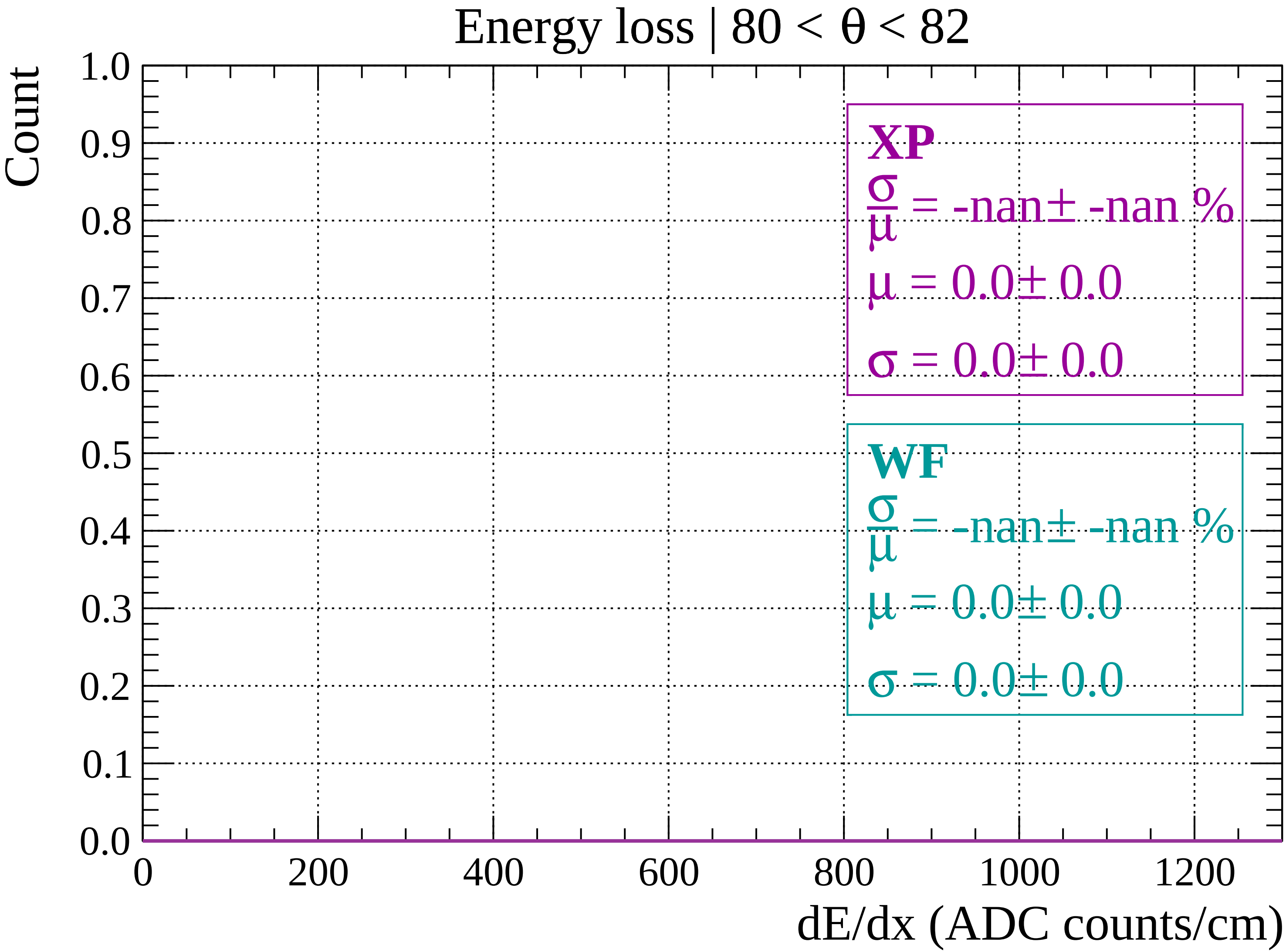
Energy loss | 74 < θ < 76
Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 237.55± 988.07 %
μ = 313.9± 881.2
σ = 745.7± 1008.5
WF
σ/μ = 182.33± 831.67 %
μ = 383.8± 1009.5
σ = 699.8± 1351.5



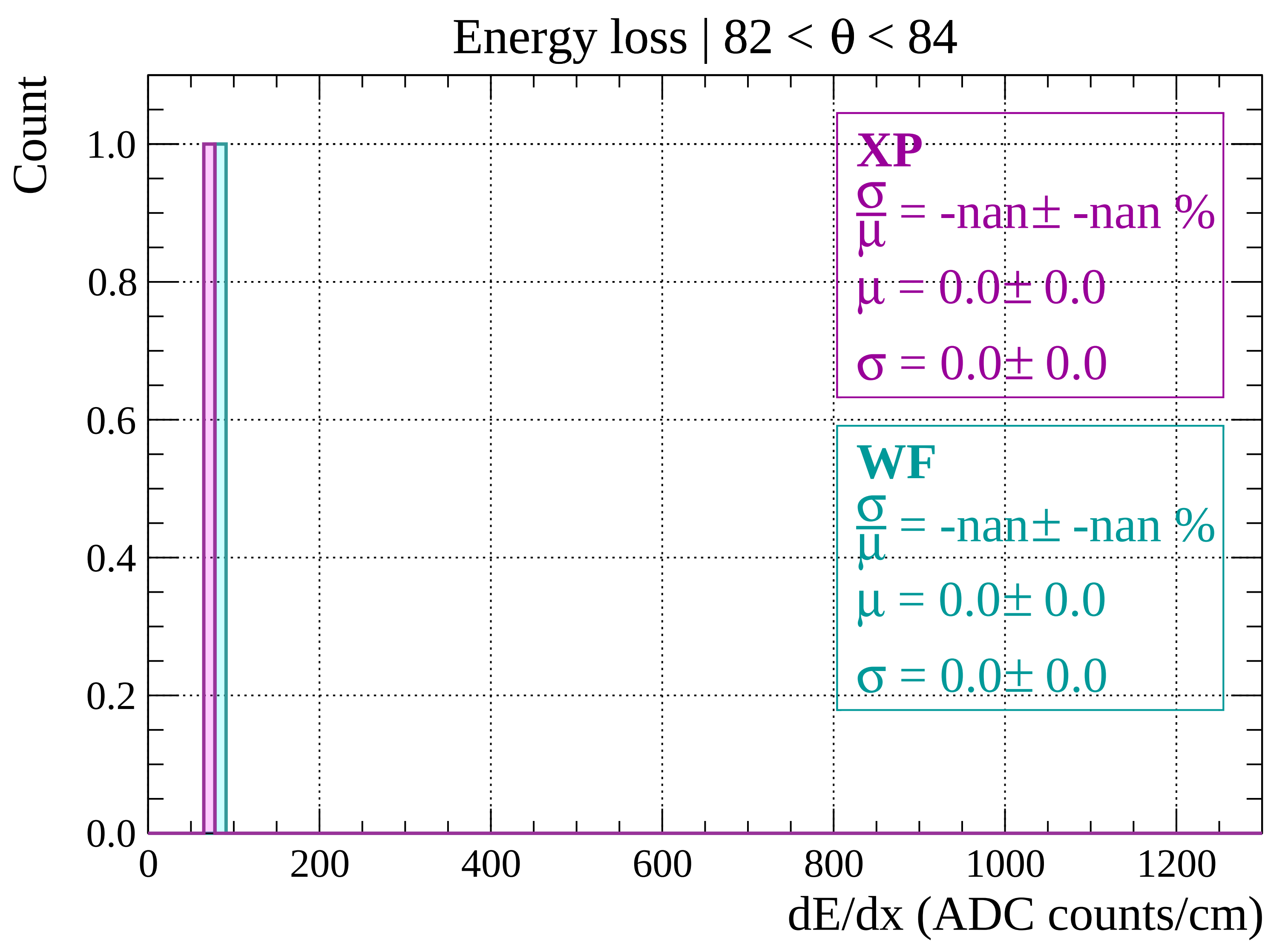
Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.49± 7.47 %
μ = 162.5± 396.1
σ = 0.8± 10.2
WF
$\frac{\sigma}{\mu}$ = 0.16± 3.81 %
μ = 123.4± 2.1
σ = 0.2± 4.7



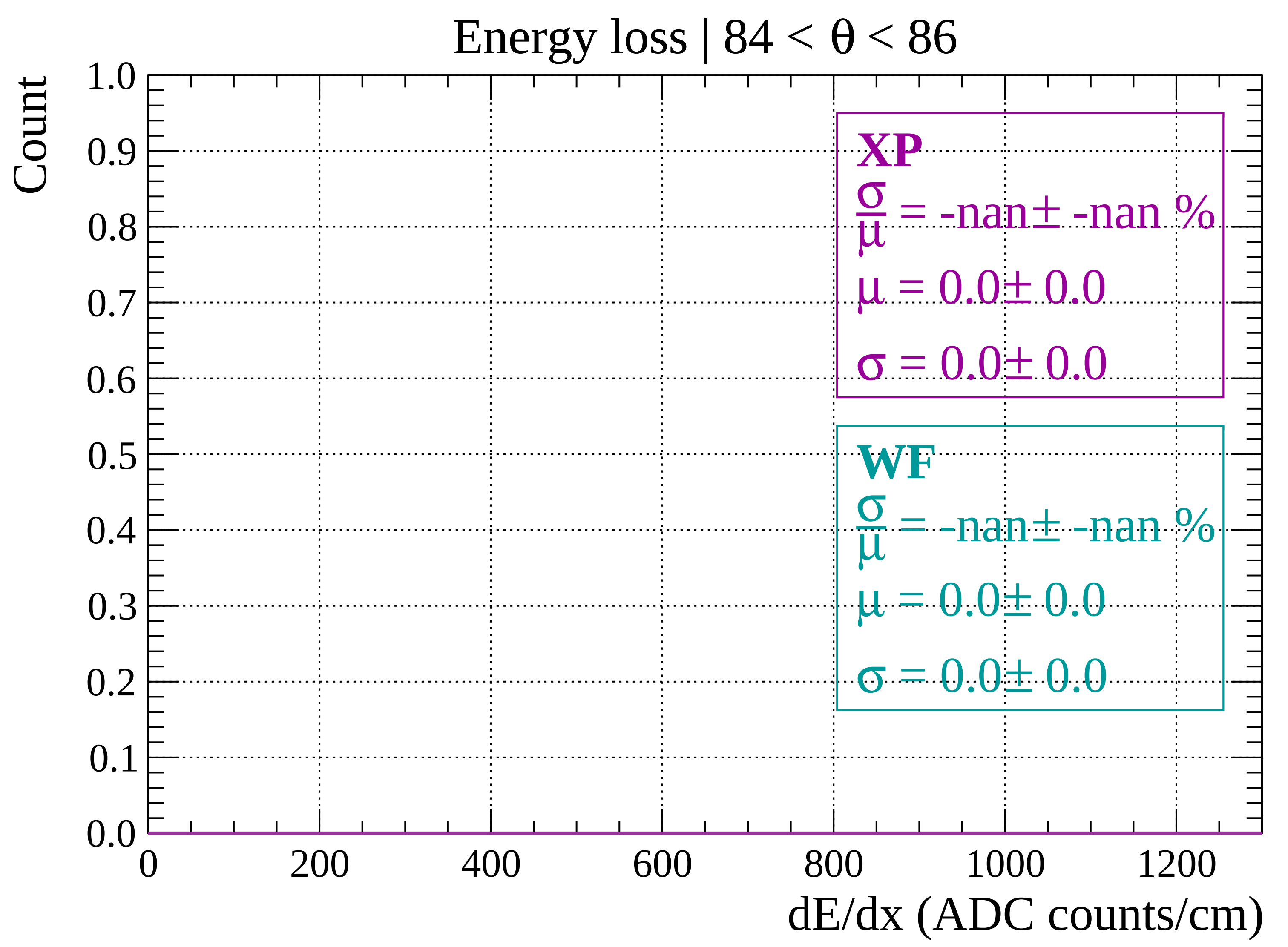
Energy loss | 78 < θ < 80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 68.03± 60.96 %
μ = 86.2± 25.5
σ = 58.6± 35.2
WF
σ/μ = 540.10± 8717.42 %
μ = 196.7± 3038.7
σ = 1062.4± 736.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200

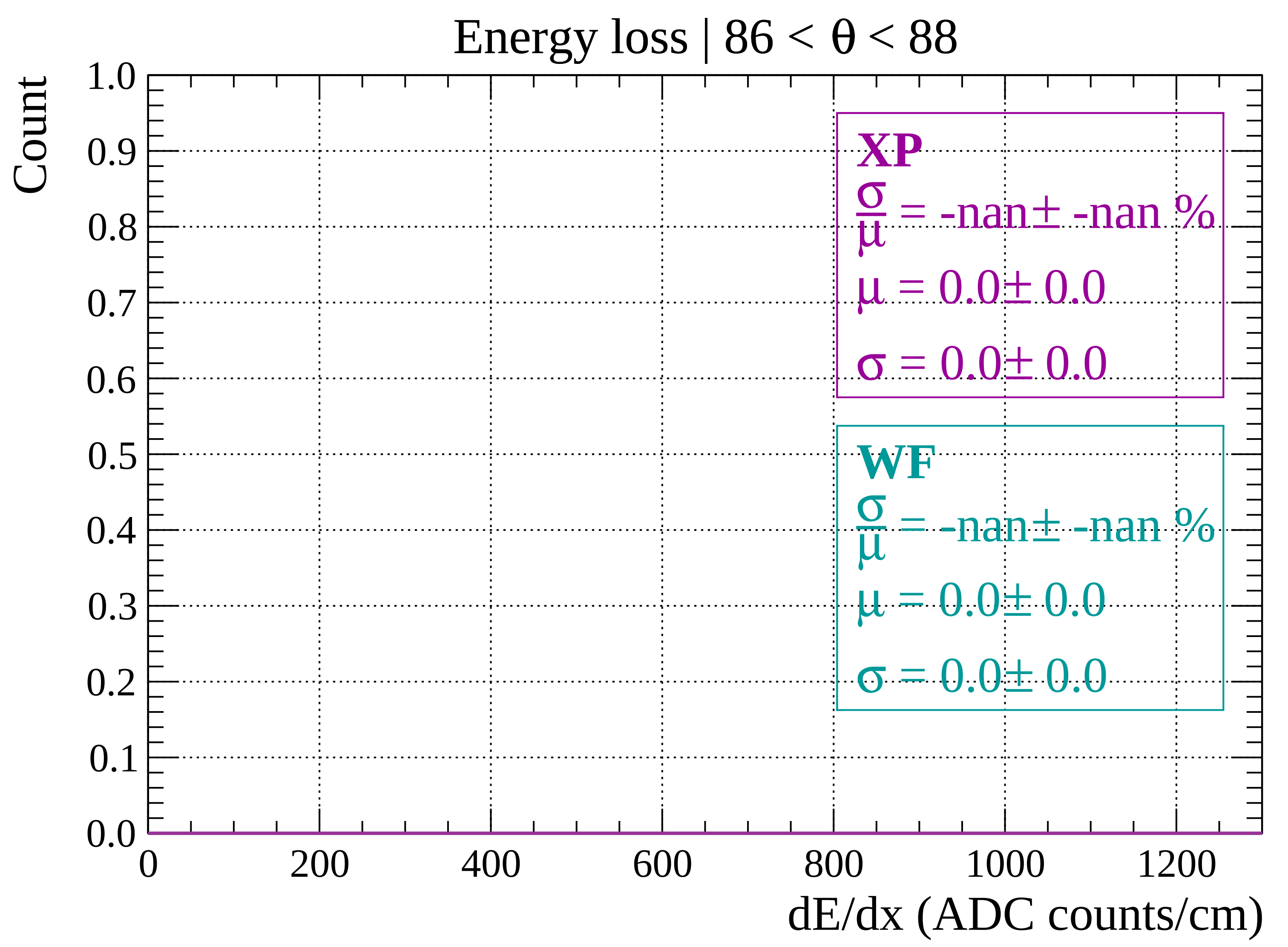


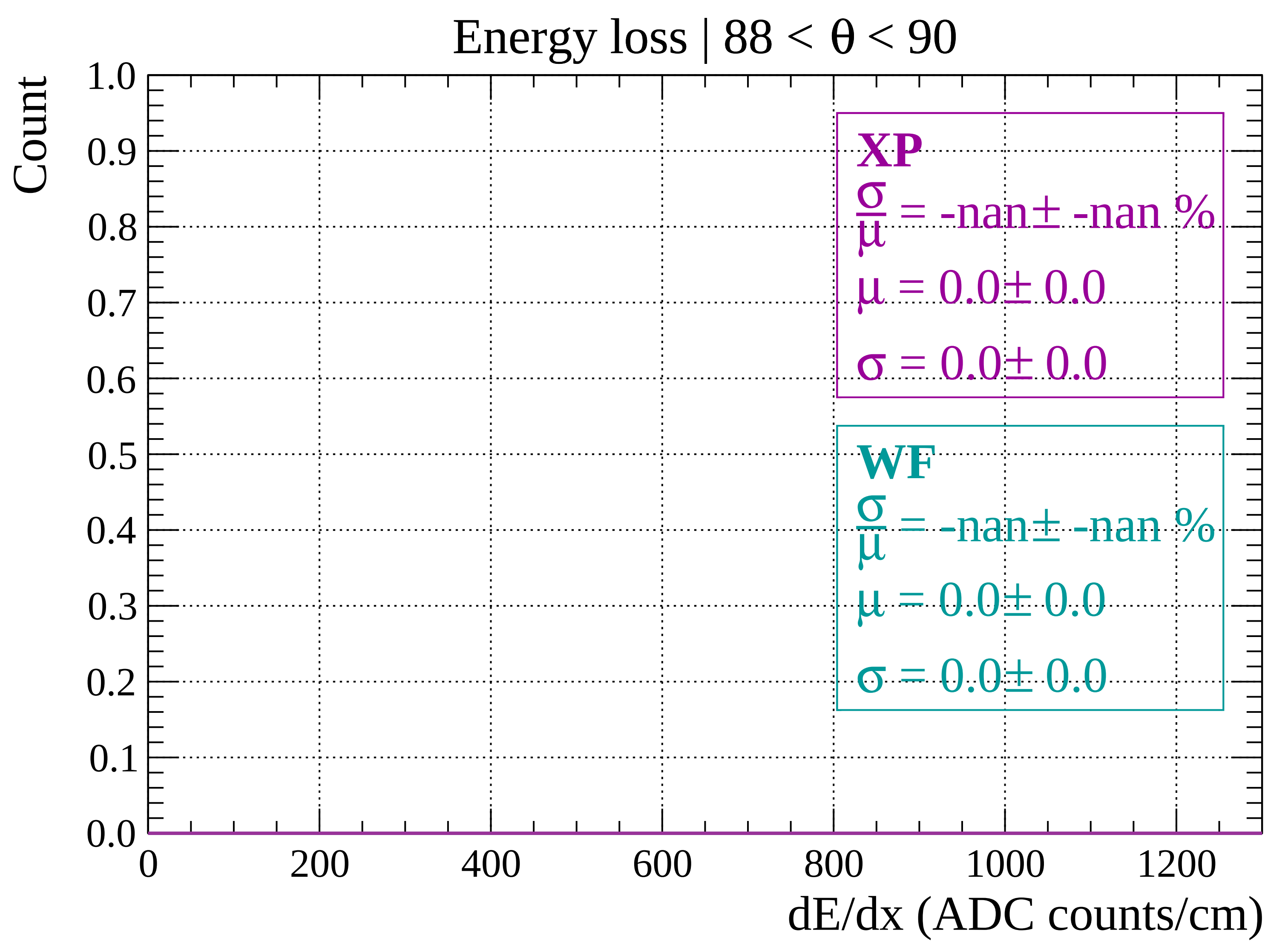
Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



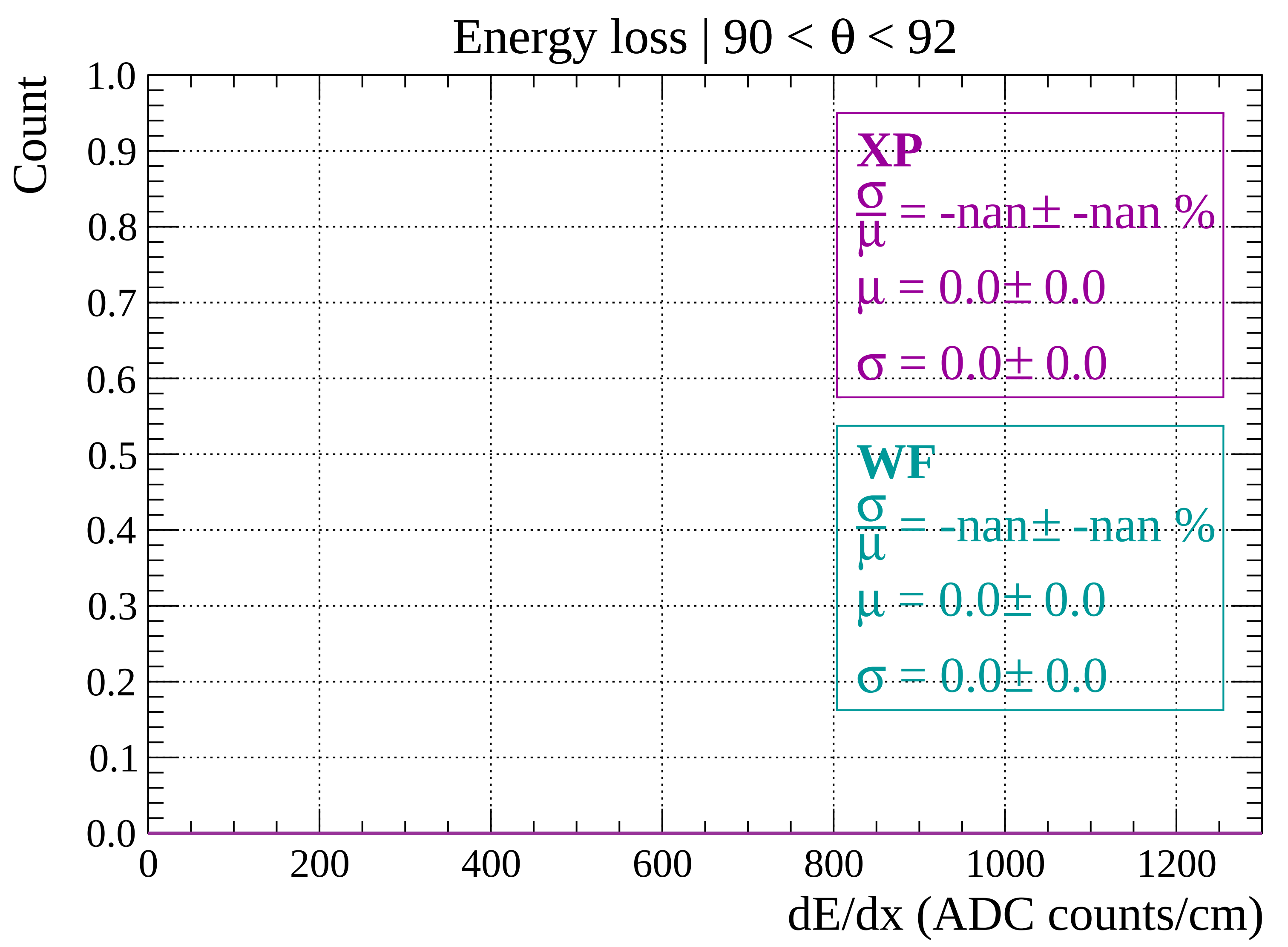
Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

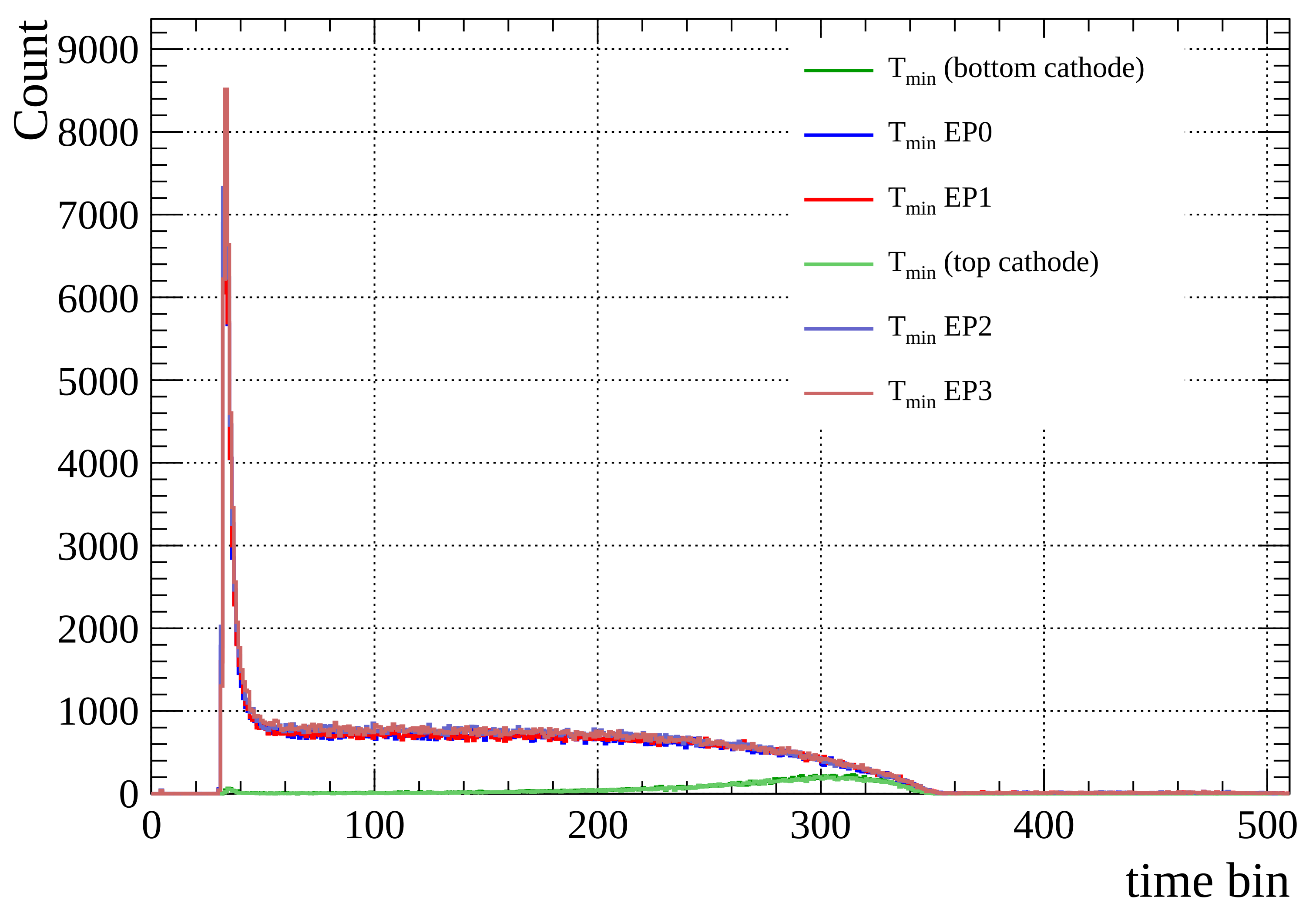

Energy loss | 84 < θ < 86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

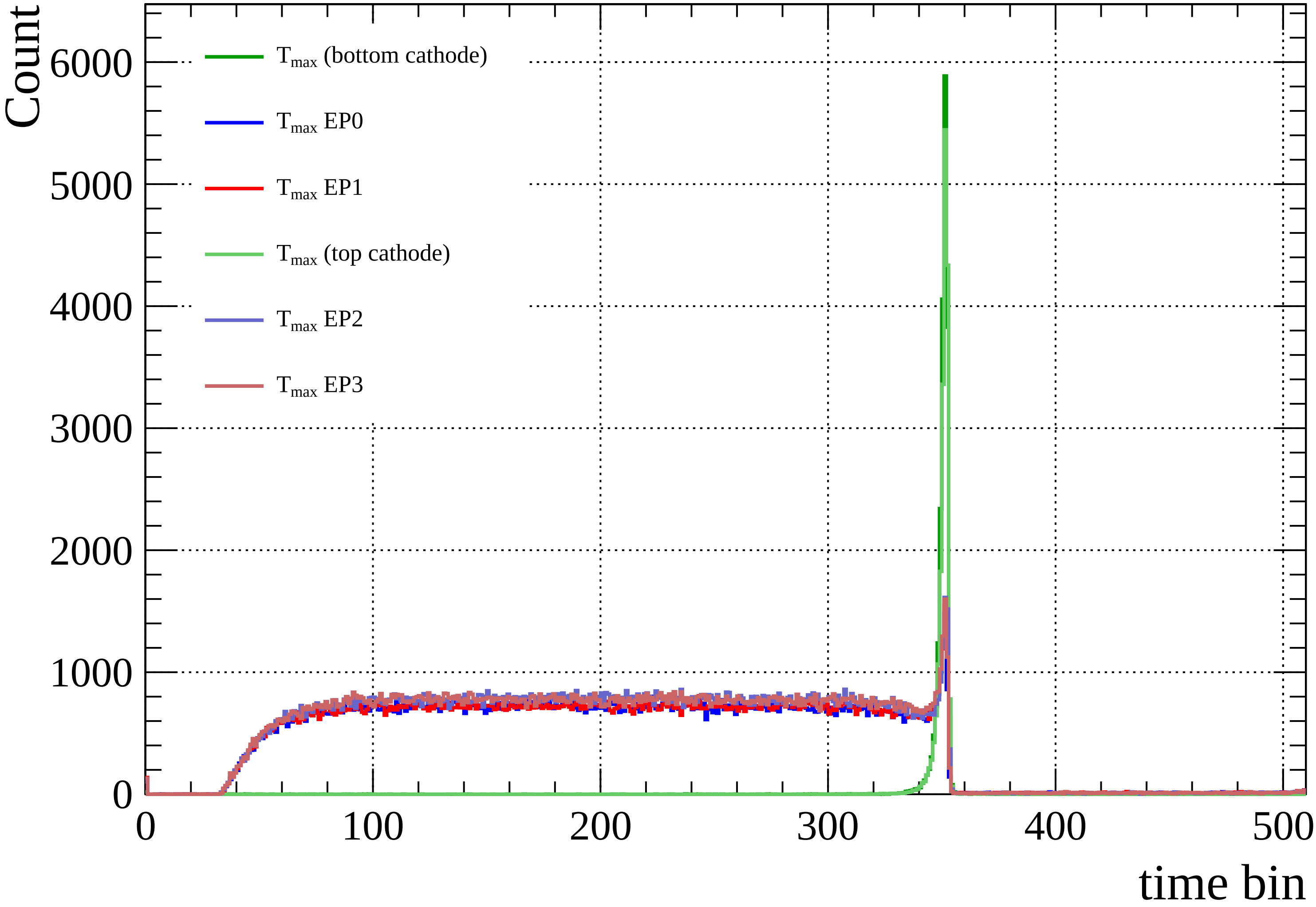


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
100
200
300
400
500
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
6000
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3